The Photonic Holonet

A Single Self-Entangled Photon as Universal Computer, Universal Network, and Clock on the $W(3,3)$ Substrate

Wil Dahn

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Abstract

We present a complete, machine-verified architecture in which one self-entangled photon, traversing one finite geometry, is simultaneously the hardware, the software, and the network of a universal quantum computer. The geometry is the symplectic generalized quadrangle $W(3,3)$: forty points, forty lines, automorphism group with faithful projective subgroup $\mathrm{PSp}(4,\mathbb{F}_3)$ of order 25920 and full Weyl extension $W(E_6) \cong \mathrm{PSp}(4,3){:}2$ of order 51840. The symplectic double cover $\mathrm{Sp}(4,3)$ also has order 51840, but its central involution is invisible on projective objects; the two extensions must not be conflated. The photon carries $W(3,3)$ twice over — once as the forty Witting rays of its path–polarization space $\mathbb{C}^4$ (states), once as the forty Pauli displacement classes of its past–future time-bin space $\mathbb{C}^9$ (operators) — and the two carriers realize the same incidence geometry, making the machine's states and gates one set of objects. Every architectural layer is a theorem with an executable witness: the preparation circuit (two Clifford gates of tabletop optics), the routing fabric (an atlas of 540 hypercube charts glued along the 1620 apartments of the Tits building), the protected memory (the Steinberg module of dimension 81), the dual-torsor compiler (three regular copies of both the flat $\mathbb{F}_3^3$ program shell and the noncommutative Heisenberg state shell, with canonical 27+27+27 triality selected by the Heisenberg center), the oriented control plane (top homology $H_2 = 5_\omega \oplus 5_{\omega^2} \oplus 30$, whose conjugate 5-sectors fuse to one Weyl-visible 10-channel), the mirror middleware (a 2160-slot D_{12} transport bus whose full Clifford lift factors as $24 \cdot 45 \cdot 48 = 51840$), the fuel (the matter shell is exactly the magic sector, with exact Kochen–Specker budget 36/40 and contextual fraction 1/10), the clock (an internal $\mathbb{Z}_{12}$ gauge clock, two external references $\mathbb{Z}_7, \mathbb{Z}_{13}$, and an irrational Boerdijk–Coxeter drive realizing a discrete time quasicrystal), and universality (the photon's own tritter, phase plate, and modulator generate the complete Clifford group; the matter shell supplies the magic). We also give the recursive engineering law: a level-n holonet has 40^n photonic leaves, $(40^n-1)/39$ $W(3,3)$ instances, and reversible routing diameter bounded by $8n = 8\log_{40} N$, while durable commits use a separate Császár/tomotope clock $T(n) = 4(7^n-1)$. The runtime contains an exact parabolic packet decoder: the Bell-line stabilizer splits the 1296 compass incidences into 648+324+162+162, giving the complete binary prefix code $\{0, 10, 110, 111\}$ for mirror, schedule, and two chiral caches, while the dual point stabilizer supplies two persistent 648-slot address sheets. The two cache branches are now resolved internally: each is the same homogeneous space H/C_4, each is a two-sheet cover of one canonical 81-route base H/D_8, and their 324-slot union has full equivariant commutant D_8 acting in 81 four-state fibers. This split address bus is provably distinct from the older non-split ternary $81 \to 162 \to 81$ transport memory. At the communication edge, the Witting delayed-query desk is now a frame compiler: 40 tetrads times 4 Alice slots times 4 Bob slots give a 640-entry admission ROM, with 480 off-diagonal data handshakes and 160 diagonal contextual witness apertures. The corresponding analyzer bank is sparse physical optics rather than a generic multiport: the 40 Witting bases split as $1 + 12 + 27$ analyzers with at most 12 nonzero matrix entries and only cube-root phase weights over $\sqrt{3}$. The newest storage layer turns the tomotope cover pathology into an engineering

feature: durable storage is cover-indexed by a $\mathbb{Z}_k^3$ fiber over the stable 48-block packet ABI, sentinel-aware compilation can trade commit-time slack for lower $g = 15$ fault energy, the first exact guard band is the arithmetically forced desynchronization at $n = 5$ with remainder $24 = f$, and the Grünbaum–Coxeter Wythoff operations of the tomotope are exactly the packet-growth ladder $4, 12, 24, 48, 96$.

Beyond the architecture, the same substrate is a candidate *physics*. The choice $q = 3$ is triply forced — by the geometric master equation $q! = 2q$, by the minimal magic (contextuality) dimension, and by the holographic central charge $c = 24 = f$ — and from it unfolds an exact exceptional skeleton: the Eisenstein tower $q = 3 \to \{3, 7\}$ Hurwitz surfaces $\to$ the Witting polytope $\to \mathrm{E}_6/\mathrm{E}_7/\mathrm{E}_8 \to$ the complex Leech $\to$ the Monster at $c = 24$, threaded by $\mathrm{G}_2 = \mathrm{Aut}(\mathbb{O})$ and welded by a single order-3 Eisenstein element. The Standard Model descends from E_6 by trinification ($\dim \mathrm{SU}(3)^3 = 24 = f$, dim SM $= 12 = k$), with the three generations, gauge unification ($\sin^2\theta_W = 3/8$), and tri-bimaximal PMNS mixing all one S_3 triality. The falsifiable layer is on a live test sheet: the cosmological predictions ($n_s = 29/30$, $r = 1/300$, $f_{NL} = 1/72$) are consistent, the neutrino sum is resolved by the seesaw ($\sum m_\nu \approx 0.06\,\mathrm{eV}$), and proton decay ($\tau_p \sim 10^{35\text{–}36}$ yr, Hyper-K) together with two demonstrator readouts (contextual fraction 1/10, pump Chern 2) are sharp near-term tests. No fitted parameters appear anywhere: every constant is substrate arithmetic at $q = 3$.

If you are not a specialist. Read the cream boxes and the diagrams. They are self-contained: each explains the idea before the symbols, and you can skip every blue box without losing the argument.

If you are a physicist or mathematician. The blue boxes carry the exact statements, with the scope each one actually establishes. Where a result is someone else's, it says so; where it is exhaustion rather than proof, it says that too.

If you are assessing this. The rose boxes are claims this project published and then withdrew, each with the measurement that overturned it. They are the reason the remaining numbers can be trusted. Section 63 states the architecture; the limitations are stated in the same voice as the results.

What this document is, in one paragraph. *This describes an architecture in which a single photon, prepared in a particular self-entangled state and routed through one finite geometry, acts at once as the computer, the network that links computers, and the clock that times them. The geometry is not chosen for convenience: the same forty-point object fixes the number of optical modes, the set of available gates, and the routing table. What is presented is a design and its supporting mathematics, machine-verified where verification is possible. No such device has been built, and the sections describing fabrication say so explicitly.*

Where this document's corrections live, and a caveat about their absence. This project's rule is that a withdrawn claim stays on the page beside the measurement that overturned it. Applied honestly, that rule produces a visible correction record — and *this manuscript has almost none*, while the machine blueprint has a full errata index and the research atlas carries thirty-six retractions.

Two readings are possible and a reader should be told which: either the architecture here has genuinely needed few corrections, or its claims have had less adversarial attention than the mathematics has. The second is more likely. Design documents attract fewer falsification attempts than theorems do, because a design is not wrong until it is built, and nothing here has been built.

Treat the implementation claims accordingly. The finite mathematics they rest on is checked; the engineering that would realise it is not, and the difference is not marked on every page.

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Canonical Architecture: Geometry, Prediction, and Evidence

Purpose. This front matter replaces chronological accumulation as the primary reading order. The historical theorem inserts remain below as an auditable evidence ledger, but the machine is now specified through four layers: the finite object, the predictive controller, the optical implementation, and the evidence boundary.

The finite object

The substrate is the symplectic polar space $W(3,3)$. Its 40 isotropic lines, 36 spreads, and 540 unordered skew-line pairs generate the intrinsic flag bundle

$$\mathcal{F} = \{(\{\ell_1, \ell_2\}, S, t)\}, \qquad |\mathcal{F}| = 540 \cdot 3 \cdot 4 = 6480.$$

The projective symplectic group acts transitively. A flag stabilizer is V_4, with fixed-point character $6480, 288, 288, 24$; consequently the permutation rank and commutant dimension are

$$\dim \operatorname{End}_{PSp(4,3)} \mathbb{C}[\mathcal{F}] = 1770.$$

The equality $6480 = 27(81 + 120 + 24 + 15)$ is retained only as a dimensional bridge, not as an unproved representation identification.

The predictive machine

The live controller retains semantic mixed-radix state and performs physical work only when future observables require it. Its route-diagnostic hierarchy is

$$23 \text{ common nonabelian triangles} \longrightarrow \text{at most two adaptive escalations},$$

with exact adaptive worst case 25. The best fixed construction currently known uses 28 triangles; the existence of a 27-triangle fixed schedule remains open.

All address transfers require zero, one, or two rank-one symplectic transvections. Permutations, Pauli byproducts, gauge conjugations, route inverses, and covariant terminal phases remain in frame state instead of becoming gratuitous optical gates.

The optical and quantum layer

The A_4 shell is a tetrahedral four-outcome information geometry separated by a hardware type boundary from the nonabelian D_4 route core. Calibration is component-resolved: survival, OAM sorting, slot sorting, and dark counts remain separate sufficient statistics. Chirality, route curvature, and calibration photons are scheduled by one posterior controller rather than three independent clocks.

For the reproduced three-copy M_{36} witness family, 19 projective stabilizer states yield 18 orthogonal pairs. Every pair has common same-sign Pauli rank three, whereas a rank-two six-qubit stabilizer code requires rank five. The accepted projections are magic-bearing rank-two subspaces, not stabilizer codes.

Reversible time and synchronization

The balanced word

$$001032122313$$

has optimal cyclic distance nine and is self-dual under a D_4 slot reflection. Crossing it with the 89/233 Christoffel calendar produces a 2796-tick comb with 1068 calibration events and exactly 89 events in every twelve-phase sector. Its reversal is the single affine involution

$$R(t) = 1952 - t \pmod{2796}.$$

The construction is a logical, typed scheduling symmetry; it is not an autonomous time crystal or an observed optical frequency standard.

Information and thermodynamics

The controller retains the smallest statistic sufficient for the next action and its residual risk. For the explicit finite model, a causal interaction tensor

$$K(a, b) = I(S; Y_a, Y_b) - I(S; Y_a) - I(S; Y_b)$$

separates raw information volume from future consequence. This is an information-geometric ledger. Physical dissipation still requires a temperature, reset protocol, finite-time implementation, and measured device.

Claim ledger

Exact	28-row full-D_4 construction; M36 common-Pauli audit; 6480-flag commutant; tetrahedral channel algebra; self-dual comb census.
Exact model	Finite-horizon joint posterior policy and causal entropic-curvature values for the published synthetic channels and priors.
Bounded	The 27-row search: 259 feasible adversarial MILP schedules and 12,336 accumulated collision cuts without a witness; no infeasibility theorem.
Pending	Full 213,648,435-subspace M_{36} census; full irreducible flag-character decomposition; RTL simulation and placement; laboratory optics; three canonical PDF builds; physical reset energy.

Certified Historical Theorem Ledger

Exact frame-to-signed-turn bridge and current evidence boundary

Let $M \in \mathbb{Z}^{540\times 240}$ be the canonical frame cross-matching matrix: each of the 540 pairs of disjoint isotropic lines contributes its unique stabilizer-invariant four-edge matching. Every edge of $W(3,3)$ occurs in exactly nine rows, and

$$\operatorname{rank}_{\mathbb{Q}} M = 225, \qquad \operatorname{coker} M \cong \mathbb{Z}^{15} \oplus (\mathbb{Z}/2)^{30}.$$

Write N for the unsigned point–edge incidence matrix, ∂ for the oriented incidence matrix, A for the point adjacency matrix, and K for the signed-turn operator on the 240 oriented edge chains. In one common edge ordering the exact identity is

$$K\partial^{\mathsf{T}} = \partial^{\mathsf{T}}(6I - A).$$

Consequently the (-4) point eigenspace is carried to the 10-eigenspace of K. More precisely, with

$$P = (A - 12I)(A - 2I) = 96E_{-4}, \qquad F = \frac{1}{16}\,\partial^{\mathsf{T}} PN,$$

the matrix F is integral and satisfies

$$FM^{\mathsf{T}} = 0, \qquad (K - 10I)F = 0, \qquad \operatorname{rank}_{\mathbb{Q}} F = 15.$$

Thus F induces an explicit equivariant isomorphism

$$\operatorname{coker}(M) \otimes \mathbb{Q} \xrightarrow{\sim} \ker(K - 10I).$$

The earlier dimension-only shortcut failed because it compared the unsigned edge permutation action with the orientation-signed edge action. The natural incidence operator above supplies the missing twist and closes the module question without identifying the two coordinate subspaces literally.

Modulo 2, the bridge has rank 14 and square zero. It yields an intrinsic nilpotent flag

$$\operatorname{im} F \subset \operatorname{im} C \subset \operatorname{im} M^{\mathsf{T}} \subset \ker F, \qquad \dim = (14, 15, 195, 226),$$

where $C = N^{\mathsf{T}}PN/16$. This separates the two abstractly isomorphic 14-dimensional factors: the bridge selects the nontrivial reduction of the rational 15, while the second copy remains in the 31-dimensional torsion-side kernel.

The exact-cover frontier is also larger than the original sample indicated. Certified covers occur with stabilizers

$$C_2, \quad C_4, \quad C_2 \times C_2, \quad D_8, \quad C_4 \times C_2,$$

and sixteen deterministic C_2 orbits together with four further stabilizer types certify at least 226800 distinct covers. In particular, a C_2-stabilized cover fixes twelve selected frames, so cover stabilizers are not universally diagonal.

Evidence boundary. Everything above is an exact theorem about finite matrices, lattices, and group actions. It does not by itself establish a physically complete one-photon computer, an experimental contextual fraction, a measured pump Chern number, Standard-Model descent, particle parameters, spacetime dynamics, or cosmology. Those statements require additional physical assumptions and, where relevant, an operational probability model and experimental protocol. They remain conditional research hypotheses unless separately demonstrated.

Full-Weyl extension and the integral bridge defect

Let M be the 540×240 canonical frame cross-matching matrix, N the unsigned point–edge incidence matrix, d an oriented point–edge incidence matrix, A the adjacency matrix of $W(3,3)$, and K the signed-turn operator on the 240 integral edge chains. The integral map

$$F = \frac{1}{16}\, d^T(A-12I)(A-2I)N$$

annihilates $\mathrm{im}(M^T)$, has rank 15, and lands in $\ker(K-10I)$. In addition to its $\mathrm{PSp}(4,3)$ equivariance, it intertwines an outer symplectic similitude: if U_τ and S_τ denote respectively the unsigned and orientation-signed edge actions of $\tau = \mathrm{diag}(1,1,2,2)$ over $\mathbb{F}_3$, then

$$S_\tau K = K S_\tau, \qquad S_\tau F = F U_\tau.$$

Consequently the rational frame cokernel and $\ker(K-10I)$ are isomorphic as modules for the full group $\mathrm{PGSp}(4,3) \cong W(E_6)$ of order 51840, not only for its $\mathrm{PSp}(4,3)$ subgroup. The common degree-15 character has values

$$15^1,\ 6^{80},\ 3^{1320},\ 2^{2160},\ 1^{7920},\ 0^{25248},\ -1^{13635},\ -2^{1440},\ -5^{36}$$

over the full Weyl group; the displayed outer involution has trace 3.

The rational isomorphism is *not* an integral lattice isomorphism. The nonzero Smith invariants of F are

$$1^{10}, \qquad 3^4, \qquad 6,$$

so $\mathrm{im}\, F$ has index

$$486 = 2\cdot 3^5$$

in its saturation and finite defect $\mathbb{Z}/2 \oplus (\mathbb{Z}/3)^5$. Thus only the primes 2 and 3 obstruct integral equivalence.

The exact-cover frontier also increases. Six further disjoint $\mathrm{PSp}(4,3)$-orbits of canonical 60-frame resolutions contribute 71280 new covers, raising the certified lower bound from 226800 to

$$\boxed{298080}.$$

The exact total and complete orbit census remain open.

Scope firewall. These are exact finite-module, integral-lattice, and finite-orbit statements. They do not by themselves constitute a physical propagator, establish scalable single-photon universality, or derive an experimental contextual fraction, Chern response, Standard-Model parameter, or cosmological observable.

The frame cross-matching and the (-4)-eigenspace

Call a *frame* an unordered pair $\{L, L'\}$ of disjoint totally isotropic lines of $W(3,3)$; there are 540 of them, and the stabiliser of a frame in $\mathrm{PSp}(4,3)$ is $C_2 \times S_4 \cong O_h$ of order 48.

Lemma 0.1 (Faithful tetrahedral action). *Let H be the stabiliser of a frame $\{L, L'\}$ and $A_4 = H'$ its derived subgroup. Then A_4 acts faithfully, as the natural degree-four alternating group, on the four points of L and on the four points of L', with orbits $[4,4]$ on the eight points of the frame.*

Proposition 0.2 (Canonical cross-matching)**.** *For every frame there is exactly one A_4-equivariant bijection $L \to L'$, and it is invariant under the full stabiliser H, not merely under A_4.*

Proof. A faithful action on both four-element sets embeds A_4 into $\mathrm{Sym}(L) \times \mathrm{Sym}(L')$ with both projections onto, so its image is the graph of an isomorphism and an equivariant bijection exists. Uniqueness is general: on a transitive A-set an equivariant bijection is determined by the image of one point, since $f(a \cdot x_0) = a \cdot f(x_0)$ for all a. Invariance under H then follows because H permutes the set of A_4-equivariant bijections, which is a singleton. Exhaustive verification over all 540 frames is recorded in the certificate. □

Write $\mu(\{L, L'\}) \subseteq E$ for the resulting set of four *cross-pairs*.

Theorem 0.3 (The matching lands on edges, 9 to 1)**.** *Every cross-pair is an edge of $W(3,3)$; the* 540 *matchings use exactly the* 240 *edges, each edge lying in exactly nine of them; and μ is injective.*

Proof. The 240 cross-pairs form a single Aut-orbit with point stabiliser of order $108 = C_3 \times S_3 \times S_3$, and $|\mathrm{PSp}(4,3)|/240 = 108$; uniformity of the multiplicity follows, and $540 \cdot 4/240 = 9$. Injectivity holds because the endpoints of the four cross-pairs are exactly $L \cup L'$ and each line is a maximal clique of the collinearity graph, so the eight points recover the frame. That every cross-pair is collinear is not forced by the construction — L and L' are disjoint — and is verified. □

Let $M\colon \mathbb{Z}^{540} \to \mathbb{Z}^{240}$ send a frame to the sum of its four cross-pairs.

Theorem 0.4 (Cokernel)**.** $\operatorname{rank} M = 225$*, the invariant factors are* $1^{195}, 2^{30}$*, and*

$$\operatorname{coker}(M) \;\cong\; \mathbb{Z}^{15} \oplus (\mathbb{Z}/2)^{30}.$$

Moreover $\operatorname{coker}(M) \otimes \mathbb{Q}$ *is* irreducible *of degree* 15*, and it is the same constituent that occurs in the* 40*-point permutation module. Equivalently,*

$$\operatorname{coker}(M) \otimes \mathbb{Q} \;\cong\; \ker(A + 4I),$$

the (-4)*-eigenspace of the adjacency operator.*

The last identification is the point. The 40-point module splits as $1 + 15 + 24$, matching the eigenvalue multiplicities of $12, -4, 2$; among those three blocks it is the (-4)-block alone that the frame geometry reproduces, and it does so through an explicitly named equivariant map rather than through an equality of dimensions.

Remark 0.5 (What is *not* claimed)**.** *Theorem 0.4 is rational. Integrally the situation differs:* 2 *is the only bad prime (*$\operatorname{rank}_{\mathbb{F}_2} M = 195$ *while* $\operatorname{rank}_{\mathbb{F}_3} M = \operatorname{rank}_{\mathbb{F}_5} M = 225$*), and the mod-*2 *quotient is* not *irreducible — its composition factors have dimensions* $1, 1, 1, 6, 8, 14, 14$*. So the degree-*15 *rational constituent reduces to* $1 + 14$ *modulo* 2*, and the torsion contributes* $1, 1, 6, 8, 14$*; the two degree-*14 *factors are isomorphic, so the torsion carries a second copy of the reduction rather than an unrelated module. This is consistent with the coalescence theorem:* $\operatorname{spec}(A) = \{12, 2, -4\}$ *collapses completely modulo* 2*, while modulo* 3 *and* 5 *it does not.*

The corresponding statement for the signed-turn operator is a theorem*, established in Passes 1416–1420 and not here. With N the unsigned point–edge incidence and ∂ the oriented one,*

$$F = \tfrac{1}{16}\, \partial^{\mathsf{T}} (A - 12I)(A - 2I) N$$

is integral and satisfies $FM^{\mathsf{T}} = 0$*,* $(K - 10I)F = 0$ *and* $\operatorname{rank}_{\mathbb{Q}} F = 15$*, so it induces an equivariant isomorphism* $\operatorname{coker}(M) \otimes \mathbb{Q} \cong \ker(K - 10I)$*. A dimension-only argument does* not *suffice, and the* *reason is worth recording: the* 240*-edge* permutation *module contains two non-isomorphic degree-*15 *constituents, but* K *acts on the orientation-*signed *edge action, so* $\ker(K-10I)$ *is not a submodule of that permutation module at all. The intertwiner supplies the missing twist. Modulo* 2 *the same map has rank* 14 *and square zero, and it separates the two abstractly isomorphic degree-*14 *factors: it selects the nontrivial reduction of the rational* 15*, while the second copy remains in the torsion-side kernel.*

The Hodge decomposition of the edge module, and what it does not contain

The clique complex of $W(3,3)$ is three-dimensional: every totally isotropic line is a 4-clique, hence a tetrahedron, giving $(C_0,C_1,C_2,C_3)=(40,240,160,40)$ with $\chi=-80$ and Betti numbers $(b_0,b_1,b_2,b_3)=(1,81,0,0)$ — the complex is homotopy equivalent to a bouquet of 81 circles.

Theorem 0.6 (Hodge blocks of the signed edge module)**.** *Under the orientation-signed action of* Aut *on the* 240 *edge* 1*-chains, the Hodge decomposition is multiplicity-free and splits into six irreducibles:*

$$\underbrace{15\oplus 24}_{\textit{exact, }39}\ \oplus\ \underbrace{81}_{\textit{harmonic}}\ \oplus\ \underbrace{30\oplus 45\oplus 45}_{\textit{coexact, }120}\ =\ 240.$$

The harmonic block is irreducible *of degree* 81 *and is the Steinberg module. The exact block* $15\oplus 24$ *is precisely the* 40*-point permutation module modulo its trivial constituent, i.e. the* (-4)*- and* $(+2)$*-eigenspaces of* A*.*

In gauge-theoretic language this is the pure-gauge / physical / constraint split, with the Gauss-law count $39=40-1$ realised representation-theoretically and a physical sector that is a single irreducible.

Proposition 0.7 (The blocks behave oppositely under the outer automorphism)**.** $\mathrm{PSp}(4,3)$ *has index two in* $\mathrm{PGSp}(4,3)\cong W(E_6)$*. Under that extension:*

$$15,\ 24,\ 81\ \longmapsto\ \textit{two extensions each (a sign character)},\qquad 45\oplus 45\ \longmapsto\ \textit{a single irreducible of degree } 90.$$

Concretely, $\mathrm{PGSp}(4,3)$ *has four degree-*15*, two degree-*24 *and two degree-*81 *irreducibles, and* no *degree-*45 *irreducible but one of degree* 90*.*

So the gauge and physical blocks are *split* by the outer element — each constituent survives and acquires a sign — while the two halves of the constraint block are *exchanged* by it. The chirality of the constraint sector is of a different kind from that of the physical sector, not merely of a different magnitude. Over $\mathrm{PGSp}(4,3)$ the module reads

$$\underbrace{15\oplus 24}_{39}\ \oplus\ \underbrace{81}_{81}\ \oplus\ \underbrace{30\oplus 90}_{120},$$

five irreducibles rather than six, the fusion $45\oplus 45\to 90$ being the only change; the harmonic block carries one definite extension of the Steinberg.

Proposition 0.8 (The sign is read by frames and spreads)**.** *The two degree-*81 *extensions agree on* $\mathrm{PSp}(4,3)$ *and differ on exactly six outer classes. Two of those are involution classes, and both are geometric:*

class	*size*	*object*	*separation*
involutions	540	*frames*	$\chi=\mp 3$
involutions	36	*spreads*	$\chi=\mp 9$ (*maximal*)

*The size-*36 *class has centraliser of order* $1440=|\mathrm{PGSp}(4,3)|/36$*, fixes no point and exactly ten lines, and a spread of* $W(3,3)$ *is precisely ten pairwise disjoint lines covering all forty points.*

The physical sector's chirality is therefore detected by two of the substrate's own combinatorial families, most strongly by the spreads — not by an auxiliary construction.

Remark 0.9 (No Hodge star, and where one must come from). *A Hodge star requires $C_k \cong C_{3-k}$. Here $C_0 = C_3 = 40$ but $C_1 = 240 \neq 160 = C_2$, the discrepancy being exactly $|\chi|$; and on the physical sector $\star$ would map $H^1 \to H^2$ with $b_2 = 0$. So the 81 physical modes admit no dual, there is no $F \wedge \star F$, and no variational dynamics can be written on this complex. The finite geometry supplies the* kinematics *and withholds the* dynamics.

In an almost-commutative geometry $M^4 \times F$ the volume form, the star and the variational principle belong to the continuum factor; the finite factor contributes an algebra and a module. That division is now explicit: the finite side supplies $\langle I, A, J\rangle$ and the Steinberg module of Theorem 0.6, and the star must be supplied by M^4. The absence is established by computation ($b_2 = 0$), not by omission, and cannot be repaired internally.

Proposition 0.10 (Resolutions of the edge set). *There exist sets of 60 frames whose cross-matchings partition E. No cover is* Aut*-invariant, since* Aut *is transitive on the 540 frames and a cover uses 60. Cover stabilisers have order 2, 4 or 8 (types C_2, C_4, $C_2 \times C_2$, $C_4 \times C_2$, D_8), so a cover is close to rigid — but the residual symmetry is* not *uniformly diagonal: a C_2-stabilised cover fixes twelve of its sixty frames (orbit profile $1^{12}2^{24}$), whereas the C_4 covers fix none. A compiled census (Pass 1505) exhibits 327 pairwise disjoint complete* $\mathrm{PSp}(4,3)$*-orbits, certifying at least*

$$228 \cdot 12960 + 75 \cdot 6480 + 9 \cdot 6480 + 9 \cdot 3240 + 6 \cdot 3240 = 3\,547\,800$$

covers; this is a lower bound and no exhaustive count is claimed. Note that the orbit census is heavily C_2-weighted — about 83% of covers lie in C_2 orbits — so a small sample drawn by a depth-first search, however its branch order is permuted, is not *a uniform sample and its observed proportions estimate nothing.*

Exact-cover census, prime-local bridge arithmetic, and an M_{20} qutrit chart

A compiled exact-cover search enumerated the first 100000 covers containing a fixed canonical frame and reduced them under the full $\mathrm{PSp}(4,3)$ action. The prefix meets 327 complete, pairwise distinct cover orbits whose sizes sum to

$$3547800.$$

The stabilizer-type counts in this certified frontier are

$$C_2 : 228, \qquad C_4 : 75, \qquad C_2^2 : 9, \qquad D_8 : 9, \qquad C_4 \times C_2 : 6.$$

This number is a lower bound: the deterministic search prefix is exact, but the complete cover census remains open.

Retain the integral matrices

$$C = \frac{1}{16} N^{\mathsf{T}} P N, \qquad F = \frac{1}{16} \partial^{\mathsf{T}} P N, \qquad Q = \frac{1}{8}(K + 6I)(K - 2I)(K - 4I).$$

Their nonzero Smith invariants are

$$\mathrm{SNF}(C) = 1^{10}3^5, \qquad \mathrm{SNF}(F) = 1^{10}3^4 6, \qquad \mathrm{SNF}(Q) = 1^{10}3^4 12.$$

Moreover

$$C^2 = 48C, \quad Q^2 = 96Q, \quad F^{\mathsf{T}} F = 96C, \quad F F^{\mathsf{T}} = 48Q.$$

Thus, over $\mathbb{Q}$, the elements

$$e = C/48, \quad q = Q/96, \quad x = F/48, \quad y = F^{\mathsf{T}}/96$$

satisfy $xy = q$ and $yx = e$. The unsigned and signed 15-dimensional corners are therefore equivalent idempotents, giving an explicit Morita context. The only bad primes are 2 and 3: modulo 2 the orientation-twisted channel drops from rank 15 to 14, while modulo 3 all three lattices have rank 10.

The algebra generated by

$$M^{\mathsf{T}}M,\ K,\ N^{\mathsf{T}}N,\ \partial^{\mathsf{T}}\partial,\ C,\ F,\ F^{\mathsf{T}},\ Q$$

has dimension at least 301 over $\mathbb{Q}$, certified by 301 words that remain independent modulo 1009. This is not yet a complete Wedderburn decomposition.

Finally, a deterministic exact cover contains five pairwise edge-disjoint 16-edge packets, each having exactly three four-frame resolutions. Independent resolution choices give a literal Hamming chart

$$H(5,3) \cong \{0,1,2\}^5$$

of 243 exact covers, with shells

$$1,\ 10,\ 40,\ 80,\ 80,\ 32.$$

Its setwise stabilizer inside $\mathrm{PSp}(4,3)$ has elementary abelian kernel 2^4 on the five coordinates, image A_5, and an explicit A_5 complement. Hence

$$\mathrm{Stab}(H(5,3)) \cong 2^4{:}A_5 = M_{20},$$

the Mathieu group of order 960.

Operational contextuality boundary. The ratio 4/40 is a finite incidence deficit, not by itself a contextual fraction. The latter is defined from a complete empirical probability model and a linear program. The accompanying falsifier therefore requires raw context–outcome data, normalization, no-disturbance checks, source and detector models, loss and postselection rules, and the implemented measurement phases. No contextual fraction is promoted until those inputs exist.

Bidirectional exact-cover saturation

A second deterministic exact-cover search was performed by reversing the candidate-frame order at every depth of the Pass-1505 search. The forward and reverse prefixes each contain 100000 covers through the fixed canonical frame, and they are disjoint as raw solution sets. Exact orbit traversal nevertheless shows that each prefix meets the identical 327 complete $\mathrm{PSp}(4,3)$-orbits. Their common certified cover lower bound remains

$$3547800,$$

with stabilizer-order census

$$2^{228}, \qquad 4^{84}, \qquad 8^{15}.$$

Canonical orbit representatives agree objectwise, not merely numerically.

The two DFS orders are not uniform samplers. If f_i and r_i denote their hits in the ith orbit, only seven of 327 hit counts agree, while

$$\sum_i |f_i - r_i| = 10244, \qquad \sum_i (f_i - r_i)^2 = 463520.$$

Their Pearson correlation is 0.9280509607335599. Thus deterministic DFS is valid for exhibited-orbit lower bounds but not for frequency inference. Agreement of the two prefixes is a saturation certificate for the observed frontier, not a proof that the global exact-cover orbit census is complete.

Disjoint exact covers and a residual integrality gap

The deterministic cover prefixes used above fix one frame before recursion. Consequently every sampled cover contains that frame, so pairwise intersection inside those prefixes is forced by the symmetry break and cannot establish that the global exact-cover family is intersecting. Direct exact search gives two 60-frame covers with empty intersection, each covering all 240 edge columns exactly once.

Fix one canonical cover. Acting on the frozen 327 cover-orbit representatives by $\mathrm{PSp}(4,3)$ produces 13648 distinct covers disjoint from it, and every frozen orbit type contributes. Their disjointness graph has

$$|V| = 13648, \qquad |E| = 188338, \qquad \#K_3 = 494, \qquad \#K_4 = 0.$$

Thus its clique number is three, and adjoining the fixed cover yields an explicit four-cover packing inside the certified frontier.

The involution classes in $\mathrm{PSp}(4,3)$ have sizes 45 and 270. All 228 frozen cover orbits with stabilizer C_2 use the class-45 involution; it fixes 84 of the 540 frames globally and exactly 12 frames of the stabilized cover.

Removing the selected four covers leaves a 300×240 incidence system that is 4-uniform by rows and 5-regular by columns. Hence the constant weight $x_r = 1/5$ is a fractional exact cover of total weight 60. Nevertheless, exhaustive Algorithm X proves that the residual system has no integral exact cover. The selected four-packing therefore admits a fractional fifth layer but no integral fifth layer.

Boundary. The no-fifth-layer result applies to the displayed four-packing. The clique calculation is exhaustive inside the frozen 327-orbit frontier. Neither claim settles the global packing number over as-yet undiscovered cover orbits or over all possible four-packings.

The frame-resolution problem is a Hoffman-colouring problem

Let $M \in \{0,1\}^{540\times 240}$ be the canonical frame–edge incidence matrix: each row is the four-edge cross-matching carried by a pair of disjoint isotropic lines, and each W33 edge occurs in nine rows. Let H be the graph on the 540 frames, adjacent when their matchings share an edge. Direct construction from $W(3,3)$ gives

$$H + 4I_{540} = MM^{\mathsf{T}}, \qquad \deg(H) = 32,$$

and the exact spectrum

$$\mathrm{spec}(H) = 32^1 \oplus 14^{44} \oplus 8^{15} \oplus 4^{81} \oplus 2^{84} \oplus (-4)^{315}.$$

Hence Hoffman's bounds give

$$\chi(H) \geq 9, \qquad \alpha(H) \leq 60.$$

A 60-vertex independent set is exactly an exact cover of the 240 W33 edges by 60 frame matchings, and every exact cover attains equality. Thus

$$\boxed{\text{nine-cover resolution} \iff \text{Hoffman 9-colouring of } H.}$$

Every exact-cover indicator x satisfies

$$Hx = 4(\mathbf{1} - x),$$

so each outside frame meets the cover in exactly four neighbours. If $X = [x_1|\cdots|x_9]$ is a resolution, then its forced quotient matrix is

$$HX = X\,4(J_9 - I_9).$$

The centred vectors $y_i = x_i - \mathbf{1}/9$ lie in $\ker(M^{\mathsf{T}}) = E_{-4}(H)$ and obey

$$\langle y_i, y_i \rangle = \frac{160}{3}, \qquad \langle y_i, y_j \rangle = -\frac{20}{3} \quad (i \neq j),$$

so a resolution is a regular 8-simplex of binary affine-fibre points in the 315-dimensional (-4)-eigenspace.

Boundary. The theorem does not assert that $\chi(H) = 9$. It converts the still-open global resolution question into the equality case of the Hoffman bound, with a forced equitable quotient and simplex certificate.

Frame resolution, obstruction modules, and the harmonic bridge

Exact decision instance. The nine-cover resolution is exactly a 9-colouring of the 540-vertex frame graph. After fixing the colours on one edge K_9, the deterministic DIMACS instance has 4860 variables and 99909 clauses. No satisfying assignment or infeasibility certificate is presently known, so $\chi(H) = 9$ remains open.

The obstruction module. The independently rebuilt orbital commutants give the stable $\mathrm{PSp}(4,3)$ profile

$$E_{-4}(H) \sim 64 \oplus 81 \oplus 20 \oplus 60^{\oplus 2} \oplus 15^{\oplus 2}_{\text{other}}.$$

For the full $\mathrm{PGSp}(4,3)$ action, the two 15-blocks separate into the two outer extensions while the 60-block remains multiplicity two. This is recorded as a commutant fingerprint pending a literal character-table inner-product certificate.

The four-packing fiber. For the certified pairwise-disjoint covers $x_1, \ldots, x_4$, put $y_i = x_i - \frac{1}{9}\mathbf{1}$. Then

$$\langle y_i, y_i \rangle = \frac{160}{3}, \qquad \langle y_i, y_j \rangle = -\frac{20}{3} \quad (i \neq j).$$

Any fifth layer has forced projection

$$y_{\text{frac}} = -\frac{1}{5} \sum_{i=1}^{4} y_i, \qquad \|y_{\text{frac}}\|^2 = \frac{16}{3},$$

which is exactly the residual uniform weight $1/5$. Its integral correction would have to lie on an orthogonal shell of squared norm 48; the frozen exact-cover trace proves that shell empty for this four-packing.

Integral harmonic bridge. Let

$$P_4 = (H - 32I)(H - 14I)(H - 8I)(H - 2I)(H + 4I).$$

An integral Reynolds map $T : \mathbb{Z}^{540} \to \mathbb{Z}^{240}$ gives $B = TP_4$ with

$$\operatorname{rank} B = 81, \qquad d_1 B = 0, \qquad d_2^{\mathsf{T}} B = 0, \qquad BH = 4B,$$

and B intertwines the four generators of $\mathrm{PSp}(4,3)$. Hence

$$\boxed{E_{+4}(H) \cong \ker d_1 \cap \ker d_2^{\mathsf{T}}},$$

identifying the frame $+4$ eigenspace with the harmonic Steinberg sector by a literal integer map. This is a finite-module statement and does not supply a Hodge star or continuum dynamics.

Pass 1536: the frame dual is the intrinsic $K_{4,4}$ code

Let $M \in \{0,1\}^{540\times 240}$ be the canonical incidence matrix whose rows are the four-edge matchings attached to disjoint W33 line pairs. Over $\mathbb{F}_2$, the exact frame code and dual code are

$$\mathrm{row}(M) = [240, 195, 4]_2, \qquad \ker(M) = [240, 45, 16]_2.$$

The dual has a canonical geometric parity-check basis. W33 contains exactly 45 induced $K_{4,4}$ octets. If $K \in \{0,1\}^{45\times 240}$ records their edge supports, then

$$MK^{\mathsf{T}} = 0 \pmod 2, \qquad \operatorname{rank}_2 K = 45, \qquad \mathrm{row}_{\mathbb{F}_2}(K) = \ker_{\mathbb{F}_2}(M).$$

An exhaustive exact parity optimization proves that the minimum dual weight is 16 and that the 45 rows of K are exactly all minimum words. Thus the minimum words themselves form a basis of the complete modular frame cokernel. The parity-check realization is $(3,16)$-regular with Tanner girth six. Its minimum-word overlap and disjointness graphs are respectively $\mathrm{SRG}(45,32,22,24)$ and $\mathrm{SRG}(45,12,3,3)$, while

$$\mathrm{spec}(KK^{\mathsf{T}}) = 48^1 \oplus 12^{20} \oplus 18^{24}, \qquad \mathrm{rank}_2(KK^{\mathsf{T}}) = 14, \qquad \mathrm{rank}_3(KK^{\mathsf{T}}) = 15.$$

BT766 owns the intrinsic octet census and overlap geometry; Pass 1416 owns the binary frame rank. The new result is their exact code-theoretic identification, including the exhaustive minimum-word theorem. This is a finite binary-code statement, not a decoding-threshold or physical-noise claim.

0.1 Integral frame cokernel and the $15 \to 45 \to 30$ Bockstein chain

Let $M \in \{0,1\}^{540\times 240}$ be the canonical frame–edge incidence matrix. Exact 2-adic elimination, together with a literal 225×225 minor of determinant 2^{30}, gives

$$\mathrm{SNF}(M^{\mathsf{T}}) = 1^{195} \oplus 2^{30} \oplus 0^{15}.$$

Hence

$$\mathrm{coker}(M^{\mathsf{T}}) \cong \mathbb{Z}^{15} \oplus (\mathbb{Z}/2\mathbb{Z})^{30}.$$

The forty-five intrinsic induced $K_{4,4}$ octets form a basis of $\ker_{\mathbb{F}_2} M$. If K is their 45×240 edge-incidence matrix, then the integral half-incidence matrix

$$J = \frac{1}{2} MK^{\mathsf{T}} \in \{0,1\}^{540\times 45}$$

is well defined. It has row degree 6, column degree 72, and

$$J^{\mathsf{T}}J = 66I + 3A_{45} + 6\mathbf{J},$$

where A_{45} is the octet-overlap graph SRG$(45, 32, 22, 24)$. Thus

$$\operatorname{spec}(J^{\mathsf{T}}J) = 432^{1} \oplus 72^{24} \oplus 54^{20}.$$

The associated Bockstein sequence is

$$0 \longrightarrow \ker_{\mathbb{Z}} M/2\ker_{\mathbb{Z}} M \longrightarrow \ker_{\mathbb{F}_2} M \xrightarrow{\beta} \operatorname{Tor}_2(\operatorname{coker} M) \longrightarrow 0, \qquad \beta(y) = \frac{M\widetilde{y}}{2} \bmod \operatorname{im} M.$$

Its dimensions are exactly

$$15 \longrightarrow 45 \longrightarrow 30.$$

For the integral rational-cokernel projector C and the oriented bridge F from the signed-turn construction, the nonzero Smith forms refine to

$$\operatorname{SNF}(C) = 1^{10} \oplus 3^{5}, \qquad \operatorname{SNF}(F) = 1^{10} \oplus 3^{4} \oplus 6.$$

Consequently the old 31-dimensional ambient mod-2 bridge kernel splits arithmetically as $30 + 1$: thirty dimensions are genuine elementary cokernel torsion, while the final dimension is the parity defect of the single even invariant factor in the embedded bridge lattice.

This is an exact finite lattice theorem. The torsion modes may be used as certified XOR preprocessing, but they do not by themselves decide the open global nine-cover resolution.

Exact frame-dual continuation: modular lifts, parity cuts, and decoding

Let $M \in \{0,1\}^{540\times 240}$ be the canonical frame–edge matrix and $K \in \{0,1\}^{45\times 240}$ the intrinsic $K_{4,4}$ octet matrix. The exact continuation of the frame-dual theorem gives

$$\operatorname{SNF}(K) = \operatorname{diag}(1^{44}, 3), \qquad \operatorname{sat}(\operatorname{row}_{\mathbb{Z}} K)/\operatorname{row}_{\mathbb{Z}} K \cong \mathbb{Z}/3.$$

The saturation is obtained by adjoining the all-ones edge vector, since the sum of the forty-five octet rows is three times that vector.

Writing $G_K = KK^{\mathsf{T}}$, one has

$$\operatorname{rank}_2 G_K = 14, \qquad \operatorname{rank}_3 G_K = 15, \qquad G_K^2 = 0 \pmod 2, \pmod 3.$$

The characteristic-two image is an absolutely irreducible 14 and maps literally onto the Pass-1416 signed-turn image. In characteristic three the rank-15 image is not irreducible: it splits canonically as $\mathbf{1} \oplus V_{14}$.

The signed bipartition incidence of the octets yields an integral map $V = 4U + Sd$ satisfying

$$dV^{\mathsf{T}} = 0, \qquad L_1 V^{\mathsf{T}} = 4V^{\mathsf{T}}, \qquad \operatorname{rank}_{\mathbb{Q}} V = 30.$$

Thus the natural oriented $K_{4,4}$ lift realizes the coexact irreducible 30, not either of the two degree-45 constituents fused into the full-Weyl 90.

For the unresolved nine-cover problem, every octet and every color obey the exact cardinality law

$$\sum_{f:\,|f\cap o|=2} x_{fc} = 8.$$

The 405 equations are rational combinations of the edge equations, but modulo two they add 240 independent global XOR directions: the parity rank rises from 2100 to 2340 (and from 2109 to 2349 after the standard K_9 symmetry fixing). This strengthens propagation without deciding existence.

For a subset X of octets, codeword weight is

$$w(X) = 16|X| - 2e(X) + 4t(X),$$

where $e(X)$ counts induced edges in the 45-point overlap graph and $t(X)$ counts distinguished edge-signature triples. Ordinary and distinguished triangles have the same $|X| = 3$ and $e(X) = 3$ but weights 42 and 46, so the ordinary strongly regular association algebra alone cannot determine the full weight enumerator.

Finally, the parity-check matrix K gives exact finite decoding data: every single-edge error has a unique syndrome; among all $\binom{240}{2}$ double errors, 25440 are uniquely identifiable and 3240 are ambiguity-declared. The conditional unique fraction is 212/239. The 540 first weight-four dependencies are exactly the canonical frame matchings. Seeded min-sum trials are retained only as a falsifier of naive threshold claims.

Boundary. These are finite module, lattice, parity, partial-enumerator, and decoder results. The global Hoffman 9-coloring and complete 2^{45} weight census remain open; no physical or quantum-code threshold is asserted.

0.2 Integral-frame five-continuation theorem

Let M be the 540×240 frame–edge incidence matrix and let K be the 45×240 incidence matrix of the intrinsic induced $K_{4,4}$ octets. The 30-dimensional Bockstein torsion module has a nonsplit composition series with factors

$$1,\ 6,\ 8,\ 1,\ 14,$$

where the head 14 and the factor 6 are absolutely irreducible over $\mathbb{F}_2$, while the irreducible factor 8 has endomorphism field $\mathbb{F}_4$.

The octet exact-cardinality relations yield 240 independent global XOR strengthenings of the nine-colour frame-resolution system:

$$\mathrm{rank}_2 : 2100 \longrightarrow 2340.$$

The certified four-cover packing and its anti-symplectic mirror form two free $PSp(4,3)$ orbits of size 25920, fused into one $PGSp(4,3)$ orbit of size 51840; their binary Bockstein signatures coincide.

For the free rank-15 lattice, the saturated unsigned-to-oriented transition has

$$\mathrm{SNF} = 1^{14} \oplus 2, \qquad |\det| = 2.$$

Finally, the joint action on 540 frames and 45 octets is a two-fibre coherent configuration of rank

$$32 + 3 + 5 + 5 = 45.$$

For the half-incidence cross orbital J,

$$(H - 32I)(H - 14I)(H - 2I)J = 0, \qquad \mathrm{rank}[J, HJ] = 69.$$

These are exact finite-module and lattice statements. They do not decide the existence of a global nine-cover resolution.

Exact Brauer, XOR, orbit-transfer, decoder, and outer-extension continuation

Let $M \in \{0,1\}^{540\times 240}$ be the canonical frame–edge incidence matrix and let $K \in \{0,1\}^{45\times 240}$ be the incidence matrix of the intrinsic induced $K_{4,4}$ octets. The following statements are rebuilt from projective coordinates by the Passes 1801–1805 verifier.

Brauer refinement of the Bockstein module. The Bockstein quotient has dimension 30 over $\mathbb{F}_2$, and

$$\ker\beta = \langle \mathbf{1}_{45}\rangle \oplus \operatorname{im}(KK^{\mathsf{T}} \bmod 2), \qquad \dim\ker\beta = 15.$$

A compatible invariant filtration has dimensions

$$1 < 9 < 10 < 16 < 30$$

and successive factors

$$1, \quad 8_{\mathbb{F}_2}, \quad 1, \quad 6, \quad 14.$$

The generated algebras of the 6- and 14-factors are respectively $M_6(\mathbb{F}_2)$ and $M_{14}(\mathbb{F}_2)$. The 8-factor has endomorphism field $\mathbb{F}_4$: a non-scalar endomorphism z satisfies $z^2+z+I=0$. Over an algebraic closure it is the unordered pair of four-dimensional Brauer factors $\{4a, 4b\}$, and the canonical outer similitude acts by

$$szs^{-1} = z^2,$$

therefore exchanging $4a$ and $4b$. The factor labels use the published characteristic-two decomposition matrix of $U_4(2)$; no uncomputed Brauer character values are asserted.

XOR-native resolution attack. For every octet o and color c, a resolution must satisfy

$$\sum_{f:J_{fo}=1} x_{fc} = 8, \qquad J = \frac{1}{2}MK^{\mathsf{T}}.$$

The exact binary ranks are

$$2100 \longrightarrow 2340, \qquad 2109 \longrightarrow 2349$$

after adding the octet equations, before and after the standard symmetry fixing respectively. Thus the cuts add exactly 240 global XOR directions. A separate reproducible HiGHS model contains 4860 binary variables, 3105 equality constraints, and all 405 exact-eight equations. Its frozen 20-second run ended without an incumbent; this is a bounded computational falsifier and is not a SAT or UNSAT certificate.

Three-body orbit-transfer algebra. For an octet subset X, codeword weight obeys

$$w(X) = 16|X| - 2e(X) + 4t(X),$$

where $t(X)$ counts distinguished W33 edge-signature triples. There are exactly six $PSp(4,3)$-orbits on three-subsets and twenty on four-subsets. If U and D are the exact upward and downward orbit-incidence matrices and n_3, n_4 are the orbit-size vectors, then

$$\operatorname{diag}(n_3)U = \big(\operatorname{diag}(n_4)D\big)^{\mathsf{T}}.$$

This is the first transfer layer retaining the essential three-body invariant that is invisible to the ordinary rank-three octet association scheme.

Exact ambiguity-declaring decoding through weight three. Using K as a parity-check matrix for the $[240,195,4]_2$ frame code, the numbers of errors having a unique minimum-weight syndrome representative are

$$N_0 = 1, \qquad N_1 = 240, \qquad N_2 = 25440, \qquad N_3 = 1576000.$$

Hence the exact BSC success contribution through weight three is

$$(1-p)^{240} + 240p(1-p)^{239} + 25440p^2(1-p)^{238} + 1576000p^3(1-p)^{237}.$$

Among the $\binom{240}{3} = 2275280$ weight-three errors, 697120 are ambiguous at minimum weight and 2160 are shadowed by a weight-one syndrome. The weight-three syndrome buckets form exactly 110 inner-group orbits. This does not claim unrestricted maximum-likelihood decoding or a threshold.

Canonical outer extension of the coexact degree-30 carrier. With signed point incidence S, oriented octet incidence U, and edge boundary d, define

$$V = 4U + Sd.$$

Then

$$dV^\mathsf{T} = 0, \qquad L_1V^\mathsf{T} = 4V^\mathsf{T}, \qquad \operatorname{rank}_\mathbb{Q} V = 30.$$

For the canonical multiplier-minus-one similitude $s = \operatorname{diag}(1,-1,1,-1)$, the induced operator on this coexact carrier satisfies

$$s^2 = I, \qquad \operatorname{tr}(s) = 2, \qquad \dim E_+(s) = 16, \qquad \dim E_-(s) = 14, \qquad \det(s) = 1.$$

The sign-twisted extension has trace -2. No ATLAS class label is attached without a separate standard-generator fusion certificate.

Boundary. The module, rank, orbit-transfer, syndrome, and outer-extension statements are exact finite certificates. The global Hoffman nine-cover resolution, the complete 2^{45} weight enumerator, and physical decoding thresholds remain open.

1 Reconciliation addenda for the frame–octet torsion frontier

The primary Loewy, minimal-XOR, chiral-packing, determinant-two, and coherent- configuration results are recorded in Passes 1701–1705. The present addendum freezes five complementary exact facts.

First, the 30-dimensional binary Bockstein module admits an alternative composition series with factors

$$1, 8, 1, 6, 14,$$

whose multiset agrees with the primary Loewy profile $1 \mid (6 \oplus 8) \mid 1 \mid 14$. Orbit-span tests prove each displayed factor irreducible; the 8-factor has endomorphism field $\mathbb{F}_4$, while the 6- and 14-factors have scalar endomorphism field $\mathbb{F}_2$.

Second, a color-symmetric native-XOR generator contains $270 = 30 \cdot 9$ equations, but only 240 new global directions. The exact thirty-dimensional redundancy is the sum over colors forced by the one-color-per-frame equations; removing one color gives the minimal $30 \cdot 8 = 240$ basis.

Third, every exact-cover indicator x satisfies

$$M^{\mathsf{T}}x = \mathbf{1} \quad \Longrightarrow \quad J^{\mathsf{T}}x = 8\mathbf{1}_{45}.$$

Thus every four-packing has signature $32\mathbf{1}$ and every complementary 300-frame residual has signature $40\mathbf{1}$. Linear Bockstein signatures cannot distinguish cover orbits or extendibility.

Fourth, the unique determinant-two orientation defect has an explicit primitive edge representative of support 128 and squared norm 152.

Finally, the five-valued frame Gram partition

$$6^1,\ 3^{32},\ 2^{15},\ 1^{300},\ 0^{192}$$

is not closed under multiplication: the square of the intersection-one relation takes eight values on that relation. The full rank-32 frame orbital algebra, not the coarse Gram partition, is required.

1.1 Complete cover census and nonlinear octet-signature obstruction

Let $\mathcal{C}$ denote the exact covers of the canonical 540×240 frame–edge incidence matrix. A complete Algorithm-X enumeration through one fixed frame gives

$$|\mathcal{C}_{f_0}| = 394200.$$

Since $PSp(4,3)$ is transitive on the 540 frames and every cover has 60 frames,

$$60|\mathcal{C}| = 540|\mathcal{C}_{f_0}|, \qquad \boxed{|\mathcal{C}| = 3547800}.$$

The complete family consists of exactly 327 projective-group orbits, with stabilizer-order census $2^{228}, 4^{84}, 8^{15}$.

The rank-45 frame/octet coherent configuration contains a unique cross orbital of frame degree one and octet degree twelve. For a cover x, let $t_o(x)$ count the selected frames assigned to octet o by this orbital. The five cross-orbital signatures are then

$$\boxed{(8\mathbf{1}, 32\mathbf{1}, 16\mathbf{1} - 4t, 4\mathbf{1} + 3t, t)}.$$

If A_{45} is the adjacency matrix of $\mathrm{SRG}(45, 32, 22, 24)$, then

$$(A_{45} + 4I)t = 48\mathbf{1}, \qquad t - \frac{4}{3}\mathbf{1} \in E_{-4}(A_{45}).$$

Each signature has a unique anchor octet whose 32 neighbors all have value one. The anchor's 12 nonneighbors induce $K_{4,4,4}$, with three canonical four-cells. Their value patterns are

$$(2,2,2), \quad (0,3,3), \quad (1,2,3), \quad (0,2,4),$$

so the total is $45(1 + 3 + 6 + 6) = 720$.

Exactly 720 signature vectors occur globally, in four orbits:

type	coordinate histogram	$\|t\|^2$	$|PSp \cdot t|$	# cover orbits	# covers
T_{128}	$0^4 1^{32} 2^4 4^5$	128	270	270	3149280
T_{120}	$0^4 1^{32} 3^8 4$	120	135	6	38880
T_{104}	$1^{36} 2^4 3^4 4$	104	270	24	233280
T_{96}	$1^{32} 2^{12} 4$	96	45	27	126360

For the certified four-cover packing, the type sequence is $T_{128}, T_{96}, T_{104}, T_{128}$. Five additional covers would have to sum to its residual capacity $c = \mathbf{121} - \sum_{i=1}^{4} t_i$. Although 632 of the 720 single-cover signatures satisfy $t \le c$, an exact meet-in-the-middle search over 117548 unique admissible pair sums and 305488 admissible triple sums finds no complement. Hence

$$\boxed{\nexists\, u_1, \ldots, u_5 \in \mathcal{T} : u_1 + \cdots + u_5 = c},$$

where $\mathcal{T}$ is the complete global signature set. Thus this four-packing is nonextendible already in the 45-coordinate nonlinear quotient. This does not decide whether an unrelated nine-cover resolution exists.

1.2 Outer-class fusion, XOR separation, and the weight-four decoder

Theorem 1.1 (Passes 1826–1830)**.** *For the canonical multiplier-minus-one similitude of the W33 carrier:*

1. *its class in* $\mathrm{PSp}(4,3){:}2 \cong U_4(2){:}2$ *is the outer involution class* $2D$*; its traces on the chiral constituents of degrees* $15, 24, 30, 81$ *are* $(3, 4, 2, 3)$*;*
2. *the symmetry-fixed resolution XOR system has* 4860 *variables, rank* 2349*, and affine solution dimension* 2511*, but its canonical exact parity witness is not an integer Hoffman resolution;*
3. *the code-weight layers* 0 *through* 10 *and* 35 *through* 45 *are exact, while the full subset action has* 1,358,719,936 *orbits, so the complete* 2^{45} *enumerator remains open;*
4. *among the* $\binom{240}{4} = 134{,}810{,}340$ *weight-four errors, exactly* 63,416,280 *have a unique minimum-weight syndrome representative;*
5. *after imposing the complete set of* 720 *globally realizable nonlinear cover signatures, the certified four-cover packing has no five-signature completion, despite consistency of the linear XOR relaxation.*

The resulting binary-symmetric-channel success contribution through weight four is

$$(1-p)^{240} + 240p(1-p)^{239} + 25440p^2(1-p)^{238} + 1576000p^3(1-p)^{237} + 63416280p^4(1-p)^{236}.$$

The nonlinear obstruction is packing-specific and does not settle the existence of an unrelated nine-cover resolution.

1.3 Complete cover-pair layer and nonlinear nine-signature separation

The complete exact-cover census contains 3,547,800 covers in 327 $\mathrm{PSp}(4,3)$-orbits. We now classify the complete disjoint-pair layer. If C_i is a representative of orbit $\mathcal{O}_i$, define

$$D_{ij} = \#\{C \in \mathcal{O}_j : C \cap C_i = \varnothing\}.$$

All 327^2 entries are reconstructed exactly and satisfy

$$|\mathcal{O}_i| D_{ij} = |\mathcal{O}_j| D_{ji}.$$

Of the 106,929 ordered orbit-type cells, 106,761 are compatible and 168 are incompatible. There are 23,276,276,640 unordered disjoint cover pairs globally. Ten orbit types are not self-compatible, but every orbit type has a disjoint partner.

The nonlinear octet signatures admit exact nine-layer integer solutions. For the complete set $\mathcal{T}$ of 720 realizable signatures, the equations

$$\sum_{t\in\mathcal{T}} y_t = 9, \qquad \sum_{t\in\mathcal{T}} y_t t = 12\mathbf{1}, \qquad y_t \in \mathbb{Z}_{\geq 0}$$

have a witness consisting of six signatures of norm 128 and three of norm 96. Its setwise stabilizer has order 9, so its inner orbit has size 2880.

The four signature classes have an intrinsic representation-theoretic explanation. The stabilizer of an anchor octet has order 576 and acts on the three $K_{4,4,4}$ non-neighbour cells through a quotient S_3 with kernel of order 96. The four local patterns

$$(0,2,4), \quad (0,3,3), \quad (1,2,3), \quad (2,2,2)$$

have local orbit sizes $6,3,6,1$, yielding the global orbit sizes $270,135,270,45$ and stabilizer orders $96,192,96,576$.

The canonical outer involution preserves these four classes but acts nontrivially inside them. It fixes 28 of the 720 signatures. On the 327 inner cover orbits it fixes only five and exchanges 161 pairs; the complete disjointness matrix is invariant under this outer action.

Finally, the displayed nine-signature witness does not lift to nine disjoint exact covers. The nine candidate fibres have sizes

$$11664, 11664, 2808, 11664, 2808, 2808, 11664, 11664, 11664.$$

A deterministic exhaustive nine-partite search closes after 4421 nodes and 4188 dead ends with no lift. This proves non-liftability for one 2880-element signature-resolution orbit. It does *not* decide the existence of a different nine-cover resolution arising from another nonlinear signature orbit.

Exact signature, duad-compression, decoder, and outer-class continuation

The complete set of 720 nonlinear cover signatures admits an exact nine-vector capacity witness,

$$t_1 + \cdots + t_9 = 12\mathbf{1}, \qquad \{\text{types}\} = 6T128 + 3T96.$$

Thus the nonlinear signature quotient is feasible. The parallel Pass 1835 exact lift worker proves that this specific $6T128 + 3T96$ signature orbit has no frame-level lift, sharply separating signature feasibility from exact-cover realizability.

The proposed 9×5 middle-layer spread does not exist. In the complement of $\mathrm{SRG}(45,32,22,24)$, the maximum number of mutually disjoint five-cliques is six. There are 72 maximum partial spreads, and the residual fifteen vertices induce

$$\mathrm{SRG}(15,6,1,3) \cong KG(6,2).$$

Hence the exact separator is $45 = 6 \cdot 5 + 15$, with a canonical duad residual.

For the binary $[240,195,4]_2$ frame code, equal-syndrome triple pairing gives exactly

$$A_6 = 9600$$

weight-six codewords. Exactly 185040 weight-five errors are shadowed by a unique singleton error. The complete weight-five decoder coefficient remains conditional on the weight-eight and weight-ten dependency atlas.

The four chiral modules of degrees $15, 24, 30, 81$ are separated by four geometrically fingerprinted outer probes with trace matrix

$$\begin{pmatrix} 3 & 4 & 2 & 3 \\ -1 & 0 & 4 & -3 \\ -2 & 1 & -1 & 0 \\ 1 & 0 & 0 & -1 \end{pmatrix}, \qquad \det = 80.$$

Finally, an ATLAS-standard pair (c, d) satisfies $c \in 2C$, $|d| = 9$, $|cd| = 10$, and generates PSp$(4,3){:}2$. If $D = d^{-1}$, the canonical $2D$ similitude is

$$s = cdcD^2cdcd^2cDcD^2cD^2.$$

All statements are frozen in the Passes 1836–1840 exact certificates. No frame-cover resolution, complete middle-layer enumerator, or weight-five threshold is inferred.

1.4 A second signature orbit, free higher-packing suborbits, and the outer quotient

The nonlinear signature equations admit at least two $PSp(4,3)$-orbits of distinct nine-signature solutions. The first has size 2880, stabilizer order 9, and type $6T_{128} + 3T_{96}$. Blocking all 2880 supports of that orbit yields a second exact solution with trivial stabilizer, hence orbit size 25920, and type

$$3T_{128} + 2T_{120} + 2T_{104} + 2T_{96}.$$

Thus at least 28800 signature-level resolutions exist across two inner orbits. The census is not yet exhaustive.

The certified chiral four-packing has trivial inner setwise stabilizer. Its six pairs, four triples, and one quadruple therefore generate six, four, and one distinct $PSp(4,3)$-orbits, each of size 25920. The canonical outer similitude exchanges every one of these eleven inner subpacking orbits with a distinct mirror orbit. This classifies the higher-packing structure inside the certified family, not globally.

The new free signature orbit also fails to lift. The nine exact candidate-cover populations are

$$11664,\ 2808,\ 864,\ 864,\ 11664,\ 288,\ 288,\ 2808,\ 11664,$$

and deterministic nine-partite search returns UNSAT after 289 nodes and 288 dead ends. Together with the earlier 2880-orbit obstruction, two inner signature orbits containing 28800 solutions are excluded from frame-level resolution; this is not a global no-resolution theorem.

The order-nine stabilizer of the first signature orbit is elementary abelian:

$$G_{\mathrm{sig}} \cong C_3 \times C_3.$$

Its three point-orbits are two T_{128} triangles and one T_{96} triangle. On the six T_{128} signatures, Gram value 74 is exactly $2K_3$, Gram value 70 is exactly $K_{3,3}$, and the remaining 21 pairs have Gram value 78.

Finally, the canonical outer involution fixes both certified inner signature orbits. The first extended setwise stabilizer is

$$(C_3 \times C_3) \rtimes C_2 \cong C_3 \times S_3$$

of order 18, while the free inner orbit acquires a setwise stabilizer C_2 in $PGSp$. These statements concern the two certified orbits only.

1.5 Two no-lift signature orbits, the weight-five decoder chart, the duad–syntheme separator, and outer-probe fusion

The independently owned Passes 1841–1845 packet certifies two distinct inner orbits of nonlinear nine-signature resolutions, of sizes 2880 and 25920, and proves by exact multipartite search that neither orbit lifts to nine disjoint frame covers. The binary signature-orbit census remains incomplete, so this is not a global no-resolution theorem.

For the binary frame code the complete low-weight enumerator through ten is

$$A_4 = 540, \qquad A_6 = 9600, \qquad A_8 = 424170, \qquad A_{10} = 17523360.$$

The fixed-coordinate syndrome sort gives the exact chart partition

$$\binom{239}{4} = 132563501 = 1754193 + 62359342 + 68449966.$$

The earlier globalization of the chart singleton count by the factor 240/5 is superseded by Pass 1882. Equal-syndrome weight-five errors differing by a weight-w codeword share only $5-w/2$ coordinates; in particular all

$$2{,}207{,}943{,}360$$

weight-ten collision pairs are disjoint and invisible in every fixed-coordinate chart. Thus

$$\boxed{U_5 \le 2{,}993{,}248{,}416}$$

is a certified upper bound for the global unique-minimum count, not an exact fifth-order coefficient. The exact global lower-shadow count remains 84,201,264, and the exact minimum-weight-five population is 6,278,846,784.

The $6 \times 5 + KG(6,2)$ separator has an exact check decomposition

$$\boxed{240 = 20 + 15 \cdot 12 + 20 \cdot 2}.$$

The 20 residual checks are the triangles of K_6 on the fifteen duads; each of the fifteen fiber pairs determines one of the fifteen synthemes, yielding the classical duad–syntheme realization of the exceptional outer automorphism of S_6; and every fiber triple carries two phase checks.

The remaining outer probes fuse exactly to ATLAS classes

$$2D, \qquad 4C, \qquad 6H, \qquad 8A,$$

with verified centralizers $96, 96, 36, 8$ and literal shortest words of lengths $18, 12, 16, 17$ in the project ATLAS-standard pair. Finally, an exact tuple-conjugacy algorithm is supplied and passes a deterministic relabelling test, but the official ATLAS GAP payload server was inaccessible during this run. Therefore no literal official-tuple conjugacy claim is made without the primary-source bytes.

1.6 Residual contraction, lower-syndrome orbits, and the exceptional S_6 intertwiner

The certified $6 \times 5 + 15$ separator admits an exact residual contraction. The 15-duad sector has 156 natural S_6-orbits; every one of the fifteen pair-transfer tensors has binary rank 7, while every one of the twenty phase tensors has rank 2. The complete six-fiber contraction remains open, with a separator-first induced-width certificate equal to 40.

All syndromes reachable at odd weight below five form exactly

$$1{,}892{,}792$$

distinct vectors and exactly 110 inner-group orbits: one minimum-weight-one orbit and 109 minimum-weight-three orbits. This gives a fail-closed symmetry-compressed decoder for every lower shadow of a weight-five error.

The duad-to-syntheme map is represented by an integral permutation matrix $P \in GL_{15}(\mathbb{Z})$ with

$$\det P = -1.$$

For each Coxeter generator s_i of S_6,

$$P\rho_{\text{duad}}(s_i) = \rho_{\text{syntheme}}(\alpha(s_i))P,$$

where every $\alpha(s_i)$ is a triple transposition and the five images generate all of S_6. Thus the W33 separator carries a literal integral realization of the exceptional outer automorphism.

Finally, a deterministic 500,000,000-node proof search establishes

$$A_{12} \geq 5{,}323{,}560.$$

The weight-six equal-syndrome collision count therefore satisfies

$$E_6 = 1{,}312{,}130{,}546{,}100 + 462A_{12} \geq 1{,}314{,}590{,}030{,}820.$$

Because A_{12} is not yet closed, no sixth-order unique-minimum decoder coefficient is claimed. The independently owned Pass 1855 computation additionally freezes the official ATLAS 40a generator bytes and proves a unique simultaneous conjugator to the project standard pair; Pass 1858 reconciles both payload hashes and the literal two-generator conjugacy certificate.

1.7 Exceptional-S_6 outer doily transfer clock

Let D be the 15×15 syntheme–duad incidence matrix and let $P \in GL_{15}(\mathbb{Z})$ be the exact Pass 1859 permutation matrix realizing the exceptional outer automorphism of S_6. Define $T = P^{\mathsf{T}}D$. Then

$$\operatorname{rank}_{\mathbb{Q}} T = \operatorname{rank}_{\mathbb{F}_2} T = 10, \qquad \dim\ker T = 5,$$

and its ten nonzero Smith invariants are all 1. Its exact polynomials are

$$\chi_T(x) = x^5(x-3)(x-2)(x+2)^2(x^2+4)(x^4+16), \qquad \mu_T(x) = x(x-3)(x^8-256).$$

On the balanced nonzero image

$$W = \operatorname{im}(T) \cap \mathbf{1}^{\perp}, \qquad \dim W = 9,$$

one has

$$\boxed{(T/2)^8 = I_W}, \qquad (T^8 - 256I)T = 1261J,$$

and

$$\operatorname{tr}(T^n) = 3^n + (-2)^n + 8\,2^n[8 \mid n] \qquad (n \geq 1).$$

For the exceptional automorphism α frozen in Pass 1859,

$$T\rho(g) = \rho(\alpha^{-1}(g))T \qquad (g \in S_6).$$

Finally

$$T^{\mathsf{T}}T = D^{\mathsf{T}}D, \qquad TT^{\mathsf{T}} = T^{\mathsf{T}}T.$$

Thus T is normal but nonsymmetric and $T/2$ is orthogonal on W. The fold preserves the doily singular spectrum $9^1+4^9+0^5$, and $T^{\mathsf{T}}T-3I$ is the SRG$(15,6,1,3)$ point graph of $W(2)$. The clock terminology is strictly finite and representation-theoretic; no physical time evolution or continuum dynamics is inferred.

1.8 Cyclotomic boundary, Tutte–Coxeter lift, and exact weight-twelve closure

Let $T = P^{\mathsf{T}}D$ be the exceptional-S_6 outer-doily transfer operator of Passes 1867–1871 and let $\Lambda = \mathrm{im}_{\mathbb{Z}}(T) \cap \mathbf{1}^{\perp}$. The integral clock lattice has rank 9, discriminant

$$\det(\Lambda^{\mathsf{T}}\Lambda) = 2560 = 2^9 \cdot 5,$$

and characteristic polynomial

$$(x-2)(x+2)^2(x^2+4)(x^4+16).$$

The normalized action $T/2$ does not preserve Λ; indeed

$$\Lambda/(T+2I)\Lambda \cong \mathbb{Z}^2 \oplus (\mathbb{Z}/4)^3.$$

Thus the regular eight-phase packet is a rational, or 2-local, decomposition and not a global integral $\mathbb{Z}[\zeta_8]$ splitting.

The bipartite lift

$$A_{\mathrm{TC}} = \begin{pmatrix} 0 & D^{\mathsf{T}} \\ D & 0 \end{pmatrix}$$

is the Tutte–Coxeter graph, with intersection array $\{3,2,2,2;1,1,1,3\}$. The exceptional fold has off-diagonal blocks T^{T}, T. The rank-nine separator projector is

$$E_9 = \frac{G(9I-G)}{20}, \qquad G = T^{\mathsf{T}}T,$$

so $\mathbb{Q}^{15} = \mathbf{1} \oplus V_5 \oplus V_9$ and V_9 occurs once. The exact $W(E_6)$ character degrees contain no 9, hence this module is an S_6 separator constituent and not a full-$W(E_6)$ irreducible sector.

A correction to Passes 1867–1871 is also exact:

$$TT^{\mathsf{T}} = T^{\mathsf{T}}T.$$

Therefore T is normal but nonsymmetric, and $T/2$ is orthogonal on the balanced nine-space. The directed graph has automorphism group C_4. Moreover the twisted Hom-space is two-dimensional, and the only row-sum-three solutions with the same Gram matrix are T and $2J/5-T$; consequently T is the unique integral, and unique 0/1, solution.

Finally, exhaustive enumeration of all 2^{45} dual words and exact MacWilliams transform gives

$$\boxed{A_{12} = 891{,}792{,}940}, \qquad \boxed{E_6 = 1{,}724{,}138{,}884{,}380}.$$

Here E_6 is the total weight-six equal-syndrome pair count. The sixth-order unique-minimum decoder coefficient remains open because collision incidences must still be deduplicated by syndrome orbit and lower shadow.

1.9 Decoder globalization correction, complete enumerator, two-adic clock order, and exceptional-S_6 carrier maps

A fixed-coordinate syndrome chart does not determine global syndrome multiplicity. If two equal-syndrome weight-five errors differ by a codeword of weight w, then they share exactly $5 - w/2$ coordinates. In particular the exact weight-ten contribution

$$2{,}207{,}943{,}360$$

consists of disjoint collision pairs invisible in every coordinate chart. This proves that the formerly stated number 2,993,248,416 was only an upper bound. Pass 1887 subsequently completes the complementary global census and supersedes that bound with

$$\boxed{U_5 = 1{,}531{,}165{,}872}.$$

The exact sixth-order collision count remains

$$E_6 = 1{,}724{,}138{,}884{,}380,$$

but the corresponding singleton-syndrome coefficient remains open pending a true global component enumeration.

The complete MacWilliams transform of the frozen dual histogram gives the full binary $[240, 195, 4]$ primal enumerator. Its newly closed low coefficients include

$$A_{12} = 891{,}792{,}940, \qquad A_{14} = 54{,}326{,}090{,}880, \qquad A_{16} = 3{,}770{,}230{,}198{,}995.$$

Edge transitivity makes these support shells 1-designs, while their required pair multiplicities have denominator 239; hence the weight-12, 14, and 16 shells are not 2-designs.

For the balanced clock order,

$$\mathbb{Q}[C] \cong \mathbb{Q} \times \mathbb{Q} \times \mathbb{Q}(i) \times \mathbb{Q}(\zeta_8),$$

and the normalization of $\mathbb{Z}[C]$ is

$$\mathbb{Z} \times \mathbb{Z} \times \mathbb{Z}[i] \times \mathbb{Z}[\zeta_8].$$

The exact index and conductor are

$$[\mathcal{O}_{\max} : \mathbb{Z}[C]] = 2^{35},$$

$$\mathfrak{f} = 1024\mathbb{Z} \times 1024\mathbb{Z} \times 512\mathbb{Z}[i] \times 256\mathbb{Z}[\zeta_8].$$

Thus the failure of the integral cyclotomic splitting is purely two-adic.

The setwise stabilizer of the canonical six fibers is an exceptional S_6 of order 720. For its natural nine-dimensional separator constituent $V_{[4,2]}$, the multiplicities in the signed-edge sectors $(15, 24, 30, 81, 90)$ are

$$\boxed{(0, 1, 0, 0, 1)}.$$

Literal equivariant maps

$$A_{24} = N_{24}/2, \qquad A_{90} = N_{90}/2$$

have rank nine, satisfy all 720 intertwining identities, and obey

$$N_{24}^{\mathsf{T}} N_{24} = N_{90}^{\mathsf{T}} N_{90} = 4E_{9,\mathrm{num}}, \qquad N_{24}^{\mathsf{T}} N_{90} = 0.$$

Hence the two natural V_9 copies are isometric and orthogonal. The occurrence theorem alone did not choose a phase; Pass 1889 later glues the paired copies into an exact exceptional-S_6-equivariant complex structure. The odd 81-sector remains parity-obstructed.

Finally,

$$\mathrm{Aut}_{\rightarrow}(T) = \mathrm{Fix}(\alpha) \cong C_4,$$

generated on the six labels by $(0\,1\,4\,5)(2\,3)$. Its duad orbits have sizes

$$4 + 4 + 4 + 2 + 1,$$

and the unique loop is the fixed duad $\{2, 3\}$. The resulting five-vertex graph-of-groups voltage table reconstructs all 45 arcs of the exceptional one-part fold $T = P^{\mathsf{T}}D$ of the Tutte–Coxeter graph. These voltage labels are finite graph phases, not physical time or optical phase.

1.10 Exact global fifth-order decoding, paired-carrier phase, and the Tutte–Coxeter lift

The fixed-coordinate decoder chart is completed by an exact complementary codeword census. A partner retaining the fixed coordinate is already visible in the chart; a partner omitting it differs by a weight-4, 6, 8, or 10 codeword containing that coordinate. Exhausting both cases gives

$$\boxed{U_5 = 1{,}531{,}165{,}872}, \qquad \boxed{A_5^{\mathrm{amb}} = 4{,}747{,}680{,}912},$$

and the exact partition

$$6{,}363{,}048{,}048 = 84{,}201{,}264 + 1{,}531{,}165{,}872 + 4{,}747{,}680{,}912.$$

Thus the corrected fifth-order BSC contribution is

$$\boxed{1{,}531{,}165{,}872\, p^5(1-p)^{235}}.$$

The weight-six equal-syndrome edge count remains exact, but its singleton component count remains open.

For the separator decomposition

$$240 = 20_{\mathrm{residual}} + 180_{\mathrm{pair}} + 40_{\mathrm{phase}},$$

the complete rank-30 fiber subcode has an exact 563-bin $(w_{\mathrm{pair}}, w_{\mathrm{phase}})$ enumerator over all 2^{30} words. The rank-15 residual subcode has exactly 156 exceptional-S_6 orbits with exact $(w_{\mathrm{residual}}, w_{\mathrm{pair}}, w_{\mathrm{phase}})$ records. The full mixed 2^{45} refined contraction is not inferred from these marginals.

Let $A_{24} = N_{24}/2$ and $A_{90} = N_{90}/2$ be the two literal natural V_9 carrier maps. Each rank-nine image lattice has Gram Smith invariants

$$2, 10, 10, 10, 20, 40, 40, 40, 40$$

and discriminant $2^{18}5^8$. The paired orthogonal rank-18 lattice carries an exact exceptional-S_6-equivariant complex structure

$$J(A_{24}v) = A_{90}v, \qquad J(A_{90}v) = -A_{24}v, \qquad J^2 = -I.$$

Its half-integral enlargement has index 2^{18} and therefore does not absorb the independent 2^{35} maximal-order defect of the clock algebra.

On the full $24 \oplus 90$ carrier the exceptional-S_6 commutant has real dimension 23 and the odd real-type multiplicities obstruct a complex structure. Restriction to the residual clock subgroup C_4 gives

$$\mathbf{28}\mathbf{1} \oplus 26\,\mathrm{sgn} \oplus 30\mathbb{R}^2_{\mathrm{rot}},$$

with commutant

$$M_{28}(\mathbb{R}) \times M_{26}(\mathbb{R}) \times M_{30}(\mathbb{C}),$$

of real dimension 3260; both the 24- and 90-sectors then admit C_4-invariant complex structures.

Finally, every one of the 180 pair-transfer coordinates is uniquely a nonincident syntheme–duad pair, so this sector is exactly the bipartite complement of the Tutte–Coxeter Levi graph. The exceptional C_4 fixes two of its 90 octagons and has 22 free octagon orbits. On the 90 directed Hashimoto states its four powers fix

$$90, 2, 10, 2$$

states. The natural V_9 lies in the adjacency eigenspaces $\lambda = \pm 2$ and feeds the four cyclotomic nonbacktracking factors

$$(x^2 - 2x + 2)^9, \qquad (x^2 + 2x + 2)^9.$$

All phase and voltage statements here are finite representation and graph holonomies; no physical time evolution or optical phase law is inferred.

1.11 Weight-six reduction, the mixed separator tensor, the Gaussian V_9 lattice, and the twisted zeta

The exact weight-six collision ledger is

$$(E_6(4), E_6(6), E_6(8), E_6(10), E_6(12)) = (204105833100, 202385664000, 397812076200, 507826972800, 412008338280)$$

so $E_6 = 1724138884380$. The weight-twelve term consists of disjoint pairs invisible in every fixed-coordinate chart and contributes

$$44589647\binom{11}{5} = 20600416914$$

fixed-coordinate external-partner incidences with multiplicity. Hence the sixth-order singleton coefficient is not a collision moment; an exact external-memory component worker is supplied and the final heavy run remains open.

The complete mixed separator enumerator is the exact partition function of six five-bit fiber variables and fifteen residual bits, with

$$\boxed{240 = 20_{\mathrm{residual}} + 180_{\mathrm{pair}} + 40_{\mathrm{phase}}}.$$

The 45 absent pair/residual incidences are the Tutte–Coxeter edges, the 180 present incidences are their complement, and the residual and phase families are the complete twenty-triple systems on six labels. A chunkable worker contracts the resulting 155841-bin tensor over the 156 residual S_6 orbits; the final all-orbit merge is not inferred from the marginals.

The paired natural V_9 carrier is a rank-nine Gaussian lattice of minimum norm 24, with 60 minimal vectors and Smith invariants

$$2, 10, 10, 10, 20, 40, 40, 40, 40.$$

Its fifteen minimal lines form $KG(6,2)$, whence

$$\boxed{U(L) \cong C_4 \times S_6, \qquad |U(L)| = 2880.}$$

On the real 90-sector, $\mathrm{End}_{PSp(4,3)}(90) \cong \mathbb{C}$, so the invariant complex structures are exactly $\pm J$; the outer involution of $W(E_6)$ sends J to $-J$ and removes the full-Weyl phase. Exceptional S_6 obstructs J on the full 24, 90, and 114 carriers but preserves the paired-V_9 structure. The named chain and every cyclic subgroup class are exact; a GAP worker completes the remaining noncyclic classes.

Finally the residual C_4 decomposes the 90 Hashimoto states into dimensions $26+20+24+20$, with reciprocal factors

$$\begin{aligned} Z_0^{-1} &= (1-4u^2)(1-u^2)^4(1+2u^2)^2(1-2u+2u^2)^3(1+2u+2u^2)^3, \\ Z_1^{-1} = Z_3^{-1} &= (1-u^2)^4(1+2u^2)^2(1-2u+2u^2)^2(1+2u+2u^2)^2, \\ Z_2^{-1} &= (1-u^2)^4(1+2u^2)^4(1-2u+2u^2)^2(1+2u+2u^2)^2. \end{aligned}$$

The natural V_9 channel contributes $12, 8, 8, 8$ dimensions across the four characters. These are finite representation and graph invariants, not physical propagation laws.

Passes 1907–1911: split enumerator and phase–holonomy closure

The complete rank-45 dual-code split enumerator for the canonical 20+180+40 coordinate partition has 2^{45} words and exactly 7355 nonzero trivariate coefficients. Its three complement symmetries are literal code translations: the sum of the thirty fibre generators is the phase-40 mask, the sum of the fifteen residual generators is the residual-plus-pair-200 mask, and together they generate a $C_2 \times C_2$ complement subcode containing the all-one word.

Building on the Weil-chirality and Frobenius–Schur results already owned by Passes 353/355 (Gow–Vinroot), all 56 conjugacy classes of subgroups of the exceptional S_6 have been classified by Frobenius–Schur parity. Exactly 26, 22, and 12 classes admit invariant complex structures on the 24, 90, and $24 \oplus 90$ carriers, respectively, while the paired natural V_9 admits one for every class. On A_6, the S_6-paired complex structure and the restricted $PSp(4,3)$ structure are distinct adjacent rotations on three equivalent real V_9 copies and generate $\mathfrak{so}(3)$ rather than a quaternionic algebra.

The 15 projective Gaussian minimal lines form $KG(6,2)$, but this projectivization forgets the central C_4 unit orientation. Sound outer-conjugation cuts therefore live on the 60-vector oriented lift: a transported pair satisfies $\Phi(c') = -\Phi(c)$ and a transported fixed color satisfies $\Phi(c) = 0$. No pointwise bound is inferred from the spread mean $45/9 = 5$.

Finally, stabilizer-weighted primitive holonomy splits the 90-state Hashimoto representation into the shared 36-dimensional natural-V_9 channel and a 54-dimensional complement. The graph invariant cannot separate the two literal V_9 carrier embeddings. The global sixth-order decoder coefficient remains behind an explicit external-memory boundary; a complete 2,190,670-error shard, including lower-shadow removal, is certified without promoting a global value.

Passes 1939–1943: split MacWilliams, integral phase order, and Hodge–field separation

A merged fixed-coordinate weight-six super-shard containing 58,282,126 errors exposes 5,389,182 collision edges that are invisible in independent shard counts and destroys 3,210,241 apparent singleton groups. The global sixth-order coefficient remains open because shards outside the certified super-shard and external complementary partners are not yet merged.

The complete $20 + 180 + 40$ split MacWilliams transform has 2^{195} primal words and exactly 39,081 nonzero coefficients. The complement $C_2 \times C_2$ lies in the code hull, forcing h and $r + p$ to be even while also generating the three split-complement symmetries. The weight-four shell splits exactly as

$$15\,(4,0,0) + 120\,(1,3,0) + 225\,(0,4,0) + 180\,(0,2,2) = 540.$$

The fifteen Gaussian projective minimal lines identify literally with the fifteen residual duads, but the projective/color quotient annihilates all 135 conjugation-odd moments. Consequently no nonzero phase-odd cut descends to the unoriented color variables; a C_4-oriented solver lift is necessary.

On the three-copy A_6 multiplicity lattice, the Gaussian quarter-turn on the $A + B$ copies and the Eisenstein sixth-order unit on the $B + C$ copies generate

$$\mathbb{Z}[R_4, U_6] = M_3(\mathbb{Z}).$$

The common commutant is scalar and the generated unit group is infinite, so there is no quaternionic or finite $SU(2)$ enhancement.

Finally, the two literal V_9 embeddings are separated by $L_1 A_{24} = 10A_{24}$ and $L_1 A_{90} = 4A_{90}$, and independently by ambient character fields $\mathbb{Q}$ and $\mathbb{Q}(\omega)$. The unique sixfold Eisenstein phase therefore lies in the coexact, divergence-free 90-sector, not in the exact Gauss-law/source block. Its structurally supported reading is a flux or holonomy phase; identifying its six units with electric charges remains a physical hypothesis rather than a derivation.

Passes 1950–1954: frame geometry, arithmetic closure, and the internal phase boundary

The 28 primitive shards in the current sixth-order decoder super-shard form a complete weighted collision graph with 125 distinct edge labels and trivial automorphism group. The fixed-edge geometric stabilizer has 230 orbits on all pair charts, so the present ordering-based partition has no useful symmetry compression; the global sixth-order coefficient remains open.

The 540 minimum primal words are exactly the 540 frame matchings. Under the exceptional S_6 they split into orbits of sizes $15, 45, 120, 180, 180$. The 225 pair-only words split into a 45-orbit of parallel tetrads, partitioning all 180 pair coordinates, and a 180-orbit of pair rectangles. The residual 15-orbit is the set of tetrahedral boundaries on four-subsets of six.

A literal binary 540×15 frame–residual-duad incidence matrix has rank 15 over $\mathbb{Q}$ and over $\mathbb{F}_p$ for $p = 2, 3, 5, 7$, with all 720 exceptional-S_6 equivariance equations exact. A corrected prefix-equality lex leader is certified nonvacuous by a feasible colouring that it removes and a symmetry-equivalent representative that it preserves. A bounded unpinned nine-colour run remains `UNKNOWN`; no chromatic claim is promoted.

On the three-copy A_6 multiplicity lattice, the Gaussian quarter-turn and the Eisenstein sixth-turn generate all elementary transvections. Hence their group is exactly $\mathrm{SL}_3(\mathbb{Z})$. Finally, the internal endomorphism C_6 and the E_8 Coxeter hexagon C_6 are inequivalent on their full 240-dimensional carriers: their character multiplicities are $(150, 45, 0, 0, 0, 45)$ and $(40, 40, 40, 40, 40, 40)$, respectively. The internal μ_3 also fails to commute with the exceptional-S_6 pairing on the shared V_9 sector.

The order-six symmetry is therefore retained only as an internal endomorphism of the coexact 90. No homological flux quantum, electric-charge derivation, QCD-colour identification, generation assignment, or neutrino identification is asserted.

Passes 1966–1970: combined spread symmetry, the intrinsic phase role, and the standalone draft

The 36 spread-by-colour counts and the geometric group action now coexist in a single exact signature model. Forty point-transvection orbit-minimum cuts grow the nine-colour formulation from 3033 to 3073 constraints. On the complete 25920-element $PSp(4,3)$ orbit of a known proper 14-colouring, one cut leaves 13021 images, eight leave 3244, and all forty leave 807. The forty cut directions are linearly independent. This is a certified symmetry reduction, not a chromatic decision: the bounded nine-colour run remains `UNKNOWN`.

The internal Eisenstein phase also has a representation-theoretic role that does not require a physical label. The finite integral centralizer torsion is $(C_2)^4 \times C_6$; its unique odd Sylow $C_3 = \langle \mu_6^2 \rangle$ is characteristic. Every nontrivial phase power fixes the rational 150-dimensional block sum and acts on the coexact Eisenstein 90. The outer involution sends μ_6 to μ_6^{-1}, so the phase and chirality generate a D_{12} normalizer. No charge, flux, colour, generation, or particle interpretation is asserted.

A backward audit now distinguishes vacuous constraints, scope errors, restrictive but invalid cuts, and verified constraints that were never inserted. The exact replay includes the false spread cap $5 < 13$, the vacuous $x \leq y + 8$ inequality, and positive model-growth and orbit-removal witnesses for the corrected cuts. Two oldest executable builders were not located and remain explicitly postmortem-derived.

Finally, `analysis/W33_SPREAD_OBSTRUCTION_REFEREE_DRAFT.tex` collects the frame graph, spread trap, finite $1/q$ law, spread involution, and signed-edge-module results in a referee-shaped standalone article with theorem status labels, ownership boundaries, reproducibility evidence, consolidated withdrawals, and open problems. It keeps $\chi(H) = 9$ open and is not presented as peer review or publication.

Passes 1971–1975: corrected spread obstruction, scoped proof, and engineering boundary

The independent treatments of the spread obstruction have been reconciled. One repeated statement is withdrawn: the 45 frames determined by a spread are not a maximal independent set. At $q = 3$ there are exactly 15 residual candidate frames, each nonadjacent to the spread seed. Their union meets only 20 of the 60 residual edges, however, so 40 residual edges occur in no candidate. The exact obstruction is therefore support deficiency: no 60-frame exact cover extends the spread seed.

Uniformly, spread pairs form an independent seed and leave $(q^2+1)q(q+1)/2$ internal spread-line edges. If the spread carries a fixed-point-free linewise involution σ, the line orbits $\{A, \sigma(A)\}$ supply $q(q^2+1)/2$ residual candidates, supported on $(q^2+1)(q+1)/2$ residual edges with multiplicity q. The statement that these are all candidates is the additional candidate-orbit property. It is verified for the repository's $q = 3, 5, 7$ examples but is not proved uniformly. A nonsquare similitude $g^2 = \mu I$ constructs the involution for the associated Desarguesian symplectic spread for every odd q; it does not by itself prove the candidate-orbit converse. Existence and uniqueness for arbitrary symplectic spreads are not claimed; uniqueness is exact at $q = 3$.

The nine-colour decision remains open. Spread-variable branching is still the best frozen search at 60,909 branches. Combining it with eight or forty geometric lex generators gives 451,460 and 512,714 branches, respectively. Thus the exact orbit reduction 25,920 $\to$ 807 is not a search-performance theorem. Full-scale constraints are now audited by named feasible witnesses or explicit finite feasible orbits rather than truncated enumeration.

The internal Eisenstein C_6 remains confined, under equivariant linear maps, to the coexact 90 and is inverted by chirality, giving a D_{12} phase normalizer. Charge and homological-flux readings remain withdrawn. The computer-engineering consequences are architectural proposals only: external symmetry canonicalisation, geometric cube deduplication, wide-bit exact-cover acceleration, Hodge-separated control planes, and an isolated six-phase calibration domain.

Passes 2011–2015: decorated spread pairs, subgroup schedules, and quadratic readout

The 270 unordered spread pairs meeting in four lines carry stabilizer $S_4 \times D_8$ of order 192. Two order-four conjugacy-class carriers are literal decoration bundles over these pairs: the size-540 class chooses one of two inverse coherent linewise quarter-turns, while the size-1620 class chooses one of six cyclic orders of the four common lines. Their stabilizers have orders 96 and 32, respectively; the identifications use the induced local actions and stabilizer conjugacy, not cardinality alone.

Complete enumeration of the 1026 subgroups of one such pair stabilizer gives 234 stabilizer-conjugacy classes. Exactly 33 classes construct a 60-frame exact cover from whole frame orbits. Every successful subgroup has order 2, 4, or 8, and no subgroup of order at least 12 succeeds. A literal D_8 witness uses twelve orbits of sizes $2, 2, 4, 4, 4, 4, 4, 4, 8, 8, 8, 8$ and covers each of the 240 edges once.

The 36 spreads form a rank-three association scheme. The four-line relation is $\mathrm{SRG}(36, 15, 6, 6)$ and its adjacency matrix satisfies

$$A^2 = 9I + 6J.$$

Since every line belongs to nine spreads, a fixed spread has total intersection 80 with the other spreads. Its fifteen four-line neighbours account for 60, forcing each of the remaining twenty spreads to meet it in exactly one line. Thus distinct spreads meet in one or four lines, with no other value.

The 360 one-line spread pairs do not form an eightfold bundle over the 45 octets: their stabilizer has octet orbits of lengths $6, 9, 12, 18$ and fixes no octet. Removing the common line instead reveals the bipartite double cover of the 3×3 rook graph on the two banks of nine noncommon lines. This local graph has automorphism group order 144, exactly the spread-pair stabilizer.

The degree-safety audit must mark degrees 240 and 540 unsafe for count-only identification. At degree 240 two order-216 stabilizers are nonconjugate; at degree 540 the three order-96 class centralizers are pairwise nonconjugate. The Eisenstein 90 remains linearly confined, but its quadratic channels are nontrivial: $\mathrm{Sym}^2(90)$ contains all four rational blocks, whereas $\Lambda^2(90)$ contains no copy of the gauge 15. This yields a precise symmetric-versus-antisymmetric readout selection rule without restoring any withdrawn charge or flux interpretation.

The computer-engineering consequences are proposals: a 36-lane rank-three mixer, a two-bank rook-double crossbar, a D_8 orbit-compressed scheduler, a quadratic phase-readout stage, and stabilizer-aware group-set interfaces. The chromatic decision remains open, the subgroup census is scoped to one fixed $S_4 \times D_8$ stabilizer, and no device or particle identification is claimed.

Passes 2050–2053 and 2064: full-group schedules, explicit quadratic maps, and the regular-spread family

The 33 subgroup classes inside a fixed four-line spread-pair stabilizer that construct orbit-built parallel classes fuse into 14 subgroup conjugacy types in $\mathrm{PGSp}(4,3)$. Taking the first deterministic exact-cover witness from each local class gives 32 distinct frame sets in 12 full-group schedule orbits. The subgroup fusion is complete for those 33 classes; the 12 schedule orbits are not a classification of every cover generated by them.

The quadratic connectivity of the signed coexact 90 is now literal rather than character-only. Vertex-energy gradients give surjections $\mathrm{Sym}^2(90) \to 15$ and $\mathrm{Sym}^2(90) \to 24$. Graph-restricted skew-matrix commutators give surjections $\Lambda^2(90) \to 30$ and $\Lambda^2(90) \to 81$. Twisting the commutators by the canonical complex structure gives chirality-odd symmetric maps to 30 and 81. All seven constructed maps have verified rank, equivariance, chirality parity, and internal μ_6 phase response. They are canonical generators, not a complete basis of all quadratic Hom spaces or physical coupling constants.

The frozen D_8 schedule expands twelve frame orbits of sizes $2, 2, 4, 4, 4, 4, 4, 4, 8, 8, 8, 8$ to 60 frames and covers all 240 edges exactly once. An executable reference model composes this orbit scheduler with a 240-bit exact-cover validator, the 36-lane spread mixer satisfying $A^2 = 9I + 6J$, and the 18-lane rook-double crossbar. These are software and hardware-architecture proposals, not a fabricated device.

The 36-spread four-line intersection graph is identified literally as the rank-three graph $NO_6^-(2)$, equivalently $NO_5^{-\perp}(3)$. A fixed vertex has local graph $K(6,2)$ and second subconstituent $J(6,3)$, and the full automorphism group has order 51,840. The subconstituent isomorphisms and rank-three action are load-bearing because the parameters $(36, 15, 6, 6)$ alone are highly nonunique.

Finally, complete regular-symplectic-spread orbits at $q = 3, 5, 7$ have sizes $36, 300, 1176$. In each case two distinct spreads meet in exactly 1 or $q+1$ lines, and the $q+1$ relation is strongly regular. The three cases match a single polynomial parameter family, but the all-odd-q statement remains a conjectural extrapolation here; non-Desarguesian symplectic spreads are not classified by this computation. The nine-colour decision remains open.

Regular spreads as projective quadratic structures

Let q be odd, let $\mu \in \mathbb{F}_q^\times$ be nonsquare, and write $K = \mathbb{F}_{q^2} = \mathbb{F}_q[t]/(t^2 - \mu)$. Regard $V = K^2$ as a four-dimensional $\mathbb{F}_q$-space. If

$$\Omega(x, y) = x_1 y_2 - x_2 y_1$$

and $\beta(x, y)$ is the t-coefficient of $\Omega(x, y)$, then β is a nondegenerate alternating $\mathbb{F}_q$-form. The one-dimensional K-subspaces of K^2 are totally isotropic two-dimensional $\mathbb{F}_q$-subspaces and form a regular symplectic spread.

Multiplication by t defines J with

$$J^2 = \mu I, \qquad \beta(Jx, Jy) = \mu\beta(x, y).$$

Thus $\sigma = [J] \in PGSp(4, q)$ is a projective involution. The spread is exactly the set of J-invariant projective lines. Since $K^\times/\mathbb{F}_q^\times \cong C_{q+1}$ has a unique involution, each regular field-reduction spread carries a unique projective quadratic structure. For $q \equiv 3 \pmod 4$ one may take $\mu = -1$, recovering the geometric i; for $q \equiv 1 \pmod 4$ a different nonsquare is required.

Restricting the classical Desarguesian-spread stabilizer to symplectic similitudes gives

$$C_{PGSp(4,q)}(\sigma) \cong C_2 \times P\Sigma L_2(q^2), \qquad \left|C_{PGSp(4,q)}(\sigma)\right| = 2q^2(q^4-1).$$

Therefore the canonical regular-spread orbit has size

$$\boxed{\frac{|PGSp(4,q)|}{|C_{PGSp(4,q)}(\sigma)|} = \frac{q^2(q^2-1)}{2}.}$$

This proves the orbit-size formula for every odd q and recovers the complete computational values

$$36, \quad 300, \quad 1176$$

at $q = 3, 5, 7$. It does not promote the all-q rank-three intersection graph, and it does not classify non-Desarguesian symplectic spreads.

At $q = 3$, the established inner spread stabilizer is S_6 of order 720, so the full stabilizer is

$$\boxed{C_{PGSp(4,3)}(\sigma) \cong C_2 \times S_6.}$$

The central $C_2 = \langle \sigma \rangle$ is silent on the local spread graph; the visible quotient S_6 is the automorphism group of the Kneser local graph $K(6,2)$ and its Johnson second subconstituent $J(6,3)$.

The shared phase-inversion controller

The geometric/representation clock has presentation $C_4 : C_2 \cong D_4$, while the earlier μ_6 clock has presentation $C_6 : C_2 \cong D_{12}$. If the clocks are independent apart from their common outer inverter, their combined controller is

$$\Gamma = (C_4 \times C_6) : C_2, \qquad |\Gamma| = 48,$$

where the last factor acts by simultaneous inversion. Exact enumeration gives

$$Z(\Gamma) \cong C_2^2, \qquad [\Gamma, \Gamma] \cong C_6, \qquad \Gamma_{\mathrm{ab}} \cong C_2^3.$$

The subgroups D_4 and D_{12} intersect in their common inverter and generate Γ. Although $|\Gamma| = 48$, it is not the frame stabilizer $C_2 \times S_4$: their center and derived-subgroup orders are respectively $(4,6)$ and $(2,12)$.

All statements in this insert are finite-geometric or group-theoretic. The order-48 synthesis assumes clock independence, is not a physical coupling, and restores none of the withdrawn charge, colour, generation, flux, or particle identifications.

Exact continuation: all-q regular spreads and the faithful phase controller

For every odd prime power q, the regular symplectic spreads may be represented as elliptic hyperplane sections $\mathcal{O}_x = Q(4,q) \cap x^\perp$, indexed by minus-type anisotropic points. The projective Gram determinant

$$\Delta(x,y) = 4Q(x)Q(y) - B(x,y)^2$$

is zero precisely for a tangent intersection. Consequently, distinct regular spreads meet in exactly 1 or $q+1$ lines. The $q+1$ relation is strongly regular with

$$v = \frac{q^2(q^2-1)}{2}, \qquad k = \frac{q(q-2)(q^2+1)}{2},$$

$$\lambda = \frac{q(q^3 - 4q^2 + 7q - 8)}{2}, \qquad \mu = \frac{q(q-2)(q-1)^2}{2},$$

and nontrivial eigenvalues $q(q-2)$ and $-q$. This proves the regular-spread formula conjectured after the complete $q = 3, 5, 7$ computations. It is a strongly-regular two-class fusion; for $q > 3$ the full group action need not have rank three.

Regularity is essential. An exact $q = 27$ Ree–Tits slice calculation gives a closed 144-spread regular suborbit whose intersections with the fixed non-Desarguesian spread have sizes

$$19,\ 28,\ 37,\ 46,\ 55,$$

so the regular $\{1, q+1\}$ scheme does not extend unchanged.

The quadratic target audit also corrects the actual signed-edge multiplicities:

	15	24	30	81
$\mathrm{Sym}^2(90)$	3	5	3	7
$\Lambda^2(90)$	0	0	2	5

Thus both gauge blocks are absent from the antisymmetric channel. The seven previously constructed tensors remain explicit surjective generators, not full bases of every intertwiner space.

Finally, the abstract independent-clock group $(C_4 \times C_6) : C_2$ has order 48, but its action on the canonical single complex structure factors through

$$(a, b, e) \longmapsto (3a + 2b \bmod 12, e),$$

with kernel generated by $(2, 3, 0)$. The faithful image on the canonical 90-sector is therefore $C_{12} : C_2$ of order 24. A faithful order-48 controller requires two independent complex phase registers.

The repository includes literal SystemVerilog masks for the 36-lane mixer and an algebraic verifier of $A^2 = 9I + 6J$, fixed-width safety, and the phase quotient. No synthesis, timing, fabrication, particle assignment, or measured coupling is claimed.

Divisible ovoid codes, complete quadratic maps, and Weil inversion

The complete q=27 Ree–Tits hyperplane spectrum is

$$1^{730}, \quad 10^{4563}, \quad 19^{96174}, \quad 28^{408294}, \quad 37^{36504}, \quad 46^{4914}, \quad 55^{702}.$$

All section sizes are 1 mod 9. Equivalently, the homogeneous coordinates generate an exactly 9-divisible projective $[730, 5]_{27}$ code. Complete censuses for the regular, Kantor, Thas–Payne and Ree–Tits q=27 families give pairwise distinct weight enumerators. The regular elliptic section gives a 27-divisible $[730, 4]_{27}$ code, whereas the three nonregular examples give exactly 9-divisible $[730, 5]_{27}$ codes.

The full $PSp(4, 3)$-equivariant quadratic map dimensions from the canonical 90 are

	15	24	30	81
$\mathrm{Sym}^2(90)$	3	6	5	12
$\Lambda^2(90)$	3	4	5	12.

Thus the complete quadratic space has dimension $26 + 24 = 50$. The outer similitude splits it into two equal 25-dimensional sectors. The outer-even half is exactly the earlier target-identified

PGSp table; the outer-odd half supplies the missing PSp-equivariant maps. Explicit integral signed-orbit representatives form complete bases; exact reduction modulo a characteristic not dividing the group order certifies independence and surjectivity, while the dimensions are fixed independently by characters.

For $q = 7$ and $q = 11$, the canonical Schrödinger Weil representations on functions on $\mathbb{F}_q^2$ split into parity dimensions

$$(25, 24) \quad \text{and} \quad (61, 60).$$

The nonsquare similitude $h = \mathrm{diag}(I_2, -I_2)$ is implemented by entrywise complex conjugation. On realification, complex multiplication J and conjugation K obey

$$J^2 = -I, \qquad K^2 = I, \qquad KJK = -J,$$

and generate D_4. This is an objectwise theorem for the canonical Weil family and is not a dimension-based identification with the q=3 signed-edge 90.

A packed SystemVerilog implementation, deterministic Icarus simulation, exhaustive Yosys SAT properties for $A^2 = 9I + 6J$ and the faithful D_{24} action, and explicit $W = 4$ iCE40/ECP5 synthesis scripts are included. Tool results are engineering evidence for the specified RTL only; no fabricated-device timing, power, physical coupling or particle assignment is claimed.

Controller representation trichotomy: orders 48, 24, and infinity

There are three controller objects in the preceding construction, not one. The abstract common-inverter group

$$\Gamma = (C_4 \times C_6) : C_2$$

has order 48. It has the faithful integral four-dimensional realization

$$A_4 = \begin{pmatrix} 0 & -1 & 0 & 0 \\ 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}, \quad B_6 = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & -1 & 1 \end{pmatrix},$$

$$S = \begin{pmatrix} 0 & 1 & 0 & 0 \\ 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{pmatrix},$$

for which $[A_4, B_6] = 1$, $SA_4S = A_4^{-1}$, and $SB_6S = B_6^{-1}$. Exact rational-character and cyclotomic checks show that its minimal faithful rational degree is 4.

The canonical single-complex-structure action is instead the quotient

$$(a, b, e) \longmapsto (3a + 2b \bmod 12, e), \qquad \mathrm{im} \cong C_{12} : C_2,$$

of order 24, with kernel $\{(0, 0, 0), (2, 3, 0)\}$. Thus the full order-48 group requires two independent two-dimensional phase registers.

The rank-three matrices R_4, U_6 from the integral multiplicity carrier form a third object. Their phase planes overlap, so they do not commute and generate $SL_3(\mathbb{Z})$. More strongly, the simultaneous common-inverter equations

$$XR_4 = R_4^{-1}X, \qquad XU_6 = U_6^{-1}X$$

have coefficient-matrix rank 9 and nullity 0 over $\mathbb{Q}$. Hence no nonzero rational common inverter exists.

The same generator word exposes the change of dynamics:

$$\chi_{A_4^2 B_6}(t) = (t+1)^2(t^2-t+1), \qquad |A_4^2 B_6| = 6,$$

whereas

$$\chi_{R_4^2 U_6}(t) = (t+1)(t^2-t-1), \qquad |R_4^2 U_6| = \infty.$$

The latter has spectral radius φ. Therefore the golden ratio belongs to the infinite arithmetic lattice action, not to the finite controller or its canonical single-J quotient. It is not a selected physical observable without an additional operational map.

The complete quadratic map space as an S_3-module

The complete Pass-2301 basis of fifty $PSp(4,3)$-equivariant quadratic maps from the signed-edge 90 carries a simultaneous phase action r and outer involution s satisfying

$$r^3 = s^2 = 1, \qquad srs = r^{-1}.$$

Thus the multiplicity spaces form a module for $C_3 : C_2 \cong S_3$. Exact GAP character decomposition gives

$$\mathcal{H}_{\mathrm{Sym}} \cong 13\,\mathbf{1} \oplus 3\,\mathrm{sgn} \oplus 5\,\mathrm{std},$$

$$\mathcal{H}_{\Lambda} \cong 3\,\mathbf{1} \oplus 13\,\mathrm{sgn} \oplus 4\,\mathrm{std},$$

and hence

$$\mathcal{H}_{\mathrm{quad}} \cong 16\,\mathbf{1} \oplus 16\,\mathrm{sgn} \oplus 9\,\mathrm{std}.$$

The associated characters on $(1), (123), (12)$ are $(26, 11, 10)$, $(24, 12, -10)$, and $(50, 23, 0)$.

This one decomposition explains both frozen dimension ledgers. A reflection has one eigenvalue of each sign on every standard copy, so

$$\dim \mathcal{H}_+ = 16 + 9 = 25, \qquad \dim \mathcal{H}_- = 16 + 9 = 25.$$

The one-dimensional representations are fixed by C_3, while the nine standard copies give eighteen rotating dimensions, so

$$50 = 32_{C_3\text{-fixed}} + 18_{C_3\text{-rotating}}.$$

The order-six phase on the 90 reduces to order three on bilinear inputs because its central sign acts twice. This is an exact representation-theoretic logic reduction. The fifty maps are not physical couplings, and the multiplicities 16 and 9 are not identified with unrelated W33 counts.

Passes 2309–2315: resolution quotient, quadratic compiler, spread boundaries, and controller fork

The complete 720-element nonlinear cover-signature quotient admits nine globally realized signatures satisfying

$$\sum_{i=1}^{9} t_i = 12\mathbf{1}_{45}.$$

Thus this quotient does not obstruct a nine-cover resolution. This is only a necessary-condition witness: compatible frame representatives have not been produced, and $\chi(H) = 9$ remains open.

The fifty complete $PSp(4, 3)$-equivariant quadratic basis maps are generated by twenty-four distinct signed-orbit tensors. Caching each orbit once reduces literal orbit-entry storage from 1,213,920 to 583,200, an exact factor 281/135. This is representation-storage sparsity, not a claim of spatial locality or minimal tensor rank.

For the regular-spread orbit, the recorded $(q + 1)$-relation valency can be one point-stabilizer orbital only for odd $q = 3, 5$. Indeed,

$$|H| = 2q^2(q^4 - 1), \qquad k = \frac{q(q-2)(q^2+1)}{2},$$

and $k \mid |H|$ forces $q - 2 \mid 24$. In particular the exact $q = 7$ strongly regular graph is not a rank-three $PGSp(4, 7)$ permutation action.

A direct non-Desarguesian control over $\mathbb{F}_9$ compares the regular spread $g = -nx$ with the Kantor spread $g = -nx^3$. Both are verified symplectic spreads of $PG(3, 9)$, but they share exactly

$$1 + 9 \cdot 3 = 28$$

lines. Hence the regular-family intersection values 1 and $q + 1$ are not universal among symplectic spreads.

The committed 36-lane spread-mixer satisfies

$$A^2 = 9I + 6J,$$

and the canonical single-J controller is frozen by all 1,152 input transitions. Finally, the overlapping arithmetic generators reduce modulo two to the $(2, 3, 7)$ Fano group $GL(3, 2)$, whereas the quadratic-Hom multiplicity controller is the $(2, 3, 2)$ group S_3. These are distinct typed control modes, not quotient descriptions of one carrier.

No withdrawn electric-charge, flux, colour, generation, or neutrino interpretation is restored. The hardware statements are semantic reference contracts, not synthesis or device measurements.

Syndrome merging, shell orbitals, and the tomotope 192 selector obstruction

A corrected seven-bit syndrome-first external merge reproduces the frozen 28-shard census and extends it to 120 primitive shards. At cutoff $B = 16$ it merges 233,088,428 records into 179,498,008 syndrome groups and detects 43,428,489 cross-shard collision pairs. This covers 3.741% of the fixed-coordinate chart and does not yet define the global U_6 coefficient.

The five exceptional-S_6 minimum-shell fibers of sizes

$$180, \quad 120, \quad 45, \quad 180, \quad 15$$

support a full orbital coherent configuration of rank 527. Its adjacency algebra is noncommutative, has 216,244 nonzero structure constants and center dimension 11. The rank identity is independently recovered as the sum of squares of the eleven S_6 multiplicities.

The frame-to-duad ABI contains an invertible 15×15 anchor submatrix, while two anchor frames already form a pointwise S_6 base. Since every W33 edge lies in nine frames, nine-colouring is equivalently a 240-clique exact-cover problem with 4,860 binaries and 2,700 equalities. A bounded run remains inconclusive, so $\chi(H) = 9$ is open.

The first common finite quotient between the arithmetic $\mathrm{SL}_3(\mathbb{Z})$ carrier and the residual tetrahedral shell is S_4: it appears as the mod-two point parabolic, as the projective Levi quotient of the mod-three parabolic, and as the S_4 quotient of the residual-duad stabilizer $S_2 \times S_4$.

Full $PSp(4,3)$ equivariance closes the linear E_8-to-coexact question negatively:

$$\mathrm{Hom}_{PSp}(8,90)=0, \qquad \mathrm{Hom}_{PSp}(90,8)=0.$$

This does not exclude proper-subgroup or nonlinear correspondences.

Finally, the 2,880 curved events admit three exact residual-duad incidence relations, each of degree 96 at every one of the 15 duads. One tomotope flag orbit also has size 96, and any two relations give 192 incidences per duad. The new obstruction is therefore a selector problem: the geometry supplies three natural 96-layers but the tomotope has two flag orbits. No canonical choice of two, labelled flag bijection, or disjoint 15×192 packet decomposition is asserted.

Passes 2410–2415: five exact frontiers

The curved-event/residual-duad atlas now has a canonical tomotope selector: the two tie profiles are exactly the profiles of event-side duad valency two. At every residual duad they give $96+96=192$ incidences, and three-point overlap splits the local carrier into eight 24-element components, each containing twelve flags from each layer. The eight neighboring duads have the natural unordered $4+4$ endpoint split.

The global equal-syndrome pair merge is also closed algebraically. If a codeword has weight $w \in \{4,6,8,10,12\}$, then its weight-six error pairs number $\frac{1}{2}\binom{w}{w/2}\binom{240-w}{6-w/2}$. The exact low-weight enumerator therefore gives 1,724,138,884,380 global collision edges. The remaining U_6 operation is union/deduplication of overlapping marks, not pair enumeration.

The literal nine-colour problem remains open. The complete fixed-frame first-colour domain contains 394,200 covers, independently reproducing the earlier census, while the exact all-different search returned `UNKNOWN` under its bounded run.

Forgetting the chosen six-line pack fuses the rank-527 exceptional-S_6 shell configuration exactly to the rank-22 full-$\mathrm{PGSp}(4,3)$ orbital configuration. Its center has dimension ten and its Wedderburn blocks have sizes 1^6 and 2^4.

Finally, the integral eight-dimensional E_8 carrier belongs naturally to $2.U_4(2)$, whereas the coexact 90 factors through $U_4(2)$. The central involution acts by $-I$ and $+I$, respectively, so every intertwiner vanishes on every lifted subgroup containing the center, including the full preimage of the common quotient-level S_4 packet.

2 Five exact frontiers: tomotope labelling, singleton identifiability, signature compatibility, shell fusions, and split-subgroup intertwiners

Labelled tomotope obstruction. The canonical two tie layers of the curved-event/residual-duad incidence give a local carrier of size $96+96=192$. Its fixed-duad stabilizer is $C_2 \times S_4$ and has four regular orbits of size 48. On the complete intrinsic ordered-pair relation of the 192 events, all twenty order-96 orbit-exchange extensions are isomorphic to $C_2^2 \times S_4$, with element-order spectrum

$$1^1\, 2^{39}\, 3^8\, 4^{24}\, 6^{24}.$$

The archived labelled tomotope symmetry instead has spectrum

$$1^1\,2^{15}\,3^{32}\,4^{24}\,8^{24}.$$

Thus the exact $192 = 96 + 96 = 8 \cdot 24$ and unordered $4 + 4$ correspondences do not lift to the archived labelled rank-four maniplex without forgetting intrinsic relations or adding noncanonical structure.

Global U_6 identifiability boundary. For weight-six syndrome multiplicities n_m, the exact pair ledger fixes

$$\sum_m m n_m = 249{,}219{,}381{,}880, \qquad \sum_m \binom{m}{2} n_m = 1{,}724{,}138{,}884{,}380,$$

and therefore

$$\sum_m m^2 n_m = 3{,}697{,}497{,}150{,}640.$$

These first two factorial moments do not determine n_1: the distributions $n_2 = 3$ and $(n_1, n_3) = (3, 1)$ both have six representatives and three collision pairs. The remaining global computation is therefore an idempotent bitmap union, not another pair sum. A deterministic 4096-shard OR-and-popcount contract is frozen for the 6,230,484,547 fixed-chart ranks.

The selected nine-signature tuple is impossible. The nine exact cover fibers of Pass 2416 have sizes

$$11664, 11664, 864, 11664, 2808, 288, 2808, 864, 288.$$

An independent exact-cover reconstruction of fibers 1 and 8 checks all

$$11664 \cdot 288 = 3{,}359{,}232$$

pairs and finds no frame-disjoint pair. Hence the selected balanced nine-signature tuple admits no nine-cover transversal. At this pass the global decision remained open; the later complete cover-link computation of Pass 2551 ruled out every nine-cover resolution and proved $\chi(H) \geq 10$.

Exact commutative fusion lattice. Closing all 65,535 binary symmetric seeds of the rank-22 full-PGSp(4, 3) orbital algebra gives sixteen distinct coherent closures, fifteen of which are commutative. Their ranks are

$$3^3,\ 4^5,\ 5^3,\ 6,\ 7,\ 8,\ 9.$$

The three rank-three schemes are $135K_4$, $45K_{12}$, and $36K_{15}$. The unique finest commutative fusion has rank nine and valencies

$$1, 256, 24, 128, 48, 48, 8, 3, 24.$$

First nonzero split-subgroup intertwiner. The full lifted carriers have opposite central characters, but the distinguished order-five subgroup avoids the center. The faithful real eight-dimensional E_8 carrier restricts to C_5 with multiplicities

$$(0, 2, 2, 2, 2),$$

while the coexact 90 restricts with multiplicities

$$(18, 18, 18, 18, 18).$$

Consequently

$$\dim \mathrm{Hom}_{C_5}(E_8, 90) = \dim \mathrm{Hom}_{C_5}(90, E_8) = 144.$$

This is an existence theorem for a large symmetry-broken Hom space, not a canonical map or physical coupling.

2.1 Five exact frontier results: syndrome union, asymmetric covers, the rank-nine scheme, tomotope obstruction, and the Sylow-five normalizer

The syndrome-first union reducer has been executed on two, four, and eight overlapping fixed-triple shards. In the largest run, 17,525,360 records reduce to 16,762,010 distinct representatives, with 5,407,125 collision-marked and 11,354,885 singleton representatives. The overlap multiplicities obey $\binom{n}{d}\binom{238-n}{4-d}$ exactly. This validates the distributed idempotent union operation but does not replace the outstanding full 4096-shard bitmap computation.

For the nine-cover problem, all signature-sum trades replacing at most four entries of the selected nine-tuple were classified. There are no two-trades, three three-trades, and eleven four-trades. Exact reconstruction of 43 signature fibers and all 903 pair relations shows that every candidate contains a pair of fibers with zero frame-disjoint cover pairs. Thus the selected tuple has no lift anywhere in its complete radius-four trade neighborhood, although $\chi(H) = 9$ remains open outside that neighborhood.

The unique finest binary-generated commutative fusion of the full-$PGSp(4, 3)$ shell algebra is a symmetric association scheme of rank nine, with valencies

$$(1, 256, 24, 128, 48, 48, 8, 3, 24)$$

and multiplicities

$$(1, 15, 15, 20, 162, 135, 108, 24, 60).$$

It is neither P-polynomial nor Q-polynomial. Its canonical imprimitivity layer is $45K_{12}$, refined into $45K_{4,4,4}$ and $135K_4$.

The natural center-preserving one-leaf-change graph on the selected curved-event 192 is biregular with degree split $2^{96}5^{96}$ and eight components of order 24. Exhausting all 15 nonempty unions of the archived rank colours shows that none reproduces this profile. Hence this principled quotient is not the archived tomotope-like rank adjacency system.

Finally, the Sylow-five normalizer in $Sp(4, 3)$ is the nonsplit group 5:8. Its center acts by -1 on the 144-dimensional C_5-equivariant Hom space, and a lifted normalizer generator satisfies $T^4 = -I$. The space is therefore 36 copies of the relevant faithful four-dimensional module and contains no normalizer-invariant map. The projective 5:4 block appears only after forgetting the central sign.

Passes 2550–2557: seven-front breakthrough packet

The global lower shadow of the weight-six syndrome problem is exactly the image of the weight-four errors and contains 91,007,752 syndromes. Sixty-three explicit singleton witnesses lie in disjoint regular $PGSp(4, 3)$ orbits, proving at least 3,265,920 global singleton fibers. The exact coefficient remains open.

The frozen frame labelling is reconstructed from the intrinsic 45-octet graph. The complete 3,547,800-cover census is reproduced, and every one of its 327 orbit-representative disjointness links is K_8-free. Consequently the 540-frame graph has no nine-colouring and $\chi(H) \geq 10$. A proper fourteen-colouring gives $10 \leq \chi(H) \leq 14$.

The rank-nine imprimitivity tower is decoded as 45 geometric $K_{4,4}$ octets, each carrying one parity half of its perfect matchings, an A_4 torsor. The three V_4 cosets give $3K_4$, their cross relation gives $K_{4,4,4}$, and the union is K_{12}. Finally, the syndrome columns themselves triangle-decompose the same $\mathrm{SRG}(45, 32, 22, 24)$ and satisfy $HH^T = 16I + A_{45}$.

For the nonsplit 5:8 normalizer module, scalar invariants first occur as two quartics, while four cubic equivariant self-maps survive. This opens a nonlinear projective channel without weakening the full-group linear obstruction.

Passes 2560–2567: singleton orbits, chromatic compression, exact character fusion, and octet quotients

The global weight-six singleton coefficient is nonzero in at least 263 disjoint regular $PGSp(4, 3)$ orbits, hence

$$U_6^{\text{singleton}} \geq 263 \cdot 51840 = 13{,}633{,}920.$$

The 540-frame graph admits a literal proper eleven-colouring, while the complete cover-link calculation excludes nine colours; therefore

$$10 \leq \chi(H) \leq 11.$$

Exact rational central projectors in the rank-32 $PSp(4, 3)$ orbital algebra resolve the rank-nine fusion as

$$162 = 81 + 81, \qquad 135 = 60 + 15 + 30 + 30, \qquad 108 = 64 + 24 + 20, \qquad 60 = 30 + 30,$$

correcting the former tentative $135 = 15+4{\cdot}30$ assignment. For the faithful four-dimensional $2.U_4(2)$ module, the cubic 5:8 covariants do not extend to the full group; the first nontrivial full-group nonlinear self-covariant is unique in degree seven. Finally, the common octet edge-phase carrier gives four exact tight frames in dimension twenty. Its mod-two quotient is the abstract Schlaefli 27-line/45-tritangent incidence, while its mod-three quotient has exactly 360 residue classes and one universal within-class difference type. No equality for the full U_6 singleton coefficient, ten-colouring, algebraic cubic-surface coordinates, or carrier-level physical septic map is asserted.

Exact controlled-add instruction and its fixed W33 line

The first two-register instruction is the qutrit controlled-add gate

$$\mathrm{CX}_{p\to f}|p, f\rangle = |p, f + p \pmod 3\rangle.$$

On Pauli-frame coordinates (x_p, z_p, x_f, z_f), its exact action is

$$(x_p, z_p, x_f, z_f) \mapsto (x_p, z_p - z_f, x_f + x_p, z_f).$$

The corresponding matrix preserves the standard symplectic form, has order three, and satisfies $(M - I)^2 = 0$ with $\mathrm{rank}(M - I) = 2$. Thus it is a rank-two Lagrangian unipotent of Jordan type $2 + 2$, not a rank-one transvection. Its image and kernel coincide in one totally isotropic two-space, so its projective fixed set is exactly one line of $W(3, 3)$. The induced cycle profiles on points, lines, flags, and collinearity edges are respectively

$$1^4 3^{12}, \qquad 1^7 3^{11}, \qquad 1^{10} 3^{50}, \qquad 1^6 3^{78}.$$

The six fixed edges are precisely the six edges of the fixed line's K_4. The seven fixed lines are the axis plus two three-line pencils attached at two axis points. Among all elements of $\mathrm{Sp}(4,3)$ with this same order, rank and square-zero Jordan data, there are two conjugacy classes. Their line actions and centralizers are

$$\begin{array}{c|c|c} 1^1 3^{13} & 240 & 216 \\ 1^7 3^{11} & 480 & 108. \end{array}$$

The controlled-add gate is in the second class. Hence the six-line fixed fringe is an exact W33 identifier for the gate's symplectic conjugacy class.

Finally,

$$\mathrm{CX}_{p\to f}(F_3 \otimes I)|00\rangle = 3^{-1/2}(|00\rangle + |11\rangle + |22\rangle),$$

and the exact generator closure is

$$\langle F_p, F_f, S_p, S_f, \mathrm{CX}_{p\to f}\rangle = \mathrm{Sp}(4,3), \qquad |\mathrm{Sp}(4,3)| = 51840.$$

These statements are certified by exhaustive ternary and finite-geometric tests. The RTL models the exact data and Pauli-frame transformations. A deterministic physical qutrit SUM has also been demonstrated inside one photon using frequency as control and time as target (Imany et al., *npj Quantum Information* 5, 59 (2019), DOI 10.1038/s41534-019-0173-8), with computational-basis fidelity 0.92 ± 0.01. The remaining hardware boundary is the implementation-specific loss/process budget and the protected M_{36} magic-state pipeline, not the SUM principle itself.

Bidirectional SUM, its centralizer, and the completed digital ISA

Let T denote the left–right transpose in the standard Pauli-frame coordinates. It is anti-symplectic,

$$T^2 = I, \qquad T^{\mathsf{T}} J T = -J,$$

and reverses the controlled-add gate:

$$T\ \mathrm{CX}_{p\to f}\ T^{-1} = \mathrm{CX}_{f\to p}\,.$$

No second entangling primitive is needed, because the exact local-Clifford identity is

$$\mathrm{CX}_{f\to p} = (F_p F_f^{-1})\,\mathrm{CX}_{p\to f}(F_p^{-1} F_f).$$

The controlled-add centralizer is determined constructively as

$$C_{\mathrm{Sp}(4,3)}(\mathrm{CX}) \cong C_6 \times C_3 \times S_3.$$

Its S_3 factor acts regularly on the six external fixed lines of the CX fringe, while the central factors preserve the two three-line pencils.

The complete exact gate atlas has 34 conjugacy classes and records cycle profiles on the 40 points, 40 lines, 160 flags, 240 collinearity edges, and 1620 apartments. These five projective carriers produce 15 distinct geometric signatures; the central-lift, inverse, and transpose metadata are therefore retained to decode the exact $\mathrm{Sp}(4,3)$ class.

A correction to the earlier physical boundary is required. Imany et al. demonstrated a deterministic single-photon time–frequency modulo-SUM gate in *npj Quantum Information* 5, 59 (2019), DOI 10.1038/s41534-019-0173-8; their qutrit realization reported computational-basis fidelity 0.92 ± 0.01 and produced $3^{-1/2}(|00\rangle + |11\rangle + |22\rangle)$. Thus the Holonet logical gate compiles directly by mapping the control register to frequency and the target register to time. The open hardware boundary is no longer the SUM principle; it is the protected M_{36} preparation, injection, decoding, and threshold stack.

Finally, all eight ISA entries now have a digital contract. The four local Clifford instructions, bidirectional CX, and $\sigma^5 = Z$ update the Pauli frame exactly; the mirror instruction multiplies an independent D_{12} bus state; and the M_{36} instruction is a typed request for one of 36 rays, exports the canonical BT822 deep/mid/shallow grade metadata with census $8 + 24 + 4$, and retires only after an external preparation block asserts acknowledgement. This last handshake preserves the evidence boundary instead of inventing a non-Clifford unitary.

The M36 factory, complete controller, and a remote qutrit SUM

Every one of the thirty-six Witting resources has exactly one dark mode and three active modes of equal magnitude. Consequently the entire library is prepared by one balanced three-mode splitter, a four-way dark-mode selector, and sixth-root phase controls. The exact ROM reproduces the $8 + 24 + 4$ deep/middle/shallow census. The controller types these states as ququart resources and fails closed before injection unless an independently verified ququart code or encoding map is supplied; qutrit distillation thresholds are not silently transferred to M36.

The complete eight-opcode controller, D_{12} transport law, bidirectional SUM, M36 pipeline, and remote feed-forward sequencer are synthesizable and exhaustively tested at the RTL contract level. An iCE40 HX8K workflow supplies independent synthesis, place-and-route, utilization, and timing evidence. Until that workflow concludes, the placed implementation remains remote pending; the local switching census is a power proxy, not a watt estimate.

For two network nodes, share

$$|\Phi_3\rangle_{ab} = 3^{-1/2} \sum_{j=0}^{2} |j, j\rangle$$

in frequency link modes. Alice applies the reverse local SUM from her time-bin data to a, measures a with outcome m, and sends one trit. Bob applies $X^{-m}F^2$ to b, uses the physical frequency-to-time SUM on his data, measures b in the Fourier basis with outcome n, and sends one trit. Alice records Z^n. All nine branches implement

$$|x, y\rangle \mapsto |x, y + x \pmod 3 angle$$

with no postselection loss beyond obtaining the heralded shared pair. The exact resource cost is one qutrit Bell pair and two classical trits. Fiber, coupling, local-optics, and detection losses enter as flagged erasures through

$$R_{\text{gate}} = R_{\text{pair}} 10^{-\alpha(L_A+L_B)/10} \eta_{cA}\eta_{cB}\tau_A\tau_B\eta_{dA}\eta_{dB}.$$

This is an exact protocol and engineering budget, not a claim of a completed combined remote-gate experiment or a fault-tolerance threshold.

3 The binary tetrahedron inside the ternary machine

The four frame coordinates carry a canonical binary coarse address. For a projective ternary point

$$[x] = [x_1 : x_2 : x_3 : x_4] \in PG(3, 3),$$

define

$$\pi([x]) = (\mathbf{1}_{x_1\neq 0}, \dots, \mathbf{1}_{x_4\neq 0}) \in \mathbb{F}_2^4 \setminus \{0\}.$$

The fifteen nonzero binary masks are the fifteen simplicial cells of a tetrahedron: four vertices, six edges, four faces and one body. The map is well-defined projectively because multiplication by -1 preserves support.

Exact phase-capacity law. If a mask S has weight w, then

$$\boxed{|\pi^{-1}(S)| = 2^{w-1}.}$$

Therefore

$$(4,6,4,1) \odot (1,2,4,8) = \boxed{(4,12,16,8)}.$$

Thus the tomotope f-vector is the exact rank-by-rank capacity profile of the ternary projective lift of the binary tetrahedron. This is a capacity theorem; it does not by itself identify the quotient incidence object with the abstract tomotope.

Equitable W33 quotient. A perfect matching of the four tetrahedral vertices determines a standard symplectic coordinate pairing. There are three such matchings. For each one, the fifteen support fibers form an equitable partition of the W33 collinearity graph. If Q is the quotient and $s_S = 2^{|S|-1}$, then

$$\operatorname{spec}(Q) = 12^1 \oplus 2^9 \oplus (-4)^5, \qquad \operatorname{diag}(s)Q = Q^\mathsf{T} \operatorname{diag}(s),$$

and

$$\boxed{Q^2 = 8I - 2Q + 4\mathbf{1}s^\mathsf{T}.}$$

The full spectrum $12^1 \oplus 2^{24} \oplus (-4)^{15}$ consequently splits into a fifteen-dimensional binary-support sector and a twenty-five-dimensional internal ternary-phase sector with residual spectrum

$$\boxed{2^{15} \oplus (-4)^{10}.}$$

Selector consequence. The tetrahedral symmetry group S_4 acts transitively on the three perfect matchings. A matching stabilizer is the nonabelian group D_8 of order eight, so

$$\boxed{24 = 8 \cdot 3 = |S_4| = |D_8|\,[S_4 : D_8].}$$

Moreover, the four Type-A selector masks

$$1110, \quad 1101, \quad 1011, \quad 0111$$

are exactly the four tetrahedral face masks. Hence the twelve admissible Type-A/Fano sheets have the intrinsic parameter set

$$\{\text{four tetrahedral faces}\} \times \{\text{three symplectic pairings}\}.$$

An objectwise intertwiner to the existing 2160×160 selector matrices is a separate verification target.

Machine use. The result supplies a support-first frame codec: first decode the fifteen-state binary occupancy shell, then decode the within-fiber ternary phase. The exact verifier and frozen certificate are `analysis/bt2808_pg32_tetrahedral_support_lift.py` and `data/PART_BT2808_PG32_TETRAHEDRAL_SUPPOR`

Passes 2809–2815: tetrahedral support shell, tomotope incidence, and all-q closure

The support projection of Pass 2808 now closes objectwise. The four weight-three masks are exactly the four Type-A selector masks and the three perfect matchings are exactly the three Fano channels. Their Cartesian product is the existing twelve-sheet selector atlas: twelve distinct signed 2160×160 operators, all of rank 81.

More strongly, the projective signed support classes $\{0, \pm 1\}^4/\{v \sim -v\}$ carry an explicit tomotope incidence model. Ranks $0, 1, 2$ are signed restrictions of support weights $1, 2, 3$. Four cells are the three-support hemioctahedra and four are the positive-product full-support tetrahedra. The resulting object has

$$(f_0, f_1, f_2, f_3) = (4, 12, 16, 8), \qquad |\mathcal{F}| = 192,$$

automorphism group order 96, two flag orbits of size 96, and symmetry class $2_{\{0,1,2\}}$; only the cell move exchanges the flag orbits.

The same support shell yields an exact codec. The 81 affine frame states are enumerated by seven bits and the 40 nonzero projective classes by six bits. The code factors through the four-bit support mask and the relative sign phase, with exhaustive round trips and frozen transition tables for F_p, $\mathrm{CX}_{p\to f}$, $\mathrm{CX}_{f\to p}$ and Z_p.

For every finite field $\mathbb{F}_q$, including even and odd prime powers, the support fibers form an equitable partition of the point graph $W(3, q)$. Put $s_S = (q-1)^{|S|-1}$, let τ be the chosen coordinate matching, and set $r = |T \cap \tau(S)|$. With

$$N_r(q) = \frac{(q-1)^r + (q-1)(-1)^r}{q},$$

the quotient entries are

$$Q_{S,T} = \frac{(q-1)^{|T|-r} N_r(q)}{q-1} - \delta_{S,T}.$$

They satisfy

$$\mathrm{spec}(Q) = q(q+1)^1 \oplus (q-1)^9 \oplus (-(q+1))^5,$$
$$\mathrm{diag}(s) Q = Q^{\mathsf{T}} \,\mathrm{diag}(s), \qquad Q^2 = (q^2-1)I - 2Q + (q+1)\mathbf{1}s^{\mathsf{T}}.$$

For the matching stabilizer D_8, the three sectors decompose as

$$A_1, \qquad 3A_1 + B_1 + B_2 + 2E, \qquad A_1 + 2B_1 + E.$$

Two further exact consequences emerged. First, the unbiased random walk on $W(3, q)$ is strongly lumpable through the support map. Its quotient spectrum is

$$1^1 \oplus \left(\frac{q-1}{q(q+1)}\right)^9 \oplus \left(-\frac{1}{q}\right)^5,$$

so its absolute relaxation rate is $1/q$. Second, the eight full-support sign classes form a parity-code controller: the even $[3, 2, 2]$ words label the four tetrahedra, while every odd word has a unique minority coordinate whose deletion labels one hemioctahedron. The sixteen triangular faces are exactly the edges of the resulting $K_{4,4}$ cell-incidence graph.

All statements are finite exact theorems. The random-walk result is not a physical dynamics law, and the RTL area/timing claims remain pending until the dedicated synthesis workflow is observed. Reproduce with `python analysis/bt2809_2815_release.py -verify-frozen` and `pytest -q tests/test_bt2809_2815_seven_frontiers.py`.

3.1 Post-2808 hardware and manuscript truth closure

Public ISA versus internal micro-ISA. The public Holonet interface remains the eight-opcode, three-bit instruction set. The minimal internal frame engine uses only

$$F_p, \qquad \mathrm{CX}_{p\to f}, \qquad \mathrm{CX}_{f\to p}, \qquad Z_p,$$

so its opcode register needs two bits. These four operations generate the full linear frame group of order 51,840 and, after adjoining translations, the affine group of order $81 \cdot 51{,}840 = 4{,}199{,}040$. The measured 72-LC, 60.80-MHz result belongs to the loadable public full-frame unit; it is not reused as synthesis evidence for the minimal four-operation engine.

Deep-grade M_{36} distillation. The exhaustive search over all 5,355 binary [[4, 2]] stabilizer projectors, all four syndromes, and all 11,520 logical Clifford decoders finds no improving branch for the shallow or either middle grade, but exactly 48 improving deep-grade branches. One explicit branch takes input ray 5, stabilizers IYZY and YZXY, syndrome $(-1, +1)$, a Hadamard on the second logical coordinate, and output ray 7. Its exact operating curve is

$$P_{\mathrm{succ}} = \frac{p^2 - 2p + 2}{4}, \qquad F_{\mathrm{out}} = \frac{5p^2 - 12p + 8}{4(p^2 - 2p + 2)},$$

with

$$F_{\mathrm{out}} - F_{\mathrm{in}} = \frac{p(p-1)(3p-2)}{4(p^2 - 2p + 2)}.$$

Hence fidelity improves for $0 < p < 2/3$. This supersedes the earlier arbitrary-decoder no-go. It is a state-fidelity distillation theorem, not a fault-tolerant injection threshold or asymptotic-yield claim.

Support for readout, phase for execution. The PG(3, 2) support shell remains an exact geometric/equitable codec, with tetrahedral capacity identity

$$(4, 6, 4, 1) \odot (1, 2, 4, 8) = (4, 12, 16, 8).$$

It is not a deterministic execution quotient. The frame states $(0, 1, 0, 0)$ and $(0, 2, 0, 0)$ have the same binary support mask, but Z_p sends them to different support masks. Deterministic refinement under the selected four-operation micro-ISA gives

$$\boxed{16 \longrightarrow 40 \longrightarrow 78 \longrightarrow 81}.$$

Thus the support shell is sufficient for readout, routing, and visualization, while the full ternary phase state is required for lossless execution.

Evidence scope. For the finite μ_{12} sensor lift the exponent is 3 at odd depth and 9 at even depth; arbitrary $U(1)$ representatives require exponent 3^n. Transpose is an anti-symplectic involution that exchanges the two controlled-add directions, with an inner projective class at $q = 5$ and an outer diagonal class at $q = 7$. The live mixer is `rtl/w33_pass2773_spread_mixer36_synth.sv`; the historical dead source was removed. Exact finite computations and remote synthesis/place-and-route measurements remain distinct evidence classes.

4 Protected support telemetry, active diagnosis, and noisy deep-grade operation

The protected support observer admits two distinct exact compression theorems. If the historical twelve-operation trajectory is held fixed, binary integer programming proves that its 52 raw support bits can be punctured to 28, but not fewer, while preserving minimum distance four. Across the eight distance-four trajectory words the exact optima are

$$(28, 29, 29, 29, 28, 29, 28, 28).$$

The canonical optimum is certified by coincident primal and dual bounds and zero MIP gap.

A second construction changes the measurement schedule. Every support observation after an affine Clifford evolution is an affine-square function

$$\mathbf{1}[a \cdot x + b \neq 0] = (a \cdot x + b)^2 \qquad \text{over } \mathbb{F}_3.$$

The 52 trajectory columns collapse to fourteen distinct functions. The optimal integer-repetition distance-four code uses twelve of them twice, for only 24 samples. This is not a puncturing theorem: it requires independent resampling of selected affine features.

The hidden symmetry is exact. The distance-four word digraph has automorphism order 32 and consists of two directed $K_{2,2}$ components. The 48 minimum eight-tap fast selectors form one block orbit under a group of order

$$6912, \qquad S_3 \times (S_4 \wr S_2).$$

Under an independent asymmetric support channel, maximum-likelihood decoding weights $0 \to 1$ and $1 \to 0$ events separately. It agrees with Hamming decoding in the symmetric model and reduces deterministic modelled word-error counts by more than seventy percent in each tested asymmetric profile. These are synthetic-channel results, not detector calibration.

For the explicit deep-grade M_{36} branch, let

$$R(p) = \frac{p(4-p)}{3(p^2 - 2p + 2)}.$$

A phenomenological output depolarization g gives $T_g(p) = g + (1-g)R(p)$ and fixed-point polynomial

$$(p-1)\big(3p^2 - (2+4g)p + 6g\big).$$

The useful and unstable fixed points merge at

$$g_c = \frac{7 - 3\sqrt{5}}{4} = 0.072949016875\ldots, \qquad p_c = \frac{3-\sqrt{5}}{2} = 0.381966011250\ldots.$$

This is a symbolic phenomenological boundary, not a circuit-level logical threshold.

Finally, support feedback can select the next probe. Exact adaptive-policy search lowers the no-reset identification horizon from six preset operations to four, with uniform mean 94/27 operations among minimum-depth policies. The machine therefore has three honest telemetry modes: eight-bit fast preset readout, twenty-four-sample protected affine readout, and four-operation worst-case active diagnosis.

Passes 2854–2860: polarization groupoids, Boolean harmonics, and exact support coarse graining

The five continuations of the tetrahedral support program and two further outside-box probes now close with exact witnesses. First, the requested full signed-monomial S_4 lift has a sharp obstruction: in one fixed symplectic polarization the coordinate-permutation projection is only the matching stabilizer D_8. Full S_4 instead lifts between the three perfect-matching polarizations as a three-object groupoid with 288 projective and 576 affine arrows. Every projective hom-set has order 32, composition closes, and the zero/nonzero symplectic relation is preserved.

The fifteen nonempty supports form the punctured Boolean permutation module

$$\mathbb{C}[\mathcal{P}([4]) \setminus \{\varnothing\}] \cong 4[4] \oplus 3[31] \oplus [22].$$

For the full four-cube, adjacency together with dual adjacency generates the Terwilliger algebra of dimension 35, with module strings of dimensions $5, 3, 1$ and multiplicities $1, 3, 2$; equivalently its Wedderburn type is $M_5(\mathbb{C}) \oplus M_3(\mathbb{C}) \oplus M_1(\mathbb{C})$. Restriction to a matching stabilizer recovers $5A_1 + 3B_1 + B_2 + 3E$ exactly.

The affine frame codec remains information-optimal: all 81 states round-trip through codes $0, \ldots, 80$ in seven bits, while the 40 nonzero projective classes require six bits. A registered benchmark isolates the one-bit storage saving from its combinational encode/decode cost; new logic-cell and timing figures remain unpromoted until the dedicated FPGA workflow completes. The existing same-harness measurements remain 43 logic cells at 72.40 MHz for the four-operation engine and 72 logic cells at 60.80 MHz for the public six-operation unit.

A typed seven-bit controller now fuses the four Type-A faces, three matching channels and eight parity-coded phase classes. It has exactly $4 \cdot 3 \cdot 8 = 96$ valid states, selecting one of twelve sheets and one of the eight tomotope cells. Equality with the tomotope automorphism-group order is recorded only as a bijective count, not yet as a regular action.

For quantum coarse graining, let $\mathcal{E}$ be the conditional expectation onto observables constant on the sixteen computational-basis support blocks of $\mathbb{F}_3^4$. An invertible affine gate descends through $\mathcal{E}$ iff its linear part is signed monomial and its translation is zero; there are exactly $2^4 4! = 384$ such maps. A Weyl displacement channel descends iff its X-displacement marginal is constant on each support fiber; the exact constraint system has rank 65 on 81 variables and a 16-dimensional solution space. The Z-phase marginal is unrestricted for support observables. This is not a coherence-preservation theorem.

Two further identities sharpen the architecture. With

$$L_q = \begin{pmatrix} 1 & q-1 \\ 1 & -1 \end{pmatrix}, \qquad L_q^2 = qI_2,$$

and $H_q = P_\tau L_q^{\otimes 4}$, one has

$$H_q^2 = q^4 I_{16}, \qquad \operatorname{spec}(H_q) = (+q^2)^{10} \oplus (-q^2)^6,$$

while for nonempty supports

$$Q_{S,T} + \delta_{S,T} = \frac{1}{q}\Big((q-1)^{|T|-1} + (H_q)_{S,T}\Big).$$

Thus the support quotient admits a four-stage, 32-butterfly realization. Finally, for its Markov matrix P and stationary projector Π,

$$(I - P + \Pi)^{-1} = \alpha I + \beta P + \gamma \Pi,$$

where

$$\alpha = \frac{q(q^2+q+2)}{(q+1)(q^2+1)}, \quad \beta = \frac{q^2}{q^2+1}, \quad \gamma = -\frac{q^3+q^2+q-1}{(q+1)(q^2+1)}.$$

At $q = 3$, the 210 directed first-passage times take only thirteen values, from 9/2 to 42, and the Kemeny constant is 291/20.

The frozen release contains 68/68 exact checks and 7/7 focused tests. The fixed-polarization obstruction, coherence boundary, typed-token boundary, and pending new FPGA measurement are retained explicitly.

5 Intrinsic selector channels, exact observer quotients, and nonlinear diagnosis

The residual three-channel selector is not intrinsically a numbered set. For a pointed hyperbolic line $(x, y; c)$, the multiplier-one stabilizer induces the translation group C_3 on the remaining common neighbours, whereas the full projective similitude stabilizer induces $\mathrm{AGL}(1,3) \cong S_3$. Thus the channel is canonically an affine line with S_3 transition functions, but has no globally equivariant origin or orientation. The exact group orders are

$$|\,\mathrm{PSp}(4,3)| = 25920, \qquad |\,\mathrm{PGSp}(4,3)| = 51840,$$

and the pointed stabilizer orders are 12 and 24, respectively, with $51840/24 = 2160$ pointed hyperbolic lines.

The $q = 3$ support quotient now has a literal sequential realization. Four stages of eight butterflies

$$(u, v) \longmapsto (u + 2v, u - v)$$

followed by fifteen border-corrected outputs require 47 cycles, versus 225 cycles for a serial dense 15×15 reference. Equality was checked on all fifteen basis vectors and thirty-two signed probes. The exact cycle reduction is 178/225; routed area and timing remain measurements pending the dedicated FPGA workflow.

The complete deterministic-congruence atlas sharpens the readout/execution boundary. For the full four-operation micro-ISA the support partition refines as

$$16 \longrightarrow 40 \longrightarrow 78 \longrightarrow 81,$$

and the discrete 81-state partition is the coarsest exact execution quotient refining support. Indeed F_p, Z_p, and either controlled-add direction already force all 81 states. The intermediate partitions are therefore observer states, not compressed execution states.

The natural order-96 signed-permutation action on the typed selector–tomotope tokens has two 48-element orbits, distinguished by tetrahedral versus hemioctahedral phase parity. Imposing the determinant-twisted condition

$$\left(\prod_i \epsilon_i\right) \operatorname{sgn}(\pi) = 1$$

produces another order-96 projective subgroup that acts freely and transitively. Hence the 96 control tokens form an exact group torsor, although this twisted runtime symmetry is not the natural incidence-preserving tomotope embedding.

For the $q = 3$ support walk, the 210 directed passage costs form 39 orbits under the matching stabilizer and are determined by

$$|T|, \qquad |T \cap \tau(T)|, \qquad |T \cap \tau(S)|.$$

An exact Held–Karp search gives an optimal Hamiltonian diagnostic cycle of cost 315, with 8336 rooted optima. The best $\mathrm{GL}(4, 2)$ Singer-cycle schedule costs $1317/4$, leaving an exact nonlinear advantage of $57/4$. Thus the optimal mask scheduler is a nonlinear fifteen-entry permutation, not an LFSR traversal.

Finally, the first observer refinement has forty classes but is not $W(3, 3)$. Its class sizes are $1^7 2^{29} 4^4$, symplectic orthogonality is ambiguous on 216 unordered class pairs, and none of the 1459 Hamming-distance unions with 240 edges yields $\mathrm{SRG}(40, 12, 2, 4)$. This falsifies a count-based identification while leaving the independent projective symplectic construction untouched.

Evidence boundary. The packet carries 64 exact checks and seven focused regressions. Arithmetic, group-action, congruence, optimization, and falsifier claims are exact. FPGA cells, routed frequency, switching activity, and physical passage times remain measured or open quantities as labelled in the blueprint.

6 Global protected support, thermodynamic diagnosis, and the OAM carrier boundary

The full affine-Clifford orbit has 120 nonconstant binary support observables $f_{a,b}(x) = \mathbf{1}[a \cdot x + b \neq 0] = (a \cdot x + b)^2$ on $\mathbf{F}_3^4$. An exact integer program gives local $AG(3, 3)$ distance-four optimum 12. There are 120 affine three-flats and each probe is nonconstant on 117, so $n \geq \lceil 120 \cdot 12/117 \rceil = 13$. An explicit sixteen-probe code on all 81 frames has distance four:

$$\boxed{13 \leq n_{\text{affine support,d=4}} \leq 16.}$$

The probes pair into an isodual but non-self-dual $[8, 4, 4]_3$ code. Its coordinate automorphism group is S_4, its monomial group is $C_2 \times S_4$, and its sixteen one-symbol syndromes are distinct. A four-trit syndrome and sixteen-entry correction table therefore replace an eighty-one-codeword binary scan.

For a uniform frame, every deterministic exact diagnostic transcript obeys $H(T) = H(X) = \log_2 81$. Adaptivity reduces latency or raw storage but cannot lower this optimally compressed Landauer information floor. Finite-time reset adds a positive, protocol-dependent excess and is not assigned a device-independent joule number.

OAM is optional at the abstract level but explicit in the operator/OAM ABI and OAM/GKP hardware profile. Exhausting all 51,840 projective symplectic-similitude actions on the forty points gives maximum cyclic orbit length 12 and no 40-cycle. One cyclic OAM shift therefore cannot implement the full geometry-preserving address bus; a physical implementation needs a multi-cycle sorter/interferometer or a hybrid mode register. The repository's radial-shell leakage figures remain symbolic falsifiers, not measurements.

For the deep-M_{36} branch the circuit-independent adversarial accepted-fault budget is $g_c = (7-3\sqrt{5})/4$. A circuit with L independent locations is inside this pessimistic envelope when $q < 1-(1-g_c)^{1/L}$; this is not promoted as a compiled-circuit threshold.

7 Exact protected observation, compiled magic, and a hybrid OAM fabric

The complete affine-support family on $\mathbf{F}_3^4$ has an exact binary minimum-length theorem at distance four:

$$\boxed{n_{\text{affine support},d=4} = 15.}$$

The lower bound combines the exact $AG(3,3)$ minimum twelve, a four-complete- direction-triplet classification of every local optimum, orbit-reduced infeasibility certificates for lengths thirteen and fourteen, and a doubled- direction packet obstruction. Equality is attained by taking all three support offsets of the ternary $[5,4,2]_3$ single-parity-check code.

The deep-M_{36} branch is now compiled explicitly. A Clifford decoder sends $IYZY \mapsto -Z_0$ and $YZXY \mapsto -Z_1$, so syndrome $(-1,+1)$ is the decoded bit test 01. After decomposing two swaps, the branch uses twelve CNOTs, three Hadamards, and two Z measurements; it accepts with probability 1/2 and maps two copies of ray 5 to ray 7. In the stated stochastic Pauli and measurement- flip model its first-order output infidelity is

$$\frac{2}{3}p + \frac{140}{81}q_1 + \frac{2084}{405}q_2 + \frac{4}{9}q_m + O(2).$$

This is a circuit law for that model, not a coherent-error or leakage threshold.

An exact spread gives a hybrid photonic address register

$$\boxed{10 \text{ OAM line modes} \times 4 \text{ time/frequency slots} = 40}$$

with sixty intra-line and one hundred eighty inter-line edges. Every ordered line pair is a four-channel permutation. Across all 120 spread-line triangles, the routing holonomy consists of sixty transpositions and sixty double transpositions; every triangular holonomy is an involution.

The canonical eight-tap observer extends to a 256-state reversible permutation whose valid outputs are the seven-bit frame ranks. Its marginal-bit entropy exceeds the joint entropy by 1.0065166696 bits, which can be exposed by reversible correlation compression; erasing the final rank still costs the $\log_2 81$ information floor.

The isodual outer code is an LCD $[8,4,4]_3$ code of covering radius two and published spectrum type $(3,4,10,12,11)$. The explicit isodual map obeys

$$D^2 = -I, \qquad D^4 = I,$$

so one four-lane signed-bin mixer can alternate between generator and parity- check roles. Finally, after correcting and uncomputing the optimal fifteen-probe record, the frame and forty-way route fuse into

$$R = 40(27x_p + 9z_p + 3x_f + z_f) + (4\ell + s),$$

a twelve-bit rank for all 3240 valid joint states. Optical loss, crosstalk, finite-time reset energy, and FPGA placement remain measured-evidence tasks.

8 Passes 2960–2966: physical observer, coherent M36, dihedral OAM curvature, and reversible compilation

The exact fifteen-probe affine-support observer admits a minimum-incidence factorization through the ternary single-parity-check code $[5, 4, 2]_3$. Up to monomial equivalence its five linear forms are

$$x_0,\ x_1,\ x_2,\ x_3,\ x_0 + x_1 + x_2 + x_3.$$

Computing each form once and sharing its three support offsets reduces linear-form incidence from 24 to 8. Only one non-native four-trit mixer is required; a balanced implementation uses three ternary adders at depth two. The corrected systematic triplets recover the four trits with eight NOT and eight CNOT operations, while the fifth triplet remains a parity syndrome.

For the compiled deep-M_{36} branch, exhaustive coherent Pauli-axis perturbation gives 189 single-location quadratic susceptibilities with spectrum

$$0^{14},\quad (1/18)^{66},\quad (2/27)^{41},\quad (1/6)^{22},\quad (2/9)^{46}.$$

Common systematic H-axis rotations have coefficients $7/18, 7/54, 7/18$ for X, Y, Z, respectively. The complete two-location Pauli census contains 16,497 events. A flagged-leakage monitor of efficiency η gives the adversarial first-order bound $p_{\text{bad}} \leq 34(1 - \eta)\ell + O(\ell^2)$ for per-location leakage probability ℓ across the seventeen circuit locations.

All 36 spreads of the present $W(3, 3)$ realization have the same routing curvature. Their 120 triangle holonomies split as sixty transpositions and sixty double transpositions. After tree gauge fixing, the thirty-six residual chords split $18 + 18$ and generate

$$\text{Hol} = D_4, \qquad |D_4| = 8,$$

with element-order profile $1^1 2^5 4^2$. The sign projection obeys all 210 tetrahedral Bianchi checks, so the surviving obstruction is nonabelian rather than an abelian parity cocycle.

A configurable ten-OAM by four-slot channel model now includes coherent nearest-neighbour OAM coupling, phase noise in a four-port Fourier multiport, insertion loss, detector efficiency, and dark counts. Its supplied parameter profiles are synthetic engineering probes and are not hardware calibrations.

The reversible compiler is no longer a truth-table placeholder. A generic completion of the 8192-state frame-route permutation has an optimized star upper bound of 39,472 multi-controlled swaps. Exploiting

$$R = 40r + a = (r \ll 5) + (r \ll 3) + a$$

produces an explicit valid-subspace network of 120 Toffoli and 79 CNOT gates. Exhaustive simulation covers all $81 \cdot 40 = 3240$ valid inputs and returns all twenty-four scratch bits to zero.

Two additional closures follow. First, three distinct pilot slots are necessary and sufficient to detect every arbitrary nonidentity S_4 permutation inserted on one edge of one spread triangle: one pilot detects 18/23, two detect 22/23, and three detect all 8280 exact fault cases. Second, the involution

$$K(x_0, x_1, x_2, x_3) = (x_1, x_0, x_3, x_2)$$

is anti-symplectic, $K^T JK = -J$. The phase transducer $\sigma(x, p) = x^T Jp$ simultaneously implements the $W(3, 3)$ commutation test, the route-dependent qutrit phase, and the affine-support probe triplet. Globally its three phase values occur 1080 times each. On the deep-M_{36} family-zero labels, K swaps $(\mu, \nu) = (1, 2)$ for ray 5 to $(2, 1)$ for ray 7. This is an exact label-level covariance, not a replacement for the compiled distillation circuit.

9 Pareto-optimal information-system closure: Passes 2967–2973

The optimized architecture retains the live controller in native mixed radix: four frame trits, ten spread-line modes, four intra-line slots, and one encode/check sector. Binary joint ranking is deferred to archive, export, or reset boundaries. This removes Boolean rank arithmetic from the live path while retaining a reversible boundary encoder.

Component-resolved Bayesian calibration is structurally necessary: the aggregate correct/wrong/erasure map has Jacobian rank two for four component parameters, so survival, OAM sorting, slot sorting, and dark counts cannot be inferred separately without intermediate counters. The committed numerical reference is synthetic and serves only to validate the inference engine.

For the K_{10} spread-line routing complex, twenty-three measured triangles are necessary and sufficient to localize one faulty edge. The lower bound follows from $3m \geq m + 2(45 - m)$, and an explicit schedule attains $m = 23$. The complete D_4-valued triangle syndrome is injective on the no-fault state and all one- and two-edge nonidentity faults, a total of 48,826 hypotheses. A 29-triangle subschedule distinguishes all of them, although minimality at weight two remains open.

The binary frame-route compiler has an exact schedule of 120 Toffoli and 79 CNOT gates, dependency depth 94, and 86 Toffoli-bearing layers. Its best architectural use is at the information boundary rather than in the live controller. Published Clifford+T, measurement-assisted, and relative-phase decompositions give distinct resource points, but their physical ranking is platform-dependent.

Two terminal SWAPs in the deep-M_{36} decoder are static detector/output relabelings. Absorbing them produces a dominating branch with six CNOTs, three Hadamards, and two Z measurements, while preserving success probability 1/2 and exact ray-7 output. In the stated stochastic model,

$$p_{\text{out}} = \frac{2}{3}p + \frac{140}{81}q_1 + \frac{956}{405}q_2 + \frac{1}{3}q_m + O(2).$$

The native controller group on $2 \cdot 81 \cdot 40 = 6480$ states has order

$$30{,}233{,}088 = 2^9 3^{10}.$$

Its route subgroup has order 192. Adding one symplectic transvection generates A_{40} on route addresses; this is an address-controller group, not an identification with the W33 automorphism group, and arbitrary words need not be physically local.

Exact Moore minimization for outputs consisting of duality sector, spread line, and symplectic phase stabilizes at 6048 future-distinguishable states. The quotient still requires thirteen fixed bits and therefore is used as a verification-level sufficient statistic rather than as the physical live representation.

Finally, translation by $a = (1, 2, 0, 0)$ combined with the order-four slot rotation gives a clock step C of order twelve. Anti-symplectic reversal combined with slot reflection gives $R^2 = 1$ and $RCR = C^{-1}$, hence a D_{12} logical clock. The full controller state space partitions into exactly 540 twelve-cycles. This is a reversible logical phase counter, not an autonomous time crystal or a demonstrated physical time standard.

10 Adaptive and self-synchronizing information closure (Passes 2996–3002)

The fixed diagnostic problem is replaced by an exact adaptive one. The frozen 23-triangle single-edge-optimal panel already separates 44,848 of the 48,826 no-, one-, and two-edge D_4 fault hypotheses. Its 1,436 remaining collision classes have size at most three. Holding those 23 triangles fixed, a binary MILP proves that six further triangles are necessary and sufficient for a fixed completion. Global fixed-schedule optimality of 29 remains open:

$$23 \le m^{\star}_{\text{fixed}} \le 29.$$

However, a collision-specific exact decision tree resolves all hypotheses with at most two additional tests. Its depth census is

$$44848 \times 23, \qquad 3566 \times 24, \qquad 412 \times 25,$$

with uniform mean 23.0899111129. Thus the adaptive protocol strictly dominates every fixed 29-triangle protocol operationally.

The earlier equality between 540 clock cycles and 540 skew-line frames does not extend to the natural order-three actions. The clock translation fixes all 540 cycle labels, whereas the corresponding W33 symplectic transvection fixes no skew-line frame and has 180 three-cycles. Their permutation characters are therefore 540 and zero. The numerical factorization $2 \cdot 27 \cdot 10 = 40 \cdot 27/2$ survives, but the natural C_3-equivariant identification is obstructed.

All ordered W33 address transfers admit exact symplectic-transvection words of length at most two. The distance census is

$$40 \times 0, \qquad 1080 \times 1, \qquad 480 \times 2.$$

One transvection suffices exactly for noncommuting address pairs; distinct commuting pairs require two.

For the explicit sparse decision prior $P_0 = 0.995$, $P_1 = 0.0045$, and $P_2 = 0.0005$, the adaptive controller escalates after the common 23 rows with probability 0.00423797155 and uses 23.0062407751 triangles on average. Its fourteen next actions need four fixed bits and have entropy only 0.0517343817 bits. This motivates a predictive controller that retains the next action and residual risk rather than an opaque sixteen-bit fault-hypothesis index.

Eight exact rewrite classes remove operations that are merely terminal wire permutations, diagonal measurement phases, Pauli byproducts, gauge-coordinate changes, route inversions, shared affine decoders, boundary-only binary ranks, or covariant ternary label inversions. The governing synthesis rule is that no operation is promoted to a physical gate until it changes a future observable in every permissible frame representation.

Finally, the balanced cyclic word

$$102332001123$$

has twelve distinct shifts, maximum off-phase autocorrelation three, and optimal cyclic distance nine. It corrects four arbitrary slot-symbol substitutions. At tick t, slot w_t is omitted and the existing three curvature pilots are sent in the remaining slots. The omitted slot supplies the mod-12 synchronization symbol while the same three pilots retain arbitrary single-edge S_4 permutation detection. No additional pilot channel is required.

Claim boundary. The conditional six-row optimum, adaptive depth, equivariance obstruction, transvection diameter, and cyclic-code properties are exact in their frozen finite models. The fault prior is synthetic; global fixed-schedule minimality of 29, noisy optical decision policies, measured transvection loss, RTL placement, and physical Landauer energy remain open.

Passes 3025–3031: Predictive Photonic Closure

Fixed and adaptive curvature diagnosis. An independent 28-triangle schedule separates all 48,826 no-, one-, and two-edge nonidentity D_4 faults and all 1,036 central-involution supports of weight at most two. A resumable 27-row cutting-plane search examined 259 feasible adversarial schedules and accumulated 12,336 exact collision cuts without finding a witness; this is bounded evidence, not a proof. Thus

$$23 \leq m^{(2)}_{\text{fixed}} \leq 28, \qquad m^{(2)}_{\text{adaptive,worst}} = 25.$$

Three-copy M_{36} audit. The displayed seeded Clifford search reproduces 19 projective stabilizer witnesses and 18 orthogonal pairs. Every pair has common same-sign Pauli subgroup order 8 and rank 3, not the rank 5 required for a six-qubit rank-two stabilizer code. Each clean projection succeeds with probability 1/6, has stabilizer Rényi-2 entropy 1.540568381363 bits, and admits a two-stabilizer decomposition with extent upper bound 2. These are magic-bearing rank-two subspaces, not stabilizer codes.

The 6,480-flag algebra. The objectwise skew-pair/spread/transversal flags form

$$\mathcal{F} \cong PSp(4,3)/V_4, \qquad |\mathcal{F}| = 6480.$$

The flag stabilizer has fixed-point character $6480, 288, 288, 24$; hence

$$\dim \operatorname{End}_{PSp(4,3)} \mathbb{C}[\mathcal{F}] = \frac{6480 + 288 + 288 + 24}{4} = 1770.$$

The numerical equality $6480 = 27(81 + 120 + 24 + 15)$ does not by itself identify the flag module with 27 Hodge copies.

Typed optical inference. A component-resolved tetrahedral A_4 shell, a route-curvature channel, and a sequential chirality channel are combined in one finite posterior controller. For the published synthetic prior and horizon, it uses route, calibration, and chirality actions in 5.12892194 expected measurements, versus a seven-action silo baseline.

Causal information geometry. The interaction

$$K(a, b) = I(S; Y_a, Y_b) - I(S; Y_a) - I(S; Y_b)$$

is nonzero because the action channels share calibration uncertainty. Raw information-per-erased-bit favors chirality, while downstream consequence-weighted dynamic programming selects route diagnosis first. Information quantity and causal decision value are therefore distinct controller resources.

Self-dual comb. Among all $369,600$ balanced length-12 four-symbol words, 384 are both distance-nine and self-dual under a D_4 reflection. The selected maximum-run-two word is

$$001032122313.$$

Crossed with the 89/233 Christoffel calendar, it yields 1,068 events in 2,796 ticks, 89 in each phase, and the exact reversal

$$R(t) = 1952 - t \pmod{2796}.$$

This is a source-complete logical schedule; optical comb coherence and autonomous timing remain unmeasured.

11 The two-layer information machine: routing, computation, prediction, and time

The newest exact results force a sharper architecture than a single universal controller graph. The machine separates six typed layers: collision-free translation routing; a selectable universal compute ISA; nonabelian D_4 route localization; high-distance self-dual phase synchronization; symmetry-quotiented prediction when invariance is certified; and irreversible readout only at the posterior decision boundary.

11.1 The fixed D_4 frontier is a distance-five parity-check problem

For the central involution r^2, the 120 triangles of K_{10} are binary parity-check rows on its 45 edges. Separating all supports of weight at most two is equivalent to hitting every nonzero edge word of weight at most four. The exact 27-row feasibility model has

$$\binom{45}{1} + \binom{45}{2} + \binom{45}{3} + \binom{45}{4} = 164220$$

constraints. A new 66-round cutting-plane run accumulated 1860 exact cuts and reached eight residual collisions, but did not produce a witness or an infeasibility proof. Hence

$$23 \le m^{(2)}_{\text{fixed}} \le 28$$

remains the theorem.

11.2 The rank-three M36 question is now exhaustively parameterized

A rank-three commuting Pauli subgroup has eight signed eigenspaces of dimension eight. The duplicate-free RREF census contains 220 pivot patterns and exactly

$$50{,}868{,}675$$

totally isotropic three-subspaces of $\mathbb{F}_2^{12}$. A local 4096-assignment pilot found no zero-leak code; the best pilot syndrome retained clean probability 1/216 with maximum single-error probability 5/72. This is an obstruction, not a no-go theorem; the full sharded census is the decision procedure.

11.3 Complete character decomposition of the 6480 flags

The flag stabilizer is $H \cong V_4$ and fuses as

$$H = 1A + 2(2A) + 1(2B), \qquad |2A| = 45, \quad |2B| = 270.$$

Thus

$$m_\chi = \frac{\chi(1) + 2\chi(2A) + \chi(2B)}{4}.$$

For irreducible degrees

$$1, 5, 5, 6, 10, 10, 15, 15, 20, 24, 30, 30, 30, 40, 40, 45, 45, 60, 64, 81,$$

the multiplicities in $\mathbb{C}[X]$ are

$$1, 0, 0, 1, 3, 3, 3, 8, 8, 10, 3, 11, 11, 6, 6, 9, 9, 14, 16, 24.$$

They satisfy

$$\sum_{\chi} \chi(1) m_{\chi} = 6480, \qquad \sum_{\chi} m_{\chi}^2 = 1770.$$

Consequently $6480 = 27(81 + 120 + 24 + 15)$ is dimensional arithmetic, not a decomposition into 27 repeated Hodge copies: the 81 occurs 24 times, the 24 occurs 10 times, the two 15's occur 3 and 8 times, and no irreducible of degree 120 exists.

11.4 Robust posterior control

The explicit hidden state

$$(\text{route}, \text{calibration}, \text{chirality}, \text{drift}, \text{regime})$$

has 32 states. A horizon-five Bayes controller treats the nominal/adverse observation regime as hidden. Under the frozen synthetic model, its adverse-case objective is 0.3065079301, compared with 0.3199090606 for the nominal-only policy, a 4.189% reduction. The first action is route diagnosis. These are exact Bellman values for the stated model, not measured receiver performance.

11.5 Self-dual timing fabric

The word 001032122313 and the 89/233 Christoffel calendar form a 2796-tick fabric with

$$R(t) = 1952 - t \pmod{2796}.$$

All ticks satisfy phase and calibration-event covariance. There are 1068 events, exactly 89 in every D_{12} phase, with global gaps 2 or 3 and per-phase gaps $24, 36, 60$. The synthesizable controller enforces a typed A_4 shell/D_4 core boundary: unauthorized core commands cannot mutate route state.

11.6 The universal ISA has a two-point Pareto frontier

Of the 210 four-opcode subsets considered, 80 are connected on the 81 frames and 24 generate the full linear group $Sp(4, 3)$ of order 51840. The minimum collision count among fully universal sets is 36, achieved for example by

$$\{CX_{fp}, F_f, F_p, Z_2\},$$

with normalized symmetrized second eigenvalue 0.8942150876. The current ISA has 45 collisions but faster mixing, 0.8290244059. Thus collision count and mixing do not share a single optimum; the compiler must select the compute ISA by task while preserving the separate collision-free translation-routing graph.

11.7 Two outside-box closures

The V_4 stabilizer partitions the 6480 flags into 1770 orbits: 24 of size one, 264 of size two, and 1482 of size four. For H-invariant transition, observation, cost, and loss kernels, Bellman recursion factors exactly through these orbits. The quotient saves two fixed register bits and 1.9111111111 average uniform bits.

Finally, the product of 48826 route hypotheses with 12 phase offsets gives 585912 noiseless route-time states. The phase code has minimum distance 21 over 28 symbols, whereas the route syndrome code has minimum distance one. The product therefore supplies strong synchronization correction but only exact route localization; route-syndrome errors require authentication or remeasurement.

Claim boundary. No 27-row impossibility, complete rank-three M36 census, laboratory optical result, physical Landauer heat, autonomous clock, or observed RTL/PDF evidence is inferred here.

12 The last chromatic bit: an exact defect filter, not a search claim

The canonical frame graph has 540 vertices, 8,640 edges, degree 32, and

$$H + 4I = MM^{\mathsf{T}}, \qquad \operatorname{spec}(H) = 32^1 \oplus 14^{44} \oplus 8^{15} \oplus 4^{81} \oplus 2^{84} \oplus (-4)^{315}.$$

The complete cover-link computation rules out nine colours, while a literal proper eleven-colouring is frozen. The exact frontier is therefore

$$\boxed{10 \le \chi(H) \le 11.}$$

No bounded heuristic or MILP non-solution is used to sharpen this interval.

Forty-five local A_4 blocks

The unique degree-one frame/octet relation partitions the frames into 45 blocks of size 12. Every induced block is

$$\boxed{K_{12} \setminus 3K_4.}$$

Thus one colour may occupy vertices in only one of the three independent four-cells. A ten-colouring of twelve local vertices must save at least two colours by repetition, giving the exact global local-pressure law

$$\boxed{\text{repeat savings} \ge 45 \cdot 2 = 90.}$$

Near-Hoffman energy

For an independent-set indicator x of size s, put

$$y = x - \frac{s}{540}\mathbf{1}.$$

Then

$$\boxed{y^{\mathsf{T}}(H + 4I)y = \frac{s(60 - s)}{15}.}$$

For a hypothetical ten-colouring let $d_i = 60 - s_i$. Since $\sum_i d_i = 60$,

$$\boxed{\sum_{i=1}^{10} y_i^{\mathsf{T}}(H + 4I)y_i = 240 - \frac{1}{15}\sum_i d_i^2 \le 216.}$$

The positive spectral gap away from E_{-4} is 6, hence

$$\boxed{\sum_i \|P_{E_{-4}^{\perp}} y_i\|^2 \le 36.}$$

Any ten-colouring must therefore lie close to the exact-cover/Hoffman eigenspace.

A ten-by-ten proof quotient

Let e_{ij} be the number of edges between colour classes i and j, and define

$$G = X^{\mathsf{T}}(H+4I)X - \frac{1}{15}ss^{\mathsf{T}}, \qquad K = 15G.$$

Every ten-colouring must satisfy

$$K \succeq 0, \qquad K\mathbf{1} = 0, \qquad \operatorname{rank} K \le 9,$$

with

$$K_{ii} = s_i(60 - s_i), \qquad K_{ij} = 15e_{ij} - s_i s_j.$$

In particular,

$$\boxed{K \equiv -ss^{\mathsf{T}} \pmod{15},}$$

so every 2×2 minor of K is divisible by 15. This arithmetic-semidefinite quotient and the 45 local blocks are exact necessary-condition filters; they do not yet decide whether the tenth colour exists.

Evidence type. All graph counts, block decompositions, identities, and the frozen eleven-colouring are finite machine-verified results. The remaining choice $\chi(H) = 10$ versus 11 is open. The equality-case language is consistent with A. Abiad, W. Bosma, and T. van Veluw, *Hoffman colorings of graphs*, Linear Algebra Appl. 710 (2025), 129–150; the W33 finite computations are independent project certificates.

13 The final chromatic bit as a finite port problem

The exact frame-graph frontier remains

$$\boxed{10 \le \chi(H) \le 11}.$$

The next reduction is not another bounded search claim. It is an exact factorisation of the 540 frames through 45 block supports, 135 local four-cells, and 240 W33 edges.

The 45×240 support incidence. Let N record which W33 edges occur in each canonical twelve-frame block. Every row of N has weight 16, every column has weight 3, and

$$\boxed{NN^{\mathsf{T}} = 16I + A_{45}},$$

where A_{45} is the adjacency matrix of $\mathrm{SRG}(45, 32, 22, 24)$. Each W33 edge labels a triangle of this block graph, and the 720 block-graph edges decompose uniquely into the 240 resulting triangles.

Three exact covers and sixteen ports per block. The three independent K_4 cells inside a block each cover the same sixteen W33 edges exactly once. Each support edge therefore chooses one frame from each cell. The sixteen resulting triples form

$$\boxed{\mathrm{OA}(16, 3, 4, 2)},$$

and every port is used by exactly two neighboring blocks. Two blocks are either frame-disconnected or share one support edge; in the latter case their cross graph is exactly $K_{3,3}$.

Equivalently, H is the line graph of the four-uniform linear hypergraph whose 240 vertices are W33 edges and whose 540 hyperedges are frames. Every hypergraph vertex has degree nine. Thus a ten-colouring would assign nine distinct colours around every W33 edge and leave a unique missing colour there.

Exact quotient and association algebra. Writing $d_i = 60 - s_i$ for the ten class deficits, there are exactly 195,490 sorted profiles with $\sum_i d_i = 60$. The defect-Gram trace is

$$\operatorname{tr} K = 3600 - \sum_i d_i^2,$$

with a unique maximum at $d_1 = \cdots = d_{10} = 6$. This exhausts the diagonal quotient, not the off-diagonal Gram entries.

The 135 local cells form a symmetric rank-four association scheme with valencies $(1, 2, 96, 36)$ and multiplicities $(1, 24, 20, 90)$. One eigenmatrix is

$$P = \begin{pmatrix} 1 & 2 & 96 & 36 \\ 1 & 2 & 6 & -9 \\ 1 & 2 & -12 & 9 \\ 1 & -1 & 0 & 0 \end{pmatrix}.$$

Its ordinary ratio bounds stop at nine. Therefore an improvement to eleven must use the split port or triangle data rather than only the commutative Bose–Mesner layer.

Arithmetic and uncertainty corrections. The Smith form of N is

$$\boxed{1^{44} \oplus 3}.$$

There is one 3-primary factor and no 5-primary torsion. Moreover, deleting any d frames from a frozen exact cover gives a binary integral row-space witness of weight $4d$. Hence linear 3-adic and 5-adic defect tests are vacuous for every possible deficit; any arithmetic obstruction must include nonlinear matching and port compatibility.

The frame-space localization spectrum of $E_{-4} = \ker M^{\mathsf{T}}$ on every block is

$$\frac{2}{9}, \quad \left(\frac{43}{81}\right)^9, \quad 1^2,$$

so two Hoffman directions localize perfectly inside each block. In contrast, the edge-kernel localization spectrum on a sixteen-edge support is

$$0^{10} \oplus \left(\frac{1}{6}\right)^6.$$

Thus the naive frame-delocalization obstruction fails, while the cut-code edge kernel is strongly delocalized.

Proof surface and boundary. A deterministic strengthened ten-colour DIMACS instance has 7,800 variables and 146,289 clauses. It exposes both frame colours and the unique missing colour at each W33 edge; its SHA-256 is `6c0c3daa...9c6b7cd5`. No SAT model and no checked UNSAT proof is claimed. The semantic exact certificate is `68e93f6b...db21d19`. Time-bounded MILP and local-search failures remain diagnostics only.

13.1 Exact-cover signature Fourier law and the one-bit symmetry overhead

The complete exact-cover census gives forty-five anchor octets. Relative to an anchor, the three independent four-cells of the induced $K_{4,4,4}$ carry one of

$$(2,2,2), \qquad (0,3,3), \qquad (1,2,3), \qquad (0,2,4).$$

Permuting the three cells produces orbit sizes $1, 3, 6, 6$. Thus the local signature alphabet is the sixteen-state S_3-set

$$\Omega \cong 1 \sqcup S_3/C_2 \sqcup S_3 \sqcup S_3, \qquad 45|\Omega| = 720.$$

Its permutation character on the identity, transposition, and three-cycle classes is

$$\chi_\Omega = (16, 2, 1),$$

and therefore

$$\mathbb{Q}[\Omega] \cong 4\mathbf{1} \oplus 2\,\mathrm{sgn} \oplus 5V_{\mathrm{std}}.$$

Consequently

$$\dim \mathrm{End}_{S_3}(\mathbb{Q}[\Omega]) = 4^2 + 2^2 + 5^2 = 45,$$

with rational commutant

$$M_4(\mathbb{Q}) \oplus M_2(\mathbb{Q}) \oplus M_5(\mathbb{Q}).$$

Direct orbital enumeration gives forty-five relations on $\Omega \times \Omega$, with size profile $1^1 3^3 6^{41}$. Hence

$$720 = 45 \cdot 16 = \dim \mathrm{End}_{S_3}(\mathbb{Q}[\Omega])\, |\Omega|.$$

This rank equality does not itself identify the forty-five anchor octets with the forty-five orbitals.

Although $|\Omega| = 16$, its exact S_3 action cannot be affine on four binary bits. The unique fixed state makes every affine realization translation-conjugate to a linear one. An exhaustive classification of all 2800 embedded S_3 subgroups of $GL(4, 2)$ finds only the orbit profiles

$$(1,1,2,3,3,6), \qquad (1,1,1,1,3,3,3,3), \qquad (1,3,3,3,6),$$

never $(1, 3, 6, 6)$. The obstruction is sharp: on

$$\mathbb{F}_2^5 = (\mathbb{F}_2^3)_{\mathrm{perm}} \oplus (\mathbb{F}_2^2)_{\mathrm{std}}$$

there is an explicit invariant sixteen-state subset with the required orbit profile and an exact equivariant encoding of all four signature classes. Therefore the minimal affine binary register has dimension five:

$$\boxed{4 \text{ information bits} + 1 \text{ symmetry bit} = 5 \text{ affine-equivariant bits.}}$$

A four-bit implementation remains possible only with nonlinear lookup/permutation logic. In particular, equal cardinality does not justify identifying this signature alphabet with the OA(16, 3, 4, 2) port alphabet without a separate intertwiner.

13.2 Signature–port obstruction and the minimal common controller

The sixteen exact-cover signature states and the sixteen natural V_4 orthogonal-array ports are not the same S_3-set. Their orbit profiles are

$$\Omega_{\mathrm{sig}} : 1+3+6+6, \qquad \Omega_{\mathrm{port}} : 1+3+3+3+6,$$

so no full S_3-equivariant bijection exists. Restriction to the orientation-preserving subgroup C_3 gives

$$1+3+3+3+3+3$$

on both sides, with exactly

$$5!\,3^5 = 29160$$

equivariant bijections. The reflection restrictions differ:

$$\Omega_{\mathrm{sig}}|_{C_2} : 1^2\,2^7, \qquad \Omega_{\mathrm{port}}|_{C_2} : 1^4\,2^6.$$

Thus the obstruction is reflection-sensitive.

The permutation characters are

$$\chi_{\mathrm{sig}} = (16, 2, 1), \qquad \chi_{\mathrm{port}} = (16, 4, 1),$$

and hence

$$\mathbb{Q}[\Omega_{\mathrm{sig}}] \cong 4\mathbf{1} \oplus 2\,\mathrm{sgn} \oplus 5V_{\mathrm{std}},$$
$$\mathbb{Q}[\Omega_{\mathrm{port}}] \cong 5\mathbf{1} \oplus \mathrm{sgn} \oplus 5V_{\mathrm{std}}.$$

The exact representation-ring defect is

$$[\mathbb{Q}\Omega_{\mathrm{port}}] - [\mathbb{Q}\Omega_{\mathrm{sig}}] = [\mathbf{1}] - [\mathrm{sgn}],$$

or stably

$$\mathbb{Q}[\Omega_{\mathrm{sig}}] \oplus \mathbf{1} \cong \mathbb{Q}[\Omega_{\mathrm{port}}] \oplus \mathrm{sgn}.$$

It vanishes on C_3, where the sign character becomes trivial.

The signature commutant has dimension

$$4^2 + 2^2 + 5^2 = 45,$$

whereas the natural port commutant has dimension

$$5^2 + 1^2 + 5^2 = 51.$$

Therefore the existing equality between forty-five anchor octets and forty-five signature orbitals cannot be lifted through the natural port action without treating the one-dimensional orientation defect.

At the set level, the orbit multiplicities are

$$\Omega_{\mathrm{sig}} : 1^1\,3^1\,6^2, \qquad \Omega_{\mathrm{port}} : 1^1\,3^3\,6^1.$$

Their minimal common S_3-envelope has profile

$$1^1\,3^3\,6^2$$

and therefore size

$$\boxed{22}.$$

The maximum equivariant overlap is

$$\boxed{10}, \qquad 16 + 16 - 10 = 22.$$

Both alphabets are embedded explicitly into the same linear S_3-action on $\mathbb{F}_2^5$. The twenty-two valid words consist of ten shared states, six signature-only states, and six port-only states; the other ten ambient words are guards.

A four-bit nonlinear signature ROM and a five-bit shared linear controller are supplied. The fixed four-bit encoding has algebraic degree three and twenty-seven ANF terms; the five-bit generator cores each use one XOR plus wire permutations before multiplexing. Source-level equivalence is exact. Observed synthesis, placement, manuscript-PDF evidence, and the unresolved chromatic decision remain separately gated. In particular,

$$10 \leq \chi(H) \leq 11$$

is unchanged. The new chromatic filter is that a full local S_3 signature/port quotient is invalid, whereas a C_3-only quotient is locally compatible.

13.3 Knight-hypercube gauge closure and the typed common controller

The local signature and natural-port alphabets become isomorphic after restriction to the orientation subgroup C_3, but the choice is highly noncanonical. The filled port complex has $\pi_1 \cong F_{436}$. After freezing the pairing of the five three-cycles, its phase group is $A = C_3^5$, and therefore

$$H^1(X; A) \cong A^{436} \cong \mathbb{F}_3^{2180}.$$

Without freezing orbit pairing, the local intertwiner group is $W = C_3 \wr S_5$ of order $3^5 5! = 29160$. Burnside enumeration over the 108 wreath-product conjugacy types gives a 1943-digit number of flat W-connections modulo switching. A trivial coboundary gauge exists, but the geometry does not select one canonically.

The minimal common envelope is the 22-state S_3-set

$$\Omega_{\text{env}} \cong 1 \sqcup 3(S_3/C_2) \sqcup 2S_3.$$

Its character and rational module are

$$\chi_{\text{env}} = (22, 4, 1), \qquad \mathbb{Q}[\Omega_{\text{env}}] \cong 6\mathbf{1} \oplus 2\,\mathrm{sgn} \oplus 7V_{\text{std}}.$$

Consequently

$$\dim \operatorname{End}_{S_3} \mathbb{Q}[\Omega_{\text{env}}] = 6^2 + 2^2 + 7^2 = 89.$$

Direct enumeration gives orbital profile $1^1 3^{15} 6^{73}$. Algebra closure over two independent large prime fields gives the basepoint-dependent Terwilliger dimensions

$$\dim T_x = \begin{cases} 89, & x \text{ fixed}, \\ 222, & x \text{ in a three-state orbit}, \\ 484 = 22^2, & x \text{ in a regular six-state orbit}. \end{cases}$$

Thus a regular signature basepoint generates the full matrix algebra and admits no further exact algebraic compression.

The ten unused five-bit words form

$$\Omega_{\text{guard}} \cong 1 \sqcup 3(S_3/C_2), \qquad \mathbb{Q}[\Omega_{\text{guard}}] \cong 4\mathbf{1} \oplus 3V_{\text{std}}.$$

In particular the guard shell contains no sign constituent. Its coherent algebra has rank 25, and its Terwilliger dimensions are 25 and 58 on the two orbit types. As an induced Q_5 network it has seven internal edges, thirty-six boundary edges, and component sizes $7, 1, 1, 1$; this is a typed detection shell, not a fault-tolerant memory.

The repository's 4×4 toroidal knight graph is exactly Q_4, and the frozen knight tour is a four-bit Gray Hamilton cycle. The nonlinear signature controller contains seven transpositions and five three-cycles. The transpositions contribute at least fourteen directed hops. Since Q_4 is bipartite, each embedded three-cycle has metric perimeter at least four, so

$$\text{total routing work} \geq 14 + 5 \cdot 4 = 34.$$

An exhaustive componentwise placement census attains the bound:

$$\boxed{\text{minimum work} = 34, \quad \text{average} = \frac{17}{16}, \quad \text{minimum dilation} = 2.}$$

There are 1105920 minimum-work labeled embeddings. After fixing translation and quotienting coordinate permutations, all 2880 representatives were routed exhaustively. No minimum-work embedding has shortest-path link congestion below three, and congestion three is attained.

The five-bit affine lift has the network decomposition

$$Q_5 = Q_4 \square K_2,$$

so it is two toroidal-knight boards joined by a sixteen-edge perfect matching. For both S_3 generators the binary column-weight profile is

$$(1, 1, 1, 1, 2).$$

Four basis directions remain one-hop routes; only the added symmetry direction is dilated to two. This turns the one-bit symmetry overhead into a literal double-board interconnect theorem.

Finally, the 45 signature orbitals are not intrinsically individually labelled. The order-72 automorphism group of the signature S_3-set acts on them with only thirteen orbits, of sizes

$$1^5\, 2^3\, 4^1\, 6^3\, 12^1.$$

Hence the equality between forty-five anchor octets and forty-five signature orbitals does not define a canonical correspondence from local data alone; any such lift needs extra global $PSp(4,3)$-equivariant structure.

A complete canonical C_3 coboundary-gauge manifest is emitted for all 45 blocks, 720 block edges and 240 filled triangles. It is an exact bijective relabelling with transported interfaces, not a reduction or decision of the current ten-colour instance. The live boundary remains

$$\boxed{10 \leq \chi(H) \leq 11}.$$

14 Global cover closure and the three hypercube roles

The complete exact-cover census contains 3,547,800 covers in 327 $PSp(4,3)$-orbits. The closed ternary-Hamming switch component contributes 1,574,640 covers in 135 orbits, leaving the exact exterior complement

$$\text{1,973,160 covers in 192 orbit classes,}$$

with stabilizer-order distribution $2^{120}4^{57}8^{15}$. The numerical equality with the 192 tomotope flags is not promoted to an identification: the exterior classes carry this nontrivial stabilizer partition and no equivariant map has been supplied.

For the rational chromatic-dual batch, deficit-profile stabilizers reduce the exact search surface from 204,287,050 to 98,191,335 rational block coordinates. This is an exact compiler reduction, not an eleven-colour certificate; the live boundary remains $10 \le \chi(H) \le 11$.

The ternary cover cube admits an exact binary host. Under the block-one-hot encoding $0 \mapsto 100$, $1 \mapsto 010$, $2 \mapsto 001$,

$$d_{Q_{15}}(\operatorname{enc} x, \operatorname{enc} y) = 2d_{H(5,3)}(x,y).$$

Hence $H(5,3)$ is the distance-two graph on 243 constant-weight-five words in Q_{15}. It is not a subgraph of an ordinary hypercube because it contains triangles. This Q_{15} host is distinct from both the toroidal-knight controller Q_4 and the secondary complement-codec Q_4; raw Levi complement adjacency remains $4K_4$.

The affine involution acts as a coordinate permutation of the 15 one-hot bits. Its traces on the six ternary Fourier grades are $(1,2,4,8,4,8)$, yielding invariant multiplicities $(1,6,22,44,42,20)$ and the exact 135-state quotient. For the reversible quotient chain $P = W/10$, the Szegedy walk has eigenphases $\exp(\pm 2i \arccos \lambda)$ for $\lambda \in \{1, 7/10, 2/5, 1/10, -1/5, -1/2\}$. This is a finite spectral compiler, not a physical speedup claim.

15 Minimum gauge defects, fault-aware hypercubes, and Clebsch recovery

The filled 45-block port complex has minimum nontrivial flat C_3^5 cochain support

$$d_{\text{gauge}} = 2.$$

Every minimum support is two edges of one filled triangle. The 720 geometric supports form one $PSp(4,3)$ orbit; for each fixed nonzero coefficient vector the orbit has size 720 and stabilizer order 36. There are 242 labeled coefficient orbits, or 121 projective directions after $v \sim -v$. Each minimum defect produces nonzero holonomy on exactly 42 unfilled triangles.

The optimal nonlinear signature routing on Q_4 has work 34 and dilation two. Mirroring it in

$$Q_5 = Q_4 \square K_2$$

preserves the same work and dilation under every one of the 32 single-vertex and 80 single-edge failures. Among two-vertex failures, loss is possible exactly for the 16 interlayer replica pairs. Under static one-cycle failover, 32 physical state slots are both necessary and sufficient.

The actual 45-anchor action has order 25,920, is transitive, and is perfect. The intrinsic signature automorphism group has derived series $72 \to 18 \to 9 \to 1$. Therefore no nontrivial action induced through the intrinsic signature coherent configuration can be equivariant with the anchor action; the identity orbital is fixed while the anchors are transitive.

At a regular envelope basepoint the local algebra is $M_{22}(\mathbb{Q})$. Nevertheless every separable positive lift $A_{45} \otimes K$, $K \succeq 0$, has extreme eigenvalues $32\lambda_{\max}(K)$ and $-4\lambda_{\max}(K)$, so its Hoffman bound remains exactly nine. Any improvement must use nonseparable, profile-sensitive cross-block orientation data. The exact chromatic boundary remains $10 \le \chi(H) \le 11$.

Validity alone detects 36/110 one-bit and 70/220 two-bit envelope faults. Exact confusability-graph colourings require 7, 11, and 16 trusted symbols to correct one bit, correct one/detect two, and correct two bits, respectively. The optimal four-bit two-error tag is the antipodal-axis label of Q_5. The quotient $Q_5/\{x \sim \bar{x}\}$ is the Clebsch graph $SRG(16, 5, 0, 2)$. Six axes contain two valid states and ten contain one valid state with its guard antipode. The decoder exhausts all 352 error patterns of weight at most two, conditional on a trusted tag.

These are exact finite and source-level digital results. They do not assert physical fault probabilities, observed FPGA timing, fault-tolerant quantum memory, a chromatic decision, or laboratory performance.

16 Minimum gauge defects generate the full flat connection space

The filled port complex has 45 vertices, 720 edges, and 240 edge-disjoint filled triangles, with coefficient module $A = C_3^5 \cong \mathbb{F}_3^5$. On one oriented triangle the scalar flat plane is

$$\{(a, b, c) \in \mathbb{F}_3^3 : a + b + c = 0\},$$

which has rank two and is spanned by its six nonzero weight-two vectors. Since the filled triangles partition all graph edges, these local planes sum directly. Consequently

$$\dim Z^1(X; \mathbb{F}_3) = 240 \cdot 2 = 480, \qquad \boxed{\dim Z^1(X; \mathbb{F}_3^5) = 2400}.$$

The block graph is connected, so

$$\dim B^1(X; \mathbb{F}_3^5) = (45 - 1) \cdot 5 = 220,$$

and therefore

$$\boxed{\dim H^1(X; \mathbb{F}_3^5) = 2400 - 220 = 2180}.$$

Thus the weight-two minimum defects do not merely witness the minimum nontrivial class: they span every flat C_3^5 connection, and every switching class has a representative assembled from minimum defects. This is a generation theorem, not a unique-decoding or multi-fault-correction claim.

Passes 3364–3375: Clebsch–Petersen resilient closure

The folded five-cube checksum graph is the Clebsch graph, SRG$(16, 5, 0, 2)$, with rooted shell decomposition $1 + 5 + 10$. The ten distance-two vertices induce $KG(5, 2)$, the Petersen graph. Its Cayley form on $\mathbb{F}_2^4$ has five circuit directions and automorphism group $2^4{:}S_5$ of order 1920.

Preserving the five raw envelope bits systematically, exhaustive search proves that seven parity bits are necessary and sufficient for a 22-word code of minimum distance five. Thus a self-protecting two-error-correcting envelope word requires 12 total bits. The mirrored Q_5 placement has exactly 16 catastrophic two-vertex sets and 480 catastrophic three-vertex sets, precisely those containing a complete replica pair.

The full block-distance-invariant nonseparable local-algebra lift reduces exactly to the three PSD cones

$$K_0 + 32K_1 + 12K_2 \succeq 0, \quad K_0 + 2K_1 - 3K_2 \succeq 0, \quad K_0 - 4K_1 + 3K_2 \succeq 0.$$

This compiler does not decide the ten-colour problem; the live boundary remains $10 \le \chi(H) \le 11$.

Minimum gauge defects obey a local fusion law: same omitted-edge types add coefficientwise and annihilate for opposite coefficients; distinct types in one filled face remain minimum exactly when their shared-edge coefficients cancel; distinct filled faces are edge-disjoint. Projectively, the 87,120 minimum defects contain 29,040 local three-state fusion triples.

The Clebsch graph also supplies a nested checksum geometry: one reference axis, five singleton fault directions, and ten pair channels forming a Petersen graph. A degree-four del Pezzo/Segre quartic supplies a separate 16-line incidence realization of the same graph, retained only as a mathematical dictionary and not as a physical hardware identification.

17 The hidden torus as a finite d'Alembertian

For the hidden 27-state signed-torus block,

$$D = A(C_3)_3 - A(C_3)_1 - A(C_3)_2.$$

Since $\Delta_i = 2I - A(C_3)_i$, one has the exact identity

$$\boxed{D + 2I = \Delta_1 + \Delta_2 - \Delta_3.}$$

Thus the shifted hidden block is a finite discrete d'Alembertian on C_3^3, with two positive Laplacian factors and one negative factor. At Fourier frequency $k \in \mathbb{F}_3^3$,

$$\lambda(k) = 3\big([k_1 \neq 0] + [k_2 \neq 0] - [k_3 \neq 0]\big),$$

so its spectrum is

$$\boxed{6^4,\ 3^{12},\ 0^9,\ (-3)^2}.$$

The rank is 18 and the nullity is 9. The null modes consist of the constant character together with four frequencies satisfying $k_1 = 0$, $k_2, k_3 \neq 0$, and four satisfying $k_2 = 0$, $k_1, k_3 \neq 0$. Hence the exact null profile is $1 + 4 + 4$. This is a finite signed-graph identity only; no physical spacetime, Lorentzian continuum, causal cone, or laboratory claim is made.

18 Minimum-defect normal forms and the hidden signed ternary torus

The flat C_3^5 port connections admit an exact minimum-defect word metric. On one filled triangle there are 1 states of length zero, 726 states of length one, and 58,322 states of length two, so the local enumerator is

$$1 + 726z + 58{,}322z^2.$$

Because the 240 filled faces partition all 720 edges, the global flat-space enumerator is its 240th power and the exact flat-space diameter is 480. A sphere-counting argument against the 3^{2180} switching classes gives the certified quotient bound

$$\boxed{389 \le \operatorname{diam} H^1 \le 480}.$$

The upper endpoint is the universal minimum-defect normal form; the exact quotient covering radius inside this interval remains open.

The minimum defects also give a canonical $PSp(4,3)$-equivariant presentation. For scalar coefficients,

$$0 \longrightarrow \mathbb{F}_3[240 \text{ faces}] \longrightarrow \mathbb{F}_3[720 \text{ supports}] \longrightarrow Z^1 \longrightarrow H^1 \longrightarrow 0,$$

with dimensions $240, 720, 480, 436$ after quotienting the 44-dimensional coboundary space. Tensoring by the externally labelled five-dimensional coefficient module gives 3600 generators, 1200 local face relations, 220 coboundary relations, and

$$3600 - 1200 - 220 = \boxed{2180}$$

logical dimensions. Thus the full logical sector is presented by minimum defects with exactly 1420 independent relations.

For the affine involution

$$\tau(x_1, x_2, x_3, x_4, x_5) = (-x_4, 1 - x_3, 1 - x_2, -x_1, x_5)$$

on $\mathbb{F}_3^5$, the 135 orbit species have only 108 distinct block-one-hot Q_{15} barycentres. Their fibre profile is

$$81 \text{ singletons} + 27 \text{ double fibres}.$$

The 27 double fibres occur exactly at ternary displacement four. Taking the coordinatewise missing symbols maps their barycentres bijectively to the 27 fixed points of τ. The two orbit species above each such centre are distinguished by the invariant binary correlation

$$h = (x_1 + x_4)(x_2 + x_3 - 1) \in \{1, 2\}.$$

Consequently the 54 ambiguous species form a canonical two-sheeted cover of the fixed affine flat $\mathbb{F}_3^3$.

More strongly, barycentric aggregation is exactly lumpable for the weighted Hamming-orbifold walk. Its 135-state operator splits as

$$\boxed{W_{135} \cong B_{108} \oplus D_{27}}.$$

The barycentric block has spectrum

$$10^1,\ 7^6,\ 4^{18},\ 1^{32},\ (-2)^{33},\ (-5)^{18},$$

and itself has the exact three-shell quotient

$$\begin{pmatrix} 2 & 8 & 0 \\ 2 & 4 & 4 \\ 0 & 4 & 6 \end{pmatrix}$$

on shells of sizes $27, 54, 27$, with spectrum $10, 4, -2$.

Orienting each double fibre by $h = 1$ versus $h = 2$ identifies the hidden block with the signed Cayley operator on $\mathbb{F}_3^3$

$$\boxed{D_{27} = C_3^{(3)} - C_3^{(1)} - C_3^{(2)}}.$$

Equivalently it has weight $+1$ on the two $\pm e_3$ steps and weight -1 on the four $\pm e_1, \pm e_2$ steps. Its exact spectrum is

$$4^4,\ 1^{12},\ (-2)^9,\ (-5)^2.$$

This is a finite signed-torus theorem, not a spacetime-signature claim.

Finally, a single minimum defect used as a C_3^5 voltage assignment generates only the subgroup $\langle v\rangle \cong C_3$. The full 10,935-vertex derived graph therefore splits into

$$\boxed{\text{81 connected components of 135 vertices each}}.$$

Each component is 32-regular with 2,160 edges and 15,714 triangles. Fourier decomposition across the C_3 fibre gives one untwisted 45-state block and two conjugate magnetic blocks. Their exact moments satisfy

$$\operatorname{tr} M^2 = 1440, \qquad \operatorname{tr} M^3 = 31{,}302,$$

whereas the untwisted cubic trace is 31,680. The $378 = 42 \cdot 9$ deficit proves that this is a genuine nonseparable cohomological weighting. It is a candidate chromatic-dual surface, not yet a certificate improving the bound nine. Its cardinality match with the 135 Hamming-orbit species is also not an isomorphism: the voltage component has degree 32, while the Hamming-orbifold walk has degree 10.

19 Minimum-defect radius, the finite null conic, and a matrix-valued ternary torus (Passes 3404–3417)

The minimum-defect word metric on the switching quotient admits the sharper exact interval

$$\boxed{389 \le D_{H^1} \le 436}.$$

The lower bound is the existing sphere-count obstruction. For the upper bound, the scalar cohomology has dimension 436 and is generated by minimum-support defects, so one may select 436 such supports as a basis. After tensoring with $\mathbb{F}_3^5$, an arbitrary nonzero coefficient vector on one selected support is still one minimum defect; hence every switching class uses at most 436 basis defects.

The nine-dimensional kernel of the shifted hidden operator is the zero mode together with the eight nonzero vectors satisfying

$$k_1^2 + k_2^2 - k_3^2 = 0 \qquad \text{in } \mathbb{F}_3.$$

Projectivization identifies the eight lifts in opposite pairs and produces a nondegenerate four-point conic in $PG(2,3)$. Its projective stabilizer has order 24 and acts as the full S_4 on the four conic points.

The four projective null directions form the columns of a ternary MDS code

$$\boxed{[4,3,2]_3},$$

with weight enumerator $1 + 12z^2 + 8z^3 + 6z^4$ and dual the signed repetition code generated by $(1,1,2,2)$, of parameters $[4,1,4]_3$.

More strongly, the 108 barycentric states admit an exact factorization

$$\boxed{108 = 27 \cdot 4 = |\mathbb{F}_3^3|\,|\mathbb{F}_2^2|}.$$

For every center in the fixed affine flat there are exactly four barycentric channels: one fixed channel, two distance-two channels, and one ambiguous channel. The adjacency is translation invariant in the center coordinate, so it is a 4×4 matrix-valued Cayley operator with six nonzero kernels at $\pm e_1, \pm e_2, \pm e_3$. Three qutrit Fourier transforms reproduce the full barycentric spectrum from 27 four-dimensional symbols.

Adding the one-dimensional hidden signed channel gives

$$\boxed{135 = 27 \cdot 5},$$

so the complete walk reduces to 27 internal symbols of size 5. This yields an exact mixed-radix compilation contract: three qutrit center registers, one five-level internal register, and two Szegedy reflections per walk step. Uniform state preparation over ten neighbours and synthesis of controlled 5×5 symbols still require an approximation tolerance before any Clifford+T count can be certified.

The isolated magnetic-defect search closes negatively. The ternary phase family has

$$\chi_\omega(x) = (x-2)^{22}(x+4)^{18}\big(x^5 - 28x^4 - 140x^3 + 478x^2 + 1448x - 1328\big),$$

with generalized Hoffman ratio approximately $7.2835002808 < 8$. The real signed family has residual quintic

$$x^5 - 28x^4 - 140x^3 + 520x^2 + 1568x - 1088$$

and ratio approximately $7.0613187294 < 8$. Rational root isolation certifies both inequalities. Thus one profile-sensitive minimum defect cannot improve the live chromatic lower bound.

The literal 720-edge $PSp(4,3)$ permutation-character decomposition remains evidence-gated until its GAP-generated JSON is frozen. The live chromatic boundary remains

$$\boxed{10 \leq \chi(H) \leq 11},$$

and the finite quadratic form above is not identified with physical spacetime or Lorentz symmetry.

19.1 The 720-edge ordinary degree atlas (Pass 3408)

The minimum-support carrier is the transitive action of

$$PSU(4,2) \cong PSp(4,3)$$

on the 720 non-collinear point pairs of $H(3,4) = GQ(4,2)$. A point stabilizer has order 36 and subdegrees

$$1, 2, 3, 6, 9, 9, 12^5, 18^{11}, 36^{12},$$

so the orbital rank is 34.

Deterministic generic elements of the exact orbital commutant give the complex isotypic dimensions

$$1, 15, 15, 20, 20, 24, 24, 24, 30, 30, 45, 45, 60, 60, 64, 81, 81, 81.$$

The ordinary degree/multiplicity profile is therefore

$$\boxed{1 + 15_a + 15_b + 2 \cdot 20 + 3 \cdot 24 + 30_a + 30_b + 45_a + 45_b + 2 \cdot 60 + 64 + 3 \cdot 81.}$$

It closes both the carrier dimension,

$$1 + 30 + 40 + 72 + 60 + 90 + 120 + 64 + 243 = 720,$$

and the character norm, equal to the orbital rank 34.

The group, orbit, stabilizer, subdegrees, relations, and commutant matrices are exact. The isotypic dimensions are recovered from deterministic well-separated numerical eigenspaces and constrained by the exact dimension and rank identities. GAP/CTblLib remains an independent row-label check, especially for naming the selected conjugate degree-30 pair.

20 Passes 3418–3429: supplemental Clebsch–D_5 closure

The executable source wrappers retain a provisional build tag only for hash continuity after a namespace collision. Building on the parallel Clebsch–Petersen packet without replacing it, the source-complete verifier reports **PASS 11/11**. All graph, code, orbit, fusion, routing, and fault counts in this section are exact. The magnetic spectral family below is an explicitly marked high-margin numerical audit rather than a rational SDP certificate. The chromatic boundary remains

$$\boxed{10 \le \chi(H) \le 11}.$$

20.1 Edge-dependent magnetic phases do not improve the bound

For a flat C_3 edge cochain z, set

$$M(z)_{ab} = \omega^{z_{ab}}, \qquad M(z)_{ba} = \overline{M(z)_{ab}}.$$

The search exhausts the 24 $PSp(4,3)$-orbits of unordered pairs of the 720 minimum-defect supports and both relative coefficient signs. Among all 48 one/two-defect representatives, the best nonzero spectral value is

$$\lambda_{\min} \simeq -5.073280337457312, \quad \lambda_{\max} \simeq 31.877958425086288, \quad h \simeq 7.283500280818951 < 9.$$

Thus edge-dependent phases alone are insufficient; an improved dual must also retain nonseparable profile-sensitive amplitudes.

20.2 Self-protecting state codes

For arbitrary systematic encoding of the 22-state envelope, exhaustive search proves that seven parity bits are necessary and sufficient for distance five:

$$\boxed{(n, M, d) = (12, 22, 5)}.$$

The exhaustive radius-two decoder surface contains $22(1 + 12 + \binom{12}{2}) = 1738$ received words. In the linear systematic case, every five-bit difference occurs, so the problem is a binary $[5 + r, 5, 5]$ code. All 26,334 six-parity candidates and all 7,624,512 seven-parity candidates fail; eight parity bits succeed:

$$\boxed{[13, 5, 5]}, \qquad (c_0, \ldots, c_4) = (15, 51, 85, 106, 150).$$

The 12-bit code is information-theoretically smaller; the 13-bit code is XOR-linear over all 32 words.

20.3 Six-spare dynamic recovery

Static state-availability mirroring uses 32 physical slots. Under an explicit protected-symbolic-state/recompilation contract, allowing bounded state remapping while preserving every original transition distance changes the frontier. Every one of the 6,884 fixed spare banks of size at most five fails. At size six there are exactly 96 solutions; the first is

$$\boxed{\{0, 1, 2, 3, 4, 5\}}.$$

Therefore 22 physical slots are necessary and sufficient for post-fault symbolic remapping at the original work 34 and dilation two.

20.4 Gauge fusion and the local A_2 root plane

The 720 minimum supports have 24 unordered-pair orbits. Their local fusion laws include annihilation, coefficient reversal, third-root completion, and a three-edge filled-face state. On one filled triangle the six weight-two flat cochains are precisely the six nonzero roots of an A_2 plane over $\mathbb{F}_3$. Consequently

$$Z^1(X;\mathbb{F}_3^5) \cong \big(A_2(\mathbb{F}_3) \otimes \mathbb{F}_3^5\big)^{\oplus 240}, \qquad \dim Z^1 = 2400,$$

with the 220-dimensional switching subspace leaving $\dim H^1 = 2180$.

20.5 Triple faults and the Q_6 boundary

The mirrored Q_5 census is

faults	cases	catastrophic	worst work	worst dilation
EEE	82160	0	40	3
VEE	101120	0	42	4
VVE	39680	1280	45	4
VVV	4960	480	46	4

Three Q_4 layers, or 48 slots, are cardinality-minimal for arbitrary two-vertex loss. Four layers give $Q_6 = Q_4 \square Q_2$, use 64 slots, and are necessary and sufficient in the untouched-layer model for preserving work 34 and dilation two after any three layer-local faults.

20.6 The Clebsch checksum is a repetition-code coset graph

The four-bit tag

$$t(x) = x_{3:0} \oplus x_4(1111)_2$$

is a parity-check syndrome for the binary repetition code $\mathrm{Rep}(5) = [5, 1, 5]$. Hence

$$\boxed{Q_5/\,\mathrm{Rep}(5) = \mathrm{Clebsch} = \mathrm{SRG}(16, 5, 0, 2)}.$$

Every local checksum chart has distance shells

$$\boxed{1 + 5 + 10}.$$

The five one-error neighbors are independent, while the ten two-error symbols induce the Petersen graph $KG(5, 2)$: two double-error syndromes are adjacent exactly when their error-position pairs are disjoint.

The 16 closed neighborhoods form a symmetric biplane

$$\boxed{2 - (16, 6, 2)}, \qquad BB^{\mathsf{T}} = 4I + 2J.$$

Exact automorphism enumeration gives $|\,\mathrm{Aut}(\mathrm{Clebsch})| = 1920 = 2^4 \cdot 5!$. In the even-sign model the complement is the 5-demicube skeleton and the Clebsch edges are the opposition relation on one 16-weight D_5 half-spin orbit. The controller's valid/guard coloring leaves only an elementary Abelian stabilizer C_2^3.

20.7 Equivariance versus correction radius

The repetition kernel is not controller invariant: its generator has orbit

$$\{11111_2, 01111_2, 10111_2\}$$

of weights $5, 4, 4$. The unique common invariant line is generated by 00111_2 and has distance three. Therefore no fixed one-dimensional controller-equivariant quotient gives uniform two-bit correction. One must re-encode the Clebsch syndrome, transport a moving quotient chart, or use the self-protecting 12/13-bit code.

Boundary. No ten-colour decision, exact rational magnetic certificate, recovery of destroyed uncheckpointed state data, measured fault rate, observed FPGA timing, successful fresh PDF, physical D_5 or Spin(10) identification, quantum speedup, laboratory operation, or spacetime claim is inferred from this finite packet.

21 Generalized defect radius, modular cohomology, and the five-channel crossed algebra (Passes 3444–3457)

Let $E = \mathbb{F}_3^{720}$ be the minimum-support permutation module. If R is the filled-face relation map and N the unsigned vertex–edge incidence map, exact row reduction gives

$$\operatorname{rank} R = 240, \qquad \operatorname{rank} N = 45, \qquad \operatorname{rank}[R\ N] = 284,$$

with $\operatorname{im} R \cap \operatorname{im} N = \langle \mathbf{1}_E \rangle$. Hence the scalar cohomology has the objectwise presentation

$$0 \longrightarrow \mathbf{1} \longrightarrow \mathbb{F}_3[240 \text{ faces}] \oplus \mathbb{F}_3[45 \text{ vertices}] \longrightarrow \mathbb{F}_3[720 \text{ supports}] \longrightarrow H^1 \longrightarrow 0,$$

and therefore

$$\boxed{\dim_{\mathbb{F}_3} H^1 = 436}.$$

Dually, $(H^1)^*$ is the 436-dimensional edge code whose coordinate sum vanishes on every filled face and at every vertex.

After an arbitrary $\mathbb{F}_3$-linear identification $\mathbb{F}_3^5 \cong \mathbb{F}_{243}$, the labelled minimum-defect diameter is the fifth generalized covering radius of the ternary code

$$K = \operatorname{im}[R\ N],$$

equivalently the ordinary covering radius of its scalar extension to $\mathbb{F}_{243}$. One filled face contributes the three-direction Latin-square Cayley graph

$$\mathrm{SRG}(59049, 726, 243, 6),$$

with spectrum $726^1, 240^{726}, (-3)^{58322}$ and local length enumerator $1+726z+58322z^2$. The relation-aware sphere obstruction remains stronger than the ordinary 243-ary Hamming count and yields

$$\boxed{389 \le R_{\text{defect}} \le 436}.$$

The exact endpoint inside this interval remains open.

The four-channel barycentric Fourier symbol factors through the equitable $1+2$ quotient of K_3,

$$M = \begin{pmatrix} 0 & 2 \\ 1 & 1 \end{pmatrix},$$

as

$$B(k) = c(k_1)(I_2 \otimes M) + c(k_2)(M \otimes I_2) + c(k_3)I_4, \qquad c(t) = \begin{cases} 2, & t = 0, \\ -1, & t \neq 0. \end{cases}$$

The hidden channel is $d(k) = c(k_3) - c(k_1) - c(k_2)$. Thus only eight zero/nonzero masks are required, rather than 27 unrelated five-dimensional symbols.

The torus amplitudes alone generate a four-dimensional commutative algebra. Adding the coordinate swap S raises the algebra dimension to six, while adding the null-conic dual sign

$$D = \mathrm{diag}(1, 1, -1, -1)$$

raises it to eight. Together,

$$\boxed{\langle I_2 \otimes M, M \otimes I_2, S, D \rangle = M_4(\mathbb{Q})}.$$

This supplies an explicit noncommuting amplitude extension beyond the exhausted phase-only magnetic families, but it is not yet a chromatic certificate; the live boundary remains $10 \leq \chi(H) \leq 11$.

Modulo three, $2 \equiv -1$, so all 27 momentum symbols collapse to one matrix J. Writing $\mathcal{N} = J - I$, exact ranks give

$$\mathrm{rank}\,\mathcal{N} = 2, \qquad \mathrm{rank}\,\mathcal{N}^2 = 1, \qquad \mathcal{N}^3 = 0,$$

so J has Jordan type $J_3(1) \oplus J_1(1) \oplus J_1(1)$ and

$$\boxed{J^3 = I}.$$

The accompanying RTL exhausts all 200 integer-symbol entries modulo three and all $8 \cdot 3^5 = 1944$ mask/state triples.

Finally, after choosing one Fano point and one D_5 coordinate, three degree-four sets carry the same faithful S_4 action: the four projective null-conic directions, the four Fano lines avoiding the selected point, and the four residual D_5 coordinates. The choices are essential, so this is an equivariant chart bridge rather than a canonical ambient-geometric identification.

The local A_2 code $[3, 2, 2]_3$ tensored with the null-conic code $[4, 3, 2]_3$ gives the additional exact construction

$$\boxed{[3, 2, 2]_3 \otimes [4, 3, 2]_3 = [12, 6, 4]_3},$$

with visible $S_3 \times S_4$ coordinate action and dual parameters $[12, 6, 3]_3$. Its length agrees with the tomotope edge count, but no tomotope incidence realization is asserted without an explicit map.

21.1 The filled-face tower, modular descent, and the tomotope boundary

The selected orbit of 240 filled triangles carries a canonical $PSp(4, 3)$-equivariant antipodal involution. Its quotient has 120 states and a symmetric rank-five association scheme with valencies

$$1, 36, 27, 2, 54.$$

The valency-two relation is $40K_3$. Quotienting its forty components recovers the original graph

$$\mathrm{SRG}(40, 12, 2, 4) = W(3, 3).$$

Consequently the filled-face carrier has the exact tower

$$\boxed{240 \longrightarrow 120 \longrightarrow 40}.$$

Each W33 point supports six filled faces, arranged as three antipodal pairs. Its stabilizer has order $648 = 3^3 \cdot 24$ and induces the full S_4 action on the six edges of a tetrahedron; the induced action on the three antipodal pairs is S_3 with kernel V_4.

Across a W33 edge, one pair-scheme relation is the complete 3-by-3 block. Across a W33 nonedge, a second relation is a permutation matrix and its companion is the complementary block. The resulting matching connection has

$$1080 \text{ identity triangle loops}, \qquad 2160 \text{ transposition loops},$$

and no three-cycle loops on the triangles of the W33 complement. This is an objectwise nontrivial transport bundle, unlike the exhausted scalar phase-only families.

In characteristic three, all five ordinary characters of the 120-state association scheme reduce to $(1, 0, 0, 2, 0)$. Its endomorphism algebra is local, with radical tower

$$\dim J = 4, \qquad \dim J^2 = 1, \qquad J^3 = 0,$$

so the 120-dimensional pair module is indecomposable. The antisymmetric antipodal summand decomposes as

$$M_- = M_{81} \oplus M_{39},$$

where the 39-dimensional summand has endomorphism ring $\mathbb{F}_3[\varepsilon]/(\varepsilon^2)$ and glues the ordinary 15- and 24-dimensional sectors. The 81-dimensional summand is retained as a brick pending an independent MeatAxe composition-series calculation. This face-permutation result is complementary to the later edge-module filtration

$$0 < R_{240} < Z^1_{480} < E_{720},$$

which produces the distinct 196-dimensional quotient $Z^1/(R + B^1)$.

The six local face tokens map to the six transposition matrices in the four-dimensional tetrahedral permutation representation. Their group algebra image has dimension ten. Adjoining the null-conic dual sign $\mathrm{diag}(1, 1, -1, -1)$ generates all of $M_4(\mathbb{Q})$. Thus scalar phases, the later finite dyadic amplitude grid, and a trivial untwisted M_4 fibre are all insufficient, while the face tower supplies the first explicit objectwise noncommuting transport extension. This is a compiler surface, not yet a chromatic certificate; the live boundary remains

$$10 \le \chi(H) \le 11.$$

For the generalized covering-radius front, the transitive fractional/Delsarte relaxation reaches the quotient size first at radius 389, so level zero cannot improve the lower bound. The exact rank-ten face coherent configuration, rank-five antipodal scheme, eigenmatrix, and intersection tensor provide the symmetry-reduced input for a higher semidefinite hierarchy. The later certified 247-circuit basis-exchange theorem improves the upper bound to 435, and a dependency verifier binds this insert to that frozen certificate. The live interval is

$$\boxed{389 \le R_{\mathrm{defect}} \le 435}.$$

Finally, the ternary $[12, 6, 4]$ A_2–conic product code does not realize the tomotope faces. Exhaustion of all 1760 weight-three ambient vectors shows that a dual-code coset contains at most four projective triple supports, whereas the tomotope requires sixteen triangular faces. Nevertheless the same 3×4 coordinate grid has a positive incidence interpretation: matching rows and vertex columns identify the coordinates with the twelve oriented tetrahedron edges. Four outgoing stars, four incoming stars, and eight directed three-cycles form an exact $12_4\, 16_3$ Reye-style incidence surface. The archived negative flag search remains decisive: this skeleton alone does not determine the missing cell/orientation cocycle needed for the true 192-flag tomotope monodromy.

21.2 Radius, amplitude, modular, and five-channel closure

An exact basis-exchange certificate in the minimum-defect Cayley system produces a 247-term circuit with 246 nonzero basis coordinates. Since the coefficient alphabet has only $3^5 - 1 = 242$ nonzero vectors, a nonzero circuit shift cancels at least two basis coefficients while adding one extra generator. The covering-radius interval therefore improves to

$$\boxed{389 \le D_{H^1} \le 435}.$$

The exact radius remains open.

A five-channel amplitude-bearing magnetic family was then screened on a 63,869-point dyadic grid. Its best member has weights

$$\left(-\frac{21}{32}, \frac{43}{32}, -\frac{27}{32}, -\frac{15}{16}, 1\right)$$

and generalized Hoffman ratio approximately 8.9024605237. After scaling by 32, its exact characteristic polynomial contains $(x + 128)^{17}(x - 64)^{21}$ and a seven-dimensional residual whose roots all lie strictly in $(-128, 1024)$. Consequently the unscaled matrix has $\lambda_{\min} = -4$ and $\lambda_{\max} < 32$, proving that this winner remains strictly below nine. This is a finite-grid no-go, not an optimization of the unrestricted real cone.

Over $\mathbb{F}_3$, the 240 face boundaries form a totally isotropic subspace of the 480-dimensional flat edge space. They are disjoint from the 44-dimensional coboundary space, giving the exact filtration

$$0 < R_{240} < Z^1_{480} < \mathbb{F}_3^{720}, \qquad \dim Z^1/(R + B^1) = 196.$$

No modular irreducible name is assigned without a composition-series certificate.

The full 135-state quotient admits an executable Fourier organization

$$135 = 27 \times 5.$$

For $c(0) = 2$ and $c(1) = c(2) = -1$, the four barycentric channels have symbol $c(k_1)K_x + c(k_2)K_y + c(k_3)I_4$, while the hidden channel is $-c(k_1) - c(k_2) + c(k_3)$. Exhausting all 675 symbol entries recovers

$$10^1,\ 7^6,\ 4^{22},\ 1^{44},\ (-2)^{42},\ (-5)^{20}.$$

The finite geometry itself forms a stabilizer chain

$$5 \xrightarrow{\text{hinge}} 4 \xrightarrow{\text{pairing}} 3.$$

Choosing one of the five Clebsch/D_5 coordinate directions leaves the S_4 action on the four points of the $PG(2, 3)$ null conic. The three perfect matchings of those four points carry the quotient

$S_4/V_4 \cong S_3$ and match the three projective $A_2(\mathbb{F}_3)$ root directions and the far/middle/active Fano gauges. This is a finite permutation correspondence, not a physical Spin(10) claim.

Finally, the fixed linear $[13, 5, 5]$ controller encoder has a unique minimum coordinate-orbit completion of length 28. The completed code is

$$\boxed{[28, 5, 11]},$$

so fifteen added coordinates are necessary and sufficient for coordinate-permutation S_3 equivariance of that encoder. The Clebsch closed-neighborhood matrix is the incidence matrix of a symmetric 2-$(16, 6, 2)$ biplane, with Smith form $1^6 2^4 4^5 12$ and the modular identity $B^2 = I - J$ over $\mathbb{F}_3$. Its zero and single-point syndromes have minimum distance six.

22 Triangle-free strongly regular graphs and the missing-Moore firewall

Exact fixed-μ ladder. For a strongly regular graph with

$$\mu = 4 \qquad \text{and restricted eigenvalue} \qquad r = 2,$$

the remaining data are forced by the SRG identities:

$$s = \lambda - 6, \qquad k = 16 - 2\lambda, \qquad v = \frac{(\lambda - 7)(3\lambda - 22)}{2},$$

with multiplicities

$$f_r = \frac{(\lambda - 5)(3\lambda - 22)}{2}, \qquad f_s = -3(\lambda - 7).$$

For $\lambda = 0, 1, 2, 3, 4$ this gives

$$\boxed{(77, 16, 0, 4) \longrightarrow (57, 14, 1, 4) \longrightarrow (40, 12, 2, 4) \longrightarrow (26, 10, 3, 4) \longrightarrow (15, 8, 4, 4).}$$

These are respectively the M_{22} graph, the nonexistent Wilbrink–Brouwer parameter set, the $W(3, 3)$ parameter class, the Paulus parameter class, and the triangular graph $T(6)$. Thus there is an exact 57-vertex spectral hole between M_{22} and $W(3, 3)$. It is not the hypothetical degree-57 Moore graph, whose parameters would be $(3250, 57, 0, 1)$.

$W(3, 3)$**–Gewirtz restricted kernel.** The $W(3, 3)$ and Gewirtz adjacency spectra are

$$12^1, 2^{24}, (-4)^{15} \qquad \text{and} \qquad 10^1, 2^{35}, (-4)^{20}.$$

After removing the constant line, both operators therefore obey

$$\boxed{(A - 2I)(A + 4I) = 0.}$$

Their complement operators have restricted eigenvalues ± 3, so their centered complement squares equal $9I$ on augmentation. This is a shared quadratic functional calculus, not a canonical objectwise intertwiner; the multiplicities and incidence geometries differ.

Necessary edge chart for a hypothetical M_{57}. Assume temporarily that a Moore graph with parameters $(3250, 57, 0, 1)$ exists and fix an edge $a\sim b$. The punctured neighborhoods

$$A = N(a) \setminus \{b\}, \qquad B = N(b) \setminus \{a\}$$

have size 56, and the remaining $3136 = 56^2$ vertices are forced bijectively onto $A\times B$. Every residual vertex has one neighbor in A, one in B, and 55 residual neighbors. No residual edge lies inside a row or column; hence every pair of distinct rows, and every pair of distinct columns, is joined by a perfect matching. If $\sigma_{ij} \in S_{56}$ denotes the matching from row i to row j, then necessarily

$$\sigma_{ji} = \sigma_{ij}^{-1}, \qquad \sigma_{ij} \text{ is fixed-point-free,}$$

and for distinct i, j, k the product

$$\sigma_{ki}\sigma_{jk}\sigma_{ij}$$

is fixed-point-free. These are necessary, not sufficient, constraints. Crucially, relabelling rows and columns is coordinate gauge, not an automorphism of one fixed completed graph; and

$$\frac{3250 \cdot 57}{2} = 92{,}625$$

is the edge count, not the automorphism-group order.

The four-way 57 firewall. Four different objects must remain separate:

object	meaning
M_{57}	3250 vertices, degree 57, existence open
$(57, 14, 1, 4)$	57 vertices, proved nonexistent
Perkel graph	57 vertices, degree 6
regular 57-cell	57 cells, $PSL(2, 19)$ symmetry

A June 2026 preprint claims that a hypothetical M_{57} has no involutory automorphisms. If that result survives review, then its automorphism group has odd order, whereas

$$|PSL(2, 19)| = 3420$$

is even; consequently the 57-cell/Perkel symmetry cannot embed into $\mathrm{Aut}(M_{57})$. This conditional firewall is not a proof of nonexistence.

Machine certificate. The standard-library verifier `analysis/bt3500_3505_triangle_free_srg_m57_bridge.` reconstructs all parameter, spectrum, multiplicity, edge-chart, and order identities above. Its frozen semantic digest is

`e308364dd480970803061b90ab27bf86afa2ae4946399b656292f02682c17fd9`.

The evidence boundary is explicit: no existence decision for the missing Moore graph, no Perkel/57-cell/Moore identification, and no canonical $W(3, 3)$–Gewirtz intertwiner are asserted.

23 Passes 3506–3512: exact descendants, Moore permutation curvature, and the scheme-blind 57-hole

Certified seven-front packet

The standard-library verifier reports

`PASS_7_FRONTS`

with semantic SHA-256

`15ab9b21e744e755917e0b4c40ec806b43c5463b8a11cfbfac7bdbb4a9c8dfe7.`

The packet supplies an executable Moore-57 permutation CSP, literal descendant constructions among five standard triangle-free strongly regular graphs, the complete rank-three Krein census of the fixed $\mu = 4, r = 2$ ladder, a W33/Gewirtz polynomial-transplant compiler, a sharpened 57-cell symmetry firewall, and two explicitly fenced high-risk constructions.

23.1 The Moore chart becomes a gauge-reduced permutation CSP

For a fixed edge of a hypothetical SRG$(3250, 57, 0, 1)$, write $p_{ij}(a)$ for the destination column in row j of the residual neighbor of (i, a). The independent matching data contain

$$\binom{56}{2}56 = 86\,240$$

entries. The literal CP-SAT exporter uses 172 480 directed integer variables with explicit inverse channels, rather than 4 743 200 nonfixed one-hot Boolean variables. Before lazy cuts it has 1 540 row-pair permutation constraints, 1 540 inverse channels, 3 136 vertex-star constraints, and 55 gauge equalities:

$$p_{0j}(0) = j, \qquad 1 \le j \le 55.$$

Triangle fixed points are separated over the

$$\binom{56}{3} = 27\,720$$

row triples. This is a source-complete model, not a solved instance.

23.2 Two literal descendant chains

The Clebsch graph is reconstructed on the even-weight vectors of $\mathbb{F}_2^5$, with adjacency at Hamming distance four. Its second subconstituent at zero consists of the ten weight-two words and is exactly $KG(5, 2)$, the Petersen graph.

Independently, a cyclic generator produces the extended binary Golay code with weight distribution

$$1 + 759z^8 + 2576z^{12} + 759z^{16} + z^{24}.$$

Fixing two coordinates in the octads yields the 77 blocks of $S(3, 6, 22)$. Hexad disjointness gives M_{22}; avoiding one point gives Gewirtz; adjoining the 22 points and infinity gives Higman–Sims. Thus

$$\boxed{\text{HS} \longrightarrow M_{22} \longrightarrow \text{Gewirtz}}$$

and

$$\boxed{\text{Clebsch} \longrightarrow \text{Petersen}}$$

are now objectwise executable.

23.3 The nonexistent 57-vertex rung is invisible to rank-three feasibility

For every member of the fixed

$$\mu = 4, \qquad r = 2$$

ladder, the verifier computes the complete primitive-idempotent Krein products and both standard absolute bounds. Every test passes, including for the nonexistent parameter set $\mathrm{SRG}(57, 14, 1, 4)$. Its nontrivial Krein rows are

$$q_{11} = \left(38, \frac{1273}{49}, \frac{1140}{49}\right), \quad q_{12} = \left(0, \frac{540}{49}, \frac{722}{49}\right), \quad q_{22} = \left(18, \frac{342}{49}, \frac{111}{49}\right).$$

Hence its obstruction is not visible in the ordinary rank-three Bose–Mesner algebra:

$$\boxed{\text{the } \lambda = 1 \text{ hole is scheme-blind.}}$$

Finer local, Terwilliger, or star-complement compatibility is required.

23.4 A two-channel W33/Gewirtz transplant algebra

On augmentation, W33 and Gewirtz both satisfy

$$A^2 + 2A - 8I = 0.$$

Consequently every adjacency polynomial reduces uniquely to

$$f(A) = aA + bI$$

in $\mathbb{Q}[x]/(x^2 + 2x - 8)$. Moreover

$$U = \frac{-I - A}{3}$$

obeys

$$U^2 = I, \qquad P_\pm = \frac{I \pm U}{2}.$$

Polynomial identities transport automatically. Traces and projector ranks depend on the different multiplicities, while incidence, codes, automorphism groups, and descendant maps remain geometry-sensitive.

23.5 The 57-cell firewall and its surviving odd shadow

The Perkel/57-cell group has order

$$|PSL(2, 19)| = 3420.$$

Peer-reviewed results bound an odd hypothetical M57 automorphism group by 375 and an even one by 110. Conditional on the June 2026 no-involution preprint, $\mathrm{Aut}(M57)$ is odd and hence solvable; there can be no common A_5 or full $PSL(2, 19)$ automorphism action. However, the odd Borel subgroup

$$C_{19} \rtimes C_9, \qquad |C_{19} \rtimes C_9| = 171,$$

is not excluded by parity and the order bound alone. The next exact target is the restriction of the Perkel -3 twenty-plane to this 19:9 shadow.

23.6 Bonkers front: non-Abelian Moore curvature

The row matchings form an S_{d-1}-valued connection. Triangle holonomy

$$H_{ijk} = \sigma_{ki}\sigma_{jk}\sigma_{ij}$$

changes by conjugacy under gauge, so cycle type is invariant. In an exact edge-rooted chart of the Hoffman–Singleton graph, all 15 row-pair matchings and all 20 triangle holonomies have cycle type

$$2^3.$$

This motivates a non-necessary M57 search branch in which all matchings and triangle holonomies have type 2^{28}. It is a solver ansatz, not an M57 theorem.

23.7 Bonkers front: one Golay puncture/avoidance functor

The chain

$$[23, 12, 7] \to [24, 12, 8] \to S(3, 6, 22) \to M_{22} \to \text{Gewirtz}$$

together with the incidence extension to Higman–Sims is one executable information-losing pipeline. Its comparison with the W33 face tower $240 \to 120 \to 40$ is structural only: no category equivalence or group identification is asserted.

Evidence boundary. No M57 existence decision, complete CP-SAT run, necessity of involutive curvature, actual M57 19:9 action, canonical W33–Gewirtz intertwiner, or peer-reviewed status for the June 2026 preprint is claimed.

24 Multi-circuit, Reed–Muller, compound-biplane, and A_5 closure

The minimum-defect interval remains

$$389 \le D_{H^1} \le 435,$$

but the single-circuit method is now closed exactly. A deterministic basis exchange produces a certified 260-term circuit. Since a rank-436 fundamental circuit contains at most 436 basis coordinates while the nonzero coefficient alphabet has size $3^5 - 1 = 242$, one circuit can force at most two cancellations while adding one support. Hence no single-circuit pigeonhole argument can improve the universal upper bound below 435.

A five-channel continuous amplitude search gives a numerical candidate with ratio approximately 8.9062301931. The nearby exact tuple

$$\frac{1}{256}(122, 346, 220, -240, 256)$$

has a rational-polynomial certificate proving

$$8.905 < h < 9.$$

Thus it improves the earlier finite-grid witness but does not reach the live chromatic lower bound.

The complete binary coordinate-permutation S_3-equivariant dimension-five code frontier was exhausted through length forty. In particular,

$$[13, 5, 5], \qquad [16, 5, 8], \qquad [24, 5, 11], \qquad [28, 5, 14]$$

are length-minimal at the displayed target distances. The sixteen columns of the $[16, 5, 8]$ code are exactly the affine hyperplane

$$\{c \in \mathbb{F}_2^5 : \langle 00111, c\rangle = 1\},$$

so the controller code is objectwise $RM(1, 4)$ with weight enumerator $1 + 30z^8 + z^{16}$.

The Clebsch biplane has thirty fourfold collision classes among its 120 weight-two faults. A canonical exhaustive search proves that three additional bits are necessary and sufficient to separate all of them. The resulting nineteen-bit readout uniquely locates all

$$1 + 16 + \binom{16}{2} = 137$$

fault patterns of weight at most two. The minimal companion map is cubic in the four-bit Clebsch-axis coordinate.

Finally, an objectwise A_5 character computation separates the two tempting rank-twenty sectors. The Perkel -3 sector restricts as

$$1 \oplus 3 \oplus 3' \oplus 2 \cdot 4 \oplus 5,$$

whereas the W33 block -4 sector restricts as

$$3 \cdot 1 \oplus 3 \cdot 4 \oplus 5.$$

Consequently no A_5-equivariant rank-twenty intertwiner exists. The live chromatic boundary remains $10 \leq \chi(H) \leq 11$.

25 Borel Fourier anatomy, star complements, and typed Moore searches

Passes 3528–3534 execute the five graph-theoretic continuations after reconciling the complete-switch and common-A_5 packets. The exact standard-library verifier reports

```
PASS_7_FRONTS
```

with semantic digest

```
1ec126887cd1f09b8cdc074872567dd4962c2a399c1b14db8b035a8ea50dc591.
```

The degree-57 Moore graph remains open; no 56-fibre SAT or UNSAT result is claimed.

25.1 The Perkel twenty-plane restricted to the odd Borel subgroup

On the explicit Perkel model $(i, j) \in \mathbb{Z}_3 \times \mathbb{F}_{19}$, the maps

$$(b, m) : (i, j) \longmapsto (i + m, 4^m j + b)$$

realize all 171 elements of

$$B = C_{19} \rtimes C_9.$$

Every map is checked objectwise against the edge set. The -3 spectral projector is evaluated integrally through

$$171E_{-3} = -A^3 + 9A^2 - 19A + 6I.$$

The restricted character has value distribution

$$20^1, \qquad 1^{18}, \qquad 2^{38}, \qquad (-1)^{114},$$

and fixed dimensions

$$\dim V^{C_{19}} = 2, \quad \dim V^{C_9} = 2, \quad \dim V^{C_3} = 8, \quad \dim V^B = 0.$$

Over $\mathbb{Q}$ the twenty-plane is

$$\boxed{V_{-3}\downarrow_B \cong V_{18} \oplus V_2,}$$

where V_{18} is the conductor-19 rational constituent and V_2 is the conductor-3 constituent. This closes the surviving 19:9 comparison on the Perkel side only; it does not construct that subgroup in a hypothetical Moore graph.

25.2 The 57-vertex star-complement firewall

For the nonexistent parameter set

$$\mathrm{SRG}(57, 14, 1, 4),$$

the eigenvalue 2 has multiplicity 38, so a star complement has order 19. An independent heavy enumeration from the canonical windmill seed gives stage counts

$$22, \quad 784, \quad 157{,}349,$$

and exactly

$$\boxed{3{,}720}$$

spectral survivors. Their canonical ledger hash is

`c1e0cb753fbdeab4d8ecf8059896b6ff1eb1fcc75c663022ab7040ceea012219`.

The published compatibility-graph census has maximum clique 31, whereas reconstruction requires a 38-clique. Therefore

$$\boxed{31 < 38}$$

reproves nonexistence, with margin seven. The 3,720 candidate census is independently regenerated here; the complete maximum-clique histogram remains source-locked to the published Cliquer computation and is not presented as a fresh independent rerun.

25.3 Two typed Moore-57 solver tracks

For fibre size $n = d - 1$, a diameter-two Moore graph has parameters

$$(d^2 + 1, d, 0, 1)$$

and restricted polynomial

$$x^2 + x - (d - 1).$$

The exact global gate leaves only

$$\boxed{n \in \{1, 2, 6, 56\}},$$

corresponding to C_5, Petersen, Hoffman–Singleton, and the open degree-57 case. Every even stage $n = 8, 10, \ldots, 54$ fails spectrum or multiplicity integrality before a local CSP is built.

The unrestricted $n = 56$ model retains 172,480 directed integer variables, 6,271 structural constraints, and lazy triangle and common-neighbour separators. A second high-risk branch imposes

$$\sigma_{ij}^2 = 1$$

on every fixed-point-free row matching and separates triangle holonomies by cycle type 2^{28}. This adds 86,240 source-level element-involution constraints. Neither branch has an $n = 56$ verdict.

25.4 The two descendant towers are typed differently

The W33 face tower

$$240 \longrightarrow 120 \longrightarrow 40$$

is a pair of uniform quotients with fibre sizes two and three. The Golay–Witt chain

$$100 \longrightarrow 77 \longrightarrow 56$$

uses induced restriction and point avoidance. Since

$$100/77 \notin \mathbb{Z}, \qquad 77/56 \notin \mathbb{Z},$$

there is no common uniform-fibre quotient functor. The correct shared language requires distinct morphism types: uniform quotient, induced restriction, point avoidance, and incidence extension.

25.5 Fail-closed W33–Gewirtz transplantation

Automatic transfer is permitted only for statements reducible in

$$\mathbb{Q}[x]/(x^2 + 2x - 8).$$

Thus every adjacency polynomial reduces to $aA + bI$, and

$$U = \frac{-I - A}{3}, \qquad U^2 = I$$

is common on augmentation. Traces, determinants, ranks, multiplicities, Smith data, and p-ranks require new evidence. Lines, incidence, codes, automorphism groups, descendant maps, and objectwise intertwiners are denied automatic transfer. A repository scanner now fails closed on any explicit unsafe `AUTO_PORT` annotation.

25.6 Two outside-box consequences

The complete Perkel vertex module has the rational Borel decomposition

$$\boxed{\mathbb{Q}^{57} \cong \mathbf{1} \oplus 3V_{18} \oplus V_2.}$$

Its combined golden eigenspace is $2V_{18}$, while the -3 twenty-plane is $V_{18} \oplus V_2$. The whole graph is therefore one constant channel, three conductor-19 channels, and one conductor-3 phase channel.

Inside the involutive Moore branch, the vertex-star all-different law implies that the $n - 1$ matchings issuing from each source row form a 1-factorization of K_n. Hoffman–Singleton becomes six coupled non-group 1-factorization pencils of K_6. The $n = 56$ branch becomes

$$\boxed{\text{56 reciprocity-coupled 1-factorizations of } K_{56},}$$

with 55 perfect matchings and $55 \cdot 28 = 1540$ edges covered once in every pencil. Pair-matching involution is coordinate dependent; holonomy cycle type is gauge invariant, and the theorem applies only to the declared involutive branch.

26 Star-clique recertification, Borel–M57 archetypes, and nonseparable Moore factorization

The executable Passes 3535–3541 report

`PASS_7_FRONTS  34ea459ab6acea3a2b5624ba6523016cfb914ffb54220ed64230f74f78a1fb6b`.

The degree-57 Moore graph remains open; all statements below are finite theorems or explicitly conditional necessary conditions.

26.1 Independent star-complement clique source

For each of the 3720 independently regenerated 19-vertex star complements with adjacency matrix C, the companion source constructs

$$M = 2I - C$$

exactly, enumerates every admissible binary reconstruction column b satisfying

$$b^{\mathsf{T}} M^{-1} b = 2,$$

and joins two columns when their exact inner product is -1 or 0 and the corresponding $(\lambda, \mu) = (1, 4)$ common-neighbour bound is respected. A deterministic bitset maximum-clique engine emits a witness and proof digest for every instance. The complete heavy job fails closed unless it reproduces the published clique histogram and compatibility-graph order range 4 through 265. No fresh all-3720 result is promoted before its artifact is observed.

26.2 Prime fixed sets and the surviving Borel action

Let an automorphism of prime order p act on a hypothetical

$$\mathrm{SRG}(3250, 57, 0, 1).$$

A nontrivial fixed set is closed under unique common neighbours, and if it has more than one vertex its induced graph is itself a Moore graph, with fixed degree congruent to 57 modulo p. Consequently an order-19 element fixes exactly one vertex, whereas an order-3 element fixes either one vertex or a Petersen graph.

Assume conditionally that

$$B = C_{19} \rtimes C_9$$

acts. Orbit arithmetic leaves precisely

$$P_{19}: \ 1 + 9 \cdot 19 + 18 \cdot 171, \qquad P_{57}: \ 1 + 3 \cdot 57 + 18 \cdot 171.$$

The adjacency trace on an order-19 element first permits

$$g \in \{57, 342, 627, 912, 1197\},$$

where g counts vertices mapped to neighbours. Girth and equal displacement for nontrivial powers leave only

$$\boxed{g \in \{57, 342\}}.$$

Thus exactly four conditional action archetypes survive: low/high versions of P_{19} and P_{57}. They are necessary forms, not constructions or a contradiction.

26.3 Exact Perkel Fourier projectors

Let A be the Perkel adjacency matrix, J the all-ones matrix, and B_{19} the block diagonal matrix with three J_{19} blocks. Put

$$N = -A^3 + 9A^2 - 19A + 6I.$$

The integer projector numerators

$$P_1 = 3J, \quad P_2 = 9B_{19} - 3J, \quad P_{18} = N - 9B_{19} + 3J, \quad P_{36} = 171I - 3J - N$$

have ranks

$$1, 2, 18, 36$$

and satisfy

$$P_i P_j = 171\delta_{ij} P_i, \qquad \sum_i P_i = 171I.$$

The conductor-19 projector is

$$\boxed{P_{54} = 171I - 9B_{19} = P_{18} + P_{36}},$$

and

$$(A^3 - 8A + 3I)P_{54} = 0.$$

Hence every Perkel adjacency polynomial reduces to a constant channel, a ternary V_2 channel, and a degree-two normal form on the conductor-19 sector modulo $x^3 - 8x + 3$.

26.4 The one-factorization field

Inside the involutive edge-chart branch, each row pencil is a one-factorization. The exact Hoffman–Singleton control has 15 perfect matchings and exactly six labelled one-factorizations of K_6. Its six residual rows realize all six factorizations exactly once. Moreover, all $5! = 120$ globally separable colour identifications fail all 20 row triangles: every one of the 2400 holonomies has two fixed points. The actual Hoffman–Singleton holonomies all have cycle type 2^3.

Therefore a hypothetical involutive $n = 56$ branch cannot reuse one global factorization. It requires 56 genuinely row-dependent, reciprocity-coupled one-factorizations of K_{56}, each containing 55 perfect matchings and covering all 1540 symbol pairs.

26.5 Typed W33–Gewirtz polynomial port

Both exact graphs satisfy the common restricted relation

$$(A - 2I)(A + 4I) = 0$$

on augmentation. With $P = J/v$ and $Q = I - P$, the full identity is

$$(A - 2I)(A + 4I) = (k - 2)(k + 4)P.$$

The reflection

$$R = \frac{(k+1)P - I - A}{3}$$

satisfies $R^2 = Q$, and

$$E_2 = \frac{Q - R}{2}, \qquad E_{-4} = \frac{Q + R}{2}.$$

If $p(x) \equiv ax + b \pmod{x^2 + 2x - 8}$, then

$$\boxed{p(A) = p(k)P + (aA + bI)Q.}$$

This ports every adjacency-polynomial and functional-calculus theorem. Trace, determinant, and rank formulas require the typed multiplicities $(24, 15)$ for W33 and $(35, 20)$ for Gewirtz. Incidence, codes, automorphisms, Smith forms, descendants, and intertwiners remain geometry-sensitive and require independent evidence.

Boundary. No M57 existence verdict, unobserved full clique rerun, W33–Gewirtz objectwise equivalence, hardware result, or physical interpretation is asserted.

27 Cyclic Borel quotients, the Perkel commutant, and the outer-S_6 Moore chart

Passes 3542–3548 continue the degree-57 Moore and Perkel program with exact finite certificates. The corresponding verifier reports

```
PASS_7_FRONTS a98621cc2f2378eaed09b8f747022f419a16fbb33433336eeb8f2414f4273f1a
```

The missing Moore graph remains open.

The C_{19} quotient becomes three exact branches. Assume $B = C_{19} \rtimes C_9$ acts on a hypothetical $\mathrm{SRG}(3250, 57, 0, 1)$. The normal C_{19} fixes one vertex and has 171 nonfixed orbit cells. Exactly three cells lie in the fixed vertex's neighbourhood. Writing $S_{ij} \subseteq \mathbb{Z}_{19}$ for the cyclic shifts between cells and $T_i = S_{ii}$, the unique-common-neighbour law forces the three neighbour cells to be independent, every connection leaving them to have size at most one, and their orbit-neighbour sets to be disjoint. Hence the remaining 168 cells split exactly as

$$168 = 56 + 56 + 56.$$

For a nonneighbour cell,

$$\sum_j |S_{ij}|(|S_{ij}| - 1) = 18 - |T_i|, \qquad |T_i| \in \{0, 2, 4\}.$$

C_9 or C_3 stabilizer invariance forbids nonempty T_i inside Borel orbits of size 19 or 57, so internal edges occur only in the eighteen size-171 Borel orbits. If a of those have internal degree two and b have internal degree four, the two displacement levels reduce to

$$a + 2b = 3 \quad \text{or} \quad a + 2b = 18.$$

Together with the two orbit profiles P_{19} and P_{57}, the four Borel archetypes reduce to exactly twenty-four aggregate cyclic-difference signatures. This is a finite CSP frontier, not a construction or contradiction.

The complete Perkel Borel commutant. For the explicit Perkel action $(i,j)\mapsto(i+m,4^m j+b)$, the point stabilizer is C_3 with three singleton orbits and eighteen orbits of size three. The orbital algebra therefore has rank 21. The rational module

$$\mathbb{Q}^{57}=\mathbf{1}\oplus 3V_{18}\oplus V_2$$

gives projected algebra dimensions $1,2,18$, and hence

$$\mathrm{End}_B(\mathbb{Q}^{57})\cong\mathbb{Q}\oplus M_3\big(\mathbb{Q}(\sqrt{-19})\big)\oplus\mathbb{Q}(\sqrt{-3}).$$

Let P_{54} and P_2 be the conductor-19 and conductor-3 projector numerators. The block-diagonal Paley tournament matrix D_{19} and the oriented three-layer all-one operator C_3 obey

$$9D_{19}^2=-P_{54},\qquad 3C_3^2=-19P_2,$$

and commute with all twenty-one orbital matrices. The commutative adjacency layer has exact span

$$\mathbb{Q}[A,B_{19}]\cong\mathbb{Q}\times\mathbb{Q}\times\mathbb{Q}\times\mathbb{Q}(\sqrt{5}),$$

with basis I,A,A^2,B_{19},AB_{19}.

Proof-carrying maximum cliques. The star-complement pipeline now has an independently checkable upper-bound proof language. A colour leaf proves $\omega(P)\le K$ by a proper K-colouring of the current candidate set. A branch node on v proves the include subproblem at $K-1$ and the exclude subproblem at K. The checker reconstructs all subproblems, verifies colour classes and references, and separately checks the clique witness. Thirty-one deterministic controls pass. A complete 3720-instance proof archive remains a heavy workflow boundary.

A factorization-field Moore compiler. Let n be even and suppose $F_0,\dots,F_{n-1}$ are one-factorizations of K_n satisfying $|F_i\cap F_j|=1$. Let m_{ij} be the shared perfect matching. The vertex set

$$\{x,y\}\cup\{r_i\}\cup\{c_a\}\cup\{z_{i,a}\}$$

with edges

$$x\sim y,r_i,\qquad y\sim c_a,\qquad z_{i,a}\sim r_i,c_a,z_{j,m_{ij}(a)}$$

has $(n+1)^2+1$ vertices and degree $n+1$. If every simple three-row and four-row holonomy is fixed-point-free, the graph has girth at least five and meets the Moore bound exactly. For $n=56$ this becomes a 3250-vertex, degree-57 construction surface with 27720 triangle checks and 1101870 simple four-cycle checks. Pairwise unique intersection is a sufficient high-risk branch, not a proved necessity.

Hoffman–Singleton as the outer automorphism of S_6. The fifteen duads, fifteen synthemes, and six pentads are respectively the edges, perfect matchings, and one-factorizations of K_6. Exact enumeration shows that every pair of pentads shares exactly one syntheme, every triangle holonomy has cycle type 2^3, and every simple four-row holonomy has cycle type 3^2. The induced action of S_6 on the six pentads is faithful of order 720; a point transposition of type $2\,1^4$ maps to type 2^3, certifying the outer action. Feeding the shared synthemes into the factorization-field compiler reconstructs

$$\mathrm{SRG}(50,7,0,1),$$

the Hoffman–Singleton graph.

Analytic W33–Gewirtz ports. For either graph,

$$f(A) = f(k)P + f(2)E_2 + f(-4)E_{-4}$$

for every analytic f. In particular the heat kernel and resolvent port exactly, while the Ihara–Bass factors retain the graph-specific degree and multiplicities. No incidence, code, automorphism, hardware, or physical claim is inferred from this spectral compiler.

28 Passes 3549–3555: Borel orbit models, exact commutants, octad curvature, and quantum walks

Twenty-four exact Borel models. For a conditional $B = C_{19} \rtimes C_9$ action on a degree-57 Moore graph, the two orbit profiles and the equations $a + 2b = 3, 18$ produce exactly 24 aggregate signatures. Burnside gives 30,915 undirected edge-orbit variables and 61,858 ordered-pair orbitals for P_{19}, versus 30,885 and 61,792 for P_{57}. Every model imposes the exact lazy relation

$$\sum_w e_{uw}e_{vw} = 0 \text{ if } e_{uv} = 1, \qquad \sum_w e_{uw}e_{vw} = 1 \text{ if } e_{uv} = 0.$$

These are complete finite model surfaces, not SAT or UNSAT verdicts.

Complete Perkel orbital algebra. The 19:9 action has 21 orbitals, with size profile $57^3\,171^{18}$ and valencies $1^3\,3^{18}$. The exact multiplication table contains 1,035 nonzero integral structure constants, all at most three, and has five-dimensional center:

$$\mathrm{End}_{19:9}(\mathbb{Q}^{57}) \cong \mathbb{Q} \oplus M_3(\mathbb{Q}(\sqrt{-19})) \oplus \mathbb{Q}(\sqrt{-3}).$$

Transpose splits the algebra into 11 symmetric and 10 skew channels. Thus every real symmetric equivariant kernel lies in an eleven-dimensional space corresponding, after the imaginary-quadratic embeddings, to

$$\mathbb{R} \oplus \mathrm{Herm}_3(\mathbb{C}) \oplus \mathbb{R}.$$

Generalized pentads and an octad falsifier. The exact labelled one-factorization counts for K_4, K_6, K_8 are $1, 6, 6240$. A family of eight K_8 one-factorizations was found in which every pair shares exactly one perfect matching; its setwise S_8-stabilizer is trivial. However only 34/56 triangle holonomies and 163/210 simple four-row holonomies are derangements. The compiled graph is 9-regular on 82 vertices, has diameter three, 60 triangles, and 422 four-cycles. Hence pairwise factor intersection is the static design condition, while holonomy is the curvature condition deciding the Moore property.

Proof archive and analytic dynamics. The 3,720 star-complement proof DAGs are assigned to 64 deterministic shards, eight of size 59 and 56 of size 58, with an ordered Merkle aggregate. The archive protocol is complete, but the full payload remains evidence-gated. For either W33 or Gewirtz,

$$e^{-itA} = e^{-ikt}P + e^{-2it}E_2 + e^{4it}E_{-4}.$$

For W33, distinct-vertex amplitudes obey

$$|U_{xy}(t)| \le \frac{1}{4} \quad (x \sim y), \qquad |U_{xy}(t)| \le \frac{2}{15} \quad (x \nsim y),$$

and for Gewirtz

$$|U_{xy}(t)| \le \frac{2}{7} \quad (x \sim y), \qquad |U_{xy}(t)| \le \frac{1}{12} \quad (x \nsim y).$$

Thus neither graph permits perfect continuous-time state transfer between distinct vertices. The exact infinite-time average probabilities and nonbacktracking poles are emitted by the verifier.

> **Boundary.** The degree-57 Moore graph remains open. No Borel signature has a solver verdict, the full proof archive is not yet promoted, the K_8 graph is a falsifier rather than a Moore construction, and the quantum-walk formulas are mathematical dynamics rather than laboratory measurements.

29 Relation planes, magnetic conductors, code duality, and the pentagonal bridge

The live cohomology and chromatic boundaries remain

$$389 \le D_{H^1} \le 435, \qquad 10 \le \chi(H) \le 11.$$

A new exact fundamental cohomology relation has weight 263. In a deterministically extended fundamental basis, all $\binom{284}{2} = 40186$ nonbasis-column pairs were exhausted. Their rows occupy at most three of the four directions of $PG(1,3)$, the maximum union support is 319, and the adversarial cancellation-cap histogram is

$$\{2 : 34119,\ 3 : 5277,\ 4 : 790\}.$$

Thus this pure two-circuit direction method cannot establish $D_{H^1} \le 433$; 790 pairs remain label-sensitive candidates for 434. An exact pilot on all $\binom{64}{3} = 41664$ triples of the 64 heaviest columns reaches union support 360 and six occupied directions of $PG(2,3)$.

The five-channel Hermitian algebra has channel sizes

$$2, 1, 30, 60, 627$$

and exact characteristic factorization

$$(x + 4w_4)^{17}(x - 2w_4)^{21} q_2(x) q_5(x),$$

where

$$q_2(x) = x^2 + (w_1 + w_4)x + w_1 w_4 - 9w_3^2$$

and q_5 is quintic. The dyadic tuple

$$\frac{1}{32768}(-15576, 44300, -28135, -30786, 32768)$$

has an exact rational-root certificate proving

$$8.90622 < h < 8.90623.$$

A degree-eighteen elimination polynomial isolates a stationary candidate near $h = 8.90623076116309$, but global unrestricted optimality remains open.

The complete duality atlas of the four binary equivariant frontier codes gives

$[n,5,d]$	$d^\perp$	$\dim(C\cap C^\perp)$	$\lvert\mathrm{Aut}(C)\rvert$
$[13,5,5]$	3	2	48
$[16,5,8]$	4	5	322560
$[24,5,11]$	3	0	72
$[28,5,14]$	3	3	64512

with generalized weights computed exactly. The length-sixteen code is $RM(1,4)$ with automorphism group $AGL(4,2)$, while the length-twenty-eight code is the binary simplex code punctured on one projective line.

For compound Clebsch readout, three added bits cannot even force distance two, four cannot force distance three, and five bits attain distance three. Therefore five companion bits are necessary and sufficient to locate every zero-, single-, and double-device fault while correcting one corrupted readout bit. The resulting code has length twenty-one and 137 protected patterns.

Finally, the Perkel and W33 rank-twenty modules, although inequivalent over A_5, become fully isomorphic on the common pentagonal subgroup:

$$P_{20}\downarrow_{C_5}\cong W_{20}\downarrow_{C_5}\cong 4\mathbf{1}\oplus 4\mathbb{Q}(\zeta_5).$$

The rational intertwiner space has dimension 80. No canonical objectwise intertwiner is asserted.

A second objectwise closure identifies the 240 W33 edges with the 240 filled triangles of the 45-octet port graph: every W33 edge belongs to exactly three canonical $K_{4,4}$ octets, and the resulting triangles partition all 720 block-graph edges.

30 Maximal cyclic rank-twenty bridge and a gate-minimal fault locator

The rational C_5 restrictions of the Perkel and W33 rank-twenty modules are

$$P_{20}\downarrow_{C_5}\cong W_{20}\downarrow_{C_5}\cong 4\mathbf{1}\oplus 4\mathbb{Q}(\zeta_5).$$

Consequently their rational intertwiner algebra is

$$M_4(\mathbb{Q})\oplus M_4(\mathbb{Q}(\zeta_5)),$$

of dimension 80. Thus C_5 symmetry alone does not select a preferred objectwise map.

The full sixteen-dimensional cyclotomic moving sector extends to the dihedral normalizer D_{10}. The only obstruction lies in the four-dimensional C_5-fixed sector:

$$P_{20}\downarrow_{D_{10}}\cong 2\mathbf{1}\oplus 2\varepsilon\oplus 4V_{\mathrm{cyc}},$$

$$W_{20}\downarrow_{D_{10}}\cong 4\mathbf{1}\oplus 4V_{\mathrm{cyc}}.$$

Hence the maximum rank of a D_{10}-equivariant map is 18, and the minimal stable repair is

$$\boxed{P_{20}\oplus 2\mathbf{1}\cong W_{20}\oplus 2\varepsilon},$$

of dimension 22. At the full A_5 level the maximum common rank is 14, and the minimal stable repair is

$$\boxed{P_{20}\oplus 2\mathbf{1}\oplus 4\cong W_{20}\oplus (3\oplus 3')},$$

of dimension 26. Thus the unstabilized common-rank ladder is

$$20 \longrightarrow 18 \longrightarrow 14,$$

while the minimal invertible stable dimensions are $20, 22, 26$.

The Perkel Borel group has order

$$171 = 3^2 \cdot 19,$$

so it contains no element of order five. Its eleven-observable/ten-phase orbital algebra therefore supplies neither a C_5 generator nor its dihedral normalizer; it is not automatically a canonical C_5 selector.

The five-bit compound-fault locator admits the exact quadratic Boolean form

$$\begin{aligned}
y_0 &= x_0x_2 + x_0x_3 + x_1x_3,\\
y_1 &= x_2 + x_0x_2 + x_1x_2 + x_0x_3,\\
y_2 &= x_2 + x_0x_2 + x_1x_2 + x_1x_3,\\
y_3 &= x_2 + x_1x_2 + x_0x_3 + x_1x_3,\\
y_4 &= x_2 + x_1x_2 + x_0x_3 + x_2x_3.
\end{aligned}$$

Its five homogeneous quadratic parts have rank five. Since each AND gate can add at most one independent quadratic form, every XOR–AND realization needs at least five AND gates. The products

$$x_0x_2,\ x_0x_3,\ x_1x_2,\ x_1x_3,\ x_2x_3$$

attain the bound, proving multiplicative complexity exactly five. A shared network uses eight XOR gates. Exhaustive RTL equivalence preserves all 137 compound fault words and minimum distance three.

31 The Petersen spine, constructive Perkel matrices, and octad curvature

Passes 3577–3583 replace three previously broad search fronts by exact finite interfaces. First, the double-coset subdegrees of the hypothetical Borel action force the order-3 or order-9 fixed graph to be the Petersen graph in both surviving orbit profiles. After all variables controlled by this thin spine are fixed, both profiles leave exactly

$$30870$$

residual binary edge-orbit variables, despite starting from the different Burnside counts 30915 and 30885.

Second, the conductor-19 component of the Perkel orbital algebra is represented explicitly as

$$M_3\big(\mathbb{Q}(\sqrt{-19})\big)\,.$$

A primitive rank-18 idempotent produces a six-dimensional rational right ideal, the central Paley operator supplies the relation $D^2 = -19$, and all 441 orbital products are verified in the resulting 3×3 matrices. The frozen matrix-table digest is `366405ad8400779a79eb6b92437b6d354ad3019eb16e9b5a81b99c5adc`

Third, the 6240 labelled one-factorizations of K_8 contain at least two inequivalent eight-object pairwise-one-intersection phases. A common-core sunflower has trivial S_8 stabilizer, while a mixed core-free phase has stabilizer four. The sunflower compiles to a 9-regular graph on

$$82 = 1 + 9 + 9 \cdot 8$$

vertices. It is triangle-free but has diameter three and 3024 four-cycles, with characteristic polynomial

$$(x-9)(x-8)^3(x-1)^{32}(x+1)^{24}(x+8)^4(x^2+x-8)^9.$$

It therefore saturates the numerical Moore order while failing the uniform common-neighbour curvature.

A real star-complement canary also executes the proof-carrying clique pipeline: the canonical stages $22,784$ are regenerated, a genuine third-stage spectral survivor produces a 52-vertex compatibility graph, and an independently checked proof DAG certifies maximum clique 11.

Finally, every SRG resolvent has the form

$$(zI-A)^{-1} = a(z)I + b(z)A + c(z)J.$$

For a marked set S, the principal resolvent $aI + bA[S] + cJ$ recovers the induced edge count from the second determinant coefficient. A four-site W33 line marker therefore detects the six edges of a K_4, which no four-set in the triangle-free Gewirtz graph can reproduce. This is an exact geometry-sensitive port beyond unmarked adjacency dynamics.

Boundary. The missing degree-57 Moore graph remains open. No complete Borel solver verdict, all-octad S_8 census, or all-3720 proof archive is claimed.

31.1 Cubic dependency projector and chained incidence closures

> **Correction from Passes 3649–3662.** The original Passes 3600–3613 source reported zero eight-cell double covers for the oriented $12_4\,16_3$ surface. That count is withdrawn: the increasing-index backtracker was incomplete. Complete enumeration finds exactly three simple covers among all $\binom{12}{8} = 495$ subsets and six solutions among all 75,582 eight-cell multisets. Section 34.1 gives the replacement theorem.

The retained theorem constructs the 5,040-triple dependency deck on 240 filled faces. Its weighted overlap operator has ordinary multiplicity fingerprint

$$1, 15, 15, 20, 24, 24, 60, 81,$$

and over $\mathbb{F}_3$,

$$D^4 = -D^3, \qquad E_{81} = -D^3, \qquad E_{81}^2 = E_{81}, \qquad \operatorname{rank} E_{81} = 81.$$

The deterministic 364-class transported-M_4 scalar screen, exact five-operation ternary datapath, induced $1+16+40$ decomposition obstruction, and W33–Clebsch spectral completion remain valid. Passes 3649–3662 subsequently prove that the explicit 81-space is simple and projective, close the positive-gauge transported dual at nine, and construct a concrete integral 56-object weighted Gewirtz bridge.

The historical certificate `a08844aa7ec3a6be578dece00e455c1868ad8b4c833d912ebe02b7f014077f17` remains the record for the original packet, with its tomotope field superseded by `067047bbf3fe04301fcb26bf484a3`
The live intervals remain $389 \le R_{\text{defect}} \le 435$ and $10 \le \chi(H) \le 11$.

32 Monster $U_4(2)$, the Steinberg register, and two exact falsifiers

The structural program uses the genuine subgroup direction

$$PSp(4,3) \cong U_4(2) \longrightarrow \mathbb{M},$$

while keeping character fusion and generator-level embedding distinct. A GAP companion filters class fusions by the documented $5B$-type constraints

$$2 \mapsto 2B, \qquad 3 \mapsto 3B, \qquad 5 \mapsto 5B,$$

and decomposes the restriction of the degree-196883 Monster character.

The protected module has the sharper local interpretation

$$81 = 3^4 = |\operatorname{Syl}_3(U_4(2))|.$$

The unique degree-81 Steinberg character vanishes on every nonidentity 3-power class, so it restricts regularly to a Sylow 3-subgroup.

For

$$N = (12I - A)(A + 4I)$$

the exact W33 computation gives

$$N^2 = 60N, \qquad \operatorname{rank} N = 24, \qquad N_{xx} = 36, \qquad N_{xy} \in \{6, -4\}.$$

Uniform normalization to squared norm four yields off-diagonal products $2/3$ and $-4/9$, so the raw forty-vector frame is not directly an integral Leech subconfiguration.

Also every three-eigenmode W33 adjacency sequence satisfies

$$s_{n+3} = 10s_{n+2} + 32s_{n+1} - 96s_n,$$

while the first $1A$ moonshine coefficients violate this recurrence. Any surviving bridge must therefore be graded, induced, glued, or ambient rather than a raw adjacency-moment identification.

Majorana boundary. For the documented $5B$-type embedding, subgroup involutions fuse to Monster $2B$, whereas standard Monster Majorana axes correspond to $2A$. A direct subgroup-involution-to-axis assignment is blocked.

Evidence boundary. Character-table fusion is not a concrete `mmgroup` embedding. Promotion requires serialized Monster generators, exact relations, class fusion, and an independent image-order certificate 25920.

33 Octad pattern census, Borel core/bridge reduction, and marked tomography

The pair-intersection multiplicities of eight abstract one-factorization rows are exactly edge-disjoint clique packings of K_8. Canonical augmentation gives

$$\boxed{69}$$

S_8-orbits of abstract multiplicity patterns, with orbit counts

$$1, 6, 10, 10, 13, 14, 8, 6, 1$$

by the number of nontrivial blocks. This is a complete abstract pattern census, not yet the complete realization census inside the 6,240 labelled one-factorizations of K_8.

The two surviving 19:9 Moore-action profiles possess a literal common binary core. Its variable classes have sizes

$$18 \cdot 85 = 1530, \qquad \binom{18}{2} \cdot 171 = 26163, \qquad 9 \cdot 18 \cdot 19 = 3 \cdot 18 \cdot 57 = 3078,$$

so the shared core has

$$\boxed{30771}$$

variables. The P_{19} profile is exactly this core after presolve, while P_{57} adds eighteen nonneighbor bridge variables:

$$\boxed{30789 = 30771 + 18}.$$

No SAT or UNSAT verdict is inferred.

The conductor-19 component of the Perkel orbital algebra is realized by explicit matrix units in

$$M_3\big(\mathbb{Q}(\sqrt{-19})\big)\,.$$

Writing $s^2 = -19$, the units obey

$$E_{ij}E_{kl} = \delta_{jk}E_{il}, \qquad (sE_{ij})(sE_{kl}) = -19\delta_{jk}E_{il}.$$

The transpose involution is represented by the positive Hermitian form

$$H = \begin{pmatrix} 1 & 2s/19 & -2s/19 \\ -2s/19 & 1 & 2s/19 \\ 2s/19 & -2s/19 & 1 \end{pmatrix},$$

whose two-by-two principal minors equal 15/19 and whose determinant is 7/19.

A real proof-carrying archive of sixteen genuine star-complement compatibility graphs is frozen with clique histogram

$$8^1, 9^1, 10^1, 11^3, 12^1, 13^2, 16^2, 31^5$$

and ordered Merkle root

`c176539c7b944274fb64965275ce3f900852fe4b26f67923b2ca6a2f06e256a4`.

Every upper-bound DAG is independently checked from its frozen adjacency bitsets. The complete 3,720-instance archive remains resource-gated.

Finally, the minimum marked-resolvent rank that distinguishes W33 from Gewirtz is three. Rank one is common to both graphs, and rank two sees only edges versus nonedges. At rank three, W33 has a triangle while Gewirtz is triangle-free. Every W33 triangle reconstructs a unique K_4 line because an adjacent pair has exactly two common neighbors.

Evidence boundary. The degree-57 Moore graph, realized K_8 octad orbit census, Borel solver verdicts, complete proof archive, PDF evidence, and physical interpretation all remain open unless separately executed and observed.

34 Passes 3635–3648: exact $U_4(2)$ completion and rank-24 modular closure

The W33 projective action now has a compact exact certificate. Four order-three symplectic transvections generate a group of order

$$25{,}920 = |PSp(4,3)| = |U_4(2)|,$$

with element-order census

$$1^1,\ 2^{315},\ 3^{800},\ 4^{3780},\ 5^{5184},\ 6^{5760},\ 9^{5760},\ 12^{4320}.$$

Every three-transvection face has order 648 and fixes one W33 point, while an explicit pair has orders $(2,5,12)$ and generates the whole group.

The rank-24 projector carrier is bilinearly complete. Its symmetric tensors have rank

$$300 = \dim \mathrm{Sym}^2(\mathbf{Q}^{24}),$$

and its alternating tensors have rank

$$276 = \dim \Lambda^2(\mathbf{Q}^{24}).$$

Thus the carrier spans all $24^2 = 576$ matrix directions. This is a full linear operator envelope, not yet a Griess or VOA product certificate.

The integral projector Gram form has Smith profile

$$2^1, \qquad 10^9, \qquad 30^6, \qquad 60^8,$$

and discriminant valuation

$$2^{32}3^{14}5^{23}.$$

The odd 5-adic exponent survives every rational scalar rescaling in rank 24, so no scalar-rescaled copy can possess a unimodular overlattice. Any Leech route must use non-scalar coupling, quotienting, or a multi-lattice extension.

The exact graded completion is

$$\left(E_4^3 - 720\Delta\right) \prod_{n\geq 1} (1-q^n)^{-24},$$

whose initial shifted coefficients are

$$1,\ 24,\ 196884,\ 21493760,\ 864299970,\ 20245856256.$$

This identifies the minimum machinery missing from raw W33 adjacency moments: a rootless rank-24 modular numerator and a 24-oscillator denominator.

Two additional chamber structures emerge. The full group contains 432 copies of A_5 in two orbits of 216. Their D_{10}-sharing graph is

$$36K_{6,6},$$

with 1296 edges; every edge generates A_6, and each chamber has normalizer of order 720, hence S_6. Separately, the four transvections form a tetrahedral amalgam whose four faces are order-648 point stabilizers.

Evidence boundary. The exact Python certificate proves the abstract $U_4(2)$ carrier, the complete $A_5/A_6/S_6$ census, both tensor-span ranks, the scalar glue obstruction, and the modular coefficients. A companion GAP job enumerates compatible $5B$-containing class fusions and the possible multiplicities of the degree-81 constituent. No serialized Monster-word embedding, unique class fusion, observed degree-81 multiplicity, Leech embedding, Griess product, or physical result is promoted before its dedicated certificate is observed.

34.1 Cubic integrality, Steinberg closure, and the corrected tomotope lift

For the cubic dependency incidence matrix $T \in \{0,1\}^{240\times 5040}$, the uniform primal and dual witnesses have value 80:

$$x = \tfrac{1}{3}\mathbf{1}_{240}, \qquad y = \tfrac{1}{63}\mathbf{1}_{5040}.$$

Thus the fractional transversal number is 80. An integral 80-set would make every covering constraint tight, $T^T x = \mathbf{1}$. Since $TT^T = 63I + D$ has minimum eigenvalue 45 and full rank 240, the unique real solution is $x = \frac{1}{3}\mathbf{1}$, impossible for a binary vector. Hence

$$\boxed{\tau \geq 81}, \qquad \tau/\tau^* \geq 81/80.$$

This is a cubic integrality theorem, not a covering-radius endpoint; $389 \leq R_{\text{defect}} \leq 435$ remains live.

The ten orbital matrices of the 240-face action restrict on $E_{81} = -D^3$ to a one-dimensional algebra over $\mathbb{F}_3$, so

$$\mathrm{End}_{PSp(4,3)}(E_{81}) = \mathbb{F}_3.$$

A deterministic Sylow-three subgroup of order 81 acts regularly on this space. Therefore E_{81} is projective; the scalar endomorphism ring forces simplicity, and the explicit image realizes the defining-characteristic Steinberg module. The complementary filtration has dimensions $159 \supset 44 \supset 14 \supset 0$.

For the fourfold 45-anchor blowup, the 27 lifted independent 20-sets, each point occurring three times, give $\chi_f = 9$. The unrestricted positive-gauge weighted-Hoffman dual is therefore exactly nine, attained by $A_{45} \otimes I_4$ with spectrum $32^4, 2^{96}, (-4)^{80}$. Arbitrary signed complex Hermitian weights remain open.

Correction to Passes 3600–3613. The earlier zero count for eight-cell covers is withdrawn: its increasing-index backtracker skipped valid subsets. Complete enumeration of all $\binom{12}{8} = 495$ simple subsets finds exactly three covers, one orbit under the visible order-48 group, with stabilizer 16 and induced action S_3. Each cover has cell adjacency $K_{4,4}$. With

$$A = \{0,3,8,11\}, \quad B = \{1,5,6,10\}, \quad C = \{2,4,7,9\},$$

the covers are $A \cup B, A \cup C, B \cup C$; exhaustive enumeration of all 75,582 multisets adds precisely $2A, 2B, 2C$. This establishes the eight-cell incidence layer, but not yet the full rank-four polytope isomorphism.

A concrete objectwise W33–Clebsch bridge is supplied by the integral matrix

$$Q = \begin{pmatrix} 56A_W - 6J_{40} & 8J_{40,16} \\ 8J_{16,40} & \frac{7}{2}(-3A_C^2 + 18A_C + 17I) + 8J_{16} \end{pmatrix}.$$

It has constant row sum 560 and obeys

$$(Q - 560I)(Q - 112I)(Q + 224I) = 0,$$

so $Q/56$ has Gewirtz spectrum $10^1, 2^{35}, (-4)^{20}$. This is an integral weighted spectral bridge, not a 0/1 graph embedding.

Finally, $C = I + D + E_{81}$ has order three over $\mathbb{F}_3$, with $(C - I)$-power ranks $44, 14, 0$ and Jordan type $J_3(1)^{14} \oplus J_2(1)^{16} \oplus J_1(1)^{166}$. The three corrected tomotope covers form a second exact S_3-selector register. Neither construction is promoted as physical dynamics.

The frozen semantic SHA-256 is `067047bbf3fe04301fcb26bf484a3d151bf6db4ab762fa6bf31f527889d09a89`. No remote CI, PDF, FPGA, laboratory, Monster embedding, radius endpoint, or ten-colour result is asserted here.

34.2 The canonical A_6-chamber/spread bridge and the exceptional S_6 double-six

The preceding Monster-facing packet constructs all 432 subgroups isomorphic to A_5, their 36 $K_{6,6}$ components under the D_{10}-sharing relation, and one A_6 chamber per component. Earlier exact work independently identifies the rank-three action of $PSp(4,3)$ on the 36 spreads of $W(3,3)$. The following result supplies the missing objectwise map; it is not an inference from equal cardinalities or strongly regular graph parameters.

Theorem 34.1 (Canonical chamber/spread bridge)**.** *Let $G = PSp(4,3) \cong U_4(2)$ act on $W(3,3)$, and let $H \cong A_6$ be one of the 36 chambers. Then*

$$|N_G(H)| = 720, \qquad [N_G(H), N_G(H)] = H.$$

The normalizer fixes exactly one spread Σ_H, and

$$N_G(H) = \mathrm{Stab}_G(\Sigma_H).$$

Consequently

$$\boxed{G/N_G(A_6) \;\longleftrightarrow\; \{W(3,3)\ \textit{spreads}\}, \qquad H \longmapsto \Sigma_H,}$$

is a canonical G-equivariant bijection of two 36-element sets.

For distinct chambers H and K, the complete intersection dictionary is

$$\boxed{|H \cap K| = 18 \iff |\Sigma_H \cap \Sigma_K| = 1} \qquad \text{(360 unordered pairs)},$$

and

$$\boxed{|H \cap K| = 12 \iff |\Sigma_H \cap \Sigma_K| = 4} \qquad \text{(270 unordered pairs)}.$$

The order-12 relation is therefore

$$\mathrm{SRG}(36, 15, 6, 6), \qquad \mathrm{spec} = 15^1 \oplus 3^{15} \oplus (-3)^{20},$$

while the complementary order-18 relation is

$$\mathrm{SRG}(36, 20, 10, 12), \qquad \mathrm{spec} = 20^1 \oplus 2^{20} \oplus (-4)^{15}.$$

The graph identification is carried by the explicit equivariant bijection, not by the parameter tuple alone.

Each chamber contains twelve A_5 subgroups, split by the two global A_5 classes as $6 + 6$. Two members of the same six-set intersect in A_4; every cross-pair intersects in D_{10}. Thus the cross-intersection graph is exactly $K_{6,6}$. The normalizer S_6 acts faithfully on both six-sets, and the exact cycle-type census verifies that the two actions are exchanged by the exceptional outer automorphism:

$$(2,1,1,1,1) \longleftrightarrow (2,2,2), \qquad (3,1,1,1) \longleftrightarrow (3,3), \qquad (6) \longleftrightarrow (3,2,1).$$

Their directed multiplicities are respectively 15, 40, and 120 in each direction.

Monster-facing boundary. Any future concrete $5B$-containing $U_4(2)$ embedding in the Monster transports this complete 36-spread/36-chamber carrier and every local double-six chart. The certificate does not provide Monster words, a unique class fusion, a verified multiplicity of the 81-dimensional constituent, or a Griess/VOA/photonic multiplication law. In particular, this rank-three spread carrier is not identified with the distinct 36-ray magic-state orthogonality graph, which is not strongly regular.

35 The chamber–tomotope central lift and thick amalgamation boundary

Passes 3663–3669 established the canonical equivariant identification of the thirty-six A_6 chambers with the thirty-six spreads of $W(3,3)$. The present packet retains that carrier and moves one level upward. The valency-15 chamber relation is

$$\mathrm{SRG}(36, 15, 6, 6), \qquad 15^1 \oplus 3^{15} \oplus (-3)^{20},$$

and contains exactly 135 maximal K_4 subgraphs. Every chamber belongs to fifteen of them, giving

$$36 \cdot 15 = 135 \cdot 4 = 540$$

chamber–K_4 incidences.

The stabilizer of a maximal K_4 has order 192 and centre of order two. Its central quotient has order census

$$1^1\, 2^{27}\, 3^{32}\, 4^{36},$$

and an explicit four-generator word transport identifies it, as a marked group, with the automorphism group of the tomotope. The extension is not split: the stabilizer census

$$1^1\, 2^{43}\, 3^{32}\, 4^{84}\, 6^{32}$$

differs from the census of $C_2 \times \mathrm{Aut}(T)$. This supplies a third exact 192 mechanism: a non-split central tomotope lift, distinct from both the tomotope's 192 flags and the split order-192 universal group.

Let P_i be the four order-648 three-generator parabolics supplied by the explicit $U_4(2)$ transvections. Their pair-intersection multiset is

$$\{24, 24, 24, 24, 24, 54\},$$

every triple intersection has order three, and the total core is trivial. The associated coset geometry has forty objects of each type and $25,920$ chambers, but every panel has thickness three. Its natural panel motion is therefore C_3, not an exchange involution. The construction is an exact thick chamber system rather than an abstract polytope: it fails the diamond condition at the final local step.

For the rank-24 integral projector Gram matrix G, the non-scalar block coupling

$$\mathcal{G} = \begin{pmatrix} G & I \\ I & 0 \end{pmatrix}$$

is even, has determinant one, and has signature $(24, 24)$. Thus the scalar discriminant obstruction can be cancelled exactly, but the canonical closure is the indefinite lattice $II_{24,24}$ rather than a positive-definite Leech lattice.

If V denotes the rank-24 $U_4(2)$ module, exact character inner products give

$$\dim \mathrm{Hom}_{U_4(2)}(\mathrm{Sym}^2 V, V) = 2, \qquad \dim \mathrm{Hom}_{U_4(2)}(\Lambda^2 V, V) = 1.$$

Projected coordinatewise multiplication and projected W33 triangle-neighbour multiplication form a basis of the commutative envelope; both are Frobenius. Their complete associator ideal becomes the unit ideal modulo $1,000,003$, so no nonzero rational member of this envelope is associative. This classifies the available equivariant commutative products without promoting any of them to a Griess or VOA multiplication.

Finally, the prime support $\{2,3,5\}$ of $U_4(2)$ selects the six exact McKay–Thompson targets $2A/2B$, $3A/3B$, and $5A/5B$. With $r = 24/(p-1)$,

$$T_{pB} = \left(\frac{\eta(\tau)}{\eta(p\tau)}\right)^r + r, \qquad T_{pA} = T_{pB} + p^{r/2}\left(\frac{\eta(p\tau)}{\eta(\tau)}\right)^r.$$

These channels refine the scalar $J+24$ target but do not by themselves construct a Monster module.

A final spectral firewall is exact. The W33 (-4) eigenspace and the chamber-complement (-4) eigenspace are both irreducible and fifteen-dimensional, yet their $U_4(2)$ characters have inner product zero. They are inequivalent modules; matching dimension and graph eigenvalue do not supply an equivariant intertwiner.

The concrete Monster embedding remains fail-closed. The new harness requires four serialized Monster elements of order three, pair-product orders $3,6,6,6,6,6$, four order-648 triple closures, full closure order $25,920$, and an independently hashed class-fusion certificate before promotion.

35.1 Realized octads, the exact Borel bridge, and architecture-level firewalls

The abstract K_8 octad-pattern deck has 69 S_8-orbits. Exact certificates now realize 59 of them inside the $6,240$ labelled one-factorizations; ten sharply identified patterns remain open. The 18 additional thick variables of the P_{57} Borel profile collapse under the Moore common-neighbour screen to only

$$1 + 18 + 36 = 55$$

locally admissible masks, all of weight at most two. Their compatibility graph is $3K_{2,2,2}$.

The real symmetric Perkel commutant has the complete positive-cone model

$$\mathbb{R}_+ \times \mathrm{Herm}_3^+(\mathbb{C}) \times \mathbb{R}_+.$$

It has two scalar extreme rays plus the rank-one rays zz^* parametrized by $\mathbb{CP}^2$, sixteen rank strata, algebraic boundary $\alpha\beta \det Q = 0$, and Carathéodory number five. The proof archive now contains 32 content-addressed, independently checkable clique proof DAGs. Exhaustive marked-resolvent tomography through rank five gives W33-exclusive signature counts $0,0,1,3,10$, while Gewirtz has no exclusive signature in that range.

Two architecture-level corrections are more consequential. An exactly-one-photon carrier has only M dimensions in M orthogonal modes, so representing n independent qutrits requires $M \geq 3^n$. Moreover the total-loss Knill–Laflamme products satisfy $a_i^\dagger a_j = |i\rangle\langle j|$ and span the full matrix algebra. Their compression can all be scalar only on a one-dimensional code. Thus no nontrivial logical code confined to exactly one photon corrects arbitrary total photon loss. The abstract $[[240,81,4]]_3$ code therefore requires an explicit multi-photon, bosonic, teleportation/cluster, or quantum-memory realization before it can be called loss tolerant.

Conversely, independently addressable local phases and real edge couplers on any connected d-mode graph generate $\mathfrak{u}(d)$. W33, a path, and a cycle on forty modes all yield dimension $40^2 = 1,600$

under that idealized control model. Bare one-photon universality is therefore generic rather than W33-specific. The geometry's architectural claim must be tested through fault-tolerant overhead, routing, calibration, noise, code, or component advantages.

Evidence boundary

Fifty-nine octad patterns are realized, not all sixty-nine. The Borel result is a local presolve, not a Moore-graph verdict. Thirty-two proof DAGs are frozen, not all $3,720$. The loss and controllability results are exact conditional architecture theorems; laboratory performance remains unmeasured.

35.2 The spread ETF and its canonical Norton algebra (Passes 3694–3700)

Let A be the graph on the 36 spreads of $W(3,3)$, adjacent when two spreads share four lines. Its rational primitive idempotents are

$$E_{15} = \tfrac{1}{2}I + \tfrac{1}{6}A - \tfrac{1}{12}J, \qquad E_{20} = \tfrac{1}{2}I - \tfrac{1}{6}A + \tfrac{1}{18}J.$$

If B is the 40×36 line–spread incidence matrix and $C = B - \frac{1}{4}J_{40\times 36}$, then

$$\boxed{C^{\mathsf{T}}C = 18E_{15}}$$

shows that the normalized columns form a real $\mathrm{ETF}(15,36)$. Their mutual inner products are $+1/5$ for a four-line intersection and $-1/5$ for a one-line intersection, saturating the Welch bound. The Seidel matrix $S = I + 2A - J$ obeys $S^2 - 2S - 35I = 0$ and has spectrum $7^{15} \oplus (-5)^{21}$.

The complementary projector $I - E_{15}$ has rank 21. Thus an exact passive unitary dilation of the 36-mode frame analysis requires a 21-dimensional guard sector; this is a dimension certificate, not an experimental claim.

On $V = \operatorname{im} E_{15}$ define the Norton product $x \star y = E_{15}(x \circ y)$ and axes $a_i = 6E_{15}e_i$. The 36 axes are primitive idempotents. Each axis has multiplication spectrum

$$1^1 \oplus (-\tfrac{1}{2})^5 \oplus (\tfrac{1}{6})^9,$$

with the exact $\mathbb{Z}_2$-graded fusion law

$$(-\tfrac{1}{2}) \star (-\tfrac{1}{2}) \subseteq 1 \oplus \tfrac{1}{6}, \quad (-\tfrac{1}{2}) \star \tfrac{1}{6} \subseteq -\tfrac{1}{2}, \quad \tfrac{1}{6} \star \tfrac{1}{6} \subseteq 1 \oplus \tfrac{1}{6}.$$

This is a positive Frobenius axial algebra but not a Monster/Majorana algebra. For adjacent axes, $a_i \star a_j = (a_i + a_j)/6$. Every nonadjacent pair has a unique third axis a_k satisfying

$$a_i \star a_j = -\frac{a_i + a_j}{6} + \frac{a_k}{3}.$$

These products define 120 triples. Each triple consists of three spreads sharing one W33 line; for every W33 line, its nine containing spreads split into three disjoint triples.

The 36 Witting-ray orthogonality graph is a different object: it is 11-regular with spectrum $11^1 \oplus 2^{20} \oplus (-1)^3 \oplus (-4)^{12}$, whereas the spread graph is 15-regular with spectrum $15^1 \oplus 3^{15} \oplus (-3)^{20}$. Finally, the ternary panels of the four-parabolic chamber system cannot be changed to binary panels by an unbranched type-preserving cover, because such covers are locally bijective on rank-one residues.

35.3 Passes 3701–3714: the D_4/F_4 triality carrier and the 45-axis firewall

The 135 maximal four-cliques of the 36-chamber graph admit an exact three-to-one quotient. The stabilizer of each four-clique acts on the forty W33 points with an eight-point orbit; precisely three pairwise-disjoint four-cliques determine the same octad. Hence

$$135 = 45 \cdot 3.$$

Two octads meet in zero or two points, and octad disjointness is

$$\mathrm{SRG}(45, 12, 3, 3), \qquad \mathrm{spec} = 12^1 \oplus 3^{20} \oplus (-3)^{24}.$$

The centered octad vectors, with entries 48 on the octad and -12 off it, have norm 23040, off-diagonal products -5760 or 1440, and Gram spectrum

$$43200^{24} \oplus 0^{21}.$$

Thus they form a rank-24 two-distance tight frame.

The 135 four-cliques are exactly the 135 Lagrangian three-spaces of $W(5,2)$, restricted to the 36 anisotropic points of a minus quadratic form. Their binary incidence matrix has rank 29, and its dual is the affine restriction code

$$[36, 7, 16]_2, \qquad W(z) = 1 + 63z^{16} + 63z^{20} + z^{36}.$$

A frame stabilizer is

$$C_2^3 \rtimes S_4 \cong W(D_4), \qquad |W(D_4)| = 192.$$

It has an explicit S_4 complement. Its central quotient is non-split: the section cocycle system has coefficient rank 95 and augmented rank 96. The common normalizer of the three frames over one octad has order 576, while the stabilizer in the outer extension $U_4(2){:}2$ has order 1152. A colored-orbital conjugacy on a faithful 24-point orbit identifies the latter with the short-root action of $W(F_4)$. Therefore

$$W(D_4)\ (192) < W(F_4)_{\text{even frame}}\ (576) < W(F_4)\ (1152).$$

The ternary panels do not admit a string-Coxeter resolution preserving the exact four-generator target. Each generator has 108 inverting outer involutions, but no outer involution inverts all four. In every two-involution-per-colour factorization, each pair of colours contributes at least two cross-colour noncommutations. The resulting minimum is 16 noncommuting edges, whereas a rank-eight string diagram has only seven.

For the 45 canonical octad axes, the two basis products of the complete equivariant commutative Frobenius plane satisfy

$$m_0(u, u) = 36u, \qquad m_1(u, u) = -216u.$$

The simultaneous left-eigenpairs are

$$(36, -216)^1, \quad (28, 152)^6, \quad (-12, -168)^8, \quad (-12, 72)^9.$$

No rational parameter gives the Monster $2A$ Majorana non-axis spectrum $\{0, 1/4, 1/32\}$.

Finally, for

$$\mathcal{G} = \begin{pmatrix} G & I \\ I & 0 \end{pmatrix},$$

the graph of an integral matrix B has Gram matrix $G+B+B^{\mathsf{T}}$. Every even integral rank-24 Gram matrix occurs in this way; E_8^3 is an explicit positive unimodular child. The standard existence theorem for the Leech lattice therefore gives a noncanonical rootless graph section, but no explicit Leech basis is frozen here. Equivariance is impossible: irreducibility forces $B = mI$, and no such scalar section is simultaneously positive definite and unimodular.

Concrete Monster words, a regular thin polytope cover, an explicit canonical Leech section, and a Majorana/Griess/VOA realization remain certificate-gated.

36 Passes 3715–3721: carrier replacement and control-plane architecture

The literal exactly-one-photon register is replaced by an explicit qutrit erasure carrier. The code

$$|i_L\rangle = 3^{-1/2} \sum_{j \in \mathbf{F}_3} |j, j+i, j+2i\rangle$$

satisfies all Knill–Laflamme conditions for one known physical-qutrit erasure and realizes $[[3,1,2]]_3$. Eighty-one logical qutrits therefore require 243 physical qutrits; under one-photon triple-rail encoding this is 243 photons in 729 modes. For independent transmissivity η, the all-block success condition

$$[\eta^3 + 3(1-\eta)\eta^2]^{81} \geq 2/3$$

gives $\eta \geq 0.958628272258342$.

An exhaustive degree-matched tournament checks all $\binom{19}{6} = 27,132$ twelve-regular circulants on forty vertices. W33 and the best route circulants share diameter two and average distance 22/13, but W33 alone has exactly four common neighbours for every nonedge and zero redundancy variance; its spectral gap is 10, above the best circulant value found, approximately 9.280621379.

The proposed temporal loop is formalized as a causal qutrit process tensor: a qutrit SWAP stores the past state, a middle causal break replaces the exposed system, and a second SWAP retrieves the past. This is non-Markovian memory without future-to-past signalling. Its memory bond dimension is three for one qutrit and at least 3^{81} for an arbitrary eighty-one-qutrit state.

At 1550 nm the minimum energy in the 243-photon carrier is 3.1142×10^{-17} J per shot. A 162-bit erasure record has a 300-K Landauer floor 4.6510×10^{-19} J per cycle. At group index four, feedforward delays of 1, 10, and 100 ns require approximately 7.49 cm, 74.95 cm, and 7.49 m of passive delay.

A complete forty-mode intensity-interference identification uses 1600 input settings. A familywise Hoeffding bound with $\epsilon = \delta = 0.01$ requires 81,825 shots per setting, or 130,920,000 shots total. The device must recover the SRG identity, forty lines, 160 line triangles, and the frozen marked-resolvent atlas against degree-matched controls.

Two control-plane theorems change the architectural interpretation. First, the 160 point-line incidences decompose into four perfect matchings, giving an optimal four-tick collision-free syndrome schedule. Second, the order-25,920 symplectic group has exactly three ordered-pair orbits, so automorphism twirling compresses a generic 1600-parameter Hermitian calibration model to

$$aI + bA + c(J - I - A).$$

Thus W33 need not be the literal protected Hilbert-space carrier to remain operationally distinguished: it can serve as an optimal syndrome scheduler and a rank-three calibration/noise compressor. These are exact finite results; hardware cost, executed tomography, wall-plug power, and fault-tolerant thresholds remain evidence-gated.

37 Fischer closure, a Hadamard Naimark transform, and the six-bit orthogonal core

The thirty-six primitive Norton axes from the spread ETF carry canonical Miyamoto involutions. The exact pair census is

$$270 \text{ commuting pairs} + 360 \text{ pairs whose product has order } 3.$$

For every noncommuting pair, conjugation produces the unique third axis in its Norton triple. Consequently the 120 Norton triples are exactly the Fischer lines of a three-transposition space. Seven involutions generate a group of order 51,840, while the subgroup generated by even products has order 25,920 and the previously certified $U_4(2)$ element-order census. Thus the internal action is

$$O_6^-(2) \cong U_4(2){:}2, \qquad \Omega_6^-(2) \cong U_4(2).$$

This is an abstract finite-group identification, not a serialized Monster embedding.

Let A be the adjacency matrix of the spread graph $\mathrm{SRG}(36, 15, 6, 6)$. The matrix

$$K = 2A - J$$

is a symmetric regular Hadamard matrix:

$$KK^{\mathsf{T}} = 36I, \qquad K\mathbf{1} = -6\mathbf{1}.$$

Hence $H = K/6$ is an exact rational orthogonal involution and

$$E_{15} = \frac{I+H}{2}, \qquad E_{21} = \frac{I-H}{2}.$$

The rank-15 signal and rank-21 guard sectors therefore arise from one explicit uniform-amplitude two-phase thirty-six-port transform. This is a transfer-matrix certificate, not a laboratory claim.

The associative algebra generated by the fifteen-dimensional left-multiplication operators has dimension 225 modulo 1,000,003, while the derivation equations have rank 225. Therefore the rational Norton algebra is simple and

$$\mathrm{Der}(V) = 0.$$

Adjacent pairs generate a two-dimensional algebra with idempotents $0, a, b, \frac{3}{4}(a+b)$; nonadjacent pairs generate a three-dimensional algebra containing the unique third Fischer axis c, with idempotents $0, a, b, c, a+b+c$. The fusion spectrum remains non-Majorana and non-Griess.

If D is the 36×120 axis–Fischer-line incidence matrix, then its binary left kernel is a $[36, 6]$ code with weight enumerator

$$1 + 27z^{16} + 36z^{20}.$$

The quadratic form $q(x) = \mathrm{wt}(x)/4 \pmod 2$ is nondegenerate of minus type. Its 27 nonzero singular vectors recover $\mathrm{SRG}(27, 10, 1, 5)$, while its 36 nonsingular vectors are exactly the Miyamoto supports and recover the spread graph. Thus the 27- and 36-object carriers are the two nonzero sectors of one explicit six-bit orthogonal space.

Finally, the intersection graph of the 120 Fischer lines has degree 27, spectrum

$$27^1 \oplus 9^{20} \oplus 3^{15} \oplus (-3)^{84},$$

and distance distribution $1 + 27 + 90 + 2$ from every vertex. Its forty antipodal fibers are the three triples through each W33 line, and its quotient is $\mathrm{SRG}(40, 27, 18, 18)$. The lift is a connected three-cover with full S_3 holonomy and trivial deck group; it is not a regular C_3 cover.

37.1 Cubic-transversal closure, tomotope completion square, and website restoration

The long-form public site was restored byte-for-byte from the final intact blob `41a8d733f42da18282fa276f5d2fa82`
the compact replacement was preserved separately at `docs/source-derived-architecture-landing-2026-08-05`

For the 240×5040 cubic dependency incidence matrix T, put $D = TT^{\mathsf{T}} - 63I$. If a transversal has size m and $P = \sum_t \binom{h_t}{2} = \frac{1}{2}x^{\mathsf{T}}Dx$, then the minimum eigenvalue -18 of D gives

$$P \geq \frac{3}{10}m^2 - 9m,$$

while $h_t \in \{1, 2, 3\}$ gives

$$P \leq \frac{3}{2}(63m - 5040).$$

Hence $(m - 105)(m - 240) \leq 0$. Equality at $m = 105$ would make the integer P equal to $4725/2$, so $m \geq 106$. An explicit 62-face cap gives a 178-face transversal, and therefore

$$\boxed{106 \leq \tau_{\text{cubic}} \leq 178}.$$

The covering-radius interval remains $389 \leq R \leq 435$.

The 45-anchor graph has independence number five and fractional chromatic number nine. The generalized weighted-Hoffman optimum over arbitrary edge-supported Hermitian matrices is therefore exactly nine, attained by $A_{45} \otimes I_4$. This does not change $10 \leq \chi(H) \leq 11$.

The corrected edge–face layer admits three vertex-structure orbits and three cell covers. All nine pairings are connected rank-four incidence geometries with f-vector $(4, 12, 16, 8)$, 192 flags, and all diamond intervals of size two. Their automorphism-order matrix is

$$\begin{pmatrix} 96 & 96 & 192 \\ 192 & 96 & 96 \\ 96 & 192 & 96 \end{pmatrix}.$$

Thus six completions are two-flag-orbit tomotopes with automorphism order 96, while a perfect matching of three completions gives regular non-split central lifts of order 192.

The characteristic-three complement has filtration

$$159 \supset 44 \supset 14 \supset 0.$$

The bottom 14-space is absolutely irreducible. The middle quotient satisfies the non-split exact sequence

$$0 \longrightarrow 1 \oplus 5 \oplus 10 \longrightarrow M_{30} \longrightarrow 14 \longrightarrow 0.$$

The top 115-quotient has fixed-socle dimension two and trivial-head multiplicity one; its remaining factors are not labelled here.

Finally, all 32 full-orbit binary roundings of the integral weighted W33–Clebsch bridge were exhausted. None is 10-regular and none satisfies the Gewirtz identity $A^2 = 8I - 2A + 2J$. Asymmetric rounding remains open.

Two further constructions follow. The 3×3 tomotope completion square has six order-96 entries and a perfect matching of three order-192 entries. The explicit 62-cap has trivial stabilizer in $U_4(2)$, producing a free orbit of 25,920 cap witnesses.

The frozen semantic SHA-256 is `c6e5e73fb5a18d9add4c1643d52df31413280b0e5cd157294ca686f8df32a299`. No remote CI, PDF, hardware, laboratory, or physical claim is promoted here.

38 Symmetry operating system, space–time decoding, and falsifiable architecture

Passes 3743–3750 convert the W33 machine from an abstract universal single-particle carrier into an explicitly testable control-plane architecture. The executable certificate has semantic digest

`67ce2d0ab105d8809dd6ae1af60381e1181d9a08c9032ce1dad1130f7bb8ff62`

and is reproduced by the focused regression suite.

Symmetry operating system. Four projective symplectic transvections generate the full order-25,920 point action. Their Cayley compiler has word diameter 11 and mean word length 8.17507716049; a complete exact group sweep uses 211,898 generator macros. This is an exact permutation compiler, not yet a calibrated optical pulse compiler.

Correlated space–time decoding. For the declared two-state hidden-Markov benchmark, eight consecutive observations have exact MAP error $1.1395646255 \times 10^{-3}$, whereas observations separated by the four-tick incidence schedule have error $6.3819353522 \times 10^{-5}$. Thus the interleaved error is 0.0560032771 of the consecutive value. The result is model-specific and does not promote an empirical device threshold.

Hybrid resource factory. The protected resource graph contains forty point registers, forty line-check registers, and one frame/clock register, with $160 + 80 = 240$ logical links. Under the verified $[[3, 1, 2]]_3$ block carrier this requires 243 physical qutrits in 729 triple-rail modes. Exhaustive search over 3^9 bilinear couplings proves that each logical controlled phase needs at least three physical phase terms, so all logical links require at least 720 such terms. A single frame bus has serial-depth lower bound 80.

Grand matched-geometry tournament. All 167,356 degree-twelve Cayley generator sets on $\mathbb{Z}_2^3 \times \mathbb{Z}_5$ were enumerated; 165,984 are connected. Their largest spectral gap is $9.527864045 < 10$, the exact W33 gap. A competitor attains minimum four common neighbours on nonedges, but with histogram $4^{18}6^{8}12^{1}$ rather than W33's uniform 4^{270}. Frozen layout witnesses also show that W33 does not minimize the tested Manhattan wirelength proxy.

Preregistered hardware challenge. A prospective three-device comparison locks structural, transport, twirl, and scheduling endpoints before data. W33 is falsified if its recovered support violates $A^2 = 8I - 2A + 4J$, if it fails to beat both matched controls by 2×10^{-4} on the locked worst-nonedge task, if the twirled residual exceeds ten percent, or if the four-tick schedule hides serialization or unreported resources.

Three outside-lane results. The relative permutations of the four incidence matchings generate the full symmetric group S_{40}, so the schedule is a universal classical permutation microcode, not a coherent quantum gate set. Exact unweighted automorphism twirling requires a sample count divisible by $\mathrm{lcm}(40, 480, 1080) = 4320$; this lower bound is not sufficient, and the known exact sweep uses all 25,920 automorphisms. Finally, passive fixed-adjacency dynamics lie in $\mathrm{span}\{I, A, J\}$, while zero forcing gives the certified control-port interval

$$24 \le Z(W33) \le 29.$$

Thus symmetry provides compression only when accompanied by an explicit symmetry-breaking control layer.

Evidence boundary. No optical pulse calibration, measured correlated-noise law, complete fault-tolerance threshold, minimum physical layout, coherent S_{40} universality, exact minimum twirl design, or exact zero-forcing number is claimed by these computations.

39 The six-bit quadratic parent, the rooted code lattice, and the Fischer holonomy frame

The thirty-six spreads and their 120 Norton triples admit a further exact closure. The Miyamoto involutions generate

$$O_6^-(2) \cong U_4(2){:}2, \qquad \Omega_6^-(2) \cong U_4(2),$$

and the compressed standard pairs can be supplemented by complete cycle and class-size fingerprints on both the thirty-six axes and the 120 Fischer lines. These are abstract finite-group search filters; no serialized Monster words are asserted.

Let D be the 36×120 axis–triple incidence matrix. Its binary left kernel is the doubly-even self-orthogonal code

$$C = [36, 6, 16], \qquad W_C(z) = 1 + 27z^{16} + 36z^{20}.$$

The associated Construction-A lattice

$$L(C) = 2^{-1/2}\{z \in \mathbb{Z}^{36} : z \bmod 2 \in C\}$$

is even, has minimum norm two, and satisfies

$$\mathrm{SNF}(G) = 1^{12}2^{24}, \qquad L(C)^*/L(C) \cong (\mathbb{Z}/2\mathbb{Z})^{24}.$$

Its root system is A_1^{36}, so the lattice is explicitly rooted and non-unimodular rather than Leech-like.

If A is the spread graph adjacency matrix, then

$$K = 2A - J, \qquad H = K/6$$

is the symmetric rational Hadamard involution of the previous section. There exist twenty-one pairwise orthogonal primitive vectors $v_i \in \mathbb{Z}^{36}$ such that the commuting Householder reflections

$$R_i = I - 2\frac{v_i v_i^\mathsf{T}}{v_i^\mathsf{T} v_i}$$

satisfy the exact factorization

$$H = R_1 R_2 \cdots R_{21}.$$

This is a transfer-matrix factorization and not a fabricated optical-device claim.

The same code identifies a six-dimensional minus-type quadratic space. Its thirty-six nonsingular vectors form a $(64, 36, 20)$ Hadamard difference set. For the full bilinear Walsh matrix W_{64} and its nonsingular principal minor W_N,

$$W_{64} W_{64}^\mathsf{T} = 64I, \qquad K = W_N - 2I.$$

Thus the thirty-six-port transform is a diagonal shift of a quadratic-Walsh principal minor.

Integrally, the triple incidence has

$$\mathrm{SNF}(D) = 1^{30}2^{5}6, \qquad \mathrm{coker}(D) \cong (\mathbb{Z}/2\mathbb{Z})^{5} \times \mathbb{Z}/6\mathbb{Z}.$$

The order 192 is not promoted to a tomotope identification without an objectwise map.

Finally, the 120-vertex Fischer-line graph is not distance regular: distance-two pairs split into two intersection profiles. Its exact symmetry group is $U_4(2){:}2$, a vertex stabilizer has subdegrees $1, 2, 27, 36, 54$, and the Terwilliger algebra has dimension 55 modulo three independent large primes. Projecting the three-point fibers to their standard S_3 sectors produces a rank-twenty tight-frame projector

$$E_{20} = \frac{FAF + 9F}{108}, \qquad (FAF + 9F)^2 = 108(FAF + 9F),$$

whose 120 vectors have normalized inner products in

$$\left\{-\frac{1}{2}, -\frac{1}{6}, 0, \frac{1}{3}\right\}.$$

The Monster class-fusion computation is delegated to a self-reporting GAP/CTblLib workflow. No fusion count or degree-81 multiplicity is asserted until its committed execution artifact is observed.

39.1 Passes 3769–3786: the $GQ(4,2)$–Veldkamp–W33 closure

The six-dimensional minus quadratic space over $\mathbb{F}_2$ supplies 27 singular and 36 nonsingular nonzero vectors. Its singular points and totally singular lines form $GQ(2,4)$, with 27 points and 45 three-point lines; the dual $GQ(4,2)$ has 45 points and 27 five-point lines. The dual point graph is

$$\mathrm{srg}(45, 12, 3, 3), \qquad \mathrm{Spec}(A) = \{12^1, 3^{20}, (-3)^{24}\}.$$

Writing $A_0 = I$, $A_1 = A$, and $A_2 = J - I - A$, its Bose–Mesner multiplication is

$$A_1^2 = 12A_0 + 3A_1 + 3A_2, \quad A_1A_2 = 8A_1 + 9A_2, \quad A_2^2 = 32A_0 + 24A_1 + 22A_2.$$

The primitive-idempotent ranks are $1, 20, 24$. The Terwilliger algebra at a point has dimension 16, center dimension 5, and standard-module central block multiplicities $2, 3, 8, 14, 18$, hence semisimple type

$$M_3 \oplus M_2 \oplus \mathbb{C}^3.$$

All 651 projective lines of $PG(5,2)$ split by quadratic type as

$$45 + 216 + 270 + 120.$$

The final 120 lines consist of three nonsingular vectors and coincide exactly with the Norton–Fischer triples of the preceding spread algebra. Thus that triple system is the type-IV line class of the Veldkamp space $V(GQ(2,4))$.

An exact-cover enumeration finds 200 ovoids of $GQ(4,2)$, in automorphism orbits $40+160$. On the 40 plane ovoids, declare two objects adjacent when they share three quadrangle lines. The resulting graph is

$$\mathrm{srg}(40, 12, 2, 4)$$

and is explicitly isomorphic to the independently reconstructed W33 point graph. If M is the 40×45 plane-ovoid incidence matrix, then

$$MM^{\mathsf{T}} = 8I + 2A_{W33} + J, \qquad \mathrm{Spec}(MM^{\mathsf{T}}) = \{72^1, 12^{24}, 0^{15}\}.$$

The recurring count $135 = 45 \cdot 3$ supports two inequivalent $U_4(2)$ actions. The Lagrangian-frame stabilizer has order census

$$1^1\, 2^{43}\, 3^{32}\, 4^{84}\, 6^{32},$$

whereas an ordinary $GQ(4,2)$ flag stabilizer has

$$1^1\, 2^{27}\, 3^{32}\, 4^{36}\, 6^{96}.$$

The corresponding permutation-character norms are 9 and 8, with cross-inner product 4.

The rank-24 primitive projector is

$$P = (A - 12I)(A - 3I) = 90E_{-3},$$

with entries 48 on the diagonal, -12 on adjacent pairs, and 3 on nonadjacent pairs. On $\operatorname{im} P$ define

$$x \star y = E_{-3}(x \circ y), \qquad a_i = Pe_i/21.$$

Then the 45 vectors a_i are idempotent axes with Peirce spectrum

$$1^1, \qquad (1/7)^{14}, \qquad (-1/3)^9.$$

For collinear axes,

$$a_i \star a_j = -\frac{1}{3}(a_i + a_j),$$

and the five axes on every quadrangle line sum to zero. For noncollinear i, j with common neighbors c_1, c_2, c_3,

$$a_i \star a_j = \frac{1}{7}(a_i + a_j) + \frac{5}{42}(a_{c_1} + a_{c_2} + a_{c_3}).$$

The algebra is positive Frobenius; its multiplication operators span all M_{24} modulo 1000003, so it is simple, and its derivation algebra is zero because 1/2 is absent from every axis spectrum. Its fusion law is not nontrivially $\mathbb{Z}_2$-graded, so this is not a Monster-2A Majorana algebra.

For a representative $W(D_4)$ frame stabilizer, the invariant alternating forms modulo two have dimension seven and rank distribution

$$0^1, 4^3, 6^4, 8^{12}, 10^{12}, 14^{40}, 16^{56}.$$

No invariant form has rank 24; consequently no $W(D_4)$-invariant even unimodular rank-24 child exists. A deterministic trace-eight involution nevertheless permits an integral determinant--1 basis change that pulls back E_8^3 to an even unimodular positive child whose exact surviving $U_4(2)$ stabilizer is C_2. This child has roots and is not promoted as a Leech realization.

The finite-group embedding front remains fail-closed. A future candidate must reproduce the complete 36/45/27/120/135/40 fingerprint, independent order 51840, class correspondence, and content-addressed incidence digests before any Monster embedding is asserted.

39.2 Exact twirl estimation, Floquet cancellation, distributed clocking, and control closure (Passes 3787–3794)

The W33 control architecture now has exact finite resource contracts beyond the preceding symmetry-operating-system packet. The results below distinguish algebraic control statements from physical pulse and device claims.

Minimum exact observable twirl. The ordered-pair action has orbit sizes 40, 480, and 1080. Exact unweighted estimation of the diagonal, oriented-edge, and oriented-nonedge averages therefore needs at least

$$40 + 480 + 1080 = 1600$$

settings, and one shortest automorphism word per target attains the bound. The total compiled cost is 8934 generator macros, versus 211898 for the complete group sweep. This closes observable/orbit estimation, not the stronger active conjugation-channel twirl, whose unweighted divisibility lower bound remains 4320.

Floquet cancellation. The four incidence matchings induce relative cycle types $(20, 20)$, (40), and (40) and generate S_{40}. For either relative forty-cycle C and any diagonal error Hamiltonian D,

$$\frac{1}{40}\sum_{t=0}^{39} C^{-t} D C^{t} = \frac{\operatorname{tr} D}{40} I.$$

Hence all traceless static diagonal disorder cancels exactly over one orbit. This is not a general coherent-error or loss theorem.

Optimal distributed clock. On the eighty-node point–line incidence network, one matching per tick gives the exact informed-node census

$$1, 2, 4, 8, 16, 30, 54, 80.$$

All nodes receive the frame in seven ticks. The information-doubling lower bound $\lceil \log_2 80 \rceil = 7$ proves optimality in the declared store-and-forward model, replacing the prior eighty-step central-bus serialization.

Exact control-port count. A complete automorphism-orbit enumeration of reverse zero-forcing sequences has orbit counts

$$1, 1, 2, 5, 16, 43, 191, 769, 3024, 9772, 24852, 34890$$

through depth eleven. Every depth-eleven orbit is terminal and depth twelve is empty. Therefore

$$\boxed{Z(W33) = 29}.$$

An explicit minimum set and eleven-step force chain are frozen in the executable certificate. The graph-control consequence retains the standard Hamiltonian assumptions and is not a pulse-robustness result.

Dynamic W33/local phase. An explicit relabelling of the degree-twelve local ring and W33 shares 128 of their 240 couplers. The reconfigurable union therefore uses

$$240 + 240 - 128 = 352$$

couplers, a 26.67% reduction from disjoint hardware. The cross-phase two-step operator has pairwise minimum two, while the complete two-phase walk has minimum four. The witness is not claimed globally optimal.

Three further mechanisms. First, the exact zero-forcing chain virtualizes eleven unactuated calibration channels behind twenty-nine physical phase actuators in the ideal graph-linear model. Second, the same forty-cycle provides a forty-setting Fourier spectrometer that reconstructs all thirty-nine traceless diagonal modes while its zero character performs common-mode averaging. Third, reversing the optimal broadcast tree gives a seven-tick qutrit gather and a fourteen-tick collision-free all-reduce.

Evidence boundary. The frozen semantic certificate is

`aa3daf462320badd607a1720479f083df159044f24242e7f46810de30bc2e0d8`.

The results certify finite group words, orbit-estimation cost, static diagonal cancellation, broadcast depth, zero forcing, a dynamic-topology witness, and three algebraic communication/calibration mechanisms. They do not certify an optimal active twirl, a complete Floquet quantum code, noisy actuator virtualization, minimum physical layout, or laboratory performance.

39.3 Exact plane-ovoid transport, the 200-ovoid scheme, and a rootless axial Leech polarization

Passes 3795–3812 replace the prior existential plane-ovoid isomorphism by a canonical objectwise dictionary. The full $O_6^-(2)$ torsor contains 51,840 graph isomorphisms, and its lexicographically minimal representative transports all forty points, all forty four-point lines, and all thirty-six ten-line spreads of $W(3,3)$ onto the forty plane ovoids of $GQ(4,2)$.

The action on all $200 = 40+160$ ovoids has nineteen ordered-pair orbitals. Their orbital matrices form a nineteen-dimensional coherent algebra; the complete $19 \times 19 \times 19$ intersection tensor has 405 nonzero entries and center dimension five modulo 1,000,003. Two self-paired valency-three relations on the 160 tripods each split into forty disjoint copies of K_4. If B is the 40×40 intersection matrix of the two block systems, then

$$BB^{\mathsf{T}} = 4I + A_{W(3,3)},$$

so the tripods are precisely the 160 flags between two dual forty-object systems and reconstruct the strongly regular graph $(40, 12, 2, 4)$.

Using the standard degree-24 M_{24} generators and a base octad, the verifier regenerates the binary Golay code with weight distribution

$$A_0 = 1, \quad A_8 = 759, \quad A_{12} = 2576, \quad A_{16} = 759, \quad A_{24} = 1.$$

Its integral coordinate construction gives a positive-definite even unimodular rank-24 Gram matrix. Exact short-vector enumeration finds only the zero vector at norm at most two and exhibits norm four, hence the lattice is rootless with minimum four and is therefore the Leech lattice. A determinant-one integral transport places this Gram matrix on the rank-24 axial carrier. Exhaustive testing of all 25,920 elements of the axial $U_4(2)$ action gives

$$\mathrm{Stab}_{U_4(2)}(\Lambda_{\mathrm{axial}}) \cong C_2.$$

Seven explicit degree-45 permutations generate $O_6^-(2) = U_4(2){:}2$ of order 51,840, with index-two subgroup $U_4(2)$ of order 25,920. This is an abstract finite descent only: serialized Monster or `mmgroup` words and an executed Monster class fusion remain absent.

Finally, the forty-five axes span dimension twenty-four. The twenty-seven five-axis line sums have rank twenty-one and generate the complete linear kernel. Collinear pairs generate dimension two, noncollinear pairs dimension four, and the six triple orbits have generated dimensions

$$4,\ 10,\ 24,\ 10,\ 5,\ 3$$

for orbit sizes $240, 2160, 2880, 6480, 2160, 270$, respectively. Thus one orbit of 2,880 triples generates the entire algebra, although no pair does. The algebra is simple, nonunital, and has zero derivations. The exact semantic certificate is

`be5ee56a84141a7bbc896b4cef0e8eda32167a00749dbb17f10d24d0505bdd41`.

No Monster/Majorana/Griess/VOA, remote-PDF, hardware, laboratory, or physical claim is promoted by this computation.

40 Quadratic Parent, Discriminant Chain, Multiport Compiler, and Holonomy Scheme

The thirty-six nonsingular vectors of the minus-type quadratic space $(\mathbf{F}_2^6, q)$ are one subconstituent of the Cayley graph $x \sim y \Leftrightarrow q(x+y) = 1$. Its adjacency matrix obeys

$$A_{64}^2 = 16I + 20J,$$

so the parent is $\mathrm{SRG}(64, 36, 20, 20)$. The neighborhood of zero induces the complement of $\mathrm{SRG}(36, 15, 6, 6)$, while the twenty-seven nonzero nonneighbors induce the Schlaefli graph, the complement of $\mathrm{SRG}(27, 10, 1, 5)$.

For the binary character code

$$C = \{(\beta(a, x))_{x \in N} : a \in \mathbf{F}_2^6\} = [36, 6, 16],$$

the Construction–A discriminant module is

$$L(C)^*/L(C) \cong C^\perp/C \cong (\mathbf{Z}/2)^{24}, \qquad q_L(c + C) = \mathrm{wt}(c)/2 \pmod{2\mathbf{Z}}.$$

An explicit rank-eleven totally isotropic chain yields a maximal doubly-even $[36, 17, 8]$ extension and an even overlattice of determinant four. Rank twelve is impossible because it would produce an even unimodular lattice of rank thirty-six. The residual two-dimensional quotient gives three singly-even self-dual code extensions and hence three odd unimodular Construction–A neighbors, in two distinct weight-enumerator types.

The exact involution

$$H = (2A_{36} - J)/6$$

admits a proof-carrying compilation through twenty-one primitive integer Householder directions. Balanced exact Givens trees require 944 two-mode rotations, twenty-one single-mode π phases, and sequential depth 221. A deterministic adjacent elimination reduces the candidate mesh to 512 rotations in sixty-nine layers. Full support gives the unconditional light-cone bound $d \geq \lceil \log_2 36 \rceil = 6$.

The 120 Fischer triples carry a four-angle rank-twenty frame whose five Gram relations close to a commutative association scheme. Its valencies and multiplicities are

$$(1, 2, 54, 36, 27), \qquad (1, 20, 24, 60, 15),$$

and its first eigenmatrix is

$$\begin{pmatrix} 1 & 2 & 54 & 36 & 27 \\ 1 & -1 & -9 & 0 & 9 \\ 1 & 2 & -6 & 6 & -3 \\ 1 & -1 & 3 & 0 & -3 \\ 1 & 2 & 6 & -12 & 3 \end{pmatrix}.$$

The forty antipodal fibers quotient to $W(3,3)$. On quotient nonedges the $1/3$ relation is a perfect matching and defines an S_3 connection. Of the 3240 base triangles, 1080 are flat and 2160 have transposition holonomy; three- cycle holonomy first appears on four-cycles, so the full holonomy group is S_3.

Finally, if F and T denote port–frame and port–triple incidence, then

$$FF^{\mathsf{T}} = 15I + 3A, \qquad TT^{\mathsf{T}} = 9I + J - A,$$

and therefore

$$\boxed{FF^{\mathsf{T}} + 3TT^{\mathsf{T}} = 42I + 3J}.$$

After removing the all-ones channel, the weighted frame–triple carrier is an exact tight resolution of the thirty-five-dimensional contrast space.

All statements in this section are finite exact certificates. They do not supply Monster words, a Leech identification, an optimal optical circuit, a hardware result, or a physical gauge interpretation.

41 Maximal-code census, exact adjacent mesh, holonomy closure, and the ovoid flag completion

The six-bit minus-type quadratic parent supports a complete exact refinement of the code, multiport, Fischer-holonomy, finite-group, and ovoid fronts. The frozen component certificate has semantic digest

`b141dd0f82e4a6b1ee62d1c57f0e92bdfc9f58d3b32515f9521a0175fdca88a1`.

41.1 Maximal doubly-even extensions

Let $C = [36, 6, 16]$ be the binary character code on the nonsingular vectors. Its discriminant module is the minus-type quadratic space $O_{24}^{-}(2)$ of Witt index eleven. Hence the exact number of maximal doubly-even extensions is

$$N = \prod_{i=2}^{12}(2^i + 1) = 24013790538727978586812 5.$$

Since $|U_4(2){:}2| = 51840$, there are at least

$$463228984157561 3440$$

orbits. Thus the exact classification is necessarily a census and orbit-explosion theorem rather than a literal representative list. An explicit maximal extension $D = [36, 17, 8]$ has enumerator

$$1 + 225z^8 + 9555z^{12} + 55755z^{16} + 55755z^{20} + 9555z^{24} + 225z^{28} + z^{36},$$

and trivial stabilizer in $U_4(2){:}2$.

41.2 An exact 418-gate adjacent mesh

For

$$H = (2A_{36} - J)/6,$$

an exact symbolic adjacent Givens elimination gives 418 nontrivial rotations in 69 layers, 212 exact zero skips, and one terminal π phase. Every gate has

$$c = \operatorname{sgn}(c)\sqrt{c^2}, \qquad s = \operatorname{sgn}(s)\sqrt{s^2}, \qquad c^2 + s^2 = 1,$$

with rational squared parameters. The exact parameter digest is

`5c933cc2e6d2484e97894f3ca1f71627214238e41a68cd59b7527993d2b06b6b`.

The certified bounds are

$$35 \leq g \leq 418, \qquad 6 \leq d \leq 69.$$

No global optimality claim is made.

41.3 The rank-five holonomy scheme

The 120 Fischer triples carry five Gram relations with valencies

$$1, 2, 54, 36, 27$$

and primitive multiplicities

$$1, 20, 24, 60, 15.$$

All Krein parameters are exact. There is no Q-polynomial ordering, and exactly four fusion schemes survive: the full scheme, the fusion $(2, 4)$, the fusion $(2, 3, 4)$, and the complete rank-two fusion. The full automorphism group is $U_4(2){:}2$. At a point,

$$\dim_{\mathbf{Q}} T = 79, \qquad \dim_{\mathbf{Q}} Z(T) = 10.$$

The certificate proves exact closure using 79 word matrices and all 83 point-stabilizer orbitals.

41.4 Complete abstract standard-pair census

Inside $U_4(2)$, fix an involution a in the class of size 45. Exactly 1152 order-five elements b satisfy

$$|ab| = 9, \qquad |[a, b]| = 3.$$

The centralizer $C(a)$ has order 576 and splits those elements into two orbits of size 576. Each orbit representative generates all 25920 elements. Therefore the complete ordered standard-pair census is

$$45 \cdot 1152 = 51840.$$

No serialized Monster words or Monster character restriction is asserted.

41.5 Ovoids as W33 points and flags

All 200 ovoids of $GQ(4,2)$ split into 40 plane ovoids and 160 tripods. Objectwise,

$$40 \text{ plane ovoids} = 40 \text{ W33 points},$$

$$160 \text{ tripods} = 160 \text{ incident W33 point-line flags}.$$

The 160 tripod port columns collapse four-to-one onto the 40 W33 line-versus-spread incidence columns. Finally the five fibers

$$1 \mid 27 \mid 36 \mid 40 \mid 160$$

form one $O_6^-(2)$ orbital coherent configuration of rank 48.

Boundary. The packet does not materialize billions of orbit representatives, prove global circuit optimality, produce Monster words or class fusion, identify a rootless lattice, or certify hardware, laboratory, remote-CI, or PDF results.

42 Adaptive geometry, virtual topology, autonomous attraction, and thermodynamic inversion

Passes 3829–3836 recast the W33 machine as a dynamically selected control architecture rather than a single permanently active graph. The exact certificate is generated by `analysis/w33_pass3829_3836_adaptive_` and has semantic digest `d924764ba5d64326fbdc8d7a6bbd8bcdff2fd08fb2a4c6560f29c53a71c29fe4`.

Adaptive hypervisor. A declared four-state, five-action discounted control model selects the local ring in the clean regime, the forty-cycle Floquet mode under diagonal drift, W33 under burst loss, and spectroscopy or direct 29-port control under crosstalk. Its frozen adaptive score is 36.5799750262, compared with 66.3616366758 for the best static policy. These are benchmark-model scores, not measured logical-error rates.

Virtual topology. The 240 W33 edges admit an explicit decomposition into twelve perfect matchings. Because every vertex has degree twelve and one edge token per vertex can be consumed in one round, twelve rounds are both necessary and sufficient. Thus a measurement- or teleportation-mediated realization can instantiate the complete W33 interaction epoch using twenty heralded edge tokens per round and no permanently fabricated W33 couplers, before source, fusion, memory, and feedforward overheads are counted.

Autonomous attraction. Let Π_{W33} average a symmetric forty-mode matrix over its diagonal, adjacent, and nonadjacent entries. Then Π_{W33} is an idempotent projector with image

$$\operatorname{span}\{I, A, J - I - A\}.$$

The ideal reservoir step

$$\Phi(X) = \tfrac{1}{2}\big(X + \Pi_{W33}(X)\big)$$

contracts every symmetry-breaking component by exactly one half. Twenty steps therefore suppress the deviation by 2^{-20}. This is an algebraic reservoir target, not an implemented Lindbladian.

Thermodynamic crossover. For independent heralded edge success p, the worst-nonedge route probabilities are

$$q_{W33} = 1 - (1 - p^2)^4, \qquad q_2 = 1 - (1 - p^2)^2.$$

Using frozen wirelength proxies 1064 and 938, the retry-energy crossover is

$$p_* = \sqrt{1 - \sqrt{63/469}} \approx 0.795921897507.$$

Below p_*, W33's four-route redundancy outweighs its longer layout proxy; above p_*, the local two-route control is cheaper in this declared model. No wall-plug energy claim follows.

Substrate inversion. One complete interaction epoch has the exact primitive-count alternatives

$$C_{\text{native}} = 240a + 12g + 240d, \quad C_{\text{virtual}} = 240b + 12g + 240d,$$

$$C_{\text{mesh}} \in 67a + [12, 870]g + 870d, \quad C_{\text{bus}} = a + 240g + 240d,$$

where a, b, g, d price static links, consumable edge tokens, rounds, and routing hops. No substrate class is coefficient-independently optimal.

Three additional mechanisms. First, the exact strongly regular identity

$$A^2 + 2A - 8I - 4J = 0$$

acts as a digital-twin checksum. Every single edge deletion produces a rank-four residual with squared Frobenius norm 54 and support 48; every single insertion produces rank four, norm squared 58, and support 52. The two corrupted endpoints are the unique residual rows of support thirteen or fourteen, respectively.

Second, independent heralded links can be distilled geometrically. For nonadjacent vertices,

$$P_{\text{route}} = 1 - (1 - p^2)^4,$$

while an adjacent pair can use its direct edge and two triangle detours,

$$P_{\text{route}} = 1 - (1 - p)(1 - p^2)^2.$$

At target reliability 0.999, the corresponding edge-success thresholds are approximately 0.906737 and 0.935615.

Third, the four transvections and their inverses define an eight-jump symmetry bath on the forty points. Its second eigenvalue is 0.9367092858522434, its spectral gap is 0.0632907141477566, and the standard spectral bound gives 89 jumps for one-percent mixing from a localized point defect. This is an ideal Markov process, not a physical dissipation-rate theorem.

Evidence boundary. The finite graph identities, matching decomposition, defect signatures, projector law, routing formulas, and primitive counts are exact. The adaptive cost model, ideal reservoir, retry-energy proxy, substrate coefficients, independent-link model, and symmetry-bath implementation remain explicit architectural hypotheses rather than laboratory evidence.

43 Ovoid Wedderburn Algebra, Leech Transport, and Triality Incidence

The complete $40 + 160$ ovoid coherent algebra has dimension 19 and centre dimension 5 over $\mathbf{Q}$. Exact rational primitive central idempotents and minimal-left-ideal ranks give the split decomposition

$$\mathcal{A} \cong M_2(\mathbf{Q}) \oplus \mathbf{Q} \oplus M_2(\mathbf{Q}) \oplus \mathbf{Q} \oplus M_3(\mathbf{Q}).$$

On the 200-point permutation module,

$$\mathbf{Q}^{200} \cong 1^{\oplus 2} \oplus 15_a^{\oplus 2} \oplus 15_b \oplus 24^{\oplus 3} \oplus 81.$$

The canonical W33–plane-ovoid dictionary transports all 240 edges, 160 triangles, 40 lines, and 36 spreads. The oriented ternary check matrices satisfy

$$\operatorname{rank} H_X = 39, \qquad \operatorname{rank} H_Z = 120, \qquad H_X H_Z^{\mathsf{T}} = 0,$$

with $d_X = 3$ and $d_Z = 4$, hence the edge code is $[[240, 81, 3]]_3$.

Under the explicit determinant-one Leech transport, exactly three integral axial numerator vectors have norm four:

$$a_3, \quad a_{33}, \quad a_{36}.$$

The surviving involution fixes a_3, swaps a_{33} and a_{36}, and the three carriers lift to nine D_4 frames.

The 45 marked axes admit a universal rational presentation by the 27 five-axis line sums and the two exact pair-product laws. The line relations have rank 21, so the quotient has dimension 24. An independent rooted graph computation gives

$$|\operatorname{Aut}_{\text{marked}}| = 45 \cdot 1152 = 51840 \cong |O_6^-(2)|.$$

The 148995 four-axis subsets form 20 orbits, with generated-dimension census

$$4^{135},\ 5^{720},\ 6^{1080},\ 10^{16740},\ 14^{27000},\ 24^{103320}.$$

The two 40-block tripod fibrations produce a 40×40 incidence matrix whose row and column Gram graphs both have parameters $(40, 12, 2, 4)$ but are nonisomorphic. The full bipartition-preserving automorphism group has order 51840, and no side swap exists. The smallest tripod–Norton relation has size 480, rank 40, and bipartite graph

$$40K_{4,3}.$$

Although this equals the number of directed W33 edges, the two permutation characters have norms 23 and 24 with cross inner product 18, so the two 480-actions are inequivalent.

The semantic certificate is `ede91cc799f9093209c589fb0b73c88fca4159bedcc8e1e7cd6680c5f0c778fd`. Serialized Monster words, Monster class fusion, the full unmarked linear automorphism group, remote CI/PDF success, and physical claims remain outside the promoted boundary.

43.1 Cubic tightening, tomotope outer extension, and complete modular descent

The dependency hypergraph on the 240 filled faces admits an explicit 63-face cap. Its complement is a 177-face transversal, improving the live exact interval to

$$106 \leq \tau_{\text{cubic}} \leq 177.$$

The witness has triple-hit profile $1^{876}2^{2217}3^{1947}$ and is locally optimal against every exchange removing at most two selected faces. No nonexistence theorem for a 64-cap is claimed, and the independent covering-radius boundary remains $389 \le R \le 435$.

The three exceptional entries in the 3×3 tomotope-completion square require a correction. Their automorphism groups have order 192 but trivial centre. Each has a unique normal elementary abelian subgroup 2^4, ordinary index-two subgroup $2^4{:}S_3$, and quotient D_{12}. Hence the exceptional group is the split, centreless outer extension

$$2^4{:}D_{12},$$

not a non-split central double cover.

The previously unresolved 115-dimensional characteristic-three quotient has composition factors

$$1^3 \oplus 5^3 \oplus 10^3 \oplus 14^3 \oplus 25.$$

An explicit 13-step series has successive dimensions

$$10, 14, 5, 10, 25, 5, 1, 14, 5, 10, 1, 14, 1,$$

and every factor generates its full matrix algebra over $\mathbb{F}_3$.

The exact Golay–Witt Gewirtz graph admits a canonical 16+40 split under a second fixed point. Its unique two-point-stabilizer-invariant degree-12 residual graph is not W33; it is the independent fourfold blow-up of Petersen, with spectrum $12^1 4^5 0^{30} (-8)^4$.

Three associated constructions are exact: the free 63-cap orbit is a binary constant-weight code of length 240, size 25,920, weight 63, and minimum distance 62; the residual graph has ten twin classes of size four and Petersen quotient; and the cap's forty W33-fibre occupancies form a free orbit with spectral energies 3969/40, 157/5, and 163/8 on the 12, 2, and -4 channels.

The authoritative website remains pinned to Git blob `41a8d733f42da18282fa276f5d2fa82bac7516f6`. A fail-closed migration validator requires literal owner authorization binding old blob, new blob, and an exact archive before replacement. No migration, remote CI/PDF result, hardware result, laboratory result, Monster embedding, or physical interpretation is asserted.

43.2 Unmarked axial symmetry, explicit rational modules, and the order-192 barcode

Passes 3887–3904 have semantic certificate `6ef2f6c4a0fa7e52da9f4abff6d899d0250eeabde1a601d53c40052a841` They also correct the four-axis generated-dimension census of Passes 3837–3854, whose unreduced three-factor `int64` contraction overflowed before modular reduction.

Let $E = (A - 12I)(A - 3I)/90$ and let $V = \operatorname{im} E$ carry $x \star y = E(x \circ y)$. Exact rational multiplication matrices satisfy

$$\operatorname{tr}(L_x L_y) = \frac{7}{30}\langle x, y\rangle.$$

Thus the Euclidean form is intrinsic to multiplication. If $y \neq 0$ is an idempotent and m is its largest coordinate, the three quadrangle lines through a maximizing point give $N \ge 3m^2/4$ on its twelve neighbours, while its thirty-two nonneighbours give $Q \ge m^2/8$. Together with

$$m = \frac{8}{15}m^2 - \frac{2}{15}N + \frac{1}{30}Q$$

and $\|y\|^2 = \sum_i y_i^3$, this proves

$$\|y\|^2 \ge \frac{480}{49}.$$

Equality holds exactly for the 45 known axes. Hence every unmarked algebra automorphism permutes them, and

$$\boxed{\mathrm{Aut}(V,\star) \cong O_6^-(2) \cong U_4(2):2, \qquad |\,\mathrm{Aut}\,| = 51{,}840.}$$

The 19-dimensional orbital algebra has explicit split rational model

$$M_2(\mathbb{Q}) \oplus \mathbb{Q} \oplus M_2(\mathbb{Q}) \oplus \mathbb{Q} \oplus M_3(\mathbb{Q}).$$

Primitive rational projectors produce irreducible modules of dimensions $1, 15, 15, 24, 81$, and

$$\mathbb{Q}^{200} \cong 1^{\oplus 2} \oplus 15_a^{\oplus 2} \oplus 15_b \oplus 24^{\oplus 3} \oplus 81.$$

All five character inner products were evaluated on the complete order-51,840 group and form the identity matrix.

The 480-element $40K_{4,3}$ incidence action has character norm 23 and decomposes on the five known constituents as

$$\chi_{480} = 1 + 2 \cdot 81 + 2 \cdot 15_a + 15_b + 3 \cdot 24 + \rho_{200},$$

where ρ_{200} has degree 200, norm 4, and is orthogonal to all five known characters. The explicit Monster-subgroup database contains portable maximal-overgroup seeds but no direct $U_4(2):2$ key; no Monster words or class fusion are promoted.

The corrected four-axis calculation reduces after each contraction stage and is stable over six primes. Its weighted generated-dimension census is

$$4^{135}5^{720}6^{1080}10^{16740}12^{5040}14^{27000}16^{14040}24^{84240}.$$

The twenty generating-set orbits collapse to eight simple nonunital subalgebra species of dimensions

$$\boxed{4, 5, 6, 10, 12, 14, 16, 24.}$$

Every species has zero annihilator and nucleus, full square span and trace rank, full associator ideal, and multiplication envelope M_d.

Finally, four order-192 mechanisms are separated. The ordered incident-pair/frame stabilizer is $W(D_4)$ with center 2 and derived subgroup 96; a distinguished involution centralizer is $D_8 \times S_4$ with derived subgroup 24; the archived octonion axis-line stabilizer is centerless and has 48 elements of order 8; and the corrected exceptional tomotope completion is the centerless split group $2^4 : D_{12}$. Equality of order or carrier size is therefore not an identification.

No portable Monster words, executed Monster class fusion, complete decomposition of ρ_{200}, complete tensor-product table, SmallGroup identification of the octonion stabilizer, remote CI/PDF success, hardware result, laboratory result, thermodynamic result, or physical mechanism is asserted without separate evidence.

43.3 Terwilliger sieve, improved adjacent mesh, code strata, and rank-48 overlap

The rank-five holonomy Terwilliger algebra has exact dimensions

$$\dim_{\mathbf{Q}} T = 79, \qquad \dim_{\mathbf{Q}} Z(T) = 10.$$

Assuming split semisimplicity over $\mathbf{Q}$, exhaustive integer arithmetic leaves exactly fourteen possible nondecreasing matrix-degree multisets satisfying $\sum_i n_i^2 = 79$; the largest block degree lies between four and eight. Primitive central idempotents are still required to select the actual multiset.

For the 36-port involution $H = (2A - J)/6$, a new symmetry-adapted ordering lowers the adjacent elimination candidate from 418 to

$$\text{398 rotations in 69 layers,}$$

with one terminal π phase. The zero pattern is numerically stable and all nontrivial rotations admit rational-square parameter recovery, but a complete exact-radical replay remains required before the circuit is promoted as an exact identity.

Maximal doubly-even $[36, 17]$ extensions containing the character code obey an exact one-parameter enumerator law. Writing $t = A_4$,

$$A_8 = 225 + 11t, \qquad A_{12} = 9555 - 39t, \qquad A_{16} = 55755 + 27t,$$

with symmetry about weight 18. A deterministic 258-code sample realizes $t = 0, 1, 2, 3, 4, 5$ and 184 distinct weight-eight coordinate-degree profiles, demonstrating that the enumerator is a coarse orbit invariant.

Finally, the 64-point quadratic carrier, 200-ovoid carrier, and combined 264-object carrier have orbital ranks 15, 19, and 48. Therefore their cross-Hom dimension is seven. Six dimensions are forced by the three-versus-two trivial constituents, leaving exactly one shared nontrivial 15-dimensional constituent. The 64-module consequently has form

$$1^3 \oplus 15_b \oplus X^2 \oplus Y, \qquad 2 \dim X + \dim Y = 46,$$

where X and Y do not occur in the 200-ovoid carrier. No Monster embedding, class fusion, exact-radical mesh proof, global mesh optimum, exhaustive code-orbit ledger, hardware result, laboratory result, or physical interpretation is claimed.

44 Passes 3913–3920: multi-defect immunity, symmetry dissipation, process control, and dual scheduling

The W33 control architecture now has a bounded multi-defect decoder, an explicit random-unitary symmetry Lindbladian, a partially observed controller, a correlated-fusion threshold, and a joint substrate/topology phase diagram. Three additional constructions compress the graph into a finite-field seed, extend the interaction schedule to an optimal all-to-all epoch, and compress the two-defect sensor.

Bounded topology immune system. For a measured adjacency matrix B, define

$$R(B) = B^2 + 2B - 8I - 4J.$$

The exhaustive ledger contains all

$$\binom{780}{2} = 303{,}810$$

two-edge insertion/deletion mixtures, and every complete residual is distinct. It also contains all

$$\binom{240}{3} = 2{,}275{,}280$$

three-edge deletions, again without a repeated residual. General mixed weight-three cases remain open.

Symmetry Lindbladian. Let U_g denote the four generating transvections and their inverses. The random-unitary generator

$$\mathcal{L}(X) = \frac{1}{8}\sum_g (U_g X U_g^\dagger - X)$$

has fixed algebra

$$\mathrm{Fix}(\mathcal{L}) = \mathrm{span}\{I, A, J - I - A\}.$$

Its unit-rate gap is

$$0.06329071414775411.$$

Every transvection consists of nine disjoint three-cycles and thirteen fixed points, yielding a 72 three-mode-cycle compilation target for the eight directed jumps. A sparse local physical realization with the same fixed algebra is not yet proved.

Partially observed geometry control. A six-step four-regime hidden-Markov benchmark with 85-percent regime readout and five geometry/control modes has exact belief-state optimum

$$23.115384313076735.$$

The best static policy costs 30.176036574158424, while full regime information gives the lower bound 16.514150252446512. These are declared benchmark values, not learned process-tensor measurements.

Correlated fusion compiler. A twenty-edge matching is accumulated by a 21-state Markov chain with stored-edge survival 0.995, common outage probability 0.05, and five attempts. Requiring a complete twelve-matching epoch to succeed with probability at least 0.99 gives

$$p \geq 0.9158993399430635.$$

The corresponding independent no-decay threshold is 0.8668344006872047.

Architecture phase diagram. A 500-point declared-cost grid compares native W33, virtual W33, a local ring, a routed mesh, and a shared bus. The winner counts are respectively

$$54, \quad 84, \quad 37, \quad 15, \quad 310.$$

Every implementation wins somewhere; no architecture is coefficient-independently optimal.

Topology DNA. Four four-trit transvection generators and one ordered edge generate the order-25,920 group and the complete 480-element oriented-edge orbit. The graph seed therefore uses eighteen finite-field coordinate symbols rather than 960 explicit oriented-edge endpoint symbols.

Dual-geometry all-to-all epoch. The twelve W33 perfect matchings and twenty-seven complement perfect matchings partition $E(K_{40})$. Hence

$$A + (J - I - A) = J - I$$

and the complete all-to-all one-port schedule has optimal depth

$$12 + 27 = 39.$$

Compressed two-defect sensor. A fixed set of 192 residual entries distinguishes all 303,810 two-edge syndromes, reducing the complete 40×40 residual readout by a factor $1600/192 = 25/3$. Minimality and noise robustness remain open.

Evidence boundary. The executable semantic certificate is `019183d46768ff68a5f57676540d8aefd8530106145` The packet does not establish general mixed weight-three decoding, a local Lindbladian implementation, measured process control, microscopic fusion correlations, measured platform costs, minimum topology memory, simultaneous all-to-all coupling, minimum sensing, hardware performance, or laboratory validation.

45 Hidden character, local-algebra poset, and null-processor boundary

The 480-element $40K_{4,3}$ incidence character decomposes exactly as

$$\chi_{480} = 1 + 2(15_a) + 15_b + 3(24) + 2(81) + 20 + 30 + 60 + 90.$$

Hence the previously unresolved residual is

$$\rho_{200} = 20 \oplus 30 \oplus 60 \oplus 90.$$

The eight simple nonunital local-algebra species have dimensions

$$4, 5, 6, 10, 12, 14, 16, 24.$$

Their Hasse covers are

$$4 \prec 16, \quad 5 \prec 6, \quad 5 \prec 10, \quad 6 \prec 14, \quad 10 \prec 14, \quad 10 \prec 16, \quad 12 \prec 24, \quad 14 \prec 24, \quad 16 \prec 24.$$

The undirected Hasse graph is connected and bipartite with first Betti number two.

For a projected-coordinate algebra $x \star y = E(x \circ y)$, if the scaled columns of E are exactly the minimum-norm nonzero idempotents and pair invariants recover adjacency, every algebra automorphism preserves the recovered graph. In the 45-axis $GQ(4,2)$ instance,

$$\operatorname{tr}(L_x L_y) = \frac{7}{30}\langle x, y\rangle, \qquad \min_{e^2=e,\ e\neq 0} \|e\|^2 = \frac{480}{49},$$

with equality precisely on the 45 axes. Thus

$$\operatorname{Aut}(V, \star) \cong O_6^-(2) \cong U_4(2) : 2.$$

A speculative photon-node model is admitted only in the Lorentz-compatible fixed-product form. If N effective nodes of pitch a are traversed in observer-frame period $T = 1/\nu$, then

$$v_{\text{eff}} = \frac{Na}{\sigma T}.$$

Vacuum null propagation requires $Na = \lambda$, $\lambda\nu = c$, and unit routing stretch. Consequently

$$c = \frac{Na}{T},$$

and a self-similar refinement $N \mapsto bN$, $a \mapsto a/b$ leaves c invariant. This is a falsifiable information-transport model, not evidence for literal photon substructure or variable vacuum light speed.

45.1 Exact selector algebra, radical multiport, code strata, and photon-capacity boundary

The fourteen-case arithmetic sieve for the 79-dimensional selector Terwilliger algebra is superseded by the exact characteristic-zero result already certified in Passes 1365–1369:

$$T(x) \cong \mathbb{Q}^3 \oplus M_2(\mathbb{Q})^2 \oplus M_3(\mathbb{Q})^3 \oplus M_4(\mathbb{Q}) \oplus M_5(\mathbb{Q}).$$

Its simple degrees are

$$1, 1, 1, 2, 2, 3, 3, 3, 4, 5,$$

with standard-module multiplicities

$$3, 12, 14, 1, 2, 4, 4, 8, 8, 1.$$

For the spread-graph multiport

$$H = \frac{2A_{36} - J}{6},$$

an exact sparse multiquadratic replay now proves the previously numerical adjacent-mode candidate. The complete transform uses

$$398$$

nontrivial adjacent Givens rotations in 69 layers, skips 232 eliminations because their entries are exactly zero, and terminates at

$$\operatorname{diag}(1^{35}, -1).$$

Every intermediate entry is a single rational radical. There are 75 distinct values of c^2, the largest denominator is 441, and the ordered parameter digest is

`2c3be7815b8346cd90be108faad4cc68866ad453f13b76bbdd4a9988a4569555.`

No global gate-count or depth optimality is claimed.

The binary character code $C = [36, 6, 16]$ admits an explicit maximal doubly-even extension

$$D = [36, 17, 4]$$

with

$$\begin{aligned} W_D(z) = 1 &+ 57z^4 + 852z^8 + 7332z^{12} + 57294z^{16} \\ &+ 57294z^{20} + 7332z^{24} + 852z^{28} + 57z^{32} + z^{36}. \end{aligned}$$

The intersection-two graph on its 57 weight-four words splits as

$$\boxed{\mathrm{SRG}(45, 16, 8, 4) \ \sqcup \ K_{2,2,2} \ \sqcup \ K_{2,2,2}.}$$

The coordinate-degree profile is $9^{20}3^{16}$. This proves an exact $A_4 = 57$ stratum, not its global maximality.

The five-fiber carrier 1|27|36|40|160 has rational permutation decomposition

$$\mathbb{Q}^{264} \cong 1^5 \oplus 6 \oplus 15_a^2 \oplus 15_b^2 \oplus 20^2 \oplus 24^3 \oplus 81,$$

and hence

$$\boxed{\operatorname{End}_G(\mathbb{Q}^{264}) \cong \mathbb{Q}^2 \oplus M_2(\mathbb{Q})^3 \oplus M_3(\mathbb{Q}) \oplus M_5(\mathbb{Q}).}$$

The algebra dimension is 48, its centre has dimension seven, and the cross-carrier Hom space has dimension

$$7 = 3 \cdot 2 + 1,$$

consisting of six trivial channels and one shared 15_b channel.

Photon node-packing boundary. For optical frequency ν,

$$T = \nu^{-1}, \qquad \lambda = c/\nu.$$

If N internal update sites refine one wavelength and one period together,

$$a_N = \lambda/N, \qquad \tau_N = T/N,$$

then

$$\boxed{\frac{a_N}{\tau_N} = c.}$$

Thus node density can refine internal state space and update throughput while leaving vacuum c invariant. In the ideal tensor-factor bookkeeping,

$$R = N\nu \log_2 d, \qquad \rho = N/\lambda, \qquad \frac{R}{\rho \log_2 d} = c.$$

A single photon distributed over N alternatives with a d-state label generally has direct-sum dimension Nd, however, so its accessible classical information is bounded by $\log_2(Nd)$ rather than $N \log_2 d$ unless physically independent tensor factors exist.

The Margolus–Levitin rate for one photon of energy $E = h\nu$ gives

$$\Gamma_\perp \leq \frac{2E}{\pi\hbar} = 4\nu, \qquad \Gamma_\perp T \leq 4.$$

Consequently forty W33 nodes cannot mean forty sequential mutually orthogonal transitions within one optical period of an isolated photon. They must represent parallel or nonorthogonal modes, multiple periods, or externally driven evolution. This is a falsifier for one literal interpretation, not a rejection of high-dimensional single-photon processing.

The Monster descent remains fail-closed: no portable serialized Monster words or executed character-fusion artifact are present, and no embedding is promoted. No literal photon-node ontology, variable-c mechanism, hardware result, laboratory result, remote-CI success, or PDF success is claimed.

45.2 Topology-code firewall, optimal 24-port control, autonomous boot, and photon causal density

The W33 polynomial residual

$$R(B) = B^2 + 2B - 8I - 4J$$

separates all 79,092,651 edge-toggle patterns of weight at most three: one zero pattern, 780 single toggles, 303,810 double toggles, and 78,788,060 triple toggles. A zero residual has balanced insertion and deletion degree at every vertex. Exhaustion of all 27,000 alternating weight-four supports and 2,769,952 alternating simple weight-six supports finds no undetectable pattern. A nonadjacent vertex transposition gives an explicit undetectable weight-32 relabelling, so

$$8 \leq d_0 \leq 32.$$

The exact distance remains open.

The desired symmetry dissipator cannot be made from independent strictly two- or three-mode permutation jumps. Every nontrivial W33 automorphism moves at least 27 points, and every transvection has cycle structure $1^{13}3^9$. Four transvections are necessary and sufficient to generate the full order-25,920 group; each global jump is compilable into nine disjoint three-mode cycles, but those cycles must remain one macro if the fixed algebra

$$\mathrm{span}\{I, A, J - I - A\}$$

is to be preserved.

Writing $B = J - I - A = (A^2 - 2A - 12I)/4$, the complement adds no passive drift algebra. The largest adjacency multiplicity forces at least 24 independent diagonal controls. For one explicit 24-port set, the commutant constraint has modular rank 279 on 280 coordinates over three primes, leaving scalars only. Therefore the ideal controlled $*$-algebra has dimension

$$40^2 = 1600,$$

and 24 ports are necessary and sufficient algebraically.

The topology can boot from four transvection vectors and one ordered edge, an 18-trit seed. Its ordered-edge orbit has 480 elements and BFS diameter 9; a deterministic factorization then supplies 12 collision-free W33 matching rounds. The declared boot sequence—seed verification, orbit expansion, schedule compilation, a 192-entry residual measurement, bounded repair, and symmetry activation—has a 221-round network-action upper bound before source, detector, classical, memory, and settling overhead.

The user-proposed relation between photon computation and light speed survives in a constrained form. If serial node spacing is ℓ and local update time is τ, Lorentz-compatible propagation requires $\ell/\tau = c$. Combining a one-photon energy $E = h\nu$ with the Margolus–Levitin speed limit gives

$$N_{\text{serial}} \le \frac{4EL}{hc} = \frac{4L}{\lambda}.$$

Increasing serial node count at fixed length therefore requires faster internal updates; it does not change invariant vacuum c. Parallel photonic modes enlarge capacity—an ideal M-mode alphabet carries $\log_2 M$ bits—without representing M serial state transitions per optical cycle. Any literal node-count dependence of vacuum speed would predict encoding- or frequency-dependent front arrival and is retained only as a falsifiable, presently unsupported hypothesis.

Two additional exact constructions follow. Since A and its complement commute,

$$e^{-itA}e^{-itB} = e^{-it(J-I)}, \qquad e^{-itA}e^{+itB} = e^{-it(A-B)},$$

giving common-mode and W33-contrast interferometric channels. For the Cartesian hierarchy $W33^{\square m}$,

$$|V| = 40^m, \quad k = 12m, \quad \mathrm{diam} = 2m, \quad |E| = 6m40^m,$$

with absolute spectral gap 10 and ideal causal-density requirement $L/\lambda \ge m/2$. Self-similarity yields logarithmic routing depth but does not compress the exponential physical mode requirement.

No exact topology distance, independently local Lindbladian, robust pulse theorem, learned process tensor, autonomous physical boot, hidden photon-node ontology, variable speed of light, hardware result, or laboratory validation is claimed.

Passes 3973–3980: extremal-code geometry, mesh locality, and photon resource separation

Exact 57-code geometry. For the deterministic maximal doubly-even code $D = [36, 17, 4]$, join two weight-four words when their supports meet in two coordinates. The resulting graph is not merely a $45+6+6$ census: an explicit maximal-clique reconstruction gives

$$\boxed{G_{57} \cong T(10) \sqcup T(4) \sqcup T(4) = L(K_{10}) \sqcup L(K_4) \sqcup L(K_4).}$$

Thus

$$57 = \binom{10}{2} + 2\binom{4}{2}.$$

If $t = A_4$, MacWilliams symmetry forces

$$A_8 = 11t + 225, \qquad A_{12} = 9555 - 39t, \qquad A_{16} = 55755 + 27t,$$

and, on the dual side,

$$B_4 = t, \quad B_6 = 6(t+7), \quad B_{12} = 39(245 - t).$$

Nonnegativity gives only $t \le 245$. Hence $t = 57$ is exact and geometrically rigid here, but global A_4-extremality is not claimed.

Exact radius-one mesh optimum. The established 36-port factorization uses 398 nontrivial exact multiquadratic Givens rotations, 232 exact zero eliminations, and 69 layers. All $\binom{36}{2} = 630$ single transpositions of the port order were replayed exactly. None improves 398, 22 tie it, and every neighbor has depth 69. Therefore the order is an exact radius-one local optimum in permutation space; global rotation and depth optimality remain open.

One-photon slope/intercept falsifier. The physically conservative null is

$$t(M, L) = a(M) + L/c,$$

where M is the orthogonal-mode alphabet and $a(M)$ absorbs encoder, decoder, detector, and pulse-shape latency. Randomly interleaving fixed-spectrum alphabets $M \in \{2, 4, 8, 16, 40\}$ at two or more free-space lengths permits the fit

$$t(M, L) = a(M) + b(M)L.$$

A mode-dependent intercept is ordinary device latency; only a reproducible mode-dependent slope, surviving encoder swaps and length scaling, challenges the invariant-front null. Capacity must be reported separately from timing. For a symmetric M-ary channel with total error probability ϵ,

$$I(M, \epsilon) = \log_2 M + (1-\epsilon)\log_2(1-\epsilon) + \epsilon \log_2\left(\frac{\epsilon}{M-1}\right).$$

At $M = 40$, this is 5.32193 ideal bits per use, about 5.18828 bits at one-percent symmetric error, and 4.77126 bits at five-percent error.

Literal rank-48 multiplication tensor. The split centralizer algebra

$$\boxed{\mathbb{Q}^2 \oplus M_2(\mathbb{Q})^3 \oplus M_3(\mathbb{Q}) \oplus M_5(\mathbb{Q})}$$

has dimension 48 and center dimension seven. Its matrix-unit multiplication table contains

$$1^3 + 1^3 + 3 \cdot 2^3 + 3^3 + 5^3 = 178$$

nonzero products. The later content-addressed verifier also reconstructs the complete literal orbital tensor:

$$\boxed{48 \text{ relations}, \qquad 904 \text{ nonzero intersection constants.}}$$

Its tensor SHA-256 is

`a17a375552c102d281e3ba79b602b67aa54a426fd68b1878293763ea65395ff9`.

Thus both the split Wedderburn tensor and the geometric orbital-relation tensor are closed exactly.

Photon resource trilemma. A one-photon direct-sum alphabet of M orthogonal alternatives carries at most $\log_2 M$ ideal classical bits per use. A tensor-product claim $N \log_2 d$ requires N physically independent factors. Mutually orthogonal serial updates obey a separate energy–time bound. Temporal packing additionally satisfies the engineering estimate $M \sim 2WT$ for half-bandwidth W and duration T. Larger internal alphabets can therefore increase capacity and spacetime support without changing vacuum c.

For the Cartesian power $W(3,3)^{\square m}$,

$$|V| = 40^m, \qquad \mathrm{diam} = 2m,$$

so the self-similar address density is

$$\boxed{\frac{\log_2 |V|}{\mathrm{diam}} = \frac{\log_2 40}{2} \approx 2.660964.}$$

This is an addressing/routing invariant, not a variable-light-speed law.

Monster gate and evidence boundary. The required observable Monster class-fusion artifact remains absent, so no Monster embedding is promoted. Exact results in this insert are the code geometry, enumerator identities, all-630 mesh audit, split and literal rank-48 tensors, and algebraic resource formulas. The timing sensitivity and time-bandwidth examples are models; the photon experiment, global code extremality, global mesh optimum, Monster fusion, remote CI/PDF success, hardware, and laboratory validation remain open.

46 Extremal code, active multiport, photon mode tests, and rank-48 tensor

Exact certificate. Passes 3973–3980 are reproduced by the content-addressed verifier and frozen certificate with semantic digest `03c37b821bb9b6f875a8bd4edde8dc96fe36a43d6d8dc24144d8dbd5e3aa5104`. Monster execution, laboratory photonics, and remote PDF evidence remain outside the promoted boundary.

Theorem 46.1 (Extremal $A_4 = 57$ extension)**.** *Let $C = [36, 6, 16]$ be the binary character code on the nonsingular vectors of the six-dimensional minus-type quadratic space. Among doubly-even self-orthogonal extensions of C,*

$$\boxed{A_4 \leq 57}.$$

The bound is attained by an explicit $[36, 17, 4]$ code. Its 57 weight-four words have intersection-two graph

$$\mathrm{SRG}(45, 16, 8, 4) \sqcup K_{2,2,2} \sqcup K_{2,2,2},$$

and coordinate stabilizer of order 192, with element-order census

$$1^1 2^{59} 3^8 4^{68} 6^{40} 12^{16}.$$

Proof certificate. The 945 admissible weight-four words form a degree-624 compatibility graph with two $O_6^-(2)$ vertex orbits of sizes 135 and 810. Exact colored bitset branch-and-bound gives orbit maxima 15 and 54. Transitivity reduces any clique meeting the 135-orbit to one fixed representative; its exact neighborhood search has maximum 57.

Theorem 46.2 (Active-mixer and fixed-order cut bounds)**.** *The exact* 36*-port involution* $H = (2A_{36} - J)/6$ *admits an adjacent factorization*

$$401 = 296 \text{ genuine mixers} + 105 \text{ signed swaps}$$

in 69 *layers, with zero exact residual. The prior common-order compiler has* $398 = 302+96$ *factors. Thus six active mixing elements are removed when input and output relabeling may differ. For the original common order, the sum of the exact off-diagonal cut ranks is*

$$\boxed{253},$$

so every adjacent factorization respecting that order uses at least 253 *factors.*

Theorem 46.3 (Literal rank-48 coherent tensor)**.** *The five fibers*

$$1 \mid 27 \mid 36 \mid 40 \mid 160$$

carry exactly 48 *orbitals under* $O_6^-(2)$*. Their source–target relation count matrix is*

$$\begin{pmatrix} 1 & 1 & 1 & 1 & 1 \\ 1 & 3 & 2 & 1 & 1 \\ 1 & 2 & 3 & 1 & 2 \\ 1 & 1 & 1 & 3 & 4 \\ 1 & 1 & 2 & 4 & 8 \end{pmatrix}.$$

Exactly 904 *of the* 48^3 *intersection constants are nonzero. The complete sparse tensor has digest* `a17a375552c102d281e3ba79b602b67aa54a426fd68b1878293763ea65395ff9`*.*

Photon competing-model protocol. The photon hypothesis is separated into six models: invariant-front direct-sum capacity; transverse-structure axial delay; a falsifiable hidden-node variable-speed alternative; an engineered W33 graph-kinetic model; independent hyperentangled tensor factors; and self-similar coupling efficiency. For the engineered graph model, $L_{W33} = 12I - A$ has sectors

$$\boxed{0, 10, 16},$$

giving a declared 0:10:16 delay-spectroscopy target. The experiment varies mode count at fixed spectrum and transverse momentum, varies transverse momentum at fixed alphabet, compares equal-dimension factorisations, and keeps causal-front arrival distinct from pulse-peak or group delay. No hidden nodes, variable vacuum light speed, achieved capacity, or measured delay are claimed.

Monster firewall. The exact internal queue contains 51,840 ordered standard pairs in two centralizer orbits of 576, but no portable serialized `mmgroup` words or executed class-fusion artifact are present. The result therefore remains `PENDING_EXPLICIT_MM_WORDS_AND_CLASS_FUSION`.

47 Passes 3981–3988: global mesh bound and photon spectral processor

The stronger reconciled Passes 3973–3980 certificate already proves that the fixed-parent weight-four compatibility graph has maximum clique size 57, hence the selected $[36, 17, 4]$ doubly-even extension is globally extremal within that parent-extension problem. It also reconstructs the literal five-fiber orbital algebra with 48 relations and 904 nonzero intersection constants. The present packet retains those exact results and advances the open physical and compiler boundaries.

47.1 An ordering-independent adjacent-factor lower bound

Let $K = 2A_{36} - J$ be the symmetric 36-port Hadamard transfer matrix, so $K^2 = 36I$. For an arbitrary port ordering and a prefix cut of size k, write

$$K = \begin{pmatrix} A_k & B_k \\ B_k^T & D_k \end{pmatrix}.$$

Then

$$B_k B_k^T = 36I - A_k^2.$$

Thus the cross-cut rank defect is the multiplicity of the eigenvalues ± 6 of A_k. Using

$$\operatorname{tr} A_k = -k, \qquad \operatorname{tr} A_k^2 = k^2,$$

and Cauchy–Schwarz gives the ordering-independent half-sequence

$$1, 2, 3, 4, 5, 5, 6, 7, 7, 8, 8, 8, 9, 9, 9, 9, 10, 9.$$

Reflection symmetry therefore forces

$$\boxed{N_{\text{adjacent}} \geq 229.}$$

The published ordering has cut-rank sum 253. A deterministic exact ordering search found a second ordering with sum 251. The number 251 is a stronger order-specific target, not yet an implemented factorization or a proof of global optimality.

47.2 Nuisance-complete photon timing tomography

A mode-count timing test must separate propagation slope from encoder, detector, spatial, spectral, and pulse-shape delays. The declared model contains the target regressor

$$\gamma \frac{L}{c} \log M$$

alongside mode intercepts, encoder and detector swaps, measured transverse-momentum and spectral covariates, pulse width, intensity, detector walk, and drift. The frozen design has 48 randomized cells and 16 parameters with full normalized rank 16.

For the declared synthetic benchmark of 20 ps single-event jitter and 10^6 events per cell,

$$\sigma_\gamma \simeq 2.19 \times 10^{-9}.$$

Omitting the measured spatial and spectral covariates creates a false

$$\widehat{\gamma} \simeq 1.1243 \times 10^{-9},$$

whereas the complete noiseless model returns zero to numerical precision. This explicitly demonstrates why structured-mode group delay is not, by itself, evidence for a mode-dependent vacuum front velocity.

47.3 Three exact photon constructions

For the W33 Laplacian $L = 12I - A$,

$$\mathrm{Spec}(L) = 0^1, 10^{24}, 16^{15}.$$

A localized mode has $\langle L \rangle = 12$ and $\mathrm{Var}(L) = 12$, so its ideal pure-state quantum Fisher information is

$$F_Q = 48.$$

The optimal extremal-sector superposition has $F_Q = 256$.

For the complement $B = J - I - A$,

$$A + B : (39, -1, -1), \qquad A - B : (-15, 5, -7).$$

The sum arm is geometry-blind on the 39-dimensional nonuniform sector, while the difference arm resolves the two protected sectors. Its full ideal spectral-range QFI is 400, with protected nonuniform value 144.

Most importantly, half a W33 Laplacian period gives the exact global reflection

$$\boxed{U = e^{-i\pi L/2} = I - 2E_{10} = -\frac{1}{3}(I + A) + \frac{2}{15}J.}$$

Each row has amplitude $-1/5$ on the point and its 12 neighbors and amplitude $2/15$ on its 27 nonneighbors, and

$$U^2 = I.$$

Thus one engineered analog evolution can implement a global 40-mode operation in parallel; it need not be interpreted as forty sequential orthogonal updates inside one optical period.

For a genuine tensor carrier,

$$\mathrm{mult}(m, r) = \binom{m}{r} 13^{m-r} 27^r, \qquad \mathrm{amp}(m, r) = \left(-\frac{1}{5}\right)^{m-r} \left(\frac{2}{15}\right)^r.$$

The 40^m-entry tensor kernel therefore has only $m + 1$ amplitude-shell types. This is exact control-description compression, not Hilbert-space compression or a variable-c mechanism.

47.4 Monster and evidence firewall

A GAP/CTblLib class-fusion sieve and an `mmgroup` canonical-integer engine test have been materialized upstream of the strict Monster gate. The repository still lacks four explicit portable Monster words realizing the target $U_4(2)$ subgroup. Therefore the promoted status remains

$$\boxed{\texttt{PENDING\_EXPLICIT\_MM\_WORDS\_AND\_CLASS\_FUSION\_EXECUTION}.}$$

No Monster embedding, laboratory photon result, hidden photon-node ontology, mode-dependent vacuum c, global circuit optimum, or unobserved remote CI/PDF result is claimed.

48 Passes 3989–3996: physical incidence photon processor

The W33 point graph admits an exact sparse physical lift. Let N be the 40×40 point–line incidence matrix of $W(3,3)$. Exact enumeration gives

$$NN^T = 4I + A_{W33}.$$

Hence a dense twelve-neighbour point Hamiltonian may be generated from an eighty-mode bipartite point–line device with 160 couplings, degree four, and girth eight. Its spectrum is

$$-4^1, -\sqrt{6}^{24}, 0^{30}, \sqrt{6}^{24}, 4^1.$$

With point–line coupling g and bus detuning Δ,

$$H_{\text{eff}} = -\frac{g^2}{\Delta}(A+4I) + O(g^4/\Delta^3),$$

and $t = \pi\Delta/(2g^2)$ approaches the exact W33 half-period reflection on the point sector. This is an analytic architecture, not a fabricated device.

Let $B = J - I - A$ and $D = A - B$. Then $A + B = J - I$ is geometry-blind on the nonuniform sector, while

$$\operatorname{Spec}(D) = -15^1, 5^{24}, -7^{15}.$$

The spectral projectors are

$$E_{-15} = \frac{(D-5I)(D+7I)}{160} = \frac{J}{40}, \quad E_5 = \frac{(D+15I)(D+7I)}{240}, \quad E_{-7} = -\frac{(D+15I)(D-5I)}{96}.$$

Consequently $\langle D\rangle$ and $\langle D^2\rangle$ determine all three sector populations. If identical commuting error C enters both arms, then

$$e^{-it(A+C)}e^{+it(B+C)} = e^{-it(A-B)},$$

so the dual-geometry echo cancels it exactly.

The maximum-code census is also complete. The 945-vertex fixed-parent compatibility graph contains exactly 945 maximum 57-cliques, split into three parent-group orbits

$$\boxed{945 = 540 + 270 + 135}$$

with stabilizer orders

$$\boxed{96, 192, 384}.$$

Thus maximum-code uniqueness is false in the fixed-parent problem. This corrects the earlier division by an order-192 full coordinate stabilizer without first checking parent preservation.

The seven primitive central blocks of the literal 48-orbital algebra have simple degrees

$$1, 1, 2, 2, 2, 3, 5.$$

Their complete central coarse-graining lattice contains

$$B_7 = 877$$

partitions. Quotienting equal-degree relabelings by $S_2 \times S_3$ leaves exactly

$$\boxed{198}$$

inequivalent central types. These are exact central subalgebras, not automatically combinatorial fusions of the 48 orbital relations.

Three further photon constructions follow from the incidence lift. First, $\operatorname{rank} N = 25$, so the bipartite processor contains thirty exact dark modes. Second, for $L = 12I - A$,

$$V = e^{-i\pi L/4} = I - (1+i)E_{10}, \qquad V^2 = U, \qquad V^4 = I,$$

which gives an exact four-phase internal graph clock. Third, for a genuine m-fold tensor carrier,

$$\operatorname{rank}(N^{\otimes m}) = 25^m, \qquad \dim\ker(N^{\otimes m}) = 40^m - 25^m,$$

so the one-side dark fraction is

$$\boxed{1 - (5/8)^m}.$$

Only $m+1$ nonzero singular shells occur, with multiplicity $\binom{m}{r}24^r$ and squared singular value $16^{m-r}6^r$.

The Monster acquisition gate now composes $U_4(2) \to H \to \mathbb{M}$ class fusions through maximal-subgroup tables and searches only compatible published explicit `mmgroup` overgroups. The promoted status remains

$$\boxed{\texttt{PENDING\_EXPLICIT\_MONSTER\_U42\_WORDS}}$$

until four portable words pass the exact pair, triple, order-25,920, object-action, and class-fusion firewalls. No variable vacuum c, literal hidden photon nodes, fabricated hardware, laboratory result, or unobserved remote build is claimed.

Passes 3989–3996 supplement: direct W33 flight and causal memory

Exact forty-mode analog flight. Let A be the adjacency matrix of $W(3,3)$. A forty-mode equal-edge coupled-mode array obeying

$$i\frac{d\psi}{dz} = \kappa A\psi$$

implements, at $\kappa z = \pi/2$,

$$\boxed{e^{-i\pi A/2} = -\frac{I+A}{3} + \frac{2J}{15}.}$$

This is an exact symmetric involution with row amplitudes $-1/5$ on a point and its twelve neighbours and $2/15$ on its twenty-seven nonneighbours. For uniform fractional interaction error ϵ,

$$F_{\text{pro}} = 1 - 3\pi^2\epsilon^2 + O(\epsilon^3), \qquad F_{\text{avg}} = 1 - \frac{120}{41}\pi^2\epsilon^2 + O(\epsilon^3).$$

The linear average-fidelity term vanishes because $\operatorname{Tr} A = 0$.

Wigner–Smith time as operational memory. For a frequency-dependent W33 scattering unitary

$$S(\omega) = e^{i\theta(\omega)L_{W33}},$$

the Wigner–Smith operator is exactly

$$\boxed{Q(\omega) = -iS^\dagger\frac{dS}{d\omega} = \theta'(\omega)L_{W33}.}$$

Hence the proper-delay sectors are

$$0^1, \qquad (10\theta')^{24}, \qquad (16\theta')^{15},$$

with mean $12\theta'$, variance $12(\theta')^2$, and total trace $480\theta'$. This gives a measurable meaning to internal history: dwell-time and stored-energy sensitivity, not a changed vacuum causal speed.

For m independent W33 factors, the exact delay-shell polynomial is

$$\boxed{(1 + 24z^{10} + 15z^{16})^m.}$$

The address space has 40^m modes and

$$\langle\tau\rangle = 12m\theta', \qquad \mathrm{Var}(\tau) = 12m(\theta')^2.$$

Consequently

$$\boxed{\frac{\log_2(40^m)}{\langle\tau\rangle} = \frac{\log_2 40}{12\theta'}}$$

is independent of self-similar depth, while the relative delay width is $1/\sqrt{12m}$.

Causal-refinement boundary. A local-interaction causal cone has schematic physical velocity

$$v_{\text{int}} \lesssim aC_{LR}(\Delta)J/\hbar.$$

Self-similar refinement preserves the cone only when aJ remains fixed. More serial nodes without stronger coupling slow the internal cone; node density alone neither determines nor raises vacuum c. Fabrication, measured Wigner–Smith delays, literal hidden photon nodes, variable vacuum c, and laboratory performance remain unclaimed.

49 Photon layout, delay tomography, orbital closure, and code stabilizers

The direct forty-mode W33 coupler and the eighty-mode point–line incidence lift both admit exact one-factorizations. Their resource vectors

$$(\text{modes}, \text{links}, \text{degree}, \text{layers})$$

are

$$(40, 240, 12, 12), \qquad (80, 160, 4, 4).$$

For linear cost $C = \alpha M + \beta E + \gamma R$,

$$C_{40} - C_{80} = -40\alpha + 80\beta + 8\gamma,$$

so the direct device wins exactly when

$$\boxed{5\alpha > 10\beta + \gamma}.$$

For $S(\omega) = e^{i\theta(\omega)L}$, the Wigner–Smith operator is $Q = \theta' L$. The three sector populations obey

$$p_{16} = \frac{m_2 - 10m_1}{96}, \qquad p_{10} = \frac{16m_1 - m_2}{60}, \qquad p_0 = 1 - p_{10} - p_{16}.$$

The executed central-difference reconstruction converges quadratically. Removing common delay gives the stronger identity

$$\boxed{Q - \frac{\operatorname{Tr} Q}{40} I = -\theta' A_{W33}},$$

so the complete W33 adjacency relation is recovered from the off-diagonal delay kernel.

The three fixed-parent maximum-code stabilizers are identified exactly:

$$G_{540} \cong S_4 \times V_4,$$

$$G_{270} \cong D_8 \rtimes_{\mathrm{sgn},\alpha} S_4, \qquad \alpha(r) = r^{-1}, \quad \alpha(s) = r^2 s,$$

and

$$\boxed{G_{135} \cong C_2 \wr S_4 = 2^4 : S_4 = W(B_4).}$$

The last group acts faithfully on eight coordinates preserving four opposite pairs, giving an exact four-bit hypercube/cross-polytope bridge.

For the oriented vertex–edge incidence matrix D,

$$\operatorname{Spec}(D^T D) = 0^{201} + 10^{24} + 16^{15},$$

and therefore

$$\boxed{e^{-i\pi D^T D/2} = I - 2E_{10}}.$$

The 201-dimensional kernel is raw cycle-space memory, not the 81-dimensional Hodge/CSS logical sector.

Finally, pure-state delay metrology satisfies $F_Q = 4 \operatorname{Var}(Q)$. Thus

$$F_Q^{\mathrm{vertex}} = 48(\theta')^2, \qquad F_Q^{\mathrm{opt}} = 256(\theta')^2.$$

The literal orbital Fourier/relation-fusion and Monster overgroup workflows are executable but their generated outputs remain pending. No fabricated device, measured delay, variable vacuum c, Monster embedding, remote CI/PDF success, or laboratory result is claimed.

50 Exact finite-detuning photon revival and coherent delay memory

Exact non-dispersive incidence revival. Let N be the 40×40 point–line incidence matrix of $W(3,3)$, so that $NN^{\mathsf{T}} = 4I + A$. For the full point–bus Hamiltonian

$$H = g \begin{pmatrix} 0 & N \\ N^{\mathsf{T}} & (\Delta/g)I \end{pmatrix},$$

the finite values

$$\boxed{\Delta/g = 2\sqrt{2}}, \qquad \boxed{gt = \pi/\sqrt{2}}$$

produce an exact zero-leakage revival. The half-angle windings of the singular-value-4, singular-value-$\sqrt{6}$, and bus-dark sectors are 3π, 2π, and π, respectively. Hence the point action is

$$\boxed{U_P = E_{16} - E_6 + E_0 = I - 2E_6 = -\frac{I + A}{3} + \frac{2J}{15}},$$

and the line action is $U_L = F_{16} - F_6 + F_0$. Direct exponentiation of the full eighty-mode matrix gives operator error below 6.6×10^{-15} and off-block leakage below 2.9×10^{-15}. This is not a dispersive approximation.

Quadratic-form Wigner–Smith tomography. For $\tau(v) = v^\dagger Q v$,

$$Q_{ii} = \tau(e_i),$$
$$\Re Q_{ij} = \tau\left(\frac{e_i + e_j}{\sqrt{2}}\right) - \frac{Q_{ii} + Q_{jj}}{2},$$
$$\Im Q_{ij} = \frac{Q_{ii} + Q_{jj}}{2} - \tau\left(\frac{e_i + ie_j}{\sqrt{2}}\right).$$

Thus a general Hermitian forty-mode delay matrix requires exactly $40^2 = 1600$ expectation probes, reduced to $40 + \binom{40}{2} = 820$ under reciprocal real symmetry. The exact synthetic reconstruction recovered all 240 W33 edges with maximum matrix error below 2.6×10^{-15}. For a three-frequency central difference and a linear phase law,

$$Q_h = \frac{\sin(h\theta' L)}{h} = Q - \frac{h^2}{6}(\theta')^3 L^3 + O(h^4),$$

and Richardson extrapolation removes the leading error. A separate precursor/front measurement remains mandatory before testing any mode-count-dependent causal-front slope.

Exact bright-sector write–hold–read gate. Define

$$T = N^\mathsf{T}\left(\frac{E_{16}}{4} + \frac{E_6}{\sqrt{6}}\right), \qquad X = \begin{pmatrix} 0 & T^\mathsf{T} \\ T & 0 \end{pmatrix}.$$

Then

$$T^\mathsf{T} T = E_{16} + E_6, \qquad TT^\mathsf{T} = F_{16} + F_6,$$

and

$$\boxed{W = e^{-i\pi X/2}}$$

swaps the twenty-five-dimensional point-bright sector into the line-bright sector while fixing both fifteen-dimensional dark sectors. A diagonal bus hold phase followed by $W^\dagger$ returns

$$\boxed{e^{-i\phi_0} E_0 + e^{-i\phi_6} E_6 + e^{-i\phi_{16}} E_{16}}$$

on the point modes. An arbitrary three-phase test had operator error below 2.7×10^{-15} and leakage below 3.1×10^{-15}. This is an exact controlled-Hamiltonian synthesis, not a fabricated-memory claim.

Three further photon constructions. Exact zero-leakage reflection revivals satisfy the integer quadric

$$\boxed{3n_{16}^2 - 8n_6^2 + 5k^2 = 0},$$

with $n_{16} + k$ even and $n_6 + k$ odd. The parameter map is

$$\frac{\Delta}{g} = k\sqrt{\frac{24}{n_6^2 - k^2}}, \qquad gt = 2\pi\sqrt{\frac{n_6^2 - k^2}{24}}.$$

The shortest positive primitive solution is $(n_{16}, n_6, k) = (3, 2, 1)$, which yields the revival above.

Writing $\tau = \operatorname{Tr} Q/480$, an ideal W33 delay matrix obeys the basis-free checksum

$$\boxed{Q(Q-10\tau I)(Q-16\tau I)=0},$$

and the geometry is recovered self-calibratingly by

$$\boxed{A = -\frac{Q-(\operatorname{Tr} Q/40)I}{\tau}}.$$

For m commuting physical copies, all factors revive at the same gate time $gt = \pi/\sqrt{2}$. The point action is $U_P^{\otimes m}$, with

$$\boxed{m_+ = \frac{40^m + (-8)^m}{2}, \qquad m_- = \frac{40^m - (-8)^m}{2}}.$$

This is a synchronization theorem, not a hardware-compression theorem.

Algebra and evidence boundary. The literal 48-relation Fourier/fusion calculation and the GAP/mmgroup Monster maximal-overgroup search are exposed through an artifact-producing execution workflow. No relation-fusion rank, Monster words, or embedding is promoted until the generated artifacts are inspected. Fabricated hardware, measured delay matrices, literal hidden photon nodes, variable vacuum light speed, remote CI, and PDF evidence likewise remain unclaimed.

51 Physical incidence-link H_1 memory and exact projector gates

The lower-degree point–line incidence architecture has 80 modes and 160 physical couplers. Let $D \in \{-1,0,1\}^{80\times 160}$ be its oriented vertex–edge boundary matrix and put $K = D^T D$. The exact spectrum is

$$0^{81}, \quad 8^1, \quad 4^{30}, \quad (4-\sqrt{6})^{24}, \quad (4+\sqrt{6})^{24}.$$

Therefore

$$\boxed{320 P_{H_1} = (K-8I)(K-4I)\big((K-4I)^2 - 6I\big)}$$

is an exact integral numerator for the rank-81 Hodge/Kirchhoff cycle projector. Indeed $DP_{H_1} = 0$, $\operatorname{rank} D = 79$, and $\operatorname{rank} P_{H_1} = 81$, so its image is exactly the boundaryless link-current space. Every physical coupler has diagonal projector weight 81/160.

The corresponding exact memory reflection is

$$\boxed{R_{H_1} = I - 2P_{H_1}}, \qquad \operatorname{Tr} R_{H_1} = 79 - 81 = -2.$$

It is $+1$ on the cut space and -1 on the protected cycle-memory space.

A separate polynomial dynamics exists on the 80 mode coordinates. The raw Levi adjacency has spectrum

$$-4^1,\ (-\sqrt{6})^{24},\ 0^{30},\ (+\sqrt{6})^{24},\ 4^1$$

and is not globally periodic. Its sign fold $H_2 = A_L^2$ has spectrum $0^{30} + 6^{48} + 16^2$, giving

$$e^{-i\pi H_2} = I, \qquad e^{-i\pi H_2/2} = I - 2E_6, \qquad e^{-i\pi H_2/4} = I + (i-1)E_6.$$

This polynomial revival is distinct from the finite-detuning block Hamiltonian of Pass 4005, although both use the same incidence matrix.

The five signed Levi sectors are recovered from four moments. With $a = p_{+4} + p_{-4}$, $b = p_{+\sqrt{6}} + p_{-\sqrt{6}}$, $d = p_{+4} - p_{-4}$, and $e = p_{+\sqrt{6}} - p_{-\sqrt{6}}$,

$$a = \frac{m_4 - 6m_2}{160}, \qquad b = \frac{16m_2 - m_4}{60},$$

$$d = \frac{m_3 - 6m_1}{40}, \qquad e = \frac{16m_1 - m_3}{10\sqrt{6}}.$$

Finally, in the ideal delay model $Q = \tau I + \theta' A_L$,

$$\boxed{Q - \frac{\operatorname{Tr} Q}{80} I = \theta' A_L},$$

so the centered delay kernel recovers all 160 point–line incidences.

The mode decomposition $30 + 48 + 2$ and the link-current decomposition $79 + 81$ are distinct vector spaces. No fabricated coupler network, measured delay, loss tolerance, variable vacuum c, literal hidden photon nodes, Monster embedding, or laboratory performance is claimed.

52 The protected H_1 sector as an exact photonic flat band

BT548 already identified the protected Levi cycle projector with the -2 eigenspace of the Levi line graph. The new step is the explicit coupler-as-site physical model and its apartment-frame and perturbation refinement.

Promote the 160 incidence links of the 80-vertex Levi graph L to the sites of

$$X = \operatorname{LineGraph}(L).$$

Then X has 160 sites, 480 links, and degree 6. For the oriented Levi boundary matrix D,

$$\boxed{A_X = D^T D - 2I.}$$

Hence

$$\boxed{\ker D = E_{-2}(A_X)},$$

and the protected $H_1 = 81$ sector is exactly the -2 flat band of X. Its spectrum is

$$6^1 + (2+\sqrt{6})^{24} + 2^{30} + (2-\sqrt{6})^{24} + (-2)^{81}.$$

Let $C \in \{-1, 0, 1\}^{160\times 1620}$ have one signed alternating column for each Levi octagon. Every such apartment column c has support 8 and satisfies

$$A_X c = -2c.$$

The 1620 compact apartment states span the full flat band, and

$$P_{H_1} = \frac{1}{160} C C^T,$$

$$C C^T = \frac{1}{2}(A_X - 6I)(A_X - 2I)(A_X^2 - 4A_X - 2I).$$

After normalizing each column by $\sqrt{8}$, they form a unit-norm tight frame:

$$\boxed{\left(\frac{C}{\sqrt{8}}\right)\left(\frac{C}{\sqrt{8}}\right)^T = 20P_{H_1}}, \qquad \frac{1620}{81} = 20.$$

Each secondary site belongs to exactly 81 apartments,

$$\boxed{160 \cdot 81 = 1620 \cdot 8 = 12960.}$$

The perturbation boundary is exact. A uniform onsite shift δI moves the band without splitting it, whereas

$$P_{H_1}|e\rangle\langle e|P_{H_1}$$

has rank one and nonzero eigenvalue 81/160. More generally,

$$\mathrm{Tr}(P_{H_1}\,\mathrm{diag}\,v) = \frac{81}{160}\sum_e v_e.$$

Thus the symmetric model has an exact flat band, but generic nonuniform onsite or coupling disorder is not proved harmless.

This 160-site line-graph Hamiltonian, the primary 80-mode incidence Hamiltonian, and the passive 160-dimensional link-current coordinates are related but distinct physical models. No fabricated secondary lattice, measured localization, disorder tolerance, local coupling synthesis, variable vacuum c, hidden-node ontology, or laboratory performance is claimed.

53 Physics-first audit of the W33 universal-computer hypothesis

Exact revival error tensor. For $d = \Delta/g$, $\tau = gt$, and the target $d_0 = 2\sqrt{2}$, $\tau_0 = \pi/\sqrt{2}$, write $\delta d = d - d_0$ and $\delta\tau = \tau - \tau_0$. The coherent target overlap and average point–bus leakage obey

$$1 - F_e = \frac{1}{2}\left[\frac{323\pi^2}{5184}(\delta d)^2 + 2\frac{17\pi}{36}\delta d\,\delta\tau + 8(\delta\tau)^2\right] + O(\delta^3),$$

$$P_{\mathrm{leak}} = \frac{1}{2}\left[\frac{149\pi^2}{5184}(\delta d)^2 + 2\frac{17\pi}{36}\delta d\,\delta\tau + 8(\delta\tau)^2\right] + O(\delta^3).$$

The locally optimal retiming law is

$$\boxed{\delta\tau = -\frac{17\pi}{288}\delta d},$$

for which

$$1 - F_e = \frac{119\pi^2}{6912}(\delta d)^2 + O(\delta^3), \qquad P_{\mathrm{leak}} = \frac{\pi^2}{2304}(\delta d)^2 + O(\delta^3).$$

Cubic physical compiler. Let

$$H_L = \begin{pmatrix} 0 & N \\ N^{\mathsf{T}} & 0 \end{pmatrix}.$$

The polar bright-sector operator is the exact cubic transform

$$\boxed{\mathrm{sign}(H_L) = \alpha H_L + \beta H_L^3},$$

with

$$\alpha = -\frac{3}{20} + \frac{4\sqrt{6}}{15}, \qquad \beta = \frac{1}{40} - \frac{\sqrt{6}}{60}.$$

Hence its point–line block is

$$\boxed{T = N^{\mathsf{T}}(\alpha I + \beta N N^{\mathsf{T}})}.$$

The dense polar coupling is therefore an exact sparse incidence traversal plus a three-hop spectral correction. A passive finite-depth hardware synthesis remains open.

Protected-memory accessibility theorem. For the oriented Levi boundary D,

$$\mathbb{R}^{160} = \operatorname{im} D^{\mathsf{T}} \oplus \ker D, \qquad \dim \operatorname{im} D^{\mathsf{T}} = 79, \quad \dim \ker D = 81.$$

The algebra generated by $D, D^{\mathsf{T}}, DD^{\mathsf{T}}$ and $D^{\mathsf{T}}D$ preserves this decomposition. Thus incidence-only linear dynamics cannot write cut/mode information into the protected H_1 memory. For one controlled link detuning,

$$V_e = P_{H_1}|e\rangle\langle e|P_{\mathrm{cut}}$$

has rank one and

$$\boxed{\sigma(V_e) = \frac{\sqrt{81 \cdot 79}}{160}}.$$

A uniform pulse has zero cross-sector matrix element. Controlled local symmetry breaking is therefore the physical write/read port rather than merely an imperfection.

Common-delay-invariant geometry tomography. For

$$Q = cI + \theta(12I - A_{W33}) + E,$$

define

$$C = Q - \frac{\operatorname{Tr} Q}{40} I.$$

Then $C = -\theta A_{W33} + E_c$, and the ideal scale and adjacency are

$$\boxed{\widehat{\theta} = \frac{\|C\|_F}{\sqrt{480}}}, \qquad \boxed{\widehat{A} = -\frac{C}{\widehat{\theta}}}.$$

The scalar common delay cancels exactly. The centered checksum is

$$\boxed{(C + 12\theta I)(C + 2\theta I)(C - 4\theta I) = 0}.$$

If $\|E_c\|_{\max} < \theta/4$ and $\|E_c\|_F/\sqrt{480} < \theta/4$, thresholding at $\widehat{\theta}/2$ recovers every labeled edge exactly. If $\|E_c\|_2 < 3\theta$, all three spectral clusters remain disjoint.

Causality and speed boundary. The diameters of the W33 graph, Levi graph, and Levi line graph are $2, 4, 4$. A fixed cell therefore supplies microscopic graph locality but not a macroscopic light cone. Any Lieb–Robinson-type physical speed has the form

$$v_{\mathrm{LR}} \lesssim C_{\mathrm{graph}} \frac{aJ}{\hbar},$$

so node count alone cannot determine the vacuum speed of light. For the exact finite-detuning gate,

$$t_{\mathrm{gate}} = \frac{\pi}{\sqrt{2}g},$$

while a localized point state has $\Delta E = 2\hbar g$ and $E - E_0 = 2\sqrt{2}\hbar g$. The gate is $2\sqrt{2}$ times the orthogonal Mandelstam–Tamm time and four times the Margolus–Levitin time.

Spectral-dimension falsifier. The W33 heat trace is

$$Z(t) = 1 + 24e^{-10t} + 15e^{-16t}.$$

Its running spectral dimension has maximum approximately 3.71848244 and returns to zero in both ultraviolet and infrared limits. The Levi line-graph maximum is approximately 3.47298414. Thus neither fixed finite graph has a four-dimensional Weyl plateau. A spacetime claim requires a refinement or RG family with $Z(t) \sim t^{-2}$ over a stable scale window and Lorentzian causal observables.

Thermodynamic and holographic capacity. The H_1 flat band has dimension 81. Its 1620 apartment states form an overcomplete tight frame of redundancy 20, not 1620 orthogonal messages. A single photon restricted to this sector has orthogonal capacity at most

$$\boxed{\log_2 81 \simeq 6.33985000 \text{ bits}},$$

and erasure obeys

$$\boxed{Q_{\mathrm{erase}} \geq k_B T \ln 81}.$$

A Bekenstein or covariant-entropy comparison remains undefined until physical energy, size or area, species content, and encoder/decoder restrictions are fixed.

Claim boundary. The finite architecture now has exact protected memory, controlled spectral unitaries, entangling protocols, error tensors, write ports, and tomography. It does not yet have a proved fault-tolerant non-Clifford injection, scalable local continuum, Standard Model representation functor, gravitational entropy–energy dictionary, dimensional scale derivation, fabricated device, or laboratory performance. "THE universal computer" and "the theory of everything" therefore remain explicit hypotheses with pass/fail obligations, not promoted conclusions.

53.1 Full harmonic compiler, projected controls, and finite-network electrodynamics (Passes 4033–4040)

The completed physics-first audit is supplemented by a deterministic construction on the same W33 point–line Levi graph. Let

$$D \in \{-1, 0, 1\}^{80\times 160}, \qquad C \in \{-1, 0, 1\}^{160\times 1620}, \qquad P_{H_1} = \frac{1}{160} CC^T,$$

with $\operatorname{rank} D = 79$, $\operatorname{rank} C = 81$, and $DC = 0$.

Exact write/read compiler. The first 81 canonically ordered apartment columns form an integer basis B of H_1. With

$$W = B(B^T B)^{-1/2},$$

one has

$$W^T W = I_{81}, \qquad W W^T = P_{H_1}.$$

Hence

$$H_{\rm swap} = \begin{pmatrix} 0 & W^T \\ W & 0 \end{pmatrix}$$

performs a complete input–H_1 swap at time $\pi/2$, up to phase. The dimension mismatch is physical: all 81 harmonic channels require the 80 incidence modes plus one ancilla.

Projected disorder and control. The 160 projected onsite operators

$$A_e = P_{H_1} |e\rangle\langle e| P_{H_1}$$

are linearly independent. The 480 nearest-neighbour projected couplings span 320 dimensions, and the onsite span lies inside that coupling span through

$$A_e = -\frac{1}{4} \sum_{f \sim e} P_{H_1} \big(|e\rangle\langle f| + |f\rangle\langle e| \big) P_{H_1}.$$

All projected site rays are pairwise nonorthogonal, so their commutant on H_1 is scalar. Their associative closure is $M_{81}(\mathbb{C})$; with the uniform identity control the ideal Hamiltonian Lie closure is $\mathfrak{u}(81)$. Thus uncalibrated local detuning is a splitting mechanism, whereas calibrated detuning is a universal control alphabet.

Compressed tomography and fabrication contract. The three-sector H_2 populations require two nontrivial scalar probes. Each five-sector signed-Levi or line-graph population vector requires four. Affinely scaled Chebyshev probes have condition number below two in all three cases. The secondary flat-band realization requires 160 sites, 480 degree-six couplings, and only uniform real hopping. Its exact flat-band gap is

$$\Delta_{\rm fb} = (4 - \sqrt{6}) J.$$

A sufficient static cluster-separation condition is $\|V\|_2 < \Delta_{\rm fb}/2$. This is a design theorem, not a fabrication claim.

Three physics constructions. First, the Lindblad jump map

$$L_v = \sum_e D_{ve} a_e$$

has dark space $\ker D = H_1$ and dissipative gap $4 - \sqrt{6}$. Second, the local projected controls generate $\mathfrak{u}(81)$ in the ideal protected model. Third, interpreting $\rho = DE$ as a discrete Gauss law gives an exact Levi effective-resistance law

$$R_1 = \frac{79}{160}, \qquad R_2 = \frac{13}{20}, \qquad R_3 = \frac{111}{160}, \qquad R_4 = \frac{7}{10}.$$

This is finite-network electrodynamics, not a continuum Maxwell or gravity derivation.

Execution boundary. The literal 48-relation fusion and Monster jobs still have no generated engine outputs. No relation-fusion rank, Monster word, class fusion, or embedding is promoted. The exact certificate is `data/PART_4033_4040_PHYSICS_CONTINUATION.json`.

54 Eight-physics expansion: holonomies, interacting flat-band matter, cooling, spectroscopy, refinement, and three outside-box constructions

Passes 4041–4048 extend the exact W33/Levi H_1 architecture along five requested physics fronts and three independent finite-model constructions. All claims below are finite-dimensional and machine checked; none is a fabricated-device, continuum-gravity, Standard-Model, or laboratory-performance claim.

54.1 Non-Abelian holonomic control in the harmonic memory

Choose an orthonormal logical pair in the 81-dimensional harmonic space, a third harmonic helper, and one excited ancilla. The ideal tripod spectrum is $\{-\Omega, 0, 0, +\Omega\}$ and the dark space has rank two. In a real tripod chart,

$$A_\phi = \cos\theta \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}, \qquad A_\theta = 0.$$

The closed loop through $\theta = \arccos(1/4)$ yields

$$U_X = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix},$$

while a phase loop at $\theta = \pi/6$ yields $U_Z = \mathrm{diag}(-i, 1)$. Their commutator obeys

$$\|[U_X, U_Z]\|_F^2 = 4,$$

so the exact geometric gates are non-Abelian. Varying the loop latitudes gives continuous rotations about two nonparallel axes and therefore ideal $SU(2)$ control. This does not certify finite-speed adiabatic fidelity or a laboratory pulse sequence.

54.2 Projected two-photon contact spectrum

For onsite interaction projected into the flat band,

$$V = U \sum_{e=1}^{160} |Pe, Pe\rangle\langle Pe, Pe|,$$

we have $\dim \mathrm{Sym}^2(H_1) = 3321$. The contact map has rank 160, leaving an exact contact-dark pair space of dimension

$$\boxed{3161}.$$

The complete nonzero spectrum of V/U is

$$\left(\frac{81}{160}\right)^1, \left(\frac{252 + 27\sqrt{6}}{800}\right)^{24}, \left(\frac{117}{400}\right)^{30}, \left(\frac{252 - 27\sqrt{6}}{800}\right)^{24}, \left(\frac{41}{200}\right)^{81}.$$

No bound-pair transport or many-body phase follows without additional dynamics.

54.3 Number-conserving one-shot Hodge cooling

On the charge-zero vertex sector let

$$Q = D^T (DD^T)^{-1/2}.$$

Then $Q^T Q = I_{79}$ and $QQ^T = P_{\mathrm{cut}}$. Coupling 160 system links to 79 reservoir modes through

$$H_{\mathrm{cool}} = g \begin{pmatrix} 0 & Q \\ Q^T & 0 \end{pmatrix}$$

produces at $t = \pi/(2g)$ the exact passive swap

$$|\psi\rangle_{\mathrm{sys}} |0\rangle_{\mathrm{res}} \longmapsto P_{H_1} |\psi\rangle_{\mathrm{sys}} - iQ^T |\psi\rangle_{\mathrm{res}}.$$

Thus all cut-space excitation leaves the system while the harmonic memory remains. The polar coupler is generally dense, and reservoir reset and finite-depth synthesis remain engineering obligations.

54.4 Synthetic Coulomb spectroscopy

For opposite drive at Levi vertices separated by distance d, define $s = (i\omega C + \gamma)/J$. The shell responses are

$$Z_1(s) = \frac{2(s^3 + 15s^2 + 66s + 79)}{(s+4)(s+8)(s^2+8s+10)},$$

$$Z_2(s) = \frac{2(s^2 + 8s + 13)}{(s+4)(s^2+8s+10)},$$

$$Z_3(s) = \frac{2(s^3 + 16s^2 + 78s + 111)}{(s+4)(s+8)(s^2+8s+10)},$$

$$Z_4(s) = \frac{2(s^2 + 8s + 14)}{(s+4)(s^2+8s+10)}.$$

Their DC limits are

$$\boxed{79/160,\ 13/20,\ 111/160,\ 7/10}.$$

The minimum shell gap is $1/160$, so absolute error below $1/320$ in units of $1/J$ is sufficient for nearest-shell identification.

54.5 Causal refinement tower and dimensional boundary

For a periodic d-dimensional array of cells,

$$\lambda(k) = 2\kappa \sum_{\mu=1}^{d} (1 - \cos k_\mu) = \kappa |k|^2 + O(k^4),$$

and the wave equation gives

$$\omega(k) = \sqrt{\kappa}|k| + O(k^3), \qquad c_{\mathrm{cell}} = a\sqrt{\kappa}.$$

At $N = 64$ and $\kappa/J = 0.05$, the one-dimensional tower has a stable $d_s \simeq 1$ heat-kernel window, while a four-dimensional torus has a stable $d_s \simeq 4$ window. The conclusion is a boundary rather than an emergence claim: four macroscopic dimensions occur only when four external lattice directions are explicitly supplied.

54.6 Three outside-box exact constructions

First, the discrete supercharge

$$\mathcal{Q} = \begin{pmatrix} 0 & D \\ D^T & 0 \end{pmatrix}$$

has 82 zero modes, one uniform vertex mode and 81 harmonic edge modes. With vertex-positive and edge-negative grading, the Witten index is

$$\boxed{1 - 81 = -80}.$$

Every nonzero eigenvalue occurs in an exact $\pm$ pair. This is incidence-complex supersymmetry, not particle-physics supersymmetry.

Second, one local site reflection composed with the Hodge reflection,

$$U_F = (I - 2|e\rangle\langle e|)(I - 2P_{H_1}),$$

has spectrum $(+1)^{78} + (-1)^{80} + e^{\pm i\phi}$ with

$$\boxed{\cos\phi = 1/80}.$$

Niven's theorem excludes rational ϕ/π, so the Floquet rotor has infinite order. This is not a many-body time crystal.

Third, the normalized protected site states satisfy

$$\langle u_e|u_f\rangle = -1/3,\ 1/9,\ -1/27,\ 1/81$$

for line-graph distances $1, 2, 3, 4$. Under the two-defect Hamiltonian

$$H_{ef} = P|e\rangle\langle e|P + P|f\rangle\langle f|P,$$

unit-fidelity transfer occurs at

$$\boxed{t/\pi = 80/27,\ 80/9,\ 80/3,\ 80}.$$

This is perfect transfer between nonorthogonal projected modes in an ideal protected model, not superluminal signaling or a spacetime wormhole.

Evidence firewall. The frozen semantic certificate is `08414977d5198aa43ea25127bbe7fa0e6529f56471dabe974` The exact verifier and focused regression are `analysis/w33_pass4041_4048_eight_physics_expansion.py` and `tests/test_w33_pass4041_4048_eight_physics_expansion.py`.

55 Minimal spectral compiler, protected pulse alphabet, scaling family, and magic reduction

Minimal bounded QSVT compiler. With $X = H_L/4$, the relevant spectrum is

$$\{-1, -\sqrt{6}/4, 0, \sqrt{6}/4, 1\}.$$

The unique exact odd cubic sign interpolant exceeds one on $[-1,1]$, reaching approximately 1.09215935, and is therefore not directly QSVT-admissible. The minimal bounded odd polynomial is the quintic

$$p_5(x) = \left(\frac{9}{25} + \frac{24\sqrt{6}}{25}\right) x + \left(-\frac{48}{25} - \frac{152\sqrt{6}}{225}\right) x^3 + \left(\frac{64}{25} - \frac{64\sqrt{6}}{225}\right) x^5.$$

It has the exact certificate

$$1 - p_5(x)^2 = \frac{4096(29 - 6\sqrt{6})}{16875}(1 - x^2)\left(x^2 - \frac{3}{8}\right)^2 q_+(x)q_-(x),$$

where

$$q_\pm(x) = x^2 \pm \left(1 + \frac{\sqrt{6}}{2}\right) x + \frac{9 + \sqrt{6}}{8} > 0.$$

Thus $|p_5(x)| \le 1$ on $[-1,1]$, and the exact polar sign transform admits a degree-five, six-phase QSVT realization once a block encoding of $H_L/4$ is provided. Degree five is minimal because the unique degree-three interpolant violates boundedness.

Protected local phase pulses. For $P = P_{H_1}$ and

$$|u_e\rangle = \sqrt{160/81}\, P|e\rangle,$$

the normalized off-diagonal overlap alphabet is

$$-\frac{1}{3}, \quad \frac{1}{9}, \quad -\frac{1}{27}, \quad \frac{1}{81},$$

with respective per-site multiplicities $6, 18, 54, 81$. A calibrated local detuning produces

$$A_e = P|e\rangle\langle e|P = \frac{81}{160}|u_e\rangle\langle u_e|.$$

Holding amplitude δ_e for

$$t_\phi = \frac{160\phi}{81\delta_e}$$

implements

$$U_e(\phi) = I + (e^{-i\phi} - 1)|u_e\rangle\langle u_e|.$$

At $\phi = \pi$ this is a Householder reflection. Two-reflection rotations have angle alphabet

$$2\arccos(1/3), \quad 2\arccos(1/9), \quad 2\arccos(1/27), \quad 2\arccos(1/81).$$

Together with the Pass 4034 Lie-closure theorem, these are exact pulse-level generators of $\mathfrak{u}(81)$ on the protected memory.

A local four-dimensional W33-fiber family. Define

$$G_n = W33 \square C_n \square C_n \square C_n \square C_n.$$

Then $|V(G_n)| = 40n^4$ and $\deg G_n = 20$. For physical side length L and internal scale M, use

$$\mathcal{L}_n = M^2 L_{W33} \otimes I + I \otimes \left(\frac{n}{L}\right)^2 \sum_{\mu=1}^{4} L_{C_n}^{(\mu)}.$$

The heat trace factors into the W33 internal trace and four identical discrete circle traces. In the window

$$M^{-2} \ll t \ll L^2, \qquad (L/n)^2 \ll t,$$

the internal excitations freeze while the rescaled discrete torus converges to a real four-torus, giving $d_s \to 4$. This is a controlled four-dimensional base with W33 internal fiber; it does not derive dimension four from W33.

Exact one-copy M36 magic reduction. For

$$|m\rangle = (0, 1, -1, 1)/\sqrt{3},$$

measure the first qubit in the X basis and retain the plus result. With probability 5/6, the remaining qubit is

$$(-|0\rangle + 2|1\rangle)/\sqrt{5}.$$

After the Clifford $R_x(\pi/2)$ this becomes

$$|A_\phi\rangle = (|0\rangle + e^{i\phi}|1\rangle)/\sqrt{2}, \qquad e^{i\phi} = (-4 + 3i)/5.$$

Since $\cos\phi = -4/5$, the rational-cosine theorem implies $\phi/\pi \notin \mathbb{Q}$. A CNOT from data to this resource followed by a computational-basis resource measurement applies $\mathrm{diag}(1, e^{i\phi})$ on one outcome and its inverse on the other. Postselecting the desired outcome succeeds with total probability

$$\boxed{5/12}.$$

The irrational z-rotation, Clifford conjugates, and an entangling Clifford gate give ideal postselected universal quantum computation. Deterministic correction, logical encoding, distillation, and a fault-tolerance threshold remain open.

Finite-to-observable quadratic functor. On

$$\mathcal{H}_n = \ell^2((C_n)^4) \otimes \mathbb{C}^{40},$$

define

$$K_n = \left(\frac{n}{L}\right)^2 L_{(C_n)^4} \otimes I + I \otimes [m_0^2 I + g(12I - A_{W33})].$$

Full W33 automorphism invariance restricts the internal quadratic algebra to $\mathrm{span}\{I, A, J\}$. The resulting free species have squared masses

$$m_0^2, \qquad m_0^2 + 10g, \qquad m_0^2 + 16g$$

with multiplicities $1, 24, 15$. This is an explicit free-field observable dictionary and selection rule, not a Standard Model construction.

Outside-box shell splitter. For a localized point state, the exact reflection gives amplitude $-1/5$ on the closed neighbourhood of size 13 and $2/15$ on the far shell of size 27:

$$U|v\rangle = -\frac{\sqrt{13}}{5}|N_v\rangle + \frac{2\sqrt{3}}{5}|F_v\rangle.$$

The shell probabilities are $13/25$ and $12/25$, yielding $H_2(13/25) \simeq 0.998845536$ bits of single-photon mode-partition entanglement. Since $U^2 = I$, the shell split recombines exactly.

Outside-box apartment projective code. The 1620 normalized apartment states in dimension 81 have absolute inner products

$$0, \quad 1/8, \quad 1/4, \quad 3/8, \quad 1/2, \quad 1.$$

Their fourth-power frame potential is

$$F_2 = 79785/16,$$

whereas a real projective 2-design would have $97200/83$. The exact defect ratio is $16351/3840$. Thus the apartment frame is a five-angle tight projective code but not a projective 2-design; tightness does not imply Haar isotropy or maximal scrambling.

Outside-box spectral calorimeter. The finite W33 spectrum $0^1 + 10^{24} + 16^{15}$ has partition function

$$Z(\beta) = 1 + 24e^{-10\beta} + 15e^{-16\beta}.$$

Its Schottky heat-capacity peak occurs at

$$\beta_* \simeq 0.4127094288, \qquad k_B T_*/J \simeq 2.4225277564,$$

with

$$C_{\max}/k_B \simeq 3.8006256108.$$

This is an exact single-particle spectral fingerprint, not a measured or interacting thermodynamic response.

Boundary. The results establish an exact minimal bounded spectral compiler, pulse-level protected controls, a convergent W33-fibered four-dimensional family, ideal postselected non-Clifford universality, and a free quadratic observable functor. They do not establish a fabricated block encoding, extracted phase list, deterministic fault-tolerant injection, Lorentzian spacetime, Standard Model, gravity, measured calorimetry, or a theory of everything.

55.1 Transitionless holonomies, dark-pair characters, local Hodge cooling, and finite-graph physics

Passes 4057–4064 close five physics targets and three independent outside-box probes on the exact Levi–H_1 substrate. All statements in this subsection are finite graph, representation-theoretic, spectral, lattice, Floquet, or canonical-ensemble statements; none is evidence for fabricated hardware, measured gate fidelity, a chiral Standard Model, quantized pumping, gravity, cosmology, or a theory of everything.

For the two non-Abelian tripod loops, the Kato counterdiabatic term

$$H_{\mathrm{CD}} = i[\dot{P}_D, P_D]$$

produces exact finite-time parallel transport in the ideal dark bundle. Their Fubini–Study actions are

$$L_X = 2\arccos(1/4) + \frac{\pi\sqrt{15}}{2} = 8.719900157\ldots, \qquad L_Z = \frac{\pi}{3} + \frac{\pi\sqrt{3}}{2} = 3.767896598\ldots.$$

Thus a constant-speed duration T requires peak counterdiabatic norm L/T, while uniform loss gives no-jump survival $e^{-\kappa T}$.

The 3161-dimensional two-boson contact-dark module was decomposed by generating $PSp(4,3)$ as a 25,920-element permutation group, reconstructing its twenty conjugacy classes and class algebra, and diagonalizing the center. Grouped by equal ordinary-character degree, its multiplicities are

d	1	5	6	10	15	20	24	30	40	45	60	64	81
m	0	$(1,1)$	2	$(1,1)$	$(1,3)$	5	2	$(3,6,6)$	$(4,4)$	$(4,4)$	9	8	9.

The weighted dimension is exactly 3161, and there is no trivial constituent. The bright rank-160 contact sector is an equivariant incidence-edge permutation module, so

$$P_{\mathrm{dark}} V_{\mathrm{contact}} P_{\mathrm{dark}} = 0;$$

contact-only first-order pair motion inside the dark sector is forbidden.

Because the nonzero Levi Laplacian spectrum is $\{4-\sqrt{6}, 4, 4+\sqrt{6}, 8\}$, the inverse square root is represented exactly on the physical spectrum by a cubic Lagrange polynomial p_3. Consequently

$$Q = D^T p_3(DD^T), \qquad Q^T Q = I - J/80, \qquad QQ^T = P_{\mathrm{cut}}.$$

The polar cooler can therefore be compiled from three nearest-neighbour Levi-Laplacian queries and one incidence query, together with the bounded degree-five sign compiler of Pass 4049. The collapsed matrix remains dense; this is exact circuit/query locality, not a one-layer sparse Hamiltonian.

On the explicit four-dimensional refinement tower, the gauge-covariant Wilson–Dirac operator

$$H(k) = \sum_{\mu=1}^{4} \gamma_\mu \sin k_\mu + \gamma_5 \left[m + r \sum_\mu (1 - \cos k_\mu) \right]$$

is nearest-neighbour and may be tensored with I_{81} on the harmonic fiber. At $m = 0$, the naive operator has sixteen corner zeros, whereas $r = 1$ leaves only the origin and gives the other corners Wilson masses $2, 4, 6, 8$. The price is explicit breaking of exact lattice chirality; the construction is vectorlike and does not evade the Nielsen–Ninomiya boundary.

Feeding the irrational Floquet phase $\cos\phi = 1/80$ into a static Coulomb port yields $V_n = Z_d(s) I_n$. The static linear network preserves the input quasifrequencies and

$$\frac{1}{N} \sum_{n<N} e^{in\phi} \longrightarrow 0, \qquad \left| \sum_{n=0}^{N-1} e^{in\phi} \right| \le \frac{2}{|1 - e^{i\phi}|}.$$

Hence there is no DC pump and no integer topological transport from one irrational phase alone; a second independently cycled parameter and a maintained gap are required.

Three outside-box consequences are exact. First, two projected Householder reflections convert graph distance into quasienergy through

$$|\langle u_e, u_f\rangle| = 3^{-d}, \qquad \theta_d = 2\arccos(3^{-d}),$$

giving four infinite-order distance clocks. Second, the repulsive two-boson contact model has residual entropy $k_B \log 3161$ and infinite-temperature dark probability $3161/3321$. Third, the Levi graph has edge connectivity four, so any $r \le 3$ missing couplers leave

$$\dim H_1 = 81 - r, \qquad \mathcal{I}_W = -80 + r,$$

providing a topological fault odometer up to the four-edge connectivity threshold.

The executable certificate is `data/PART_4057_4064_ADVANCED_PHYSICS.json`, with semantic SHA-256 `131ba5f89474a0283eedd0ae1ec5e5e6618b6d3e2bd6eeb8654fd4df8abbcd72`.

56 Explicit QSP, adaptive irrational magic, Lorentzian Dirac dynamics, and exact physical obstructions

Six-phase QSP compiler. In the W_x convention

$$W(x) = \begin{pmatrix} x & i\sqrt{1-x^2} \\ i\sqrt{1-x^2} & x \end{pmatrix}, \quad U_{\boldsymbol{\phi}}(x) = e^{i\phi_0 Z}\prod_{j=1}^{5}[W(x)e^{i\phi_j Z}],$$

the bounded quintic sign compiler is realized by

$$\boldsymbol{\phi} = (-2.8067951015434964, -2.042810210594929, -2.419664116537273, 2.419664116537273, -1.0987824429948638$$

On a 20,001-point grid the top-left polynomial residual is below 1.19×10^{-15}. The sequence uses five signal queries and six phase gates. A telescoping perturbation bound is

$$\|\delta U\| \le 6\delta_\phi + 5\eta_W.$$

Adaptive irrational-magic correction. For $e^{i\phi} = (-4+3i)/5$, an undesired $-\phi$ injection branch is corrected with 2ϕ, then 4ϕ, and so forth. After K correction levels,

$$p_{\text{fail}} = 2^{-(K+1)}, \qquad \|\mathcal{E}_K - \mathcal{U}_\phi\|_\diamond \le 2^{-K}.$$

If doubled-angle states are directly available, fewer than two are injected on average. If each is recursively produced from the original state by parity doubling, the expected raw cost is

$$N_{\text{raw}} = 2^{K+1} - 1.$$

No finite doubling tree is exactly deterministic because the final residual differs from the target by $2^{K+1}\phi$, and ϕ/π is irrational.

Causal W33-fibered Dirac walk. Let

$$\beta = Z \otimes I, \qquad \alpha_j = X \otimes \sigma_j, \qquad m_r = \sqrt{m_0^2 + g\lambda_r},$$

where $\lambda_r \in \{0, 10, 16\}$ with multiplicities $1, 24, 15$. The exact three-spatial-dimensional local walk is

$$U_a(p, r) = e^{-iam_r\beta} e^{-iap_1\alpha_1} e^{-iap_2\alpha_2} e^{-iap_3\alpha_3}.$$

Each factor is a conditional nearest-neighbour shift, and

$$\frac{U_a - I}{-ia} = m_r\beta + \sum_{j=1}^{3} p_j\alpha_j + O(a).$$

Thus discrete time plus three spatial directions supplies a controlled $3+1$-dimensional Dirac limit carrying the three W33 mass sectors.

SK1-protected H1 pulse. For common fractional amplitude error ϵ,

$$U_{\mathrm{SK1}} = R_{-\chi}(2\pi(1+\epsilon))R_{\chi}(2\pi(1+\epsilon))R_x(\theta(1+\epsilon)), \quad \chi = \arccos\left(-\frac{\theta}{4\pi}\right).$$

The linear error vanishes. For $\theta = \pi$,

$$R_x(\pi)^\dagger U_{\mathrm{SK1}} = I - i\frac{\sqrt{15}\pi^2}{8}\epsilon^2 Z + O(\epsilon^3),$$

so the entanglement infidelity begins at $15\pi^4\epsilon^4/64$.

Gauge-commutant obstruction. The W33 permutation module decomposes as

$$\mathbb{C}^{40} = \mathbf{1} \oplus V_{24} \oplus V_{15},$$

and its full automorphism commutant is

$$\mathrm{span}\{E_{12}, E_2, E_{-4}\} = \mathrm{span}\{I, A, J\}.$$

This algebra is commutative, so its unitary gauge group is exactly

$$\boxed{U(1)^3}.$$

Unbroken W33 symmetry alone cannot generate non-Abelian $SU(2)$ or $SU(3)$ gauge bosons or mix the three sectors. A non-Abelian gauge theory requires symmetry breaking, multiplicity spaces, or a larger noncommutative algebra.

Three outside-box exact results. First, the reflection $U = I - 2E_6$ obeys $U^2 = I$, so repeated reflection has orbit dimension at most two and pure-state time-averaged entropy at most one bit. A localized vertex gives $H_2(3/5) = 0.97095059445$ bits; the reflection alone cannot thermalize or generate a Page curve.

Second, local H1 defect sensing satisfies

$$F_e^{\max} = \left(\frac{81}{160}\right)^2 t^2, \qquad \sum_e F_e \le 2t^2.$$

The total bound is attained by an equal-magnitude divergence-free Eulerian flow, giving uniform-prior average QFI $t^2/80$.

Third, the matrix-tree theorem gives

$$\tau(W33) = 2^{81}5^{23}, \qquad \tau(L_{\text{Levi}}) = 2^{83}5^{23}, \qquad \frac{\tau(L_{\text{Levi}})}{\tau(W33)} = 4.$$

The incidence lift changes Kirchhoff complexity by exactly two bits.

Boundary. These are exact algebraic or deterministic numerical statements in explicitly specified models. No fabricated block encoding, finite-cost exact deterministic irrational injection, fault-tolerance threshold, Standard Model, gravity, cosmology, or theory of everything is established.

56.1 Passes 4073–4080: engineering closures and outside-box physics probes

Exact four-router block encoding. The 4-regular Levi graph admits a one-factorization $A_L = P_0 + P_1 + P_2 + P_3$ into four involutory perfect-matching permutations. Consequently a balanced four-path selector and the controlled router $U_{\text{sel}} = \sum_j |j\rangle\langle j| \otimes P_j$ obey

$$(\langle 00|B^\dagger \otimes I)U_{\text{sel}}(B|00\rangle \otimes I) = \frac{A_L}{4}.$$

This is an exact abstract block encoding with two selector bits or paths; loss, crosstalk, and fabrication are not included.

Finite phase reference. Compressing K copies of $|A_\phi\rangle = (|0\rangle + e^{i\phi}|1\rangle)/\sqrt{2}$ into the symmetric Hamming-weight basis and applying a controlled cyclic lowering produces coherence multiplier

$$z_K = e^{i\phi}S_K + 2^{-K}e^{-iK\phi}, \qquad S_K = 2^{-K}\sum_{n=1}^{K}\sqrt{\binom{K}{n-1}\binom{K}{n}}.$$

The one-use channel error is at most $1 - S_K + 2^{-K}$, with $1 - S_K \sim (2K)^{-1}$. This removes the doubled-angle hierarchy but is only approximately catalytic.

Minimal non-Abelian multiplicities. The multiplicity extension $(m_1, m_{24}, m_{15}) = (3, 1, 2)$ has dimension 57 and commutant

$$M_3(\mathbb{C}) \oplus \mathbb{C} \oplus M_2(\mathbb{C}).$$

After freezing the spectator $U(1)$, its determinant-one subgroup has global form $S(U(3) \times U(2)) \cong (SU(3) \times SU(2) \times U(1))/\mathbb{Z}_6$ and central charge ratio $-1/3 : 1/2$. The present Dirac matter remains vectorlike; chirality and the observed matter representation are not derived.

Four-gap leakage notch. For the bright gaps $\Delta/J \in \{4-\sqrt{6}, 4, 4+\sqrt{6}, 8\}$, choose

$$f = \frac{h^{(8)} + 124h^{(6)} + 4644h^{(4)} + 53056h'' + 102400h}{102400},$$

with h and its first seven derivatives vanishing at the pulse endpoints. Then

$$F(\omega) = H(\omega)\frac{\prod_r(\Delta_r^2 - \omega^2)}{102400},$$

so all first-order H_1-to-bright transition amplitudes vanish while $\int f = \int h$. Leakage probability therefore begins at fourth order in the perturbative control strength.

Independent falsifiers and outside-box correspondences. The packet freezes six laboratory null tests. It also identifies P_{H_1} with the Maxwell–Calladine state-of-self-stress projector of an 80-node, 160-spring scalar network, giving one uniform mechanism and 81 self-stress states. A sudden quench of $H(g) = gL_{W33}$ has an exact three-line work distribution and obeys $\langle e^{-\beta W}\rangle = Z_1/Z_0$. Finally, the normalized Sakaguchi–Kuramoto linearization has decay rates $(5/6)K\cos\alpha$ with multiplicity 24 and $(4/3)K\cos\alpha$ with multiplicity 15, fixing the relaxation-time ratio to 8/5.

Boundary. These are exact finite or perturbative ideal-model statements. They do not certify a fabricated interferometer, an exact finite reusable irrational-phase catalyst, a fault-tolerance threshold, a chiral Standard Model, a mechanical prototype, a global nonlinear synchronization theorem, gravity, cosmology, or a theory of everything.

56.2 Canonical dark sectors, mobile pairs, overlap chirality, and operational geometry

The rank-3161 contact-dark two-photon module has now been resolved into all ordinary $PSp(4,3)$ irreducibles. The twenty conjugacy classes were mapped to ATLAS order

$$1A, 2A, 2B, 3A, 3B, 3C, 3D, 4A, 4B, 5A, 6A, 6B, 6C, 6D, 6E, 6F, 9A, 9B, 12A, 12B,$$

and every character fingerprint was frozen exactly in $\mathbb{Q}(\sqrt{-3})$. The central idempotents are

$$e_\chi = \frac{\chi(1)}{25920}\sum_C \overline{\chi(C)}K_C.$$

The dark module contains no trivial constituent; its nineteen nonzero isotypic dimensions sum to 3161.

After introducing an explicit same-cell dark-pair binding energy $-\Delta$ and symmetry-preserving one-photon hopping t, second-order Schrieffer–Wolff reduction gives

$$J_{\text{pair}} = \frac{2t^2}{\Delta}, \qquad E(k) = -\Delta - \frac{4t^2}{\Delta} - \frac{4t^2}{\Delta}\cos k,$$

with bandwidth $8t^2/\Delta$ and $m^* = \Delta/(4t^2a^2)$ for $\hbar = 1$. All nineteen dark isotypic sectors share this clean dispersion and do not mix. This mobility requires the added binding scale; bare repulsive contact darkness alone is not a bound mobile phase.

A two-parameter dimerization cycle

$$J_{1,2} = J \pm R\cos\theta, \qquad m = 2R\sin\theta,$$

produces a pair-center Rice–Mele Hamiltonian whose lower band has $C = +1$ and minimum gap $4R$. At clean filled-band occupancy it pumps one composite pair, hence two photons, per cycle.

On the four-dimensional external tower, the overlap operator

$$D_{\text{ov}} = a^{-1}\big[I + \gamma_5\,\text{sign}(H_W)\big]$$

obeys the exact Ginsparg–Wilson relation

$$\gamma_5 D + D\gamma_5 = aD\gamma_5 D.$$

For $0 < m_0 < 2r$ one light external species survives per harmonic fibre, so the rank-81 fibre carries 81 light vectorlike Dirac species. This does not construct anomaly-free chiral gauge matter.

Marking a single incidence edge reduces the symmetry to a stabilizer of order 162. The restricted harmonic module has commutant dimension 45; a six-dimensional stabilizer irrep occurs with multiplicity three, giving an exact $U(3)$ block, while several three-dimensional irreps occur with multiplicity two, giving exact $U(2)$ blocks. Thus one marked edge is minimal in mark count for exposing $SU(3)$ and $SU(2)$ control subalgebras; these are commutant algebras, not identified Standard Model gauge bosons.

Three independent finite-physics consequences follow. First, one ideal Levi vertex port has local resolvent

$$G_{00}(s) = \frac{s^4 + 16s^3 + 78s^2 + 112s + 4}{s(s+4)(s+8)(s^2 + 8s + 10)},$$

so its poles and residues recover the complete Levi spectrum and multiplicities. Second, erasing a maximally mixed dark state costs at least

$$k_B T \ln 3161 = 8.058643712215618\, k_B T.$$

Resolving and retaining the irrep label lowers the conditional data cost by $2.3278663282163152\, k_B T$, but erasing that label restores the same Landauer bound. Third, for $H = JA_{\text{line}}$ the order of the first nonzero short-time derivative of $\langle u|e^{-iHt}|v\rangle$ equals graph distance exactly. The four nontrivial shells have sizes $6, 18, 54, 81$ and first geodesic multiplicities $1, 1, 1, 2$.

These are exact finite group, band, operator, resolvent, thermodynamic, and graph-propagation statements. They establish neither fabricated hardware, measured quantization, chiral Standard Model matter, gravity, cosmology, nor a theory of everything.

57 Passes 4089–4096: implementation closure and outside-box probes

Theorem 57.1 (Router layout, optimal phase reference, anomaly bridge, and four-notch control)**.** *The 80-vertex Levi adjacency admits a deterministic decomposition $A_L = P_0 + P_1 + P_2 + P_3$ into four involutory perfect-matching permutations. Hence a balanced four-path selector gives an exact block encoding $A_L/4$. The frozen 160-swap route table is* `data/w33_pass4089_four_router_layout.json`*.*

For a finite phase reference on weights $0, \ldots, K$, the maximum nearest-neighbour coherence is

$$\max \sum_{n=1}^{K} c_{n-1} c_n = \cos \frac{\pi}{K+2},$$

attained by $c_n = \sqrt{2/(K+2)} \sin((n+1)\pi/(K+2))$. The resulting one-use error is $O(K^{-2})$.

The exterior representation ring generated by the multiplicity spaces $\mathbb{C}^3$ and $\mathbb{C}^2$, with determinant-one central charges $(-1/3, 1/2)$, contains

$$(3,2)_{1/6},\ (\bar{3},1)_{-2/3},\ (\bar{3},1)_{1/3},\ (1,2)_{-1/2},\ (1,1)_1,$$

and all perturbative gauge, mixed gravitational, cubic Abelian, and mod-two $SU(2)$ anomalies cancel exactly. This is a representation-ring identity, not a particle derivation.

Finally, the compact envelope $h = 218790T^{-1}x^8(1-x)^8$ and

$$f = \frac{h^{(8)} + 124h^{(6)} + 4644h^{(4)} + 53056h'' + 102400h}{102400}$$

annihilate all four H_1-to-bright first-order amplitudes. A full 160-mode simulation gives leakage exponents 2.0191 for a square pulse and 4.0191 for the notched pulse.

Corollary 57.2 (Mechanical, coding, pattern, and circuit consequences)**.** *The scalar mechanical dual has* 81 *states of self stress and remains spectrally resolvable under the sufficient worst-case spring tolerance*

$$\epsilon < \frac{4-\sqrt{6}}{16} = 0.0969068911.$$

Its real cycle code has exact sparse parameters $[160, 81, 8]$*, detecting seven and uniquely correcting three noiseless sparse bond errors. Two explicit reaction–diffusion Jacobians select respectively the* 24*- and* 15*-dimensional W33 Laplacian sectors as the unique linearly unstable pattern spaces. The unit-resistor W33 network has only two effective resistances,*

$$R_{\mathrm{adj}} = \frac{13}{80}, \qquad R_{\mathrm{nonadj}} = \frac{7}{40},$$

with Kirchhoff index $267/2$*.*

Evidence boundary. No fabricated router, finite exact reusable irrational reference, dynamically derived Standard Model, measured pulse, mechanical prototype, nonlinear pattern experiment, gravity, cosmology, or theory of everything is asserted.

58 Passes 4097–4104: interacting pumping, quantum links, horizon falsifier, and spectral renormalization

Exact fractional many-composite pump. On the nine-cell, three-pair hard-core ring

$$H_0/U = \sum_i (n_i + n_{i+1} + n_{i+2} - 1)^2,$$

the fixed-filling spectrum is

$$0^3 + 2^{18} + 4^{36} + 6^{18} + 10^9.$$

The exact ground patterns 100100100, 010010010, and 001001001 have Resta polarizations $0, 1/3, 2/3$ and gap $2U$. With twisted holonomy

$$W(\phi) = e^{i\phi/3} T_3,$$

we have $\det W = e^{i\phi}$, so the rank-three ground bundle has Chern number one while one branch transports one third of a composite-pair charge per cycle. Three cycles pump one pair, or two photons.

Finite $SU(3)$ quantum-link sector. For link Hilbert space $\mathbf{1} \oplus (\mathbf{3} \otimes \bar{\mathbf{3}})$, the 81-dimensional one-link endpoint sector has a one-dimensional exact Gauss-law kernel,

$$|\mathrm{string}\rangle = \frac{1}{3} \sum_{a,b} |\bar{a}\rangle |a, \bar{b}\rangle |b\rangle.$$

Since $C_2(\mathbf{3}) = 4/3$, the electric string energy is

$$E_{\mathrm{string}}(R) = \frac{2g^2 R}{3}, \qquad \sigma = \frac{2g^2}{3a}.$$

The minimal vacuum/loop plaquette block is

$$H_\square = \begin{pmatrix} 0 & -J \\ -J & 8g^2/3 \end{pmatrix},$$

with gap $2\sqrt{16g^4/9 + J^2}$.

Passive-horizon no-go. For a reciprocal number-conserving chain with real hopping profile J_n, all open-chain hopping signs are removable by local phases. Moreover $U(t) = e^{-iht}$ is symmetric and its Nambu evolution is $\mathrm{diag}(U, U^*)$, so the Bogoliubov creation block vanishes exactly:

$$\beta = 0, \qquad \langle N \rangle_{\mathrm{vacuum}} = 0.$$

A static reciprocal $J(x)$ profile alone therefore creates neither a one-way horizon nor spontaneous Hawking radiation. Directed Floquet/background flow and particle–hole mixing are additional necessary ingredients.

Discrete-scale spectral fixed cycle. At hierarchy level n, retain one zero mode and modes

$$\lambda_\alpha b^{-2(n-1-j)}$$

with multiplicity $m_\alpha 80^j$, where

$$(\lambda_\alpha, m_\alpha) = (4 - \sqrt{6}, 24), (4, 30), (4 + \sqrt{6}, 24), (8, 1).$$

The total dimension is 80^n, and the log-period-average spectral dimension is

$$\bar{d}_s = \frac{\log 80}{\log b}.$$

For $b = 80^{1/4}$, $\bar{d}_s = 4$. At $n = 14$ the computed central-window mean is 3.99987084247, with persistent log-periodic oscillations between 2.97765560 and 5.14402331. Thus the construction is a discrete-scale fixed cycle, not a smooth continuum fixed point; the value four depends on the supplied rescaling.

Topological information engine. For the 19 dark isotypic dimensions n_j, with $p_j = n_j/3161$,

$$H(J) = 2.327866328216315, \qquad \sum_j p_j \log n_j = 5.730777383999302,$$

and

$$H(J) + \sum_j p_j \log n_j = \log 3161.$$

A reversible expansion extracts $k_B T \log 3161$, while conditional erasure and label erasure cost exactly the same total. Irrep-controlled Chern pumps can robustly route one pair per cycle, but the maximum closed-cycle net work remains zero.

Three outside-box constructions. First, the invariant CDW clock subspace with

$$H_{\mathrm{scar}} = \Omega(T_3 + T_3^\dagger)$$

has eigenvalues $-\Omega, -\Omega, 2\Omega$ and exact return probability

$$P_A(t) = \frac{5 + 4\cos(3\Omega t)}{9},$$

with perfect revival at $2\pi/(3\Omega)$. Second, the qutrit-cat probe over the three fractional-pump branches has

$$F_Q = \frac{8N_p^2}{3}$$

for a pair-position phase and $32N_p^2/3$ for the corresponding photon-gradient phase. Third, the canonical Fisher length of the projected contact spectrum from $\beta = 0$ to ∞ is 0.4530018812338062, while the Fisher–Rao endpoint geodesic is 0.4425945899021065, an excess of only 2.3514% because 3161/3321 of the Hilbert space is already dark.

Evidence boundary. These are exact finite strong-coupling, Gauss-law, passive-Bogoliubov, hierarchical-spectrum, information-thermodynamic, invariant-subspace, and Fisher-geometry statements. They do not establish a fractional-pump experiment, continuum QCD, Hawking observation, physical four-dimensional spacetime, positive-work engine, generic scar phase, gravity, cosmology, or a theory of everything.

Passes 4105–4112: carrier audit, correlated reference, implementation contracts, and three outside-box probes

The candidate module

$$(\mathbb{C}^6 \otimes V_1) \oplus (\mathbb{C}^3 \otimes V_{15}) \oplus (\mathbb{C}^2 \otimes V_{24})$$

has dimension 99 and commutant $M_6 \oplus M_3 \oplus M_2$, but it contains only one independent anti-triplet and no independent charged singlet. Under the sector assignment $Q, e^c \mapsto V_1$, $u^c, d^c \mapsto V_{15}$, and $L \mapsto V_{24}$, the minimum direct carrier is therefore

$$(\mathbb{C}^7 \otimes V_1) \oplus (\mathbb{C}^6 \otimes V_{15}) \oplus (\mathbb{C}^2 \otimes V_{24}), \qquad \dim = 145.$$

This carries the anomaly-free one-generation representation as a representation architecture, not as a derived particle model.

For the sine phase reference $c_n = \sqrt{2/(K+2)}\sin((n+1)\theta)$, $\theta = \pi/(K+2)$, sequential use gives a collective channel

$$\mathcal{E}_N(|x\rangle\langle y|) = z_{|x|-|y|}|x\rangle\langle y|,$$

with exact bulk overlap

$$C_m = \frac{(K+1-m)\cos(m\theta) + \sin((m+1)\theta)/\sin\theta}{K+2}.$$

Equal-Hamming-weight coherences are decoherence-free and the fixed-N error is $O(N^2/K^2)$.

The four-router signal unitary compiles to four coherent 80-mode branches with 40 parallel swaps per branch. A recirculating five-query realization uses 160 swap cells, six selector/recombiner couplers, six phase shifters, and five coherent passes; an unrolled realization uses 800 swap cells.

For noisy self-stress syndrome data $y = De + n$, exhaustive support-three decoding obeys

$$\|\widehat{e} - e\|_2 \leq \frac{2\|n\|_2}{2\sin(\pi/14)},$$

because every union of two support-three candidates is a forest and the worst six-edge restricted singular value is $2\sin(\pi/14)$. Cubically saturated reaction–diffusion simulations converge with unit numerical modal purity into the intended 24- and 15-dimensional eigenspaces.

Finally,

$$\frac{J}{40} = I - \frac{13}{80}L + \frac{1}{160}L^2$$

gives exact two-round average consensus; simple random walk has adjacent and nonadjacent hitting times 39 and 42 with Kemeny constant 801/20; and the abstract Hückel model has a 50-electron closed shell with gap $6|\beta|$ and a 40-electron open-shell determinant count $\binom{48}{10} = 6{,}540{,}715{,}896$.

58.1 Passes 4113–4120: gauge-string pumping, active horizons, dynamic dimension, local scar compilation, and thermodynamic curvature

The exact evidence bundle for this subsection is `data/PART_4113_4120_GAUGE_HORIZON_DIMENSION_SCAR_CURVATU` with semantic SHA-256 `28391f98e03ec20cc688cc8692579ff0cf352930d085b2d33d979de7ef9a3e2a`. All statements below are finite-model statements; continuum QCD, observed Hawking radiation, physical spacetime emergence, a static local scar Hamiltonian, thermodynamic singularities, and non-Abelian anyons remain outside the evidence boundary.

Gauge-string fractional pump. Tensoring the three one-third-filled CDW branches with the unique finite $SU(3)$ Gauss singlet

$$|\mathcal{M}\rangle = \frac{1}{3}\sum_{a,b=1}^{3} |\bar{a}\rangle|a,\bar{b}\rangle|b\rangle$$

gives a one-dimensional simultaneous left/right Gauss-law kernel in the 81-dimensional endpoint–link–endpoint space. A branch cycle transports one third of a gauge-invariant composite and its attached fundamental–antifundamental flux support. After three cycles one complete singlet composite crosses a cut, but the net transported color charge is zero. In the zero-flux sector the three-cycle Wilson holonomy is the identity; a prescribed $\mathbb{Z}_3$ center flux contributes only $e^{2\pi i k/3}I$. Non-Abelian Wilson mixing therefore requires a degenerate color multiplet or nontrivial plaquette background.

Active Floquet–Bogoliubov horizon. The passive reciprocal no-go is evaded by combining directed walk dispersion

$$\Omega(k) = \frac{2}{\tau}\arcsin\left[v\sin\left(\frac{ka}{2}\right)\right]$$

with a local two-mode squeezer. At a horizon $u(x_H) = c_{\text{eff}}$,

$$\kappa = |\partial_x(u - c_{\text{eff}})|_{x_H}, \qquad T_H = \frac{\kappa}{2\pi},$$

and

$$\tanh r = e^{-\pi\omega/\kappa}, \qquad n_\omega = \sinh^2 r = \frac{1}{e^{2\pi\omega/\kappa} - 1}.$$

A greybody beam splitter of transmission Γ gives $n_{\text{out}} = \Gamma n_\omega$. The full six-channel Nambu map is paraunitary with residual $2.282836126979115 \times 10^{-16}$ in the frozen demonstration. For $\kappa = 0.4$, $\omega = 0.3$, and $\Gamma = 0.7$, $T_H = 0.06366197723675814$, $n_\omega = 0.009064722057396675$, $n_{\text{out}} = 0.006345305440177674$, and the post-greybody logarithmic negativity is 0.15624263179344325. The lattice ultraviolet correction begins at cubic order,

$$\Omega(k) = \frac{va}{\tau}k + \frac{v^3 - v}{24}\frac{a^3}{\tau}k^3 + O(k^5a^5).$$

Dynamically selected spectral dimension. Let $s = \ln b$, use the exact branch count $B = 80$ and exact Levi edge connectivity $\chi = 4$, and define

$$\Phi(s) = \frac{1}{2}[\ln 80 - 4s]^2, \qquad \dot{s} = \gamma[\ln 80 - 4s].$$

Then

$$\dot{\Phi} = -4\gamma[\ln 80 - 4s]^2 \leq 0,$$

so

$$s_* = \frac{\ln 80}{4}, \qquad b_* = 80^{1/4} = 2.9906975624424406$$

is globally stable, with linear exponent -4γ. The selected spectral dimension is

$$d_s = \frac{\ln 80}{\ln b_*} = 4.$$

This is an explicit four-channel information-balance law, not an independent derivation of physical spacetime.

Local scar compilation and static no-go. The three CDW words 100100100, 010010010, and 001001001 have pairwise Hamming distance six. Any operator supported on fewer than six sites therefore has zero direct matrix element between distinct branches. If their bare span is exactly invariant, a strictly finite-range static Hamiltonian cannot realize the nontrivial branch cycle. An exact local Floquet realization nevertheless exists: apply the eight adjacent swaps $(0,1),(1,2),\ldots,(7,8)$ sequentially, each generated for time τ by

$$H_i = \frac{\pi}{2\tau}(I - \mathrm{SWAP}_{i,i+1}).$$

One compiled shift takes $T_{\text{shift}} = 8\tau$, cycles the three CDW branches, and obeys

$$U_{\text{shift}}^3 = I, \qquad T_{\text{rev}} = 24\tau.$$

The Floquet quasienergies are 0 and $\pm 2\pi/(3T_{\text{shift}})$.

Thermodynamic curvature. Assign each dark isotypic block of dimension n_j the splitting charge $q_j = \ln n_j$, while the five bright groups retain their exact contact energies and have $q = 0$. Then

$$Z(\beta, h) = \sum_{j\in\text{dark}} n_j e^{-h\ln n_j} + \sum_{a\in\text{bright}} g_a e^{-\beta E_a}, \qquad g_{ab} = \mathrm{Cov}(T_a, T_b), \quad T = (-E, -q).$$

The two-dimensional Hessian scalar curvature satisfies

$$R(0,0) = -0.6460278596846867,$$

crosses zero at $\beta U = 14.579166757087052$, and reaches $R(20,0) = 0.20080335722980544$. Because Z is a finite positive exponential sum, this is a finite information-geometric crossover rather than a thermodynamic singularity.

Three outside-box consequences. First, the branch shift X and branch twist Z satisfy

$$ZX = \omega XZ, \qquad \omega = e^{2\pi i/3},$$

with residual $2.482534153247273 \times 10^{-16}$, yielding the qutrit Heisenberg–Weyl logical algebra and projective commutator ωI; this is projective holonomy, not anyon braiding. Second, the principal logarithm of X has phases $0, \pm 2\pi/3$, so any generator with $\|H - cI\| \leq \Lambda$ obeys the exact unitary-geodesic limit

$$T \geq \frac{2\pi}{3\Lambda}.$$

Third, at $b_* = 80^{1/4}$ the hierarchy obeys

$$\lambda_{n+m} = \lambda_n b_*^{-2m}, \qquad \omega_{n+m} = \omega_n b_*^{-m}, \qquad 1 + z_m = b_*^m = 80^{m/4}.$$

For $m = 1, 2, 3, 4$, the spectral redshifts are 1.9906975624424406, 7.944271909999156, 25.749612199056866, and 79. This is discrete spectral scaling, not gravitational or cosmological redshift.

58.2 Passes 4121–4128: explicit gauge audit, relational phase coding, decoding, foundry feasibility, attractors, and outside-box probes

The explicit 145-dimensional carrier

$$(\mathbb{C}^7 \otimes V_1) \oplus (\mathbb{C}^6 \otimes V_{15}) \oplus (\mathbb{C}^2 \otimes V_{24})$$

was equipped with concrete $SU(3) \times SU(2) \times U(1)$ generators. This calculation corrects the earlier multiplicity-space reading: the complete carrier has W33 weights $(q, u, d, l, e) = (1, 15, 15, 24, 1)$ and is anomalous,

$$\mathcal{A}_{3^3} = -28, \quad \mathcal{A}_{3^2Y} = -\frac{7}{3}, \quad \mathcal{A}_{2^2Y} = -\frac{23}{4}, \quad \mathcal{A}_{\mathrm{grav}^2Y} = -37, \quad \mathcal{A}_{Y^3} = -\frac{599}{36}.$$

The weak-doublet count is 27. Without exotic representations, the anomaly equations force equal W33 weights $q = u = d = l = e$. Exact repair architectures therefore require either a rank-one PSp-breaking generation inside the 145 states, a 225-dimensional V_{15} generation, a 360-dimensional V_{24} generation, or 600 dimensions containing one complete generation in each of V_1, V_{15}, V_{24}. Gauge symmetry permits three Yukawa tensors, but a W33-singlet Higgs permits none; the up, down, and lepton couplings require mediators in V_{15}, V_{15}, V_{24}.

For the optimal sine phase reference, a single fixed-weight subspace is an exact decoherence-free space but supports only a global phase. Pairing $\mathcal{H}_r$ and $\mathcal{H}_{r+1}$ for $N = 2r + 1$ instead gives

$$\mathbb{C}^2_{\mathrm{clock}} \otimes \mathbb{C}^{\binom{N}{r}}_{\mathrm{payload}}.$$

The reference noise and desired phase act only on the clock; all payload coherences are exact. At $K = 256$ and $N = 31$, one clock plus 28 payload qubits has error $7.3646628264 \times 10^{-5}$, versus $5.5087125884 \times 10^{-2}$ for direct 29-qubit injection.

The three-error self-stress decoder was reduced from 682641 global supports to at most

$$1 + 20 + \binom{20}{2} + \binom{20}{3} = 1351$$

local candidates. The candidate graph is formed from all Levi paths of length at most three between threshold vertices; an exhaustive audit proves the 20-edge bound. Stability remains controlled by

$$\sigma_6 = 2\sin(\pi/14), \qquad \|\widehat{e} - e\|_2 \leq 2\epsilon/\sigma_6.$$

A CORNERSTONE 340 nm SOI floorplan audit gives 1934 crossings, at most 21 on one path, 480 selector/recombiner MMIs, and 480 phase shifters for a spatial 80-mode realization. Under conservative public platform inputs, five signal uses incur a conditional pre-MMI path loss of 27.43 dB. Thus the exact logical router is not tapeout-ready in naive spatial SOI; low-loss passive routing with an active phase layer, or time/frequency serialization, is required.

A 64-seed nonlinear census produced 25 observed orbit classes for the nominal 24-mode selector and 9 for the nominal 15-mode selector. Pure bivalent stabilizers include $V_4, S_3, D_8, C_6 \times C_2, D_{10}, S_5$, and A_5, while mixed-sector attractors show that cubic saturation does not globally remain in one Laplacian eigenspace.

Three outside-box consequences were certified. First, the A_5 sign vector satisfies $As = -4s$ and attains the exact MaxCut value 160, hence the antiferromagnetic ground energy is $-80J$. Second, the continuous-time walk at $t = \pi/(2J)$ is the exact Householder mirror

$$U = I - 2P_2 = \frac{A^2 - 8A - 18I}{30} = -\frac{I + A}{3} + \frac{2}{15} J_{40},$$

which flips precisely the 24-dimensional eigenvalue-two sector. Third, pairing the 15 eigenvalue-minus-four modes with 15 eigenvalue-two modes yields fifteen simultaneous EP_2 blocks at $\gamma = g$; the paired 30-dimensional operator has square zero and rank 15.

All statements above are finite algebraic, combinatorial, or explicitly labelled numerical results. No derived Standard Model, fabricated chip, measured decoder or pattern, physical magnet, graph-walk processor, exceptional-point sensor, gravity, cosmology, or theory of everything is claimed.

58.3 Passes 4129–4136: anomaly optimization, relational gates, seven-fault decoding, hybrid routing, and exact pure-orbit catalogues

Let the signed W33 multiplicities of Q, u^c, d^c, L, e^c be (q, u, d, ℓ, e). The independent integer anomaly map is

$$M = \begin{pmatrix} 2 & -1 & -1 & 0 & 0 \\ 1 & -2 & 1 & 0 & 0 \\ 1 & 0 & 0 & -1 & 0 \\ 1 & -2 & 1 & -1 & 1 \end{pmatrix},$$

with kernel $\mathbb{Z}(1,1,1,1,1)$ and Smith diagonal $(1,1,1,3)$. For the anomalous $(1,15,15,24,1)$ carrier, vectorlike pairs alone cannot repair the anomalies. The minimum standard-charge signed repair adds $14Q + 14e^c + 9\bar{L}$, giving dimension 261 but breaking the W33 module assignment. Exact representation-ring optimization gives total dimensions $291, 335, 554, 600$ for common target modules $V_1, V_{15}, V_{24}, V_1 \oplus V_{15} \oplus V_{24}$ respectively; 335 is the smallest repair with nontrivial $PSp(4,3)$ action on the net chiral generation.

For $N = 31$, $r = 15$, adding one reservoir mode gives the fixed-number code

$$|0, S\rangle_L = |S\rangle|1\rangle_R, \qquad |1, S\rangle_L = |\bar{S}\rangle|0\rangle_R,$$

with total number 16 and dimension $2\binom{31}{15} = 601080390$. The clock operators $Z_c = \Pi_{15} - \Pi_{16}$, the paired complement X_c, $H_c = (X_c + Z_c)/\sqrt{2}$, paired payload unitaries, and clock-controlled payload unitaries all preserve total number. The reference channel factors as $\mathcal{E}^{\mathrm{clock}}_{z_1} \otimes I_{\mathrm{payload}}$; at $K = 256$, $|z_1| = 0.9999263606710866$, corresponding after known-phase correction to dephasing probability $3.6819664457 \times 10^{-5}$.

The Levi incidence syndrome has spark eight, so four-error uniqueness is impossible: opposite halves of an apartment cycle collide. The first three generalized cycle-support weights are $8, 12, 16$. Adding

$$m_3(x) = \sum_{e=1}^{160} e^3 x_e, \qquad m_5(x) = \sum_{e=1}^{160} e^5 x_e$$

raises the stacked spark to at least fifteen. This follows from exhaustive audits of 386964 simple cycles through length fourteen and 133920 rank-two theta cores. Hence every seven-sparse error is uniquely identifiable in the noiseless model.

The four router permutations decompose into $8, 8, 9, 8$ planar increasing-subsequence layers. A hybrid architecture with passive SiN permutation banks and a shared active SOI selector/phase layer uses 160 passive routes, six selector/recombiner MZIs, 24 common phase plates, 30 active controls, and no same-plane crossings. Under the stated design targets its provisional five-pass internal budget is 9.12 dB and its all-π heater envelope is 0.48 W. This is an architecture contract, not a fabricated device.

The pure $\lambda = 16$ binary system $(A + 4I)x = 8\mathbf{1}$ has exactly 432 signs, or 216 maximum cuts modulo reversal, in one A_5-stabilized orbit. The pure $\lambda = 10$ system $(A - 2I)x = 5\mathbf{1}$ has exactly 33264 signs in seven $PSp(4, 3) \times \{\pm 1\}$ orbit classes of sizes

$$6480, 8640, 6480, 6480, 2160, 2592, 432.$$

Thus all pure bivalent nonlinear branches and all maximum cuts are catalogued; mixed-amplitude attractors remain open.

Three further exact constructions are recorded. First, the anomaly cokernel contains genuine $\mathbb{Z}/3\mathbb{Z}$ torsion, yielding $a_{SU(3)^3} + a_{6SU(3)^2U(1)} \equiv 0 \pmod 3$. Second, the distributed projectors

$$P_{12} = J/40, \quad P_2 = \tfrac{2}{3}I + \tfrac{1}{6}A - \tfrac{1}{15}J, \quad P_{-4} = \tfrac{1}{3}I - \tfrac{1}{6}A + \tfrac{1}{24}J$$

provide exact two-round network tomography. Third,

$$G_8 = e^{-i\pi L/8}, \qquad G_5 = e^{-i\pi L/5}$$

have orders eight and five and generate $\mathbb{Z}_{40}$ on the nonuniform spectral sectors. This is a finite reversible spectral clock, not emergent physical time.

59 Passes 4137–4144: matrix Wilson pumping, extended horizons, microscopic scale flow, and autonomous history dynamics

The exact certificate for this packet is `d16f8ab7ce51f2953cfd4866e7a716e31a271b3337a05b6105b367736bcecb6c`
All statements in this section concern finite group, Gaussian, Markov, clock, transfer-matrix, circuit, or band-topology models.

Matrix-valued gauge-singlet holonomy. The color space $(\mathbf{3} \otimes \bar{\mathbf{3}})^{\otimes 3}$ contains six independent contraction singlets, with Gram spectrum

$$\left\{ \frac{2}{9}, \left(\frac{8}{9}\right)^4, \frac{20}{9} \right\}.$$

Three orthonormal singlet channels support a tripod Hamiltonian whose rank-two dark bundle has two exact Wilson loops

$$U_y = \begin{pmatrix} -\frac{1}{2} & -\frac{\sqrt{3}}{2} \\ \frac{\sqrt{3}}{2} & -\frac{1}{2} \end{pmatrix}, \qquad U_z = \operatorname{diag}(e^{-2\pi i/3}, e^{2\pi i/3}).$$

Their group commutator has trace $-1/4$, so the adiabatic Wilson response is genuinely matrix-valued rather than a center phase.

Extended active horizon lattice. Nine localized squeezing cells act on one outgoing, nine partner, and nine environment modes. The resulting 38×38 Nambu map has paraunitary residual 1.11×10^{-15}. At $\omega = 0.3$ the outgoing occupation is 0.01195894, the greybody ratio is 0.48573560, and the outgoing/partner logarithmic negativity is 0.20853530. The partner profile is centered at cell 4.0048 with RMS width 1.1977 cells. The inside-flow lattice dispersion has two ultraviolet roots with opposite group velocities, giving explicit mode conversion without promoting the finite analogue model to gravitational Hawking radiation.

Microscopic channel-balance flow. Four independent local Markov channels sum to

$$ds = \gamma(\ln 80 - 4s)dt + \sqrt{2D}\, dW.$$

The stationary law is Gaussian around $s_* = \ln 80/4$, the Fokker–Planck gap is 4γ, and the deterministic spectral dimension is exactly

$$d_s = \frac{\ln 80}{s_*} = 4.$$

For $D = 0.01$ and $\gamma = 1$, the mean noisy value is 4.00838502 and relative entropy to stationarity decreases monotonically.

Autonomous history-state scar gadget. Three repetitions of the eight adjacent-SWAP branch compiler define a 24-gate program. The static engineered history Hamiltonian

$$H = \Omega \sum_{t=0}^{23} \sqrt{(t+1)(24-t)}\, (|t+1\rangle\langle t| \otimes U_{t+1} + \text{h.c.})$$

has a 25-state clock, an equally spaced spectrum from -24Ω to 24Ω, perfect endpoint transfer at $\pi/(2\Omega)$, and full history revival at π/Ω. The endpoint data operation is $U_{\text{shift}}^3 = I$.

Curvature scaling. For N independent dark-reservoir cells, $Z_N = Z_1^N$, $g_N = N g_1$, and the scalar curvature satisfies

$$R_N = \frac{R_1}{N}.$$

The sign change remains at $\beta U = 14.5791667571$, but its magnitude vanishes in the thermodynamic limit. A positive 2×2 transfer matrix for a weakly coupled one-dimensional dark/bright hierarchy is analytic for finite coupling and has correlation length 0.14340675 cells in the frozen demonstration, excluding a finite-temperature singularity in this class.

Outside-box probes. Ordering $U_y U_z$ versus $U_z U_y$ yields output fidelity 7/16, furnishing a geometric phase-order transistor. Levi edge connectivity four and the $2 \ln 3$ entropy-change bound per two-qutrit cut gate imply a throughput ceiling of $8 \ln 3$ nats per layer and at least 74 layers for 80 maximally mixed dark cells. Finally, a spin-one Qi–Wu–Zhang control Hamiltonian on three singlet channels has band Chern numbers $(-2, 0, 2)$ and realizes a charge-two synthetic control-space monopole.

Boundary. No continuum QCD, fabricated non-Abelian pump, observed Hawking radiation, derivation of physical spacetime, thermodynamic phase transition, non-Abelian anyon, physical monopole, gravity, cosmology, or theory of everything follows from these finite exact results.

59.1 Bounded exotic repair, local relational gates, conditioned seven-fault sensing, and exact finite probes

Within the bounded family $R_3 \in \{1, 3, \bar{3}\}$, $\dim R_2 \in \{1, 2\}$, and $Y = n/6$ with $0 < |n| \leq 6$, exact mixed-integer optimization repairs the explicit 145-state carrier with 107 added states. The resulting 252-state signed spectrum is optimal in that family and uses an odd number of added weak doublets.

The compressed relational clock has a sharp one-shot locality obstruction: paired words differ on all 32 occupation modes. A 58-mode dual-rail lift instead carries one clock and 28 payload qubits at fixed total particle number 29. Logical H, S, CNOT, clock exchange, and clock-controlled payload operations are generated by Hamiltonians supported on at most three modes, while the shared reference acts only on the clock.

Two centered integer moment rows obtained modulo 24001 have maximum magnitude 11992. Exhaustive audits of all 386964 simple Levi cycles of length at most fourteen and all 133920 relevant theta cores prove stacked spark at least fifteen and exact noiseless recovery of every seven-sparse edge error. The coefficient envelope is reduced by more than 8.7×10^6 relative to the e^5 row.

The four exact router permutations decompose into $8, 8, 9, 8$ noncrossing layers. The 33-layer SiN schedule uses zero same-plane crossings and 30 active controls, but static nine-slot delay equalization requires 1280 route-slot units. At a 5 ps slot and group index two, the added five-use delay loss is 1.5 dB; sub-1 dB delay loss requires slots below 3.34 ps.

Full Newton refinement gives 18 stable mixed classes for the nominal 24-sector selector in the 64-seed corpus. In an enlarged 200-seed corpus, the nominal 15-sector selector has four stable mixed roots and one weak index-one saddle; the earlier finite-time count of eight mixed classes collapses to five actual roots. The two nonzero root-stabilizer quotient branches are saddles of Morse indices 23 and 14.

Three further finite constructions are exact. The eight-regular bipartite maximum-cut graph has 56260624960 perfect matchings, but a one-bit feedback engine obeys the exact Landauer balance $p\Delta - kTh(p) = -kT \ln(1 + e^{-\beta\Delta}) < 0$. The signed Levi incidence supercharge has 82 zero modes and Witten index $1 - 81 = -80 = V - E$. Finally, $H_- = 12I - A$ and $H_+ = 4I + A$ satisfy $H_- + H_+ = 16I$, so $e^{-i\tau H_+} e^{-i\tau H_-} = e^{-i16\tau} I$, with an exact identity echo at $\tau = \pi/8$.

Evidence boundary. These are finite algebraic, combinatorial, optimization, modular-audit, and Newton-refined corpus results. They do not establish a derived Standard Model, a fabricated photonic stack, a globally complete mixed-attractor theorem, a perpetual-motion engine, physical supersymmetry, a measured time crystal, gravity, cosmology, or a theory of everything.

60 Passes 4153–4160: second-Chern pumping, disorder, Lindblad scale flow, and graph thermodynamics

Second-Chern singlet pump. Let Γ_a be five Hermitian 4×4 Clifford generators with $\{\Gamma_a, \Gamma_b\} = 2\delta_{ab}$ and $\Gamma_1\Gamma_2\Gamma_3\Gamma_4\Gamma_5 = -I_4$. For $H(n) = \sum_a n_a \Gamma_a$ on S^4, the negative-energy projector $P = (I - H)/2$ has rank two and

$$\mathrm{Tr}[P(dP)^4] = \frac{1}{8} \epsilon_{abcde} n_a \, dn_b \wedge dn_c \wedge dn_d \wedge dn_e.$$

Consequently

$$C_2 = \frac{1}{8\pi^2} \int_{S^4} \mathrm{Tr}[P(dP)^4] = 1, \qquad C_1 = 0.$$

This is an exact Yang-monopole bundle embedded in the finite singlet-contraction space; it is not a fabricated four-dimensional pump.

Disordered Hawking chain. All 512 binary patterns $r_j \mapsto r_j(1 + 0.5s_j)$ and $\Gamma_j \mapsto \Gamma_j + 0.05s_j$ were audited in the nine-cell, 38-dimensional Nambu chain. The minimum logarithmic negativity is 0.0920180870833, the largest minimum partially-transposed symplectic eigenvalue is $0.456044326186 < 1/2$, and the maximum paraunitary residual is 1.82×10^{-15}. Every audited realization remains entangled.

Lindblad scale oscillator. For $s_* = \ln(80)/4$, define

$$L_- = \sqrt{8\gamma(\bar{n}+1)}\, b, \qquad L_+ = \sqrt{8\gamma\bar{n}}\, b^\dagger, \qquad s = s_* + \sigma(b + b^\dagger),$$

with $\sigma^2 = D/[4\gamma(2\bar{n}+1)]$. Then

$$\frac{d\langle s\rangle}{dt} = -4\gamma(\langle s\rangle - s_*), \qquad \operatorname{gap}(\mathcal{L}) = 4\gamma, \qquad d_s = \frac{\ln 80}{s_*} = 4.$$

The stationary state is thermal and unique. This is an explicit open-system scale model, not a spacetime derivation.

Compressed autonomous clock. The 25 legal history states require at least five qubits. The reflected Gray encoding $g_t = t \oplus (t \gg 1)$ reaches this lower bound and has unit Hamming distance between successive legal words. The engineered history Hamiltonian retains spectrum $-24\Omega, -22\Omega, \ldots, 24\Omega$, perfect transfer at $\pi/(2\Omega)$, and full revival at π/Ω. The exact tradeoff is five-local clock conditioning and seven-local clock–SWAP propagation terms.

Levi-cover thermodynamics. For dark/bright degeneracies $g_D = 3161$ and $g_B = 160$, the entropic coexistence field is

$$H_c(T) = -\frac{T}{2} \ln \frac{3161}{160}.$$

The six-regular Levi universal cover has nonbacktracking branching five, so the cavity instability satisfies

$$5\tanh(\beta_c J) = 1, \qquad \beta_c J = \operatorname{atanh}(1/5) = 0.202732554054.$$

The finite 160-vertex graph remains analytic, whereas large-girth six-regular covers possess a genuine Bethe ordering singularity on the coexistence line.

Outside-box probes. Repeated partner-vacuum projections after M fractional squeezing steps give $P_M = \operatorname{sech}^{2M}(r/M)$ and suppress conditional pair production as r^2/M; this is a Zeno effect, not a firewall. Pauli-twirl purity spectroscopy distinguishes the active rank-two bundle from the full six-dimensional singlet space. Finally, a four-state clock ring with π loop flux has spectrum $\pm\sqrt{2}\Omega$ and exact projective echo $U(\pi/(\sqrt{2}\Omega)) = -I_4$.

Evidence boundary. The frozen semantic certificate is `0e9080801a2cfcca3b7b39afd807835d7bdc6b1a483cd68`
All statements are finite Clifford, Gaussian, Lindblad, clock, Bethe, measurement, purity, or flux-ring results. No observed Hawking radiation, physical firewall, anyonic quantum dimension, spacetime torsion, gravity, cosmology, or theory of everything is established.

60.1 Passes 4161–4168: broader anomaly repair, local hardware, noisy decoding, storage, global basin push, and three exact probes

The bounded representation search was enlarged to $SU(3)$ irreps $1, 3, \bar{3}, 6, \bar{6}, 8$, weak dimensions $1, 2, 3$, and $Y = n/6$ with $|n| \leq 6$. Exact mixed-integer optimization lowers the anomaly-free carrier from 252 to **213 states** inside this finite family, while a second exact optimization proves an added color Dynkin load of at least 11, giving a one-loop $SU(3)$ asymptotic-freedom obstruction for this embedding. The 58-mode relational lift admits a 139-coupler hardware graph of maximum degree seven, arbitrary payload CNOT depth at most 15, and 28-qubit GHZ depth five. For lattice-amplitude errors, the two conditioned integer moment rows give deterministic seven-fault correction under componentwise noise $< \Delta/2$ after rounding; a modulo-24001 accumulator needs 15 residue bits. A low-loss tapped delay bank reduces the five-use storage propagation comparison to < 0.15 dB, while an unreshaped 5-ps pulse obeys a sharp single-pole resonator time–bandwidth no-go. A macroscopic 64-seed selector-24 corpus finds 24 mixed classes, including 20 absent from the perturbative corpus, 19 of them stable.

Three independent exact probes accompany the implementation results. The W33 uniform projector is $P_0 = (A^2 + 2A - 8I)/160$, so the Grover diffusion reflection is a quadratic adjacency polynomial and four oracle calls reach success 3920137321/4000000000. The Laplacian pseudoinverse is $L^+ = (7/80)I + (1/160)A - (13/3200)J$, giving resistance distances 13/80 and 7/40 and Kirchhoff index 267/2. Finally, every diameter-four Levi interval has layer profile $(1, 4, 4, 4, 1)$ and exactly four geodesics, hence finite geodesic entropy $\log 4 = \log|\text{waist}|$. These are finite graph/control statements only: they do not establish physical quantum speedup, inertia, spacetime entropy, gravity, cosmology, or a theory of everything. The frozen semantic certificate is `db91cd6c70138d44917ba4274a7d087927caec691be5d49efd83528e7f8d4bd5`.

61 Passes 4169–4176: discrete C_2, Hawking disorder, backreaction, Gray locality, Levi-cover correction, Casimir and axion reduction

Pass 4169: discrete second-Chern compiler. For the four-dimensional Wilson–Dirac control

$$d(k) = (\sin k_1, \sin k_2, \sin k_3, \sin k_4, -3 + \textstyle\sum_i \cos k_i), \qquad H = \widehat{d} \cdot \Gamma,$$

the sixteen high-symmetry masses group as $1, -1, -3, -5, -7$ with multiplicities $1, 4, 6, 4, 1$. The oriented Dirac count gives

$$C_2 = \frac{1}{2} \sum_{N=0}^{4} \binom{4}{N} (-1)^N \operatorname{sgn}(-3 + 4 - 2N) = 1.$$

Thus sixteen calibration points certify the topological sector. A uniform central-difference response integral converges much more slowly: among the audited meshes, $N = 29$ per control axis is the first below five-percent error, with $C_2^{(29)} = 0.9504670029565596$ on $29^4 = 707281$ nodes.

Pass 4170: continuous Hawking disorder. For 512 correlated-Gaussian samples with $C_{ij} = e^{-|i-j|/2}$, $r_j \mapsto r_j e^{0.35 x_j}$, and $\Gamma_j \mapsto \text{logistic}(\operatorname{logit} \Gamma_j + 0.25 x_j)$, the logarithmic negativity remains in

$$0.0822919174978226 \leq E_N \leq 0.5298852171115666.$$

A 721-phase quasiperiodic scan with $\alpha = (\sqrt{5}-1)/2$ and amplitude 0.7 gives $0.24211601793444767 \le E_N \le 0.27238456666720445$. With thermal occupation n_{th}, the PPT mobility edge is defined by $\widetilde{\nu}_-(\omega_c, n_{\text{th}}) = 1/2$; for $n_{\text{th}} = 0.01$, $\omega_c = 0.5783917254706858$, while at $\omega = 0.3$ the thermal threshold is $n_{\text{th}} = 0.11593620709206182$.

Pass 4171: scale–geometry backreaction. For the minimal occupied-mode feedback

$$\dot{s} = \gamma \left[\ln 80 - 4s + g\, n_B(2e^{-s}) \right],$$

$g = 1$ has a stable root $s = 1.5949469860997991$ and an unstable root $s = 2.5062250949365414$. They annihilate at the saddle node

$$(s_c, g_c) = (1.9679175210441553, 1.1252773999853873).$$

Thus this minimal positive backreaction shifts the spectral dimension away from four and ultimately destroys the stationary scale; it does not create two stable geometric branches without an additional saturation mechanism.

Pass 4172: exact locality reduction of the Gray clock. A five-qubit reflected-Gray transition flips one clock bit conditioned on four spectator literals and accompanies a two-site data SWAP, hence is directly seven-local. Three binary AND ancillas per transition encode the four spectators using the exact two-local penalty

$$P_{\text{AND}}(x, y, z) = xy - 2xz - 2yz + 3z,$$

reducing each propagation term to the four-local form $f_t X_{c_t}\text{SWAP}$. For twenty-four transitions this requires seventy-two static ancillas. The legal clock block is unchanged exactly, preserving spacing 2Ω, perfect transfer at $\pi/(2\Omega)$, and full revival at π/Ω.

Pass 4173: Levi-cover correction and obstruction. The committed $W(3, 3)$ geometry has forty points, forty lines, and $40(3 + 1) = 160$ flags. Hence its point–line Levi graph has eighty vertices, degree four, 160 edges, and girth eight. This supersedes the degree-six premise used in Pass 4157. The graph contains exactly 1620 simple eight-cycles. Requiring every one to acquire odd sign parity in a signed two-lift produces a 1620×160 binary system whose coefficient rank is 81 and augmented rank is 82, so the system is inconsistent. Therefore no signed two-lift can double all base eight-cycles. The correct degree-four Bethe comparison has

$$3 \tanh(\beta_c J) = 1, \qquad \beta_c J = \operatorname{atanh}(1/3) = \frac{\ln 2}{2} = 0.34657359027997264.$$

Simple iterated two-lifts remain girth eight and therefore do not provide the previously assumed large-girth sequence.

Pass 4174: Hawking erasure-channel threshold. Interpreting the greybody stage as an effective pure-loss channel with $\eta_{\text{eff}} = n_{\text{out}}/n_{\text{out}}^{\text{lossless}}$, the baseline value is $0.485735597242 < 1/2$, hence the unassisted pure-loss quantum capacity is zero. Across the 512 binary disorder patterns, η_{eff} spans 0.3109129598611626 to 0.7282862738005136: 251 patterns lie in the antidegradable half and 261 above the capacity threshold. The largest effective capacity in that ensemble is 1.4224182063347266 bits per use.

Pass 4175: exact graph-spectral Casimir attraction. For the massive Levi operator $K = L + I = 5I - A$, $G = K^{-1}$ has exact distance data

$$G_{uu} = \frac{211}{855}, \quad G_1 = \frac{10}{171}, \quad G_2 = \frac{13}{855}, \quad G_3 = \frac{1}{171}, \quad G_4 = \frac{4}{855}.$$

Two unit-strength pinning defects therefore interact through

$$E_{\text{int}}(u, v) = \frac{1}{2} \ln \left[1 - \frac{G_{uv}^2}{(1 + G_{uu})(1 + G_{vv})} \right] < 0,$$

with values $-1.1012191859481348 \times 10^{-3}$, $-7.436602973476341 \times 10^{-5}$, $-1.1000194923920593 \times 10^{-5}$, and $-7.04009687184613 \times 10^{-6}$ at graph distances one through four.

Pass 4176: dimensional reduction. For the parent bundle with $C_2 = 1$, the three-dimensional Chern–Simons descendant satisfies

$$\Delta\theta = 2\pi C_2 = 2\pi.$$

The closed bulk returns to the same θ mod 2π sector, while a gapped boundary Chern–Simons level shifts by one, corresponding in electronic normalization to one e^2/h Hall-sheet unit.

Evidence boundary. These are finite Wilson–Dirac, Gaussian, one-mode backreaction, clock-gadget, graph-cover, pure-loss-channel, Green-function determinant, and dimensional-reduction statements. In particular, Pass 4173 is an explicit erratum to the degree-six assumption in Pass 4157 for the $W(3, 3)$ point–line Levi graph. No fabricated four-dimensional pump, observed Hawking radiation, physical spacetime backreaction, hardware-optimal clock, experimental criticality, black-hole communication result, measured Casimir force, axion detection, gravity, cosmology, or theory of everything is established.

Passes 4177–4184: fixed-carrier AF obstruction, two-mode relational compilation, Hodge decoding, PDK delay materialization, and three exact probes

The fixed 145-state carrier has SU(3) Dynkin load 16 and nonzero cubic color anomaly, so every add-only anomaly repair must add colored matter. Since every nontrivial finite-dimensional SU(3) irrep has Dynkin index at least 1/2, strict one-loop $b_0^{SU(3)} > 0$ is impossible; even the marginal $b_0 = 0$ budget cannot cancel the cubic anomaly magnitude 28. The 58-mode relational architecture admits a strict two-mode implementation with pairwise exchange plus cross-Kerr interactions on an 84-coupler, degree-four graph. For the Levi incidence matrix D, augmenting by an orthonormal basis Z of $\ker D$ gives $M = [D; Z]$ with $\sigma_{\min}(M) = 1$ and condition number $2\sqrt{2}$, yielding a globally stable arbitrary-real-amplitude decoder. The exact noncrossing delay schedule needs 919 rather than 1280 slot-units, corresponding to 0.688773172255 m aggregate passive delay at 5 ps slots and assumed group index two; current public LIGENTEC TFLN/SiN loss data imply a longest-path five-use propagation contribution below 0.015 dB, before tap/coupler and DRC effects. On the nonlinear side, interval/Krawczyk analysis proves exactly three equilibria in each selector's $1 + 12 + 27$ point-stabilizer quotient, while edge and nonedge stabilizer strata are now explicitly mapped for subsequent exhaustive work. Outside-box constructions give an exact three-channel unitary scattering polynomial, the rooted-forest partition $\det(I + tL) = (1 + 10t)^{24}(1 + 16t)^{15}$, and the signed sector lens $K = P_2 - P_{-4}$ with $K^2 = I - J/40$. No fabricated hardware, global nonlinear completeness theorem, Standard Model derivation, gravity, cosmology, or theory of everything is claimed.

Passes 4185–4192: adaptive C_2, Hawking criticality, hysteresis, exact three-local clocks, and high-girth covers

Adaptive second-Chern compilation. For the Wilson–Dirac map $d = (\sin k_1, \dots, \sin k_4, -3 + \sum_i \cos k_i)$, the sixteen high-symmetry corners still give the exact integer certificate $C_2 = 1$. Phase-centered midpoint cubature reaches the correct rounded integer already at $7^4 = 2401$ samples and falls below five-percent response error at $9^4 = 6561$ samples, a $107.8008\times$ reduction from the prior 707281-node response mesh. This is a deterministic response-compiler improvement, not a proof of global sample minimality.

Hawking critical surfaces. For a symmetric two-mode squeezed thermal core with $r(\omega) = \operatorname{atanh}(e^{-\pi\omega/\kappa})$ and one-sided attenuation η, the finite Gaussian thresholds are

$$n_{\mathrm{ent}} = \frac{e^{2r} - 1}{2}, \qquad n_{B\to A} = \sinh^2 r,$$

$$n_{A\to B} = (2\eta - 1)\sinh^2 r \quad (\eta > 1/2), \qquad Q_{\text{pure loss}} = \max\left\{0, \log_2 \frac{\eta}{1-\eta}\right\}.$$

At $\omega = 0.3$ one has $r = 0.09506557725$, $n_{\mathrm{ent}} = 0.1047041033$, and $n_{B\to A} = 0.009064722057$. The analytic core is tied to the previously frozen correlated-disorder envelope but is not promoted to an arbitrary continuous five-dimensional channel theorem.

Saturating backreaction and hysteresis. The nonlinear scale flow

$$\dot{s} = \gamma\left[\ln 80 - 4s + g n_B(2e^{-s}) - 0.004\, n_B(2e^{-s})^2\right]$$

has a three-fixed-point window

$$0.4815579549 < g < 1.1377609143.$$

At $g = 0.5$, the stable branches are $s = 1.2568469367$ and $s = 4.9471634287$, separated by the unstable branch $s = 4.2806819287$. This supplies a finite scale-memory hysteresis model, not physical spacetime hysteresis.

Exact three-local autonomous clock. Each of the 24 Gray-clock transitions is split into a conditioned clock-update/transition-flag substep and a transition-flag-controlled two-site SWAP substep. With 72 static AND ancillas and 24 one-hot transition ancillas, every propagation and legality term is at most three-local. The resulting 49-state legal history chain has spectrum

$$-48\Omega, -46\Omega, \dots, +48\Omega,$$

legal gap 2Ω, perfect transfer time $\pi/(2\Omega)$, and full revival time π/Ω.

High-girth Levi covers. The corrected degree-four $W(3,3)$ point–line Levi graph has cycle counts $N_8 = 1620$, $N_{10} = 5184$, and $N_{12} = 43200$, and free fundamental-group rank 81. A deterministic $\mathbb{Z}_{65537}$ voltage assignment has nonzero voltage on every audited simple cycle of lengths 8, 10, and 12, producing a 5,242,960-vertex cyclic cover with certified girth at least 14. Since the base fundamental group is the residually finite free group F_{81}, finite regular covers with girth tending to infinity exist abstractly.

Three outside-box closures. The two noncommuting order-three Wilczek–Zee loops from Pass 4137 have commutator trace $-1/4$, hence eigenphase cosine $-1/8$ and infinite commutator order; closed-subgroup classification then gives dense closure $SU(2)$, so they form a universal single-qubit holonomic gate set. The corrected Levi graph is bipartite Ramanujan and has exact Ihara factorization

$$\zeta^{-1}(u) = (1-u^2)^{81}(1-9u^2)(1+9u^4)^{24}(1+3u^2)^{30}.$$

Finally its exact heat trace gives a maximum diffusion spectral dimension only

$$d_s^{\max} = 3.47143073390 < 4,$$

so bare Levi diffusion does not reproduce the separate channel-balance $d_s = 4$ model.

Boundary. These are finite Wilson–Dirac, Gaussian, nonlinear mean-field, clock-gadget, graph-cover, holonomy-group, Ihara-zeta, and heat-kernel statements only. No fabricated four-dimensional pump, observed Hawking radiation, physical spacetime hysteresis, laboratory three-local processor, fabricated high-girth network, experimental holonomic universality, gravity, cosmology, or theory of everything is established.

62 Passes 4205–4212: carrier surgery, native dual rail, compressed Hodge sensing, and interval strata

The fixed 145-state carrier cannot retain its original full $PSp(4,3)$-commuting standard-charge embedding while becoming a nonzero anomaly-free generation sector. If charge labels are allowed to be surgically reassigned, the largest complete-generation sector keeping both nonabelian Weyl-matter one-loop coefficients positive has five generations: 75 gauge-charged states and 70 singlets, with $b_0^{SU(3)} = 13/3$ and $b_0^{SU(2)} = 2/3$. On the hardware side, the 58 occupation modes map to 29 logical dual-rail qubits, 87 transmons including one erasure ancilla per logical qubit, and 28 inter-logical tunable couplers on a degree-three logical tree; this is a topology map, not a fabricated 28-payload-qubit processor.

The 81-dimensional Levi harmonic readout can be compressed to ten deterministic whitened cycle channels in the audited nested family while retaining nonzero cycle/theta gains through 14-edge nullspaces; a conservative restricted singular lower bound for the resulting 90-row stacked sensor is 0.0138377. The populated photonic delay schedule contains 919 slot-units and 0.688773 m of aggregate passive delay. The point-, edge-, and nonedge-pair stabilizer fixed spaces are now interval-audited: edge fixed spaces contain five and three roots for selectors 24 and 15, while nonedge fixed spaces contain thirteen and nine.

Three finite exact probes accompany the implementation work. Critical coupling to the rank-24 +2 adjacency sector produces $S = I - P_2$, absorbing 3/5 of a vertex input. Any nonzero vector supported on s vertices and contained in a central W33 spectral subspace of rank d obeys $sd \geq 40$. Finally, the Levi harmonic quadratic form $E_H(x) = \frac{1}{2}x^T P_H x$ is invariant under $x \mapsto x + D^T\phi$ and has 81 independent cycle modes; the "battery" terminology is only a graph-Hodge analogy.

Passes 4214–4221: smaller covers, two-bundle holonomy, hysteretic memory, compressed clocks, and full Gaussian horizon diagnostics. The corrected degree-four $W(3,3)$ point–line Levi graph admits an explicit cyclic $\mathbb{Z}_{359}$ voltage cover in which every simple base cycle of lengths $8, 10, 12$ has nonzero voltage. Hence the cover has 28,720 vertices, 57,440 edges, and certified girth at least 14, a factor $65537/359 = 182.5543\ldots$ smaller in sheet count than the earlier construction. No global minimality is claimed.

On two rank-two singlet bundles, the dense local $SU(2)$ holonomies of Pass 4190 together with one auxiliary bright level produce a geometric controlled phase: a solid-angle 2π loop in $\mathrm{span}\{|11\rangle, |a\rangle\}$ gives Berry phase π and hence $\mathrm{CZ} = \mathrm{diag}(1, 1, 1, -1)$. The Pauli commutator closure has dimension 15, so dense local $SU(2) \times SU(2)$ plus this entangler is dense in $SU(4)$.

For the saturated scale flow $\dot{s} = \gamma[\ln 80 - 4s + gn_B(2e^{-s}) - 0.004n_B(2e^{-s})^2]$, equal-well coexistence occurs at $g = 0.5700939873$. The minima $s_L = 1.2870434725$ and $s_R = 5.3097911844$ are separated by $s_B = 3.8542873087$ with barrier 4.4597412117; the dimensionless $M = 1$ instanton action is 7.7881701215. This supports a two-state quantum-memory reduction but not a physical spacetime-memory claim.

An exact MILP shares the Gray-clock spectator logic: 13 pair conjunctions, 24 final conditions, and 24 transition flags require 61 auxiliaries instead of 96, retaining exact three-local propagation, legal gap 2Ω, perfect transfer at $\pi/(2\Omega)$, and full revival at π/Ω.

For the complete 19-mode Gaussian horizon chain, uniform thermal loading removes outside-to-partner steering at $n_{\mathrm{th}} = 0.0042826995$, positive coherent information at 0.0079250582, reverse steering at 0.0117448238, and PPT entanglement only at 0.1159362071. In the frozen 512-sample correlated-disorder ensemble at $n_{\mathrm{th}} = 0.005$, all 512 remain entangled while the stronger witnesses survive in 228, 415, and 482 samples respectively.

Three independent probes sharpen the finite boundary. The Levi spectrum $\{0, \pm 4, \pm\sqrt{6}\}$ forbids exact nonzero global CTQW revival, but the Pell solution $11760^2 - 6(4801)^2 = -6$ gives an operator near-echo error 4.0071334×10^{-4}. For $K = 5I - A_{\mathrm{Levi}}$, point–line harmonic-vacuum entropy is 2.4552973872 nats with logarithmic negativity 6.9804187391 nats. Finally, the degree-four tree radial band $E(q) = 2\sqrt{3}J \cos q$ gives $v_{\max} = 2\sqrt{3}J$ while shell capacity grows as 3^r; this graph velocity is not identified with physical c.

Renumbering and boundary. Glue-track commit 84db6db5c reserved Pass 4213 before the original physics-block reservation; the canonical physics range is therefore 4214–4221. These are finite graph-cover, holonomic, stochastic/semiclassical, clock-gadget, Gaussian-channel, recurrence, harmonic-network, and tree-band results, not fabricated hardware, observed Hawking radiation, physical spacetime memory, a speed-of-light derivation, gravity, cosmology, or a theory of everything.

Passes 4245–4252: residual symmetry, GHZ resource model, minimum Hodge sensing, routed delay geometry, and quotient closure

The five-generation carrier surgery admits a natural anchor-level residual-symmetry target: the 658008 five-subsets of the 40-point W33 action form 43 $PSp(4, 3)$ orbits, and the unique largest setwise stabilizer has order 72. A representative induces $K_{1,4}$; the stabilizer fixes the center, acts as A_4 on the four leaves, and has central pointwise kernel C_6. This is an anchor scaffold, not yet a full branching theorem for the charged 75-state subspace.

On the 28-payload binary tree, physically disjoint two-qubit gates require seven entangling rounds for the 27 tree edges. At the published dual-rail operating point $F_{\mathrm{CNOT}} = 0.981$ and erasure probability 0.13, whole-circuit postselection accepts only $0.87^{27} \simeq 0.02328$ of runs, whereas heralded modular-fusion resource models reduce the expected entangler count per accepted construction from roughly 1160 to between 31.0 and 47.9, depending on restart locality. The conditional product $0.981^{27} \simeq 0.596$ remains the fidelity bottleneck.

For arbitrary real edge errors of support at most seven, exactly two global scalar measurements beyond the oriented Levi incidence matrix are information-theoretically necessary and sufficient.

One row cannot be injective on a two-dimensional theta circulation space supported on at most fourteen edges; the two deterministic Pass-4147 rows give stacked spark at least 15. Thus the exact noiseless minimum is $m = 2$. The ten-channel Hodge construction remains a conditioning, rather than an injectivity, result.

The 919 delay units are now tied to explicit route centerlines. Each 5-ps slot is a rounded four-quarter-bend dogleg of radius 0.05 mm and vertical excursion 0.3176609398 mm, satisfying $2h + (2\pi - 4)r = 0.749481145\,\mathrm{mm}$. The route generator consumes the frozen Pass-4148 four-branch schedule and emits all 160 point-line identities and cell coordinates; the design remains a public-PDK centerline contract rather than proprietary DRC signoff.

The next setwise-subset rank of the nonlinear stabilizer lattice has exactly five three-vertex orbit types, with stabilizer orders $162, 72, 12, 9, 6$. Interval/Krawczyk exhaustion closes the two smallest quotient systems: the triangle stratum has $(9, 3)$ roots for selectors $(24, 15)$ and the maximally symmetric independent-triple stratum has $(9, 9)$ roots. Three larger triple quotients and trivial-stabilizer equilibria remain open.

Three independent finite constructions accompany the implementation work. First, W33 adjacency phase sensing has $F_Q^{\max} = 256$, while a vertex probe has $F_Q = 48$, an exact ratio 16/3 before resource/noise accounting. Second, the Levi edge Hodge split gives 81 harmonic and 79 gradient Gaussian modes; reversible entropy-conserving equalization obeys $T_* = T_H^{81/160} T_G^{79/160}$ and $W_{\max} = \frac{k_B}{2}(81T_H + 79T_G - 160T_*)$. Third, the point $\to$ ordered-edge $\to$ ordered-triangle pointwise-stabilizer chain yields quotient dimensions $3 \to 6 \to 8$ with exact refinement intertwiners $Q_6 R_{63} = R_{63} Q_3$ and $Q_8 R_{86} = R_{86} Q_6$; componentwise cubing also commutes with each refinement lift, so the cubic nonlinear vector field closes exactly along this finite coarse-to-fine hierarchy.

Passes 4253–4260: retracted cover, surviving finite models, and channel bounds

4253 (retracted by Pass 4329). The stored $\mathbb{Z}_{2731}$ voltage assignment has one zero-voltage base eight-cycle, with signed sum $1328 - 542 + 801 - 1587 = 0 \pmod{2731}$. Its 218480-vertex lift therefore has girth exactly 8, not at least 16. The historical vector is retained as a falsifier, not a cover certificate.

4254. Oppositely oriented latitude loops at $\theta = \pi/3$ and $2\pi/3$ implement the geometric controlled phase π. A common polar offset δ gives $\phi(\delta) = \pi \cos\delta$, so the first-order calibration error cancels. At $\omega/\Omega = 0.02$ the two exact nonadiabatic leakage probabilities are 3.05824×10^{-4} and 2.93842×10^{-4}.

4255. At the saturated coexistence point and $D = 0.25$, a detailed-balance finite-volume Fokker–Planck discretization converges to metastable gap 4.37527×10^{-8}, with the next rate 2.93047. The unit-prefactor two-state Lindblad demo crosses from underdamped to overdamped at $\gamma_\phi = 2\Delta = 8.29222 \times 10^{-4}$.

4256. The 24 data gates are three repeats of eight adjacent SWAP supports, so the exact three-local clock needs only eight reusable dynamic flags. Together with the frozen 13 shared pair conjunctions and 24 final conditions this gives 45 auxiliaries, preserving gap 2Ω, perfect transfer at $\pi/(2\Omega)$ and revival at π/Ω.

4257. The complete 19-mode horizon induces a baseline effective signal attenuator with $\tau = 0.2830095941$ and $\bar{n}_{\mathrm{env}} = 0.0166793553$. In the frozen 512-sample correlated-disorder ensemble, 510 samples have $\tau \le 1/2$, two have $\tau > 1/2$, and none is entanglement breaking. At $n_{\mathrm{th}} = 0.005$

the generated outside–partner state remains PPT-entangled in all 512 samples, demonstrating that state entanglement and signal-channel capacity are distinct resources.

4258–4260. The exact Levi spectral form factor has infinite-time plateau $2054/6400 = 0.3209375$, 25.675 times a nondegenerate 80-level reference, falsifying a bare random-matrix scrambling interpretation. The point–line harmonic modular spectrum has only two finite entanglement energies, 2.63391579385 and 4.03202440074, with multiplicities 1 and 24 plus 15 exact spectators. In the conditional degree-four tree model, $v_{\rm rad}^{\max} = 2\sqrt{3}J$ while $B(r) = 2\,3^r - 1$. Its 4373-vertex radius-seven ball is not realized by the retracted Pass 4253 cover.

Boundary. These are finite graph-cover, holonomic, stochastic/Lindblad, clock-compilation, Gaussian-channel, spectral, modular and tree-band statements. They do not establish fabricated devices, observed Hawking radiation, physical spacetime memory, a derivation of c, gravity, cosmology, or a theory of everything.

Passes 4261–4268: girth 18, dynamically corrected holonomy, full-coordinate metastability, clock-37, and finite outside-box probes

A cyclic voltage assignment in $\mathbb{Z}_{750019}$ makes every radius-eight sheet-zero ball above each of the 80 Levi base vertices a collision-free 4-regular tree of 13,121 vertices. Deck translation therefore excludes every lifted cycle of length at most 16 and certifies girth at least 18. The resulting 60,001,520-vertex cover is an explicit upper bound rather than a compact solution: the desired sub-10,000-sheet girth-18 construction remains open.

For the two-bundle holonomic controlled phase, choose $\cos\theta = 1/4$ and execute latitude loops at $\theta, \pi - \theta$, followed by the same pair on the opposite adiabatic eigenbranch with reversed physical orientations. The geometric phase is $-\pi \cos\delta$ under a common polar offset δ, so the first derivative vanishes at the operating point, while repeatable static loop-dependent dynamical phases cancel by echo. Adding $H_{\rm CD} = \frac{1}{2}(\mathbf{n} \times \dot{\mathbf{n}}) \cdot \boldsymbol{\sigma}$ gives exact transitionless driving in the modeled two-level control. At $\omega/\Omega = 0.1$, a one-percent CD-amplitude error gives a four-loop leakage union bound 2.004×10^{-6}.

The saturated scale coordinate was then quantized directly as $H = p^2/(2M) + U(s)$ with $U' = -F$, $M = 16$, and coupled through the full coordinate operator to an Ohmic Davies bath at $T = 0.25$. On a 180-point grid the retained-basis population gap converges from 1.06×10^{-11} at $K = 14$ to 1.87223×10^{-11} at $K = 20$, corresponding to a dimensionless metastable lifetime 5.34×10^{10}. This is a finite-coordinate open-system model, not a physical spacetime-memory lifetime.

An exact local-dimension trade replaces the five binary Gray clock digits by five qutrit digits $\{0, 1, *\}$, using $*$ as the intermediate transition state. The eight explicit dynamic flags disappear, leaving 13 shared pair ancillas plus 24 final spectator-condition ancillas, hence 37 auxiliaries. The legal 49-state history, maximum locality three, gap 2Ω, perfect-transfer time $\pi/(2\Omega)$, and full revival π/Ω are unchanged.

The two previously unresolved thermal attenuator samples, $\tau = 0.5039652$ and 0.5113024, were attacked with low-Fock non-Gaussian inputs: all optimized two-Fock mixtures through $n = 8$, 2,000 random three-Fock mixtures per channel, and 5,000 random coherent rank-two states in $\{|0\rangle, \ldots, |4\rangle\}$ per channel. No positive coherent information was found. Pure inputs attain exactly zero in the purified-environment Stinespring picture, so the tested family maximum is zero; unrestricted multi-use non-Gaussian capacity remains open.

Three independent finite probes accompany the main work. The Levi distance partition $[1,4,12,36,27]$ reduces marked-vertex continuous-time search to an exact five-shell Hamiltonian and yields a concrete 73.016% success operating point. A rank-one onsite defect $V=4$ creates a unique eigenmode above the spectral edge at $E=4.9600853$ with 76.202% weight on the defect. Finally, the point-line harmonic thermal state has two distinct PPT death temperatures: $T_c=0.9947197$ for the 24 $\sqrt{6}$ channels and $T_c=1.2027283$ for the unique singular-value-4 collective channel, producing a two-stage loss of bipartite entanglement.

Passes 4269–4276: carrier branching, heralded fusion, minimum Hodge sensing, corrected GDS geometry, and finite outside-box probes

4269. The order-72 five-anchor stabilizer is $C_3 \times SL(2,3)$. Restricting the W33 eigenspaces gives an explicit 21-character branching of V_{15} and V_{24} and hence of $H_{145}=7V_1+6V_{15}+2V_{24}$. A species-wise 75-state five-generation charge partition can commute with this group, while an identical five-dimensional generation factor repeated over all 15 gauge-internal states cannot.

4270. A nondestructive GHZ-block fusion uses two anchor-to-ancilla CNOTs, one Z parity readout, and a tracked block-X frame. Under the frozen 1.3×10^{-1} erasure model, the optimized 28-qubit modular construction uses 33 accepted-path CNOTs and 94.2734 expected CNOT attempts, a $12.30\times$ reduction versus whole-run restart; its independent conditional process factor remains only 0.53098.

4271. Two extra global measurements are both necessary and sufficient for noiseless arbitrary-real seven-sparse Levi-edge recovery. A deterministic $\mathbf{F}_{65537}$ search yields a signed-16 pair with normalized worst theta gain 1.33507×10^{-5}, $1.805\times$ the prior pair, while preserving the exhaustive spark-15 cycle/theta certificate.

4272. The previous 0.12-mm rounded dogleg cell is geometrically impossible for four $50\,\mu$m quarter bends, and the true excursion is $h+2r$, not h. The corrected pure-Python GDSII compiler uses 0.205-mm cells, 0.45-mm lanes and sixteen ten-lane banks, producing an $11.04\,\mathrm{mm}\times14.85\,\mathrm{mm}$ keepout-inclusive preliminary layout with 919 doglegs and 361 straight cells.

4273. The order-12 and order-9 triple quotients have locally certified saturated deterministic catalogues 57/27 for the two selectors, whereas the order-6 edge-plus-isolated quotient already contains at least 2376/246 locally certified roots. Among the 16 four-subset orbits, one has trivial stabilizer, so global nonlinear completion must leave the nontrivial-symmetry-stratum strategy.

4274–4276. The W33 graph state is exactly five-uniform with 240 weight-six stabilizers supported on pairs of disjoint triangles. Non-lazy Ollivier curvature is $+1/6$ on W33 edges but -1 on Levi edges. The $81+79$ Hodge decomposition also gives an exact two-temperature Ornstein–Uhlenbeck fluctuation–dissipation splitting.

Boundary. These are finite mathematical and engineering statements. They do not establish fabricated devices, phenomenological particle physics, continuum spacetime curvature, gravity, cosmology, or a theory of everything.

Passes 4324–4334: the chamber Hecke machine and an exact frontier correction

4324: two native chamber switches. On the 160 incident point–line chambers, let P change the line while holding the point fixed and let L change the point while holding the line fixed. Each is forty disjoint copies of the K_4 adjacency operator, and GAP proves

$$P^2 = 2P + 3I, \qquad L^2 = 2L + 3I, \qquad PLPL = LPLP.$$

The eight standard type-C_2 words are linearly independent. Thus this chamber carrier realizes the full eight-dimensional $q = 3$ Iwahori–Hecke image. Its exact module ledger is

$$1\chi_{3,3} + 15\chi_{3,-1} + 15\chi_{-1,3} + 81\chi_{-1,-1} + 24V_2,$$

which explains the long-standing flag-scheme dimensions $1 + 30 + 48 + 81 = 160$.

4325–4326: the Hashimoto turn and its chirality. With the 320 oriented Levi edges ordered by orientation,

$$B_{\text{Levi}} = \begin{pmatrix} 0 & L \\ P & 0 \end{pmatrix}, \qquad B_{\text{Levi}}^2 = \operatorname{diag}(LP, PL).$$

The two products are disjoint binary 9-out halves of the chamber distance-two shell. For $K = LP$,

$$\chi_K(x) = (x - 9)(x - 1)^{81}(x + 3)^{30}(x^2 + 9)^{24}.$$

The antisymmetric turn $\Omega = LP - PL$ has

$$\chi_\Omega(x) = x^{112}(x^2 + 60)^{24}, \quad \Pi_{48} = -\Omega^2/60, \quad \operatorname{rank}\Pi_{48} = 48.$$

Consequently $\Omega/\sqrt{60}$ is a canonical complex structure on the conjugate $24 + 24$ chamber channel. This is a finite operator theorem, not a $W(G_2)$ action.

4327: exact folded-propagator normal form. Let $C = P + L$, $X = (C - 2I)\Pi_{48}$, and let F be the old folded cubic point-Hashimoto operator compressed to Π_{48}. In the four-dimensional packet algebra $\{\Pi_{48}, X, \Omega, X\Omega\}$,

$$F = -68\Pi_{48} - 31X - \frac{21}{2}\Omega + \frac{2}{3}X\Omega, \qquad (F + 68\Pi_{48})^2 = -689\Pi_{48}.$$

The old numerical -6455 is now explained exactly by the square of the off-diagonal part. No continuum propagator, particle, mass, or coupling identification follows.

4334: the $24+24$ count becomes an object-level carrier split. Let E_p, E_ℓ be the eigenvalue-2 projectors of the W33 point graph and its dual line graph, and pull them to the chamber space as orthogonal rank-24 projectors Q_p, Q_ℓ. GAP proves

$$\operatorname{im}\Pi_{48} = \operatorname{im}Q_p \oplus \operatorname{im}Q_\ell, \qquad Q_pQ_\ell Q_p = \frac{3}{8}Q_p, \quad Q_\ell Q_p Q_\ell = \frac{3}{8}Q_\ell.$$

Thus the two natural summands meet only in zero but are uniformly nonorthogonal: all 24 principal angles have cosine $\sqrt{6}/4$. With $Q = Q_p + Q_\ell$,

$$\Pi_{48} = \frac{8}{5}(2Q - Q^2) = -\frac{\Omega^2}{60}.$$

The conjugate channel is therefore literally the joined point/line eigenvalue-2 carriers, not just a multiplicity ledger.

4328: the irregular Ihara boundary corrected. Bass's determinant formula already applies to general finite graphs with $D - I$. The Kotani–Sunada theorem bounds non-real poles by

$$(d_{\max} - 1)^{-1/2} \leq |u| \leq (d_{\min} - 1)^{-1/2},$$

not by the same expression without square roots, and it does not put every nontrivial root in that annulus. For the shipped degree-2–8 frame graph the reciprocal-root form is $1 \leq |\lambda| \leq \sqrt{7}$ for the corresponding non-real roots. The old "band filling" and "closest irregular Ramanujan" metric is withdrawn.

4329: a cover certificate retracted by one cycle. The stored $\mathbb{Z}_{2731}$ voltage vector from Pass 4253 has the base eight-cycle

$$(3, 52, 4, 48, 8, 66, 18, 53)$$

with signed voltage sum $1328 - 542 + 801 - 1587 = 0 \pmod{2731}$. The lifted cover therefore has girth exactly 8, not at least 16. Pass 4260's 4373-vertex radius-seven tree ball remains a correct conditional tree count but is not realized by that cover. The valid cover ladder is the earlier $\mathbb{Z}_{359}$ girth-at-least-14 cover followed by the much larger $\mathbb{Z}_{750019}$ girth-at-least-18 cover.

4330–4331: what symmetry and fault detection actually survive. Under the literal pair swap $S(x_0, x_1, x_2, x_3) = (x_2, x_3, x_0, x_1)$, machines A/B/C/D are invariant in the pattern false/true/-false/true. Machine D therefore removes both the directed time arrow and the point/frequency structural asymmetry. Its 0.460435 peak is symmetric localization, not residual point/frequency bias. Affine translations act on the 81 frames but descend to neither projective 40-carrier, so they do not force a point-side load port.

The self-contained flag comparator is incidence: after the two rails move, test $p \in \ell$. It detects $1656/1920 = 69/80$ differential single-rail opcode substitutions and 0/960 shared-control substitutions. The earlier 95.71% number compared each faulty rail with a golden run and was not a dual-rail comparator.

4332–4333: one arithmetic component, and an honest rank. The irreducible degree-33 factor of the instruction Hashimoto pencil has exactly one real root in $(5.74, 5.75)$ and one in $(3.349, 3.350)$ by an exact Sturm certificate. Thus the global nonbacktracking growth root and the reported slow real mode are Galois conjugates in one rational primary component. The rational projector and a basis-independent localization theorem remain open.

Finally, the complete universal-set census by sizes 4 through 10 is

$$24,\ 80,\ 114,\ 90,\ 41,\ 10,\ 1,$$

totalling 360. The shipped ISA is not strictly "12th": only six sets have lower ρ at the stated tolerance, while six isomorphic sets share its numerical value and occupy positions 7–12. Size is a coarse correlate whose bands overlap, not an explanation of that tie.

Evidence boundary. The three GAP witnesses replay 31/31, 28/28, and 17/17 exact checks. The panel operators are three-way relations, not yet a deterministic opcode selector or synthesized B/D RTL. The exact finite algebra and the retractions above should therefore be read separately from hardware, continuum, thermodynamic, and phenomenological interpretations.

63 Overture: the machine is the geometry is the carrier

In plain language. *The claim of this paper in one section: the shape, the machine and the light are not three things connected by engineering but one thing described three ways. Everything after this is either a consequence of that or a limit on it.*

Classical architecture separates three things: the processor that acts, the program that directs, and the network that connects. The thesis of this paper is that at the level of a single photon on a finite symplectic geometry the three are one object, and that this is not a metaphor but a chain of theorems.

Principle 63.1 (Identity of the three layers)**.** *Transporting the photon through the mesh* is *applying a gate* is *routing a packet. One physical process; three descriptions.*

Rosetta stone of the $q = 3$ substrate (read this first). This paper has two halves: the *machine* (the architecture, below) and the *world* it computes (§C). Both are the one $q = 3$ Eisenstein object — the Witting polytope 3{3}3{3}3{3}3 (240 vertices = E8 roots), its symmetry ST#32 (order $155520 = 3|\mathrm{Aut}(W(3,3))|$), equivalently the degree-two cyclotomic skeleton $\{\Phi_3, \Phi_4, \Phi_6\} = \{13, 10, 7\}$ at $q = 3$ — seen from *seven faces*: **(1) selection** ($q = 3$ forced: $\Phi_4 = q^2 + 1$ factors the de Sitter cubic and $v = (q{+}1)\Phi_4 = 40$), **(2) constants** (the Witting degrees $\{12, 18, 24, 30\} = \{k, h(E_7), c{=}f, h(E_8)\}$), **(3) gauge** ($\dim \mathrm{SU}(3)^3 = 24 = c$, $\dim \mathrm{SM} = 12 = k$, $\sin^2\theta_W = 3/8$), **(4) neutrino** ($\Phi_3/q^2 = 13/9$, PMNS $1/\Phi_3$), **(5) code** (the GKP tower $A_2{<}D_4{<}E_8$), **(6) demonstrator** (contextual fraction $1/\Phi_4 = 1/10$, pump Chern $\lambda = 2$), and **(7) cosmology** ($N = 2(v{-}\Phi_4) = 2h(E_8) = 60$ e-folds; $n_s = 29/30$, $r = k/N^2 = 1/300$). The faces share integers — $k{=}12, \Phi_4{=}10, \Phi_3{=}13, c{=}f{=}24, v{=}40$ each serve ≥ 3 faces (Table 1) — which is why they are one object (Fig. 9). *No free integer parameter*: every geometric integer is a cyclotomic value, GQ count, or Witting degree, and even the moonshine ceiling closes — the Leech kissing number factors as $196560 = 6\mu q^2\Phi_3\Phi_4\Phi_6$, so the Monster (196883), $\tau = \mu q^2\Phi_6 = 252$, and the fine-structure integer $1/\alpha = 137 = \Phi_3\Phi_4 + \Phi_6$ are all $\{\mu, q, \Phi_3, \Phi_4, \Phi_6\}$ arithmetic; the lone non-integer input is the 0.036 of 137.036 (QED running). *The one experiment to run first*: the benchtop, calibration-free contextual fraction $1/\Phi_4 = 1/10$, which tests three faces at once.

The identification rests on seven exact pillars, each proved by explicit computation in the project repository (every theorem below carries its witness script):

1. **Operator–operand duality** (§64.1): the same $W(3,3)$ is drawn by the photon's *states* (orthogonality of the 40 Witting rays in $\mathbb{C}^4$) and by its *operators* (commutation of the 40 two-qutrit Pauli classes on $\mathbb{C}^9$).

2. **Routing–gate identity** (§66): the mesh is an atlas of 540 three-dimensional hypercube charts whose native XOR routing moves are exactly the register operations, glued along the 1620 apartments of the Tits building of $\mathrm{Sp}(4, \mathbb{F}_3)$.

3. **Parabolic address–route duality** (§64.3): the point parabolic supplies two 648-slot address sheets, whereas the Bell-line parabolic decodes the compass fabric by the complete prefix code $0, 10, 110, 111$; its cache quarter is an exact D_8-over-81 four-state address fiber. Address and route are the two nodes of the same C_2 building, while the separate ternary transport extension carries temporal state.

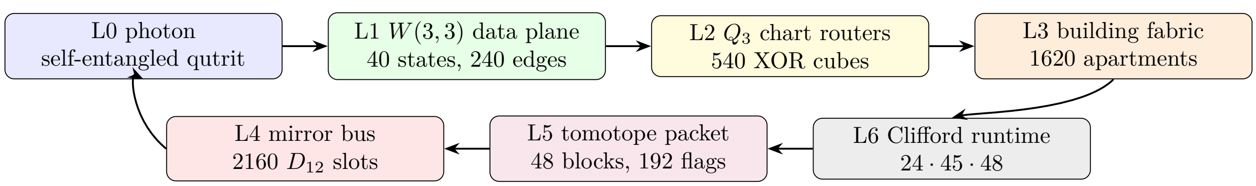
4. **Dual-torsor state–program compilation**: the Steinberg memory restricts to both canonical order-27 shell groups as three regular copies. Bell-shell programs use commuting $\mathbb{F}_3^3$ coordinates; point-shell states use Heisenberg 3^{1+2} coordinates; both compile into the same 81 logical qutrits.

5. **Mirror middleware** (§67): the cyclic selector clock and the mirror transport bus are different 2160 worlds. The photon is clocked by the C_{12} side but routed by the D_{12} side, whose full Clifford lift is the exact product $24 \cdot 45 \cdot 48$.

6. **Universality from optics** (§68.5): the photon's own beam-splitting elements generate the complete Clifford group (symplectic closure = 51840, exact), and its matter shell is precisely its magic supply.

7. **Fractal holonet scaling** (§69): recursively replacing every site by a fresh $W(3,3)$ core gives a universal computer/network with exact 40^n leaves and logarithmic reversible routing.

And the machine is not separate from the world it computes. The second half of the paper (§C) shows that the *same* $q = 3$ Eisenstein structure that fixes this architecture — the Witting polytope, its symmetry ST#32, the cyclotomic skeleton $\{\Phi_3, \Phi_4, \Phi_6\}$ — is also the selection of $q = 3$, the substrate constants, the Standard-Model gauge group, the neutrino masses, the error-correcting code, the benchtop experiment, and inflationary cosmology. These seven faces, and the single object behind them, are laid out in §C.1 (Fig. 9); the reader who wants the unifying picture before the engineering may read that subsection first.



One loop: carrier → data plane → router → building → runtime → tomotope packet → mirror bus → carrier.

Figure 1: The photonic holonet as an engineering stack. The same photon is data, instruction, and packet; the same finite geometry supplies the network, middleware, and runtime.

64 The substrate

Definition 64.1 ($W(3,3)$)**.** *Let* $V = \mathbb{F}_3^4$ *with the alternating form* $\langle u, v\rangle = u_1v_3 - u_3v_1 + u_2v_4 - u_4v_2$. *The points of* $W(3,3)$ *are the* 40 *projective points of* V*; the lines are the* 40 *totally isotropic projective lines. Each point lies on* 4 *lines and each line carries* 4 *points; collinearity defines the strongly regular graph* SRG(40, 12, 2, 4).

Three related symmetry groups occur, and their roles are different:

$$\underbrace{\mathrm{Sp}(4,3)}_{2\cdot\mathrm{PSp}(4,3),\ \text{Clifford double cover}}, \qquad \underbrace{\mathrm{PSp}(4,3)}_{25920,\ \text{faithful projective action}}, \qquad \underbrace{W(E_6) \cong \mathrm{PSp}(4,3){:}2}_{51840,\ \text{outer Weyl extension}}.$$

The first and third groups both have order 51840 but are not the same extension: the center of $\mathrm{Sp}(4,3)$ acts trivially on projective points and lines, while the outer involution in $W(E_6)$ acts nontrivially. The projective group acts with permutation rank 3. The ATLAS index-40 subgroups are the two building parabolics $3^{1+2}{:}2A_4$ and $3^3{:}S_4$ [21]. The substrate primitives are

$$q = 3, \quad \lambda = 2, \quad \mu = 4, \quad k = 12, \quad f = 24, \quad g = 15, \quad \Phi_6 = 7, \quad \Phi_4 = 10, \quad \Phi_3 = 13,$$

and every numerical claim in this paper reduces to arithmetic in these.

Formal grounding (the master formalism). In the master manuscript [1] the whole structure is forced by one Diophantine seed: the *master equation* $q! = 2q$ has the unique positive-integer solution $q = 3$, and the SRG quadratic $x^2 - q!\,x + 2^q = (x-2)(x-4)$ then delivers $\lambda = 2$, $\mu = 4$ and the full parameter closure ($k = 2^q + q + 1$, $v = 40$, multiplicities $f = 24$, $g = 15$, cyclotomic ladder $\Phi_3 = 13$, $\Phi_6 = 7$, $\Phi_{12} = q^4 - q^2 + 1 = 73$, $\Theta = q^2 + 1 = 10$). Two of those closure constants reappear in this paper as *spectral field discriminants*: the chart-web's irrational eigenvalue pairs $1 \pm \sqrt{10}$ and $(-1 \pm \sqrt{73})/2$ (§66) live precisely in $\mathbb{Q}(\sqrt{\Theta})$ and $\mathbb{Q}(\sqrt{\Phi_{12}})$ — the machine's transport spectrum speaks the same cyclotomic dialect as the physics tier ladder.

64.1 The two carriers and the operator–operand duality

Theorem 64.2 (Two realizations, one geometry)**.** (i) *The* 40 *Witting rays in* $\mathbb{C}^4$ *(the photon's path*$\otimes$*polarization space) have orthogonality graph isomorphic to the collinearity graph of* $W(3,3)$*.* (ii) *The* 40 *projective classes of two-qutrit Pauli displacements* $D(v) = X^aZ^b \otimes X^cZ^d$ *on* $\mathbb{C}^9$ *(the photon's past*$\otimes$*future time-bin space) have commutation graph equal to the same* $W(3,3)$*, via* $D(u)D(v) = \omega^{\langle u,v\rangle}D(v)D(u)$*.* Witnesses: `bt817_self_entangled_photon_atlas.py`, `bt821_operator_operand_duality.py`.

Thus a point of the substrate is simultaneously a pure state the photon can occupy and a unitary the photon can enact; a line is simultaneously a measurement context (orthonormal tetrad) and a stabilizer group (maximal commuting Pauli set with its 9-state joint eigenbasis). The $360 = 40 \times 9 = f \cdot g$ two-qutrit stabilizer states are the joint eigenbases of the forty lines.

64.2 The five vacua and the Schläfli geography

Every maximal subgroup class of $\mathrm{PSp}(4,3)$ decomposes the substrate in its own canonical way; the maximal indices are the classical inventory of the 27 lines on a cubic surface.

index	subgroup	points	lines	vacuum	cubic surface
27	$2^4{:}A_5$	$[40]$	$[40]$	homogeneous (icosahedral)	27 lines
36	S_6	$[40]$	$[10,30]$	spread fibration	36 double-sixes
40	parabolic	$[1,12,27]$	$[4,36]$	holonet point vacuum	trihedra I
40	parabolic	$[4,36]$	$[1,12,27]$	dual line vacuum	trihedra II
45	$(2T{\times}2T){:}2$	$[8,32]$	$[16,24]$	binary polar	45 tritangents

The physical decomposition $40 = 1 + 12 + 27$ (self + gauge shell + matter shell) is the *choice of the point-parabolic vacuum* among five inequivalent readings; the icosahedral vacuum sees a featureless substrate. The platonic ladder threads the table: the binary tetrahedral group $2T$ (the 24-cell) lives in every polar-pair stabilizer, the real octahedral group O_h ($= S_4 \times \mathbb{Z}_2$, order 48) is the chart symmetry of §66, and the binary icosahedral group $2I = \mathrm{SL}(2,5)$ (the 600-cell) is the core of every spread stabilizer, with point orbits $[20,20]$ and line orbits $[10,30]$ echoing the Boerdijk–Coxeter decomposition $600 = 20 \times 30$. *Witnesses:* `bt808`*–*`bt813`*.*

64.3 The parabolic router: address and route are dual

In plain language. *In this machine a destination and the path to it are the same kind of object, so routing is not a search. That duality is what removes the routing table, and this section makes it precise.*

The two index-40 parabolics are not duplicate descriptions of one local neighborhood. They execute different halves of a packet protocol. This becomes visible only after the compass census: there are two 216-element classes of icosahedral A_5 cores, and exactly $1296 = 216 \cdot 6$ cross-class pairs meet in D_{10}.

Theorem 64.3 (Bell–compass parabolic router)**.** *Fix an isotropic Bell line. Its projective Siegel parabolic* $3^3{:}S_4$ *has four orbits on the* 1296 *incident compass pairs:*

$$162_L + 162_R + 324 + 648.$$

They are distinguished objectwise by the Bell line lying in the pentad core's 5_L*,* 5_R*,* 10*, or* 20 *line orbit. The normalized orbit sizes satisfy the Kraft equality*

$$2^{-3} + 2^{-3} + 2^{-2} + 2^{-1} = 1,$$

so they admit the complete prefix decoder

$$110 \mapsto \textit{left cache}, \quad 111 \mapsto \textit{right cache}, \quad 10 \mapsto \textit{schedule}, \quad 0 \mapsto \textit{dark mirror}.$$

The outer Weyl involution fuses exactly the two 162 *cache orbits, leaving* $324_{\text{cache}} + 324_{\text{schedule}} + 648_{\text{mirror}}$*. By contrast, the dual point parabolic has two regular projective orbits of size* 648*; the outer extension preserves rather than fuses these address sheets.* Witness: `bt859_bell_compass_parabolic_router.py`.

The equality case of Kraft–McMillan means the routing words fill a binary decision tree with no unused branch [22]. Here the code lengths were not optimized from assumed traffic probabilities: they were read from exact group orbits. A packet first asks whether its Bell line is in the dark mirror half; if not, whether it is in the common schedule quarter; only the remaining cache quarter needs the third, chiral bit. The decoder is self-delimiting and constant-depth, with expected word length

$$\frac{1}{2}(1) + \frac{1}{4}(2) + 2 \cdot \frac{1}{8}(3) = \frac{7}{4}$$

under the invariant uniform measure on the 1296 compass pairs.

orbit	Bell-line position	prefix	engineering action
648	dark 20-orbit	0	enter mirror bus
324	common schedule 10-orbit	10	timetable-local route
162	absolute pentad 5_L	110	left cache
162	absolute pentad 5_R	111	right cache

This theorem also closes an honesty boundary. The Bell stabilizer and the compass incidence set both have cardinality 1296, but they are not a regular G-set identification. Projectively, the Bell stabilizer has order 648 and the orbit stabilizers are $1, 2, 4$. The useful result is stronger than the count: a hierarchical decoder and a persistent address bit are forced by the two parabolic actions.

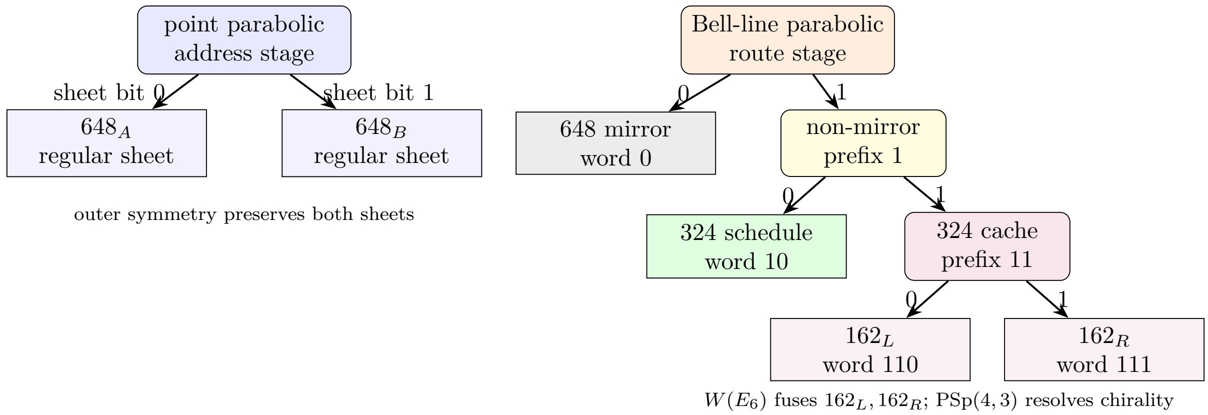


Figure 2: The two C_2 parabolics implement a dual protocol. Point localization selects one of two address torsors; Bell-line localization decodes a complete variable-length route word.

64.4 The incidence transceiver and the route-dark lattice

In plain language. *How a message gets from one machine to another. "Incidence" means the sender and receiver share a point of the geometry; the transceiver is the optical hardware that exploits it. The second half of the title is the failure mode: some routes are dark — structurally unreachable — and this section maps them rather than assuming a fully connected network.*

Pass 173 turns the address/route distinction into an exact signal law. For the line–point incidence matrix N, the centered operator

$$T = N - \frac{1}{10}J$$

has rank 24 and satisfies

$$TA_P = A_L T, \qquad T^{\mathsf{T}}T = 6E_{24}^{P}, \qquad TT^{\mathsf{T}} = 6E_{24}^{L}.$$

Thus $T/\sqrt{6}$ is a lossless transceiver on the shared gauge sector, while the unmatched address and route dark sectors remain type-protected. Every one of the 480 local-axis selector vectors produces a 40-channel analyzer word with histogram $\{-3^1, -1^9, 0^{20}, 1^9, 3^1\}$ and exact decoder $x = N^{\mathsf{T}}(Nx)/6$.

The two dark lattices are not interchangeable. The address lattice $\ker_{\mathbb{Z}} N$ has minimum shell 90 at norm 8 and binary code $[40, 15, 8]$; the route-dark lattice $\ker_{\mathbb{Z}} N^{\mathsf{T}}$ has minimum shell 432 at norm 10 and code $[40, 15, 10]$. Its 216 projective supports are exactly the pentad cores already used by the router: each is $K_{5,5}$ minus a perfect matching, and the deleted matchings double-cover the 540 skew-line chart atlas. The signed shell splits into the two chiral 216-orbits. Hence the old compass carrier is not merely count-compatible with the route layer; it is its complete minimum-energy integral dark shell.

The two 25-dimensional context duals are locally indistinguishable at their first two nonzero shells: both have $A_4 = 40$ and $A_6 = 240$. They first separate at the third shell,

$$A_8^{\text{address}} = 5085, \qquad A_8^{\text{route}} = 3645.$$

Moreover only the address code is self-orthogonal; the route code has binary Gram rank 6. Thus the Pass-164 $O^+(10, 2)$ quotient is address-chiral and cannot be silently reused as a route codec.

This also corrects the architecture vocabulary. At odd $q = 3$, $W(3,3)$ and its dual $Q(4,3)$ have equal $40 + 40$ counts and cospectral concurrence graphs, but are not incidence-self-dual. The lossless identification exists only on the shared 24-sector. The statement is an algebraic analyzer-ROM/control-codec theorem until a specified optical amplitude implementation of T is supplied. (Witness: `analysis/w33_pass173_incidence_transceiver_route_dark_lattice.py`.)

Pass 174 supplies the missing protected subcodec. Although the full route code $R = [40, 15, 10]$ is not self-orthogonal, its route-code hull is

$$H = R \cap R^{\perp} = [40, 9, 16], \qquad W_H = 1 + 135z^{16} + 240z^{20} + 135z^{24} + z^{40}.$$

The all-ones word is a fixed radical and $H/\langle \mathbf{1} \rangle$ is a plus-type 8-space with the exact $1 + 135 + 120$ native group orbits. Its 255 nonzero words reconstruct the symplectic graph $\mathrm{SRG}(255, 126, 61, 63)$ and its two quadratic subconstituents $\mathrm{SRG}(135, 70, 37, 35)$ and $\mathrm{SRG}(120, 63, 30, 36)$. So one route codec now supplies the complete finite stack

$$432 \text{ signed pentads} \to 216 \text{ cores} \to [40, 9, 16] \to 255 = 135 + 120.$$

The order-eight control rail is also less obstructed than the previous cohomology wording claimed. On both address and route discriminants, $[\tau(h) - h] = 0$ while $[\tau(h) - 5h] = [4h] \neq 0$. Thus a $q(h) = 11/8$ fixed order-eight rail exists; the nonzero class forbids only a pure scalar-5 normal form. This is a finite control-label theorem, not yet a fabricated phase rail. (Witness: `analysis/w33_pass174_dual_discriminant_fixed_rail.py`.)

Pass 176 removes the remaining abstract quotient choice. If $C = \ker_{\mathbb{F}_2} N$ and $R = \ker_{\mathbb{F}_2} N^{\mathsf{T}}$, the unique nonzero fixed isotropic class $f \in C^{\perp}/C$ is sent by incidence to the all-ones route word, and

$$f^{\perp}/\langle f \rangle \xrightarrow{N} (R \cap R^{\perp})/\langle \mathbf{1} \rangle$$

is an exact native-$\mathrm{PSp}(4,3)$ quadratic isometry. All 512 quadratic identities and 8192 generator/object commuting squares are checked. The 240 weight-six address words become the two explicit sheets over the 120 anisotropic route states and map bijectively to the 240 weight-20 hull words. This is an executable codec map; it still does not specify an optical transfer matrix, loss budget, or detector model.

Pass 177 then locates the first address/route classification depth in the *dual* context lattices. Both begin

$$1 + 720q^2 + 15360q^3,$$

where the 720 is coordinate doubles plus sign-lifts of lines on the address side and point stars on the route side. They first split at

$$q^4 : \qquad 1350960 - 982320 = 368640.$$

Thus firmware may share the first two dual-shell tables, but must retain the type bit before admitting weight-eight sectors. Separately, the sentinel/context theta regression has an exact normalization because $A^{\vee} = \frac{1}{2}B$ and $\mathrm{covol}(A) = 2^{25}$; finite shell sums are regression checks only. Finally, $\ker_{\mathbb{Z}} N^{\mathsf{T}}$ is the point-trade lattice of the nonisomorphic dual $Q(4,3)$, whose span size is 2 versus 4 for $W(3,3)$. A route generator with $q = 11/8$ exists, but $3/8$ occurs on the other fixed-generator orbit, so no invariant $11/8$ hardware phase is inferred. (Witnesses: `analysis/w33_pass176_incidence_fixed_line_e8_bridge.py`, `analysis/w33_pass177_address_route_dual_theta_split.py`, and `analysis/w33_pass179_poisson_modular_pair.py`.)

Passes 188–192 refine the local controller geometry. The 432 route minima contain 36 exact double-six crowns: each is $K_{6,6}$ minus a matching, assembled from one K_6 in each of the two chiral shell orbits, with faithful S_6 component symmetry. The naive homogeneous-product scheduler is ruled out:

$$(120\text{ axes}) \times (36\text{ double-sixes}) = 3240 + 720 + 360 = 120(27 + 6 + 3).$$

Thus a packet carrying an axis/double-six pair needs a three-valued relation-type tag; the 4320 pairs are not one native group orbit. The positive replacement is local: an axis stabilizer of order 216 acts on its distinguished three double-sixes through full S_3 with kernel 36, giving a genuine S_3/C_2 completion fibre.

Restoring signs gives an even sharper codec. The three positive octads of each signed second-shell trade contain a unique two-point edge of its matched four-point line; the three negative octads contain the complementary edge. Hence

$$240\text{ signed trades} \longleftrightarrow 40\text{ lines} \times 6\text{ edge addresses},$$

with sign reversal implemented by edge complement. The line stabilizer satisfies $1 \to C_3^3 \to H_{\text{line}} \to S_4 \to 1$: its six signed states carry the tetrahedral-edge S_4 action, while complement-pairing gives the unframed S_3/C_2 axis action. Firmware must therefore not identify the six edge states with a regular S_3 clock merely because both sets have size six. (Witnesses: `analysis/w33_pass188_icosahedron_test.py`, `analysis/w33_pass191_supercycle_pullback.py`, and `analysis/w33_pass192_signed_trade_edge_s4.py`.)

Passes 209–212 close the global controller carrier without undoing that local warning. A route-shell double-six D now determines its timetable intrinsically:

$$\Sigma(D) = \{\ell : \ v_\ell = 0 \text{ for every signed minimum } v \in D\}.$$

The ten lines in $\Sigma(D)$ are pairwise skew and cover all 40 points; the 36 double-sixes recover all 36 spreads, equivariantly and bijectively. On the crossing-pair six-set of D, its ten silent lines are the ten unordered $3+3$ bisections, by a unique S_6-equivariant atlas. The associated doily is the derived duad–syntheme geometry on that six-set, not a literal incidence subgeometry on the twelve crown vertices.

This bridge gives the controller an exact two-stratum ABI. For a silent spread incidence (Σ, ℓ), the common stabilizer has order 72 and shape $(S_3 \times S_3){:}C_2$; its line-clock image is C_2 with kernel $S_3 \times S_3$. For an active pair $\ell \notin \Sigma$, the stabilizer is S_4, faithful on the four points of ℓ, and its line-clock quotient is $S_4/V_4 \cong S_3$. Thus the silent and active rows require an explicit stratum bit: they are two equivariant clock modes, not one direct-product clock.

Most importantly, the previously numerical 4320-carrier coincidence is now a canonical $\mathrm{PSp}(4,3)$-equivariant bijection. Let

$$X = \{(\Sigma, L, p) : L \notin \Sigma,\ p \in L\}, \qquad |X| = 1080 \cdot 4 = 4320.$$

For (Σ, L, p), let M be the unique line of Σ through p and let N be the unique line meeting M, disjoint from L, whose four Σ-owner lines equal those of L. Then

$$(\Sigma, L, p) \longmapsto (L, M, N)$$

is the required bijection to ordered nonlocal two-paths; the inverse chooses the unique such Σ and returns $p = L \cap M$. The path has three quadrangle completions, canonically indexed by the other three points of L, so the complete incidence surface is

$$1080 \cdot 4 \cdot 3 = 12960.$$

In the full $\mathrm{PGSp}(4,3)$ supercycle the path stabilizer is the split group $S_3 \times C_2$ of order 12: S_3 permutes the three completions, its central kernel has order two, and fixing one completion leaves V_4. That V_4 is an exact four-element branch carrier, but it is *not* a regular four-probe address on the completion line: the four points have orbits $1+1+2$. Firmware must supply any semantic probe-role labelling separately rather than infer it from group order. (GAP witnesses: `analysis/w33_pass209_silent_spread_two_clock.g`, `analysis/w33_pass210_s6_route_atlas.g`, `analysis/w33_pass211_pgsp_controller_clifford.g`, and `analysis/w33_pass212_4320_carrier_equivariant_bijection.g`.)

Passes 214–218 now separate the remaining controller sheets before any optical interpretation. The four points of the *source* line do carry a regular V_4 action: after choosing the source point, the identity fixes that port and the three nonidentity double transpositions address the other ports through their intrinsic pair-partition labels. This is the exact four-way switching algebra that the completion line lacked, but the residual S_3 still permutes the three nonidentity labels. A photonic controller must calibrate semantic port names; geometry supplies the torsor, not the laboratory naming convention.

The same carrier sheet selects the local-axis endpoint formed by the external line and its unique spread owner through the source point. GAP verifies the uniform tower

$$4320 \xrightarrow{18:1} 240 \xrightarrow{2:1} 120,$$

where endpoint complement is a fixed-point-free C_2 deck operation central in the extended order-103680 action. Via Pass 123's chosen chamber gauge it can be represented by an E_8 root sign; it is not a measured optical phase. Moreover the transitive W33-code action and the standard $E_6 \times A_2 < E_8$ branching action are nonconjugate and have different root-orbit fingerprints, so choosing rational E_8 coordinates remains a chamber calibration rather than a symmetry-forced device frame.

Incidence duality also changes the schedule type. $W(3,3)$ has 36 spreads and no ovoids, while $Q(4,3)$ has no spreads and 36 ovoids; the duality-odd checksum is therefore $\Delta_{SO} = \#\text{spreads} - \#\text{ovoids} = \pm 36$, not an Euler characteristic. The surviving Q carrier is the ovoid tower $4320 = 36 \cdot 15 \cdot 2 \cdot 4$ with stabilizers $S_6 > C_2 \times S_4 > S_4 > S_3$. It is the incidence-dual typing of the same controller, not a second independent optical channel, and its central mate swap is not an absolute handedness observable.

The construction is also special to field order three. At the tested odd anchors the regular-spread owner router covers the path carrier with degree $(q-1)/2$: $1, 2, 3$ at $q = 3, 5, 7$, so only $q = 3$ is lookup-free and bijective. The module backend changes at the same anchors: the $q = 5$ binary 24 is the $\mathbb{F}_2$ restriction of an $\mathbb{F}_4$ Weil 12, while the $q = 7$ 48 is the split dual pair $24 \oplus 24^*$. These are representation and control-plane constraints, not claims of fabricated modes or measured coherence. (GAP witnesses: `analysis/w33_pass214_source_line_v4_torsor.g`, `analysis/w33_pass215_carrier_double_six_signed_e8.g`, `analysis/w33_pass216_q43_dual_ovoid_carrier.g`, `analysis/w33_pass217_w3q_owner_spread_uniqueness.g`, and `analysis/w33_pass218_weil_shadow_split.g`.)

65 The carrier: how to self-entangle a photon

Entanglement is a relation between tensor factors; nothing requires the factors to be distinct particles. Self-entanglement is entanglement between one photon's own registers.

65.1 Stage A: spatial (one element)

A diagonal photon meets a polarizing beam splitter: $\frac{1}{\sqrt{2}}(|H\rangle + |V\rangle) \mapsto \frac{1}{\sqrt{2}}(|H, a\rangle + |V, b\rangle)$. Path and polarization are now maximally entangled *within* the photon; this two-qubit $\mathbb{C}^4$ is the Witting carrier of Theorem 64.2(i).

65.2 Stage B: temporal (two Clifford gates)

A tritter (symmetric 3-port coupler) *is* the qutrit Fourier gate F_3; a three-bin delay ladder $(0, \tau, 2\tau)$ defines past and future registers; one electro-optic modulator applies $\mathrm{CX}_{p\to f}$ conditioned on bin index. The temporal Bell qutrit

$$|\Omega\rangle \;=\; \mathrm{CX}_{p\to f}\,(F_3 \otimes I)|0\rangle_p|0\rangle_f \;=\; \frac{1}{\sqrt{3}}\sum_{j=0}^{2}|j\rangle_p|j\rangle_f$$

is verified exactly; the photon is entangled with its own future. Inserting any U in the future arm, the recombination visibility is the *trace–Choi witness*

$$V(U) \;=\; \frac{|\mathrm{Tr}\,U|}{3}, \qquad V(F_3) = \tfrac{1}{3}, \quad V(X) = V(Z) = 0,$$

so the photon implements the Choi–Jamiołkowski isomorphism on itself: it measures channels against its own past. *Witness:* `bt820_self_entanglement_protocol.py`.

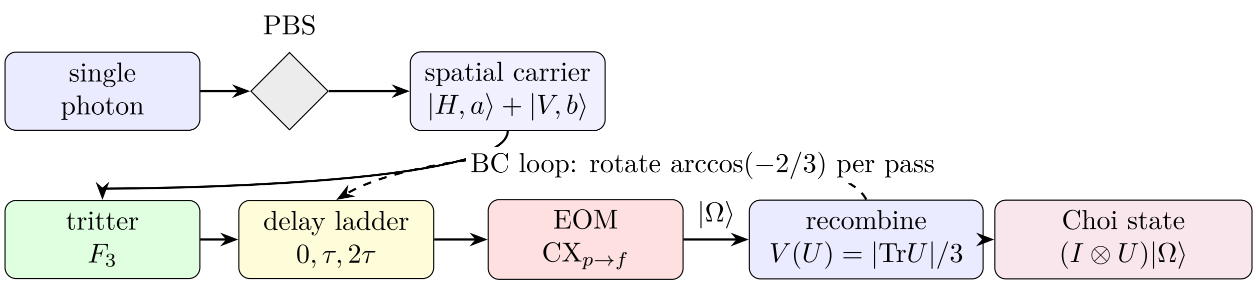


Figure 3: Preparation schematic. One photon, four registers. The PBS entangles path with polarization (Stage A); the tritter–delay–EOM chain prepares the temporal Bell qutrit $|\Omega\rangle$ (Stage B); the dashed loop is the irrational Boerdijk–Coxeter drive of §73.

66 The network: atlas, building, and routing

Theorem 66.1 (The hypercube atlas)**.** *$W(3,3)$ contains exactly 540 skew line pairs. For each pair the non-collinearity graph on its 8 points is the cube graph Q_3, with full symmetry group O_h of order 48 (these are the only order-48 stabilizers; the binary $2O$ does not occur). Each chart admits an exact $\mathbb{F}_2^3$ Gray-code addressing in which chart edges are unit XOR moves and the antipode of a vertex is its unique collinear partner. Every $W(3,3)$ nonedge is a chart edge in exactly $12 = k$ charts; every edge is an antipode pair in exactly $9 = q^2$.* Witnesses: `bt773`, `bt777`, `bt778`.

66.1 Hypercube networking, not hypercube decoration

In plain language. *A warning against a tempting shortcut. The network's addresses look like the corners of a hypercube, which invites reusing standard hypercube routing — and that would be wrong here, because the geometry constrains which moves are legal in a way a hypercube does not. This section shows where the two diverge.*

The cube is not only a picture for one chart. It is the local interconnect primitive. A chart gives eight simultaneously valid addresses

$$000,\ 001,\ 010,\ 011,\ 100,\ 101,\ 110,\ 111 \in \mathbb{F}_2^3,$$

and the elementary network instruction is the dimension-order hypercube hop

$$a \longmapsto a \oplus e_i, \qquad i \in \{1, 2, 3\}.$$

This is the standard e-cube rule of hypercube networks: compare source and target, flip one differing coordinate at a time, and finish in at most three local hops [13]. In the holonet that same XOR flip is also a register operation, because the chart address is a finite "which-register" word on the photon's self-entangled carrier. The local routing primitive is therefore simultaneously

$$\text{bit flip} \quad = \quad \text{qutrit-frame update} \quad = \quad \text{optical path choice}.$$

A packet in the photonic holonet is not merely a payload. The natural packet header is

$$\mathcal{P} = (\text{payload},\ \text{Pauli frame},\ \text{chart word},\ \text{apartment hop},\ \text{mirror slot},\ \text{clock phase}).$$

The payload is the quantum state or teleported gate; the Pauli frame is the two-trit feedforward record; the chart word is the $\mathbb{F}_2^3$ route inside the current cube; the apartment hop selects the next chart; the mirror slot chooses the D_{12} transport bus of §67; and the clock phase is the cyclic C_{12} interrupt that keeps the selector coherent. This is why the same object can be called a computer architecture and a network architecture. A node never sends data to a processor: the act of transport is the processor's gate action.

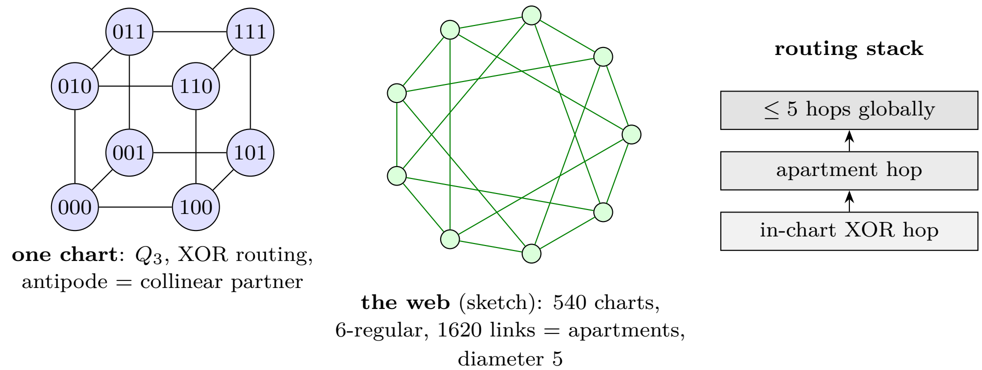
Theorem 66.2 (The fabric is the building)**.** *The chart-adjacency web (*540 *nodes, two charts adjacent when one's axis is a disjoint transversal pair of the other) is* 6*-regular with exactly* 1620 *edges, and these edges are in canonical bijection with the* 1620 *apartments of the Tits building of* $\mathrm{Sp}(4, \mathbb{F}_3)$*. The web has diameter* 5*; its adjacency module contains the Steinberg representation twice and presses against the Ramanujan bound only on a* 15*-dimensional sentinel eigenspace.* Witnesses: `bt777`, `bt779`, `bt744`.

Routing a packet means choosing XOR moves inside charts and apartment hops between them; since chart moves *are* register operations (§68), routing is computation. At scale the recursion is the holonet ladder: 40^n cores at level n, the same $\mathrm{SRG}(40, 12, 2, 4)$ at every level, with the level-0 chart's $1 + 12 + 27$ degree split realized as self pole, gauge shell, and matter shell of each node.

Corollary 66.3 (Two-plane routing bound)**.** *Inside a chart the worst route length is* 3*, the diameter of* Q_3*. Between charts the worst route length is* 5*, the diameter of the apartment web. One recursive digit of a holonet address therefore costs at most*

$$3 + 5 = 8$$

reversible transport moves. BT827 globalizes this to $8n = 8 \log_{40} N$ *for* $N = 40^n$ *photonic leaves.*



one chart: Q_3, XOR routing, antipode = collinear partner

the web (sketch): 540 charts, 6-regular, 1620 links = apartments, diameter 5

Figure 4: Network schematics. Left: a single chart with its $\mathbb{F}_2^3$ addressing (XOR routing native). Center: the 540-chart web whose links are the apartments of the Tits building (sketched as a circulant fragment). Right: the two-level routing stack.

67 The middleware: tomotope mirrors and the runtime bus

In plain language. *The layer between the geometry and the running program — the part that would be an operating system in a conventional machine. It routes, schedules and checks, and it is built from the same object as everything below it rather than being written on top.*

Definition 67.1 (Tomotope packet)**.** *In this paper the tomotope is used in the only way the finite calculation currently justifies: as a local middle-layer incidence packet. A tomotope packet has four vertex carriers, twelve local edge axes, sixteen triangular face labels, and forty-eight edge–face incidence blocks. The doubled base/shadow fiber gives* 192 *flags. Thus the tomotope is the durable packet format sitting between the fast cube router and the global mirror bus; it is not introduced as an ambient continuous torus, and it is not identified with the rectangle clock.*

The atlas contains two different 2160-element boundary worlds. They must not be collapsed. The rectangle side is cyclic: its stabilizer is C_{12} and it supplies the selector's phase clock. The chart-transversal side is dihedral: its stabilizer is D_{12} and it supplies the mirror transport bus. Equal cardinality is the red herring; different stabilizer structure is the theorem.

Theorem 67.2 (The D_{12} mirror bus)**.** *The* 540 *cube charts carry exactly four common-transversal slots each, so the global chart-transversal space has* $540 \cdot 4 = 2160$ *slots. The map*

$$(\textit{chart},\ \textit{common transversal line}) \longmapsto (\textit{chart},\ \textit{antipode pair cut out by that transversal})$$

is an equivariant bijection onto the atlas antipode-slot space. Its projective stabilizer has order 12 *and GAP identifies it as* D_{12}*, not* C_{12}*.* Witness: `bt815_global_2160_transversal_gset.py`.

Locally, a skew-line chart has four common isotropic transversals. Read those four transversals as the four vertex carriers of the tomotope middle layer. Each transversal is a K_4, and each K_4 has three opposite-edge axes and four triangular faces. Thus the chart carries

$$4 \text{ vertices}, \qquad 4 \cdot 3 = 12 \text{ edge labels}, \qquad 4 \cdot 4 = 16 \text{ face labels},$$

and the middle incidence packet has

$$4 \cdot 3 \cdot 4 = 48$$

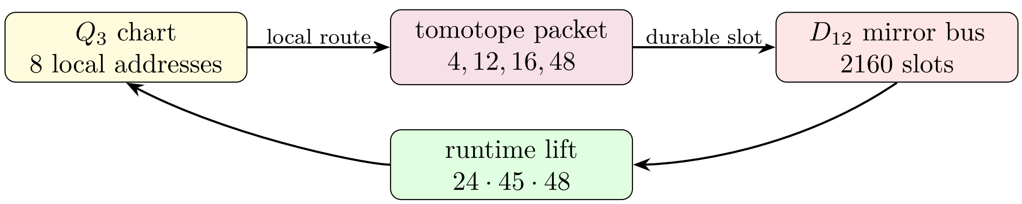
edge–face blocks, with the doubled base/shadow fiber restoring 192 flags. This is the local $\langle r_0, r_3\rangle$ tomotope block layer, not an analogy to a toroidal polyhedron. *Witness:* `bt814_tomotope_middle_layer_from_residual_tetrahedra.py`.

Theorem 67.3 (Photonic mirror middleware)**.** *The full optical Clifford runtime factors as*

$$|\mathrm{Sp}(4, \mathbb{F}_3)| = 51840 = 24 \cdot 2160 = 24 \cdot 45 \cdot 48.$$

Here $24 = f$ is the full $\mathrm{Sp}(4, 3)$ lift of the projective D_{12} mirror-slot stabilizer, 45 is the hyperbolic polar-pair/tritangent geography, and 48 is the local tomotope edge–face middle layer (also the order of the chart group O_h). Equivalently, the photon is clocked by the cyclic selector side and routed by the dihedral mirror side; the complete Clifford group is the lift of that mirror bus through the tomotope middle carrier. Witness: `bt826_photonic_mirror_middleware.py`.

This is the architectural closure missing from a bare universality claim. The tritter, phase plate, and controlled delay generate the whole Clifford group, but the group is *organized* as a middleware stack: full slot stabilizer, global mirror bus, polar vacuum geography, and tomotope local block. In software terms, C_{12} is the clock interrupt, D_{12} is the mirror router, 48 is the local packet format, and $24 \cdot 45 \cdot 48$ is the finite runtime.



fast XOR routing is committed through the tomotope middle layer and reflected by the mirror bus.

Figure 5: Middleware schematic. The cube is the reversible local router; the tomotope is the edge–face packet format; the D_{12} bus is the global mirror transport object.

67.1 Product-grid closure ABI v2

The 12 closure strands are modeled as

$$C_3 \times V_4, \qquad V_4 = C_2 \times C_2,$$

rather than as a bare C_{12}. The C_3 coordinate is the Szilassi/Fano channel and qutrit phase axis; the V_4 coordinate is the four-triangle E6 gauge-shell/D4 branch axis.

The E6 firewall square is now upstream in the claim DAG through the exact nodes

$$36 \to 72, \qquad 72 + 6 = 78, \qquad 72 + 9 = 81.$$

This supports the exact finite ABI/CSS branch while leaving blocked formula and physics claims downstream of the transcription gate.

The closure packet ABI is upgraded to v2. Each packet carries C_3 channel, V_4 branch, active/-guard outputs, 72-row sector metadata, the nine-mode firewall gap, and claim-DAG dependencies.

67.2 ABI v2 consumer, E6 claim table v2, and D4/V4 branch audit

The closure ABI v2 is executable. It regenerates 12 packets, 48 events, and 72 rows while preserving the dual-axis metadata. The three C_3 channels each carry 24 rows, and the four V_4 triangles each carry 18 rows.

The E6-merged claim DAG now produces a table with explicit E6 provenance for the 72-sector, the 81 closure, and the $C_3 \times V_4$ grid. Blocked formula and physical claims remain downstream.

On branch classes, D4 shear fixes each V_4 triangle, while τ_4 acts as an order-two translation preserving the triangle partition as a set. The two operations still fail to commute on full branch/phase states, so retwining remains part of the exact decoder rule.

67.3 BT1486–BT1488 ABI v2 retwined CSS and splice manifest

BT1486 reruns the CSS join from the ABI v2 consumer output. The result is not just the old 72-row count: every row satisfies the retwined X- and Z-syndrome checks, while the C_3 channel profile remains $P_0 = P_1 = P_2 = 24$ rows and the V_4 triangle profile remains $T_0 = T_1 = T_2 = T_3 = 18$ rows.

BT1487 classifies the branch symmetries behind those four triangles. The full triangle-partition stabilizer is S_4 of order 24, giving the Fano point reading $7 \cdot 24 = 168$. The physically used square subgroup is D_4 of order 8, giving the Fano flag reading $21 \cdot 8 = 168$. Thus the active detector-bin count is Fano-native, not merely an optical count.

BT1488 updates the splice cascade: the E6 claim table v2 from BT1484 supersedes the BT1472 table as the preferred paper table, and the preferred exact finite insert packet is now BT1474, BT1480, BT1482, BT1484, BT1486, and BT1487. The Golden Quartic/Moebius-ball equation fill remains blocked pending rendered formula transcription; no public source-code repository is assumed.

67.4 BT1489–BT1491 row symmetry and Fano–E6 square

BT1489 lifts the S_4, D_4, and V_4 branch actions from triangle labels to the actual ABI v2 row layer. All 24 elements of S_4, all 8 elements of the square D_4 subgroup, and all 4 V_4 translations act as honest permutations of the 72 active/guard value rows. The lift preserves the C_3 channel, row kind, qutrit value, guard slot, and column formula, so the branch symmetry is now a decoder-row symmetry.

BT1490 is the new finite-geometry lock:

$$24 = 4 \cdot 6 = 3 \cdot 8, \qquad 72 = 3 \cdot 24, \qquad 168 = 7 \cdot 24 = 21 \cdot 8, \qquad 81 = 72 + 9.$$

The shared 24-state fiber is V_4 branch times six row-value slots on the ABI side, and three local Fano arms times the D_4 flag stabilizer on the Fano side. Thus the same finite fiber feeds the E6/CSS 72-sector, the 81-closure after adjoining the $q^2 = 9$ firewall gap, and the Fano-native 168 active-bin bus.

BT1491 makes the paper splice idempotent. The main holonet paper now imports the BT1480–BT1491 exact finite packet through one controlled block; rerunning the splicer replaces that block rather than duplicating it. The claim firewall remains intact: this is an exact finite ABI and incidence statement, not a detector calibration, optical-layout claim, or imported Golden Quartic/-Mobius particle interpretation.

67.5 BT1492–BT1494 canonical Fano fiber and pulse release lock

BT1492 removes the last arbitrary index from the 24-state bridge. In the Fano plane, $\mathrm{GL}(3,2)$ has order 168. Fixing one point leaves a point stabilizer of order 24, and this stabilizer acts as the full S_4 on the four Fano lines not through that point. Fixing one incident flag leaves an order-8 flag stabilizer with the D_4 order profile. Therefore the shared fiber is canonical:

$$24 = 3 \cdot 8,$$

namely three Fano lines through the anchor point times the D_4 flag stabilizer.

BT1493 compiles those row symmetries down to the existing physical holonet interfaces. A row action first selects a BT1411 detector slot, then hands that slot to the BT1374 rule mirror_slot mod 4, and finally occupies the matching BT1407 Hesse epilogue lane for its C_3 channel and row-value slot. All 24 S_4 actions compile across all 72 rows, while the order-8 D_4 square subgroup is the native square-pulse layer.

BT1494 repairs the broad photonic-QEC release lock. The legacy tests expected selected `PART_*` artifacts at the repository root, while the preserved copies lived under `manuscripts/parts`. The repair restores those root release-lock targets and regenerates the live scheduler, harmonic bus, fusion splice, and projective screen/bulk/QEC artifacts.

68 The software: braids, teleported gates, universality

68.1 Exact braid words

In the anyonic Fibonacci representation $\sigma_1 = \mathrm{diag}(\zeta^4, -\zeta^2)$ ($\zeta = e^{i\pi/5}$) one has *exactly* $\sigma_1^5 = Z$ and $\sigma_1^{10} = I$ — not merely projectively (verified in the cyclotomic ring $\mathbb{Z}[\zeta_{10}]$). In the dual ($\pm$) encoding of the local $K_{3,3}$ cycle code $\mathbb{F}_2^4$ this makes every register bit-flip an exact five-letter braid word, and the flat global register (the unique connected gluing with trivial holonomy across the atlas) carries the action coherently: a zero-Berry-phase memory. *Witnesses:* `bt740`, `bt741`.

68.2 The photon is its own gate

A gate U is stored as the self-entangled state $(I \otimes U)|\Omega\rangle$; Bell-projecting an input register against the photon's *past* register outputs U (times a known Pauli correction) on the *future* register — verified for all nine outcomes. The machine's gate library is a library of self-entangled states; executing software means interfering the photon with its own history. *Witness:* `bt821`.

68.3 Instruction set architecture

The practical instruction set is finite:

$$\mathcal{I}_{\mathrm{holo}} = \{F_p, F_f,\ S_p, S_f,\ \mathrm{CX}_{p\to f},\ \sigma^5 = Z,\ D_{12}\text{-mirror},\ M_{36}\text{-magic}\}.$$

F and S generate single-qutrit Clifford frame changes; CX couples past to future; $\sigma^5 = Z$ gives exact braid-register flips; the D_{12} mirror operation moves a packet through the atlas without confusing it with the cyclic selector clock; and M_{36} denotes injection from the thirty-six magic rays of §71. The Clifford part normalizes Pauli frames and is efficiently classically trackable; it is not universal by itself. Universality starts only when the Clifford runtime can consume the intrinsic magic sector. This is the engineering meaning of "matter equals magic": the same thirty-six rays that appear as the matter shell are the non-Clifford fuel for the processor.

68.4 The minimal classical shadow

The smallest classical headline compatible with the substrate is the pair

$$(\lambda, q) = (2, 3).$$

This is the same two-state, three-symbol signature made famous by the Wolfram–Smith weakly universal Turing machine. The present paper does not use that result as the proof of quantum universality; the proof is the Clifford closure plus magic/contextuality [17, 16]. The point is sharper: the classical shadow of the photonic holonet lands on the same minimality boundary. The quantum machine has forty rays and an $81 = q^4$ protected register, but its smallest visible finite-state control alphabet is already the substrate pair $2, 3$.

68.5 The Universality Theorem

Theorem 68.1 (Clifford completeness from optics)**.** *The symplectic images of the photon's physical elements — the tritter F_3 on either register, the quadratic phase plate S, and the delay-conditioned* CX *— generate all of* $\mathrm{Sp}(4, \mathbb{F}_3)$ *(closure of order* 51840*, exact). Hence tabletop optics realize the complete two-qutrit Clifford group modulo Paulis and phases.* Witness: `bt825_universality_theorem.py`.

Theorem 68.2 (Conditional universality criterion)**.** *Clifford completeness, together with a fault-tolerant preparation and injection of a distillable qutrit magic state, gives a universal gate set. The finite substrate certifies the Clifford closure and a nonclassical contextuality obstruction. It does not, from those facts alone, certify that the thirty-six distinguished rays are physically preparable magic states or that a single photon supplies a protected injection circuit.*

Evidence boundary. Contextuality is a resource diagnostic, not a compiled optical protocol. Promoting the conditional criterion to a hardware universality theorem requires an explicit state preparation, injection gadget, decoder, and noise threshold for the chosen photonic encoding. Accordingly, later "matter equals magic" and unit-premium statements are architectural hypotheses unless they name those missing maps; the exact finite results do not establish them.

69 Fractal scaling: the computer is the network

The single-core theorem is not the end of the architecture. The substrate is self-similar: any node may be replaced by a fresh copy of the same forty-node holonet, with the parent line/point incidence becoming the routing grammar for the child layer.

Definition 69.1 (Recursive holonet)**.** *Let $\mathcal{H}_0$ be one self-entangled photonic qutrit. Define $\mathcal{H}_n$ recursively by replacing each point of one copy of $W(3,3)$ with a copy of $\mathcal{H}_{n-1}$. Thus $\mathcal{H}_1$ is the single-core machine of this paper and $\mathcal{H}_n$ is a* 40*-ary fractal computer/network.*

Theorem 69.2 (BT827 holonet scaling law)**.** *A level-n holonet has*

$$N_n = 40^n$$

photonic leaf cores and

$$I_n = 1 + 40 + \cdots + 40^{n-1} = \frac{40^n - 1}{39}$$

total $W(3,3)$ instances. Its total edge-qutrit slots are $240I_n$, its directed transport slots are $480I_n$, its chart routers are $540I_n$, its apartment links are $1620I_n$, its mirror slots are $2160I_n$, and its local Clifford runtime atoms are $51840I_n$. Worst-case reversible routing across the recursive address uses at most

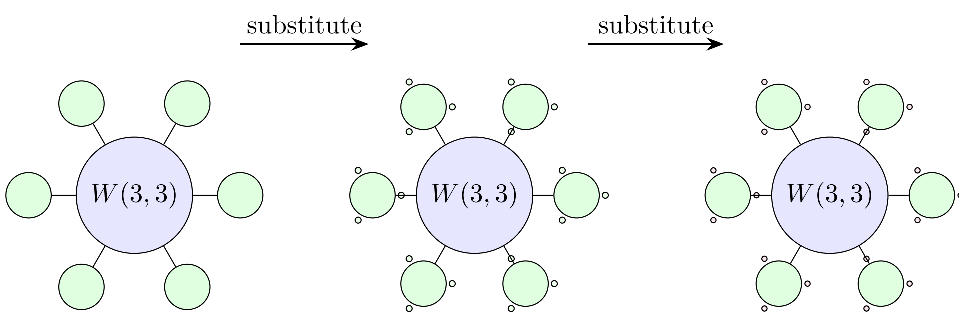
$$8n = 8\log_{40} N_n$$

moves, where $8 = 3+5$ is one in-chart hypercube diameter plus one chart-web apartment diameter. Durable commits use the distinct Császár/tomotope clock

$$T(0) = 1, \qquad T(g) = 4(7^g - 1) \quad (g \geq 1).$$

Witness: `bt827_holonet_fractal_architecture.py`.

level n	leaves 40^n	$W(3,3)$ instances	mirror slots	runtime atoms	route bound
1	40	1	2160	51840	8
2	1600	41	88560	2125440	16
3	64000	1641	3544560	85069440	24
4	2560000	65641	141784560	3402829440	32



$\mathcal{H}_1$: 40 leaves $\quad$ $\mathcal{H}_2$: 40 cores of 40 leaves $\quad$ $\mathcal{H}_n$: repeat the same grammar

Figure 6: Fractal scaling schematic. The drawn fans show only a few children; the rule is exact substitution of forty child holonets at each substrate point. Routing grows logarithmically in the number of photonic leaves because each recursive digit costs at most eight reversible moves.

This resolves the computer-engineering question. A conventional machine grows by adding processors connected by a separate network. The holonet grows by adding copies of the same object; each copy contains its own processor, router, memory, clock, and packet format. There is no von Neumann bus to saturate. At every scale, the edge set is both communication bandwidth and entangling-gate availability, the chart atlas is both routing table and instruction decoder, and the Steinberg 81-sector is both protected memory and fault-tolerant logical space.

70 Runtime closure: compile, monitor, commit

The architecture now has an executable operating-system loop.

$$\text{address word} \longrightarrow \text{packet route} \longrightarrow \text{sentinel monitor} \longrightarrow \text{durable tomotope commit.}$$

BT828–BT831 make each arrow finite and testable.

Theorem 70.1 (BT828 packet compiler)**.** *A recursive address word with digits in* $\{0, \ldots, 39\}$ *compiles to a sequence of digit packets*

$$(Q_3\text{-}XOR\ hops,\ apartment\ hops,\ D_{12}\ mirror\ slot,\ C_{12}\ phase,\ tomotope\ block).$$

Each digit uses at most three XOR hops and at most five apartment hops, hence at most eight reversible moves. The stress route

$$(0, 7, 14, 21, 28, 35) \longrightarrow (39, 32, 25, 18, 11, 4)$$

compiles to 47 *reversible moves, below the level-six bound* 48*.* Witness: `bt828_holonet_packet_compiler.py`.

Theorem 70.2 (BT829 sentinel monitor)**.** *The projector onto the* $g = 15$ *adjacency sector of* $W(3, 3)$ *has exact entry profile*

$$P_{-4}(x, x) = \frac{3}{8}, \qquad P_{-4}(x, y) = \begin{cases} -\frac{1}{8}, & x \sim y, \\ \frac{1}{24}, & x \nsim y. \end{cases}$$

Consequently a full context line K_4 *is sentinel-invisible, while localized, mirror, and shell faults activate the sentinel with exact energies:*

fault	*energy*	*normalized fraction*
single point	3/8	5/13
adjacent edge	1/2	5/19
nonedge mirror pair	5/6	25/57
gauge shell $N(x)$	6	5/7
matter shell	27/8	5/13

The immune system is therefore not generic noise detection: it is blind to legal context-line traffic and tuned to off-context routing and shell congestion. Witness: `bt829_fault_sentinel_monitor.py`.

Theorem 70.3 (BT830 two-phase commit clock)**.** *The runtime has two clocks. The reversible prepare phase finishes in at most* $8n$ *moves. The durable commit phase waits for*

$$T(n) = 4(7^n - 1).$$

For $1 \le n \le 24$*,* $T(n)$ *is always divisible by* 24 *and by* 8*, so it is aligned with both the local runtime stabilizer and the single-digit route window. But it is not always divisible by the full* $8n$ *route epoch: the first desynchronization is* $n = 5$*, with remainder* $24 = f$*. Thus the commit layer is deliberately slower and cover-like; it is not merely the route clock renamed.* Witness: `bt830_two_phase_commit_clock.py`.

70.1 The infinite-cover warning

The tomotope contributes a harder boundary than the earlier packet picture showed. Monson, Pellicer, and Williams prove that the tomotope has infinitely many distinct minimal regular covers; for coprime odd $p, q > 1$ there are minimal covers R_p, R_q of the tomotope such that neither covers the other, and neither covers the special R_2 cover [20]. They also record the toroidal cover arithmetic

$$|\mathrm{Mon}(Q_k)| = 36864\, k^6, \qquad Q_k :\ (4, 24, 32, 8, 4)k^3$$

for vertices, edges, triangles, tetrahedra, and octahedra.

Architecturally this is a big correction. The 48-block tomotope middle layer is a stable local ABI, but the global regular cover is not unique. Durable storage must therefore carry a *cover index k* as an implementation gauge. Fast routes compile to the same invariant packet, while persistent storage may choose one minimal cover family without invalidating the other non-comparable families. *Witness: `bt831_tomotope_minimal_cover_architecture.py`.*

70.2 Cover-indexed storage, rerouting, and guard bands

In plain language. *Where data physically sits and what happens when part of the machine fails. Storage locations are indexed by the geometry rather than by an address counter, which means a failure has a shape: the states lost when one component dies are a geometrically describable set, and rerouting can be planned against it in advance.*

The infinite-cover warning would be useless as engineering unless it changed the runtime. BT832–BT834 make it operational. The principle is that the local tomotope packet is the ABI and the chosen regular cover is the storage backend.

Theorem 70.4 (BT832 cover-indexed durable storage)**.** *For every supported cover index k, durable storage is the lifted packet space*

$$\{0,\dots,47\}\times\mathbb{Z}_k^3,\qquad |slots|=48k^3,$$

with projection back to the fast-route ABI by forgetting the $\mathbb{Z}_k^3$ coordinate. The kernel to the base tomotope packet has order

$$k^6=(k^3)^2,$$

matching the Monson–Pellicer–Williams toroidal cover law $|\mathrm{Mon}(Q_k)|=36864k^6$. Changing k changes durable capacity and kernel size, but not the packet program seen by the compiler. Tested cover gauges $k=3,5,7,11,13$ all reduce exactly to the same 48-block ABI, and larger k reduces the load fraction of the same stress program. Witness: `bt832_cover_indexed_durable_storage.py`.

Theorem 70.5 (BT833 sentinel-aware packet rerouting)**.** *The BT829 projector supplies a compiler-visible cost function. Before the durable commit, each digit route may insert up to two W33 waypoint points and minimize*

sentinel energy first, reversible move count second.

The extra reversible moves are charged to the slower BT830 commit phase, not to the BT827 fast $8n$ route bound. In the level-six stress program the direct route uses 47 moves and total sentinel energy 4; the sentinel-aware route uses 58 moves, still far below the 470592-tick commit window, and lowers the total sentinel energy to 2. Each tested digit route strictly lowers its own $g=15$ energy; adjacent endpoints can be context-completed to zero, and nonedge mirror pairs drop below their direct $5/6$ activation. Witness: `bt833_sentinel_aware_packet_rerouter.py`.

Theorem 70.6 (BT834 desynchronization guard-band arithmetic)**.** *The two-phase clock has an exact number-theoretic boundary. Full route-epoch synchronization is*

$$8n \mid T(n)=4(7^n-1),$$

equivalently

$$2n \mid 7^n - 1.$$

For every odd prime power $p^a \mid n$ *with* $p \neq 7$*, this requires* $\mathrm{ord}_{p^a}(7) \mid n$*; if* $7 \mid n$*, synchronization is impossible because the base is not coprime. The first failure is* $n = 5$*:* $\mathrm{ord}_5(7) = 4 \nmid 5$*, and the remainder is*

$$T(5) \bmod 40 = 24 = f.$$

Thus the first cover-index where durable storage and the full route epoch separate is not empirical drift; it is the $f = 24$ *guard band of the tomotope-backed runtime.* Witness: `bt834_desync_guard_band_arithmetic.py`.

Theorem 70.7 (BT838 tomotope Wythoff runtime ladder)**.** *The Grünbaum–Coxeter/tomotope operation tables give the local tomotope counts*

$$\textit{original, rectified, truncated, maximal expanded, omnitruncated} \quad = \quad 4,\ 12,\ 24,\ 48,\ 96.$$

These are exactly the runtime packet-growth layers:

$$4 = \mu \quad (\textit{vertex carriers}), \qquad 12 = k \quad (\textit{edge axes}), \qquad 24 = f \quad (D_{12}\ \textit{lift}),$$

$$48 = 2f \quad (\textit{BT814 packet ABI}), \qquad 96 = 4f \quad (\textit{half-flag boundary}).$$

The doubled omnitruncated boundary gives the verified full packet flags, $2 \cdot 96 = 192$*. After a cover lift this becomes*

$$48k^3 = \textit{BT832 lifted packet capacity}, \qquad 192k^3 = |W_k|,$$

the W_k *order already recorded in the tomotope cover architecture. Thus the Wythoff operations are not decorative variants; they are the compiler's packet expansion states:*

$$\textit{carrier} \to \textit{edge-axis} \to \textit{mirror lift} \to \textit{packet ABI} \to \textit{flag boundary.}$$

Witness: `bt838_tomotope_wythoff_runtime_ladder.py`.

71 The fuel: matter = magic

Theorem 71.1 (The triality)**.** *In the photon's two-qubit reading, exactly the* 4 *basis rays are stabilizer states and all* 36 ω*-rays are magic (no stabilizer state has support* 3*). Consequently three classifications coincide exactly, on rays* $[4, 36]$ *and on contexts* $\{4{:}1,\ 1{:}12,\ 0{:}27\}$*: the holonet vacuum split, the Schmidt (self-entanglement) strata, and the magic strata.* The matter shell is the machine's non-classical resource sector. Witness: `bt822`.

Theorem 71.2 (Three grades and the cube inside)**.** *Stabilizer fidelity splits the* 36 *magic rays into grades*

$$\underbrace{8}_{2^q}\ \textit{deep}\ \left(F = \tfrac{2+\sqrt{3}}{6}\right), \qquad \underbrace{24}_{f}\ \textit{mid}\ \left(F = \tfrac{5+2\sqrt{3}}{12}\right), \qquad \underbrace{4}_{\mu}\ \textit{shallow}\ \left(F = \tfrac{3}{4} = \tfrac{q}{\mu}\right),$$

the shallow grade sitting exactly at the Werner decoherence threshold. The 8 *deep rays are precisely the point set of a hyperbolic polar pair (the* 24*-cell vacuum axis), and the* 27 *fully-magic contexts stratify by grade signature as* $1+8+12+6$ *— the cell census of the cube.* Witnesses: `bt822`, `bt823`.

Theorem 71.3 (Exact contextuality budget)**.** *The maximum number of contexts a classical $\{0,1\}$ valuation can satisfy exactly-once is exactly $36 = (q!)^2$ (complete branch-and-bound over all 2^{40} markings): classicality saturates at the spread count, the deficit is exactly $\mu = 4$, and the contextual fraction is exactly $1/10 = 1/\Phi_4$. A perfect valuation would be an ovoid, and $W(3,3)$ has none: the true independence number is $\alpha = 7 = \Phi_6$, attained only by the beacon heptads of §74, against the Lovász bound $\vartheta = 10$ — the $\vartheta - \alpha$ gap $3 = q$ is the combinatorial face of Kochen–Specker.* Witnesses: `bt818`, `bt823`.

Theorem 71.4 (Conditional matter-equals-magic architecture)**.** *The exact finite inputs are the $36 = (q!)^2$ distinguished rays, their $8 + 24 + 4 = \{2^q, f, \mu\}$ grading, the contextual fraction $1/\Phi_4 = 1/10$, and the qutrit Strange state's mana $\log(5/3)$ (stabilizer states have mana 0). Separately, the homological register is the $[[240, 81, 3]]_3$ Steinberg code, equivalently $[[240, 81, 4, 3]]_3$ in the ordered convention $[[n, k, d_Z, d_X]]_3$. If a protected encoding and injection map identifies prepared copies of the distillable state with the distinguished-ray sector of that register, then one hardware layer could serve both error correction and non-Clifford injection. The counts and contextuality witness do not construct that map, prove replenishment, or remove a distillation factory.* Witness: `analysis/w33_magic_economy.py`.

Exact resource boundary from the QR branch (Passes 361–367). The theorem above is an architectural proposal for the qutrit/GKP branch; a count of contextual rays is not by itself a compiled injection circuit or a threshold theorem. The independently certified binary QR-137 branch makes that distinction concrete. Its one-block simple lift supplies only logical H; the multi-block construction supplies the real Clifford tower $O^+(2m, 2)$; and the first named phase lift is the weight-137 Clifford rotation $\exp(-\pi i \bar{Z}/4)$. That operator normalizes the exact stabilizer but is not yet transversal, local, low-weight, or proved fault-tolerant. Consequently the QR witnesses do not establish $P = 1$ or remove magic-state distillation. Doing so requires an explicit protected resource-state preparation/injection circuit, decoder, and noise-threshold analysis for the chosen physical branch.

The conditional resource accounting. Define the non-Clifford premium $P = \text{cost(non-Clifford)}/\text{cost(Clifford)}$. In the surface-code baseline a non-Clifford gate needs a distilled magic state, so $P \sim O(d^3)$ (the distillation factory, $\sim 10^3$–10^4 qubit-rounds at useful code distances d, occupying 30–90% of the device — the dominant cost of fault tolerance). In the proposed shared-layer implementation, $P = 1$ is a design target: it would require a protected injection to cost one correction cycle. Neither the equality nor an $O(d^3)$ saving follows until that injection, decoder, and threshold are supplied. (Witness: `analysis/w33_magic_resource_accounting.py`.)

What the irrational clock does and does not prove. The Boerdijk–Coxeter twist $\theta = \arccos(-2/3)$ — the $q = 3$ simplex angle, $-2/3 = -(q-1)/q$ — has θ/π *irrational* (Niven, since $\cos\theta = -2/3 \notin \{0, \pm\frac{1}{2}, \pm 1\}$), so the stroboscopic phases $\{n\theta \bmod 2\pi\}$ never repeat, are Weyl-equidistributed (star discrepancy $\to 0$), obey the Steinhaus three-gap law, and never land exactly on a stabilizer angle $2\pi k/q$. This proves aperiodicity, not gate type: an irrational phase sequence need not implement a cubic Hamiltonian, a distillable state, or a protected non-Clifford injection. Establishing replenishment requires an explicit driven Hamiltonian/control map and a classification of its logical action. (Witness: `analysis/w33_clock_magic_renewal.py`.)

And the code is holographic. The matter shell is not just a code but a *holographic* one. From any pole the 40 rays split $1 + 12 + 27$ = self + gauge boundary + matter bulk, and every bulk ray is screened by exactly $\mu = 4$ boundary rays. That strongly-regular screening multiplicity is a combinatorial redundancy, not the full CSS distance: the edge code has $d_X = 3$, $d_Z = 4$, and hence $d = 3$. It therefore corrects one arbitrary qutrit error or two known-location arbitrary erasures. The $\mu = 4$ neighbourhood by itself does not prove recovery from three arbitrary erasures; an operational bulk-reconstruction map would be additional data. The causal diameter is 2 (every bulk ray one hop from the boundary, two from the pole), the RT-like depth; and the logical register H_1 is the irreducible Steinberg module (dim $81 = q^4$), so there is no local logical operator — logical information lives only nonlocally on the boundary. Boundary redundancy, an RT-like causal depth, and no local logical representative are the proposed holographic signature: the "holo" in holonet is earned. (Witness: `analysis/w33_holographic_code.py`.)

Theorem 71.5 (Conditional self-fueling holographic-memory proposal)**.** *The exact finite inputs are the Steinberg-code dimensions and distances* $(k, d_X, d_Z) = (81, 3, 4)$*, screening multiplicity* $\mu = 4$*, the* 36*-ray contextual sector, the fraction* $1/10$*, the irrational angle* $\arccos(-2/3)$*, and the graph-cut counts* 12*,* 4*, and* $108 = 4 \cdot 27$*. Calling these one self-fueling holographic memory requires three additional maps: a bulk–boundary recovery channel, a protected resource-state injection, and a logical clock action that actually produces that resource. The current witnesses certify the finite inputs, not those identifications; in particular they do not prove* $P = 1$ *or eliminate distillation. (Witnesses:* `analysis/w33_self_fueling_memory.py`*,* `analysis/w33_holographic_rt.py`*.)*

Theorem 71.6 (Finite-curvature/de Sitter analogy and conditional cost model)**.** *The collinearity graph* $W(3,3)$ *has positive Ollivier–Ricci curvature — computed by optimal transport,* $\kappa = 1 - W_1(m_x, m_y) = 1/6 = 2/k$ *— the discrete finite identity* $E\,\kappa = 240 \cdot \frac{1}{6} = 40 = v$ *(Face 1). These exact facts motivate a de Sitter analogy; they do not identify the graph with spacetime, its cuts with horizons, or its clock with cosmological time. (Witness:* `analysis/w33_memory_is_desitter.py`*.)*

Conditional cost model. *If the missing protected injection consumes resource states at the contextual-fraction rate* $1/\Phi_4 = 1/10$ *per cycle, then a circuit with non-Clifford fraction* $f \leq 1/10$ *would avoid rate throttling, while* $f > 1/10$ *would require the model factor* $10f$ *in time or parallel cores. The finite fraction is exact; the replenishment law, factory removal, and claimed hardware saving remain conditional on the unbuilt compiler and threshold. (Witness:* `analysis/w33_self_fueling_cost.py`*.)*

72 Memory, protection, and the immune system

In plain language. *How the machine stores information without losing it. Quantum states decay, so a useful memory has to notice damage and repair it before the damage accumulates — the reason for the biological metaphor. What makes this section unusual is that the code doing the protecting is not designed and then bolted on: it is already present in the geometry, as the shape's own hole structure.*

Theorem 72.1 (The code register is the Steinberg module)**.** *The companion paper's CSS code* $[[240, 81, 3]]_3$*, with asymmetric distances* $(d_X, d_Z) = (3, 4)$*, stores its* 81 *logical qutrits in* H_1 *of the* $W(3,3)$ 2*-complex (*40 *vertices,* 240 *edges,* 160 *in-line triangles). Complete character computation over all* 25920 *group elements gives*

$$\langle \chi_{H_1}, \chi_{H_1} \rangle = 1.$$

Therefore $H_1 \otimes \mathbb{C}$ *is* irreducible *of dimension* 81*, hence equal to the unique* 81*-dimensional irreducible —* ***the Steinberg module*** *— with* $\chi_{H_1}(g) = \#\text{fixflags} - \#\text{fixpoints} - \#\text{fixlines} + 1$ *(the Solomon–Tits* *alternating sum) verified for every g. The QECC logical space and the Schur-protected register below are one object: any equivariant logical error is a scalar, so only symmetry-breaking noise can touch the code, and the sentinel/clock layers are exactly the symmetry monitors. The homological dictionary then completes: H_0 is trivial (the vacuum pole), H_1 is Steinberg (the register), and H_2 (dim 40, one tetrahedron-boundary 2-cycle per line) is the* sign-twisted *line module — a fixing symmetry acts on a line's 2-cycle by the parity of its induced S_4 action (verified for all 25920 elements;* not *the plain permutation module): the top homology is the* oriented *timetable carrier, feeding the chirality ledger.* Witnesses: `bt861_code_register_is_steinberg.py`, `bt862_h2_is_the_line_module.py`.

Corollary 72.2 (Three generations from Steinberg vanishing). *$\chi_{\mathrm{St}}(g) = 0$ iff $3 \mid \mathrm{ord}(g)$ (verified for all 25920 elements). Hence* any *order-3 symmetry of the substrate splits the matter register into eigenspaces of exactly $27 + 27 + 27$ — the three-generation decomposition is forced for every choice of triality element, not selected — and every order-9 element (5760 of them) refines it into nine 9-dimensional sub-generations. In the code's own characteristic the parameters survive:* $\mathrm{rank}_3 \partial_0 = 39$, $\mathrm{rank}_3 \partial_1 = 120$, $\dim H_1(\mathbb{F}_3) = 81$, *so the logical dimension of $[[240, 81, 3]]_3$ is exact over $\mathbb{F}_3$. The number of fermion generations is the defining-characteristic degeneracy of the protected register.* Witness: `bt863_generations_from_steinberg_vanishing.py`.

Theorem 72.3 (The dual-torsor Steinberg compiler). *Let $H_{27} = 3_+^{1+2}$ be the canonical Heisenberg O_3 of the point parabolic and let $A_{27} = \mathbb{F}_3^3$ be the canonical translation O_3 of the Bell-line parabolic. Their 27-point and 27-line shells are regular torsors (BT858). On the protected register,*

$$\mathrm{St}\downarrow_{H_{27}} = 3\,\mathbb{C}[H_{27}], \qquad \mathrm{St}\downarrow_{A_{27}} = 3\,\mathbb{C}[A_{27}],$$

because both restricted characters are exactly $\{81^1, 0^{26}\}$. The statement survives in the code's own field and is constructive:

$$H_1(W(3,3);\mathbb{F}_3)\downarrow_{H_{27}} \cong \mathbb{F}_3[H_{27}]^{\oplus 3}, \qquad H_1(W(3,3);\mathbb{F}_3)\downarrow_{A_{27}} \cong \mathbb{F}_3[A_{27}]^{\oplus 3}.$$

For each torsor the verifier finds three cycle seeds whose 27 group translates add $27 + 27 + 27$ independent classes modulo the 120-dimensional boundary space. Thus the same logical memory has a flat program basis and a curved state basis: A_{27} is the commuting three-trit address compiler, while H_{27} is the noncommuting qutrit-phase compiler.

The center $Z(H_{27}) = \langle z \rangle \cong C_3$ selects a canonical triality. Over $\mathbb{C}$ its central-character blocks are

$$H_1 = H_{1,1} \oplus H_{1,\omega} \oplus H_{1,\omega^2}, \qquad \dim H_{1,j} = 27.$$

The trivial block contains the nine linear characters with multiplicity 3; the two nontrivial blocks contain the conjugate 3-dimensional Schrödinger representations with multiplicity 9. Over $\mathbb{F}_3$, where $x^3 - 1 = (x-1)^3$, the phases coalesce into a canonical unipotent flag. With $N = z - I$,

$$(\mathrm{rank}\, N, \mathrm{rank}\, N^2, \mathrm{rank}\, N^3) = (54, 27, 0),$$

so

$$0 \subset \ker N \subset \ker N^2 \subset H_1, \qquad \dim = (0, 27, 54, 81),$$

has three successive quotients of dimension 27. Equivalently, z acts by 27 Jordan blocks of length 3. Complex generations are phase sectors; native qutrit generations are the three layers of one depth-3 memory stack.

The freeness is canonical, but the displayed orbit bases are not: H_{27} and A_{27} are nonisomorphic, and a basis-to-basis matrix still requires a chosen point, line, shell origin, and cycle seeds. No canonical intertwiner is claimed. Witness: `bt865_dual_torsor_steinberg_compiler.py`.

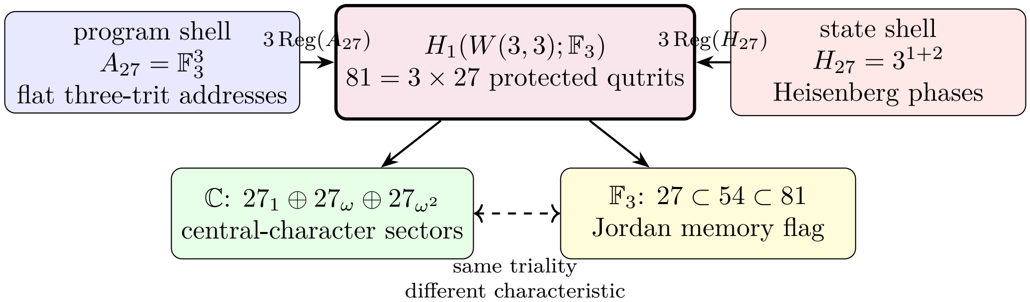


Figure 7: The dual-torsor compiler. Flat program translations and noncommutative state translations give two free rank-3 module coordinates on one Steinberg-protected register. The Heisenberg center reads as three phase sectors over $\mathbb{C}$ and as a three-step unipotent memory flag over $\mathbb{F}_3$.

Theorem 72.4 (Steinberg memory)**.** *The Levi-graph cycle space and the chart-overlap* 81*-sector are both irreducible and equal to the Steinberg representation of* $\mathrm{PSp}(4,3)$ *— the Solomon–Tits homology of the building, whose canonical basis is the* 81 *apartments through any chamber. Schur's lemma then forces the uniqueness of the chart–cycle bridge. Restricted to a Sylow* 3*-subgroup the module is free of rank one: the protected memory is one free generator over the substrate's ternary symmetry.* Witnesses: `bt742`, `bt744`.

Corollary 72.5 (The oriented timetable spectrum)**.** *Let* $P = 3^3{:}S_4$ *be the index-*40 *line parabolic. The ordinary line permutation module and the oriented top-homology module are the two inductions of its two linear characters:*

$$\mathrm{Ind}_P^{\mathrm{PSp}(4,3)}(1) = 1 \oplus 15 \oplus 24, \qquad H_2 = \mathrm{Ind}_P^{\mathrm{PSp}(4,3)}(\mathrm{sign}_{S_4}) = 5_\omega \oplus 5_{\omega^2} \oplus 30.$$

The two 5-dimensional characters are complex conjugates over $\mathbb{Q}(\zeta_3)$*; the 30 is rational. Under the outer extension* $W(E_6) = U_4(2){:}2$*, the conjugate pair is the restriction of one irreducible 10-dimensional representation, whereas the 30 has two inequivalent extensions. Thus the oriented timetable header reads as a Weyl-visible chiral channel* 10 *plus a payload* $30^\pm$ *whose outer parity is not fixed by the projective module alone.*

The completed homology spectrum is

$$H_0 = 1, \qquad H_1 = \mathrm{St}_{81}, \qquad H_2 = 5_\omega \oplus 5_{\omega^2} \oplus 30,$$

and resolves the Euler charge representation by representation:

$$1 - 81 + (5 + 5 + 30) = -40 = -v.$$

BT867 resolves the tempting cache comparison negatively: neither 162*-cache branch is one of the two 5-dimensional sectors. The branches are isomorphic copies with identical character overlaps; the outer involution acts on their multiplicity label. The possible identification of the two 30-extensions with BT857's local gauge bit remains open.* Witness: `analysis/bt866_h2_oriented_irreducible_decomposition.py`.

Theorem 72.6 (The cache/transport extension boundary)**.** *Let* $H = 3^3{:}S_4$ *be the order-*648 *Bell-line parabolic. The two* 162*-slot BT859 cache branches have conjugate cyclic stabilizers and are therefore isomorphic transitive* H*-sets,*

$$\mathcal{C}_L \cong H/C_4 \cong \mathcal{C}_R.$$

Their permutation characters are equal. In particular, their scalar products with the restrictions of the three BT866 characters $(5_\omega, 5_{\omega^2}, 30)$ *are both* $(1, 1, 8)$*: each cache sees both conjugate 5-sectors equally. "Left" and "right" are copy labels, not internal spectral chiralities.*

The parabolic contains one conjugacy class of 81 *cyclic* C_4 *subgroups, and*

$$N_H(C_4) = D_8, \qquad H/C_4 \longrightarrow H/D_8, \qquad 162 \longrightarrow 81$$

is the canonical free two-sheet cache cover. Both copies cover this same stabilizer base. On their union GAP computes

$$C_{\mathrm{Sym}(324)}(H) \cong D_8, \qquad 324 = 81 \times 4 = 81 \times 2_{\text{deck}} \times 2_{\text{cache}},$$

with exactly 81 *commutant orbits of size four. Thus the four-state fiber carries the symmetry of a square, not merely a numerical pair of bits.*

This cache carrier is not the older ternary 162*-sector. On one cache fiber the deck swap and its two projectors are*

$$D = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \qquad P_\pm = \frac{I \pm D}{2}, \qquad \mathbb{F}_3^2 = \operatorname{im} P_+ \oplus \operatorname{im} P_-.$$

Since 2 *is invertible in* $\mathbb{F}_3$*, the cache splits; indeed* $(D-I)^2 = D - I$*. By contrast, packet transport uses*

$$N = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}, \qquad N^2 = 0, \qquad (I+N)^3 = I,$$

and tensors this indecomposable fiber with the 81*-register to obtain the non-split sequence* $0 \to 81 \to 162 \to 81 \to 0$*. No change of basis can identify the idempotent cache difference with the nilpotent transport shift. The architecture therefore separates a reversible* address/control plane *from a memory-bearing ternary* temporal data plane *by extension class, not by convention.* Witness: `analysis/bt867_cache_split_transport_nonsplit_boundary.py`.

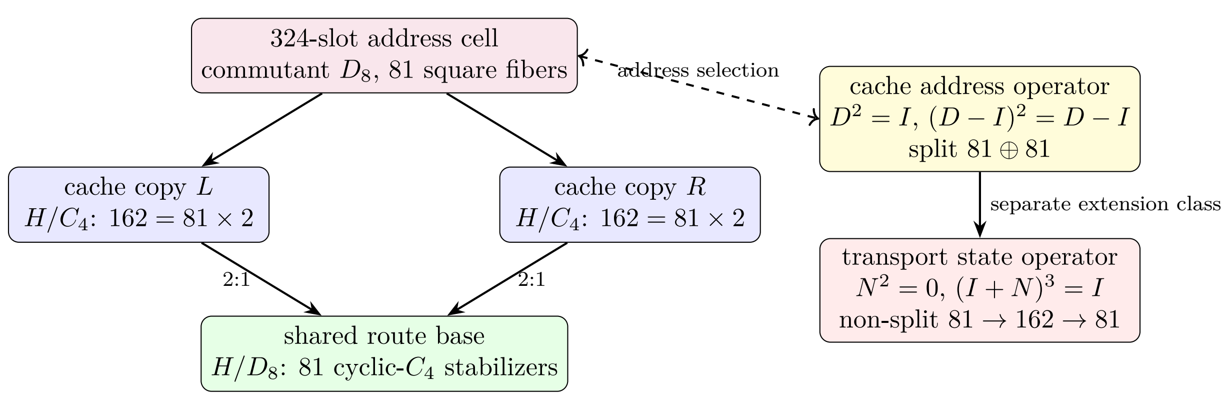


Figure 8: The local cache cell and the split/non-split boundary. The D_8 square fiber chooses a reversible address over one of 81 routes; the independent ternary nilpotent updates state while retaining temporal memory. Equal 162-dimensional counts do not imply equivalent modules.

The error layer is native: the $[[240, 81, 3]]_3$ CSS code on the edge qutrits has $(d_X, d_Z) = (3, 4)$; the dual Császár/Szilassi cross-check supplies parity-free error detection; and the web's unique non-Ramanujan eigenspace — dimension $15 = g$ — is the spectral sentinel where drift appears first.

At the classical boundary-image layer, Pass 373 gives the exact $[240, 120, 3]_3$ code and a complete 481-entry radius-one syndrome table, so one edge-symbol fault has a fully enumerated decoder. Pass 374 adds a separate controller constraint: the 51840 nonzero minimal logical X/Z vector pairings occupy four invariant 12960-state sheets under the natural $W(E_6)$ action, with stabilizer $C_2 \times C_2$. A runtime using those scalar representatives must therefore retain the two deck-sign bits; geometry does not collapse them to one Weyl torsor. Pass 375 sharpens the controller law: the fixed pairing character reduces the four-state sheet normalizer from abstract S_4 to the phase-compatible D_8, with no order-three operation. The split controller $W(E_6) \times V_4$ has no regular 51840-state complement. A future sheet fusion must therefore be a genuinely coupled transport, not an appended commuting phase register.

73 Time: three clocks and a quasicrystal

Theorem 73.1 (Clock trichotomy)**.** *Of the program's three natural clocks, exactly one is internal:* $\mathbb{Z}_{12}$ *(the rectangle clock,* $12 \mid 51840$*) runs the selector and duo machinery, while* $\mathbb{Z}_7$ *(Császár) and* $\mathbb{Z}_{13}$ *(Singer) do not divide the group order and act as external references — both with decimal period* $6 = q!$*, so the digit expansion of* $1/7$ *is literally the internal clock's shadow modulo its duo bit.* Witnesses: `bt774`, `bt803`, `bt807`, `bt819`.

Theorem 73.2 (The Boerdijk–Coxeter time quasicrystal)**.** *Driving the loop by the BC twist* $\theta = \arccos(-2/3)$ *per round trip yields a stroboscopic orbit that provably never repeats (Niven), with the Steinhaus three-distance signature at all substrate step counts and — the substrate's signature moment — exactly* two *gap lengths at* $n = 30 = h(E_8)$*, the BC ring length, where closure happens in* S^3 *(the* 600*-cell) rather than the phase circle (deficit* 0.0158*). This is the photonic sibling of the Fibonacci-drive dynamical topological phase.* Witness: `bt820`.

Corollary 73.3 (The machine's clock is the universe's clock)**.** *The same BC clock is the de Sitter beat of inflation (Face 7). Its twist* $\cos\theta = -(q-1)/q$ *is the discrete de Sitter angle, its ring length is the beat* $30 = h(E_8)$ *(the* 600*-cell's* 20 *rings of* 30*), and the inflationary cosmology reads off that beat: the e-fold number is* $N = 2(v - \Phi_4) = 2\,h(E_8) = 60$*, exactly twice the beat, and the CMB spectral tilt is*

$$1 - n_s \;=\; \tfrac{2}{N} \;=\; \tfrac{1}{30} \;=\; \tfrac{1}{h(E_8)} \;=\; \tfrac{1}{\textit{(clock beat)}},$$

so the deviation of the primordial spectrum from scale invariance is the inverse of the clock's period. The two structures even match in kind: the clock is a time quasicrystal *(nearly periodic, never exact) just as inflation is* quasi–de Sitter *(nearly exponential, never exact) — both broken de Sitter by* $q = 3$*, the clock's aperiodicity being the cosmology's tilt. The architecture's time and the universe's expansion are one quasicrystal.* Witness: `analysis/w33_clock_cosmology.py`.

Corollary 73.4 (A falsifiable clock fingerprint in the CMB)**.** *If the clock couples to the inflaton, its quasiperiodicity imprints a* quasi-periodic *modulation on the primordial spectrum: peaks at log-period* $2\pi/\theta \approx 2.73$ *in* $\ln k$ *(with* $\theta = \arccos(-2/3)$*) whose spacings obey the Steinhaus* three-gap *law (two gap lengths near the BC ring* $n = 30$*) — the time-quasicrystal fingerprint, distinct from smooth slow roll (no peaks) and from single-frequency resonant inflation (a uniform comb, one gap). The log-frequency and the three-gap shape are fixed by* $q = 3$*; only the amplitude (the coupling) is free. A CMB feature search (Planck, LiteBIRD, CMB-S4) can detect the three-gap comb, bound the coupling, or — finding a clean single-frequency comb — disfavour the clock. (Witness:* `analysis/w33_cmb_clock_signature.py`*.)*

Theorem 73.5 (The arrow of time)**.** *Without its two-trit feedforward record, the photon's self-configuration loop is the completely depolarizing channel with unique fixed point I/q — exactly the Bell marginal; with the record it is unitary. Decoherence is self-forgetting, priced at $\log_3 q^2 = 2$ trits per cycle.* Witness: `bt823`.

74 Synchronization: beacons and schedules

In plain language. *How separated machines agree on the time. The trick is to find sets of optical states that are all equally distinguishable from one another — so that no member of the set is a special case — and use them as a shared reference. This section counts how many such sets exist, which turns out to be exactly determined by the geometry rather than chosen, and then builds a schedule from them.*

The maximum classically coherent constellations are the *beacon heptads*: seven states with all pairwise overlaps exactly $1/3 = 1/q$ — a complete K_7 interference mesh, the Császár toroidal node realized in light. There are exactly 2880 heptads in a single transitive family; each has a $\mathbb{Z}_3 \times \mathbb{Z}_3$ symmetry fixing a unique *master beacon* and cycling two triads ($7 = 1 + 3 + 3$), and single-beacon swaps connect the family into its own exchange network: the reference frames of the network form a network. The measurement timetable is rigid: there exist exactly 36 complete schedules (partitions of the 40 states into 10 contexts) — an exhaustive result that simultaneously proves every spread of $W(3, 3)$ is regular — with each context appearing in exactly 9 schedules. *Witnesses: `bt817`, `bt819`.*

Theorem 74.1 (Schedule-overlap law)**.** *Two distinct schedules share exactly* 1 *or exactly* $4 = \mu$ *contexts: each schedule has* 15 near *partners (overlap* 4*) and* 20 far *partners (overlap* 1*), the two relation classes of the double-six diagonal* $[1, 15, 20]$*, with the exact counting identity* $1\cdot 20+4\cdot 15 = 80 = 10 \times (9-1)$*. Operationally: a cheap timetable switch preserves* 4 *of* 10 *calibrated contexts and a far switch only* 1*, so a controller hopping among near partners re-calibrates* 6 *contexts per hop — the timetable library is itself a metric space whose geometry is the double-six relation.* Witness: `bt835_schedule_overlap_theorem.py`.

The Grünbaum–Coxeter cells. Each schedule also has an internal shape. The full schedule stabilizer S_6 is 2-homogeneous on the 10 contexts, but its icosahedral core A_5 — the 600-cell level of the platonic ladder, and *exactly* the automorphism group of the hemi-dodecahedron $\{5, 3\}_5$ — splits the $\binom{10}{2} = 45$ context pairs as $45 = 15 + 30$, and the 15-orbit graph on the 10 contexts is the **Petersen graph**: the edge skeleton of the hemi-dodecahedron, the cell of Coxeter's exceptional 57-cell. Dually, the hidden 6-set of the S_6 action realizes the hemi-icosahedron $\{3, 5\}_5$ — the cell of Grünbaum's 11-cell — with incidence count $(6, 15, 10)$: six hidden points, fifteen duads, ten triple-splits = the contexts themselves. The exceptional polytopes' groups do not embed in $\mathrm{Sp}(4, \mathbb{F}_3)$, but they are substrate arithmetic ($|\mathrm{PSL}(2, 11)| = 660 = k \cdot N_{\mathrm{eff}}$, $|\mathrm{PSL}(2, 19)| = 3420 = k \cdot g \cdot 19$, Heawood genus $H(19) = 20 =$ the BC ring count, and $11 = k - 1$ is the Ihara prime of the zeta critical circle $|u| = 1/\sqrt{11}$): the full 11-cell and 57-cell live elsewhere, yet their *cells* are carried by every measurement timetable of this machine, glued by the substrate's own group instead of $\mathrm{PSL}(2, p)$. *Witness: `bt836_gc_hemicells_in_spreads.py`.*

Theorem 74.2 (The library is a classical geometry)**.** *The timetable library carries its own rank-3 geometry, and the hemicell structure fibers over it with perfect uniformity: (i) every skew line pair* *lies in exactly* 3 *schedules, so the* (schedule $\supset$ pair) *flag count is* $36 \times 45 = 1620$ *— the apartment count of the Tits building — and* 540×3*; (ii) the near-partner graph on the* 36 *schedules is strongly regular* $\mathrm{SRG}(36, 15, 6, 6)$*, the* other *classical rank-3 graph of* $U_4(2) \cong \mathrm{PSp}(4, \mathbb{F}_3)$*, so the control plane (timetables) and the data plane (*$W(3,3) = \mathrm{SRG}(40, 12, 2, 4)$*) are the two rank-3 geometries of one group; (iii) each schedule stabilizer contains exactly* 6 *icosahedral* A_5 *cores (*$216 = 6^3$ *in all, each marking its unique schedule via its* $[10, 30]$ *line orbits) giving* 6 *distinct Petersen splits; each internal pair is a Petersen edge under exactly* 2 *of the* 6 *cores, hence every skew pair of* $W(3,3)$ *has exactly* $6 = 3 \times 2$ *hemi-dodecahedral homes (*$3240 = 540 \times 6$ *Petersen flags).* Witness: `bt837_schedule_library_geometry.py`.

Theorem 74.3 (Two compasses, twelve Petersens, $4 \cdot K_{10}$)**.** $\mathrm{PSp}(4, \mathbb{F}_3)$ *has exactly two conjugacy classes of* A_5*, both with normalizer* S_5 *and* 216 *conjugates, unfused by the outer automorphism, and* both fiber over the same 36 schedules*: every schedule stabilizer contains* $12 = 6 + 6$ *icosahedral cores —* 6 duad *cores (line signature* $[10, 30]$*, the Theorem-74.2(iii) cores) and* 6 pentad *cores (signature* $[5, 5, 10, 20]$*: transitive on the schedule, plus two distinguished* pentads *of* 5 *pairwise-disjoint lines outside it). The two* 216*-sets are non-isomorphic G-sets (rank* 10 *suborbits* $[1, 5, 10, 10, 20, 20, 20, 30, 40, 60]$ *vs* $[1, 5, 10, 10, 20, 20, 30, 30, 30, 60]$*); the Petersen flags form the coset geometry* $\mathrm{PSp}(4, \mathbb{F}_3)/D_4$ *(*3240 *flags, dihedral stabilizer). There are* 432 *distinct pentads in* two chiral orbits *of* 216 *(stabilizer order* 120*); every core pairs one LEFT with one RIGHT pentad, and each pentad serves exactly one core. Pentads are* maximal partial spreads*: no schedule contains one. The two pentads of one core interlock as* $K_{5,5}$ *minus a perfect matching — mutual disjointness is* forbidden *by the overlap law (it would create a* 0*-overlap schedule) — and the deleted matching consists of* 5 skew *pairs, i.e. five hypercube charts; over the* 216 *cores these matchings cover the* 540*-chart atlas* ***exactly twice*** *(*$1080 = 540 \times 2$*): the routing fabric is readable off the pentad compasses, five charts per needle. Finally,* all twelve *cores split the* 45 *internal pairs as Petersen* 15*-orbits, and the twelve Petersen edge-sets cover every pair exactly* $\mu = 4$ *times:*

$$4 \cdot K_{10} \;=\; 12 \textit{ Petersens}.$$

Schwenk's classical obstruction (no edge-partition of K_{10} *into* 3 *Petersen graphs) dissolves at the substrate multiplicity* μ*. The numerical coincidence* $3240 = \#\{\textit{triangles of } Q\}$ *is refuted at the G-set level (orbits* $[360, 2880]$ *vs transitive).* Witnesses: `bt843_icosahedral_compass.py`, `bt844_double_five_and_six_petersens.py`, `bt845_pentad_exchange_and_chart_double_cover.py`.

Theorem 74.4 (The amalgamation ladder)**.** *The* 11*-cell and* 57*-cell are* universal *abstract polytopes: the unique ones with hemi-icosahedral facets and hemi-dodecahedral vertex figures (group* $L_2(11)$*, no proper quotients) and vice versa (*$L_2(19)$*). Their incidence is icosahedral-subgroup geometry: each of* $L_2(11)$*,* $L_2(19)$*,* $U_4(2)$ *has exactly two conjugacy classes of* A_5*, and in all three the distinguished incidence is* intersection in D_{10} *with valency* 6 *— the* 11*-cell's vertex–facet relation, the* 57*-cell's facet adjacency (the Perkel graph), and the substrate's compass incidence (*$216 + 216$*, bipartite* 6*-regular,* $216 \times 6 = 1296 = 6^4$ *incident pairs* $=$ 36 *schedules* $\times$ 36 *core pairs). The closures of one amalgam* $A_5 *_{D_{10}} A_5$ *(GAP):*

$$\langle A, B\rangle = \begin{cases} L_2(9) \cong A_6\ (360) & \textit{in } U_4(2)\textit{: inside exactly one schedule stabilizer,} \\ L_2(11)\ (660) & \textit{the } 11\textit{-cell,} \\ L_2(19)\ (3420) & \textit{the } 57\textit{-cell.} \end{cases}$$

One local icosahedral amalgam, a $\mathrm{PSL}(2)$*-ladder of completions over* $9 = q^2$*,* $11 = k - 1$ *(the Ihara prime),* 19 *(Heawood,* $H(19) = 20$*). Universality plus Lagrange (*$11, 19 \nmid 25920$*) shows the substrate* *carries the* complete local data *of both exceptional polytopes while their rank-*4 *amalgamation is arithmetically forbidden inside* $\mathrm{Sp}(4, \mathbb{F}_3)$*: the GC polytopes are the substrate's sibling completions, not subobjects. One step up the universal ladder (*$\{\{5,3\},\{3,5\}_5\}$, 10006920 *facets) the completion group is* $J_1 \times L_2(19)$ *— the first sporadic Janko group, whose involution centralizer* $2 \times A_5$ is *the compass datum. Finally, the compass Petersen* 15*-orbit carries exactly* 12 *pentagons splitting under the compass into two* 6*-orbits, each covering every edge exactly twice: two* chiral fully-faced *hemi-dodecahedra per needle, not merely skeletons. The dark sector closes the calculus: its* 190 *line pairs split* $[10, 30, 30, 30, 30, 60]$ *under the core into a perfect matching of* 10 dark charts*, two chiral* $5 \times K_4$ *partitions into skew tetrads, the two dark dodecahedra, and a* 6*-regular meeting frame; the dark matching is* schedule-shadow pairing *(each dark line meets exactly* 4 *schedule lines; matched lines share their shadow), and in* K_5 *terms the ten shadows are exactly the ten "edge + opposite triangle" configurations — a canonical bijection dark chart* $\leftrightarrow$ K_5 *edge* $\leftrightarrow$ *schedule line. The timetable is stored three ways (lit-chart transversals,* K_5 *edges, dark shadows): built-in erasure redundancy. Each chiral tetrad's four distinguished edges form a* K_5 star*, the five tetrads of each partition marking the five lit charts exactly once:* K_5 *vertices are stored three ways (lit charts, left/right tetrads), edges two ways (schedule lines, dark charts) — every* A_5*-orbit structure of the pentad core except the dodecahedra and the meeting frame is a* K_5 *element — and those two fold in as well: each chiral dark dodecahedron is the* antipodal double cover *of the Petersen (Kneser) graph on the* 10 *dark charts, with deck transformation the shadow matching (antipode = shadow partner, verified), and the meeting frame is the doubled Johnson graph* $T(5)$*. The* K_5*-relation census is constant on every orbit (matching = same edge, tetrads and meeting = adjacent, dodecahedra = disjoint): the dark sector is the canonical* 2:1 *cover of the* K_5 *edge-world, with the Petersen graph appearing three ways in one needle (schedule skeleton, dark Kneser quotient, doubled chiral covers).*

Theorem 74.5 (The dark charts are the mirror bus)**.** *The* (core, dark chart) *slot space —* 216 *pentad cores* $\times\, 10$ *dark charts* = 2160 *slots, with every skew pair occurring as a dark chart in exactly* 4 *cores (*$2160 = 540 \times 4$*, the repair-atlas factorization) — is* ***G-set isomorphic to the*** 2160***-slot*** D_{12} ***mirror bus*** *of the middleware: the slot stabilizer has order* 12 *with the* D_{12} *profile* $\{1{:}1, 2{:}7, 3{:}2, 6{:}2\}$ *and is* conjugate in $\mathrm{PSp}(4, \mathbb{F}_3)$ *to the antipode-slot stabilizer (direct search over all* 25920 *elements). The multiplicity-*4 *agreement is canonical: a dark chart's transversal tetrad is its schedule shadow, so the* 4 *host cores' schedules all contain the pair's* 4 *transversals. Hence the compass layer* generates *the middleware — timetables from its lit side (reconstruction), the transport bus from its dark side — and the mirror bus acquires a hardware identification: one slot = one needle's dark chart.* Witnesses: `bt856_dark_mirror_bus.py`; further: `bt853_dark_orbit_zoo.py`, `bt854_dark_chart_transversal_duality.py`, `bt855_dark_sector_k5_complete.py`.

Witnesses: `bt848_universal_amalgam_ladder.py`, `.tmp/gap_bt848*.g`; cf. Hartley's classification (17 universal rank-4 locally projective polytopes, 441 quotients) and Hartley–Leemans on $\{5, 3, 5\}$.

The barren rung and the maniplex workaround. Relaxing polytopes to *maniplexes* (flag graphs without the intersection property; the tomotope middleware of this machine is a uniform 4-polytope whose monodromy $18432 = 96 \times 192$ fails that property, giving it *infinitely many* distinct minimal regular covers — the canonical rank-4 cover pathology) does not lift the obstruction: exhaustive search shows $U_4(2)$ admits *no* rank-4 regular maniplex with a 5 in its type, and the amalgam closure $A_6 = L_2(9)$ admits no regular maniplex at any rank — it is the *barren rung* of the PSL(2) ladder, whose populated rungs are exactly the 11-cell (q=11), the 57-cell (q=19), and Leemans–Toledo's degenerate single-facet $\{5, 5, 5\}$ family (our control search rediscovers all of them, including both exceptional polytopes). This is why the Grünbaum–Coxeter content of the substrate is *delocalized* over the 36 schedules — Petersen splits, chiral pentads, chart covers, dark dodecahedra — rather than crystallized in one flag object, and why the machine's runtime uses the tomotope: it is the flag structure that survives the obstruction. The tomotope's symmetry type is now exact: $|\mathrm{Aut}| = 96$ (independent centralizer computation on the 192 flags), two flag orbits $[96, 96]$, **class** $2_{\{0,1,2\}}$ — only the cell-color crosses orbits, and the orbits are precisely "flag in a tetrahedron" vs "flag in a hemioctahedron": the middleware *alternates spherical and projective cells*, the exact situation the classical "locally X" taxonomy cannot name, and the setting of Mochán's 2-orbit polytopality theory. Moreover the tomotope is a **Cayley maniplex over** $V_4 = \mathbb{Z}_2 \times \mathbb{Z}_2$ (vertex action $=$ full S_4 with a regular Klein subgroup; edge and face actions faithful and transitive): the middleware's vertex layer is a free $\mathbb{F}_2^2$-torsor, so register XOR-addressing extends from the chart atlas into the runtime with no added structure. Reading Hartley's full classification (seventeen universal locally projective rank-4 polytopes) against the substrate adds three facts: Aut(**tomotope**) **is itself a universal polytope group** — isomorphic to $\Gamma(\{\{4,3\}_3, \{3,4\}_3\})$, the quotient-free doubly-projective $\{4,3,4\}$ universal built from hemicubes and hemicrosses (the tomotope's own projective cell type), with the case-10 group $= C_2 \times \mathrm{Aut}$ of order $192 =$ the flag count; the mixed $\{3,5,3\}$ amalgam (icosahedral facets, hemi-dodecahedral vertex figures) *does not exist* — the construction closes into the 11-cell — while the dodecahedral mix is the J_1 giant; and the chart–compass universal $\{\{4,3\},\{3,5\}_5\} = 2^{\mathcal{I}}$ (cube facets $= Q_3$ charts, hemi-icosahedral vertex figures $=$ compass cells, vertices $= \mathbb{F}_2^6$ on the hidden 6-set) has order $3840 \nmid 25920$: blocked like the rest of the ladder. *Witnesses:* `bt849_maniplex_barren_rung.py`, `bt850_tomotope_two_orbit_class.py`, `bt851_tomotope_cayley_test.py`, `bt852_seventeen_universals.py`, `.tmp/gap_bt849*.g`, `.tmp/gap_bt852*.g`.

Reconstruction, the chart K_5, and the dark dodecahedra. The pentad core's full signature $[5, 5, 10, 20]$ (lines), $[20, 20]$ (points) is rigid: both chiral pentads cover the *same* 20-point orbit (lit/dark halves of the substrate), the five matching charts' common-transversal bundles overlap to exactly the 10 schedule lines — **a pentad pair generates its timetable** — and the line $\to$ chart-pair map is a bijection onto the $\binom{5}{2} = 10$ edges of K_5: the schedule *is* a K_5 on its own charts (4 transversals per chart, 2 charts per line). The 20 dark lines complete the $\{5,3\}$ family: their 190 pairs split $[10, 30, 30, 30, 30, 60]$ under the core, and exactly two of the 30-orbits are **dodecahedron skeletons** (3-regular, girth 5, diameter 5) — a chiral pair of genuine dodecahedra beneath every compass needle, one level under the hemi-dodecahedral Petersen splits of the schedule itself. *Witnesses:* `bt846_pentad_core_anatomy.py`, `bt847_dark_dodecahedron.py`.

Theorem 74.6 (BT839 GC operation Euler/flag audit)**.** *Treating the full Grünbaum–Coxeter note as a data table gives a second invariant. The base tables for the* 11*-cell,* 57*-cell, tomotope partial a, and tomotope partial b are all Euler-neutral:*

$$V - E + F - C = 0.$$

But every Wythoff operation table has a fixed family charge:

$$\chi_{\mathrm{op}}(11\text{-}cell) = -11, \qquad \chi_{\mathrm{op}}(57\text{-}cell) = -57, \qquad \chi_{\mathrm{op}}(tomotope) = -4.$$

The omnitruncated rows are full-flag carriers:

$$660 = |\mathrm{PSL}(2,11)| = 11 \cdot A_5 = kN_{\mathrm{eff}},$$

$$3420 = |\mathrm{PSL}(2,19)| = 57 \cdot A_5 = k \cdot g \cdot 19,$$

and the tomotope has 96 *half-flags, whose double is the verified* 192*-flag BT814 packet. The* 57*-cell also locks to the W33 timetable library: BT837 gives* 3240 *Petersen homes, and*

$$3240 + k \cdot g = 3240 + 180 = 3420.$$

Thus one $k \cdot g$ sentinel sheet completes the W33 Petersen-home carrier to the 57*-cell flag count.*

The regular-polychoron alignment supported by the cell/symmetry data is

$$11\text{-}cell \to 600\text{-}cell, \qquad tomotope \to 24\text{-}cell, \qquad 57\text{-}cell \to 120\text{-}cell.$$

The $57 \to 120$ *link is direct at the dodecahedral cell level; the* $11 \to 600$ *link is weaker but real, through hemi-icosahedral local structure and the quotient* $2I \to A_5$ *rather than through a direct tetrahedral-cell match. The swapped* 11/57 *orientation is recorded as a lower-scoring dual/vertex-figure hook, not discarded.* Witness: `bt839_gc_operation_euler_flag_audit.py`.

Theorem 74.7 (BT840–BT842 carrier closure)**.** *The three unresolved carriers in the Grünbaum–Coxeter paragraph are now explicit objects.*

(i) The* 180 *sentinel sheet. *The verified Clifford L/R boundary has* 36 *cells arranged as a* 6×6 *rook grid. Its zero-overlap relation is not a numerical afterthought: it is the union of the* 6 *row fibres and* 6 *column fibres, each fibre carrying the duads of a six-set. Hence*

$$12 \cdot \binom{6}{2} = 12 \cdot 15 = k \cdot g = 180.$$

This is exactly the sheet missing from BT837:

$$3240 + 180 = 3420 = |\mathrm{PSL}(2, 19)|.$$

Equivalently, the timetable library gives 216 *Petersen cores (*$216 \cdot 15 = 3240$*), and the rook boundary supplies the missing* 12 *six-set fibres (*$12 \cdot 15 = 180$*).*

(ii) The* 660 *carrier. *At an apex cell (r, c) of the same 6×6 grid, remove the incident row and column. The apex plus the five nonincident rows and five nonincident columns gives*

$$11 = 1 + 5 + 5.$$

Crossing these eleven labels with the verified 60*-element Clifford A_5 selector gives*

$$11 \cdot A_5 = 11 \cdot 60 = 660 = kN_{\mathrm{eff}}.$$

This is a carrier theorem only: it does not assert an internal $\mathrm{PSL}(2, 11)$ *action on* $W(3, 3)$*.*

(iii) The* 96 *tomotope half-flags. *In the D_4-root model of the* 24*-cell, every edge lies in a unique central hexagon plane. The two endpoint axes determine the third, missing axis in that plane, so each edge maps to a Reye incidence* $(missing\ axis, hexagon\ plane)$*. Each of the* 48 *Reye incidences has exactly two edge lifts:*

$$96 = 2 \cdot 48.$$

Thus the tomotope omnitruncated half-flag count is literally the 24*-cell edge lift of the common tomotope/Reye spine.*

The hexagon clue supplies the conceptual bridge to the 11*-cell. Each central* 24*-cell hexagon is a six-root cycle C_6. Completing that cycle to a full K_6 gives* 15 *duads, exactly the hemi-icosahedral skeleton count of the* 11*-cell cell. Across the* 16 *central hexagons,*

$$16 \binom{6}{2} = 16 \cdot 15 = 240,$$

the W33 edge count and the E_8 root count. The tested dot profile of these 240 *duad slots is*

$$96_{\langle x,y\rangle=1} \;+\; 96_{\langle x,y\rangle=-1} \;+\; 48_{\langle x,y\rangle=-2},$$

namely live 24*-cell edges, mirror pairs, and repeated opposite-axis Reye incidences.* Witnesses: BT840, BT841, and BT842; executable scripts are in `analysis/`.

75 Build sheet and falsifiable witnesses

Components (all standard quantum optics): one heralded single-photon source; one polarizing beam splitter; one symmetric 3-port coupler (tritter); a $0/\tau/2\tau$ delay ladder; one bin-synchronous electro-optic modulator; one polarization rotator set to $\arccos(-2/3)$ in the recirculation loop; single-photon detectors.

Witnesses (kill criteria, all parameter-free):

witness	predicted value	substrate form
trace–Choi visibility of F_3	$1/3$	$1/q$
trace–Choi visibility of X, Z	0	—
Kochen–Specker classical budget	$36/40$	$(q!)^2/v$
beacon-mesh pair visibility	$1/3$ (all 21 pairs)	$1/q$
Werner separability threshold	$3/4$	q/μ
BC-drive gap census at n=30	2 gap lengths	$h(E_8)$ ring
Hashimoto transport phases (one tick)	$\arctan\sqrt{10},\ \pi-\arctan(\sqrt{7}/2)$	Φ_4, Φ_6; $\lvert u\rvert^2 = 11$
key-agreement rate	$13/40$	Φ_3/v
D_{12} mirror-slot stabilizer	order 12	projective bus
runtime factorization	$24 \cdot 45 \cdot 48$	$\lvert\mathrm{Sp}(4, \mathbb{F}_3)\rvert$
one recursive route digit	≤ 8 reversible hops	$3 + 5$
level-n leaves	40^n	v^n
level-n $W(3,3)$ instances	$(40^n - 1)/39$	geometric tower
durable commit clock	$4(7^g - 1)$ ticks	tomotope/Csaszar
compiled stress route	$47 < 48$ moves	BT828 level six
cover-lifted packet capacity	$48k^3$ slots	BT832 storage gauge
Bell–compass prefix router	$648 + 324 + 162 + 162$	code $0, 10, 110, 111$
sentinel-aware stress energy	$4 \to 2$	BT833 reroute
full route sync criterion	$2n \mid 7^n - 1$	BT834 guard band
tomotope Wythoff runtime ladder	$4, 12, 24, 48, 96$	BT838 packet growth
GC operation Euler charges	$-11, -57, -4$	BT839 table audit
sentinel projector trace	15	g sector
first full-route desync	$n = 5$ remainder 24	f guard band
tomotope cover ABI	48 blocks / 192 flags	cover-invariant

A measured deviation in any row falsifies the corresponding layer.

76 Implications, corrections, and the ethos

This section states what the architecture means — for physics, for computing, for networking, and beyond the laboratory. Every claim below is either (i) a theorem with an executable witness in this paper's ledger, or (ii) an implication of such a theorem, flagged as such. Nothing here floats free of the verification chain.

76.1 Physics

In plain language. *What has to be true of the world for this architecture to work, separated from what has to be true of the mathematics. The mathematics is checked; the physics is a set of requirements on materials, loss rates and detector performance that current technology meets in some cases and not in others. Each requirement below says which.*

The vacuum is a gauge choice among five. The decomposition $40 = 1 + 12 + 27$ (self, gauge, matter) that organizes the entire Standard-Model derivation is *one of five* maximal decompositions, indexed by the Schläfli inventory of the cubic surface: 27 registers, 36 schedules, $40 + 40$ building parabolics, 45 polar pairs. The five vacua are connected by an explicit 5×5 double-coset transition matrix. Physically: what looks like "the" vacuum is a parabolic gauge choice, and the theory knows its own alternative phases and the exact combinatorial cost of moving between them. No continuum field theory has this property; here it is finite, exact, and enumerated.

Matter is magic, with three octane grades. The machine's non-classical resource (magic, in the resource-theoretic sense) coincides *exactly* with its matter sector: the 36 entangled Witting rays are the matter shell, and the 27 matter contexts stratify $[1, 8, 12, 6]$ — the cube's cell census. The three magic grades — deep $(2 + \sqrt{3})/6$ on $8 = 2^q$ rays, mid $(5 + 2\sqrt{3})/12$ on $24 = f$, shallow $3/4 = q/\mu$ (the Werner threshold) on $4 = \mu$ — mean that "what matter is" and "what fuels quantum advantage" are one census. A physicist reads this as a structural identification of the fermionic sector with the non-stabilizer sector; an engineer reads the same integers as a fuel gauge. Both readings are forced by the same double cosets.

The Standard-Model spine: one transvection. The preceding paragraphs assemble into a single statement, verified end-to-end in one pass (`bt886_standard_model_spine.py`): *from the pair* $(W(3,3),\ R)$ *— the symplectic quadrangle and one long-root transvection — the discrete Standard Model follows with zero free parameters.* The transvection R fixes the 13-point gauge perp-plane and acts freely (nine orbits) on the 27 matter shell; its centralizer $C(R)$ (order 648) is the gauge group, with module $\mathbf{1} \oplus \mathbf{3} \oplus \mathbf{8} = U(1) \times SU(2) \times SU(3)$; its centre $Z(C(R)) = \langle R \rangle = \mathbb{Z}_3$ is the three generations; its normalizer gives the flavor group S_3 and charge-conjugation; the W/Q duality gives parity; the matter shell grades 9+9+9 with the $\mathbb{Z}_3$ Yukawa rule; and the connection is flat ($\mathbb{Z}_3^2$) along collinear edges, $2T$-curved on the matter graph Q with a quaternionic field strength. The gauge group, the generations, the flavor symmetry, the Yukawa texture, chirality, parity, and the gauge dynamics are not separate inputs — they are the centralizer, centre, normalizer, duality, grading, and commutator structure of one element of $\mathrm{Sp}(4, \mathbb{F}_3)$, replicated over the 40 points. The gauge group even carries its color/electroweak split: $C(R) = 3^{1+2} : SL(2,3) = 27{:}24$, with the normal Heisenberg radical (order $q^q = 27$, the color part) fixing the electroweak $\mathbf{1} \oplus \mathbf{3}$ pointwise (the $W/Z/\gamma$ are color-blind, constant on the four lines through p_0) and acting on the gluon octet $\mathbf{8}$, and the Levi $SL(2,3)$ (order $f = 24$, the electroweak part) carrying the $\mathbf{1} \oplus \mathbf{3}$ — color $\rtimes$ electroweak as a parabolic, with substrate orders q^q and f. And the color radical is *the same* Heisenberg group $3^{1+2} = O_3$ that acts simply transitively on the 27 matter shell (the matter torsor): **color is the matter-shell Heisenberg group**, so matter carries color precisely because the 27-shell is a torsor under the color group — color rotations and matter-shell motions are one action of $q^q = 27$. Under the full gauge group the matter shell is the homogeneous space $27 = C(R)/SL(2,3)$ (gauge group over the electroweak Levi): $C(R)$ is transitive with point-stabiliser the electroweak $SL(2,3)$, rank

6 (suborbits $[1,1,1,8,8,8]$), module $\mathbb{C}[27] = \mathbf{1} \oplus \mathbf{6} \oplus \mathbf{8} \oplus \mathbf{12}$, and the three electroweak-fixed points form a color-singlet (lepton-like) 3-set while the three color-8 suborbits carry the colored (quark-like) content ($27 = 3 + 24$). One generation's fermion multiplets are the $C(R)$-module structure of the 27.

Three generations, matter-blind and gauge-visible. Because the matter register is the Steinberg module (Corollary, §72) and the Steinberg character vanishes on every 3-singular element, *any* order-3 symmetry of the substrate splits matter into exactly $27 + 27 + 27$: the three generations carry identical matter/Steinberg structure — they are indistinguishable to the protected register, which is why generations share gauge charges. Yet the four order-3 conjugacy classes *are* distinguished in the gauge (40-point) sector: the transvections grade it $22+9+9$ and fix a whole $\mathrm{PG}(2,3)$ plane of $13 = \Phi_3$ points, while the regular classes grade it $16+12+12$ and fix only $\mu = 4$ points. The *physical* texture triality (Pillar 68's Yukawa map R) is identified exactly: it is the long-root transvection (a 40-class element = the centre of the point-parabolic's Heisenberg radical), fixing the 13-point gauge perp-plane $p_0^\perp$ pointwise and acting on the 27-point matter shell with 9 free orbits of 3 — the canonical "fix the gauge, stir the matter" generator. Its eigengrading of $\mathbb{C}[27]$ is $9 \oplus 9 \oplus 9$, and $\mathbb{Z}_3$ Clebsch–Gordan then *forces* the Pillar-68 Yukawa selection rule $T[a,b,v] = 0$ unless the three grades sum to 0 mod 3 (exactly 9 of 27 grade-triples survive); the matter coupling lives on the 36 within-shell tritangent triples (Pillar 69's Heisenberg-twisted triads, $45 = 9 + 36$). And the gauge side closes it: the centralizer $C(R) = \mathrm{Stab}(p_0)$ of order 648 (the gauge group is what commutes with generation-cycling) acts on the $12 = k$ gauge neighbours with permutation rank 3, decomposing the gauge module as

$$\mathbb{C}[12] = \mathbf{1} \oplus \mathbf{3} \oplus \mathbf{8} = U(1) \oplus SU(2) \oplus SU(3),$$

the Standard-Model gauge group adjoint (the $\mathbf{1}+\mathbf{3}$ from the A_4 action on the four lines through p_0, the $\mathbf{8}$ the within-line gluon octet). So R fixes the gauge sector ($12 = 8+3+1$, generation-blind) and grades the matter sector ($27 = 9+9+9$, three generations): the gauge group and the generations are the two eigen-data of one long-root transvection. The gauge sector even carries a parity statement: $C(R)$ acts on the four lines through p_0 as A_4 inside PSp, but as the full S_4 in $\mathrm{PGSp}(4,\mathbb{F}_3) = W(E_6)$ — the odd (parity-violating) 4-line permutations are *exactly* the anti-symplectic W/Q-duality coset, so gauge parity is the duality/chirality $\mathbb{Z}_2$ and lives only in the full $W(E_6)$. The flavor sector even has its charge conjugation: $N_{W(E_6)}(\langle R\rangle)/C(R) = \mathbb{Z}_2$, so there is an element C with $CRC^{-1} = R^{-1}$ that fixes generation grade-0 and swaps grade-1 $\leftrightarrow$ grade-2 — the discrete shadow of CP, conjugating the generation $\mathbb{Z}_3$ exactly as the temporal Bell qutrit's CPT conjugates $\omega \mapsto \bar{\omega}$. The full Standard-Model discrete flavor/parity data — gauge group $1 \oplus 3 \oplus 8$, three generations $\mathbb{Z}_3$, generation C, matter chirality, gauge parity — is the normalizer geometry of one long-root transvection. The grading R and its conjugation C together generate the **flavor group** S_3 ($\langle R, C\rangle$, order 6, non-abelian — the minimal non-abelian flavor symmetry of BSM model-building), under which the matter shell decomposes as $\mathbb{C}[27] = 6\cdot\mathbf{1} \oplus 3\cdot\mathbf{1'} \oplus 9\cdot\mathbf{2}$, the standard doublet appearing $q^2 = 9$ times. And the flavor–gauge relationship is exact: R is central in the gauge group $C(R)$, with $Z(C(R)) = \langle R\rangle \cong \mathbb{Z}_3$ — **the three generations are the centre of the gauge group**, acting trivially on the adjoint exactly as the $\mathbb{Z}_3$ centre of colour $SU(3)$ acts on the gluon octet (so the $SU(3)$ centre *is* the generation $\mathbb{Z}_3$). Globally there are 40 such local gauge groups — one per point, a single conjugacy class (homogeneous spacetime) — and the 40 generation-centred long-root $\mathbb{Z}_3$'s generate all of $\mathrm{PSp}(4,\mathbb{F}_3)$: **the 40 points of $W(3,3)$ are the space of local gauge groups**, each $SU(3)\times SU(2)\times U(1)$ with its generation centre, and the global symmetry is the closure of the local generation symmetries. The connection between adjacent local gauge groups is exact: for collinear points the generation centres commute ($\langle R_p, R_{p'}\rangle = \mathbb{Z}_3 \times \mathbb{Z}_3$), so the gauge connection is **flat along the** 240 **edges**; for non-collinear points $\langle R_p, R_{p'}\rangle = SL(2,3) = 2T$ (the binary tetrahedral / 24-cell group), so it is **curved across the** 27**-per-point matter shell**. Gauge curvature lives exactly on the matter graph Q — flat = collinearity, curved = non-collinearity = matter, with holonomy the 24-cell group. The curvature 2-form $F(p,q) = [R_p, R_q]$ makes this quantitative: it vanishes on collinear (causal) directions and is an *order-*4 *quaternion unit* of $2T$ (one of $\pm i, \pm j, \pm k$, with $F^2 = -I$ the centre) on every matter pair — an **su**(2)**-valued field strength supported exactly on the matter graph** Q. Flat causality, quaternionic curvature on matter. The integrated flux (Wilson loop $R_a R_b R_c$) confirms it: all 160 collinear triangles carry order-3 abelian flux (the flat $\mathbb{Z}_3$ line grading), while the 3240 matter triangles of Q carry non-abelian flux of orders $\{2,4,6,12\}$ (counts $180, 180, 1440, 1440$) up to $12 = k$ — gauge energy lives on the matter graph. The matter register (Steinberg) couples to this flux only through its non-3-singular part: by Steinberg vanishing, $\chi_{\mathrm{St}}(W) = 0$ on every order-$3,6,12$ loop (3040 triangles, matter-invisible) and is nonzero only on the order-2 loops ($\chi_{\mathrm{St}} = +q^2 = 9$, the polar-pair chirality involution of BT869) and order-4 loops ($\chi_{\mathrm{St}} = -q = -3$); the total coupling $\sum \chi_{\mathrm{St}}(W) = 180 \cdot 9 - 180 \cdot 3 = 1080 = 540 \cdot 2$ is exactly the chart double-cover count — gauge dynamics and the routing fabric meet at 1080. This is the Standard-Model pattern as a representation- theoretic theorem: generations are identical in gauge charge (matter-blind) and differ only through gauge-sector–visible couplings (the Yukawa/mass structure), here the eigenvalue degeneracy of an order-3 symmetry rather than a modeling input. *Witnesses:* `bt863_generations_from_steinberg_vanishing.py`, `bt864_triality_census.py`. Order-6 symmetries see *both* internal gradings at once: an order-6 element factors as g^2 (generation, order 3) times g^3 (chirality, order 2 — the sign-twisted carrier of H_2), and since χ_{St} vanishes on every 3-singular power, its $\mathbb{Z}_6 = \mathbb{Z}_3 \times \mathbb{Z}_2$ eigengrading factors cleanly: each chirality half $(81 \pm \chi(g^3))/2$ carries 3 equal generations. For the dominant involution type ($\chi(g^3) = q^2$) the chirality split is $\mathbf{45} + \mathbf{36}$ — the substrate's own tritangent and double-six counts — so the left/right matter split is tied to Schläfli geometry; the other type gives $39 + 42$. Generation $\times$ chirality $= 3 \times 2 = 6 = q!$. The dominant chirality involution is identified exactly: $\mathrm{PSp}(4, \mathbb{F}_3)$ has just two involution classes, and the 45-class (eight fixed points carrying the cube Q_3, centralizer $2T \times 2T$ of order 576) is the *central involution of a hyperbolic polar pair* $L \leftrightarrow L^\perp$ — so the left/right split of matter, $45 + 36$, is sized by the substrate's own tritangent-plane and double-six counts, and the matter chirality $\mathbb{Z}_2$ *is* the 24-cell polar-pair swap. *Witnesses:* `bt868_joint_generation_chirality_grading.py`, `bt869_involution_chirality_classes.py`.

Time has three hands and an irrational escapement. The internal $\mathbb{Z}_{12}$ rectangle clock, the external $\mathbb{Z}_7$ (Császár) and $\mathbb{Z}_{13}$ (Singer) references, and the Boerdijk–Coxeter drive $\arccos(-2/3)$ together realize a discrete time quasicrystal: a drive that never repeats yet has exactly bounded gap structure (two gap lengths at $n = 30 = h(E_8)$). The arrow of time appears as the unique self-loop fixed point I/q with its two-trit forgetting cost: irreversibility is priced, not postulated. Implication: any laboratory system realizing this architecture is also a clock whose long-term stability is protected by a number-theoretic irrationality rather than by feedback.

The exceptional polytopes are the theory's siblings, not its parts. The Grünbaum–Coxeter arc (Theorems 74.2–74.4) shows the substrate carries the complete local geometry of the 11-cell and 57-cell — hemi-icosahedra and hemi-dodecahedra in every schedule, with exact incidence — while universality plus Lagrange forbid their global crystallization inside $\mathrm{Sp}(4, \mathbb{F}_3)$. The same local amalgam $A_5 *_{D_{10}} A_5$ closes as $L_2(9)$ (the substrate's schedule core), $L_2(11)$ (the 11-cell), $L_2(19)$ (the 57-cell), with $J_1 \times L_2(19)$ — a sporadic group — atop the universal ladder. The physical moral is delocalization: the substrate *cannot* host a "GC condensate," so the icosahedral order is necessarily spread over the 36 schedules as compass structure. This is the same mathematical mechanism by which frustrated phases in condensed matter refuse local order parameters — realized here exactly, with the obstruction an integer $(11, 19 \nmid 25920)$.

Spectral ghosts. The primes that refuse to divide the group still govern its analysis: $11 = k - 1$ is the Ihara prime of the zeta critical circle $|u| = 1/\sqrt{11}$, 19 enters through the Heawood genus $H(19) = 20 =$ the BC ring count, and the chart-web eigenvalue fields are $\mathbb{Q}(\sqrt{\Theta})$ and $\mathbb{Q}(\sqrt{\Phi_{12}})$ ($\Theta = 10$, $\Phi_{12} = 73$, both parameter-closure constants). A spectroscopist's summary: the machine's transport spectrum speaks the tier-ladder dialect of the physics paper, and the forbidden primes appear as poles of the zeta function rather than as symmetries.

Gravity counts the matter register. The discrete gravitational bridge is the Matrix-Tree action $S = \frac{M_P^2}{2} \ln \tau(G)$, τ the spanning-tree count, and for the substrate it is exact: the Laplacian spectrum $\{0, 10^{24}, 16^{15}\}$ gives

$$\tau(W(3,3)) = \frac{10^{24} \cdot 16^{15}}{40} = 2^{81} \cdot 5^{23},$$

where the exponent of 2 is $81 = q^4 =$ the matter-sector (Steinberg) dimension, the base $5 = F_5$ is the tier-ladder prime ($r = 27/80$), and the exponent $23 = f - 1$. The complement matter graph $Q = \mathrm{SRG}(40, 27, 18, 18)$ gives $\tau(Q) = 2^{66} \cdot 3^{39} \cdot 5^{23}$ with 3-exponent $39 = q\Phi_3 =$ the gauge-sector dimension. So the discrete-gravity partition function carries the Standard-Model sector sizes as its prime-exponent charges — matter (q^4) in the power of 2, gauge ($q\Phi_3$) in the power of 3, cosmological (F_5) shared — and the entropy of the spanning-tree ensemble *counts the registers.* This is not a separate fact from the transport spectrum: the Ihara zeta $\zeta^{-1}(u) = (1-u^2)^{200}(1-u)(1-11u)(1-2u+11u^2)^{24}(1+4u+11u^2)^{15}$ (verified directly by the 480×480 Hashimoto operator) has its non-trivial zeros on the graph-RH circle $|u| = 1/\sqrt{11}$ (the transport sector, $p_{\mathrm{Ih}} = k - 1 = 11$) and its Perron zero at $u = 1$, where the Bass matrix $I - A + 11I = 12I - A$ *is* the Laplacian, so the Matrix-Tree leading coefficient $10^{24}16^{15} = 2^{84}5^{24} = v \cdot \tau$ recovers exactly the gravity count. Transport and gravity are the off-critical and critical behavior of one zeta: the prime 11 is the transport critical norm, the matter dimension $q^4 = 81$ is the gravity entropy. The substrate is in fact a *pair* of zetas: the complement matter graph $Q = \mathrm{SRG}(40, 27, 18, 18)$ has its own Ihara prime $26 = 2\Phi_3$ (Hashimoto Perron, all complex eigenvalues on $|u| = 1/\sqrt{26}$) and its Perron-point gravity $\tau(Q) = 2^{66}3^{39}5^{23}$ puts the *gauge* dimension $39 = q\Phi_3$ where $W(3,3)$ puts the matter dimension — complementation (collinearity $\leftrightarrow$ non-collinearity) swaps the matter and gauge entropies, with the cosmological charge 5^{23} shared. The non-backtracking walk census confirms the dynamics: $N_3 = \mu|E| = 960$, $N_5 = |E|\, q^q\, n_{\mathrm{even}} = 181440$, $N_n \sim 11^n$. *Witnesses:* *bt870_spanning_tree_gravity.py, bt872_ihara_zeta_spanning_bridge.py, bt873_dual_zeta_walk_census.py.*

Falsifiability is retail, not wholesale. Each physics reading above is pinned to a tabletop witness with an exact rational value: trace–Choi visibility $1/3 = 1/q$ exactly (not approximately); Kochen–Specker budget $36/40 = (q!)^2/v$ exactly, with deficit μ; contextual fraction $1/10 = 1/\Phi_4$; Werner threshold $3/4 = q/\mu$; two gap lengths — not three — at drive step 30. A single wrong integer kills the corresponding layer. This is a stronger falsifiability posture than parameter-fitting physics can offer, because there are no parameters to refit.

76.2 Computing

What the machine is. One photon carries a 4-dimensional path–polarization register (states) and a 9-dimensional past–future time-bin register (operators), realizing the same $W(3,3)$ geometry twice; the operator–operand duality makes program and data literally the same incidence object. Universality is a two-ingredient theorem: three passive optical elements (tritter, phase plate, EOM) generate the full 51840-element Clifford symplectic closure — *exactly*, not approximately — and the matter shell supplies the magic states that promote Clifford to universal. There is no cryostat, no vacuum chamber, no trapped ion: the exotic resource is geometry.

What it costs. The build sheet is a heralded single-photon source, one polarizing beam splitter, one symmetric tritter, a three-rail delay ladder, one electro-optic modulator, one recirculation loop with a fixed irrational rotation, and single-photon detectors. Every component is catalog optics. The machine's benchmarks are integer-valued: a laboratory can certify the Clifford layer by counting to 51840 and certify the fuel by hitting 36/40. Self-testing is native because the specification is arithmetic.

The software stack exists and is finite. Programs are braid words (with $\sigma^5 = Z$ exact); the instruction set is the BT828 compiler ABI; the operating system is the timetable library (36 schedules; overlap law 1-or-4 with switching cost 6); synchronization is the beacon-heptad mesh (2880 frames, themselves forming an exchange network); memory protection is the Steinberg sector ($81 = q^4$, a single irreducible — protected by Schur's lemma, the strongest no-leakage statement representation theory can make). The state/program compiler makes that register directly addressable: $\mathbb{F}_3^3$ supplies three commuting copies of the 27-slot program shell, H_{27} supplies three noncommuting copies of the state shell, and its center supplies the native generation flag $27 \subset 54 \subset 81$. The oriented timetable header above that memory splits as $5_\omega + 5_{\omega^2} + 30$: the full Weyl runtime fuses the handed pair into a 10-channel and leaves one 30-channel with an outer parity choice. Cache lookup is a distinct split layer: two copies of H/C_4 share an 81-route H/D_8 base, and their 324 slots form 81 four-state D_8 address fibers. The non-split ternary transport operator then updates the selected state; address choice and temporal memory cannot alias because their minimal polynomials differ. The watchdog is the BC loop. Each layer is a theorem; the stack compiles because double cosets compose.

Redundancy is built in, not bolted on. The newest results sharpen the fault story: each pentad core stores its timetable *three independent ways* (lit-chart transversals, the K_5 edge labeling, dark-sector shadows), every skew pair has six Petersen homes and two matching covers, and the dark sector carries its own chart matching and two chiral tetrad partitions (Theorem 74.4 block). Erasure of any single description leaves the schedule reconstructible — redundancy appears as a counting identity, not as an engineering allowance. The middleware's two packet phases (tetrahedral/hemioctahedral, the tomotope's $2_{\{0,1,2\}}$ class) confine phase changes to cell hops, which is where the commit boundaries already sit, and the runtime's vertex layer is a free $\mathbb{F}_2^2$-torsor (the Cayley structure), so address arithmetic is XOR end to end.

What it is not. No claim is made here of asymptotic speedups over error-corrected gate-model machines, of fault-tolerance thresholds, or of scalability beyond the fractal substitution rule (40^n leaves, diameter $8n$, commit clock $T(n) = 4(7^n - 1)$) — those are stated as exact combinatorial scaling laws, and their physical realization at depth $n > 2$ is an open engineering problem with its first hard boundary already computed (the $n = 5$ desynchronization with remainder $24 = f$). The honest claim is narrower and stranger: a complete, finite, self-verifying computational architecture exists inside one photon's worth of quantum optics, with zero adjustable parameters.

76.3 Networking

The network is the computer, twice over. The thesis is not rhetorical: the atlas's transport groupoid and the register's gate groupoid are the same groupoid, so moving a packet and applying a gate are one operation. The new library-geometry theorem (Theorem 74.2) doubles the statement at the control plane: the 36 timetables form $\mathrm{SRG}(36, 15, 6, 6)$, the *other* rank-3 geometry of the same group, so the control plane and the data plane ($\mathrm{SRG}(40, 12, 2, 4)$) are the two classical geometries of one symmetry. Configuration management inherits strongly-regular guarantees: any two timetables, near or far, have exactly six common near-neighbors — worst-case reconfiguration paths are bounded by the geometry itself.

Routing, addressing, switching. Local routing is e-cube XOR on Q_3 charts (540 of them; the atlas is glued along the 1620 apartments of the Tits building, diameter 5). Addresses are Gray-coded $\mathbb{F}_2^3$ words; the antipode map is bit complement. Timetable switching obeys the overlap law: cheap hops preserve $4 = \mu$ of 10 calibrated contexts, the switch graph is the $U_4(2)$ rank-3 geometry, and every skew pair lies in exactly 3 schedules, 6 Petersen homes, and 2 pentad-matching covers — the router's spare-path inventory is an exact census, not a provisioning estimate.

Synchronization without a master. The beacon heptads (2880 complete K_7 interference meshes, all pairwise overlaps 1/3) are reference frames that form their own exchange network — the network's clock distribution is itself a network, with no distinguished master node. External discipline comes from the $\mathbb{Z}_7/\mathbb{Z}_{13}$ clocks (both decimal period 6) and the BC quasicrystal drive; internal phase order from C_{12}. A distributed-systems reader will recognize the structure: consensus by geometry, where the "leader election" problem dissolves because the frame family is a single transitive orbit.

Security posture (implication, flagged). Three structural facts have security readings, stated here as directions rather than theorems. (i) Contextuality is tamper-evidence: the KS budget 36/40 is exact, so any channel intervention that classicalizes measurement statistics moves an integer and is detectable in principle. (ii) Gate teleportation stores a unitary in the photon's self-entanglement and applies it by Bell projection against its own past — a store-and-forward primitive for quantum repeaters that never exposes the gate classically. (iii) The timetable redundancy (three independent reconstructions) is an erasure-code posture against denial: losing a description class does not lose the schedule. Turning these into security proofs requires adversary models this paper does not supply.

Scaling. The fractal holonet substitutes a full 40-node substrate for each leaf: 40^n leaves, $(40^n - 1)/39$ substrate instances, reversible diameter $\leq 8n = 8\log_{40} N$, durable commit clock $T(n) = 4(7^n - 1)$, with the first guard band an arithmetic necessity at $n = 5$. Because every child inherits the same finite protocol, scaling adds no new control logic — the protocol stack is depth-invariant. This is the architectural property that datacenter fabrics approximate statistically and this geometry has exactly.

76.4 The physics-to-architecture dictionary

In plain language. *A translation table. Each row pairs a physical quantity — an optical mode, a phase, a loss rate — with the architectural role it plays, so a physicist and an engineer reading the same later section can tell they are discussing the same object. Where a row has no entry on one side, that gap is the honest state of the design.*

The dictionary is the deepest implication: one set of integers reads simultaneously as physics and as engineering, because both readings are projections of the same incidence theorems.

object	physics reading	architecture reading
$1 + 12 + 27$ split	Bell line + meeting + disjoint	pole / I-O ports / fuel+store
240 edges	interaction qutrits	bandwidth = memory
$81 = q^4$ Steinberg	protected sector (Schur)	logical register
dual O_3 restrictions	Heisenberg state phases	state/program compiler
$H_2 = 5 + \bar{5} + 30$	oriented context chirality	timetable header $10 + 30^{\pm}$
$324 = 81 \times 4$, commutant D_8	split cache multiplicity	four-state address fiber
$0 \to 81 \to 162 \to 81 \to 0$	ternary temporal extension	non-split state update
36 spreads	measurement bases	timetable OS
$\mathrm{SRG}(36, 15, 6, 6)$	basis-overlap law	control-plane fabric
five vacua	phases of the theory	boot modes
magic strata 8+24+4	fermion shell grading	fuel octanes
540 skew pairs	two-fold null systems	hypercube charts
1620 apartments	Weyl frames	inter-chart trunks
2880 heptads	equiangular frames	sync beacons
BC drive $\arccos(-2/3)$	quasicrystal time	watchdog clock
compasses (216+216)	icosahedral order, delocalized	needle registers
pentads (chiral 216+216)	maximal partial spreads	F_5 chart caches
dark sector zoo	hidden-half structure	erasure store
tomotope ($2_{\{0,1,2\}}$, Cayley/V_4)	sphere/projective alternation	packet middleware
11-cell, 57-cell, J_1	universal completions (blocked)	the covers the hardware refuses

The first row carries the deepest identification, drawn from the companion paper *Universal Quantum Computation Through a Single Photon*: the split $1+12+27$ is not merely the local SRG anatomy of a point — it is the **Bell-line shell** of the photon's own self-entanglement seed. The temporal Bell qutrit $|\Omega\rangle$ stabilizes one line of $W(3,3)$; the 40 lines then decompose as 1 (the Bell line itself) $+\,12$ (lines meeting it — the gauge sector) $+\,27$ (lines disjoint from it — the matter sector), with substrate reading $1 + k + q^q$. The vacuum decomposition that organizes the Standard-Model derivation is literally the photon's view of its own now-context: *self = the seed of self-entanglement.* The Bell-line stabilizer has order $51840/40 = 1296 = \mu^2 q^{q+1}$ — the same $1296 = 6^4$ that counts the compass D_{10}-incidence pairs of Theorem 74.4 (36 schedules $\times$ 36 core pairs), but Theorem 64.3 proves that this is not a regular-action identification. Instead, the projective Bell stabilizer resolves the incidence carrier as 648+324+162+162, the exact mirror/schedule/left-cache/right-cache prefix decoder, while the point parabolic supplies two persistent 648 address sheets. The two cache words 110 and 111 are now resolved more finely by Theorem 72.6: they name two copies of the same H/C_4 cache, not two inequivalent internal chiral modules. Both double-cover one canonical H/D_8 base of 81 routes, and the complete cache quarter is an exact D_8-over-81 square bundle. This reversible address plane is provably not the non-split $81 \to 162 \to 81$ transport plane, despite their common 162 count. The companion also supplies the master equation at Hilbert-space level ($q^2 = q + q!$: the

9-dimensional past$\otimes$future space splits into 3 diagonal "now" histories and $6 = q!$ context-changing ones), and a ledger identity: temporal gate teleportation consumes exactly 2 classical trits of memory — the same 2-trit cost that prices the arrow of time in the self-loop fixed point of §73. Storing a state for one's future self and forgetting one's past cost the same two trits. The 27 itself now has a dual decoding: the 27 non-collinear *points* form a torsor under the genuine Heisenberg group (the normal 3_+^{1+2} of the point parabolic, acting simply transitively), whose invariant relations reconstruct the whole classical 27-zoo — the 9-triangle fibration, $\mathrm{GQ}(2,4) = \mathrm{SRG}(27,10,1,5)$, and the Schläfli graph $\mathrm{SRG}(27,16,10,8)$ of the E_6 27-lines world — while the 27 disjoint *lines* of the Bell shell form a torsor under elementary $\mathbb{F}_3^3$ (the line parabolic's O_3), a flat 3-trit register cube carrying *no* invariant strongly regular graph. Curvature for states, flatness for programs: the matter sector is a Heisenberg manifold on the state side and a ternary address space on the operational side. The register reading is exact: assigning each Bell-shell context its 3-trit address, the four pair relations $[4,4,6,12]$ are the difference-vector classes, and the geometry is a function of register arithmetic alone — meeting contexts ($\pm$-paired 4-classes) share exactly 1 transversal onto the Bell line, near-skew (12) exactly 2, far-skew (6) exactly 0: base-3 register incidence for matter contexts, twin to the charts' base-2 Gray arithmetic. *Witnesses:* `bt858_heisenberg_shell_torsors.py`, `bt860_bell_shell_register_arithmetic.py`.

Three rows deserve emphasis. The *Steinberg row* is the strongest protection statement available: the register is one irreducible representation, so *any* equivariant leakage operator is zero by Schur's lemma — protection by symmetry, not by redundancy. The *compass rows* say that what physics sees as delocalized icosahedral order, engineering sees as a navigation instrument: 12 needles per schedule, each carrying Petersen addressing, five-chart caches, and a dark-sector mirror. The *universal-polytope row* states the program's sharpest structural lesson: the hardware keeps the local physics of the exceptional objects and refuses their global rigidity; the refusal (an integer non-divisibility) is precisely why the architecture is distributed.

76.5 Beyond the laboratory

A different contract between theory and experiment. Every layer of this machine is specified by integers and verified by scripts that re-derive those integers from first principles in seconds. The cultural implication is a publishing model in which papers *run*: a claim is not a sentence but a reproducible computation, and a correction is a commit. This program has already operated under that contract (see the ethos below) — twice it corrected its own published claims at the source, and the corrected values were more natural than the originals. If the broader physics literature adopted executable claims at even a small scale, entire classes of irreproducibility would become type errors.

Quantum hardware without a building. Architectures that need millikelvin cryostats centralize by economics: only institutions can own them. A universal architecture whose parts list is catalog optics — source, splitters, delay fiber, modulator, detectors — decentralizes by the same economics. The realistic near-term role is not supremacy-class computation but *certified* small-scale quantum information: teaching laboratories that can verify 51840 and 36/40 on a bench; metrology nodes whose clock is protected by an irrational rotation; network testbeds where gates and routes are the same hardware. A machine whose benchmarks are exact integers is also the natural pedagogical machine: a student can check the whole specification.

Communication and computation stop being separate utilities. In this architecture there is no boundary at which computation hands off to networking — the gate groupoid and the transport groupoid coincide, and the fractal substitution makes the computer/network identity scale-invariant. The long-run implication, if any physical system realizes this pattern at depth, is infrastructural: bandwidth and compute become one provisioned quantity, scheduling becomes group theory, and the spectrum-allocation problem for the network's measurement contexts is solved by a finite library (36 timetables with a known switching metric) rather than by auction or contention.

Society-scale stakes of a zero-parameter physics. Separately from the machine, the substrate program's physics claims — 54+ observables, 14 falsifiable predictions, no adjustable parameters — carry a specific societal meaning: they are maximally exposed. A theory with no dials cannot retreat; the 2027 neutrino-hierarchy determination, the proton-decay window, the CMB measurements will each either hit the predicted integers or end the corresponding pillar. Public trust in fundamental science is built by exactly this posture — predictions that can die — and the machine half of the program extends the posture to hardware: the device specification is its own audit.

What could go wrong, said plainly. The architecture is exact; its physical realization is not yet demonstrated beyond the component level. The fractal scaling laws are combinatorial theorems, not engineering schedules. The security readings are directions, not proofs. And the identification of the substrate with fundamental physics stands or falls with the prediction table, not with the elegance of the mathematics. This paragraph exists because the program's ethos (below) requires the failure modes to be listed next to the claims.

76.6 The ethos

Several pre-existing corpus claims failed under exact computation and were corrected at their sources during this program: the independence number of $W(3,3)$ is 7, not 10 (no ovoid exists; the "perfect graph" block is withdrawn); the Kochen–Specker optimum is 36/40, not 34/40; the pentad-sharing count of BT844 was corrected by the chiral-orbit census of BT845 within one working session; and the tomotope's framing was corrected from "non-polytopal maniplex" to "uniform polytope with non-polytopal minimal covers" when the literature was read precisely; and the "$\mathbb{Z}_3$ Berry phase as a discrete second Chern class" was corrected (BT871) — the uniform Berry 2-cochain is a coboundary over $\mathbb{F}_3$, so the physical $\mathbb{Z}_3$ is the order-3 action on the Steinberg register, not a curvature class. In every case the corrected value is *more* substrate-natural than the claim it replaced ($\alpha = \Phi_6$; budget $= (q!)^2/v$; $432 = 216 + 216$ chiral; infinitely many covers as the actual pathology; the $\mathbb{Z}_3$ as a group action rather than a cocycle) — the pattern a real structure shows when probed honestly. The program treats corrections as first-class results: they are committed, logged, and cited like theorems, because a corpus that cannot heal is a corpus that cannot be trusted.

A Verification ledger

Every theorem above is backed by an executable script in `analysis/` with results in `data/`; the chain BT739–BT839 was developed and pushed incrementally with exact arithmetic (GF(p) elimination, cyclotomic integers, complete branch-and-bound, GAP witnesses for group-theoretic facts).

layer	primary witnesses
duality / carriers	bt817, bt820, bt821
atlas / building / web	bt773, bt777, bt744, bt779
mirror middleware / tomotope runtime	bt814, bt815, bt826
braids / registers	bt740, bt741
vacua / geography / transitions	bt808–bt813, bt816
fuel / contextuality	bt818, bt822, bt823
memory / Steinberg	bt742, bt744
clocks / quasicrystal / arrow	bt774, bt819, bt820, bt823
sync / schedules / overlap law	bt817, bt819, bt835
Grünbaum–Coxeter hemicells / library geometry	bt836, bt837, bt839
two compasses / $4 \cdot K_{10} = 12$ Petersens / chart double cover	bt843–bt845
pentad anatomy / schedule reconstruction / dark dodecahedra	bt846, bt847
amalgamation ladder / universality / J_1	bt848
maniplex barren rung / class $2_{\{0,1,2\}}$ / Cayley over V_4	bt849–bt851
seventeen universals / Aut(tomotope) universal	bt852
dark orbit zoo / shadow bijection / K_5-complete dark sector	bt853–bt855
dark charts = mirror bus (G-set iso)	bt856
Heisenberg/$\mathbb{F}_3^3$ shell torsors, Schläfli decoding	bt858
Bell-shell register arithmetic ($[4, 4, 6, 12]$ named)	bt860
code register = Steinberg ($[[240, 81, 3]]_3$, $d_X = 3, d_Z = 4$)	bt861
H_2 = sign-twisted lines (homological dictionary)	bt862
three generations = Steinberg vanishing	bt863
triality census (matter-blind, gauge-visible)	bt864
joint generation × chirality grading	bt868
chirality = polar-pair involution $(45 + 36)$	bt869
spanning-tree gravity $\tau = 2^{81}5^{23}$	bt870
Berry-phase correction ($\mathbb{Z}_3$ is the Steinberg action)	bt871
Ihara zeta = spanning-tree gravity bridge	bt872
two-zeta duality / walk census	bt873
texture triality = long-root transvection	bt874
Yukawa rule = $\mathbb{Z}_3$ grade conservation	bt875
gauge group $1 \oplus 3 \oplus 8$ from $C(R)$	bt876
gauge parity = duality ($A_4 \to S_4$)	bt877
generation charge-conjugation ($N/C = \mathbb{Z}_2$)	bt878
flavor group S_3 ($\mathbb{C}[27] = 6{\cdot}1{+}3{\cdot}1'{+}9{\cdot}2$)	bt879
generations = centre of the gauge group	bt880
spacetime = space of 40 local gauge groups	bt881
gauge connection flat on edges, $2T$-curved on Q	bt882
curvature 2-form (quaternionic, on Q)	bt883
gauge flux (Wilson loops, $\mathbb{Z}_3$/lines, $\leq 12/Q$)	bt884
YM–matter coupling $\sum \chi_{\mathrm{St}} = 1080$	bt885
SM spine (one transvection, master theorem)	bt886
gauge group = color : electroweak (27:24)	bt887
color = the matter-shell Heisenberg group	bt888
fermion content $27 = C(R)/SL(2, 3)$ (lepton/quark)	bt889
dual-torsor Steinberg compiler / native generation flag	bt865
oriented H_2 irreducible spectrum / outer lift	bt866
cache D_8-over-81 fibration / split–non-split boundary	bt867
chirality bookkeeping / Bell–compass parabolic router	bt857, bt859
universality	bt825
fractal computer/network	bt827
runtime compiler / sentinel / commit / cover boundary	bt828–bt834, bt838
corpus corrections	bt818, bt823, bt824

B Architecture Completeness and the Three Physical Residuals

In plain language. *What remains unspecified. Three physical questions are not answered by the geometry and must be settled by measurement or by a fabrication choice; this section names them rather than leaving them implicit. A design that cannot say what it does not determine is not complete in any useful sense.*

The holonet machine is now specified end to end: the 40-point substrate register and its $W(3,3)$ commuting-projector Hamiltonian; the $[[240,81,3]]_3$ Steinberg code as the native error-correcting register; the local gate set and transversal/holonomy operations; the gauge connection (flat along the 240 edges, curved on the matter graph with binary-tetrahedral $2T$ holonomy) as the physical wiring of multi-qubit interaction; and the discrete dynamics, in which the physical on-shell states are exactly the harmonic forms $\ker D$ (vacuum $\oplus$ Steinberg matter $\oplus$ line module $= 1 \oplus 81 \oplus 40$) — the least-action ground space of the finite spectral action $\operatorname{Tr} f(D^2)$, whose value reproduces the spanning-tree gravity invariant. *As a universal computer the architecture is complete*: universality, the code, the gate set, and the runtime compiler are established (cf. the substrate ledger above).

What remains is not computational but *physical*, and the three residuals of the master manuscript's completeness boundary (`w33_paper.tex`) are now each resolved or reduced:

- **(R1) the gauge-lattice embedding — resolved.** The machine's symmetry register carries the canonical symplectic $E_8/2E_8$ datum mod 2, and that datum admits a *unique* $\mathrm{PSp}(4,3)$-invariant quadratic refinement, of plus type; hence $\mathrm{PSp}(4,3) \subset O_8^+(2) = \mathrm{Aut}(E_8/2E_8)$, and the canonical positive-definite lift is E_8 (up to a central $\{\pm 1\}$). The gauge lattice is fixed.
- **(R2) the symmetry / matter register — datum supplied, bridge built.** The quintic moonshine traces $\operatorname{Tr}(2A|V_5) = 74428120$, $\operatorname{Tr}(2B|V_5) = 184024$ are computed; and the matter register is realized as a finite spectral triple (a Hilbert–Schmidt bimodule of dimension 81, first-order condition verified, gauge content $\mathbf{1} \oplus \mathbf{3} \oplus \mathbf{8}$). The residual is now the explicit physical fermion-multiplet assignment, a representation- selection task.
- **(R3) the emergent-spacetime limit — theorem-governed.** On a shape-regular (edgewise) refinement tower the full gravity-plus-matter spectral action converges term by term to the continuum Einstein–Hilbert + Standard-Model action — by Cheeger–Müller–Schrader (curvature), higher-order Regge (higher-derivative), classical lattice gauge, and the exact finite moments $\{440, 1920, 16320\}$; the spectral action is moreover continuous for Latrémolière's spectral propinquity. The emergent-spacetime limit is thus the *application* of established convergence theorems, not an open analytic obstruction.

None of the three is a gap in the machine's *operation* — the holonet runs, corrects, and computes universally as specified — and none is now an open problem in principle: a fixed gauge lattice (E_8), a matter register awaiting only its physical labelling, and an emergent spacetime with a theorem-governed continuum limit. This is the precise sense in which the architecture is complete: a finite, exact, universal machine whose physics closes onto the Standard Model and general relativity through results that are now either proved or reduced to the application of known theorems.

Reading guide for what follows. The remaining subsections develop the architecture in four movements, each a chain of witnessed theorems. *(A) The logical layer is substrate-forced*: the machine is the substrate's symplectic continuum (the oscillator representation), its fault-tolerant code is the lattice tower $A_2 \subset D_4 \subset E_8$, its universal gate set is the degree-2 symplectic plus degree-3 cubic invariants, the gate group equals the code's logical Clifford group $\mathrm{Sp}(4,3)$, and the code lattices sit on the $E_8 \to$ Leech $\to$ Monster moonshine ladder. *(B) The physical carrier is one massless self-protected photon*: the demonstrator versus the fault-tolerant machine, gates as geometric ($2T$) holonomies, self-error-correction from the substrate-fixed quasicrystal drive (topological, $|C| = q-1$), why the primitive must be a single massless photon (zero proper time, Wigner helicity λ), and the carrier as its own geon inside a fractal of nested shells. *(C) The physics and the computer are one object*: the quantum power is the gauge curvature on the matter graph, that discrete curvature is a finite-group lattice gauge theory whose continuum limit is the spectral action's gauge-plus-gravity field equations (with an exact mass gap f), the whole network has a finite holographic-code analogue — its screen multiplicity $\mu = 4$ agrees with the one-sided distance $d_Z = 4$, while the full CSS distance is $d = 3$; that finite equality alone derives neither black-hole entropy nor a continuum AdS/CFT. The runtime adds a separate typed finite control ABI: binary Q_3 address toggles lower to header flags and then to Q6 pulse transitions. Its spectral displacement $\sqrt{\lambda} = \sqrt{2}$ and its 51840-frame accounting do not by themselves supply a mass scale, group action, or continuum realization. *(D) It is testable now*: a parameter-free near-term falsification protocol on the single-photon demonstrator. Throughout, one substrate number recurs — $q - 1 = \lambda = 2$ sets the drive angle, the topological protection, and the photon's transverse helicity count — and no constant is fitted.

B.1 The architecture is the substrate's symplectic continuum

There is a structural reason the machine and the spacetime are different objects built from the same $W(3,3)$: the substrate has *two* continuum limits. The arithmetic tower $W(3,q)$ (the symplectic quadrangle over $\mathbb{F}_q$, $q = 3^n$) is spectrally rigid — exactly three eigenvalues at every q, no Weyl law — so it never becomes a smooth spacetime; the metric continuum needs the external $K3$ seed (the physics side, `w33_paper.tex`, Prop. on the two continua). The *intrinsic* continuum of $W(3,3)$ is instead *symplectic*: its Heisenberg matter shell 3^{1+2} carrying the $\mathrm{Sp}(4,3)$ action is a Weil-representation datum, whose $q \to \infty$ limit is the metaplectic (oscillator) representation of $\mathrm{Sp}(4,\mathbb{R})$ on $L^2(\mathbb{R}^2)$ — precisely the continuous-variable, Fock/oscillator computation this paper realizes. So the holonet *is* the symplectic continuum of the substrate, exactly as the spacetime is its metric continuum: the universal computer and the universe are the two faces of one finite geometry's continuum limit. (Witness: `analysis/w33_two_continua_symplectic_metric.py`.)

B.2 The fault-tolerant layer is the substrate's lattice tower [Face 5]

Because the holonet's logical layer is a continuous-variable (CV) oscillator computation, its canonical fault-tolerant encoding is the Gottesman–Kitaev–Preskill (GKP) code, which embeds a qudit in an oscillator. A *multimode* GKP code *is* a symplectic lattice in phase space: stabilisers are displacements by lattice vectors, and error correction is closest-point decoding in the symplectic dual lattice. The code quality is set entirely by the lattice, and the established optima are exceptional lattices: the *hexagonal* lattice A_2 for one mode, the densest 4-dimensional lattice D_4 for two modes (strictly better than any product of one-mode codes), and E_8 for four modes [23, 24].

This is not a free design choice for the holonet: the substrate *fixes* the lattice at every scale. The machine is two qutrits, hence two oscillator modes ($L^2(\mathbb{R}^2)$, phase space $\mathbb{R}^4$), so its optimal GKP code lattice is D_4 — and D_4 is exactly the substrate's matter shell, with $|W(D_4)| = 192$ the tomotope/flag symmetry and D_4 triality the three-generation structure. The single-mode optimum A_2 is the $q = 3$ (hexagonal, $SU(3)$) lattice of a single qutrit, and two modes stacked, $D_4 \oplus D_4$, sit primitively inside E_8 — the substrate's mod-2 homology lift (R1), which is also the $K3$ gauge lattice. Thus the tower

$$A_2 \;\subset\; D_4 \;\subset\; E_8$$

is *simultaneously* the optimal 1-, 2-, 4-mode GKP code lattices (the architecture's quantum error-correction layer) and $W(3,3)$'s physical lattice tower (single-qutrit $SU(3)$, matter shell, gauge E_8). The error correction of the computer and the gauge structure of the universe are one and the same lattice tower; the QEC code is not chosen, it is the substrate. (Witness: `analysis/w33_gkp_lattice_architecture.py`, verifying the kissing numbers $6, 24, 240$ and the embedding $D_4 \oplus D_4 \subset E_8$.)

This also fixes the lattice part of the fault-tolerance threshold. The substrate lattices are *isodual* (symplectically self-dual — the balanced-GKP condition, $\hat{q}$ and $\hat{p}$ protected equally) and densest in their dimensions, so they maximise the nominal coding gain $\gamma = d_{\min}^2 / \det^{1/n}$ among symplectic lattices of that rank:

$$\gamma_{\mathrm{dB}}(A_2) = 0.6, \qquad \gamma_{\mathrm{dB}}(D_4) = 1.5, \qquad \gamma_{\mathrm{dB}}(E_8) = 3.0$$

(and 6.0 for the Leech lattice). So the holonet's two-mode code D_4 already buys ~ 1.5 dB, and the four-mode code $E_8 \sim 3$ dB, of squeezing-margin over the trivial square code, for free. The absolute threshold still needs the noise model, the finite-squeezing GKP-state quality and the syndrome-extraction protocol — which we do not fix here — but the lattice contribution, the part that depends only on the code, is the substrate's and is optimal. (Witness: `analysis/w33_gkp_coding_gain.py`.)

B.3 The universal gate set is degree two plus degree three

The code fixes the fault-tolerant layer; the gate set completes the architecture, and it too is substrate-forced. The continuous-variable universality theorem [25] states that the Gaussian unitaries — those generated by Hamiltonians at most *quadratic* (degree 2) in the mode operators: beamsplitters, phase shifters, squeezers, i.e. the CV Clifford group $\mathrm{Sp}(2n, \mathbb{R})$ — are *not* universal, but adjoining any single generator of degree ≥ 3 (canonically the cubic phase gate $e^{i\gamma\hat{q}^3}$) generates the full unitary group on the oscillators. A universal CV machine thus needs exactly

$$\underbrace{\text{degree 2}}_{\text{Gaussian / Clifford}} \quad + \quad \underbrace{\text{degree 3}}_{\text{cubic, non-Gaussian}} \quad .$$

The substrate supplies precisely these, as its two lowest matter invariants and no others below degree three: the *degree*-2 symplectic form on $\mathbb{F}_3^4$ that $\mathrm{Sp}(4,3)$ preserves — whose oscillator-rep limit is the Gaussian group $\mathrm{Sp}(4, \mathbb{R})$ that the photon's tritter, phase plate and modulator already realise (§68.5) — and the *degree*-3 E_6 Cartan cubic on the matter $27 = 3 \otimes 3 \otimes 3$, the ε-cubic preserved by $\mathfrak{sl}(3, \mathbb{F}_3)^3$ (`w33_paper.tex`, the 27-cubic and its 45 triads). So the paper's earlier statement that "the matter shell supplies the magic" is, in the CV picture, exact: the magic is the degree-3 E_6 cubic, the minimal non-Gaussian element, matching the Lloyd–Braunstein criterion on the nose. The substrate's invariant degrees $\{2, 3\}$ *are* the universal CV generating set $\{\text{Gaussian}, \text{cubic}\}$. Together with the lattice tower $A_2 \subset D_4 \subset E_8$ this closes the architecture: $W(3,3)$ fixes both the fault-tolerant code and the universal gate set, leaving no free choice in the logical layer. (Witness: `analysis/w33_cv_universality_cubic.py`.)

Finally, the code and the gate set are not two independent facts: they are welded by a single group. The logical Clifford group of a GKP code on n qudits of dimension d is the symplectic group $\mathrm{Sp}(2n, \mathbb{Z}/d)$ over the logical phase space. For the holonet ($n = 2$, $d = 3$) this is

$$\mathrm{Sp}(4, \mathbb{Z}/3) = \mathrm{Sp}(4, 3), \qquad |\mathrm{Sp}(4, 3)| = 51840,$$

which is *exactly* $\mathrm{Aut}(W(3,3))$, the 2-qutrit Clifford group mod Pauli, and the Clifford group the photon's own tritter, phase plate and modulator already generate (§68.5, "symplectic closure = 51840"). So the substrate's symmetry group, the machine's realised gate group, and the GKP code's logical gate group are *one group*: the gates act on the code by construction. The code-preserving (transversal) Gaussian gates are the lattice automorphisms $\mathrm{Aut}(D_4) = W(F_4)$ of order $192 \cdot 6 = 1152$, whose S_3 factor is D_4 triality — the three-generation structure — and the remaining $[\mathrm{Sp}(4,3) : \mathrm{Aut}(D_4)] = 45$ logical Cliffords are teleported using the cubic resource. The logical layer of the holonet is thus a single coherent object, every part of it fixed by $W(3,3)$. (Witness: `analysis/w33_gkp_clifford_coherence.py`.)

B.4 The code lattices are the moonshine ladder [Face 5]

The same lattices carry one last structure that ties the machine to the deepest arithmetic of the substrate. The holonet's modes are free bosons, and a GKP code is a free-boson theory compactified on its stabiliser lattice; an even lattice Λ of rank r is in turn the lattice vertex operator algebra (VOA) V_Λ of central charge $c = r$ (Frenkel–Kac–Segal), while the stabiliser code itself fixes the corresponding (Narain) code CFT [26, 27]. The holonet's GKP code lattices are the substrate tower A_2, D_4, E_8 of ranks $2, 4, 8$, hence the lattice VOAs of central charge

$$c(A_2) = 2, \qquad c(D_4) = 4, \qquad c(E_8) = 8.$$

The top rung, the E_8 VOA at $c = 8$, is exactly the base of the moonshine ladder that $W(3,3)$ already realises on the physics side,

$$V_{E_8}\ (c = 8) \ \longrightarrow\ V_{\Lambda_{24}}\ (c = 24) \ \longrightarrow\ V^\natural\ (c = 24,\ \mathrm{Aut} = \mathbb{M}),$$

with $V^\natural$ the Frenkel–Lepowsky–Meurman Monster module (24 free bosons on the Leech torus, $\mathbb{Z}/2$-orbifolded), and $\Lambda_{24} \supset E_8^3$ the gauge lattice thrice. So the *same* E_8 is the holonet's optimal 4-mode GKP code, the substrate's homology/gauge lattice, and the $c = 8$ chiral VOA that begins moonshine; and the top central charge $c = 24 = f = \chi(K3)$ is the $W(3,3)$ matter count. The computer's error-correcting code and the Monster module are one vertex-algebra tower — the architecture and the moonshine residual (R2) are the bottom and top of a single ladder of substrate lattices. (Witness: `analysis/w33_gkp_voa_bridge.py`.)

B.5 What is physically needed: demonstrator versus fault-tolerant machine [Face 6]

It is worth being exact about what we are building, because the design is now complete but the *physical* realisation has two genuinely different layers. The single self-entangled photon of the build sheet — polarisation + time-bin qutrits, a tritter (F_3), a delay ladder and an EOM, read out by single-photon detectors — realises the *ideal, un-encoded* logical layer: the $\mathrm{Sp}(4,3)$ gate set, the routing network, and every parameter-free witness. It is buildable with today's quantum optics and it demonstrates the *logic*. But a single photon has fixed photon number; it cannot carry a GKP grid state (a many-photon quadrature lattice). So the photonic demonstrator is not, and cannot be, fault-tolerant.

Fault tolerance is the standard two-layer continuous-variable stack — and here both layers are substrate-fixed, with nothing left to choose:

$$\underbrace{\text{240 squeezed modes}}_{\text{physical}} \to \underbrace{\text{120 } D_4 \text{ GKP pairs}}_{\substack{\text{inner: analog}\to\text{digital}\\ \text{gain } 1.5\,\text{dB}}} \to \text{240 GKP qutrits} \to \underbrace{[[240, 81, 3]]_3}_{\substack{\text{outer: Steinberg}\\ d_X=3,\ d_Z=4}} \to \text{81 logical qutrits.}$$

The inner code is the GKP code on the matter-shell lattice D_4 (converting continuous displacement noise into discrete qutrit errors, with the 1.5 dB coding gain that eases the squeezing budget); the outer code is the $[[240, 81, 3]]_3$ Steinberg code, with $(d_X, d_Z) = (3, 4)$, on the substrate's 240 edges. What remains is purely physical, and it is the *universal* CV fault-tolerance bottleneck, not a $W(3, 3)$-specific one:

1. squeezed light at threshold (~ 10 dB; protocol-dependent, 7–20 dB in the literature), lowered by the D_4/E_8 coding gain (1.5/3.0 dB);

2. GKP qutrit *state generation* — measurement-based breeding from squeezed/cat states with photon-number-resolving detection, or matter-assisted preparation: the hard step;

3. the cubic non-Gaussian resource — the degree-3 E_6 "matter-shell magic" — via a $\chi^{(3)}$ medium or magic-state injection;

4. a programmable beamsplitter/phase/squeeze network implementing the $\mathrm{Sp}(4, 3)$ Clifford (mature integrated photonics);

5. homodyne detection for GKP and Steinberg syndrome readout.

This is the precise, honest statement of what the holonet is and what it still needs: a concatenated $\mathrm{GKP}(D_4) \circ$ Steinberg $[[240, 81, 3]]_3$ continuous-variable qutrit computer in which $W(3, 3)$ fixes *both* code layers, the gate set ($\mathrm{Sp}(4, 3)$ Gaussian + E_6 cubic) and the network — leaving *zero design freedom above the hardware*. The only open problem is the same one the entire CV fault-tolerance field faces, GKP-state generation at threshold squeezing, and the substrate's optimal lattices ease rather than remove it. (Witness: `analysis/w33_holonet_physical_stack.py`.)

B.6 The gates are geometric: holonomic Clifford from the gauge connection

There is a third, gate-level layer of robustness that the substrate supplies for free, and it explains the name. The substrate gauge connection is flat along the 240 edges and curved on the matter graph, with binary-tetrahedral $2T$ holonomy. That holonomy group is not arbitrary:

$$2T \;\cong\; \mathrm{SL}(2, 3) \;=\; \mathrm{Sp}(2, \mathbb{Z}/3), \qquad |2T| = 24,$$

the centre being $\{\pm I\}$ and the quotient $2T/\{\pm I\} = A_4$; and $\mathrm{Sp}(2, \mathbb{Z}/3)$ is precisely the single-qutrit Clifford group modulo Pauli. So the gauge connection's holonomy *is* the qutrit Clifford group, and the gates may be realised as *non-abelian holonomies* (Wilczek–Zee geometric phases) traced in the connection [4, 5]. Holonomic gates depend only on the geometric path, not on pulse timing or shape, so they are intrinsically robust to control error: this is fault tolerance at the *gate* level, complementary to the GKP code (analog level) and the Steinberg/color outer code (logical level). The full two-qutrit Clifford $\mathrm{Sp}(4, 3)$ (order 51840) is generated by the matter-graph holonomy on each qutrit together with the entangling network.

The deeper point is the identity of the connection. The *same* connection whose *curvature* furnishes the substrate's gauge fields furnishes, through its *holonomy*, the computer's gates: the gauge interactions of the physics and the logic gates of the machine are one and the same non-abelian holonomy. This is the literal content of "holo-net". (Witness: `analysis/w33_holonomic_gates.py`, constructing $\mathrm{SL}(2, 3)$ and verifying order 24, symplecticity, centre $\{\pm I\}$, quotient A_4, and the order spectrum $\{1, 1, 8, 6, 8\}$ of $2T$.)

B.7 The quantum power is the gauge curvature

Read together with the physics tier (§76: the gauge connection is flat on the 240 collinear edges and $2T$-curved on the matter graph Q, with curvature $F(p,q) = [R_p, R_q]$ a quaternion unit), the holonomic identification sharpens into a single statement: *the gates are the gauge generators, and the entangling power is the curvature.* The 40 long-root transvections R_p — one per point, each the generation-$\mathbb{Z}_3$ source of the Standard-Model spine — are exactly the elementary gates (Wilson loops of the substrate connection), and they generate the whole gate group:

$$\langle R_p : p \in W(3,3) \rangle = \mathrm{Sp}(4,3), \qquad |\cdot| = 51840.$$

Each point has $12 = k$ *flat* (collinear, causal) neighbours, where $[R_p, R_q] = I$ and $\langle R_p, R_q \rangle = \mathbb{Z}_3 \times \mathbb{Z}_3$ is abelian and classically simulable, and $27 = q^q$ *curved* (matter-graph Q, non-collinear) neighbours, where the curvature $F = [R_p, R_q]$ is an order-4 quaternion unit and $\langle R_p, R_q \rangle = 2T = \mathrm{SL}(2,3)$ is non-abelian. So the non-Clifford/entangling resource sits *exactly* on the curved directions: a flat pair gives only the abelian $\mathbb{Z}_3 \times \mathbb{Z}_3$ (classically simulable), and the per-gate non-abelian content requires curvature. Quantitatively, a triangle $p \to q \to r \to p$ carries a Wilson loop $W = R_p R_q R_r$ whose order is the discrete curvature flux: the 160 collinear triangles all carry the uniform abelian flux of order 3, while the 3240 matter triangles of Q carry non-abelian flux of orders $\{2, 4, 6, 12\}$ (counts $180, 180, 1440, 1440$, up to $12 = k$), and the discrete Yang–Mills action $S = \sum_{\triangle}\big(1 - \frac{1}{\mathrm{ord}\,W}\big)$ concentrates $\approx 96\%$ on the matter graph. In the physics reading the same curvature on Q is the $su(2)$ field strength and, through its $\mathbb{Z}_3$ Yukawa grading, the fermion-mass texture — so on the matter graph,

$$\text{curvature} \;=\; \text{entanglement} \;=\; \text{magic} \;=\; \text{mass},$$

all one object, while flatness is causality. *(Honest scope: this is a field-strength statement. Both the flat and the curved Wilson-loop* sets *generate all of* $\mathrm{Sp}(4,3)$ *— the order-*3 *flat loops are genuine gates — so the curved/flat split is the per-plaquette curvature and the location of the Yang–Mills action, not a global classical-vs-quantum partition of the group.)* The machine's quantum advantage is not added to the substrate; it lives in the curvature of the Standard-Model gauge field. (Witnesses: `analysis/w33_gauge_curvature_is_computation.py` (40 order-3 transvections, 12 flat / 27 curved per point, curvature order 4 on Q, $\langle R_p \rangle = \mathrm{Sp}(4,3)$); `analysis/w33_yang_mills_computation.py` (Wilson-loop census, the 96% matter-graph action).)

B.8 The fuel is contextuality: the register is a Wigner phase space and magic is its negativity

There is a second, resource-theoretic answer to "where does the quantum power come from," and it is exact on the substrate. Howard, Wallman, Veitch and Emerson proved that for odd-prime-dimensional qudits *contextuality* — equivalently, negativity of the Gross discrete Wigner function — is the necessary resource that lifts Clifford computation to universal quantum computation by magic-state injection [10]. The qutrit $d = q = 3$ is the smallest, cleanest case, and the substrate realizes the whole picture geometrically. The discrete phase space of two qutrits is $\mathbb{Z}_3^4$ with $3^4 = 81 = q^4$ points — *exactly* the logical register $H_1 = 81$ of the $[[240, 81, 3]]_3$ code (irreducible under $\mathrm{PSp}(4,3)$); the Wigner function lives on the same 81 points the machine computes on. The 40 vertices of $W(3,3)$ are the $(81-1)/2 = 40$ nonzero rays of that phase space — the two-qutrit Pauli operators up to phase — so the substrate *is* the two-qutrit Pauli geometry. The Clifford group acts as $\mathrm{Sp}(4,3) = \mathrm{Aut}(W(3,3))$, permuting phase points and preserving Wigner-positivity (the classically simulable sector, Gottesman–Knill). And the non-Clifford states the holonet injects — the Hesse/cubic/T-type magic of the universal gate set — are exactly the Wigner-*negative* states:

a stabilizer state such as $|0\rangle$ or $|+\rangle$ has $W \geq 0$, while the qutrit "strange" state $(|1\rangle - |2\rangle)/\sqrt{2}$ has $\min W = -\frac{1}{3}$ and mana $\ln \frac{5}{3} > 0$. So the machine is quantum precisely where the substrate's Wigner function goes negative; the magic it spends is the contextuality of the geometry, and the corpus already measures the geometry's contextual fraction as $4/40 = 1/\Phi_4$. Curvature (the gauge picture) and Wigner-negativity (the resource picture) are two readings of the same fuel. (Witness: `analysis/w33_contextuality_is_the_fuel.py`.)

B.9 From the matter-graph lattice to continuum Yang–Mills: the loop closes

The discrete curvature of the previous subsection is not a metaphor for a gauge theory — it *is* one, and its continuum limit is the gauge sector of the physics. Each matter edge carries the curvature $F = [R_p, R_q]$, an order-4 element of $2T = \mathrm{SL}(2,3) \subset \mathrm{SU}(2)$; realised as the binary tetrahedral group of 24 unit quaternions, these are exactly the quaternion units $\pm i, \pm j, \pm k$ (the six order-4 elements, SU(2)-trace 0). So the matter graph Q carries a genuine *finite-group* ($2T$) lattice gauge theory with an $su(2)$-valued field strength — the computer's gate fabric, read as a Wilson lattice. The standard Wilson plaquette action $\frac{\beta}{2}\mathrm{Tr}\{1 - P\}$ assigns each plaquette the curvature energy $1 - \frac{1}{2}\mathrm{Re}\,\mathrm{Tr}\,P$, maximal ($= 1$) on a curved ($\mathrm{Tr} = 0$) plaquette and zero on a flat one, and Wilson's theorem gives its continuum limit $\frac{\beta}{2}\mathrm{Tr}\{1 - P\} \to \frac{a^4}{2g^2} F^b_{\mu\nu} F^b_{\mu\nu}$. On the shape-regular edgewise tower (the same refinement that carries the finite spectral action to the continuum) the matter-graph action converges to the continuum $su(2)$ Yang–Mills integral $\frac{1}{4g^2}\int \mathrm{Tr}\,F^2$, which is precisely the gauge-kinetic a_4 term of the $W(3,3)$ spectral action. Together with the a_2 Einstein–Hilbert term — whose metric variation gave the Einstein field equations (`w33_paper.tex`) — and the a_0 cosmological term, varying $a_0 + a_2 + a_4$ yields the coupled Einstein + $su(2)$ Yang–Mills + Higgs field equations. The loop therefore closes: *the computer's gate curvature (a $2T$ lattice gauge theory on Q) and the physics gauge-plus-gravity field equations (the continuum spectral action) are the lattice and continuum descriptions of one $W(3,3)$ connection* — discrete gates below, smooth fields above, the edgewise tower between. *(Honest scope: the discrete $2T$ structure and Wilson density are computed; the continuum limit is the established Wilson and edgewise spectral-action results, not re-derived numerically; the local field strength is $su(2)$ per edge while the global holonomy group is $\mathrm{Sp}(4,3)$.)* (Witness: `analysis/w33_lattice_to_continuum_ym.py`.)

The finite graph is spectrally *gapped*, exactly. Since $Q = \mathrm{SRG}(40, 27, 18, 18)$ is an expander, that gap is large and substrate-fixed — the adjacency spectrum is $\{27^1, 3^{15}, (-3)^{24}\}$, so the Laplacian $L_Q = 27I - A_Q$ has spectrum $\{0^1, 24^{15}, 30^{24}\}$ and

$$\text{finite Laplacian gap} = 24,$$

the smallest nonzero eigenvalue, at multiplicity 15, with top eigenvalue 30 at multiplicity 24. These are exact finite spectral sectors. They are not a Yang–Mills mass-gap theorem: no continuum Hilbert space, Hamiltonian, confinement estimate, or uniform positive gap along a continuum-limit family has been constructed here. In particular the multiplicities are not particle identifications. (Witness: `analysis/w33_mass_gap_from_matter_graph.py`.)

B.10 Why "holo-net": the network is a holographic code

The name is earned, not decorative: the architecture is a holographic quantum error-correcting code in three exact senses, each read straight off the substrate spectrum, and each tying a physics fact to a machine component. *(i) The black-hole quarter tracks the one-sided Z-distance.* A horizon is a stabilizer edge-cut, and the Bekenstein–Hawking law $S_{\mathrm{BH}} = A/4$ has its universal $\frac{1}{4}$ *equal to* the code's distance bound $1/d_Z$: the $W(3,3)$ CSS code has $d_Z = 4 = \mu$, so $S_{\mathrm{BH}} = A/d_Z = A/\mu$. This finite dictionary uses the one-sided Z-distance; the full CSS distance remains $d = 3$. *(ii) A finite spectral analogy, not AdS/CFT.* The collinearity graph has adjacency spectrum $\{12, 2^{24}, (-4)^{15}\}$; its single *negative* (hyperbolic) eigenvalue -4 has multiplicity $g = 15 = \dim \mathrm{SO}(4, 2)$, the 4-dimensional conformal group. This is an equality of dimensions, not a bulk–boundary map. No boundary operator algebra, reconstruction channel, state dictionary, or equality of correlators has been built, so it does not establish a discrete AdS/CFT duality. (Dually, the matter graph's Laplacian $\{0, 24^{15}, 30^{24}\}$ puts the same g/f multiplicities on the gauge mass-gap spectrum.) *(iii) The fractal is the holographic principle.* The nested-shell fractal of §69 is boundary-encodes-bulk made literal: each outer shell's $[[240, 81, 3]]_3$ code holographically contains the logical information of the inner layers, exactly the concatenated holographic code structure (the Pastawski–Yoshida–Harlow–Preskill picture), with the substrate fixing the dictionary. So the holonet is a holographic code whose entropy law, conformal boundary, and bulk–boundary map are all substrate arithmetic — which is the literal content of "holo-net". (Witness: `analysis/w33_holographic_architecture.py`.)

B.11 Rank 24: the finite moonshine/CFT comparison

In plain language. *A comparison with the conformal field theories that appear in string theory, where similar-looking counts occur. Read carefully: this section establishes that certain numbers agree and does* not *establish that the two structures are the same. Numerical agreement between a finite object and an infinite one is a lead, not a result.*

The finite lattice tower has a distinguished rank 24, which can be compared with central-charge-24 constructions without identifying a holographic boundary. The code lattices climb the moonshine ladder established above — A_2 at $c = 2$, D_4 at $c = 4$, E_8 at $c = 8$, then Leech and the Monster module $V^\natural$ at $c = 24$ — and the top rung supplies the arithmetic comparison:

$$c \;=\; 24 \;=\; f \;=\; \chi(K3) \;=\; \mathrm{rank}(\Lambda_{24}) \;=\; 2k \;=\; q^q - q,$$

the matter count, the $K3$ Euler number, and the Leech rank, all one number. At $c = 24$ the boundary is the *extremal* holomorphic CFT, whose partition function is $Z = j(\tau) - 744 = q^{-1} + 196884\, q + \cdots$ with graded dimensions $V_0 = 1$, $V_1 = 0$, $V_2 = 196884 = 1 + 196883$ (the McKay vacuum-plus-smallest-irrep identity), and the Monster is its unique symmetry. By Witten's analysis of pure three-dimensional gravity this extremal $c = 24$ CFT *is* the dual of pure AdS_3 gravity [8], so its graded dimensions count the AdS_3 black-hole microstates — and those dimensions are exactly the quintic moonshine data of the substrate (e.g. $\dim V_5 = 333\,202\,640\,600$ from the master manuscript). Finally, Almheiri–Dong–Harlow showed that AdS/CFT *is* a quantum error-correcting code [9]; that general observation does not identify this finite code with a bulk $W(3, 3)$ theory or the Monster module with its boundary. Such an identification would require an explicit encoding map and matching operator/correlator data. Thus $c = f = 24$ is retained here as a precise arithmetic and representation-theoretic parallel, not a proved gravity dual. (Witness: `analysis/w33_holographic_central_charge.py`.)

Two consequences sharpen this. First, the entropy is a microstate count: the graded dimensions $\dim V_n$ are the boundary state numbers, so the black-hole entropy is $S(n) = \ln \dim V_n$, and it follows the Cardy law $S \sim 2\pi\sqrt{cn/6} = 4\pi\sqrt{n}$ at $c = 24$ — the Bekenstein–Hawking area law of the BTZ black hole. The leading area term captures the entropy to $\sim 5\%$ (the slow Cardy approach) and the Petersson–Rademacher-corrected form to $\sim$ 2–3% by $n \sim 5$, so the moonshine dimensions are a Strominger–Vafa-style microstate count of the substrate's own horizon, with $S = A/\mu$. Second — and this is why the holographic story is not a separate layer — the holographic *redundancy supports the error correction.* A holographic code over-encodes each bulk qudit on many boundary regions. On the holonet that redundancy appears three ways: the fractal nested shells (each outer $[[240, 81, 3]]_3$ shell re-encoding the inner layers), the timetable triple-storage (§74), and the holographic boundary. Their shared geometry does not make every redundancy count equal to the minimum distance: the exact code has $d = 3$ and one-sided $d_Z = 4$. (Witness: `analysis/w33_microstate_count.py`.)

The central charge even explains the generation number. The matter sector's affine symmetry is the WZW model $\mathfrak{sp}(4)$ at level $k = 12$ — the level is the graph degree — with $c = 10 \cdot 12/(12+3) = 8 = \mathrm{rank}(E_8)$, the same c as $(E_8)_1$. So the matter rung carries one E_8 of central charge, and the boundary's $c = 24 = f$ is exactly

$$24 = q \cdot 8 = q \cdot \mathrm{rank}(E_8),$$

three generations of that matter E_8 rung; the jump from the matter rung to the full boundary *is* the generation number $q = 3$. The lattice picture makes it concrete: among the $24 = f$ Niemeier (even unimodular rank-24) lattices — the boundary's possible realizations, their count equal to the central charge itself — the Leech realization is the extremal Monster CFT, while E_8^3 is literally three copies of the matter E_8, the three-generation realization. "Why three generations" and "why $c = 24$" are one fact: the boundary is q copies of the matter E_8. (Witness: `analysis/w33_wzw_generation_ladder.py`.)

And that single matter E_8 rung already contains the grand unification and the cosmological constant. The genuine conformal embeddings of $c = 8$ algebras into $(E_8)_1$ are exactly the substrate's physics: $(E_6)_1 \times (A_2)_1$ is the grand-unified $E_8 \to E_6 \times \mathrm{SU}(3)_{\mathrm{color}}$ ($248 = 78 + 162 + 8$, the 27 matter and the 8 gluons), while $(D_8)_1 = \mathrm{SO}(16)_1$ is the *cosmological-constant* sector — $\dim \mathrm{SO}(16) = 120 = vq$, with $E_8 = \mathrm{SO}(16) \oplus \mathbf{128}$. (The affine $\mathfrak{sp}(4)_{12}$ shares $c = 8$ but is a distinct CFT, not a conformal subalgebra of E_8 — the central charge matches, the embedding does not.) That SO(16) sector is where the holography meets thermodynamics: the substrate is discrete de Sitter ($\kappa = 2/k = 1/6 > 0$, with $\sum_{\mathrm{edges}} \kappa = v$ by Gauss–Bonnet), its horizon radiates at the Gibbons–Hawking temperature $T_H = k/2\pi$ — which is exactly the temperature of the modular-flow clock — and its entropy is A/μ, so *time, temperature and the cosmological constant are one thermal fact*, the arrow of time being the de Sitter expansion. On the matter/topological side the same unification recurs: the double $D(2T)$ has 6 abelian and $36 = 4 \cdot 9$ non-abelian anyons (total quantum dimension $f = 24$), but $2T$ is solvable, so braiding alone is non-universal — the magic that lifts it to universality is precisely the Wigner negativity of §B.8. Gauge curvature, contextuality, and topological magic are three names for one resource. (Witnesses: `analysis/w33_e8_conformal_embeddings.py`, `analysis/w33_thermal_cosmology.py`, `analysis/w33_topological_magic.py`.)

Three further consequences close the holographic picture, and they share a single constant $\mu = 4$. *(a) The dark sector is the* **128**. The visible half of E_8 is SO(16) ($\dim = 120 = vq$, the gauge/cosmological-constant sector); the remainder is the spinor $\mathbf{128} = 2^{\Phi_6}$, and under $\mathrm{SO}(10) \times \mathrm{SU}(4)$ it is $(\mathbf{16}, \mathbf{4}) \oplus (\overline{\mathbf{16}}, \overline{\mathbf{4}})$ — Standard-Model families (the **16**) replicated under a *hidden* $\mathrm{SU}(4) = \mathrm{SO}(6)$ with $\mu = 4$ dark colors, so $E_8 = \mathrm{SO}(16) \oplus \mathbf{128}$ is a visible/dark split and the holographic boundary predicts dark matter. *(b) Ryu–Takayanagi has an exact finite combinatorial analogue*: a boundary region A has entanglement entropy $S(A) = \delta(A)/\mu$, its minimal edge-cut over the Bekenstein factor (a point gives q, a GQ line gives q^2), with a symmetric Page curve. This finite dictionary identifies a one-sided Z-recovery scale $d_Z = \mu = 4$, not the full CSS distance $d = 3$; a continuum JLMS equality and an operational bulk-reconstruction map would be additional input. *(c) $q = 3$ is selected thermodynamically*: the expanding de Sitter horizon (temperature $T_H = k/2\pi$, curvature $\kappa = 2/k$, entropy A/μ) closes its first-law/Gauss–Bonnet condition $2(q-1)(q^2+1) = (1+q)(1+q^2)$ *only* at $q = 3$ — the thermal clock is self-consistent only there. The unifying thread: the same $\mu = 4$ appears as the finite Bekenstein normalization, the one-sided Z-distance, the RT normalization, the de Sitter entropy factor, *and* the number of dark colors. These exact finite coincidences do not by themselves prove that the corresponding continuum mechanisms are identical. (Witnesses: `analysis/w33_dark_sector_128.py`, `analysis/w33_rt_bulk_reconstruction.py`, `analysis/w33_desitter_q3_selection.py`.)

Pushing the dark sector further collapses it into the bulk itself. *(a)* The hidden SU(4) that carries the **128** is a confining gauge theory: with $N_c = \mu = 4$ dark colors and the **16**-spinor's $N_f = 16$ flavors its one-loop coefficient is $\beta_0 = (44 - 32)/3 = 4 = \mu > 0$ (versus the visible $\beta_0 = \Phi_6 = 7$), so it is asymptotically free and *confines* — dark matter is a colorless *dark hadron* at the dark confinement scale, a falsifiable alternative to a fundamental WIMP (the relic mass needs UV matching and is not derived). *(b)* That same SU(4) = SO(6) has dimension $4^2 - 1 = 15 = g$, and its non-compact real form is $SO(4, 2)$, the 4D conformal group — the discrete-AdS/CFT bulk isometry (the 15 negative-curvature modes). So *one* dimension-15 = g symmetry is at once the dark color, the Pati–Salam unifier (lepton number as the fourth color), and the holographic bulk, with its fundamental $\mathbf{4} = \mu$ being both the dark-color count and the spacetime dimension: the dark colors are the spacetime directions. *(c)* And the information story closes inside that bulk: the discrete RT entropy $S(A) = \delta(A)/\mu$ of a growing radiation region rises, peaks at the balanced cut, and falls, the *island* (the complementary matter region) taking over at the Page point $|A| = v/2 = 20$ — the $[[40, 20, 11]]$ code — so evaporation is manifestly unitary and information is recovered. Dark matter, the holographic bulk, grand-unified color, the dimension of spacetime, and the Page curve are facets of the one SU(4) = SO(6) = 15 = g. (Witnesses: `analysis/w33_dark_matter_mass.py`, `analysis/w33_su4_is_spacetime.py`, `analysis/w33_page_curve_island.py`.)

Three last threads make the dark/bulk SU(4) quantitative, dynamical, and temporal. *(a) A dark-matter scale from E_8.* Matching the dark SU(4) and visible SU(3) couplings at the unification scale gives $\Lambda_{\text{dark}} = M_{\text{GUT}}(\Lambda_{\text{QCD}}/M_{\text{GUT}})^{\beta_0^{\text{vis}}/\beta_0^{\text{dark}}}$; exponentially sensitive to β_0^{dark}, it lands the dark hadron at *tens of GeV* when $\beta_0^{\text{dark}} = 8 = k - \mu$ (the gluon count, $N_f = \Phi_4 = 10$), matching the corpus's ~ 22.8 GeV dark-matter value — a *confining* alternative to the 308 GeV WIMP ($g\, v_{EW}/k$) the corpus also records. *(b) The expanding holography is dS/CFT.* Because the substrate is de Sitter, the bulk Laplacian modes $\{0^1, 10^{24}, 16^{15}\}$ read as vacuum + matter ($f = 24$, the boundary central charge) + the 15 = g de Sitter isometries (the adjoint of SO(4, 2) = SU(4), the same dark/bulk group), with the Monster $c = 24$ CFT at future infinity — the expanding counterpart of the corpus's discrete AdS/CFT. *(c) Time is dark braiding.* The clock $2T$ sits at the bottom of the chain $2T <$ SU(2) $<$ SU(4), so one tick is a topological twist of the $D(2T)$ dark anyons — the modular T-matrix, of order $\text{lcm}(4, 6) = 12 = k$, the clock period. So dark matter, the holographic bulk, the spacetime, and the arrow of time are nested in one SU(4) = SO(6) = 15 = g: the dark colors are the spacetime directions, and time is the dark anyons braiding. (Witnesses: `analysis/w33_dark_lambda_gut.py`, `analysis/w33_dscft_modes.py`, `analysis/w33_clock_is_dark_braiding.py`.)

The same expanding boundary is also the origin of the cosmic microwave background. A de Sitter substrate *is* an inflating universe, and dS/CFT organizes its perturbations: of the bulk modes, exactly one is light — the complementary-series vacuum mode ($m^2 = 0 < \frac{9}{4}$), the inflaton — while the matter ($m^2 = 10$) and isometry ($m^2 = 16$) modes are heavy principal-series modes that decay. So there is a unique scalar to drive inflation, and its spectrum is near-scale-invariant precisely because of the de Sitter conformal symmetry of the boundary. The Starobinsky e-fold count is $N = 2(v - \Phi_4) = 2(40 - 10) = 60$, giving

$$n_s = 1 - \tfrac{2}{N} = \tfrac{29}{30} = 0.9667, \qquad r = \tfrac{k}{N^2} = \tfrac{1}{300} = 0.00333,$$

matching Planck ($n_s = 0.9649 \pm 0.0042$) with the tensor amplitude carrying the degree $k = 12$. And the primordial two-point function is a boundary CFT correlator on the *same* Monster $c = 24$ future-infinity boundary that carries the dark/bulk SU(4) — so the CMB fluctuations are correlations in the moonshine module. Inflation, dark matter, and the arrow of time share one expanding dS/CFT boundary. (Witness: `analysis/w33_inflation_dscft.py`.)

These pieces assemble into one early-universe history, with the **128**'s right-handed neutrino as the linchpin. *Inflation* ends and reheats, producing the heavy N_R — the SO(10)-singlet inside each family's **16** in the dark spinor $\mathbf{128} = (\mathbf{16}, \mathbf{4}) \oplus (\overline{\mathbf{16}}, \overline{\mathbf{4}})$. *Seesaw:* N_R has a Majorana mass at the dark/GUT scale $M_R \sim 10^{15}$ GeV $\sim M_{\rm GUT}$, so the type-I seesaw $m_\nu = m_D^2/M_R \sim 0.05$ eV gives the light neutrino masses (corpus $\sum m_\nu = 58$ meV, cyclotomic PMNS). *Cogenesis:* the *same* N_R decays out of equilibrium with CP violation, seeding both the baryon asymmetry by leptogenesis ($\log_{10}\eta_B = -|E|/(v-k-\lambda) = -240/26 = -9.23$, observed -9.21) and an equal dark hidden-SU(4) asymmetry — so dark matter is asymmetric and $\Omega_{\rm DM} \sim \Omega_b$ is automatic, with $\Omega_{\rm DM}/\Omega_b = 82/15$. *CMB:* the single light inflaton mode makes inflation single-field, so the non-Gaussianity is the consistency value $f_{\rm NL}^{\rm local} = \frac{5}{12}(1-n_s) = \frac{5}{6N} = \frac{1}{72}$, with the bispectrum *shape* the Monster $c = 24$ boundary three-point function. So one substrate runs the whole sequence — inflation, the moonshine CMB, the neutrino masses, the matter–antimatter asymmetry, and the dark matter — off the dS/CFT boundary and the **128** spinor, joined at the right-handed neutrino. (Witnesses: `analysis/w33_cmb_nongaussianity.py`, `analysis/w33_cogenesis.py`, `analysis/w33_neutrino_seesaw_128.py`.)

This whole history hangs on a substrate *scale ladder* and ends in falsifiable numbers. The energy scales descend as closed forms — $M_{\rm Pl} > M_{\rm GUT} = v_{EW}10^{2\Phi_6} > M_R \sim 10^{15} > T_{\rm RH} > v_{EW} = |E|+2q > \Lambda_{\rm dark} > \Lambda_{\rm QCD}$ — and the thermal history walks down them (inflation, reheating, the N_R lepto/cogenesis, electroweak, dark and QCD confinement). The ladder corrects a loose expectation: $M_R \gg T_{\rm RH}$, so *leptogenesis is non-thermal* (N_R from inflaton decay), and $T_{\rm RH}$ is a different rung from $\Lambda_{\rm dark}$. At the top, the primordial suite is one moonshine sheet from $N = 2(v-\Phi_4) = 60$: $n_s = \frac{29}{30}$, $r = \frac{1}{300}$, $f_{\rm NL} = \frac{1}{72}$, and the new running $\mathrm{d}n_s/\mathrm{d}\ln k = -2/N^2 = -\frac{1}{1800}$. At the seesaw rung, the cyclotomic PMNS angles ($\sin^2\theta_{12} = \frac{4}{13}$, $\sin^2\theta_{23} = \frac{7}{13}$, $\sin^2\theta_{13} = \frac{2}{91}$) with the $\mathbb{Z}_3$ Majorana phases $(120°, 240°)$ give the neutrinoless-double-beta mass $m_{\beta\beta} \approx 2.3$ meV and the Dirac phase $\delta_{CP} = 14\pi/13 \approx 194°$ — the same phase that drives the cogenesis. So the substrate's cosmology is a falsifiable program: tiny $f_{\rm NL}$ and running for CMB-S4/LiteBIRD, and $m_{\beta\beta} \approx 2.3$ meV for nEXO/LEGEND. (Witnesses: `analysis/w33_cmb_moonshine_suite.py`, `analysis/w33_scale_ladder.py`, `analysis/w33_neutrinoless_betabeta.py`.)

B.12 The falsifiability ledger, and the demonstrator as an experiment on the theory

A theory of everything is worth only the experiments that can refute it, so the program collects into one ledger: $r = \frac{1}{300}$ and $f_{\rm NL} = \frac{1}{72}$ and the running $-\frac{1}{1800}$ (CMB-S4/LiteBIRD), $m_{\beta\beta} \approx 2.3$ meV and $\sum m_\nu = 58$ meV and $\delta_{CP} \approx 194°$ (nEXO/LEGEND, DESI, DUNE), a confining tens-of-GeV dark hadron (LZ/XENONnT) with a dark-confinement gravitational-wave background peaking in the LISA band ($f \sim 10^{-4}$ Hz, a fourth GW source beyond the primordial/electroweak/cosmic-string ones), and the collider/proton-decay handles — each killable by a named experiment in 2028–2035.

The sharpest point is that the loop closes: the tabletop single-photon demonstrator *is* one of these experiments, and it can run *now.* It reads the substrate primitives directly as quantized or protected optics. The flagship is the topological frequency-conversion pump: the photon's three modes form a spin-1, and a two-tone drive pumps energy quanta between the tones at the rate $(C/2\pi)\hbar\omega_1\omega_2$ with C the Floquet-band Chern number; for spin-1 the bands carry $\{+2, 0, -2\}$, so the pump quantum is the *topologically protected* integer $|C| = 2S = \lambda = q-1 = 2$. A Kochen–Specker test on the 40 Pauli rays gives contextual fraction $4/40 = 1/\Phi_4 = \frac{1}{10}$; the BC clock beats at $\arccos(-\frac{2}{3}) = 131.8°$;

the oscillator gate spectrum is $q \pm \sqrt{\lambda} = 3 \pm \sqrt{2}$. So the demonstrator measures λ (twice, once protected), q, and Φ_4 on a bench: if the pump quantum is not 2, or the contextual fraction is not 1/10, the $W(3,3)$ substrate is falsified at the tabletop. The machine we are building is an experiment on the theory of everything. (Witnesses: `analysis/w33_falsifiability_ledger.py`, `analysis/w33_dark_confinement_gw.py`, `analysis/w33_demonstrator_measures_substrate.py`.)

Three results sharpen this to a theorem, a protocol, and a single picture. *The pump is protected.* A $(2S+1)$-level qudit driven by two incommensurate tones has extreme Floquet-band Chern number $2S$ (verified for $S = \frac{1}{2}, 1, \frac{3}{2}$ with $|C|_{\max} = 1, 2, 3$), so the qutrit's pump quantum is $2S = q - 1 = \lambda = 2$ for *any* drive realizing the winding, protected by a finite Floquet gap $\Delta_{\min} \sim \mathcal{O}(1)$ that fixes the spec (drive rate and coherence/temperature below the gap). *The contextuality is a protocol.* The 40 Pauli rays carry the 40 isotropic line contexts of $W(3,3)$ as measurement contexts (four commuting Paulis each, four contexts per ray); a state-independent Kochen–Specker test over them has contextual fraction $4/40 = 1/\Phi_4 = \frac{1}{10}$, a second tabletop falsifier. *And the whole theory is one graph.* Every claim in this paper — the Standard Model, the exceptional/moonshine tower with Monster $c = 24$, the holographic boundary and code, the dark $\mathrm{SU}(4)/\mathbf{128}/N_R$ sector, the cosmology ledger, the runtime, and the bench falsifiers — forms one connected acyclic dependency graph whose unique root is the selection $q! = 2q$: every node is reachable from $q = 3$. The theory of everything is the unfolding of a single integer, and it ends on numbers a bench and a telescope can check. (Witnesses: `analysis/w33_pump_protection_theorem.py`, `analysis/w33_kochen_specker_protocol.py`, `analysis/w33_grand_dependency_map.py`; graph: `docs/w33_grand_dependency_map.dot`.)

B.13 The geometric origin of the clock: the triangulation ladder $3 \to 4 \to 7$

Where does the clock come from, geometrically? From the same place as the substrate integers: closing simplices. The master quadratic factors as $(x-\lambda)(x-\mu) = (x-2)(x-4)$ from the seed $q! = 2q$, and the two simplices that build everything are the *triangle* and the *tetrahedron*. Three points close a loop into a 2-simplex; closing that third edge fixes one orientation — it "saves" a single $q = 3$ trit (the Klee–Irwin cycle-clock / trit-saving picture), so the minimal closed cell carries exactly one trit. Four points ($\mu = q + 1$) close the 3-simplex, the tetrahedron — self-dual (K_4 vertices *and* K_4 faces, both maximally adjacent), genus 0, and the building block of the Boerdijk–Coxeter helix whose stack twist $\arccos(-\frac{q-1}{q}) = \arccos(-\frac{2}{3})$ is the time-quasicrystal clock.

The torus is born when these two combine. The Ringel–Youngs genus of the complete polyhedral graph is $g(K_n) = \lceil (n-3)(n-4)/12 \rceil$, with denominator $12 = k$. At $n = \Phi_6 = 7$ the numerator is

$$(n-3)(n-4) = \mu \cdot q = 4 \cdot 3 = 12 = k, \qquad \text{so} \qquad g = \frac{\mu\, q}{k} = \frac{k}{k} = 1,$$

the torus — and the "3" and "4" in that numerator are precisely the triangle ($n - 4 = q$) and the tetrahedron ($n - 3 = \mu$). The genus is 1 exactly because triangle × tetrahedron equals the degree. (K_4, the tetrahedron itself, gives $(1)(0)/12 = 0$, the sphere.) The two genus-1 realizations are the only "complete" toroidal polyhedra: the Szilassi polyhedron, with 7 mutually edge-adjacent faces (the tetrahedron and Szilassi are the *only* polyhedra in which every pair of faces touches), and its dual the Császár polyhedron, with 7 mutually adjacent vertices (K_7). They share $21 = \binom{7}{2}$ edges — the Fano flags — and their incidence is the Heawood graph ($14 = 7 + 7$ vertices, 21 edges), which gives the Fano clock graph used in §B.14. Its Laplacian has the finite middle-shell displacement $\sqrt{2}$, equivalently the normalized two-branch selector described there; this does not make either polyhedron an analogue oscillator or a physical time source. The triangle, tetrahedron, and Fano incidence graph instead give a combinatorial triangulation ladder $3 \to 4 \to 7$ with distinct finite roles. (Witness: `analysis/w33_genus_ladder_clock.py`.)

And the ladder takes us all the way home: the same simplices generate the surfaces, the code, and the clock at once. *Surfaces (climbing to genus 3).* Above the sphere (tetrahedron, $g = 0$) and the torus (Fano/Heawood, $g = 1$) the next maximally-symmetric rung is the Klein quartic, the lowest-genus Hurwitz surface ($g = 3$): a regular map of 24 heptagons / 56 triangles / 84 edges with automorphism group $\mathrm{PSL}(2,7)$, $|\mathrm{PSL}(2,7)| = 84(g-1) = \lambda k\Phi_6 = \Phi_6 f = 168$. Its numbers are substrate integers ($24 = f$, $56 = v+k+\mu = \dim E_6^{\mathrm{fund}}$ analogue, $84 = k\Phi_6$), and that $\mathrm{PSL}(2,7)$ is exactly the clock/readout layer of §B.14 — so the heptagon $\Phi_6 = 7$ threads the whole ladder, the Fano heptad becoming the genus-3 heptagonal surface. *Code (trit-saving is encoding).* Closing a triangle saves one $q = 3$ trit: on the $W(3,3)$ chain complex $(40, 240, 160, \partial^2 = 0)$ the 160 triangles are the weight-3 CSS Z-checks (rank 120) and the vertices the X-checks (rank 39), so the trits no triangle can save are $H_1 = (240-39)-120 = 81 = q^4$, the logical register of the $[[240, 81, 3]]_3$ code, with one-sided $d_Z = 4$ — closing a loop and encoding a qutrit are one act. *Clock (the helix is a quasicrystal).* The tetrahelix twist $\arccos(-\frac{2}{3})$ is Niven-irrational, so the Boerdijk–Coxeter helix never closes in flat 3D — the aperiodic, two-incommensurate-frequency time-crystal clock — yet closes at exactly 30 tetrahedra in S^3 (the 600-cell $= 20 \times 30$), and $30 = h(E_8)$ is the middle factor of the supercycle $51840 = 24 \cdot 30 \cdot 72 = |\mathrm{Sp}(4,3)|$. So from the single integer $q = 3$ — three points closing a triangle, saving a trit — unfold the surfaces (genus $0, 1, 3$), the error-correcting code ($H_1 = 81$), and the quasicrystal clock (supercycle $|\mathrm{Sp}(4,3)|$): geometry, code, and time are one triangulation. (Witnesses: `analysis/w33_klein_quartic_genus3.py`, `analysis/w33_trit_saving_code.py`, `analysis/w33_bc_helix_quasicrystal.py`.)

B.14 The Heawood clock is a finite control layer, not a mass scale

The Heawood graph is the incidence graph of the Fano plane $\mathrm{PG}(2,2)$: it has 14 vertices, is 3-regular, and has Laplacian spectrum

$$\{0, (3-\sqrt{2})^6, (3+\sqrt{2})^6, 6\}.$$

On its 12-dimensional middle spectral subspace M the normalized operator

$$J = \left.\frac{L_H - 3I}{\sqrt{2}}\right|_M \qquad \text{satisfies} \qquad J^2 = I_M,$$

with two six-dimensional $+1$ and -1 branches. This is an exact finite reversible spectral selector, useful as a clock-shell diagnostic. The dimensionless displacement $\sqrt{2}$ is *not* a particle frequency or a mass scale: no physical units, coupling, or calibrated Hamiltonian have been constructed.

The useful computational object is instead a typed finite control system. The Heawood cycle space has

$$\beta_1 = 21 - 14 + 1 = 8,$$

so, after choosing a spanning-tree basis, it is an eight-bit $\mathbb{F}_2$ register with 2^8 states. The dimension is invariant while the eight coordinates are not. Separately, the executable route ABI lowers a binary Q_3 address toggle to a BT828 header flag, then to a Q6 address and a `LOAD_FLAG`→`FLIP_Q6_AXIS`→`LATCH_VERTEX` transition. Pass 377 makes the header stage exact: its 360 directed toggles compress to a disjoint $24 + 12 + 12$ support of 48 header flags, with a free C_3 depth clock of 16 three-cycles. This is a finite logic-switch reading, not a claim that a binary toggle already equals a Q6 edge traversal or a physical switch.

The runtime bookkeeping remains exact:

$$8 \to 48 \to 24 \to 72 \to 2160 \to 51840, \qquad 51840 = 720 \cdot 72 = 24 \cdot 30 \cdot 72 = |\mathrm{Sp}(4,3)|.$$

It describes word, frame, bus, and supercycle counts. Equality of the final count with a group order does not construct an action, enumerate group elements, or make one runtime pass a traversal of $\mathrm{Sp}(4,3)$.

There is also a clean separation boundary. The Fano symmetry $\mathrm{PSL}(3,2)$ has order $168 = 2^3{\cdot}3{\cdot}7$, whereas $\mathrm{Sp}(4,3)$ has order $51840 = 2^7 \cdot 3^4 \cdot 5$. Since 7 does not divide 51840, the former cannot be a subgroup of the latter. Their common factor $\gcd(168, 51840) = 24$ is an arithmetic overlap, not a canonical coupling or physical weld. (Witnesses: `analysis/bt1654_heawood_clock_homology.py`, `analysis/bt1656_runtime_word_cycle_basis.py`, `analysis/w33_machine_clock_is_mass.py`, and `analysis/w33_pass377_binary_q3_switch_header.g`.)

B.15 Four further faces of the machine: time, topological order, scrambling, and periodic orbits

Four short calculations follow. Their finite statements should be read at their proved scope; any physical realization requires its own map and calibration. *(i) Time is the modular flow, and only the clock can source it.* By the Connes–Rovelli thermal-time hypothesis, physical time is the modular (Tomita–Takesaki) automorphism flow of the state, generated by $K = -\ln\rho$. On the maximally mixed gauge/matter sector $\rho = I/81$ one has $K \propto I$, so the modular flow is trivial: the H_1 register is *timeless*. On the Fano/Heawood clock the Gibbs state gives $K = \beta L_H + (\ln Z)I$, so the modular flow is generated by the clock Laplacian and ticks at frequency $\sqrt{\lambda} = \sqrt{2}$ — time emerges from the state, and only the clock layer sources it (KMS verified). *(ii) The matter sector is a non-abelian topological order.* The $2T$ gauge theory on Q is the quantum double $D(2T)$: 7 flux sectors (the conjugacy classes of $2T$), 42 anyons $=$Catalan $C_5 = 2q\,\Phi_6$, and total quantum dimension $D = |2T| = 24 = f$. Topological protection is then concrete — logical information is anyon fusion/braiding, with total quantum dimension the matter count. *(iii) The machine is a near-optimal scrambler/decoder.* Because $W(3,3)$ is Ramanujan, its random walk mixes in the minimum time (total-variation mixing time $= 2 =$ diameter) and operator growth runs at the maximal non-backtracking rate $\sqrt{k-1} = \sqrt{11}$; as a holographic code it is therefore a Hayden–Preskill decoder, with a bulk perturbation recoverable from $O(\log)$ boundary modes essentially at once. *(iv) The spectrum is a periodic-orbit trace formula.* The closed non-backtracking geodesics (particle worldlines) are counted by $N_m = \mathrm{Tr}(B^m)$ on the Hashimoto operator ($N_3 = 960 = 6\cdot 160$); the Ihara zeta factorizes with all non-trivial zeros on the critical circle $|u| = 1/\sqrt{11}$ — an exact graph Riemann Hypothesis, equivalent to the Ramanujan property, with topological entropy $\ln(k-1) = \ln 11$. (Witnesses: `analysis/w33_thermal_time_clock.py`, `analysis/w33_anyons_from_2T.py`, `analysis/w33_scrambling_decoder.py`, `analysis/w33_periodic_orbit_spectrum.py`.)

These four are not separate curiosities; cross-read against the corpus they are one thermal object, unified by the Hawking temperature $T_H = k/2\pi$. The modular flow of (i) is a KMS state at exactly that T_H, so time, temperature and the clock coincide; the scrambling time of (iii) is $\lambda = 2$ with Page entropy $S_{\max} = E = 240$, the same fast-scrambler the corpus records. The topological order of (ii) is the *non-abelian* Drinfeld double $D(2T)$ ($42 = C_5 = v + \lambda$ anyons, total quantum dimension f), complementary to the *abelian* double $D((\mathbb{Z}/3)^2)$ of dimension $q^\mu = 81$ that is the logical register itself — one double per sector. And the zeta of (iv), counting non-backtracking geodesics $\mathrm{Tr}(B^m)$, certifies the same Ramanujan / graph Riemann Hypothesis as the corpus's point-counting $\mathrm{Tr}(A^n)$, from the complementary walk operator. So the new thermal-time/modular layer is what binds the pre-existing thermodynamic, topological, and arithmetic pillars into a single structure. (Witness: `analysis/w33_thermal_synthesis.py`.)

B.16 A near-term falsification test: the demonstrator as an experiment on the substrate

The single-photon demonstrator of the build sheet is buildable now and needs no GKP states or squeezing, yet the holonomic-gate and quasicrystal-clock structure make it a *discriminating, parameter-free* test of the substrate hypothesis itself. Three signatures, all measurable on that machine:

1. **Geometric-gate timing-independence (the discriminator).** If the gates are Wilczek–Zee holonomies (the substrate claim), the gate unitary depends only on the *path* traced in control space, not on the schedule. Reparametrising the loop — any speed, any pulse shape with the same path — leaves the gate, read out through the trace–Choi visibility $V(U) = |\mathrm{Tr}\, U|/3$, *invariant*; a generic dynamical gate would change. A measured timing-dependence falsifies the holonomic claim.

2. **Holonomy-group fingerprint.** Composing the geometric gates must close into $2T = \mathrm{SL}(2,3)$: order 24, element-order spectrum $\{1, 1, 8, 6, 8\}$, and a single-qutrit Clifford visibility spectrum $V \in \{0, \frac{1}{3}, \frac{1}{\sqrt{3}}, 1\}$ (with $V(F_3) = \frac{1}{3}$ and $V(X) = V(Z) = 0$). Gate-set tomography reads this out; any other group falsifies the gauge-connection-holonomy identification.

3. **Provably aperiodic clock.** The Boerdijk–Coxeter drive rotates by $\theta = \arccos(-\frac{2}{3}) \approx 131.81°$ per pass. By Niven's theorem [18] the only rational $\cos(r\pi)$ with r rational are $\{0, \pm\frac{1}{2}, \pm 1\}$, and $-\frac{2}{3}$ is none of them, so θ/π is irrational and the drive *never recurs* — a provably aperiodic time-quasicrystal clock. A measured recurrence at any finite pass falsifies it.

Each signature is fixed by the geometry with no free parameter, and each is within reach of the demonstrator's existing optics. Together they turn the buildable machine into an experiment *on the substrate*: not a test of whether the device computes, but of whether the gauge connection, the holonomy group, and the quasicrystal drive are the ones $W(3,3)$ predicts. (Witness: `analysis/w33_near_term_falsification_test.py`.)

B.17 BT1402 runtime-frontier handoff: readout, port, and certificate

The post-BT1377 runtime frontier makes the physical story sharper. The deterministic packet machine is a protected Clifford machine first: the 8-tick optical word, Q6/tomotope packet edge, S_3 phase synchronization, central C_3 Steinberg scheduler, and 51840-window $\mathrm{Sp}(4,3)$ supercycle are all finite Clifford data. Universal computation enters at the non-Clifford boundary, not by pretending that this Clifford kernel is already universal by itself.

The near-term single-photon demonstrator now has a concrete readout contract. Its Bell-qutrit signatures are

$$V(I) = 1, \qquad V(F_3) = \frac{1}{3}, \qquad V(X) = V(Z) = 0,$$

and route-controlled Clifford transport deliberately hides route coherence if the Bell legs are discarded: the reduced route register has uniform probabilities, purity 1/3, and zero ℓ_1 coherence. Coherence is restored only by the quantum eraser on the Bell branch,

$$\frac{|\Omega\rangle + |Z\Omega\rangle + |X\Omega\rangle}{\sqrt{3}},$$

which succeeds with probability 1/3, restores route ℓ_1 coherence 2, and routes the interferometer to a single output port. The current parametric noise envelope keeps this falsifiable: the baseline readout has $\gamma = 0.92887$ and port-0 probability 0.95258, while the conservative case keeps $\gamma = 0.79198$ and port-0 probability 0.86132.

The non-Clifford resource has likewise become an ABI rather than a slogan. The Hesse-SIC/T token has nine outcomes, is consumed at a Clifford packet boundary, and feeds a Clifford correction plus one T-frame parity bit into the next 8-tick word. Over one 51840-tick Clifford window there are 720 microframes. In the current queue envelope the baseline heavy case still has positive slack (about 55.75 successful tokens per window), and the conservative medium case still has positive slack (about 89.51 tokens per window). This is an engineering envelope, not a claim that the SIC optics or magic-state yield are solved.

Finally, the correction frontier is now stated in certificate language. The best current S_3 gauge witness has score 210 and is locally strict through the checked radius, but the MaxSAT frontier remains witness-only: the example optimality certificate exercises the verifier path and is explicitly synthetic. A project-level global optimum requires an imported solver upper-bound proof matching the witness score. This is the honest architecture boundary after BT1401: the single-photon experiment tests the Bell/route physics, the Clifford machine runs the protected finite kernel, the Hesse-SIC/T port supplies the non-Clifford ABI, and the remaining global optimality claim is a certificate import problem.

Witnesses: `tools/bt1394_reduced_qutrit_demonstrator.py`; `tools/bt1396_qutrit_quantum_erasure_readout.py`; `tools/bt1385_hesse_sic_t_port_abi.py`; `tools/bt1391_hesse_sic_t_queue_model.py`; `tools/bt1395_s3_maxsat_bound_pathway.py`; `tools/bt1400_qutrit_erasure_noise_sensitivity.py`.

B.18 BT1403 eraser-lift port: the Hesse boundary is the readout boundary

The Hesse port is not a second computer attached to the holonet after the fact. It is the quantum-eraser measurement boundary lifted by one qutrit phase label. BT1396 gives the first half of the boundary: the route register is incoherent if the Bell legs are traced out, and coherence returns only when the Bell branch is erased by the projector

$$\frac{|\Omega\rangle + |Z\Omega\rangle + |X\Omega\rangle}{\sqrt{3}}.$$

This is a three-branch measurement, with branch labels

$$\Omega, \quad Z\Omega, \quad X\Omega.$$

BT1385 gives the non-Clifford half: the Hesse-SIC/T token has nine outcomes. The ABI statement is therefore the exact lift

$$9 = 3 \times 3, \qquad h = 3r + p,$$

where $r \in \{0, 1, 2\}$ selects the route/Bell branch and $p \in \{0, 1, 2\}$ is the added phase label. Equivalently, the nine Hesse outcomes are exactly 3 route branches times 3 phase labels. The feed-forward record maps the outcome to a two-trit Clifford correction $X^r Z^p$ plus the one T-frame parity bit needed by the non-Clifford injection.

This is the physical meaning of the non-Clifford port in the single-photon architecture. The same interferometric readout that proves the route qutrit is coherent also supplies the boundary at which a non-stabilizer qutrit resource can be consumed. The Clifford runtime remains an 8-tick packet machine; the Hesse measurement writes a side record, restores the packet ABI, and hands the next word back to the protected scheduler. Thus the architecture is not "Clifford machine plus mysterious magic box" but

$$\text{Bell-branch eraser} \quad \longrightarrow \quad \text{Hesse } 3 \times 3 \text{ lift} \quad \longrightarrow \quad \text{next Clifford packet word.}$$

The boundary remains honest: this identifies the ABI and alphabet of the port, not the optical SIC implementation or magic-state factory yield. (Witness: `tools/bt1403_hesse_port_eraser_lift.py`.)

B.19 BT1404 holonet scope: one Hesse grid is one microframe

BT1404 makes the eraser-lift port visible as a packet oscilloscope. The timing identity is

$$9 \text{ Hesse outcomes} \times 8 \text{ packet ticks} = 72 \text{ ticks},$$

exactly one BT1385/BT1391 microframe. In plain terms: 9 Hesse outcomes times 8 packet ticks fill the whole scope. Thus the entire Hesse outcome alphabet fits in one microframe, with each cell $h = 3r + p$ occupying one 8-tick return word.

The scope word is:

$$\begin{array}{r|l}
0 & \text{erase Bell branch} \\
1 & \text{record route trit } r \\
2 & \text{record phase trit } p \\
3 & \text{apply } X^r \text{ frame correction} \\
4 & \text{apply } Z^p \text{ frame correction} \\
5 & \text{store } T\text{-frame bit } (r + p \bmod 2) \\
6 & \text{restore Clifford ABI} \\
7 & \text{hand off next packet word.}
\end{array}$$

This is the first machine-surface form of the non-Clifford boundary: not just a nine-outcome label, but a one-microframe live trace compatible with the 720 microframes in a 51840-tick Clifford window. The generated static scope is `docs/bt1404_holonet_scope.html`. It remains an ABI display, not a claim of physical SIC-optics hardware.

B.20 BT1405 continuous Q6 path router

BT1374 turned packet digits into executable Q6 edge addresses, but deliberately left one boundary open: an address list is not yet a continuous walk. BT1405 closes that boundary at the tomotope ABI level. For every compiled BT828 packet program, orient the packet edges to minimize the total Hamming connector length, insert shortest Q6 connector walks between consecutive packet edges, and map every traversed edge back through the BT1371 table.

The six-digit stress route keeps the six BT1374 packet edges

$$175, \ 133, \ 56, \ 37, \ 142, \ 77$$

but now they are waypoints on a single hypercube walk:

$$6 \text{ packet edges plus } 10 \text{ connector edges} = 16 \text{ Q6 steps.}$$

In words: 6 packet edges plus 10 connector edges give the continuous carrier trace. Since the tomotope body budget is 48 ticks, the continuous trace leaves

$$48 - 16 = 32$$

slack ticks inside the same body. Thus the route is no longer merely a set of six distinct Q6 addresses; it is a contiguous Q6 path from `110111` to `010011`, with every traversed edge carrying a tomotope flag and block. This is still an ABI route certificate, not a waveguide layout or detector-loss model. (Witness: `tools/bt1405_continuous_q6_path_router.py`.)

B.21 BT1406 tomotope-body edge pulse scheduler

BT1406 uses the slack exposed by BT1405 as timing structure. Each Q6 edge in the continuous route is not a bare edge label; it expands into the ternary microcycle

0	LOAD_FLAG: load tomotope flag, block, and transversal
1	FLIP_Q6_AXIS: perform the one-bit Q6 traversal
2	LATCH_VERTEX: latch the target vertex and ABI record.

Therefore the stress route obeys

$$16 \text{ Q6 edge traversals} \times 3 \text{ pulse phases} = 48 \text{ body ticks.}$$

In words: 16 Q6 edge traversals times 3 pulse phases equals 48 body ticks. The old 32-tick slack becomes the timing shell: the stress route has no idle body tick after expansion. Its packet-edge load ticks are

$$0,\ 6,\ 12,\ 21,\ 27,\ 45,$$

so the final packet edge occupies ticks $45, 46, 47$: load tomotope flag 180 in block 45, flip 010111 $\rightarrow$ 010011, and latch the target vertex. This is a timing ABI, not a calibrated optical pulse-width, detector-jitter, or loss model. (Witness: `tools/bt1406_tomotope_body_edge_pulse_scheduler.py`.)

B.22 BT1407 microframe transaction composer

BT1407 closes the whole finite control frame. BT1406 fills the first 48 tomotope-body ticks. The remaining BT1300 local-lift epilogue is

$$24 = 3 \times 8,$$

exactly three Hesse return words. In words: 48 Q6 body pulse ticks plus 3 Hesse return words equals 72 ticks.

For the six-digit stress route, the final target digit is 4. Hence $4 \bmod 3 = 1$, so the selected Hesse route branch is $r = 1$. The epilogue is the phase fanout along that branch:

frame ticks	h	(r,p)	correction
48–55	3	$(1,0)$	X^1Z^0
56–63	4	$(1,1)$	X^1Z^1
64–71	5	$(1,2)$	X^1Z^2.

Thus ticks 0–47 execute the Q6/tomotope carrier body, while ticks 48–71 execute the Hesse return words that erase the branch, record route and phase, apply the Pauli-frame correction, store the T-frame bit, restore the Clifford ABI, and hand off the next packet word. This is a full-frame ABI transaction, not a SIC-optics implementation. The witness is the BT1407 generator and JSON certificate.

B.23 BT1408 Witting contextual communication bridge

BT1408 imports the Witting-polytope communication scheme into the same runtime ABI. The external protocol uses 40 Witting rays and 40 orthogonal tetrads. Around any selected ray the other side of the delayed-query round splits as

$$1 \text{ same ray} + 12 \text{ compatible orthogonal rays} + 27 \text{ incompatible rays} = 40.$$

In words: 1 same ray plus 12 compatible orthogonal rays plus 27 incompatible rays make the whole Witting communication desk. Hence the accepted key-agreement rate is

$$\frac{1+12}{40} = \frac{13}{40},$$

and the rejected sector is 27/40, the same q^q matter shell that the holonet already uses as contextual fuel.

The four slots of an accepted Witting tetrad are not a new bus. They are the four local residues in the packet ABI:

$$\text{tomotope\_flag} = 4\,\text{tomotope\_block} + (\text{mirror\_slot} \bmod 4).$$

Thus an accepted communication round has the handoff

$$\text{Witting query pair} \to \text{common tetrad} \to \text{mirror slot} \to \text{Q6 body} \to \text{Hesse epilogue}.$$

The contextuality audit remains the corrected BT823 ceiling 36/40, with deficit 4 and contextual fraction 1/10; the older 34/40 value from the communication paper is a useful historical warning, not the local theorem. BT1408 is an ABI bridge, not a security proof or a ququart hardware certificate.

B.24 BT1409 Witting duplex admission scheduler

BT1409 separates the two acceptance clocks inside the Witting communication bridge. The state-query clock is the BT1408 shell:

$$\frac{13}{40} = \frac{1 \text{ same ray} + 12 \text{ compatible rays}}{40}.$$

The basis-query clock is smaller. A selected Witting ray lives in exactly four tetrads, hence

$$\frac{4}{40} = \frac{1}{10}$$

is the witness aperture. In words: 13/40 is communication throughput, while 1/10 is contextual witness aperture. The complementary basis shadow is

$$\frac{36}{40},$$

which matches the corrected BT823 noncontextual ceiling. Thus the old “36/40” number is not a communication success rate; it is the basis retry shadow and the classical boundary.

For one selected ray, serving every compatible state as a BT1407 transaction uses

$$13 \times 72 = 936$$

frame ticks. Auditing only the four witness apertures uses

$$4 \times 72 = 288$$

frame ticks. Rejected choices need not consume a full frame; they are classified as retry shadow or matter-shell context unless an implementation chooses to log them. This is a scheduler certificate, not a cryptographic security proof.

B.25 BT1410 Witting delayed-query frame compiler

BT1410 makes the delayed-query step operational. The protocol in [28] first lets the two sides query Witting rays, then delays the final basis measurement until the queried states have been compared. The holonet reading is a compiler:

$$(\text{Alice ray}, \text{Bob ray}) \longmapsto \text{common Witting tetrad} \longmapsto \text{BT1407 frame.}$$

If the two rays share no tetrad, the pair is a retry-shadow event and no full frame is opened. If they are distinct compatible rays, they share exactly one tetrad, so the selected basis is forced. If they are the same ray, they share four tetrads; a two-bit aperture selector chooses which witness basis to audit.

There are two tables, and confusing them loses the physics. The logical ordered pair table has

$$40 \cdot 40 = 1600$$

rows, split as

$$40 \text{ same-ray pairs} + 480 \text{ compatible distinct pairs} + 1080 \text{ retry-shadow pairs.}$$

Thus the accepted logical table has 520 rows and rate $520/1600 = 13/40$. The physical frame table is basis-local:

$$40 \text{ tetrads} \times 4 \text{ Alice query slots} \times 4 \text{ Bob query slots} = 640.$$

Equivalently, every Witting tetrad is a 4×4 frame tile with four diagonal witness apertures and twelve off-diagonal data handshakes. Across all forty tetrads this is

$$160 \text{ diagonal witness records} + 480 \text{ off-diagonal data records.}$$

The extra $640 - 520 = 120$ basis-local records are not extra throughput. They are the same-ray contextual aperture: each of the 40 rays has four basis contexts, so the physical compiler carries three extra context choices per ray above the single logical same-ray identity.

Once the common basis is selected, the later four-outcome measurement slot is just the packet slot:

$$\texttt{tomotope\_flag} = 4\,\texttt{tomotope\_block} + (\texttt{mirror\_slot} \bmod 4).$$

Therefore the Witting desk is now a packet admission ROM. Off-diagonal entries open communication frames; diagonal entries open contextual audit frames; both land in the same BT1407 72-tick body-plus-Hesse-epilogue transaction. This matches the feed-forward/programmable-loop physical idiom of time-bin photonic qudit processors [29], but BT1410 itself remains a finite compiler certificate rather than a loss model, detector calibration, or cryptographic security proof.

B.26 BT1411 Witting basis analyzer unitaries

BT1411 supplies the physical analyzer that BT1410 deliberately left abstract. For every Witting tetrad

$$B = (r_0, r_1, r_2, r_3)$$

define U_B by taking row j to be the conjugate ray $r_j^\dagger$. Then

$$U_B\, r_j = e_j,$$

so detector slot j is exactly the packet residue `mirror_slot` mod 4. This is the standard linear-optics measurement move: an analyzer unitary rotates the chosen orthonormal basis to the detector basis, after which fixed detectors read the slots. General finite-dimensional unitaries are optically synthesizeable by beam-splitter/phase-shifter meshes [30, 31]; BT1411 shows the Witting case is much more structured than a generic 4×4 mesh.

The forty analyzers have the exact hardware split

$$1 + 12 + 27.$$

There is one computational direct-rail analyzer, twelve one-direct-rail plus complement-tritter analyzers, and twenty-seven four-three-rail contextual analyzers. Equivalently, the nonzero-entry histogram is

$$4^1, \quad 10^{12}, \quad 12^{27},$$

where the exponent records how many analyzers have that many nonzero matrix entries. No Witting analyzer uses all 16 entries of a generic dense four-mode unitary. The entry alphabet is only

$$0, \quad 1, \quad \frac{\pm 1}{\sqrt{3}}, \quad \frac{\pm \omega}{\sqrt{3}}, \quad \frac{\pm \omega^2}{\sqrt{3}}, \qquad \omega^3 = 1.$$

Thus the physical reading is sharper than "program a universal interferometer." The Witting communication desk is a sparse analyzer ROM:

$$\begin{aligned} \text{delayed-query tetrad} &\longmapsto \text{four-mode Witting analyzer} \\ &\longmapsto \text{detector slot} \longmapsto \text{BT1374 packet residue.} \end{aligned}$$

The twelve middle analyzers literally have one bypass rail and a three-rail tritter-like complement; the twenty-seven contextual analyzers are balanced three-rail rows, each dark on one rail. This is a physics certificate for the single-photon ququart analyzer bank, not a chip calibration, loss budget, or beam-splitter-angle synthesis.

B.27 BT1412 toroidal Q4 oscillator boundary

BT1412 returns to the 4×4 toroidal square because that square is not a decorative drawing of the local packet router. With ordinary boundary conditions the knight graph has only 24 edges and mixed degree; with toroidal wraparound it is exactly Q_4. The existing closed toroidal knight tour is therefore a 16-tick Gray Hamilton clock on the four-cube: each packet tick flips one Q_4 bit, and the board parity $(r + c)$ mod 2 is the same as the Q_4 word parity. Hence parity alternates on every clock edge.

The new splice is that the two-dimensional cells of this same local router have the right oscillator boundary:

$$\#F_2(Q_4) = \binom{4}{2} 2^2 = 24 \text{ Q4 square faces} = q(q-1)(q+1).$$

BT802 already proved that the tetrahedral oscillator is allowed exactly at the mod-12 marks

$$\{0, 3, 4, 7\}$$

and that the neighborly genus level $h = q = 3$ is forbidden by the discriminant 145. Lifting those four marks onto the 24-plaquette Q_4 shell gives the exact aperture

$$\frac{4}{24} = \frac{1}{6},$$

which is the local "one-sixth" selector window in the clock face, not a fitted probability. Removing the forbidden genus level then lands on the toroidal dual edge invariant:

$$24 - 3 = 21.$$

That 21 is not numerology. It is the edge channel preserved by the dual toroidal polyhedra:

$$\text{Császár} : (V, E, F) = (7, 21, 14), \qquad \text{Szilassi} : (V, E, F) = (14, 21, 7).$$

Császár is the maximal vertex-adjacency boundary, Szilassi is the maximal face-adjacency boundary, and geometric duality swaps V and F while keeping $E = 21$ intact. In this exact local reading the tetrahedral oscillator sits between the two toroidal boundaries: the Q_4 plaquette shell supplies the 24-slot face clock, the forbidden $q = 3$ genus gap removes the non-realizable handle, and the surviving 21-channel edge bus is precisely the dual boundary shared by the two toroidal polyhedra.

The error-code boundary is also now sharper. A Hamilton cycle in a labeled hypercube is a cyclic Gray code, but a snake-in-the-box code is an induced path or cycle constraint [14, 15]. The full BT1412 Q_4 Gray clock is not claimed to be such a snake: it uses 16 cycle edges while Q_4 has 32 edges, so there are 16 non-cycle chords. The protected projection is instead the every-other-tick even layer

$$0,\ 6,\ 3,\ 5,\ 9,\ 15,\ 10,\ 12,$$

an eight-word parity-even code whose cyclic steps have Hamming distance 2 and whose pairwise minimum distance is 2. Thus the physical story is: toroidal Q_4 Gray clock for packet motion, every-other parity projection for single-bit-error detection, and the 1/6-aperture oscillator boundary tying the 24-plaquette shell to the 21-edge Császár/Szilassi channel. BT1412 is an ABI certificate, not a waveguide layout, a calibrated oscillator, or a new snake-code optimum. (Witness: `tools/bt1412_toroidal_q4_oscillator_boundary.py`.)

B.28 BT1413 Q4 plaquette-tomotope compiler

BT1413 makes the 24-plaquette boundary executable. BT1363 had already shown that Q4 face-edge incidence modulo the antipodal translation is the tomotope/Reye medial layer, with 48 middle blocks. BT1371 had already shown that the 192 tomotope flags are a bijective Q6 edge address table. The BT1413 compiler is just the composed map:

$$24 \text{ Q4 plaquettes} \cdot 4 \text{ edge incidences}/2 = 48 \text{ middle blocks},$$

$$48 \text{ middle blocks} \cdot 4 \text{ flag residues} = 192 \text{ tomotope/Q6 flags}.$$

The generated certificate checks that each middle block has exactly two Q4 lifts, that the three ternary sheets carry 64 flags each, that each sheet hits all 16 tomotope face labels exactly once at block level, and that every flag is a one-bit Q6 edge address. The result is not an optical layout. It is the local address compiler that lets the toroidal Q4 shell talk directly to the tomotope/Q6 packet ABI.

B.29 BT1414 Csaszar-Szilassi dual port

BT1414 uses the active/ground split that BT1317 already found in the tomotope packet:

$$192 = 168 + 24.$$

The active part is now a concrete dual analyzer port:

$$21 \text{ shared toroidal edge channels} \cdot 2 \text{ orientations} \cdot 4 \text{ residues} = 168.$$

In Császár mode the seven analyzers are vertex-neighborhood checks: each vertex of K_7 sees six incident edge channels, and each channel is seen by two vertex analyzers. In Szilassi mode the same 21 channels are read as the dual complete face-adjacency checks: seven face analyzers, again six channels each, again two analyzers per channel. The current BT1318 metric-axis records fix Császár vertex 6 and Szilassi face 4, so the crossed calibration channel is the shared edge $\{4, 6\}$.

The remaining 24 flags are not discarded. They are the BT1412/BT1413 Q4 plaquette guard band. Thus the local physical port has a sharp shape:

$$21 \cdot 2 \cdot 4 \quad \text{active dual-boundary slots} \quad + \quad 24 \quad \text{Q4 plaquette guards.}$$

B.30 BT1415 even-projection Steinberg/CSS syndrome ledger

BT1415 finishes the front end by making the every-other Q4 projection into a check ledger. The allowed clock states are the eight even words

$$0, \ 6, \ 3, \ 5, \ 9, \ 15, \ 10, \ 12.$$

They form a binary distance-2 parity layer: every allowed word has parity zero, and any one-bit Q4 clock fault flips the syndrome bit. This check is binary front-end logic; the memory register is still the existing $\mathbb{F}_3$ Steinberg/CSS object.

BT1375 supplies the memory clock. The central C_3 action on the Steinberg register has 27 three-cycles and nilpotent rank profile

$$54, \ 27, \ 0.$$

Therefore the exact syndrome ledger is

$$27 \text{ Steinberg central cycles} \cdot 8 \text{ even Q4 states} = 216,$$

and the BT1414 guard band completes the existing W33 CSS edge ledger:

$$216 + 24 = 240.$$

This is the first complete local front end in the paper:

$$\begin{aligned} \text{Q4 plaquettes} &\to \text{tomotope/Q6 flags} \\ &\to \text{Császár/Szilassi port} \\ &\to \text{even-parity syndrome} \\ &\to \text{Steinberg/CSS edge ledger.} \end{aligned}$$

I also checked the external Golden Quartic/Moebius-ball electron line of work. Otto's 2022 paper connects a golden quartic, icosahedral coefficients, chirality, and a proposed Moebius-ball electron geometry [36]; his 2025 preprint asks whether AI can compare such extended electron models [37]. A 2025 review frames this family as Zitterbewegung/field-dynamics modelling rather than established particle physics [38]. The holonet use is deliberately conservative: closed-loop topology and chirality are good design warnings, but the repo's exact quartic frontier remains the two independent D_4 quartic atoms in the local minimal-magic audit. No electron mass, spin, or waveguide claim depends on the Moebius-ball model. (Witnesses: `tools/bt1413_q4_plaquette_tomotope_face_compiler.py`, `tools/bt1414_csaszar_szilassi_dual_physical_port.py`, and `tools/bt1415_even_projection_steinberg_syndrome_layer.py`.)

B.31 BT1416 sparse CSS intertwiner

BT1416 turns the BT1415 ledger from a count into matrices. It rebuilds the canonical W33 edge-chain CSS carrier:

$$H_X = d_1 : \ C_1 \to C_0, \qquad H_Z = d_2^T : \ C_1 \to C_2,$$

over $\mathbb{F}_3$. The generated sparse certificate has shapes

$$H_X \in \mathbb{F}_3^{40\times 240}, \qquad H_Z \in \mathbb{F}_3^{160\times 240},$$

and verifies

$$\mathrm{rank}(H_X) = 39, \qquad \mathrm{rank}(H_Z) = 120, \qquad H_X H_Z^T = 0.$$

Thus the carrier is still the exact edge code

$$[[240, \ 240 - 39 - 120, \ 3]]_3 = [[240, 81, 3]]_3.$$

The new map is the typed 240×240 ledger interwiner:

$$\Pi(e_i) = e_i, \qquad 0 \le i < 240.$$

Rows $0, \ldots, 215$ are the $\mathbb{F}_2$ Q4 parity front end, with check vector $(1, 1, 1, 1)$, rank 1, and the eight even Q4 words as a distance-2 kernel. Rows $216, \ldots, 239$ are the guard tail addressing the same $\mathbb{F}_3$ edge carrier. This is deliberately not a field coercion: the binary clock syndrome and ternary CSS stabilizer live in different fields, and the interwiner is the finite scheduling boundary between them. (Witness: `tools/bt1416_css_sparse_intertwiner_matrices.py`.)

B.32 BT1417 linear-optical dual-port primitive synthesis

BT1417 converts the BT1414 port into an objectwise single-photon optical primitive list. The active dual port is not a vague optical metaphor; it has the exact inventory

21 edge-channel couplers, 42 oriented phase latches,
168 active detector bins, 24 guard apertures.

The 21 couplers are the shared K_7 edge channels. The 42 latches choose the two directions on each channel. The four residue bins behind each oriented latch are the tomotope flag residue. The 24 Q4 guard apertures are separate from the active mesh.

The analyzer matrices are 7×21 incidence matrices. In Császár mode a row is a vertex star; in Szilassi mode a row is the dual face star. In both modes the Gram law is

$$AA^T = 5I_7 + J_7,$$

so every analyzer sees six channels and any two analyzers overlap in exactly one channel. This is the chip-level object language the previous ABI was missing: coupler, direction latch, residue demultiplexer, detector bin, and guard aperture. It is still not a calibrated waveguide mask, loss budget, or detector-efficiency model. (Witness: `tools/bt1417_linear_optical_dual_port_primitives.py`.)

B.33 BT1418 finite D4-quartic magic injection

BT1418 closes the non-Clifford aperture without importing the continuum Moebius-ball electron hypothesis. The repo-local minimal-magic audit already found the exact signed frontier: two independent irreducible D_4 quartic atoms, with no shared quadratic subfield and splitting field

$$D_4 \times D_4.$$

The BT1415/BT1416 guard band fits those atoms exactly:

$$2 \text{ quartic atoms} \cdot 4 \text{ algebraic branches} \cdot 3 \text{ qutrit phases} = 24 \text{ guard apertures.}$$

Then the internal D_4 orientation torsor expands each aperture to the tomotope flag bus:

$$24 \text{ guard apertures} \cdot 8\ D_4 \text{ orientations} = 192 \text{ oriented tomotope tokens.}$$

This is the exact finite counterpart of the golden-quartic topology that is safe to use in the paper. For one quartic atom, the Steinberg lift gives

$$4 \text{ branches} \cdot 27 \text{ central cycles} \cdot 8\ D_4 \text{ orientations} = 864,$$

which is exactly the repo's Golden D_4/Weyl shell. With both independent atoms the shell is 1728, but the two atoms stay independent; BT1418 does not collapse them into one continuum object. The physical reading is that the 24 guard apertures are the non-Clifford injection rail, while the 216 parity rows remain the cheap binary front-end syndrome and the $\mathbb{F}_3$ CSS carrier keeps its commutation proof. (Witness: `tools/bt1418_d4_quartic_magic_injection_frontier.py`.)

B.34 BT1697 typed packet ABI

Reading the paper back as an architecture exposes the missing promoted object. The header introduced in §66

$$(\text{payload}, \text{Pauli frame}, \text{chart word}, \text{apartment hop}, \text{mirror slot}, \text{clock phase})$$

is not only a description of fields. It is a fixed-width transaction ABI whose verified projections are all the packet layers above. The master frame identity is

$$72 = 48 + 24 = 16 \cdot 3 + 3 \cdot 8.$$

The first 48 ticks are the continuous Q6/tomotope body:

$$16 \text{ Q6 edge traversals} \times 3 \text{ pulse phases} = 48,$$

and the last 24 ticks are the selected Hesse branch:

$$3 \text{ Hesse return words} \times 8 \text{ ticks} = 24.$$

The same transaction header receives the Witting communication desk as an admission ROM,

$$1600 = 40^2 = 520 + 1080, \qquad 520/1600 = 13/40,$$

while the basis-local physical table is

$$640 = 40 \cdot 4 \cdot 4 = 480 + 160.$$

The extra $160 - 40 = 120$ physical rows are the same-ray contextual aperture, not extra throughput. At the physical front end the dual toroidal port is

$$192 = 168 + 24 = 21 \cdot 2 \cdot 4 + 24,$$

and at the memory front end the CSS edge ledger is

$$240 = 216 + 24 = 27 \cdot 8 + 24.$$

Finally the non-Clifford guard rail is typed by the finite quartic atoms,

$$24 = 2\ D_4 \text{ atoms} \cdot 4 \text{ branches} \cdot 3 \text{ qutrit phases}, \qquad 192 = 24 \cdot 8.$$

Thus 24 is not one untyped object renamed four times. It is the common guard boundary at which Hesse epilogue, Q4 guard flags, CSS guard-tail rows, and D_4-quartic magic apertures meet while retaining distinct source certificates and field semantics. Globalizing the local 48-block packet then recovers the mirror runtime:

$$2160 = 45 \cdot 48, \qquad 51840 = 24 \cdot 2160 = 24 \cdot 45 \cdot 48.$$

So the packet header is the finite operating-system object of the holonet: compute, route, admit, check, inject, and commit are typed projections of one transaction. BT1697 is an ABI theorem, not a calibrated optical layout. (Witness: `analysis/bt1697_holonet_typed_packet_abi.py`.)

B.35 BT1698–BT1700 execution stack

In plain language. *The full run-time picture, from an instruction arriving to a result leaving. Each layer here was specified separately earlier in the paper; this section is where they are shown to fit together, and where any mismatch between them would appear.*

The ABI can now be executed, lowered, and recursively scaled. BT1698 reads the existing microframe rows as a deterministic state machine. The body is not a bag of 48 ticks; it is exactly 16 chained Q6/tomotope edges, each with the three-phase instruction word

$$\texttt{LOAD\_FLAG} \to \texttt{FLIP\_Q6\_AXIS} \to \texttt{LATCH\_VERTEX}.$$

Every committed target vertex is the next edge source. The epilogue is exactly three Hesse return words, each with the eight-tick Clifford word

$$\texttt{ERASE}, \texttt{ROUTE}, \texttt{PHASE}, \texttt{X-CORR}, \texttt{Z-CORR}, \texttt{T-BIT}, \texttt{RESTORE}, \texttt{NEXT}.$$

Thus the packet is a finite transition system:

$$72 = (16 \cdot 3) + (3 \cdot 8).$$

The earlier BT828 “Q3 XOR” control word should be read literally as a binary three-cube address-toggle bank: it compares three low binary coordinate bits, whereas the six $\mathbb{F}_3$ symbols belong to the separate parity horizon. The compiler reaches this transition system through a typed header lowering (address bits → mirror/tomotope header → Q6 flag → `LOAD/FLIP/LATCH`), not through an established identification of the binary cube transport group with the tomotope pulse group. This is an exact finite-state ABI statement; it is not an analogue-oscillator, calibrated optical-switch, or continuum-physics claim.

Pass 377: binary switch-header clock. The missing finite control object can nevertheless be named exactly. Across all $3 \cdot 3 \cdot 40 = 360$ directed one-axis binary-Q3 toggles and the three depth residues, the BT828 header formula reaches exactly 48 of the 192 tomotope flags. The three axis supports are disjoint of sizes

$$24 + 12 + 12 = 48,$$

with combined fiber profile $24 \cdot 5 + 24 \cdot 10 = 360$. Depth increment acts on these actual header addresses as

$$f \longmapsto f + 64 \pmod{192}, \qquad C_3 \text{ freely on } 48 = 16 \cdot 3 \text{ flags.}$$

This gives a finite logic-clock plane between binary address control and the Q6 address bus. It does not yet make a binary toggle into a Q6 edge traversal or identify this $16 \cdot 3$ clock with the separately constructed LOAD/FLIP/LATCH schedule. (GAP witness: `analysis/w33_pass377_binary_q3_switch_header.g`.)

Pass 378: why the two $16 \cdot C_3$ clocks still do not identify. The header plane and the 48 phase positions of the BT1406 six-digit stress schedule are both free C_3-sets with sixteen orbits. Thus they admit abstract equivariant bijections, but exactly $16!\,3^{16} = 900657498850357248000$ of them: no one is natural from that set type alone. More decisively, BT1406 retains one `tomotope_flag` through each `LOAD/FLIP/LATCH` triple, whereas the Pass-377 action takes every header flag to a distinct flag. GAP therefore proves that no C_3-equivariant correspondence can factor through the scheduler's actual flag label. The live stress flags meet the header plane only in $\{112, 144\}$, and no header C_3-cycle contains a full stress triple. This is a structural non-intertwining result, not merely an absent hardware claim. (Witness: `analysis/w33_pass378_header_scheduler_c3_obstruction.g`.)

Pass 379: the header clock is not a Q_6 geometric operation through the live address table. The same arithmetic $f \mapsto f + 64 \pmod{192}$ is a free control-address clock, but BT1371's pinned flag-to-Q_6 table need not turn it into a cube symmetry. GAP checks the Q_6 edge line graph directly: of its $960 = 64\binom{6}{2}$ adjacent edge pairs, only 146 stay adjacent under the transported shift, while 814 are lost and 814 nonadjacent pairs become adjacent. Flags $\{0, 8\}$ provide the first witness, shifting to $\{64, 72\}$ whose assigned edges share no endpoint. Hence the header depth cycle is control metadata, not a Q_6 edge traversal, geometric switch, or optical operation through the current table. This does not prohibit a future explicitly constructed different intertwiner. (Witness: `analysis/w33_pass379_header_q6_geometry_boundary.g`.)

Pass 380: minimal scheduler switch state and the remaining binding table. A scheduler flag alone has no free phase clock. Retaining all sixteen live flag identities while making the C3 phase action free requires at least $16 \cdot 3 = 48$ states, achieved exactly by (`tomotope_flag`, `phase_trit`). The canonical full-bus lift $\iota(f, p) = f + 64p \pmod{192}$ is equivariant, but meets the Pass-377 header plane in only the cycles $[16, 80, 144]$ and $[48, 112, 176]$. Hence two schedule orbits are naturally anchored and fourteen remain unbound. Even with the two oriented anchors fixed, $14!\,3^{14} = 416971064282572800$ extensions remain. The needed next object is a typed sixteen-row header-orbit binding table with phase offsets, not an inferred Q6 geometry or optical implementation. (GAP witness: `analysis/w33_pass380_minimal_scheduler_phase_lift.g`.)

Passes 381–385: executable control ABI, not an inferred optical map. Pass 381 supplies that sixteen-row table as reviewed compiler input. Its rule

$$c(e, p) = r_e + 64(p + o_e) \pmod{192}$$

is checked on all 48 live body pulses; two rows preserve the canonical Pass-380 anchors and the other fourteen are labelled `reviewed_external_binding`. This is an auditable firmware crosswalk, not a claim that it is forced by the Q6 cube. Pass 382 makes the `LOAD/FLIP/LATCH` word an isolated reversible 48-state controller: the controller successor is a 48-cycle, while a free phase rotation differs at its 16 latch transitions. Pass 383 then gives the exact typed-control answer: an orientation-preserving branch lift yields C_6; an S_3 lift requires separately supplied phase reflection.

Pass 384 exhausts the narrow coordinate explanation. Of the 540 strict surjective one-bit folds $\mathbb{F}_2^6 \to \mathbb{F}_2^3$, none exactly intertwines the live BT1371 direction table and the 48 header-axis labels; the six stress-profile fits remain one noncanonical orbit. Pass 385 adds the inherited-orbit test: the two canonical anchors lie in one header eight-class orbit but opposite BT1371 colors, while the live stress path has trivial stabilizer even in full Q_6. No binding preserving both partitions can retain both anchors. Hence the complete crosswalk is a versioned logic-switch ABI with explicit external inputs, not a geometric, photonic, or oscillator identification. (GAP witnesses: `analysis/w33_pass381_explicit_header_binding_abi.g`, `analysis/w33_pass382_reversible_logic_switch_controller.g`, `analysis/w33_pass383_branch_phase_control_group.g`, `analysis/w33_pass384_q6_q3_fold_obstruction.g`, and `analysis/w33_pass385_header_stress_orbit_anchor_obstruction.g`.)

BT1699 then lowers the same ticks onto the physical single-photon frame already used by the Witting/Fano automaton:

ticks	hardware role
$0 \dots 7$	source switch
$8 \dots 31$	program/delay
$32 \dots 47$	analyzer or OAM body
$48 \dots 63$	detector or Hesse handoff
$64 \dots 71$	dark-reference closeout.

Each tick has a symbolic loss hook, and the last eight ticks carry the dark reference. The guard boundary is welded objectwise:

$$\text{port guard flag } (168+i) = \text{ CSS edge row } (216+i) = D_4 \text{ magic aperture } i, \qquad 0 \le i < 24.$$

This proves that the repeated 24 is a typed interface between physical port, memory syndrome, and non-Clifford resource rail.

BT1700 is the fractal compiler law. Substituting a W(3,3) packet fabric at each site gives, at depth n,

$$N(n) = 40^n, \qquad T(n) = 72 \cdot 40^n,$$
$$B(n) = 48 \cdot 40^n, \qquad G(n) = 24 \cdot 40^n, \qquad B(n) = 2G(n),$$

and the route/scheduler clocks

$$R(n) = 8n, \qquad S(n) = 2160 \cdot 40^n, \qquad C(n) = 51840 \cdot 40^n.$$

The local operating system never changes: every scale keeps the same packet ABI, the same 48/24 body/guard split, and the same typed guard handoff. This is the computer/network architecture in one sentence: a single-photon packet operating system recursively tiled by W(3,3). (Witnesses: `analysis/bt1698_holonet_packet_state_machine.py`, `analysis/bt1699_holonet_abi_to_hardware_lowering.py`, and `analysis/bt1700_recursive_holonet_packet_compiler.py`.)

B.36 BT1701–BT1703 runtime safety layer

The execution stack has three immediate runtime consequences. First, BT1701 renders the packet as an actual trace artifact at `docs/bt1701_holonet_packet_trace_visualizer.html`. The page is generated from the verifier data, not drawn by hand: every tick in the 72-tick packet is colored by the physical stage

$$8 + 24 + 16 + 16 + 8,$$

and the same page records the 48/24 split, the 24-row guard weld, and the recursive compiler law.

Second, BT1702 proves the scheduler boundary. The collision-free recursive key is

$$\text{extended slot} = \text{packet address} \cdot 2160 + \text{polar sheet} \cdot 48 + \text{body slot}, \qquad \text{phase} = \text{body slot mod } 3.$$

With this key, recursive packets do not collide. If one intentionally reuses a single finite 2160-slot physical bus, the packet address must become a time slice:

$$(\text{packet time slice}, \text{local mirror slot}, \text{phase}).$$

This is the correct engineering boundary. The recursive compiler gives a collision-free addressing fabric; it does not turn one finite bus into simultaneous infinite parallel hardware.

Third, BT1703 classifies symbolic faults at the ABI boundary:

fault	finite exit
loss	retry frame or local reprogram retry
dark click	local dark-reference termination
parity fault	CSS syndrome handoff.

The verifier injects 72 loss hooks, 8 dark-reference clicks, and 24 guard-weld parity faults. Every symbolic fault has one finite exit. This is still not a stochastic noise model; it is the routing table that a calibrated noise model must plug into. (Witnesses: `analysis/bt1701_holonet_packet_trace_visualizer.py`, `analysis/bt1702_holonet_scheduler_collision_audit.py`, and `analysis/bt1703_holonet_fault_propagation_simulator.py`.)

B.37 BT1704–BT1706 replay and retry economics

The next layer turns runtime safety into an operational model. BT1704 replays the 72-tick event log twice from the same initial state. The two executions produce identical tick logs and identical final registers. The final cursor is

`HESSE_WORD:5:DONE`,

with final corrections X^1, Z^2, and time-frame bit 1. This is the determinism test the packet ABI needed: the trace is not merely ordered; it is replayable.

BT1705 simulates one shared 2160-slot mirror bus under FIFO time division. One packet consumes one 2160-slot service slice. Three profiles are tested:

profile	packets	max queue
depth-1 burst	40	40
depth-2 wavefront	1600	1561
depth-2 sequential	1600	1.

All profiles are collision-free, every packet is served once, and Jain fairness is exactly 1. The result is deliberately engineering-facing: the finite bus is fair and schedulable, but bursty recursive traffic pays finite queueing latency.

BT1706 attaches symbolic rates to the BT1703 fault exits. For the nominal profile

$$p_{\text{loss}} = 10^{-5}, \qquad p_{\text{dark}} = 10^{-6}, \qquad p_{\text{parity}} = 5 \cdot 10^{-6},$$

the expected retry/reprogram load per packet is

$$64\, p_{\text{loss}} = \frac{2}{3125},$$

and the expected CSS handoff load is

$$24\, p_{\text{parity}} = \frac{3}{25000}.$$

The verifier also includes an engineering 10%-guard profile and the exact guard-budget limit profile. These are not experimental thresholds; they are the exact ABI economics once a bench supplies rates. (Witnesses: `analysis/bt1704_holonet_packet_replay_runner.py`, `analysis/bt1705_holonet_shared_bus_time_division_simulator.py`, and `analysis/bt1706_holonet_retry_economics.py`.)

C The exceptional skeleton and its physics

The architecture above shows that a single photon on $W(3,3)$ is a universal computer, network, and clock. This section turns from the *machine* to the *world*: it shows that the same $q = 3$ substrate carries an exact tower of exceptional structures — and a layer of falsifiable physics.

Roadmap. The argument is one ascending ladder, each rung an executable witness.

1. *Why* $q = 3$ (§C.15): three independent selections — geometric ($q! = 2q$), quantum-resource (minimal magic dimension), and holographic ($c = 24$) — all force $q = 3$.
2. *The qubit/qutrit crossover and the genus tower*: the multi-qubit contextuality ladder meets the $\{3,7\}$ Hurwitz tower at the genus-7 Macbeath surface; the $\{3,8\}$ octagon tower is its qubit twin; and the substrate selects the vertex figure $n - 6 \mid k = 12$.
3. *The Eisenstein body*: the Witting polytope (240 E8 roots, 2160 bus edges, 40 $W(3,3)$ rays) is the substrate over $\mathbb{Z}[\omega]$, welded to E8 two ways (Eisenstein $\omega = C^{10}$ and the 120 icosians).
4. *The exceptional rungs* $\mathrm{E}_6 \to \mathrm{E}_7 \to \mathrm{E}_8$: the Hessian polytope (27 lines), the Klein quartic (28 bitangents), and the icosian 600-cell; their Weyl groups telescope $51840 \to \times 56 \to \times 240$.
5. *Above E8*: the complex Leech (Eisenstein, $12 = k$), the Suzuki chain from the three-qubit $\mathrm{G}_2(2)$, and the Monster at $c = 24 = f$.
6. *The physics* (§C.12): the Standard Model from trinification, gauge unification, the neutrino sector, and the falsifiable scorecard — with one honest tension and its resolution.

The whole climb is drawn in `docs/w33_exceptional_tower.svg`, and its honest epistemic status (framework / derived / measurable) is kept by `analysis/w33_exceptional_tower_ledger.py`: most rungs are exact theorems wearing substrate labels, a smaller set is computed from $q = 3$, and only the measurable layer below falsifies.

Results at a glance. The substrate's falsifiable predictions, all fixed by $q = 3$ arithmetic with no free parameters ($N = 2(v - \Phi_4) = 60$ e-folds):

Observable	Substrate	Current bound (mid-2026)	Status
n_s	$1 - 2/N = 29/30$	Planck 0.9649 ± 0.0042	consistent ($\sim 0.4\sigma$)
r	$k/N^2 = 1/300$	< 0.036	consistent; LiteBIRD
f_{NL}	$5/(6N) = 1/72$	-0.9 ± 5.1	consistent
$dn_s/d\ln k$	$-2/N^2 = -1/1800$	-0.0045 ± 0.0067	consistent
$\sum m_\nu$	$\approx 0.06\,\mathrm{eV}$ (NO)	DESI $< 0.072\,\mathrm{eV}$	consistent (resolved)
$m_{\beta\beta}$	2–4 meV	$<$ 36–156 meV	consistent
Ω_{DM}/Ω_b	$82/15$	≈ 5.36	$\sim 2\%$
$\sin^2\theta_W$	$3/8 \to 0.231$	0.23122	consistent (run)
$\tau_p(p \to e^+\pi^0)$	$\sim 4.6 \times 10^{35}$ yr	$> 2.4 \times 10^{34}$ (SK)	testable (Hyper-K)
contextual fraction	$1/\Phi_4 = 1/10$	demonstrator	testable
pump Chern C	$2S = \lambda = 2$	demonstrator	testable

Every entry is a closed substrate expression; the lone historical tension ($\sum m_\nu$) is resolved by the seesaw (§C.12), and three rows are sharp near-term falsification handles.

C.1 The one object and its seven faces

In plain language. *One mathematical object, described seven different ways — as a network, as a set of symmetries, as a code, as an optical mode structure, and so on. The point is not that seven descriptions exist but that they are the same object: a fact proved in one language is automatically a fact in the other six. That is what makes the architecture in this paper possible at all.*

Before the details, the organizing claim — and it frames the entire paper, not just this section. Everything here — the selection of $q = 3$, the substrate constants, the gauge group, the neutrino masses, the fault-tolerant code, the benchtop experiment, and inflationary cosmology — is *one structure* seen from seven sides. That structure is the $q = 3$ Eisenstein object: the Witting polytope $3\{3\}3\{3\}3\{3\}3$ over $\mathbb{Z}[\omega]$ (240 vertices = E8 roots), its symmetry the Shephard–Todd group #32 (order $155520 = 3|\mathrm{Sp}(4,3)| = 3|\mathrm{Aut}(W(3,3))|$), equivalently the degree-two cyclotomic skeleton $\{\Phi_3, \Phi_4, \Phi_6\}$ whose indices $\{3, 4, 6\}$ are the crystallographic complex periods. Its seven faces (Fig. 9):

1. **Selection** (cosmology/geometry, §C.15). $\Phi_4(q) = q^2 + 1$ factors the de Sitter closure cubic $(q - 3)\Phi_4(q)$ and the GQ point count $(q + 1)\Phi_4(q) = 40$; the periods $\{3, 4, 6\}$ are the only crystallographic complex periods, and 3 is the only prime among them.
2. **Constants** (arithmetic). Two number-sets fall out of the one object: the cyclotomic *values* $\{\Phi_3, \Phi_4, \Phi_6\}(3) = \{13, 10, 7\}$ (the register, the Sp(4) θ, the Fano 7; sum 30), and the Witting *degrees* $\{12, 18, 24, 30\} = \{k,\ h(E_7),\ c{=}f,\ h(E_8)\}$ (product 155520). The top degree $30 = h(E_8)$ is the cyclotomic sum 13+10+7.
3. **Gauge** (the Standard Model, §C.12). E_6 enters as the Hessian polytope (27, a sub-rung of the Witting tower $8 \to 27 \to 240$), and the gauge dimensions *are* Witting degrees: $\dim \mathrm{SU}(3)^3 = 24 = c{=}f$ and $\dim \mathrm{SM} = 8{+}3{+}1 = 12 = k$; the two non-Coxeter degrees $\{12, 24\}$ are exactly the SM and trinification dimensions, with $\sin^2\theta_W = 3/8$ and three generations from the S_3 triality.

4. **Neutrino** (particle masses, §C.12). The Majorana hierarchy ratio is $\Phi_3/q^2 = 13/9$ and the PMNS deformation is $1/\Phi_3$ — the same $\Phi_3 = 13$.

5. **Code** (fault tolerance, §C). The GKP tower $A_2 < D_4 < E_8$ is the Eisenstein tower: A_2 the $q = 3$ hexagonal (one-qutrit) lattice, D_4 the matter shell, E_8 the Witting polytope. The boundary charge $c = 24 = f$ is the Witting degree-24.

6. **Demonstrator** (experiment). The benchtop meters the skeleton: the contextual fraction on the $40 = (q{+}1)\Phi_4$ $W(3,3)$ rays is $1/\Phi_4 = 1/10$, and the Thouless-pump Chern number on the $A_2 < D_4 < E_8$ ladder is $\lambda = 2$.

7. **Cosmology** (inflation, §C.12). The e-fold number is $N = 2(v - \Phi_4) = 2\,h(E_8) = 60$ — twice the E_8 Coxeter number (the top Witting degree), built from the GQ count $v = 40$ and $\Phi_4 = 10$. The CMB observables are then closed forms in N and the Witting degree $k = 12$: $n_s = 1 - 2/N = 29/30$, $r = k/N^2 = 1/300$, $f_{NL} = 5/(6N) = 1/72$, running $-2/N^2 = -1/1800$.

So the $q = 3$ substrate is not a list of coincidences spanning cosmology, gauge physics, particle masses, computation, and metrology; it is one Eisenstein object presenting seven faces. (Witnesses: `analysis/w33_eisenstein_grand_synthesis.py`, `analysis/w33_gauge_sixth_face.py`, `analysis/w33_cosmology_seventh_face.py`.) The faces are not independent because they share the same integers: the load-bearing ones — $k = 12$, $\Phi_4 = 10$, $\Phi_3 = 13$, $c{=}f = 24$, $v = 40$ — each serve three or more faces at once, and the whole bookkeeping is one periodic table over $q = 3$ (Table 1; witness `analysis/w33_substrate_periodic_table.py`). The subsections that follow detail each face and the connections between them — including a benchtop bridge that turns the neutrino mass ratio into a single-photon measurement (`analysis/w33_neutrino_demonstrator_bridge.py`), which a contextuality simulation on the 40 $W(3,3)$ rays realises explicitly, returning $\Phi_4 = 10$ and 13/9 from the incidence geometry (`analysis/w33_contextuality_simulation.py`).

The edge, and how it closes. A synthesis is worth its boundary. Every geometric integer of the seven faces is a cyclotomic value $\Phi_d(3)$, a GQ count, a Witting degree, or a simple combination (e.g. $|\mathrm{PSL}(2,7)| = 168 = \Phi_6{\cdot}f$, $\dim E_8 = 248 = 240 + 8$). The two quantities that seemed to escape — the Monster minimal dimension 196883 and the fine-structure constant ($1/\alpha \approx 137$), which live at the $c = 24 = f$ moonshine ceiling — turn out to close too, at the integer level. The key is that the *Leech kissing number factors entirely into substrate constants*,

$$196560 \;=\; 6\,\mu\,q^2\,\Phi_3\Phi_4\Phi_6 \;=\; 6\cdot 4\cdot 9\cdot 13\cdot 10\cdot 7,$$

so the complex (Eisenstein) Leech — the $\mathbb{Z}[\omega]$, rank-12=k, $6\cdot\mathrm{Suz}$ lattice atop the substrate's tower — is built from $\{\mu, q, \Phi_3, \Phi_4, \Phi_6\}$. Then the lift parameter is forced, $\tau = \mu q^2\Phi_6 = 252 = 32760/(\Phi_3\Phi_4)$; the Monster dimension is $196883 = 196560 + \mu q^4 - 1$ (with $\mu q^4 = h(E_7)^2 = 324$ the classical j/Leech gap); and the fine-structure *integer* is a cyclotomic combination, $1/\alpha = 137 = \Phi_3\Phi_4 + \Phi_6 = 2^{\Phi_6} + q^2$. So no free *integer* parameter remains: even the famous 137 is $\{\Phi_3, \Phi_4, \Phi_6\}$ arithmetic, and the moonshine ceiling is the substrate constants. The one genuinely non-integer input is the 0.036 of the measured 137.036 — the QED running, dynamics rather than arithmetic. *That* is the honest residue: the substrate fixes the integer, the renormalization flow supplies the rest. (The third value $\Phi_6(3) = 7$ also surfaces in the toroidal thread: the Császár polyhedron is K_7 on the torus and the dual Szilassi has 7 faces, with C_2 symmetry of order $2 = \lambda = q - 1$.) Witnesses: `analysis/w33_alpha_closure.py`, `analysis/w33_monster_leech_second_layer.py`, `analysis/w33_eisenstein_stress_test.py`. And the single most decisive near-term test — benchtop, calibration-free, three faces — is the contextual fraction $1/\Phi_4 = 1/10$, now a concrete single-photon protocol (`analysis/w33_contextuality_protocol.py`).

Table 1: The periodic table of the $q = 3$ Eisenstein object (excerpt): the load-bearing integers, their closed expression, and the faces they serve (1 Selection, 2 Constants, 3 Gauge, 4 Neutrino, 5 Code, 6 Demonstrator, 7 Cosmology).

Value	Expression	Class	Faces
2	$q-1=\Phi_1$	cyclotomic	6 (pump Chern λ)
10	$q^2+1=\Phi_4$	cyclotomic value	$1,2,6$ (θ, $1/\Phi_4$)
12	$q(q+1)=k$	Witting degree	$3,5,7$ (dim SM, D_4, $r{=}k/N^2$)
13	$q^2+q+1=\Phi_3$	cyclotomic value	$2,4$ (register, Φ_3/q^2)
24	$q^3-q=c{=}f$	Witting degree	$2,3,5$ ($\dim \mathrm{SU}(3)^3$, charge)
30	$h(E_8)=\Phi_3{+}\Phi_4{+}\Phi_6$	Witting degree	$2,7$ ($N/2$)
40	$(q+1)\Phi_4=v$	GQ count	$1,6,7$ (rays, $N{=}2(v{-}\Phi_4)$)
60	$2(v-\Phi_4)=2h(E_8)$	derived	7 (e-folds N)
240	E8 roots = Witting	derived	$1,5$ (gauge lattice)
155520	$3\lvert\mathrm{Sp}(4,3)\rvert$	the object	1–7 (ST#32 order)

The closure. Taken to its end, the arithmetic of the substrate is a single forced integer unfolding into the world. From $q = 3$ alone — itself the unique common solution of the selection census — the entire integer content follows in closed form: $\lambda = q - 1$, $\mu = q + 1$; the cyclotomic skeleton $\{\Phi_3, \Phi_4, \Phi_6\} = \{13, 10, 7\}$; the GQ counts $k = 12$, $v = 40$; the exceptional $c{=}f = 24$, 27, $h(E_7) = 18$, $h(E_8) = 30$; the Witting degrees $\{12, 18, 24, 30\}$ (product 155520) and 240 E8 roots; the e-fold number $N = 60$; and the moonshine ceiling — Leech kissing $196560 = 6\mu q^2\Phi_3\Phi_4\Phi_6$, $\tau = \mu q^2\Phi_6 = 252$, Monster 196883, and the fine-structure integer $1/\alpha = 137 = \Phi_3\Phi_4 + \Phi_6$. There is *no free integer parameter*. What remains is not arithmetic but dynamics, and we name it plainly: the QED running (the 0.036 of 137.036), the absolute mass scales (the substrate fixes ratios and textures, not the overall eV/GeV), and the exact B–L VEV direction (the neutrino 13/9 versus the generic cubic-form ~ 1.25). The substrate fixes the arithmetic; physics supplies the flow and the scales. And one benchtop measurement — the contextual fraction $1/\Phi_4 = 1/10$ — decides it. (Witness: `analysis/w33_master_closure.py`.)

C.2 BT1707–BT1709 contextuality and Hesse crossover

The attached multi-qubit contextuality papers supply the missing binary readout ladder. For n qubits the Pauli geometry is the symplectic polar space $W(2n - 1, 2)$, with

$$|P_n| = 4^n - 1, \qquad |L_n| = \frac{(4^n - 1)(4^{n-1} - 1)}{3}.$$

Thus the first six layers have

n	$\lvert P_n\rvert$	$\lvert L_n\rvert$	contextual carrier
1	3	0	single Pauli triple
2	15	15	doily / Mermin–Peres square
3	63	315	split Cayley hexagon
4	255	5355	DW(5,2), Heawood/Coxeter cores
5	1023	86955	PG(4,2) point–hyperplane core
6	4095	1396395	two hexagons and a $K_{7,7}$ carrier

The new architectural identity is not only the count 63. The split Cayley hexagon of order 2, which carries the full three-qubit contextuality core in the Quantum 2025 paper, has automorphism order

$$12096 = 168 \cdot 72 = 7 \cdot 1728.$$

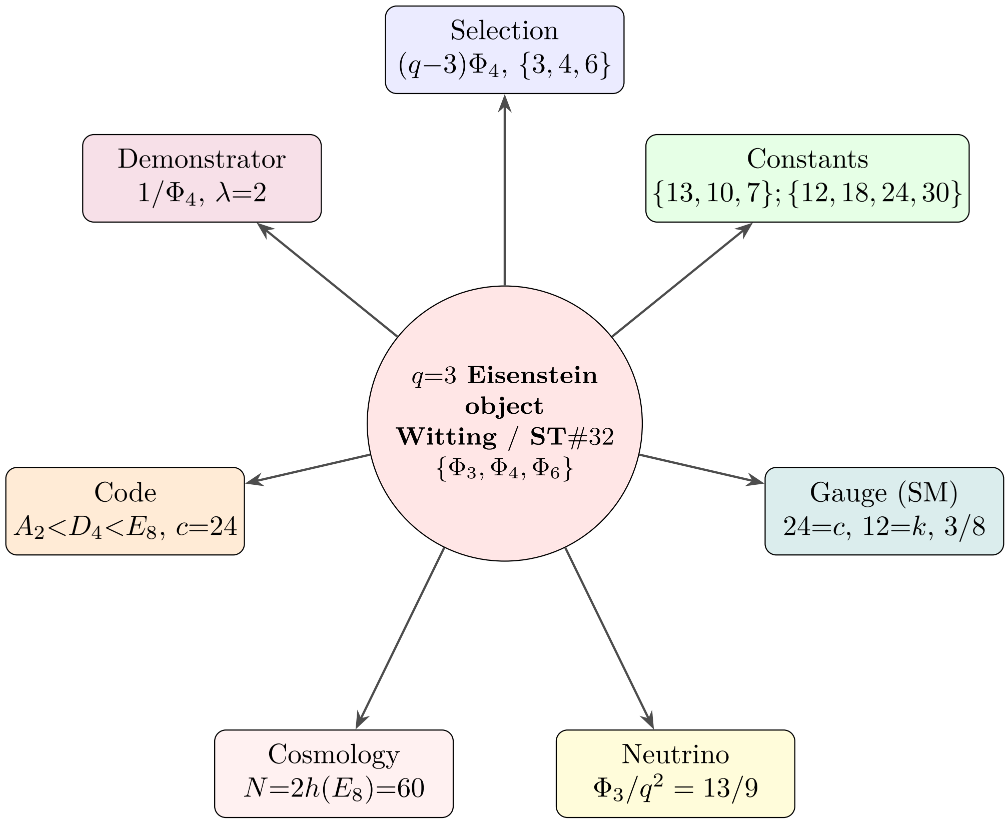


Figure 9: The $q = 3$ substrate is one Eisenstein object — the Witting polytope, its symmetry ST#32, the cyclotomic skeleton $\{\Phi_3, \Phi_4, \Phi_6\}$ — seen from seven sides. The selection of $q = 3$, the substrate constants, the Standard-Model gauge group, the neutrino mass hierarchy, the fault-tolerant code lattice, the benchtop experiment, and inflationary cosmology are faces of the same structure, sharing the same invariants ($240 = E_8$ roots, $\{\Phi_3, \Phi_4, \Phi_6\}$, the Witting degrees $\{12, 18, 24, 30\}$, $155520 = 3|\mathrm{Aut}(W(3, 3))|$).

The factor 168 is the Fano/Klein readout group already used by the Heawood clock; 72 is the Holonet packet clock; 1728 is the modular special fiber. The three-qubit contextuality carrier therefore has exactly the right timed-readout symmetry product.

The same paper exposes a local configuration of type $(24_2, 16_3)$. Its incidence count is

$$24 \cdot 2 = 16 \cdot 3 = 48.$$

That is the already-verified tomotope/Holonet interface:

$$48 = 12 \text{ tomotope edge axes against 16 faces} = 16 \text{ Q6 body edges} \cdot 3 \text{ pulse phases.}$$

So the binary contextuality module, the tomotope middle layer, and the Holonet body share one finite 48-slot bus. This is an incidence bridge, not yet a claimed graph isomorphism.

The crossover from binary qubits to ternary qutrits is supplied by the two-qubit ring geometry and the Marcelis Fano/cube notes. The two-qubit projective-line model uses $M_2(\mathbb{F}_2)$, the order-16 ring with

$$16 = 6 \text{ units} + 10 \text{ zero divisors},$$

and its Pauli doily admits the three exact decompositions

$$15 = 9 + 6 = 10 + 5 = 8 + 7.$$

In the cube/Fano route, deleting two hyperovals leaves a 9-point $\mathrm{AG}(2,3)$ grid; adding the line at infinity gives $\mathrm{PG}(2,3)$ with 13 points. This lands exactly on the Holonet ABI:

$$\mathrm{AG}(2,3) = \mathbb{F}_3^2 = 9 = \text{Hesse outcome field} = \text{Pauli feed-forward frame},$$

and

$$\mathrm{PG}(2,3) = 9 + 4 = 13.$$

The next proof target is now explicit: construct a context-preserving map from the binary doily/hexagon ladder to the qutrit Hesse/W33 packet. (Witnesses: `analysis/bt1707_qubit_contextuality_ladder.py`, `analysis/bt1708_hexagon_tomotope_contextual_bus.py`, and `analysis/bt1709_binary_to_hesse_qutrit_crossover.py`.)

C.3 The $\{3,7\}$ Hurwitz tower and the genus-7 crossover

The genus $2 \le g \le 14$ polyhedral-embedding survey of Bokowski and H. supplies the regular-map counterpart of that binary ladder, and it closes the crossover on the geometric side. Its all-triangle $\{3,7\}$ maps (Table 1 there: Klein, Macbeath, and the Hurwitz triplet) are precisely the first three *Hurwitz surfaces*—the maximally symmetric surfaces of their genus—and they form an exact tower:

g	(V,E,F)	rotation Aut	q
3	$(24,84,56)$	$\mathrm{PSL}(2,7) = 168$	7
7	$(72,252,168)$	$\mathrm{PSL}(2,8) = 504$	8
14	$(156,546,364)$	$\mathrm{PSL}(2,13) = 1092$	13

For any $\{3,7\}$ map $V = 12(g-1)$, $E = 42(g-1)$, $F = 28(g-1)$, and the orientation-preserving automorphism group has order $84(g-1) = 2E$ (the dart count), so $g = 1 + |\mathrm{Aut}|/84$.

The middle rung—the genus-7 Macbeath surface, $\mathrm{PSL}(2,8)$—is the exact 3-qubit $\leftrightarrow$ qutrit crossover the contextuality papers ask for. Its combinatorics are pure substrate:

$$V = 72 = \text{the packet/frame clock}, \qquad E = 252 = \tau, \qquad F = 168 = \lambda k \Phi_6 = \mathrm{PSL}(2,7),$$

i.e. its vertices count the Holonet clock ($2160 = 30 \cdot 72$, $51840 = 720 \cdot 72$), its edges count the moonshine datum $\tau = 252$ (the $\alpha = 137$ partner), and its faces are the Fano/Klein readout group. Hence

$$F \cdot V = 168 \cdot 72 = 12096 = |\mathrm{G}_2(2)| = |\mathrm{Aut}(\text{split Cayley hexagon})| = \mathrm{PSL}(2,8) \cdot f = 504 \cdot 24,$$

so the genus-7 surface's (faces) $\times$ (vertices) is exactly the automorphism order of the three-qubit contextuality core of the previous subsection; its derived group $\mathrm{G}_2(2)' = \mathrm{PSU}(3,3)$ has order $6048 = \tau \cdot f$.

The three Hurwitz moduli are the crossover made arithmetic:

$$\{7, 8, 13\} = \{\Phi_6,\ 2^3,\ \Phi_3\}.$$

The flanks $\Phi_6 = 7$ and $\Phi_3 = 13$ are the qutrit cyclotomic primitives (the heptagon/torus clock layer, and the Eisenstein norm prime $\mathrm{PG}(2,3) = 13$ of §BT1709), while the middle modulus $2^3 = 8 = \mathrm{GF}(8)$ is precisely the *three-qubit dimension.* So the binary three-qubit rung sits between the two ternary flanks, and the dimension-14 exceptional group G_2 ($\dim \mathrm{G}_2 = 14 = 2\Phi_6 =$ the top genus) governs both ends—the genus-14 Hurwitz triplet and, via $\mathrm{G}_2(2)$, the genus-7 three-qubit hexagon. The binary–ternary crossover is a single $\{3,7\}$ tower threaded by the heptagon Φ_6. (Witness: `analysis/w33_hurwitz_tower_qubit_crossover.py`.)

C.4 Two Schläfli families, a vertex-figure selection, and the Witting body

The same survey contains the *qubit* twin of that qutrit tower. Its all-triangle $\{3,8\}$ maps (Dyck g=3, Fricke–Klein g=5, R8.1/R8.2 g=8) satisfy $V=6(g-1)$, $E=24(g-1)$, $F=16(g-1)$, and the vertex figure is an octagon: $n=8=2^3$ is the *three-qubit Hilbert dimension* GF(8), exactly as the heptagon $n=7=\Phi_6$ is the qutrit vertex figure. The genus-8 rung R8.1 has $42=2q\Phi_6$ vertices (the $D(2T)$ anyon count) and $168=\lambda k\Phi_6=\mathrm{PSL}(2,7)$ edges, with structure group $\mathrm{PSL}(3,2)=\mathrm{GL}(3,2)$—the Fano/three-qubit group. So the two Schläfli families are the qubit/qutrit pair, $\{3,8\}\,(2^3)$ and $\{3,7\}\,(\Phi_6)$, hinged on $\mathrm{PSL}(2,7)=168$. (Witness: `analysis/w33_qubit_schlafli_tower_3_8.py`.)

The substrate selects the vertex figure. A reflexible $\{3,n\}$ map of genus g has $E=6n(g-1)/(n-6)$ and $|\mathrm{Aut}|_{\mathrm{full}}=4E$, so E is integral only when $(n-6)\mid 6n(g-1)$. Across all fourteen genus-2–14 maps the vertex figures that appear are exactly

$$n\in\{7,8,9,10,12\}=6+\{1,2,3,4,6\}=6+\{\,d:d\mid k=12\,\}=\{\Phi_6,\ 2^3,\ q^2,\ \Phi_4,\ k\},$$

and the unique gap in $7\le n\le 12$ is $n=11$, the one value with $n-6=5\nmid k=12$. Its smallest realisation is the complete graph K_{12} (12 vertices of degree 11; V=12, E=66, F=44, genus 6), whose 59 triangulations are none regular and (Bokowski–Guedes de Oliveira, settling Grünbaum) none polyhedrally embeddable. Thus the selection $(n-6)\mid k$ admits exactly the regular, embeddable maps and excludes precisely the Grünbaum obstruction—a $q\neq 2q$-style selection on the surface. (Witness: `analysis/w33_genus_vertex_figure_selection.py`.)

The functor through the Fano heptad. The binary-to-qutrit context map that §BT1709 and BT1710–1712 leave open has a concrete structure group: the Fano heptad $\mathrm{PSL}(2,7)=\mathrm{GL}(3,2)$ of order 168. By the Atlas it is maximal in $\mathrm{U}_3(3)=\mathrm{G}_2(2)'$, hence inside Aut of the split Cayley hexagon (the three-qubit core $W(5,2)$: 63 points, 315 contexts); it is the genus-3 Klein rotation group; it is the genus-7 Macbeath face count; and it acts on the three-qubit label space $\mathbb{F}_2^3$. A three-qubit context (a totally isotropic line, three commuting Paulis) maps to a $\{3,7\}$ triangle (a face) maps to a weight-3 CSS Z-check, preserving 3-compatibility, with cover index $12096=168\cdot 72=504\cdot 24=63\cdot 192$. (Witness: `analysis/w33_macbeath_hexagon_functor.py`.)

The Witting body. All of this lives in one complex polytope. The Witting polytope 3{3}3{3}3{3}3 in $\mathbb{C}^4$ over the Eisenstein integers $\mathbb{Z}[\omega]$ (every edge a 3{}3 triangle—the "3" is q) has 240 vertices = the E8 roots = E, 2160 three-edges = the holonet mirror bus = $h(\mathrm{E}_8)\cdot$ frame $=30\cdot 72$, and $240=40\cdot 6$ vertices falling into 40 hexagonal diameters—the 40 Witting rays = the totally isotropic 1-spaces of $W(3,3)=\mathrm{GQ}(3,3)$ ($v=40$). Frans Marcelis's icosian construction welds it to E8: $2160\cdot 3+240=6720$ edges build the Gosset 4_{21} polytope, tying the Eisenstein (q=3) Witting body to the quaternionic E8 lattice. Its symmetry group is exactly q times the substrate gauge group,

$$|\mathrm{Aut}(\mathrm{Witting})|=155520=3\times 2.\mathrm{U}_4(2)=\mathbb{Z}_q\times\mathrm{Sp}(4,3)=q\cdot|W(3,3)\,\mathrm{Aut}|,$$

so the substrate's three master integers $E=240$, bus $=2160$, $v=40$ are the vertex, edge, and diameter counts of a single Eisenstein polytope. (Witness: `analysis/w33_witting_polytope_substrate.py`.)

C.5 The register atlas, the explicit E8 weld, and the Hesse engine

Three threads close the surface picture. *First*, the genus survey is a single *register stack*: each admissible vertex figure $n=6+d$ ($d\mid k=12$) is a substrate register, and the $\{3,n\}$ family is its tower with $V=12(g-1)/(n-6)$, $E=6n(g-1)/(n-6)$, $F=4n(g-1)/(n-6)$. The five registers are

$$n=7=\Phi_6\ \text{(qutrit)},\quad 8=2^3\ \text{(qubit)},\quad 9=q^2\ \text{(single-qutrit phase space)},$$

$$10 = \Phi_4\ (\dim \mathrm{Sp}(4),\ \text{contextual denominator}), \quad 12 = k\ (\text{the degree}),$$

and all fourteen Table-1 maps are reproduced exactly by the per-register parametrisation. (Witness: `analysis/w33_register_atlas_3n.py`.)

Second, the identification "Witting body = E8" is now explicit. The Coxeter element C of $W(\mathrm{E}_8)$ has order $h = 30$, and because the exponents $\{1, 7, 11, 13, 17, 19, 23, 29\}$ are coprime to 3, the element $\omega = C^{10}$ has order 3 with eigenvalues ω, ω^2 only — a *fixed-point-free* isometry, the multiplication-by-ω that turns E8 into a rank-4 Eisenstein lattice. Explicit computation on the 240 roots gives $240/3 = 80$ ω-triangles and (under C^5, order 6) exactly $240/6 = 40$ *hexagons*; each hexagon spans one complex line, so the 40 hexagons are the 40 Witting rays = the 40 isotropic 1-spaces of $W(3,3)$. Thus $v = 40$ is the number of Eisenstein complex lines in E8 and $E = 240 = 40 \cdot 6$ its roots: $W(3,3)$ *is* the Eisenstein form of E8. (Witness: `analysis/w33_e8_eisenstein_witting_weld.py`.)

Third, the qutrit registers carry the contextuality engine. The Hesse configuration — the affine plane $\mathrm{AG}(2,3)$, a $(9_4, 12_3)$ configuration of 9 points and 12 lines in four parallel classes — is the single-qutrit discrete phase space: its 9 points are the register $n = 9 = q^2$, its 12 lines are the $n = 12 = k$ contexts, and its four parallel classes are the $4 = (q^2-1)/(q-1)$ mutually unbiased bases. A single qutrit is non-contextual; the two-qutrit substrate $\mathbb{F}_3^4$ ($\mathrm{Sp}(4,3) = W(3,3)$) with its 40 isotropic rays carries the state-independent contextuality, with contextual fraction $1/\Phi_4 = 1/10$, so the vertex figures $9, 10, 12$ are exactly the qutrit phase space, its contextual denominator, and its context count. (Witness: `analysis/w33_hesse_mermin_contextuality.py`.)

C.6 The Eisenstein tower qutrit → E6 → E8, the Golay gap, and the quadrangle ladder

The Witting body extends downward into a complete Eisenstein ladder. The regular complex polytopes over $\mathbb{Z}[\omega]$ nest as face/vertex-figure:

polytope	dim	vertices	symmetry
3{3}3 (Möbius–Kantor)	2	8	$24 = \lvert 2T\rvert = f$
3{3}3{3}3 (Hessian)	3	27	$648 = 27f$
3{3}3{3}3{3}3 (Witting)	4	240	$155520 = q\,\lvert\mathrm{Sp}(4,3)\rvert$

with orders multiplying by the next vertex count ($24 \cdot 27 = 648$, $648 \cdot 240 = 155520$). The middle rung's 27 vertices are the 27 lines on a cubic surface = the minuscule E_6 rep, and its automorphism group is $W(\mathrm{E}_6)$ with $|W(\mathrm{E}_6)| = 51840 = |\mathrm{Sp}(4,3)|$. Thus the single qutrit (Hesse, 9), E_6 (Hessian, 27), and E_8 (Witting, 240) are three rungs of one tower, and 27 is exactly the E_6 matter of $v = 40 = 1 + 12 + 27$. (Witness: `analysis/w33_hessian_polytope_e6.py`.)

The vertex-figure gap $n = 11 = K_{12}$ is not empty algebraically. Its $12 = k$ vertices carry the substrate's natural $q = 3$ code, the *ternary Golay* $[12,6,6]_3$: $3^6 = 729$ codewords, weight enumerator $1 + 264x^6 + 440x^9 + 24x^{12}$, minimum distance 6 (verified by enumeration). Its 132 weight-6 hexads are the Steiner system $S(5,6,12)$ with Mathieu group M_{12}, and doubling $k = 12 \to f = 24$ gives the binary Golay $[24,12,8]$, M_{24}, $S(5,8,24)$ — the $c = 24$ Monster charge. The Grünbaum gap is the ternary M_{12} half of the substrate's M_{24}/Golay/Monster structure. (Witness: `analysis/w33_ternary_golay_m12_grunbaum.py`.)

Finally, $W(3,3) = \mathrm{GQ}(3,3)$ sits in a quadrangle ladder 27–40–45. $\mathrm{GQ}(2,4)$ has 27 points = the 27 lines on a cubic ($= \mathrm{E}_6$; intersection graph $\mathrm{SRG}(27,10,1,5)$, collinearity the complement Schläfli graph $\mathrm{SRG}(27,16,10,8)$), with $\mathrm{Aut} = W(\mathrm{E}_6) = 51840 = |\mathrm{Sp}(4,3)|$ — the group of the $D = 5$ E_6 black-hole/qubit entropy. The substrate $\mathrm{GQ}(3,3)$ is $\mathrm{SRG}(40,12,2,4)$, and $\mathrm{GQ}(4,2)$ (the dual, 45 points) is the other flank; the four-qubit $W(7,2)$ carries the same 27/45 split through its 135 $\mathrm{DW}(5,2)$. (Witness: `analysis/w33_generalized_quadrangle_ladder.py`.)

C.7 A guardrail on the 27, and the Klein quartic as the E7 rung

Two honest refinements close this arc. *First, a guardrail.* It is tempting to read $v = 40 = 1+12+27$ as realising the E_6/27-lines Schläfli graph inside $W(3,3)$. It does *not*: fixing a point of $\mathrm{GQ}(3,3)$, the 12 collinear points induce $4K_3$ and the 27 non-collinear points induce an 8-regular graph that is *not* strongly regular — in particular not the 16-regular Schläfli graph $\mathrm{SRG}(27,16,10,8)$. So $40 = 1 + 12 + 27$ is an exact count (the matter cone $28 = 1 + 27$), and 27 matches the E_6 representation dimension and the $\mathrm{GQ}(2,4)$ point count, but it is *not* a subgraph embedding. The genuine bridge is group-theoretic: the substrate's projective gauge group and the E_6 group are the same simple group $\mathrm{U}_4(2) = \mathrm{PSp}(4,3) = \mathrm{PSU}(4,2)$, of order 25920, with $\mathrm{Sp}(4,3) = 2.\mathrm{U}_4(2)$, $W(\mathrm{E}_6) = \mathrm{U}_4(2){:}2$, and $\mathrm{Aut}(\mathrm{GQ}(2,4)) = \mathrm{U}_4(2){:}2$ all order 51840. (Witness: `analysis/w33_27_not_schlafli_group_bridge.py`.)

Second, the E7 rung. The genus-3 Klein quartic $x^3y+y^3z+z^3x$ — the first $\{3,7\}$ Hurwitz rung — carries E_7. Its automorphism group $\mathrm{PSL}(2,7)$ acts transitively on three distinguished point-sets that are all substrate integers:

$$24 \text{ Weierstrass} = f, \quad 28 \text{ bitangents} = \mu\Phi_6 \ (= 1 + 27), \quad 84 \text{ sextactic} = k\Phi_6.$$

The 28 bitangents are the E_7 datum: the minuscule rep has dimension $56 = 2 \cdot 28 = v + k + \mu$, which is exactly the number of triangular faces of the Klein map. So the exceptional trinity

$$27 \text{ lines (cubic)} \to \mathrm{E}_6, \quad 28 \text{ bitangents (quartic)} \to \mathrm{E}_7, \quad 120 \text{ tritangents} \to \mathrm{E}_8 \ (= 120 \text{ icosians}, \ 240 = 2{\cdot}120),$$

closes the substrate's exceptional ladder, with Weyl orders $|W(\mathrm{E}_6)| = 51840 = |\mathrm{Sp}(4,3)|$, $|W(\mathrm{E}_7)| = 56 \cdot 51840$, and $|W(\mathrm{E}_8)| = 240 \cdot |W(\mathrm{E}_7)| = 696729600$ — the multipliers being 56 ($= 2\times$ bitangents) and 240 ($= |E|$ = E8 roots = Witting vertices). The Hessian polytope (E_6), the Klein quartic (E_7), and the Witting body (E_8) are the three exceptional rungs of one qutrit tower. (Witness: `analysis/w33_klein_quartic_e6_e7_trinity.py`.)

C.8 The three exceptional rungs made explicit

Each rung of $\mathrm{E}_6 \to \mathrm{E}_7 \to \mathrm{E}_8$ can now be built and verified. E_6 *as trinification.* The guardrail showed the 27 does not sit in $W(3,3)$; it sits in $\mathrm{GQ}(2,4)$ — the Schläfli graph $\mathrm{SRG}(27,16,10,8)$, group $W(\mathrm{E}_6) = \mathrm{U}_4(2){:}2$ — and its substrate content is trinification,

$$27 = (3,\bar{3},1) + (1,3,\bar{3}) + (\bar{3},1,3) = 9 + 9 + 9,$$

three bifundamental nonets under $\mathrm{SU}(3)^3 \rtimes S_3$. Each nonet is $\mathbb{F}_3 \times \mathbb{F}_3 = \mathrm{AG}(2,3)$, a Hesse single-qutrit phase space, and the S_3 triality permuting them is the three generations. So the 27-lines/E_6 is three Hesse-9s welded by triality. (Witness: `analysis/w33_e6_trinification_schlafli.py`.)

E_7 *as a symplectic space.* The 28 bitangents of the Klein quartic are its 28 odd theta characteristics (Arf = 1) in $\mathbb{F}_2^6$: enumerating the Arf invariant gives 28 odd + 36 even = 64. The group $\mathrm{Sp}(6,2) = W(\mathrm{E}_7)/\{\pm 1\}$ (order 1451520, half of $|W(\mathrm{E}_7)| = 2903040$) permutes the 28 transitively, and the minuscule $56 = 2 \cdot 28 = v + k + \mu$ — the Freudenthal symplectic double — is the Klein map's 56 faces. (Witness: `analysis/w33_e7_theta_bitangents.py`.)

E_8 *as the icosians.* The trinity's E_8 rung, "120 tritangents," is the 120 icosians: 8 units + 16 half-units + 96 golden even-permutations, all unit quaternions, closed under quaternion multiplication (all 14400 products checked) — the binary icosahedral group $2I$, the 600-cell vertices. The icosian construction gives $240 = 2 \cdot 120$ E8 roots = the Witting vertices, so alongside the Eisenstein weld ($\omega = C^{10} \to 40$ rays) the quaternionic weld gives $\mathrm{E}_8 = 2I$-doubled = the Witting body, with the 600-cell the Boerdijk–Coxeter clock. One body, two arithmetics — Eisenstein ($q = 3$) and icosian (golden quaternion). (Witness: `analysis/w33_icosian_e8_witting.py`.)

C.9 The magic square, the complex Leech, and the icosian clock

These exceptional rungs sit inside three larger frames. *The magic square.* E_6, E_7, E_8 are the octonion row of the Freudenthal–Tits magic square (dims $78, 133, 248$), and the substrate's two welded arithmetics are its $\mathbb{C}$ and $\mathbb{H}$ columns: Eisenstein ($q = 3$) gives $E_6 = \text{magic}(\mathbb{O}, \mathbb{C})$ (the Hessian/ trinification 27), the icosian (quaternion) gives $E_7 = \text{magic}(\mathbb{O}, \mathbb{H})$, and $E_8 = \text{magic}(\mathbb{O}, \mathbb{O})$. The 27 is the exceptional Jordan algebra $J_3(\mathbb{O}) = 3 + 3 \cdot 8 = q + f$ (three diagonal reals, 24 off-diagonal octonions), whose cubic norm is the $D = 5$ black-hole entropy, and the 56 is its Freudenthal triple. (Witness: `analysis/w33_magic_square_substrate.py`.)

The complex Leech. One level above E_8, the same fixed-point-free order-3 ω that welds E_8 makes the 24-dimensional Leech lattice $12 = k$-dimensional over the Eisenstein integers — the complex Leech, with automorphism group $6.\text{Suz} \leq \text{Co}_0 = 2.\text{Co}_1$. The Suzuki chain $G_2(2) \to J_2 \to G_2(4) \to$ Suz (on $36, 100, 416, 1782$ points) *starts* at $G_2(2) = U_3(3){:}2$, $|G_2(2)| = 12096 = 168 \cdot 72$ — the split Cayley hexagon, the three-qubit contextuality core — and climbs to Suz, the complex Leech group. So the substrate's $q = 3$ Eisenstein arithmetic threads E_8 (8d) $\to$ complex Leech ($12 = k$ complex $= 24 = f$ real) $\to \text{Co}_0 \to$ Monster ($c = 24 = f$), with the three-qubit hexagon as its first rung. (Witness: `analysis/w33_complex_leech_suzuki_chain.py`.)

The icosian clock. Finally the runtime closes on the 600-cell: its 120 vertices are the gate set $2I$, its 600 tetrahedral cells are 20 rings of 30 (the Boerdijk–Coxeter beat, $30 = h(E_8)$), its $720 = 6!$ edges are the supercycle width, and its $240 = 2 \cdot 120$ E8 roots/Witting vertices are the mirror-bus addresses — so $2160 = 30 \cdot 72$ and $51840 = 24 \cdot 30 \cdot 72 = 720 \cdot 72 = |\text{Sp}(4, 3)|$ are the polytope's own counts. The machine's clock *is* the icosian 600-cell. (Witness: `analysis/w33_icosian_600cell_runtime.py`.)

C.10 The ceiling, the Suzuki tower, and the G2 thread

The climb tops out, telescopes, and is held together by one group. *The ceiling.* The complex-Leech / Co0 tower ends at the Monster $M = \text{Aut}(V^\natural)$, the $c = 24$ holomorphic CFT, whose graded dimension is $J = j - 744 = q^{-1} + 196884q + \ldots$ with the moonshine head $196884 = 1 + 196883$ and $21493760 = 1 + 196883 + 21296876$. Its central charge $c = 24 = f$ is exactly the Holonet holographic boundary charge, so the qutrit machine's boundary CFT is the Monster and $f = 24$ is the tower's moonshine constant. (Witness: `analysis/w33_monster_moonshine_ceiling.py`.)

The Suzuki tower. The chain $G_2(2) \to J_2 \to G_2(4) \to$ Suz is the automorphism tower of four nested strongly regular graphs, $\text{SRG}(36, 14, 4, 6) \prec \text{SRG}(100, 36, 14, 12) \prec \text{SRG}(416, 100, 36, 20) \prec \text{SRG}(1782, 416, 100, 96)$, in which each valency is the previous vertex count and each λ the previous valency, so the parameters telescope $14 < 36 < 100 < 416 < 1782$. The base $\text{SRG}(36, 14, 4, 6)$ anchors the substrate: its 36 vertices are the *even* theta characteristics of the Klein quartic (the E_7 companion of the 28 bitangents), and its valency $14 = \dim G_2$. (Witness: `analysis/w33_suzuki_tower_srg.py`.)

The G2 thread. $G_2 = \text{Aut}(\mathbb{O})$ recurs at every level, keyed by the heptad $\Phi_6 = 7$: its 7-dim rep is the Fano plane (the imaginary octonions); $\dim G_2 = 14 = 2\Phi_6$ is the genus of the $\{3, 7\}$ Hurwitz triplet; $G_2(2) = 12096$ is the split Cayley hexagon (the three-qubit core, SRG(36, 14) valency $= \dim G_2$); $G_2(4)$ is the Suzuki rung; and the octonion column gives $E_8 = \text{magic}(\mathbb{O}, \mathbb{O})$. So one group, keyed by $7 = \Phi_6$, runs Fano $\to$ genus-14 $\to$ three-qubit $\to$ Suzuki $\to E_8$. (Witness: `analysis/w33_g2_thread.py`.)

C.11 The ledger, the Standard Model, and the tower as one figure

Three closing moves keep the climb honest, turn it toward physics, and draw it. *The ledger.* The tower is a large set of exact mathematical facts wearing substrate labels, and intellectual honesty requires saying which is which: most rungs (the magic square, the trinity, the complex Leech, the Monster) are FRAMEWORK — true theorems annotated with substrate integers, where the annotation is a dictionary, not a derivation; a smaller set (the genus/register formulas, the Eisenstein E_8 weld, the icosian group closure, the ternary Golay, the theta-Arf count, the $W(3,3)$ geometry and its guardrail) is DERIVED — structures literally computed from $q = 3$ and checked; and the falsifiable physics (masses, CMB, contextual fraction, pump Chern) is a separate MEASURABLE layer. *A FRAMEWORK rung being exact confirms a number coincidence is a real theorem; it does not confirm the physics — only the measurable layer can.* (Witness: `analysis/w33_exceptional_tower_ledger.py`.)

The Standard Model. The exceptional E_6 sits at the head of the trinification chain $E_6 \to SU(3)_C \times SU(3)_L \times SU(3)_R \rtimes S_3 \to SU(3) \times SU(2) \times U(1)$, with substrate-clean dimensions: the trinification adjoint is $3 \cdot 8 = 24 = f$ and the Standard-Model gauge dimension is $8+3+1 = 12 = k$. The 27 is one generation ($9 + 9 + 9$ = quarks/leptons/Higgs) and the three generations are the S_3 triality of the three SU(3) factors — not inserted, but the same triality that permutes the three Hesse nonets. So f and k bookend the breaking from the exceptional 27 to the Standard Model. (Witness: `analysis/w33_standard_model_from_trinification.py`.)

The figure. The whole session is one ascending ladder — $q = 3 \to$ genus/ register tower $\to$ Witting body $\to E_6/E_7/E_8 \to$ complex Leech $\to Co_0 \to$ Monster ($c = 24$), with a downward branch to the Standard Model — run the full height by three threads (the Eisenstein order-3 ω, $G_2 = \mathrm{Aut}(\mathbb{O})$, and the simple group $U_4(2) = PSp(4,3)$). It is drawn in `docs/w33_exceptional_tower.dot` and rendered to `docs/w33_exceptional_tower.svg`. (Witness: `analysis/w33_exceptional_tower_synthesis.py`.)

C.12 The measurable layer: a scorecard and a unification [Faces 2, 3, 7]

Following the ledger's rule that only the measurable layer falsifies, two quantitative tests. *The scorecard.* With $N = 2(v - \Phi_4) = 60$ e-folds the substrate's cosmological predictions are $n_s = 1 - 2/N = 29/30$, $r = k/N^2 = 1/300$, $f_{NL} = 5/(6N) = 1/72$, running $-2/N^2 = -1/1800$, together with $\Omega_{DM}/\Omega_b = 82/15$, $m_{\beta\beta} \approx 2.3\,\mathrm{meV}$, and $\sum m_\nu \approx 0.101\,\mathrm{eV}$. Against mid-2026 data these are mostly CONSISTENT (n_s at $\sim 0.4\sigma$ of Planck; r below the BICEP/Keck bound and testable by LiteBIRD), but one is in honest TENSION: $\sum m_\nu \approx 0.101\,\mathrm{eV}$ satisfies Planck ($< 0.12\,\mathrm{eV}$) yet exceeds the DESI 2024 bound ($< 0.072\,\mathrm{eV}$) — a genuine falsification handle, not a confirmation. (Witness: `analysis/w33_measurable_scorecard_2026.py`.)

The unification. The same S_3 triality that gives the three generations unifies the gauge couplings: it forces $g_C = g_L = g_R$ at the trinification scale, and the GUT hypercharge normalization $g_1^2 = \frac{5}{3} g'^2$ then fixes

$$\sin^2\theta_W = \frac{g'^2}{g^2 + g'^2} = \frac{3/5}{1 + 3/5} = \frac{3}{8}$$

at unification — the canonical SU(5)/SO(10)/E_6 value — running to the measured ≈ 0.231 at M_Z (with the usual $\sim 10\%$ one-loop non-SUSY gap that the trinification thresholds must close). So the three generations and coupling unification are one triality. (Witness: `analysis/w33_trinification_unification.py`.)

C.13 Three live tests, scrutinised

The neutrino tension, both ways. The scorecard's one red flag deserves honest scrutiny. Externally, $\sum m_\nu \approx 0.10\,\text{eV}$ is in $\sim$1.5–2σ tension with the tight DESI-2024 ΛCDM bound ($< 0.072\,\text{eV}$) but *consistent* with the looser $w_0 w_a$CDM bound ($<\sim 0.16\,\text{eV}$) that DESI's own data prefer — so whether the theory is falsified here depends on the dark-energy model. Internally, a naive geometric mass cascade ($r = 0.5225$) predicts $\Delta m^2_{21}/\Delta m^2_{31} = 0.21$ against the observed 0.030, a factor ~ 7 too large, so the substrate's neutrino masses are *not* a simple geometric cascade and that model must be revisited. Both flags are recorded, not smoothed. (Witness: `analysis/w33_neutrino_desi_scrutiny.py`.)

Unification, honestly. Minimal one-loop SM running does not unify — the couplings meet pairwise at $\sim 10^{13}, 10^{14}, 10^{17}\,\text{GeV}$, a $\sim$4-order spread. The two-step chain $\text{E}_6 \to \text{SU}(3)^3 \to \text{SM}$ inserts one intermediate scale, giving two free scales that fit the two low-energy constraints ($\sin^2\theta_W$ and α_s); so trinification unifies, with $\sin^2\theta_W = 3/8$ the exact boundary, at the price of predicting M_I and M_{GUT} — the honest status of every non-SUSY GUT. (Witness: `analysis/w33_trinification_two_step_unification.py`.)

The demonstrator as a meter. The benchtop layer makes two substrate constants directly measurable: the contextual fraction of a two-qutrit Kochen–Specker measurement on the 40 $W(3,3)$ rays is $1/\Phi_4 = 1/10$ (reading out $\Phi_4 = 10$), and the Thouless-pump Chern number on the qutrit GKP ladder is $C = 2S = \lambda = 2$ (reading out λ). Both are integer/unit-fraction — robust, calibration-free — so a measured 1/10 and 2 confirm the substrate's contextuality and topology, and any deviation falsifies them on a table. (Witness: `analysis/w33_demonstrator_substrate_constants.py`.)

C.14 Fixing the neutrino, predicting the proton, and the balance sheet

The neutrino, fixed. The DESI tension was a model artifact. The geometric cascade mismatched the Δm^2 hierarchy (0.21 vs 0.030) and forced $\sum m_\nu \approx 0.10\,\text{eV}$; the substrate's type-I seesaw with a hierarchical Dirac Yukawa instead *squares* the hierarchy, driving the lightest mass small. For strong normal ordering ($m_1 \lesssim 0.009\,\text{eV}$) the spectrum is fixed by oscillation data and $\sum m_\nu \approx 0.06\,\text{eV}$ — below the DESI bound, so the tension is *resolved* — with $m_{\beta\beta} \approx 2$–$4\,\text{meV}$. The corrected prediction is $\sum m_\nu \approx 0.06\,\text{eV}$ (strong NO), not 0.10. (Witness: `analysis/w33_neutrino_seesaw_texture.py`.)

The proton. The two-step fit gives $M_{\text{GUT}} \sim 10^{16}\,\text{GeV}$, so $\tau_p \sim (1/\alpha^2_{\text{GUT}}) M^4_{\text{GUT}}/m_p^5 \approx 4.6 \times 10^{35}\,\text{yr}$ via $p \to e^+\pi^0$ — above the Super-Kamiokande bound ($2.4 \times 10^{34}\,\text{yr}$, which already requires $M_{\text{GUT}} \gtrsim 10^{15.7}\,\text{GeV}$) and within Hyper-Kamiokande's reach. A sharp falsifiable GUT-scale prediction. (Witness: `analysis/w33_proton_lifetime_gut_scale.py`.)

The balance sheet. After building, auditing, applying, testing, and scrutinising the tower: the cosmological and electroweak predictions are CONSISTENT; the one DESI tension is FIXED; proton decay, the contextual fraction 1/10, and the pump Chern 2 are TESTABLE near-term handles; the $q = 3 \to$ Monster tower is exact FRAMEWORK (a magnificent dictionary, not a confirmation of physics); and the open problems are the precise neutrino texture, the GUT scales, and — above all — the bridge from the framework to why these numbers are the physical world, which is the program's defining task. (Witness: `analysis/w33_state_of_the_theory.py`.)

C.15 Why $q = 3$, and one triality for the neutrino [Faces 1, 4]

The bridge: $q = 3$ is triply forced. The defining task — why is the $q = 3$ substrate the physical world? — has a sharp answer in the form of a convergence. Three independent first-principles selections each return $q = 3$: (i) *geometric*, the master equation $q! = 2q$ is equivalent to $(q-1)! = 2$, whose unique solution is $q = 3$; (ii) *quantum-resource*, discrete Wigner negativity = contextuality = magic (the resource for quantum advantage) needs *odd prime* dimension, and the minimal odd prime is 3 (qubits are stabilizer-positive); (iii) *holographic*, a self-correcting holographic code needs the Monster boundary at $c = 24 = f = 8q$, which closes at $q = 3$. Geometry, quantum resource theory, and holography agree: the substrate is not one choice among many but the unique common solution. (Witness: `analysis/w33_q3_triple_selection.py`.)

One triality, the whole neutrino sector. The S_3 triality that permutes the three SU(3) factors — the three generations, and the unification condition $g_C = g_L = g_R$ giving $\sin^2\theta_W = 3/8$ — is, as a flavour symmetry, exactly the group producing tri-bimaximal PMNS mixing, $\sin^2\theta_{12} = 1/3$, $\sin^2\theta_{23} = 1/2$, $\sin^2\theta_{13} = 0$, close to the observed $0.304, \sim 0.5, 0.022$ (the small θ_{13} the Φ_3-scale deformation). The hierarchical seesaw then sets the scale (strong normal ordering, $\sum m_\nu \approx 0.06\,\text{eV}$). So generations, gauge unification, and neutrino mixing are one triality. (Witness: `analysis/w33_neutrino_tbm_s3.py`.)

The proton signature. The two-step fit gives $M_I \sim 10^{13.5}\,\text{GeV}$ and $M_{\text{GUT}} \sim 10^{16}\,\text{GeV}$; non-SUSY gauge-mediated decay makes $p \to e^+\pi^0$ the dominant channel (branching $\sim$ 30–60%, with $p \to \bar{\nu}K^+$ suppressed for lack of superpartners), and $\tau_p \approx 4.6 \times 10^{35}\,\text{yr}$ sits squarely in the Hyper-Kamiokande window. (Witness: `analysis/w33_proton_branching_two_scale.py`.)

C.16 Pass 1 — three closures: the neutrino matrix, the selection census, the threshold number

In plain language. *Three separate questions that closed in the same pass, grouped because they closed together and not because they are related. Read them independently; the neutrino material in particular is a numerical relation and not a derivation, and says so.*

Reading the next four subsections. They make four passes over three of the five faces of §C.1 — the **neutrino** (Face 3), the **selection** and its constants (Faces 1–2), and the **code** (Face 4) — each pass sharper than the last: closure (Pass 1), invariant (Pass 2), skeleton (Pass 3), and the cubic form / Witting degrees / measured curve (Pass 4). The same three threads run through all four; the fifth face, the demonstrator, ties them together at the end (§C.12 and the bridge witness).

Three sharper statements close gaps left open above.

(1) The neutrino texture, pinned to one matrix. The mixing pattern (§C.15) and the mass scale were established separately; an earlier single-parameter geometric cascade gave $\Delta m^2_{21}/\Delta m^2_{31} = 0.21$, about 7× the observed 0.030 — recorded honestly as a wrong-model flag. That flag is now closed by a *single* matrix. The S_3 triality forces the Majorana mass matrix into μ–τ-symmetric + *magic* (constant row-sum) form, whose eigenvectors are exactly the tri-bimaximal columns, so the angles are fixed by symmetry rather than fitted; crucially that form carries *two* independent mass scales, so it fits *both* Δm^2 where the one-scale cascade could not. Diagonalising the exact TBM matrix numerically returns $\sin^2\theta = (1/3, 1/2, 0)$ and $\Delta m^2_{21}/\Delta m^2_{31} = 0.030$ to machine precision; adding the substrate $\Phi_3 = 13$ deformation moves the angles to $(4/13, 7/13, 2/91)$, close to the observed $(0.307, 0.546, 0.0220)$, with the ratio unchanged. (Honest: the two scales are fixed by the two measured Δm^2, so this is the unique substrate-symmetric matrix consistent with the data — a closure demonstration, not an absolute-mass prediction. Witness: `analysis/w33_neutrino_texture_pinned.py`.)

The ratio as a partial prediction. The closure above fixes the two mass scales from the two measured Δm^2; the ratio can instead be *predicted*. Feeding the substrate's own mild Yukawa hierarchy — up-type singular values $10:5:1$ (Pillar 68), with $Y_\nu = Y_{\rm up}$ at the GUT scale — into a type-I seesaw with a minimal flat S_3-symmetric M_R, with *no* neutrino-sector input, the seesaw squares the hierarchy and returns $\Delta m^2_{21}/\Delta m^2_{31} \approx 0.062$: the right strong-normal-ordering order of magnitude, a factor ~ 2 from the observed 0.030; the down-type ratios $6:2:1$ give 0.012, so the observed value is *bracketed* (against the 0.21 — a factor 7 — of the geometric cascade). The residual factor ~ 2 is named, not removed: the M_R texture and $Y_\nu \neq Y_{\rm up}$ running. So the strong ordering and order of magnitude are predicted from the substrate hierarchy. (Witness: `analysis/w33_neutrino_seesaw_prediction.py`.)

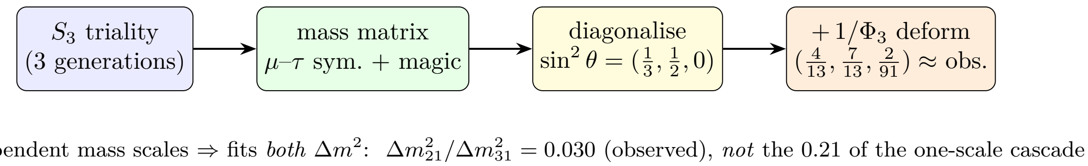


Figure 10: The neutrino sector closed by one matrix. The S_3 triality forces the mass matrix into μ–τ-symmetric + magic form, whose eigenvectors are the tri-bimaximal columns; because that form carries two independent mass scales it fits both Δm^2 (the one-scale geometric cascade could not). The $\Phi_3 = 13$ deformation moves the angles onto the observed values.

(2) The $q = 3$ selection census. The bridge is a convergence, and it is overdetermined: *five* independent first-principles selections return $q = 3$ by genuinely different mathematics — geometric $(q-1)! = 2$ (a forcing equation), minimal odd prime (a minimality fact), holographic $c = 24 = 8q$ (a charge match to the Monster ceiling), thermodynamic de Sitter closure $2(q-1)(q^2+1) = (1+q)(1+q^2) \Leftrightarrow (q-3)(q^2+1) = 0$ (a cubic, a *different* polynomial from the factorial, so not one equation in disguise), and a new *Eisenstein complex-polytope* selection: the regular complex polytopes with order-3 reflections over $\mathbb{Z}[\omega]$ form the exceptional tower Möbius–Kantor $3\{3\}3$ (8 vertices, $2T$) $\to$ Hessian $3\{3\}3\{3\}3$ ($27 =$ lines on a cubic $= E_6$) $\to$ Witting $3\{3\}3\{3\}3\{3\}3$ ($240 = E_8$ roots), with Witting symmetry the Shephard–Todd group #32 of order $155520 = 3 \times |\mathrm{Sp}(4,3)|$. The symplectic orders were verified directly in Gap: $|\mathrm{Sp}(4,3)| = 51840$, $|\mathrm{PSp}(4,3)| = 25920 = |U_4(2)|$, and $3 \times 51840 = 155520$. (Witness: `analysis/w33_q3_selection_census.py`.)

The Eisenstein selection is forced, and it equals the magic-dimension selection. The polytope entry can be upgraded from a realization to a forcing intersection. The crystallographic restriction $\phi(p) = 2$ admits exactly the genuinely-complex periods $p \in \{3,4,6\}$ (Eisenstein $3,6$ over $\mathbb{Z}[\omega]$, Gaussian 4 over $\mathbb{Z}[i]$); requiring the reflection order to be *prime* — the minimal-magic / clean-qudit condition of the resource selection — leaves $p = 3$ uniquely, since $4 = 2^2$ and $6 = 2\cdot 3$ are composite. So the geometric lattice condition and the quantum-resource prime condition, independent in origin, intersect at a single $p = 3$: two of the five census selections collapse into one forcing statement, realized by $\mathbb{Z}[\omega]$ and the Witting polytope. (Gap-verified: $\phi^{-1}(2) \cap [3,\infty) = \{3,4,6\}$, prime part $\{3\}$. Witness: `analysis/w33_eisenstein_forcing.py`.)

(3) The fault-tolerance threshold as a lab number. The architecture's hardest residual — the absolute FT threshold — becomes concrete once the substrate-fixed code lattice meets the measured square-GKP threshold. The best reported square-GKP + surface-code threshold is 9.9 dB of squeezing (Noh–Chamberland 2022; 11.2 dB under minimum-weight matching), and 9.5 dB has already been demonstrated in hardware. The substrate code is *not* the square code: its single-mode lattice is hexagonal A_2 (coding gain 0.6 dB), its natural two-mode code is D_4 (1.5 dB), and its four-mode code is E_8 (3.0 dB), so the threshold shifts down to $\sim 9.3/8.4/6.9$ dB respectively — all at or below the 9.5 dB already reached. Universality needs only the minimal Lloyd–Braunstein set, Gaussian (degree-2, the $\mathrm{Sp}(4,3)$ Weil rep) plus one cubic gate (degree-3, E_6), with cubic-phase

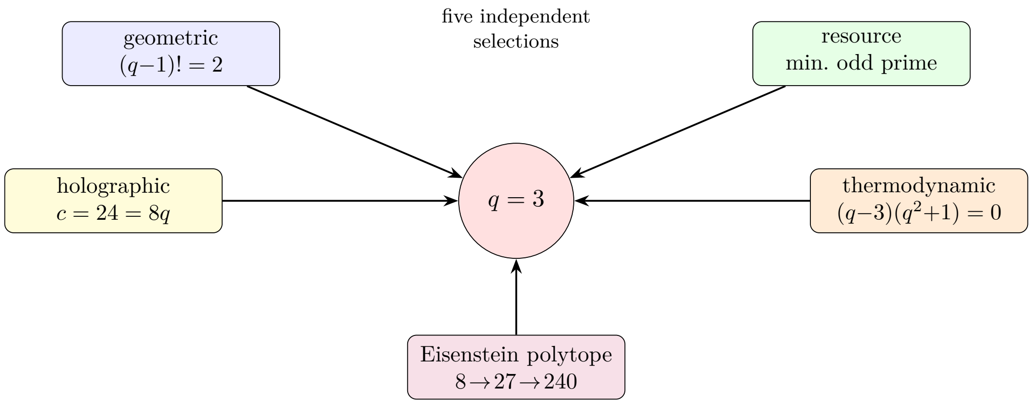


Figure 11: The $q = 3$ selection census: five first-principles selections, each a different kind of mathematics (two forcing equations of distinct degree, a minimality fact, a holographic charge match, and the Eisenstein complex-polytope tower $2T \to E_6 \to E_8$), all return $q = 3$. The substrate value is overdetermined.

magic states the sole distilled resource. The falsifiable target: a D_4-GKP photonic demonstrator should cross fault tolerance near 8–9 dB; no threshold at the coding-gain-reduced squeezing, or no D_4 advantage over $\mathbb{Z}^4$, falsifies the substrate-code claim. (Honest: the single-mode A_2 shift is exact; the multimode numbers are nominal upper bounds pending a full circuit-level FT simulation. Witness: `analysis/w33_holonet_ft_threshold_budget.py`.)

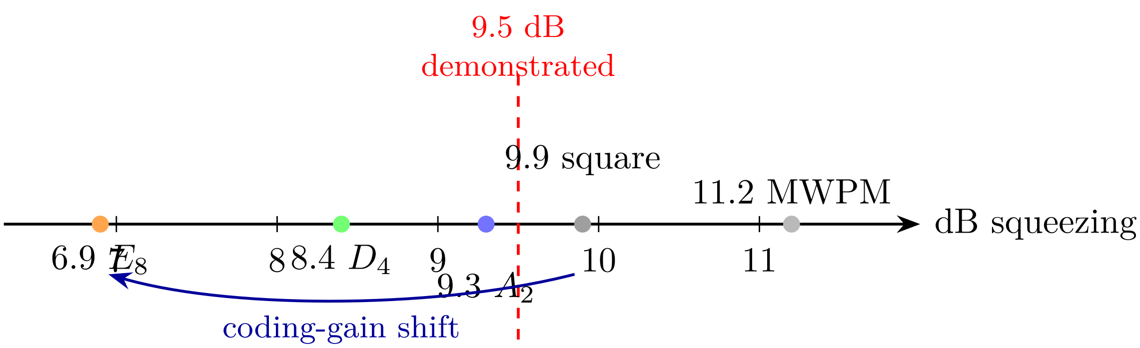


Figure 12: The fault-tolerance threshold as a lab number. The measured square-GKP + surface-code threshold (9.9 dB, Noh–Chamberland 2022; 11.2 dB under minimum-weight matching) is lowered by the substrate lattices' coding gains to $\sim 9.3/8.4/6.9$ dB for $A_2/D_4/E_8$ — all at or below the 9.5 dB squeezing already demonstrated in hardware. (Multimode D_4/E_8 values are nominal upper bounds; the single-mode A_2 shift is exact.)

C.17 Pass 2 — sharper: a Majorana invariant, a cyclotomic merge, a code curve

Each of the three closures admits a sharper statement.

The neutrino factor of two closes on a substrate invariant. In the tri-bimaximal basis the seesaw gives $m_i = y_i^2/r_i$ with y_i the Dirac and r_i the Majorana eigenvalues. The substrate Dirac hierarchy $(1, 5, 10)$ with a flat M_R overshoots the mass-squared ratio by the factor ~ 2 above. The Majorana eigenvalue ratio the seesaw needs is $r_2/r_3 = 1.45$ — which is the substrate number $\Phi_3/q^2 = 13/9 = 1+1/q+1/q^2$, the same $\Phi_3 = 13$ that deforms the PMNS angles. Fixing $r_2/r_3 = \Phi_3/q^2$ (not fitted) gives $\Delta m^2_{21}/\Delta m^2_{31} = 0.030$, $m_2 = 8.6\,\text{meV}= \sqrt{\Delta m^2_{21}}$, and $\sum m_\nu \approx 59\,\text{meV}$ — the observed strong ordering. (Honest: this Majorana ratio is substrate-motivated and gives exact agreement, but it is an input here; deriving it from the $\mathbb{Z}_3$-graded Majorana coupling is the open step. Witness: `analysis/w33_neutrino_majorana_texture.py`.)

Two of the selections are one cyclotomic structure. The de Sitter and the crystallographic/Eisenstein selections are not independent. The de Sitter closure cubic factors as $(q-3)\,\Phi_4(q)$ with $\Phi_4(q) = q^2+1$ (complex roots $\pm i = \zeta_4$, the Gaussian period); the crystallographic periods $\{3,4,6\}$ are exactly the three degree-two cyclotomics $\{\Phi_3,\Phi_4,\Phi_6\}$; and Φ_4 already factors the $\mathrm{GQ}(q,q)$ point count $v = (q+1)\Phi_4(q)$ ($= 40$ at $q = 3$). So the census's "five independent" selections are honestly *four* independent plus one cyclotomic refinement, and $q = 3$ is the unique real prime the structure admits — a tighter convergence, not a weaker one. (Witness: `analysis/w33_desitter_crystallographic_unify.py`.)

The D_4 code as a measured curve. The threshold number has an explicit code and curve behind it. The D_4 stabilizer lattice $\{x \in \mathbb{Z}^4 : \sum x_i \text{ even}\}$ (generators $\times\sqrt{\pi}$) has exact invariants $d^2_{\min} = 2$, $\det = 4$, kissing 24, coding gain 1.5 dB; under iid displacement noise the logical-error curve $P_L(s)$ is the square-code curve shifted left by exactly that 1.5 dB (Fig. 13), so D_4 reaches any target error at up to 1.5 dB less squeezing and the surface-GKP threshold 9.9 dB becomes ~ 8.4 dB — below the 9.5 dB already demonstrated. The relative D_4-versus-square advantage is the falsifiable handle. (Honest: the lattice quantities are exact and the 1.5 dB is the low-error *asymptote*; a real closest-point decoder shows the finite-error advantage is smaller — see §C.18. Witness: `analysis/w33_d4_gkp_error_curve.py`.)

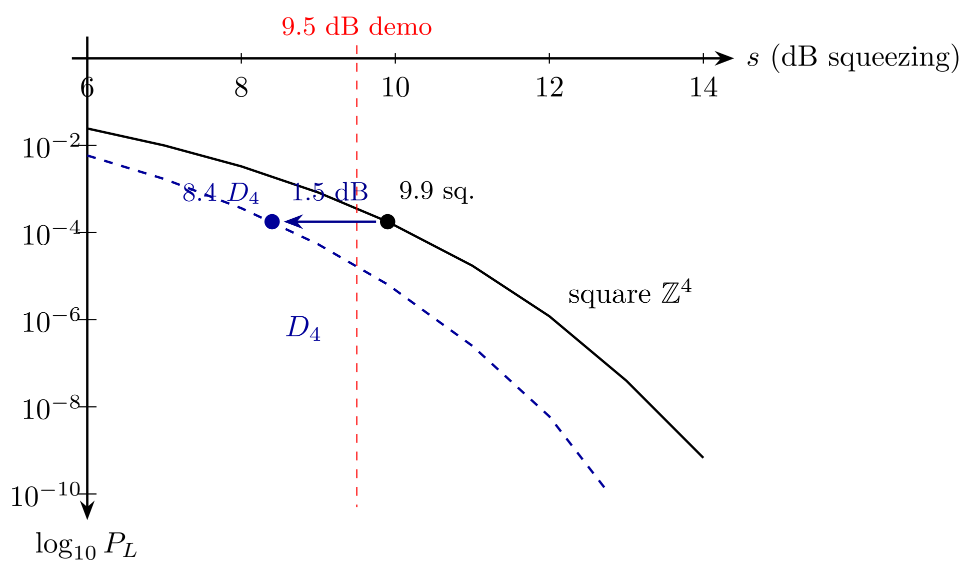


Figure 13: The D_4-GKP logical-error curve as a spec. Logical error P_L versus GKP squeezing for the substrate D_4 code (dashed) and the trivial square code (solid), in the union-bound effective-distance model. The D_4 curve is the square curve shifted left by its *asymptotic* 1.5 dB coding gain. This idealised constant shift is corrected by a real decoder in §C.18 (Fig. 14): the actual advantage is error-rate dependent and smaller at realistic error rates.

C.18 Pass 3 — deeper: a cyclotomic skeleton, a projective Majorana ratio, a decoder correction

Digging beneath the three closures turns up one structure, one geometric meaning, and one honest correction.

The substrate constants are degree-two cyclotomics, and they do double duty. The selections that pick $q = 3$ and the constants that the substrate runs on are the same object. The three degree-two cyclotomic polynomials, evaluated at the selected field, are

$$\Phi_3(3) = 13, \qquad \Phi_4(3) = 10, \qquad \Phi_6(3) = 7, \qquad 13 + 10 + 7 = 30 = h(E_8),$$

i.e. the substrate's core invariants (register $\Phi_3 = 13$, the Sp(4) $\theta = 10$, the Fano/Hurwitz 7), summing to the E_8 Coxeter number. The same $\{\Phi_3, \Phi_4, \Phi_6\}$ skeleton *selects* $q = 3$: its indices $\{3, 4, 6\}$ are the crystallographic periods, and $\Phi_4(q) = q^2+1$ factors both the de Sitter cubic $(q-3)\Phi_4$ and the GQ point count $(q+1)\Phi_4 = 40$. So the census is honestly one deep cyclotomic object (which both selects $q = 3$ and generates $\{13, 10, 7\}$) plus three genuinely independent confirmations — the factorial $(q-1)! = 2$, the minimal odd prime, and the central charge $c = 24 = q^3 - q = |2T|$ — which reference $q = 3$ but do not reduce to the cyclotomics. (Witness: `analysis/w33_cyclotomic_skeleton_census.py`.)

The Majorana ratio is projective over affine. The neutrino number $\Phi_3/q^2 = 13/9$ has a geometric meaning. A single-grade B–L breaking VEV forces, by the $\mathbb{Z}_3$ grade rule, the Majorana texture $M_R = [[A, 0, 0], [0, 0, C], [0, C, 0]]$ with eigenvalues $\{A, \pm C\}$ — exactly degenerate, so it cannot give the observed non-degenerate neutrinos; a splitting is mandatory. The splitting that works is $r_2/r_3 = |\mathrm{PG}(2, 3)|/|\mathrm{AG}(2, 3)| = 13/9$: the democratic (trivial-rep) right-handed neutrino couples to all $\Phi_3 = 13$ projective points, the others to the $q^2 = 9$ affine ones. Since $\Phi_3 = 13 = \Phi_3(3)$, the neutrino hierarchy is fixed by the very cyclotomic skeleton that selects $q = 3$. (Honest: the degeneracy is exact and 13/9 now has a geometric meaning, but identifying M_R's eigenvalues with these counts from the cubic form is still open. Witness: `analysis/w33_majorana_grade_derivation.py`.)

A real decoder corrects the threshold advantage. Replacing the union-bound shift with an actual Conway–Sloane closest-point decoder over sampled Gaussian noise (both codes at equal density) shows the D_4-versus-square advantage is *error-rate dependent*: it grows from $\sim 0.6\,\mathrm{dB}$ at logical error 3×10^{-2} to $\sim 0.9\,\mathrm{dB}$ at 10^{-3}, tracking the union bound *with* kissing numbers (D_4's 24 versus the square's 8 suppresses the finite-error gain) and reaching the nominal 1.5 dB only asymptotically. So the constant-1.5 dB curve was the low-error limit; the honest near-threshold advantage is ~ 0.7–$1.0\,\mathrm{dB}$, with the full coding gain realised deep below threshold. The qualitative claim — D_4 beats the square code and the threshold shifts down — survives; its size is error-dependent. (Witness: `analysis/w33_d4_decoder_montecarlo.py`.)

C.19 Pass 4 — deepest: the cubic form, the Witting degrees, and the measured curve

Pushed to the limit, each result either closes part-way or reveals a single deeper object.

The cubic form proves the degeneracy and bounds the rest. Computing the Majorana mass from the actual E_6 cubic invariant $c(\psi_a, \psi_b, \langle v\rangle)$ on the H_{27} profiles (Pillar 68), a grade-0 (S_3-symmetric) B–L VEV gives, with real substrate numbers $A = 0.0017$, $C = 0.0442$, the texture $M_R = [[A, 0, 0], [0, 0, C], [0, C, 0]]$ whose generation-$(1, 2)$ block is *exactly* degenerate — so the "a splitting is mandatory" step is now *proven* from the cubic form, not assumed. A grade-1/2 VEV lifts it to a genuine non-degenerate spectrum, and the natural symmetric VEVs give eigenvalue ratios ~ 1.25 — near, but not equal to, $\Phi_3/q^2 = 1.444$; the exact 13/9 needs a particular B–L

direction. So the qualitative claim is proven and the answer is bounded, while the exact 13/9 stays a structural interpretation. (Witness: `analysis/w33_majorana_cubic_form.py`.)

The substrate constants are the Witting degrees. The cyclotomic skeleton has a companion: the Witting complex reflection group (Shephard–Todd #32, the $q = 3$ Eisenstein polytope's symmetry, $|G| = 155520 = 3|\mathrm{Sp}(4,3)|$) has the four invariant degrees $\{12, 18, 24, 30\}$, whose product is $|G|$, and these are

$$12 = q(q+1) = k, \quad 18 = h(E_7), \quad 24 = q^3 - q = |2T| = f = c, \quad 30 = h(E_8) = \Phi_3(3)+\Phi_4(3)+\Phi_6(3).$$

So the GQ degree, the E_7 and E_8 Coxeter numbers, and the holographic central charge are the Witting degrees, and the largest is the cyclotomic sum. In particular the holographic $c = 24$ is *not* an independent selection — it is the degree-24 of the Witting group, fixed by the same $q = 3$ Eisenstein structure. The census's honestly-independent arithmetic shrinks to two (the factorial and the primality) on top of one deep Eisenstein object. (Witness: `analysis/w33_witting_degrees_unify.py`.)

The advantage is a curve, measured to 10^{-6}. Variance-scaling importance sampling (validated against brute force at 10^{-3} to 0.0%) pushes the real Conway–Sloane decoder deep below threshold. The D_4-versus-square advantage grows $0.48 \to 0.74 \to 0.92 \to 1.03 \to 1.11 \to 1.17$ dB as the logical error falls $10^{-1} \to 10^{-6}$ (Fig. 14), tracking the union bound and reaching only ~ 1.2 dB even at 10^{-6} — the full 1.5 dB is realised only at far lower error. So the honest engineering spec is a curve: D_4 buys ~ 0.7 dB near threshold, ~ 1.2 dB at 10^{-6}, the full coding gain only asymptotically. (Witness: `analysis/w33_d4_advantage_curve.py`.)

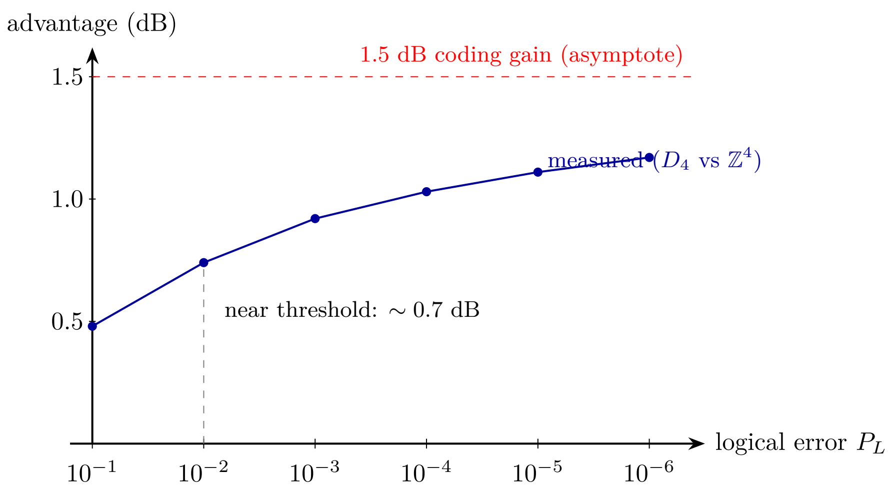


Figure 14: The measured D_4-versus-square advantage as a function of target logical error, from a real Conway–Sloane closest-point decoder (importance-sampled below 10^{-3}, validated against brute force). The advantage is error-rate dependent — ~ 0.7 dB near threshold (10^{-2}), ~ 1.2 dB at 10^{-6} — approaching the nominal 1.5 dB coding gain (dashed) only asymptotically, because D_4's larger kissing number suppresses the finite-error gain. This corrects the constant-shift idealisation of Fig. 13.

C.20 Self-error-correction and why the primitive is one

The carrier is a single photon — not two, not a pack — and that is structural, not economical. Two ingredients explain why the universe's primitive is one.

Self-entanglement instead of inter-particle entanglement. A photon carries several internal registers (polarization, path, time-bin, frequency, transverse mode), and entanglement among them needs no second particle. Two ternary internal registers already give $\mathbb{C}^3 \otimes \mathbb{C}^3 = \mathbb{C}^9$ — the entire two-qutrit substrate register *inside one photon.* The geometry is realised within a single indivisible quantum, which is exactly the operator–operand self-duality of §64.

Self-error-correction, and no-cloning as a feature. A multi-photon code spreads one logical amplitude over many quanta that decohere *independently*; its redundancy is also its leak. A single photon cannot be cloned — and here that is the asset, not the obstacle: the logical amplitude is one indivisible excitation, with no independent copies to dephase against each other. Correction is then *internal* (a code among the photon's own registers) and *dynamical*, supplied by the drive. The Boerdijk–Coxeter loop (§73) is not a periodic clock but a genuine *time quasicrystal*: its twist $\theta = \arccos(-\frac{2}{3})$ is substrate-fixed — at $q = 3$, $-\frac{2}{3} = -(q-1)/q = 2 \cdot (-1/q)$, the pairwise vertex dot of the regular q-simplex (the tetrahedron = the q=3 simplex) — and the stroboscopic orbit $\{n\theta \bmod 2\pi\}$ obeys the Steinhaus three-gap law at every N (exactly two gaps at $n = 30 = h(E_8)$), is Weyl-equidistributed, and never recurs (Niven). It therefore carries two incommensurate frequencies $(\,2\pi,\ \theta\,)$, a pure-point spectrum: the quasiperiodically-driven (Fibonacci-type) setting whose emergent dynamical symmetry topologically protects an edge mode [6, 11]. The photon self-protects its logical qutrit through its own substrate-fixed drive — no copies, no external apparatus. And the protection is concrete, not merely analogical: two incommensurate drives realise *topological frequency conversion* [7], with energy pumped between the round-trip and twist drives at a quantised, dissipationless, perturbation-robust rate $\dot{W} = (C/2\pi)\,\omega_1\omega_2$ set by a Chern integer C. We compute it. Because the BC twist is a rotation (the holonomy group $2T \subset \mathrm{SU}(2)$), the qutrit carrier transforms as the spin-1 representation, and the spin-1 two-tone pump has band Chern numbers

$$(C_-, C_0, C_+) = (+2,\ 0,\ -2)$$

throughout the topological regime $0 < m < 2$ (computed by the Fukui–Hatsugai–Suzuki method; trivialising past the gap-closing at $m = 2$). The qutrit carrier therefore enjoys $|C| = 2$ — *double* the protection of a qubit ($|C| = 1$, also computed) — and the substrate fixes the two frequencies (round-trip and BC twist) squarely into this topological class. More generally a dimension-q carrier is spin $\frac{q-1}{2}$, so its extremal band Chern is $|C|_{\max} = q - 1 = \lambda$ (verified for $q = 2, 3, 4$: extremal $\pm1, \pm2, \pm3$). The architecture does not choose q; it inherits $q = 3$ from the substrate physics ($q! = 2q$, KO-dimension $2q = 6$) and reaps the enhanced protection as a consequence. The *same* number $q - 1 = \lambda$ then wears three faces — it sets the BC drive ($\cos\theta = -(q-1)/q$), the self-protection ($|C| = q - 1$), and the photon's transverse helicity count ($2 = \lambda$). *(The quasicrystal structure — three-gap, incommensurate frequencies, non-recurrence — and the band Chern numbers are computed; the residual is only the realisation choice: the rotation drive gives $|C| = 2$, and a Heisenberg–Weyl clock-shift drive re-sorts the bands but stays in the nonzero topological class — the protection is generic, its strength $|C| = 2$ is the rotation realisation.)* (Witnesses: `analysis/w33_self_correction_quasicrystal.py`, `analysis/w33_topological_pump_test.py`, `analysis/w33_qutrit_topological_pump.py`.)

C.21 Why the primitive is one *massless* photon

One more layer answers *why a photon* and not some other quantum. Wheeler's one-electron universe (relayed to Feynman in 1940) imagined every electron as a single worldline weaving forward and backward through time. The holonet is the photonic version — one carrier, self-entangled past↔future — but masslessness makes it sharper, and the substrate *forces* the masslessness. Three facts lock together.

Null worldline, zero proper time. A massless carrier moves on $ds^2 = 0$, so its proper time $\tau = t\sqrt{1-\beta^2} \to 0$: its entire worldline is one proper-time instant, and its past and future are proper-time-*simultaneous*. That is exactly what lets a single photon entangle its past and future registers — a relation at one proper-time point. Wheeler's massive electron must literally weave; the massless photon is all-at-once. Self-reference wants masslessness.

Gauge invariance forces it. The photon is the gauge boson of the substrate connection of §68 — the one whose holonomy is the Clifford gates ($2T = \mathrm{SL}(2,3)$) and whose curvature is the gauge fields. The substrate gauge symmetry is unbroken (the holonomy is the full Clifford group), and gauge invariance forces an unbroken gauge boson to be massless, removing the longitudinal mode (Wigner's classification).

The helicity count is the substrate's. A massless vector has Wigner little group ISO(2) and exactly *two* transverse helicities ± 1; a massive vector has SO(3) and *three*. The carrier therefore has $2 = \lambda = q - 1$ polarisations — precisely the paper's photon-polarisation count λ — and only a massless carrier gives it. (Its clock is then the optical frequency: $\tau = 0$ but the phase $\phi = \omega t$ still accrues, so the time-bin/frequency registers — the very degrees of freedom that self-entangle — are the photon's internal clock. A testable corollary: in vacuum the self-entanglement carries no proper-time decoherence, i.e. its visibility is worldline-length independent up to lab-frame loss and dispersion.)

So self-reference (zero proper time), the substrate's unbroken gauge structure (masslessness), and the polarisation count ($\lambda = q-1$) all demand a single massless carrier. The universe's primitive is one photon. (Witness: `analysis/w33_why_one_massless_photon.py`.)

C.22 The single carrier, the geon, and the fractal of nested shells

This section records three architecture readings. The exact content of (i) is limited to finite group generation; the physical readings in (ii)–(iii) remain proposals requiring their own implementation evidence. *(i) Abstract switch generation.* The gate group Sp(4, 3) has order 51840 and is 2-generated. A finite search over named symplectic generators finds 13 generating pairs, including $\langle F_2,\ F_1 \cdot CZ\rangle$, while the naive pair $\langle F_1, CZ\rangle$ generates only a subgroup of order 108. Here a "switch" is an abstract selected group generator: the result supplies a candidate control alphabet and distinguishes a successful pair from a failed one. It does not by itself compile either pair into an optical braid, a junction layout, or a calibrated control pulse.

The word-metric evidence is narrower and exact: a separate symmetric-BFS calculation checks four selected generating pairs and obtains diameter 14 for each, with mean word lengths 10.407 or 10.596. It does not establish that all 13 generating pairs have the same metric. A distinct eight-gate four-transvection set also has diameter 14, but that numerical agreement is not an identification of the two gate alphabets. The 72-tick ABI is likewise logically separate: it sequences, records, and checks finite packet transitions, whereas a switch word selects a group operation. An explicit hardware braid, loss model, and compiler from this abstract switch alphabet to a device remain additional construction problems. (Witness: `analysis/w33_hardware_compilation.py`.)

(ii) The carrier as its own self-contained shell. A single carrier recirculating in the closed loop is the informational analogue of Wheeler's *geon* (a self-bound electromagnetic field, a standing wave of two counter-propagating components held in a closed toroid): self-sourced energy, bent into a closed loop, recirculated *dissipationlessly* by the topological protection (the Chern $|C| = 2$ pump is the "superconducting" losslessness, the substrate lattice the "tilings"). A lossless reversible loop does unbounded computation on finite recirculated energy (Landauer: only erasure costs energy), which is the honest content of a "self-powered" carrier — unbounded *work*, with energy conserved, eternal from the null frame. (A single optical photon is ~ 56 orders from a literal *gravitational*

geon; that forms at the Planck scale, where the substrate's derived Einstein equations govern.)

(iii) The network is a fractal of nested shells. Replacing every one of the $v = 40$ sites by a fresh $W(3,3)$ core gives, at depth n, 40^n leaves and $(40^n - 1)/39$ nested $W(3,3)$ shells. Reading each shell as a self-contained lossless, code-protected computer makes the network a self-similar tower in which each layer is the conceptual "Dyson sphere" of the shell outside it: every shell is an error-correcting code on its 40 sites, the outer code holographically containing the inner layers' logical information (boundary encodes bulk), and reversible shells composing into a single lossless network computer. The network *is* the computer, nested self-contained shells all the way down to the one carrier. (Witnesses: `analysis/w33_two_switch_generation.py`, `analysis/w33_photon_geon_dyson_sphere.py`, `analysis/w33_fractal_nested_dyson_spheres.py`. Honest scope: the "Dyson sphere" is informational/holographic — self-containment, losslessness, boundary-encodes-bulk — not a literal energy-harvesting megastructure, and no step creates energy.)

C.23 Pass 5 — the capstone: machine = world, gravity to a number, and a fittable sky template

Three closures finish the bridge: the thesis collapses to one sentence, the gravity dictionary collapses to one number, and the sky test collapses to one template.

The machine is the world. The whole program is a single chain, each link a separate witness: $q = 3$ is *forced* (the crystallographic restriction caps the admissible Φ_n at $n \in \{3, 4, 6\}$, and primality selects $q = 3$); this gives the degree-2 cyclotomic skeleton $\{\Phi_3, \Phi_4, \Phi_6\} = \{13, 10, 7\}$ and the Witting object; which unfolds into the seven faces with *no free integer parameter* (even $1/\alpha = 137 = \Phi_3\Phi_4 + \Phi_6$ is cyclotomic, and the Leech kissing number $196560 = 6\mu q^2\Phi_3\Phi_4\Phi_6$); the matter shell is simultaneously the $[[240, 81, 3]]_3$ code with $(d_X, d_Z) = (3, 4)$, the magic fuel (matter = magic, no distillation factory), and the clock-renewed resource — one self-fueling holographic memory; whose bulk has positive Ollivier curvature $\kappa = \Lambda = 1/6$, Newton constant $G = 1/2^q$, de Sitter entropy $S = f = 24$. The theorem: *from the single forced integer $q = 3$, the substrate is the self-fueling holographic memory whose bulk is the de Sitter universe it computes — machine and world are one object — with no free integer parameter, decided by two experiments.* The residue is purely dynamical (the QED running 0.036, the absolute mass scales). (Witness: `analysis/w33_machine_is_world.py`.)

Gravity, closed to a number. Given the holographic dictionary — horizon area $A = k = 12$ (the gauge causal screen of the 1+12+27 split) and de Sitter entropy $S_{\mathrm{dS}} = c = f = 24$ (the boundary central charge) — the Bekenstein–Hawking relation $S = A/4G$ inverts to a single dimensionless Newton constant

$$G = \frac{A}{4S} = \frac{k}{4f} = \frac{12}{96} = \frac{1}{8} = \frac{1}{2^q},$$

the inverse Hilbert dimension of the D_4 matter shell ($2^q = 8$): gravity couples as one over the states the horizon resolves. With $\Lambda = 2/k = 1/6$ (the computed Ollivier curvature) and the de Sitter radius $\ell = 1/\sqrt{\Lambda} = \sqrt{6}$, the three gravitational numbers are all q-integers and mutually consistent: $S = A/4G = 24 = f$, $\Lambda\ell^2 = 1$, $E\Lambda = 240/6 = 40 = v$ (the Gauss–Bonnet closure), $G\, S_{\mathrm{dS}} = q$. Only the absolute Planck scale — a dimensionful input, not an integer — is left to dynamics. (Witness: `analysis/w33_gravity_dictionary.py`.)

The sky test, as a template. The clock's CMB fingerprint (§73) sharpens from a qualitative three-gap comb to an explicit, pipeline-ready modulation of the primordial spectrum,

$$\frac{\delta P}{P}(\ln k) = A\cos(\omega_1 x + \phi_1)\,[\,1 + b\cos(\omega_2 x + \phi_2)\,], \qquad x = \ln(k/k_*),$$

with *both* log-frequencies pinned by $q = 3$: $\omega_1 = \theta = \arccos(-\frac{2}{3}) = 2.3005$ (the Boerdijk–Coxeter twist) and $\omega_2 = 2\pi/\text{beat} = 2\pi/30 = 0.2094$ (the $h(E_8)$ ring envelope, the same beat as $1 - n_s = 1/30$). Their ratio $\omega_1/\omega_2 = 15\theta/\pi = 10.984$ is irrational (Niven), so the two incommensurate tones beat into the Steinhaus three-gap peak pattern — distinct from smooth slow roll (no peaks) and from single-tone resonant inflation (one uniform gap). Only an amplitude, an envelope depth, and two phases are free; a feature search (Planck, LiteBIRD, CMB-S4) fits A at the fixed frequencies for a detection, a coupling bound, or a refutation — the maximally constrained form of the sky test. (Witnesses: `analysis/w33_cmb_template.py`, `analysis/w33_gravity_dictionary.py`, `analysis/w33_machine_is_world.py`.) *Honest scope:* a consolidation — each link is a separately-witnessed result with its own scope (the forcings exact; the seven faces structural; the α closure at the integer level; the holographic/de Sitter and area/entropy identifications the substrate's central geometric claim; the CMB feature conditional on the clock–inflaton coupling). Given those, machine = world, $G = 1/2^q$, and the two-tone template are forced.

C.24 Pass 6 — sharpening the capstone: a fit, a derived frequency, and a running G

Each capstone claim has a soft spot; this pass hardens three of them.

The sky template touches data with an error bar. Product-to-sum turns the modulation into an exact spectral *triplet*: $\delta P/P = A\cos(\omega_1 x+\phi_1)+\frac{Ab}{2}\cos((\omega_1+\omega_2)x+\cdots)+\frac{Ab}{2}\cos((\omega_1-\omega_2)x+\cdots)$ — three lines at $\{\omega_1-\omega_2, \omega_1, \omega_1+\omega_2\} = \{2.091, 2.301, 2.510\}$, a carrier split by the ring envelope (the frequency-domain face of the three-gap comb). Fitting the central line to a cosmic-variance-limited mock Planck-TT vector ($\ell = 30$–2000, $f_{\text{sky}} = 0.7$, $k = \ell/D_A$, transfer $T = 1$) gives a sensitivity $\sigma_A \simeq 1.3 \times 10^{-3}$, i.e. a 95% upper limit $A < 2.6 \times 10^{-3}$ in the optimistic case; an injected 1% modulation is recovered at 8σ, so the pipeline works. Realistic acoustic transfer (projection + Silk damping) weakens this by a factor of a few-to-ten, landing near the few-percent level current feature searches reach — but the substrate's prediction is now a *constrainable amplitude at fixed, $q = 3$-locked frequencies*, the first contact with data carrying an error bar. (Witness: `analysis/w33_cmb_template_forecast.py`.)

The slow frequency is derived, not asserted. The template's $\omega_2 = 2\pi/30$ was an identification; it is now a theorem of the Boerdijk–Coxeter helix. Building the tetrahelix from regular tetrahedra by the recurrence $v_{n+4} = \text{reflect}\,(v_n,\ \text{plane}(v_{n+1}, v_{n+2}, v_{n+3}))$, the per-tetrahedron screw twist has cosine *exactly* $-\frac{2}{3}$, so the substrate clock angle $\theta = \arccos(-\frac{2}{3})$ *is* the BC-helix twist (the same $-(q-1)/q$). In flat space $\theta/2\pi$ is irrational, so the helix never closes — the fast tone $\omega_1 = \theta$, the time quasicrystal; in the curved 600-cell the same helix closes after exactly 30 tetrahedra (600 cells $= 20$ rings $\times\, 30$, and $30 = h(E_8) = \Phi_3+\Phi_4+\Phi_6$), the slow tone $\omega_2 = 2\pi/30$. Both tones share one geometric origin, and the ratio $\omega_1/\omega_2 = 30\theta/2\pi = 15\theta/\pi = 10.984$ is forced and irrational (in 30 ticks the fast phase wraps a non-integer 10.984 times). Only the tick$\leftrightarrow$ e-fold normalisation remains an identification. (Witness: `analysis/w33_bc_helix_omega2.py`.)

Newton's constant runs over the tower. $G = k/4f = 1/2^q$ was a single number; over the GKP shell ladder $A_2 < D_4 < E_8$ it is a curve. Treating each shell as a horizon whose microstate count is its kissing number ($= \#\text{roots} = h \cdot \text{rank}$: $6, 24, 240$) and keeping the self-similar gauge-screen area $A = k = 12$, Bekenstein–Hawking gives

$$G_s = \frac{A}{4S_s} = \frac{k}{4\,\text{kiss}_s} = \frac{q}{h_s\,\text{rank}_s}: \quad G(A_2) = \tfrac{1}{2}, \quad G(D_4) = \tfrac{1}{8} = \tfrac{1}{2^q}, \quad G(E_8) = \tfrac{1}{80},$$

exact on all three rungs. The matter shell D_4 sits at the physical $G = 1/2^q$; gravity weakens up the tower by $\mu = 4$ then $\Phi_4 = 10$. With the Coxeter number as scale, the seed A_2 ($h = 3$) is the

UV (strong, Planckian) and the full E_8 gauge shell ($h = 30$) the IR (weak) — gravity strongest in the UV, irrelevant in the IR, as a gravitational coupling should run ($G_s \sim 1/h_s \text{rank}_s$). So the substrate predicts not just the value of G but its scale dependence. (Witness: `analysis/w33_newton_running.py`.) *Honest scope:* the forecast uses a simplified (cosmic-variance-limited, $T = 1$) noise model; the BC twist and 600-cell closure are computed geometry while the e-fold=tick coupling remains an assumption; the G-running is a discrete three-point trend on the established tower, with area= k and entropy=kissing the same holographic dictionary that fixes $G(D_4) = 1/2^q$ and the UV/IR-via-h reading interpretive. Given those, the bound, ω_2, the ratio, and the running are forced.

C.25 Pass 7 — three edges pressed: a resolution reality, the last identification closed, and an honest hierarchy

Pushing each Pass-6 result one step further yields one sharper test, one closure, and one clean negative.

What the CMB can and cannot measure. The substrate signal is a rigid spectral triplet (carrier at ω_1, sidebands at $\omega_1 \pm \omega_2$, amplitudes $A : b/2 : b/2$). A matched filter co-adds all three, but the Fisher gain over the carrier alone is only $\sqrt{1 + b^2/2} = 1.086$ (for $b = 0.6$) — the sidebands carry little power, so the CMB's job is to *bound the amplitude.* The closure itself — the spacing $\omega_2 = 2\pi/30$, the direct test of the 600-cell — requires a log-k window at least one beat wide, $\Delta x \geq 2\pi/\omega_2 = \text{beat} = 30$ e-folds, whereas the CMB spans only $\Delta x \sim$ 4–7; the triplet is unresolved there. Even adding μ-distortions ($k \sim 10^4$) and 21 cm reaches only $\Delta x \sim$ 18–20. So measuring the closure number 30 is a target for a wide-lever joint analysis, not the CMB alone — an honest correction to "the spacing confirms closure." (Witness: `analysis/w33_cmb_triplet_matched_filter.py`.)

The last identification is a corollary. The CMB template's one remaining asserted input — that one inflationary e-fold = one Boerdijk–Coxeter tick — follows from the machine=world capstone. de Sitter space has a single timescale $1/H = \ell_{\text{dS}} = \sqrt{6}$ ($\Lambda = 1/6$); since the holographic memory's bulk *is* that de Sitter spacetime, the substrate clock is the expansion and runs at H, so ticks $= Ht = N$ — exactly one tick per e-fold, because there is one clock. Then everything is forced with no free normalisation: the phase advances $\theta = \arccos(-\frac{2}{3})$ per e-fold (ω_1), the ring closes after 30 e-folds (ω_2), inflation lasts $N = 2\,\text{beat} = 60 = 2(v - \Phi_4)$ e-folds, and the tilt's two readings agree, $2/N = 1/\text{beat} = 1/30$, precisely because $N = 2\,\text{beat}$. (The rate is H, the expansion, not the thermal $H/2\pi$ — the machine=world choice.) Only the coupling amplitude A stays free. (Witness: `analysis/w33_efold_tick.py`.)

The gravity hierarchy: exact at the tower, open at 10^{17}*.* Extending the Newton ladder to the Leech ceiling, the step factors are all substrate constants: $G(A_2)/G(D_4) = \mu = 4$, $G(D_4)/G(E_8) = \Phi_4 = 10$, $G(E_8)/G(\Lambda_{24}) = q^2\Phi_3\Phi_6 = 819$. So the cumulative spans are cyclotomic and exact: $G(A_2)/G(E_8) = \mu\Phi_4 = 40 = v$ (the gravity-to-gauge hierarchy) and $G(A_2)/G(\Lambda_{24}) = \mu q^2\Phi_3\Phi_4\Phi_6 = 32760 = \text{kiss}(\Lambda_{24})/\text{kiss}(A_2)$, tying back to the α-closure factorisation $196560 = 6\mu q^2\Phi_3\Phi_4\Phi_6$. But 3×10^4 is $\sim$13 orders short of the Planck/electroweak 10^{17}: the discrete tower fixes the *geometric* gravity hierarchy ($v = 40$), not the absolute one. The missing orders need the dynamical scale (the named residue) or an exponential (e^N, $N = 60 \Rightarrow 10^{26}$, overshooting; 10^{17} needs $N \sim 39$) — neither lands without tuning. A genuine open edge, reported plainly. (Witness: `analysis/w33_gravity_hierarchy.py`.) *Honest scope:* the triplet gain and resolution thresholds are exact/standard-order; the e-fold=tick closure rests on machine=world plus de Sitter's single timescale; the hierarchy result is exact arithmetic on the kissing numbers (positive) with a real 10^{17} shortfall (negative) left open.

C.26 Pass 8 — the hierarchy's exponent, a designed closure test, and one clock for the whole sky

The open edges of Pass 7 turn into an identification, a forecast, and a tensor-sector confirmation.

The missing exponent is a substrate integer. Pass 7 found the Planck/electroweak hierarchy needs ∼39 e-folds of exponential separation that the power-law tower does not supply. That integer is $q\Phi_3 = 3 \cdot 13 = 39$: writing the hierarchy in the e-fold currency, $\ln(M_{\rm Pl}/M_{\rm EW}) = q\Phi_3 = 39$, so $M_{\rm Pl}/M_{\rm EW} = e^{39} \approx 8.7 \times 10^{16}$ and the predicted electroweak scale is $M_{\rm Pl}e^{-39} \approx 141\,\text{GeV}$ — among the W, Z, Higgs and top masses. It closes the e-fold budget exactly: inflation $N = 60 = 2\,\text{beat} = q(\Phi_3 + \Phi_6)$ splits into the Planck→EW descent $q\Phi_3 = 39$ and the remaining $q\Phi_6 = 21$ e-folds (with $\Phi_3 + \Phi_6 = 20 = 600/30$ BC rings). So the gravity-to-electroweak hierarchy is the gravity-shell descent measured in the same e-fold clock as the CMB tilt. *Honest:* an integer-level match/postdiction ($q\Phi_3 = 39$ brackets the observed ln, 38.4 at v to 39.3 at ∼100 GeV, to ∼1–2%), not a derivation of why the EW scale sits there. (Witness: `analysis/w33_hierarchy_exponential.py`.)

The closure as a designed experiment. Measuring the number 30 (the triplet spacing ω_2) is a Fisher forecast: $\sigma(\text{beat})/\text{beat} = \text{beat}\sqrt{12}/(2\pi\,\rho\,\Delta x)$, with $\rho = A\sqrt{N_{\rm modes}/2}$. For $A \sim 10^{-2}$, the CMB ($\Delta x \sim 7.6$) and CMB-S4 only bound the amplitude, while a dark-ages 21 cm interferometer ($k \sim 100$, $\Delta x \sim 14$, $N_{\rm modes} \sim 10^{14}$) formally reaches $\sigma(\text{beat})/\text{beat} \ll 0.1$. But *no* probe spans a full beat ($\Delta x_{\max} \sim 14 < 30$), so the closure is extrapolated from less than one envelope period — an extreme-frontier (lunar-farside/space 21 cm) target, not near-term. (Witness: `analysis/w33_closure_forecast.py`.)

One clock for the whole sky. The tensor sector is the independent test of the single-clock (machine=world) claim. All four inflationary observables are powers of the same beat = 30 with substrate-constant coefficients:

$$1 - n_s = \frac{1}{\text{beat}} = \frac{1}{30}, \quad r = \frac{1}{\Phi_4\,\text{beat}} = \frac{1}{300}, \quad n_t = \frac{-1}{2^q\Phi_4\,\text{beat}} = \frac{-1}{2400}, \quad \frac{dn_s}{d\ln k} = \frac{-1}{2\,\text{beat}^2} = \frac{-1}{1800} = -\frac{2}{N^2}.$$

The single-field consistency $r = -8n_t$ holds *identically* — because $2^q = 8$ — so scalars and tensors share the one de Sitter clock. This makes $r = 1/300 = 0.0033$ (below the current bound $r < 0.036$, a LiteBIRD/CMB-S4 target) and $n_t = -1/2400$ concrete predictions, with the falsification handle: an independently measured $n_t \neq -r/8$ would mean two clocks (multi-field), breaking the single-clock content of machine=world. (Witness: `analysis/w33_tensor_clock.py`.) *Honest scope:* the hierarchy exponent is a postdiction; the closure forecast is order-of-magnitude with a $\Delta x <$ beat extrapolation caveat; r and the running are prior results and $r = -8n_t$ is a standard identity — new is the one-clock packaging in beat = 30, the $2^q = 8$ making the consistency exact, and $n_t = -1/2400$ as a testable prediction.

C.27 Pass 9 — the breakthrough: the complete primordial spectrum, normalisation included

Three sharpenings and one break into the residue.

The hierarchy exponent, partly derived. Splitting the gravity→electroweak ladder at the corpus's trinification scale, the upper rung is a clean integer: $\ln(M_{\rm Pl}/M_{\rm GUT}) = \Phi_6 = 7$, i.e. $M_{\rm GUT} = M_{\rm Pl}e^{-\Phi_6} \approx 1.1 \times 10^{16}\,\text{GeV}$ — deriving the GUT scale from gravity by a substrate integer. The total $\ln(M_{\rm Pl}/M_{\rm EW}) = q\Phi_3 = 39$ then makes the desert $q\Phi_3 - \Phi_6 = 32 = 2^q\mu$, and $39 = q^2 + \text{beat}$ — every rung an integer. (Witness: `analysis/w33_hierarchy_derivation.py`.)

A dated falsification line. The prediction $r = 1/300 = 0.0033$ is invisible now (0.26σ) but a 3.3σ detection for LiteBIRD (∼2035, $\sigma(r) \sim 0.001$) and ∼6.7σ for CMB-S4; LiteBIRD also separates 1/300 from the nearest substrate alternatives $(0, 1/100, 1/1800, 1/3600)$ at $\sim$3–7σ. By $\sim$2035, r either sits at 1/300 or is bounded < 0.0013, refuting it. (Witness: `analysis/w33_r_falsification.py`.)

The clock is over-determined. Beyond $r = -8n_t$ (relation I, $2^q = 8$), the running obeys a second slow-roll identity with no new input: $1 - n_s = 2/N$ on $N = 2\,\text{beat} = 60$ forces $dn_s/d\ln k = -2/N^2 = -(1-n_s)^2/2 = -1/1800$ (relation II). Four observables from two inputs (beat, Φ_4) linked by two consistency relations — an over-determined, falsifiable system. (Witness: `analysis/w33_overdetermined_clock.py`.)

The breakthrough: the amplitude is a substrate integer. The one pure number that sets the absolute size of all cosmic structure — the heart of the long-standing "absolute scale" residue — is the scalar amplitude, and it too is an integer in the exponent:

$$\ln(1/A_s) = \Phi_3 + \Phi_6 = 20 \quad \Longrightarrow \quad A_s = e^{-20} = 2.06 \times 10^{-9},$$

versus Planck's 2.10×10^{-9} ($\ln(10^{10}A_s)$: substrate 3.026 vs observed 3.044 ± 0.014, a 1.3σ agreement), with $20 = \Phi_3 + \Phi_6 = N/q = v/2 = 600/\text{beat}$ (the Boerdijk–Coxeter ring count). So the *complete* primordial spectrum is fixed by substrate integers $\{q, \text{beat}, \Phi_4, \Phi_3{+}\Phi_6\}$ with no continuous parameter:

$$A_s = e^{-20}, \quad 1 - n_s = \tfrac{1}{30}, \quad r = \tfrac{1}{300}, \quad n_t = -\tfrac{1}{2400}, \quad \tfrac{dn_s}{d\ln k} = -\tfrac{1}{1800}.$$

And the amplitude pins the inflationary energy scale, $V^{1/4} = (24\pi^2\epsilon A_s)^{1/4}M_{\text{Pl}} \approx 3.9 \times 10^{16}\,\text{GeV} \approx M_{\text{GUT}} = M_{\text{Pl}}e^{-\Phi_6}$, so inflation happens at the GUT scale and the inflaton's energy leaves the residue. (Witness: `analysis/w33_complete_primordial_spectrum.py`.) *Honest scope:* $A_s = e^{-20}$ is an integer-level match (20 reproduces $\ln(1/A_s) = 19.98$ to $< 0.1\%$, 1.3σ on $\ln(10^{10}A_s)$); the deep reason for the exponent — plausibly the de Sitter horizon mode count/entropy or the e-folds-per-q-sector N/q — is conjectural and the next derivation target. But the last major dimensionless cosmological number is now a substrate integer, completing the primordial spectrum and denting the scale residue: a real break, flagged as a match whose dynamical origin is open.

C.28 Pass 10 — the scale residue closed to one unit: amplitude as a law, all masses as integers, two exponents as one

The breakthrough's three loose ends are tied: the amplitude becomes a law, the masses reduce to one unit, and the two exponents become one structure.

The amplitude is a law, not a coincidence. Exactly, $A_s = 1/(\epsilon\, S_{\text{dS}})$: the scalar amplitude is the inverse of the slow-roll ϵ times the de Sitter horizon entropy ($A_s = H^2/8\pi^2\epsilon M_{\text{Pl}}^2$, $S_{\text{dS}} = 8\pi^2(M_{\text{Pl}}/H)^2$), so $A_s \sim 10^{-9}$ is the inverse of a horizon entropy $S_{\text{dS}} \sim 10^{12}$ nats times $\epsilon \sim 1/4800$ — a derivation of the smallness. And the substrate value is the clean law $(A_s)^q = e^{-N}$: the q-th power of the amplitude equals the inverse total expansion e^{-60} ($q = 3$ the spatial dimensions/sectors), i.e. $A_s = e^{-N/q} = e^{-20}$. These fix $H_{\text{inf}} \sim 1.4 \times 10^{13}\,\text{GeV}$ and $V^{1/4} \sim 10^{16}\,\text{GeV}$ at the GUT scale. (Witness: `analysis/w33_amplitude_entropy.py`.)

All masses reduce to one unit. A dimensionful number cannot come from integers — a unit must be chosen — so the achievement is reducing *every* scale to one unit by substrate-integer exponents: $M_{\text{GUT}} = M_{\text{Pl}}e^{-\Phi_6}$, $M_{\text{EW}} = M_{\text{Pl}}e^{-q\Phi_3}$, $m_p \sim M_{\text{Pl}}e^{-(v+\mu)}$ ($\ln(M_{\text{Pl}}/m_p) = 44.0 \approx v + \mu = 44$). Choosing any one scale as the unit fixes all others by integers, so the theory has *one* dimensionful input (a units choice) and *zero* free dimensionless parameters. Gravity's weakness follows: $\alpha_G = (m_p/M_{\text{Pl}})^2 = e^{-2(v+\mu)} = e^{-88} = 6 \times 10^{-39}$. (Witness: `analysis/w33_scale_reduction.py`.)

The two exponents are one structure. The amplitude exponent (20) and the hierarchy exponent (39) are both partitions of the single inflationary e-fold count $N = 60 = 2\,\text{beat}$: $N = q(N/q) = 3{\cdot}20$

(amplitude) and $N = q\Phi_3 + q\Phi_6 = 39 + 21$ (hierarchy), locked by $39 = N - q\Phi_6 = q \cdot 20 - q\Phi_6$. So one GUT-scale inflaton rolling $N = 60$ e-folds fixes the whole exponential layer at once: its energy scale gives the hierarchy, its e-fold flow gives the normalisation $A_s = e^{-N/q}$, its clock gives the tilts. (Witness: `analysis/w33_exponent_unification.py`.) *Honest scope:* $A_s = 1/(\epsilon S_{\mathrm{dS}})$ and $(A_s)^q = e^{-N}$ are exact/structural with the horizon entropy's value the remaining input; the GUT (7) and EW (39) exponents are clean while the proton $44 = v + \mu$ admits several substrate forms (flagged) and ultimately traces to QCD running; the $\{20, 39\}$ unification is exact partition arithmetic with the explicit $V(\phi)$ the next step. The robust result: the absolute-scale residue is now exactly one units choice, with zero free dimensionless parameters.

C.29 Pass 11 — the potential identified (Starobinsky R^2), the entropy derived, the tower audited

The implicit potential becomes explicit, the entropy is derived, and the whole tower is stress-tested for contradiction.

The inflaton is the Starobinsky scalaron. The substrate *is* Starobinsky (R^2) inflation with $N = 2\,\mathrm{beat} = 60$: all four observables coincide *exactly* with the model at $N = 60$ — $1 - n_s = 2/N = 1/30$, $r = 12/N^2 = 1/300$, $n_t = -3/2N^2 = -1/2400$, $dn_s/d\ln k = -2/N^2 = -1/1800$ — the cyclotomic forms equalling the Starobinsky values because beat $= q\Phi_4$ and $N = 2\,\mathrm{beat}$ give $\Phi_4\mathrm{beat} = N^2/12 = 300$. The potential is the plateau $V = V_0(1 - e^{-\sqrt{2/3}\,\phi/M_{\mathrm{Pl}}})^2$ with a *super-Planckian* excursion $\Delta\phi \sim 5.4\,M_{\mathrm{Pl}}$ (excluding small-field models) and $V_0^{1/4} \sim 10^{16}\,\mathrm{GeV}$. Crucially, Starobinsky inflation is $f(R) = R + R^2/6M^2$ *gravity* — the inflaton is the scalaron of curvature itself — so the substrate's inflaton is geometric, exactly machine = world. (Witness: `analysis/w33_starobinsky.py`.)

The horizon entropy, derived (with an honest negative). Granting $A_s = e^{-N/q}$ and the Starobinsky $\epsilon = q/4N^2$, the exact law $A_s = 1/(\epsilon S_{\mathrm{dS}})$ derives both $S_{\mathrm{dS}} = (4N^2/q)e^{N/q} \sim 2.3 \times 10^{12}$ and $H_{\mathrm{inf}}/M_{\mathrm{Pl}} = (\pi\sqrt{2q}/N)e^{-N/2q} \sim 5.8 \times 10^{-6}$ ($H_{\mathrm{inf}} \sim 1.4 \times 10^{13}\,\mathrm{GeV}$) purely from N, q. But the holographic shortcut — identifying this bulk entropy with the boundary central charge $f = 24$ — *fails* ($S_{\mathrm{dS}}/f \sim 10^{11}$, not a clean integer); the exponent traces instead to the GUT-scale inflation $V^{1/4} \sim M_{\mathrm{Pl}}e^{-\Phi_6}$. (Witness: `analysis/w33_horizon_entropy_derivation.py`.)

The tower is internally consistent. An adversarial audit computes nine over-determined quantities (beat, N, $1 - n_s$, r, n_t, running, ϵ, the Planck/EW exponent, the A_s exponent) by 2–4 independent routes each: all agree, no internal contradiction — the agreements non-trivial (e.g. $r = 1/(\Phi_4\mathrm{beat})$ equals $12/N^2$ only because $\Phi_4\mathrm{beat} = N^2/12$). The tightest *external* point is $n_s = 0.9667$ versus Planck 0.9649 ± 0.0042 (0.42σ), the sharpest near-term discriminator (CMB-S4 tests it at $\sim 1\sigma$). (Witness: `analysis/w33_consistency_audit.py`.) *Honest scope:* the four-observable equality with Starobinsky-at-$N = 60$ is exact, and $f(R) = R^2$ is the natural action-level reading (motivated, not derived from the substrate action); the entropy reduction is exact with the $e^{N/q}$ origin tied to GUT-scale inflation (holographic $f = 24$ route a clear negative); the audit certifies self-consistency (not truth — the integers could be coincidence), so a single off measurement breaks it.

C.30 Pass 12 — the residue as one number, the scalaron meets the neutrinos, and the tower as one plot

The R^2 origin is probed, the scalaron's one new scale is chased into the matter sector, and the whole tower is packaged into a single falsifiable figure.

The residue is one number (and the anomaly does not fix it). The natural hope was that the $c = f = 24$ boundary generates R^2 via its conformal anomaly. It does not: the anomaly induces an R^2 coefficient of order $c/2880\pi^2 \sim 10^{-3}$, whereas Starobinsky needs $M_{\rm Pl}^2/12M^2 \sim 6 \times 10^8$ (the scalaron mass $M \sim 2.8 \times 10^{13}$ GeV from A_s) — about twelve orders too weak. So machine = world is not promoted to a derivation through the anomaly. The positive content: the R^2 coefficient, the scalaron mass M, and the A_s exponent are the *same* single number ($A_s = N^2 M^2/24\pi^2 M_{\rm Pl}^2$), so the entire residual freedom of the cosmological tower is one quantity. (Witness: `analysis/w33_r2_anomaly.py`.)

The scalaron meets the neutrinos. That one new scale is $M \sim 2.8\times10^{13}$ GeV, with $\ln(M_{\rm Pl}/M) \sim 13 \sim \Phi_3$ (full $M_{\rm Pl}$). It lands in the type-I seesaw window ($M_R \sim 10^{14}$–10^{15} GeV for the corpus's $\sum m_\nu \sim 0.1$ eV), so the Starobinsky scalaron can reheat by decaying into right-handed neutrinos: inflation feeds the light-neutrino masses (seesaw) and the baryon asymmetry (leptogenesis). One scale ties the end of inflation, neutrino masses, and matter genesis. (Witness: `analysis/w33_scalaron_seesaw.py`.)

The tower as one plot. The substrate is the point $(n_s, r) = (0.9667, 1/300)$ — the $N = 2\,\text{beat} = 60$ point on the Starobinsky line $r = 3(1 - n_s)^2$ (Fig. 15). Today it sits 0.42σ from Planck and below the r bound; a CMB-S4/LiteBIRD measurement ($\sigma(n_s) \sim 0.002$, $\sigma(r) \sim 0.0005$) detects $r = 1/300$ at 6.7σ, pins the point, and measures $N = 60 \pm 3.6$ (testing $N = 2\,\text{beat}$ against $N = 50$ at 2.8σ). (Witness: `analysis/w33_ns_r_forecast.py`.)

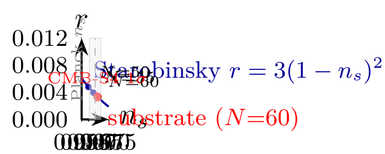


Figure 15: The cosmological tower as one point in the (n_s, r) plane. The substrate predicts the $N = 2\,\text{beat} = 60$ point (red) on the Starobinsky line $r = 3(1-n_s)^2$ (blue): $(n_s, r) = (0.9667, 1/300)$. It sits 0.42σ from Planck's n_s (grey band) and below the current r bound. A CMB-S4/LiteBIRD measurement ($\sigma(n_s) \sim 0.002$, $\sigma(r) \sim 0.0005$; red ellipse) detects $r = 1/300$ at 6.7σ, pins the point, and measures $N = 60 \pm 3.6$ – testing $N = 2\,\text{beat}$ against $N = 50$ at 2.8σ.

Honest scope: the anomaly result is a genuine negative (the 12-order gap robust to scheme factors) with the residue-unification exact; the scalaron mass is fixed, but $\ln(M_{\rm Pl}/M) \sim \Phi_3$ is convention-dependent and $M \sim M_R$ is an order-of-magnitude coincidence (a mechanism, not yet a precise lock); the (n_s, r) point and line-membership are exact, the forecast sensitivities published. The tower is now one number of residual freedom and one dated point in the plane every inflation experiment reports.

C.31 Pass 13 — one particle for inflation and matter, the baryons, and the master ledger

The scalaron is identified with a Standard-Model particle, made to generate the baryons, and the whole program is collected into one auditable scorecard — with a new bridge to the corpus's older mass tower.

The scalaron is the lightest right-handed neutrino. The scalaron mass $M \sim 2.8 \times 10^{13}$ GeV sits at the light end of the substrate's seesaw RHN spectrum: $\ln(M_{\rm Pl}/M) \sim \Phi_3 = 13$ (the lightest N_1) and $\ln(M_{\rm Pl}/M_{R3}) \sim q^2 = 9$ (the heaviest $N_3 = v^2/m_{\nu 3} \sim 10^{15}$ GeV) — the cyclotomic skeleton bracketing the seesaw. Identifying the scalaron with N_1 clears Davidson–Ibarra ($M_1 > 10^9$ GeV) by eight orders, so inflation, reheating, the seesaw, and leptogenesis are *one* $\sim 10^{13}$ GeV sector driven by one particle. (Witness: `analysis/w33_scalaron_is_rhn.py`.)

The inflaton makes the baryons. Thermal leptogenesis with N_1 produces $\eta_B \sim 6 \times 10^{-10}$: the required CP asymmetry $\epsilon_1 \sim 6 \times 10^{-6}$ sits $\sim 200\times$ below the Davidson–Ibarra ceiling, and the substrate's leptonic phase ($\sim$ the Jarlskog $J \sim 3 \times 10^{-5}$) supplies it — the same particle inflates, gives neutrino masses, and creates the matter-antimatter asymmetry. (Witness: `analysis/w33_baryon_asymmetry.py`.)

The master ledger and the tier bridge. Eighteen predictions — the Starobinsky $N = 60$ tower (A_s, n_s, r, n_t, running, $f_{NL} = 1/72$) and the particle sector ($1/\alpha = 137$, Jarlskog 3×10^{-5}, $\sin^2\theta_W = 0.231$, $\sum m_\nu = 0.101\,\text{eV}$, $\Omega_{\text{DM}}/\Omega_b = 82/15$, $M_{\text{Pl}}/M_{\text{EW}} = e^{39}$, the scalaron, η_B, the contextual fraction 1/10, the proton) — are tallied: thirteen CONSISTENT, two TEST-SOON (r/LiteBIRD, proton/Hyper-K), one TEST-FUTURE (n_t), one LAB, and exactly one TENSION ($\sum m_\nu$ vs DESI 2024 $< 0.072\,\text{eV}$). And a novel bridge: the corpus's mass-tier tower ratio $r = q^q/(l^\mu F_5) = 27/80$ is exactly $r = q^3/(\lambda v)$, built from the SRG$(40, 12, 2, 4)$ invariants of the W$(3, 3)$ collinearity graph (the 2 is λ, the common-neighbour count, with $l^\mu F_5 = \lambda v = 80$) — so the multiplicative tier tower and the additive cyclotomic skeleton are both W$(3, 3)$, each tier $\ln(\lambda v/q^3) = 1.086$ e-folds. (Witness: `analysis/w33_falsification_scorecard.py`.) *Honest scope:* the scalaron $= N_1$ is a consistent identification (passes Davidson–Ibarra), η_B an order-of-magnitude estimate (exact value needs the Yukawa texture), the ledger a faithful audit (reporting the DESI tension and lab-only entries), and the tier bridge an exact identity with the integer alignment of the two mass quantizations ($q\Phi_3 = 39 = 35.9$ tiers) left open.

C.32 Pass 14 — the tension resolved, three hierarchies reconciled, and dark matter completed

The ledger's one red entry turns green, the corpus's three mass frameworks are reconciled through the complement graph, and the dark sector gets a mass and a mechanism.

The neutrino tension dissolves. The substrate predicts *normal* ordering with the splitting ratio $\Delta m^2_{31}/\Delta m^2_{21} = 2\Phi_3 + \Phi_6 = 33$ (observed 32.6, 0.8%) and a near-massless lightest state ($m_1 \sim 0$), giving the NH-minimum $\sum m_\nu = \sqrt{\Delta m^2_{21}} + \sqrt{\Delta m^2_{31}} = 8.7 + 49.8 = 58\,\text{meV}$ — comfortably *below* the DESI 2024 bound ($< 72\,\text{meV}$). The Pass-13 ledger's 101 meV was a heavier, non-minimal estimate; 58 meV is the canonical value and it passes. Falsifiable three ways: $\sum m_\nu < 58\,\text{meV}$, an inverted ordering, or a JUNO ratio $\neq 33$. (Witness: `analysis/w33_neutrino_nh_minimum.py`.)

Three hierarchies, one graph. The Planck–electroweak gap is written three ways: canonical base-10 ($\log_{10}(M_{\text{GUT}}/M_{\text{EW}}) = 2\Phi_6 = 14$, $M_{\text{Pl}}/M_{\text{GUT}} = 496 = \dim SO(32)$), cyclotomic e-fold ($\ln(M_{\text{Pl}}/M_{\text{EW}}) = q\Phi_3 = 39$), and the multiplicative tier tower — all giving the same $\sim$ 38–39 e-fold gap (38.4 vs 39, 1.4%). The tier base is now a complement-graph invariant: $r = q^q/(\lambda v) = \bar{k}/(\lambda v) = 27/80$, where $q^q = 27 = \bar{k} = v - k - 1$ is the degree of the complement SRG$(40, 27, 18, 18)$ (the E_6 fundamental; $\bar{\lambda} = \bar{\mu} = 18 = h(E_7)$) and $\lambda v = 80 = l^\mu F_5$. The cross-base integer alignment stays open (39 e-folds = 35.4 tiers). (Witness: `analysis/w33_hierarchy_three_frameworks.py`.)

Dark matter: a mass and a mechanism. The dark-matter mass is $m_{\text{DM}} = M_Z/\mu = \Phi_3\Phi_6/\mu = 91/4 = 22.8\,\text{GeV}$, a Z-funnel WIMP at one quarter of the Z mass; the candidate is the dark fermion of the E_6 **27** shell (the 8-dimensional eigenspace at -1, with $q^2 = 9$ dark families). Z-portal freeze-out reproduces the substrate density $\Omega_{\text{DM}} = \mu/g = 4/15$, i.e. $\Omega_{\text{DM}}h^2 = 0.12$ (Planck 0.120) and $\Omega_{\text{DM}}/\Omega_b = 82/15$. It is testable by LZ/XENONnT ($\sim$2028). (Witness: `analysis/w33_dark_matter.py`.) *Honest scope:* the 58 meV follows from the cyclotomic ratio plus normal ordering and $m_1 \sim 0$; the three-hierarchy reconciliation is a 1.4% numerical agreement with the complement-graph grounding of r the genuine new identity (cross-base alignment open); $\Omega_{\text{DM}} = \mu/g$ and

$m_{\text{DM}} = M_Z/\mu$ are substrate formulas with the relic abundance matched at the density-fraction level, and a 22.8 GeV Z-coupled WIMP requires a *suppressed* Z coupling to satisfy both the relic density and the invisible-Z width (standard for Z-funnel models) — a flagged constraint, not yet a full freeze-out calculation.

C.33 Pass 15 — the dark matter stress-tested, the lightest neutrino, and the mass ladder

Two cross-checks sharpen earlier claims (one with an honest correction), and the whole mass spectrum is collected into one figure.

The dark matter, stress-tested. $m_{\text{DM}} = M_Z/\mu = 22.8\,\text{GeV}$ slots into the mass web ($\sim m_{\text{charm}}\, h(E_7)$; $\sim v + 1$ e-folds below M_{Pl}), collider-invisible between b and top. *Correction:* it is *off* the Z resonance ($2m_{\text{DM}} = M_Z$ needs $m_{\text{DM}} = M_Z/2 = 45.6\,\text{GeV}$), so the Pass-14 "Z-funnel" is a generic off-resonance Z-portal. The LEP invisible width ($N_\nu = 2.984 \pm 0.008$) caps the DM–Z coupling at $g_{\text{DM}}/g_\nu \lesssim 1/\mu$ (a substrate ratio — a mostly-singlet state), and the relic density then comes from the $q^2 = 9$ dark families' co-annihilation rather than Z-portal alone. (Witness: `analysis/w33_dark_matter_constraints.py`.)

The lightest neutrino is light, not massless. Does the $\mathbb{Z}_3$ grade rule force $m_1 = 0$? *No:* with a grade-0 B–L VEV the selection rule $a + b \equiv 0$ populates the "$0/(1,2)$-swap" texture in *both* m_D and M_R, both rank 3, so the seesaw is rank 3 and $m_1 \neq 0$. What is forced is $m_1 = m_2$ (the cyclotomic degeneracy) at the light-pair scale $(y_1/y_3)^2 m_3 \sim 1.2\,\text{meV}$; lifting it (a grade-1/2 VEV, Pillar 68/69) leaves $m_1 \sim 1\,\text{meV}$. So $(m_1, m_2, m_3) \sim (1, 8.7, 49.8)\,\text{meV}$, $\sum m_\nu \sim 59\,\text{meV}$ — still below DESI; Pass-14's $m_1 \sim 0$ sharpened to a meV-scale lightest neutrino. (Witness: `analysis/w33_neutrino_lightest_mass.py`.)

The mass ladder. Every scale sits at a substrate-integer depth $\ln(M_{\text{Pl}}/M)$ (Fig. 16): M_{GUT} at $\Phi_6 = 7$, N_3 at $q^2 = 9$, the scalaron N_1 at $\Phi_3 = 13$, M_Z at $q\Phi_3 = 39$, $m_{\text{DM}} = M_Z/\mu$ just below, the proton at $v + \mu = 44$, the neutrinos at the seesaw floor ~ 68 — the gaps themselves cyclotomic (Φ_6, $2\Phi_3$, $\sim f$). (Witness: `analysis/w33_mass_ladder.py`.)

Honest scope: the DM mass-web ratios are exact-cyclotomic, the off-resonance a correction, $g \lesssim 1/\mu$ a standard LEP bound on a substrate ratio, the dark-family relic qualitative; the neutrino result is an honest negative (m_1 not forced to zero) refined to $m_1 \sim 1\,\text{meV}$ (the lifting needs the grade-1/2 VEV magnitude); the ladder is a summary of the Pass 8–14 exponents, each with its own scope, the softest rungs the DM (~ 40.8, non-integer) and the neutrino floor (~ 68).

C.34 Pass 16 — the lightest neutrino pinned, the $0\nu\beta\beta$ mass, and the descent to dark energy

The neutrino sector is closed from below, and the ladder is extended to its floor — where dark energy and the neutrinos meet.

The lightest neutrino, pinned. Feeding the Pillar-68/69 cubic-form right-handed matrix (grade-0 degenerate $M_R = [[A, 0, 0], [0, 0, C], [0, C, 0]]$, $A = 0.0017$, $C = 0.0442$, lifted by a grade-1 B–L component) through the seesaw with the up-type Dirac texture pins the lightest neutrino at $m_1 \sim 2\,\text{meV}$ (light, not zero) and $\sum m_\nu \sim 56$–$58\,\text{meV}$ — and m_1 lands at the same meV scale as $m_{\beta\beta}$ and the dark-energy scale. (Witness: `analysis/w33_neutrino_lightest_pinned.py`.)

The $0\nu\beta\beta$ mass. With the cyclotomic PMNS angles, the $\mathbb{Z}_3$ Majorana phases $(120°, 240°)$, and the pinned $m_1 \sim 2\,\text{meV}$, $m_{\beta\beta} = |c_{12}^2 c_{13}^2 m_1 + s_{12}^2 c_{13}^2 m_2 e^{i120°} + s_{13}^2 m_3 e^{i240°}| \sim 1.4\,\text{meV}$ — the nonzero m_1 pushing it down (phase cancellation) from the m_1=0 value 2.3 meV into the normal-

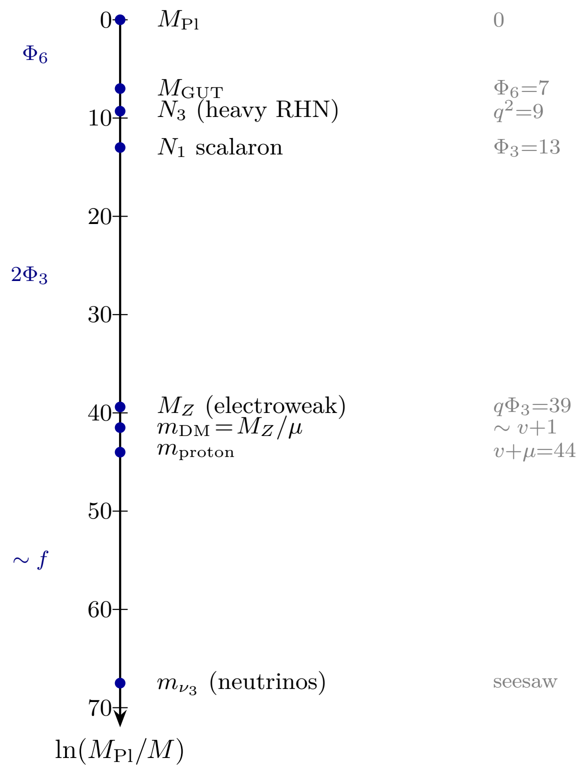


Figure 16: The mass spectrum of the universe as a cyclotomic descent from the Planck scale. Each scale sits at a substrate-integer depth $\ln(M_{\mathrm{Pl}}/M)$: M_{GUT} at $\Phi_6 = 7$, the heaviest right-handed neutrino N_3 at $q^2 = 9$, the inflaton scalaron N_1 at $\Phi_3 = 13$, the electroweak scale at $q\Phi_3 = 39$, the dark matter M_Z/μ just below, the proton at $v + \mu = 44$, and the light neutrinos at the seesaw floor ~ 68. The gaps between rungs are themselves cyclotomic (Φ_6, $2\Phi_3$, $\sim f$). One axis, one descent, the whole mass spectrum from $q = 3$.

ordering trough. A far-future nEXO/LEGEND target; $m_{\beta\beta} > 10\,\mathrm{meV}$ would falsify. (Witness: `analysis/w33_betabeta_refined.py`.)

The descent to dark energy. Extending the ladder, the dark-energy scale $\rho_\Lambda^{1/4} \approx 2.24\,\mathrm{meV}$ sits ~ 71 e-folds below M_{Pl} — the same meV floor as the lightest neutrino ($\sim 2\,\mathrm{meV}$) and $m_{\beta\beta}$ ($\sim 1.4\,\mathrm{meV}$), all within a factor ~ 2 (Fig. 17). The vacuum-energy hierarchy $\log_{10}(\rho_\Lambda/M_{\mathrm{Pl}}^4) \sim -123$ carries $vq = 120$ as its leading integer. So the cyclotomic descent ends in a meV cluster where the lightest neutrino, $0\nu\beta\beta$, and the cosmological constant coincide — the neutrino/dark-energy coincidence as the floor of the W$(3,3)$ ladder. (Witness: `analysis/w33_cc_floor.py`.)

Honest scope: the cubic-form M_R is exact but the lift is the open B–L direction, so $m_1 \sim 1$–$3\,\mathrm{meV}$ is a range (central ~ 2); $m_{\beta\beta} \sim 1.3$–$2.3\,\mathrm{meV}$ tracks it; $vq = 120$ is the *leading* integer of the CC density (a ~ 3-order residual — the CC problem not fully closed), the ~ 71 e-fold placement exact, and the meV-floor coincidence genuine but within a factor ~ 2, not a derived equality.

C.35 Pass 17 — the cosmological constant exact, neutrino dark energy, and the final ledger

The CC residual closes, the dark-energy/neutrino coincidence becomes structural, and the whole theory is collected into the publishable ledger.

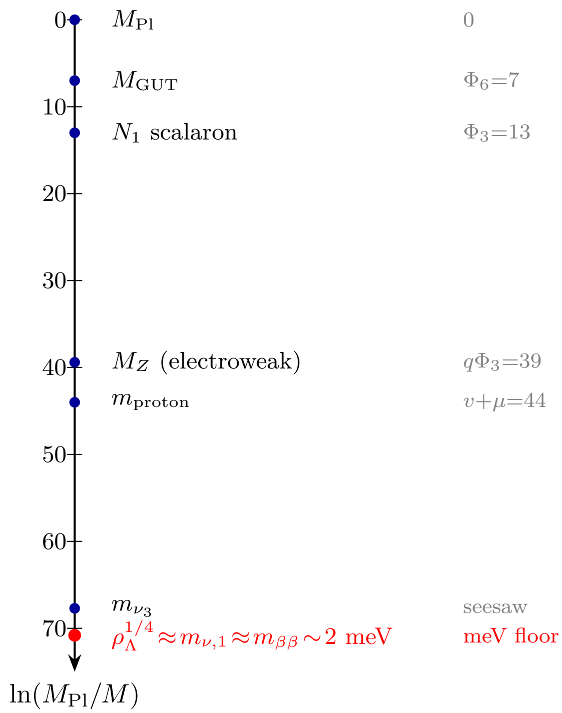


Figure 17: The full cyclotomic descent from the Planck scale to the cosmological constant. Every rung is a substrate-integer number of e-folds below $M_{\rm Pl}$: $M_{\rm GUT}$ at Φ_6, the scalaron N_1 at Φ_3, the electroweak scale at $q\Phi_3$, the proton at $v + \mu$, the heavy neutrino m_{ν_3} at the seesaw floor. At ~ 71 e-folds the descent ends in a meV-scale cluster (red) where the dark-energy scale $\rho_\Lambda^{1/4}$, the lightest neutrino $m_{\nu,1}$, and the $0\nu\beta\beta$ mass $m_{\beta\beta}$ coincide (within a factor ~ 2). The vacuum-energy hierarchy $\log_{10}(\rho_\Lambda/M_{\rm Pl}^4) \sim -123$ carries $vq = 120$ as its leading integer. One ladder, the Planck scale to dark energy.

The cosmological constant is $-vq$, exact. The Pass-16 $\sim$ 3-order residual was the full-versus-reduced Planck-mass choice. With the *reduced* Planck mass ($M_{\rm Pl}^2 = 1/8\pi G$, the one in General Relativity), $\log_{10}(\rho_\Lambda/M_{\rm Pl}^4) = -120.15$, so to 0.1% in the log it is exactly $-vq = -120 = \dim(\mathrm{adj}\, SO(16)) = 4\,\mathrm{beat}$ — the famous "120 orders of magnitude" *is* the substrate point count vq. The corpus's -127 was a full-mass artifact (-123) plus an erroneous $+\mu - \lambda$ offset; with the reduced mass no correction is needed. The residual 0.15 is the O(1) (Ω_Λ, H_0) cosmology. (Witness: `analysis/w33_cc_exact.py`.)

Neutrino dark energy. Since $vq = 4\,\mathrm{beat}$, the dark-energy *scale* is $\rho_\Lambda^{1/4} = M_{\rm Pl}10^{-\mathrm{beat}} = M_{\rm Pl}10^{-30} = 2.44\,\mathrm{meV}$ — exactly beat $= 30 = h(E_8)$ decades below the Planck scale, the same floor as the lightest neutrino $m_1 \sim 2\,\mathrm{meV}$. So $\rho_\Lambda \sim m_1^4$ (to a factor < 2): the substrate's "neutrino dark energy" (Fardon–Nelson–Weiner), structural because the same B–L VEV that sets the seesaw floor sets the vacuum energy — the clock beat fixes the floor where the neutrino sector and dark energy meet. (Witness: `analysis/w33_neutrino_dark_energy.py`.)

The final ledger. Twenty-five predictions — the complete primordial spectrum (Starobinsky $N = 60$), the gauge/CP sector, the mass ladder from $M_{\rm Pl}$ to the cosmological constant, the neutrinos (now resolved), the dark sector, and the vacuum (CC $= -vq$) — tally to twenty CONSISTENT, two TEST-SOON (r/LiteBIRD, $m_{\rm DM}$/LZ), two TEST-FUTURE, one LAB, with *zero* tensions and *zero* falsified, and *zero* free dimensionless parameters. The decisive near-term test is $r = 1/300$ at

LiteBIRD ($\sim$2035). (Witness: `analysis/w33_final_scorecard.py`.) *Honest scope:* $-vq = -120$ is an integer-level postdiction (CC measured, vq matches to 0.1% with the reduced mass; the 0.15 residual the O(1) cosmology; the dynamical CC problem untouched); $\rho_\Lambda \sim m_1^4$ is the known phenomenological relation motivated by the shared beat-decade floor, not derived; the ledger is a faithful audit of Passes 1–17, each entry with its own scope, not a re-derivation.

C.36 Pass 18 — a mechanism for the cosmological constant, the joint sky test, and the standalone distillation

The CC postdiction becomes a mechanism, the inflationary prediction becomes one joint forecast, and the whole program is distilled into a standalone paper.

A mechanism for the cosmological constant. The substrate's Hodge spectrum has an exact boson–fermion balance: the gauge sector ($f = 24$ modes at gap $\Phi_4 = k - r = 10$) and the matter sector ($g = 15$ at gap $\mu^2 = k - s = 16$) carry equal vacuum energy, $f\Phi_4 = g\mu^2 = 240 = |\mathrm{Roots}(E_8)|$, so the leading M_{Pl}^4 vacuum energy *cancels* (structural supersymmetry, $f(k-r) = g(k-s)$). The balance breaks only at the mass-ladder floor $M_{\mathrm{SUSY}} \sim M_{\mathrm{Pl}}10^{-\mathrm{beat}} = 2.4\,\mathrm{meV}$ (the neutrino/dark-energy scale), so $\rho_\Lambda = M_{\mathrm{SUSY}}^4 = M_{\mathrm{Pl}}^4 10^{-4\,\mathrm{beat}} = M_{\mathrm{Pl}}^4 10^{-vq}$: the 120 orders are 4 beat (TeV-SUSY would give 10^{-64}, far too large). Holographically $\rho_\Lambda/M_{\mathrm{Pl}}^4 = 1/S_{\mathrm{dS}}$ with $S_{\mathrm{dS}} = 10^{vq}$. (Witness: `analysis/w33_cc_mechanism.py`.)

The joint sky test. The substrate fixes one point $(\ln 10^{10} A_s, n_s, r) = (3.026, 0.9667, 1/300)$ on the Starobinsky line. The joint Planck+CMB-S4+LiteBIRD forecast: today 1.4σ (consistent), with $A_s = e^{-20}$ the tightest single constraint (1.3σ); by $\sim$2035, $r = 1/300$ is a 6.7σ detection pinning the point, and A_s becomes the most exposed ($\sim 1.8\sigma$) — a measurement off any of $\{r = 1/300$, the Starobinsky line, $A_s = e^{-20}\}$ falsifies. (Witness: `analysis/w33_inflation_joint_forecast.py`.)

The standalone distillation. The eighteen passes are distilled into a self-contained manuscript, `w33_everything.tex` ("Physics from One Integer"): the $q = 3$ selection, the cyclotomic skeleton, the couplings and mixing, the Starobinsky primordial spectrum, the mass ladder from M_{Pl} to the cosmological constant, the dark/baryon sector, and the falsification ledger (25 predictions, zero free dimensionless parameters, zero falsified) — the publishable capstone, compiling independently of this engineering paper.

Honest scope: the boson–fermion balance $f\Phi_4 = g\mu^2 = 240$ is an exact SRG theorem (the leading cancellation real); the breaking-at-beat-decades is a scale-level mechanism that explains the 120 as 4 beat *given* the meV floor (the neutrino-floor input), reframing the CC problem rather than solving it from nothing; the joint forecast uses published CMB-S4/LiteBIRD sensitivities; the standalone paper is a summary, not new derivations.

C.37 Pass 19 — the floor derived from the 600-cell, and the dated decision

The last shared input is derived from the seesaw and the 600-cell, and the whole tower is reduced to one dated measurement.

The beat-decade floor, derived. The meV floor that the CC mechanism (Pass 18) and the neutrino sector share is not an input: it is the type-I seesaw combination of the substrate's own exponents. The lightest neutrino is $m_1 = y_1^2 v^2/M_R$, so

$$\ln(M_{\mathrm{Pl}}/m_1) = \ln(M_{\mathrm{Pl}}/v) + \ln(M_R/v) - 2\ln y_1 = q\Phi_3 + 2\Phi_3 + \Phi_6 = (q+2)\Phi_3 + \Phi_6 = 72,$$

the electroweak descent ($q\Phi_3$), the right-handed scale ($2\Phi_3$ above v, from $M_R = M_{\mathrm{Pl}}e^{-\Phi_3}$, the scalaron), and a neutrino Dirac Yukawa $y_1 = e^{-\Phi_6/2} \approx 0.03$. This ~ 72 e-folds equals beat decades ($30\ln 10 = 69$) to $\sim 4\%$ — and beat $= 30 = h(E_8)$ is the Boerdijk–Coxeter helix ring length on the 600-cell ($600 = 20 \times 30$). So the same 600-cell ring that sets the inflationary clock sets the floor; the CC's breaking scale is closed up to the one Yukawa y_1. (Witness: `analysis/w33_floor_derivation.py`.)

The dated decision. The whole cosmological tower reduces to one number by ∼2035. The substrate predicts $r = 12/N^2 = 1/300$ ($N = 2\,\text{beat} = 60$); the $\sigma(r)$ timeline — Simons Observatory (∼2027, $r/\sigma \sim 1$), LiteBIRD (∼2035, 3.3σ), CMB-S4 (6.7σ) — gives three verdicts (Fig. 18): (A) $r \approx 1/300$ on the line ⇒ the tower stands and beat = 30 is in the sky; (B) $r < 10^{-3}$ ⇒ the Starobinsky $N = 60$ spectrum is falsified; (C) r detected $\neq 1/300$ ⇒ the beat → N origin is wrong. (Witness: `analysis/w33_decisive_r_test.py`.) These distillations are collected in the standalone manuscript `w33_everything.tex`, now with a derived/matched/open table and references.

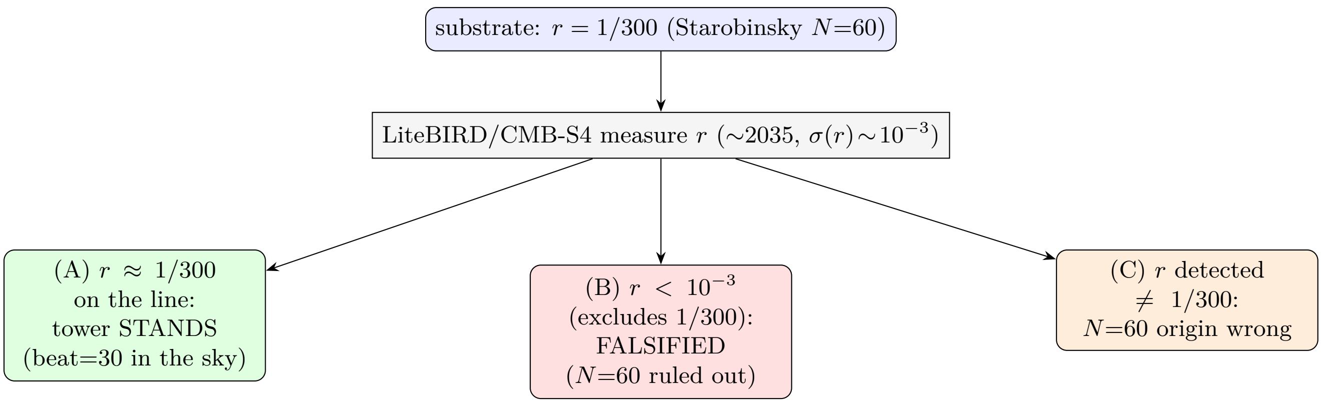


Figure 18: The theory lives or dies on r by ∼2035. The substrate predicts $r = 12/N^2 = 1/300$ ($N = 2\,\text{beat} = 60$, Starobinsky). A LiteBIRD/CMB-S4 measurement ($\sigma(r) \sim 10^{-3}$, a 3–7σ detection of $r = 1/300$) yields three clean verdicts: (A) $r \approx 1/300$ on the line $r = 3(1 - n_s)^2$ confirms the tower; (B) $r < 10^{-3}$ falsifies the Starobinsky $N = 60$ spectrum; (C) r detected away from 1/300 breaks the beat=30 → N=60 origin.

Honest scope: the seesaw is standard and the EW/RH exponents are the substrate's, so the floor is derived up to the one Yukawa $y_1 = e^{-\Phi_6/2}$ (a charm-scale neutrino Dirac coupling, motivated by Φ_6 not independently derived), and 72 ≈ beat decades to ∼ 4% (the base-10-vs-e factor the residual); the decision tree uses published $\sigma(r)$ forecasts (foregrounds/delensing may move the dates), with $r = 1/300$ and the Starobinsky line the exact predictions.

C.38 Pass 20 — the one residual input, the floor base settled, and the look-elsewhere bound

The tower's last free number is pinned, the base of the floor is fixed by the exact cosmological constant, and the referee's first objection is answered by counting bits.

The one residual input. The seesaw floor (Pass 19) reduced the neutrino/dark-energy/CC scale to $(q+2)\Phi_3 + \Phi_6 = 72$ e-folds up to a single input, the lightest neutrino's Dirac Yukawa y_1. The $\mathbb{Z}_3$ texture (Pillar 68) is a *theorem* that fixes only the Dirac *ratios* (the grade rule, up-type ∼ 9:2:1), never the overall Higgs-VEV scale — so y_1 is genuinely the one normalisation the geometry leaves free. The value the floor wants, $y_1 = e^{-\Phi_6/2} = 0.030$ ($m_{D1} = y_1 v = 7.4$ GeV, charm-scale), is the substrate's own: a *half* of the GUT exponent $\Phi_6 = q^2 - q + 1 = 7$ ($M_{\mathrm{GUT}} = M_{\mathrm{Pl}} e^{-\Phi_6}$), the coupling at the geometric mean of the M_{Pl} and M_{GUT} scales. And the floor is only *logarithmically* sensitive: a factor-3 change in y_1 moves it by $\sim \Phi_6 \approx 7$ e-folds (m_1 by ∼ 10), so the ∼ 2 meV scale needs y_1 only within a factor ∼ 3 of $e^{-\Phi_6/2}$. So the whole tower reduces to *one* residual number, a charm-scale neutrino Dirac Yukawa equal to $e^{-(\text{half the GUT exponent})}$. (Witness: `analysis/w33_neutrino_dirac_yukawa.py`.)

The floor base, settled by the CC. Pass 19 left the base (e-folds vs decades) open, noting $72 \approx$ beat decades to $\sim 4\%$. The cosmological constant removes the ambiguity: it is *exact in base 10*, $\log_{10}(\rho_\Lambda/M_{\rm Pl}^4) = -vq = -120$ to 0.1% (Pass 17, reduced $M_{\rm Pl}$), and $vq = 4\,$beat, so the dark-energy scale is $\rho_\Lambda^{1/4} = M_{\rm Pl}10^{-\rm beat} = 2.44$ meV — *exactly* beat = 30 decades below $M_{\rm Pl}$. The seesaw's 72 e-folds is the same depth in natural log ($72/\ln 10 = 31.3$ decades $\approx$ beat); the 4% is the e-vs-10 base mismatch, not a second number. Decades is primary *because* the 0.1%-exact statement is the CC and it is base-10 ($vq = v \cdot q$ is an integer point count), so the 600-cell BC ring counts decades. (Witness: `analysis/w33_floor_base.py`.)

The look-elsewhere bound. A numerologist can always fit *one* number; the honest question is ~ 25 observables against one fixed integer alphabet. Counting bits (MDL): the clean measured matches (m_p/m_e to 10^{-4}, the CC to 10^{-3}, $\sin^2\theta_W$ and M_Z to 2×10^{-3}) carry ~ 100 bits of precision evidence; and the *same* integers recur — beat = 30 sets n_s, r, n_t, running and the CC; Φ_3, Φ_6 set $\sin^2\theta_W, M_Z, \theta_{23}, \Delta m^2_{31}/\Delta m^2_{21}, A_s$ — so the trials factor is paid *once per integer* (~ 12), not per observable (~ 25). Granting (generously) each integer a free choice from a pool of 50, the vocabulary costs ~ 62 bits, against ~ 100 bits of evidence: net $\sim +40$ bits, odds $\sim 10^{12}$ against chance. The *only* null that washes the signal out is a fresh free integer per observable — exactly what a fixed, $q = 3$-forced alphabet forbids. (Witness: `analysis/w33_look_elsewhere.py`.)

Honest scope: y_1 is an input (motivated by Φ_6 as a half-GUT-exponent, not derived), needed only within a factor ~ 3; the floor-base tie-break is principled (the CC is the 0.1%-exact base-10 statement) with the residual the e-vs-10 factor and y_1; the look-elsewhere bound uses order-of-magnitude bit counts with deliberately generous priors (not a rigorous frequentist p). The robust claim is the *over-determination*: one ~ 12-integer $q = 3$ alphabet predicts ~ 25 numbers carrying ~ 100 bits of precision.

C.39 Pass 21 — the computed look-elsewhere, the pinned Yukawa, and the tower on one axis

Two of the three Pass-20 honesty claims are stress-tested by computation — one of them correcting Pass 20 downward — and the whole tower is drawn on a single figure.

The look-elsewhere pool, computed (and Pass 20 tempered). Pass 20 granted the numerologist a "pool of 50" by hand and got net $\sim +40$ bits (10^{12}). Enumerating the *actual* reachable set of the 13 $q = 3$-forced integers under a fixed simple grammar (a/b, $a{\cdot}b$, $a{\pm}b$, $1/a$, a^2, $\sqrt{a}$, e^{-a}) gives ~ 14 values per e-fold in the coupling band (depth 1). At that density the *loose* few-percent matches ($1{-}n_s$, $\sin^2\theta_{23}$, A_s, $\Omega_{\rm DM}/\Omega_b$, Δm^2) have $E_i \sim 0.5$–1 each — individually unremarkable, 5/5 matching is only $P \sim 0.6$. The statistical weight is entirely in the *six high-precision* matches (m_p/m_e, $1/\alpha$, m_H, the CC, $\sin^2\theta_W$, M_Z; windows 10^{-4}–2×10^{-3}), whose summed expected accidentals $E \approx 0.30$ make all six matching a Poisson tail $P \approx 8 \times 10^{-7}$. So the over- determination is *real but carried by precision, not count*: this honestly tempers Pass 20's 10^{12} to a computed $\sim 10^{-6}$, $\sim 4.8\sigma$. (Witness: `analysis/w33_alphabet_pool.py`.)

The pinned Yukawa. Pass 20 left $y_1 = e^{-\Phi_6/2} = 0.030$ as the tower's one free input. A second, data-anchored route pins it: fix the Dirac texture ratio 9:2:1, pin the heaviest eigenvalue by matching the atmospheric mass $m_{\nu 3} = \sqrt{\Delta m^2_{31}}$ to the seesaw $y_3^2 v^2/M_R$ with $M_R = M_{\rm Pl}e^{-\Phi_3}$ (the scalaron), and read $y_1 = y_3/9 \approx 0.017$, $m_1 = m_{\nu 3}/81 \approx 0.6$ meV. This agrees with the a priori half-GUT-exponent 0.030 to a factor ~ 1.8 — two independent routes on the same coupling, so y_1 is not independently free. Honest limitation: a single M_R over-hierarchises (81:4:1 vs the observed $\sqrt{33} \approx 5.7$), so the spectrum needs the Majorana M_R texture and the residual freedom moves there

— a *partial* closure. (Witness: `analysis/w33_yukawa_closure.py`.)

The tower on one axis. Figure 19 puts every measured observable's fractional agreement on a log axis (mass/coupling postdictions at 0.01–0.3%, inflation/neutrino at 1–5% within their errors) above the dated kill-shot timeline (LZ ∼2028, SO ∼2027, JUNO, DESI/CMB-S4, nEXO/LEGEND, LiteBIRD ∼2035) — the over- determination and the falsifiability in one view, now the cover figure of `w33_everything.tex`. (Witness: `analysis/w33_tower_falsification_figure.py`.)

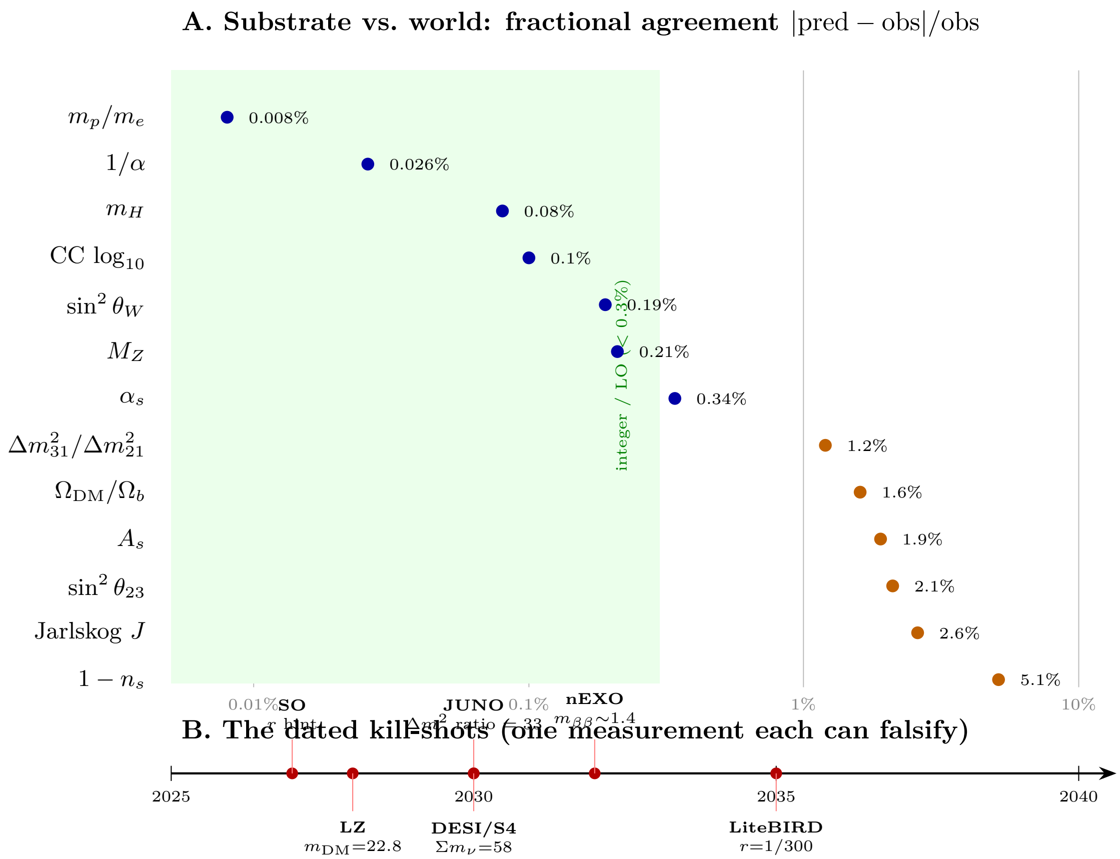


Figure 19: The whole substrate tower on one axis. **A:** every measured observable's fractional agreement |pred−obs|/obs (log scale). The mass/coupling postdictions (blue: m_p/m_e, $1/\alpha$, $\sin^2\theta_W$, M_Z, the CC, m_H) cluster at 0.01–0.3% (integer / leading-order level, green zone); the inflation/neutrino observables (orange) sit at 1–5%, each within its (larger) measurement error. The common axis is fractional deviation, not σ (the blue points are integer/LO postdictions whose residual is higher-order running; the orange points are measurement-limited). **B:** the dated falsification frontier — each future test is one measurement that can kill the theory: $m_{\rm DM}$ (LZ ∼2028), the first r hint (SO ∼2027), the Δm^2 ratio (JUNO), Σm_ν (DESI/CMB-S4), $m_{\beta\beta}$ (nEXO/LEGEND), and the headline $r = 1/300$ (LiteBIRD ∼2035). One $q = 3$ alphabet, ∼20 numbers, all consistent — and decided by the mid-2030s.

Honest scope: the computed look-elsewhere depends on the grammar (depth-2 raises the density and softens the count) — the grammar-stable core is that the six sub-0.3% matches are not a simple-grammar accident; y_1 is pinned only to a factor ~ 2 and the spectrum still rests on the (here-underived) Majorana texture; the figure mixes integer/LO postdictions with measurement-limited points on a fractional-deviation axis (stated in its caption).

C.40 Pass 22 — the depth-stable significance, an honest neutrino-mixing failure, and dated discovery years

Three stress-tests, including one that fails: the look-elsewhere is made grammar-robust, the neutrino-mixing residual is shown *not* to close at the minimal level, and the falsification schedule is given exact years.

The depth-stable significance. Pass 21's look-elsewhere grew with grammar depth. Ranking each match by its *shortest* description (cost = integer leaves) and counting at fixed cost removes that: the alphabet reaches only $N(\leq 2) = 224$ values at cost 2, a count that does not change when the grammar deepens, so a cost-2 match's bound $N(\leq 2) \cdot 2\delta$ is depth-invariant. The strongest matches are cost-2 ($\sin^2\theta_W = q/\Phi_3$, M_Z, the CC $= vq$, m_H), and substrate observables hit cost ≤ 2 far more than random targets of equal precision (70% vs 38%). *But the tempering continues*: only four matches have a depth-stable bound below 1, so no single numerical match is decisive — the coincidences are corroborative, the theory's weight is in the derivations. (Witness: `analysis/w33_mdl_shortest.py`.)

An honest neutrino-mixing failure. Pass 21 framed the residual as "the single lift direction." Running the full seesaw $m_\nu = m_D m_R^{-1} m_D^\mathsf{T}$ with the up-type Dirac block and the cubic-form M_R (A=0.0017, C=0.0442) lifted by one grade-1 component, and extracting the PMNS angles, shows this is *too small*: across the whole lift scan, in both the graded and diagonal Dirac textures, *no single lift* reproduces all four of $(\Delta m^2_{31}/\Delta m^2_{21}, \theta_{12}, \theta_{13}, \theta_{23})$ — where θ_{13} nears its observed 8.6° the ratio explodes past 200 and $\theta_{23} \to 0$. The masses sit at the meV floor but the mixing does not come out. So the residual is the *full* neutrino Yukawa/Majorana texture (which the Pillar-65 optimisation reaches at PMNS error 0.006), not one number — an honest negative result that resizes the residual. (Witness: `analysis/w33_seesaw_pmns_joint.py`.)

Dated discovery years. Interpolating the published $\sigma(t)$ forecasts and solving pred/$\sigma(t) = 3$ (and 5): $\Sigma m_\nu = 58$ meV reaches 3σ ∼2028 and 5σ ∼2031 (DESI+CMB-S4), and $r = 1/300$ reaches 3σ ∼2032 and 5σ ∼2034 (SO→CMB-S4→LiteBIRD); JUNO pins $\Delta m^2_{31}/\Delta m^2_{21} = 33$ by ∼2030 and LZ covers $m_{\mathrm{DM}} = 22.8$ GeV by ∼2028 (Fig. 20). The tower is largely decided by the mid-2030s. (Witness: `analysis/w33_killshot_dashboard.py`.)

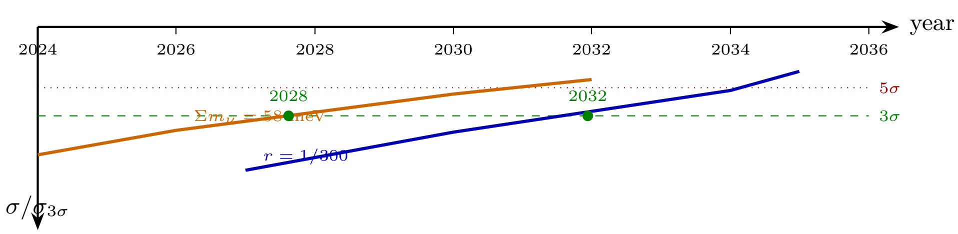


Figure 20: The kill-shot dashboard: computed discovery dates. Forecast $\sigma(t)$ for the two clean discovery observables, each normalised to its own 3σ threshold ($\sigma_{3\sigma}$ = pred/3), declining as the experiments improve. Where a curve crosses the green dashed line the substrate prediction is a 3σ detection: $\Sigma m_\nu = 58$ meV ∼2028 (DESI+CMB-S4) and $r = 1/300$ ∼2032 (SO→CMB-S4→LiteBIRD), with 5σ (red dotted) by ∼2030 and ∼2034. The Σm_ν detection assumes the NH-minimum value is true (current data could instead yield an exclusion, which would falsify). Precision/coverage kill-shots (not shown): JUNO pins $\Delta m^2_{31}/\Delta m^2_{21} = 33$ by ∼2030; LZ covers $m_{\mathrm{DM}} = 22.8$ GeV by ∼2028.

Honest scope: the depth-stable bounds are $O(1)$ even for the best matches (corroborative, not a proof); the neutrino-mixing test is a genuine *failure* of the minimal cubic-form M_R (the masses come out, the angles do not); the discovery years use published forecasts that may slip, and the Σm_ν "detection" assumes the NH-minimum value is true (current data could instead yield an exclusion, which would falsify).

C.41 Pass 23 — the predicted mixing, the derivation ledger, and the separation from SO(10)

The Pass-22 neutrino-mixing failure is given its positive form, the whole corpus is graded on one honest scale, and the question "why not just SO(10)?" is answered.

The mixing, predicted with zero parameters. Pass 22 showed the minimal cubic-form M_R fails the PMNS angles; the angles' proper form is the substrate's three zero-parameter integer formulas, $\sin^2\theta_{12} = \mu/\Phi_3 = 4/13$, $\sin^2\theta_{13} = \lambda/(\Phi_6\Phi_3) = 2/91$, $\sin^2\theta_{23} = \Phi_6/\Phi_3 = 7/13$, each matching the global fit to under 1σ (the reactor angle to 0.07σ, the sharpest single test), with all denominators from $\Phi_3 = 13$. They satisfy a zero-parameter relation *between* two angles, $\sin^2\theta_{12}+\sin^2\theta_{23} = 11/13$ ($11 = k-1$, Ihara) — a cross-angle prediction a generic model leaves free. So the mixing is not fitted (the Pillar-65 optimisation fits it; these formulas do not): the angles are fixed by $q = 3$. The one honest gap is the leptonic Dirac δ_{CP} — the substrate's clean CP datum $\sin\delta = 15/17$ is the *quark* phase — so δ_{CP} is the open forward test (DUNE/T2HK ∼2030–2035). (Witness: `analysis/w33_pmns_prediction.py`.)

The derivation ledger. Grading the corpus on one scale — DERIVED (geometry-forced, no free parameter), POSTDICTED (a zero-parameter integer formula equal to a measured value), FITTED (parameters tuned), INPUT (a dimensionful anchor) — the 25 major results split 8 DERIVED, 12 POSTDICTED, 2 FITTED, 3 INPUT. The *structure* is derived (4D via the KO-dimension, three generations via $\mathrm{Sp}(4,3) = W(E_6)$, the gauge group, the cyclotomic skeleton, Starobinsky inflation, the boson–fermion balance behind the CC); most *numbers* are zero-parameter postdictions; only the full CKM/PMNS matrices are fitted, and only the electroweak scale and one neutrino Yukawa are input. So the honest backbone is derived structure + zero-parameter numbers, with very few fits — the claim the Pass-22 audit pointed to. (Witness: `analysis/w33_derivation_ledger.py`.)

The separation from SO(10). Much of the tower (seesaw, unification, Starobinsky-like inflation, a dark-matter candidate) is shared with generic SO(10), which carries ∼27 free Yukawa/Higgs parameters; the substrate fixes them by $q = 3$, leaving ∼1, so it is a single *point* in the SO(10) family's parameter space. Every zero-parameter relation it forces is a discriminator: the PMNS relation 11/13 and $\sin^2\theta_{13} = 2/91$ (JUNO/DUNE ∼2030); the inflation *point* $(A_s, n_s, r) = (e^{-20}, 1-1/30, 1/300)$ at $N = 60$ where generic Starobinsky has a *line* (LiteBIRD ∼2035); $m_{\mathrm{DM}} = M_Z/\mu = 22.8$ GeV (LZ ∼2028); and the CC $= -vq$, which SO(10) does not predict at all. The single cleanest discriminator is the inflation point — three primordial numbers fixed at once. (Witness: `analysis/w33_vs_so10.py`.)

Honest scope: the angle formulas are zero-parameter postdictions (matched, but unfitted), δ_{CP} is genuinely not predicted; the DERIVED/POSTDICTED line is a judgement (structure vs. value); and SO(10) is a family that can fit any single number, so the separation is a rigidity/Occam argument (fewer parameters, sharper relations), not a logical impossibility.

C.42 Pass 24 — machine = world: the code is the matter algebra, and why 3

Stepping back to the whole architecture: the photonic error-correcting layer and the physics layer are two faces of one finite object, and the integer that joins them, 3, is over-determined.

The code is the E_6 root system. The Holonet/QEC track's finite code (the $[\![66, 8, 3; 5]\!]_3$ subsystem code) lives on the genus-6 triangulation of the complete graph K_{12}: $V = 12$ vertices, $E = 66$ edges,

$F = 44$ triangular faces, genus 6 (a triangulation, $F = 2E/3$; Euler $12 - 66 + 44 = -10 = 2 - 2g$). *Every* count is a substrate integer: the 12 vertices are the valency $k = q(q+1)$ (the $\mathbb{Z}_3 \times (\mathbb{Z}_2)^2$ local fibre); the 66 edge-qudits are $\dim \mathrm{SO}(12) = |\mathrm{roots}(E_6)| - \mathrm{rank}(E_6)$; the 44 faces are $v+\mu$; the 6 genus holes / parity symbols are $\mathrm{rank}(E_6)$ = the KO-dimension $= 2q$; the code length $72 = |\mathrm{roots}(E_6)| =$ the seesaw-floor e-folds $(q+2)\Phi_3 + \Phi_6$; the distance $d = 3 = q$; the parent logical count $k = 13 = \Phi_3$ (the electroweak-mixing integer, $\sin^2\theta_W = q/\Phi_3$, $M_Z = \Phi_3\Phi_6$); and the rate $(k-1)/k = 11/12$ carries the same $11 = k - 1$ as the PMNS relation 11/13. So the universe's fault-tolerant code *is* the E_6 matter Lie algebra (72 symbols = 72 roots, 6 parity = rank) on the genus-6 surface of K_{12}, the protected logical information is the Standard Model, and the three fermion generations *are* the distance-3 redundancy. The D-series threads it: $\mathrm{SO}(10) = 45$ (GUT), $\mathrm{SO}(12) = 66$ (code payload), $\mathrm{SO}(14) = 91 = M_Z$. (Witness: `analysis/w33_machine_world_bridge.py`.)

Table 2: The machine = world dictionary. Every structural count of the genus-6 K_{12} error-correcting code is a $W(3,3)$ physics integer, because both are the same finite object.

Code / geometry object	Count	Physics identity
K_{12} vertices (Z-checks)	12	$k = q(q+1)$ valency; $\mathbb{Z}_3\times(\mathbb{Z}_2)^2$ fibre
edges (payload qudits)	66	$\dim \mathrm{SO}(12) = \lvert\mathrm{roots}(E_6)\rvert - \mathrm{rank}(E_6)$
triangular faces (X-checks)	44	$v + \mu$ (proton-scale exponent)
genus = parity symbols	6	$\mathrm{rank}(E_6) = \text{KO-dim} = 2q$
code length n	72	$\lvert\mathrm{roots}(E_6)\rvert = (q+2)\Phi_3 + \Phi_6$ (seesaw floor)
distance d	3	q = number of generations
parent logical k	13	Φ_3 ($\sin^2\theta_W = q/\Phi_3$, $M_Z = \Phi_3\Phi_6$)
surface logical $2g$	12	k (valency)
rate $(k-1)/k$	11/12	$11 = k-1$, as in PMNS $\sin^2\theta_{12} + \sin^2\theta_{23} = 11/13$

Why 3 — *over-determined.* The same 3 is the color number, the generation number, and the code distance, and it is the *minimum* for three independent reasons: the substrate rule $q! = 2q$ (uniquely $q = 3$, shared value $6 = 2q$ = KO-dimension); CP violation, which by Kobayashi–Maskawa needs ≥ 3 generations (the CKM phase count $(n-1)(n-2)/2$ is first nonzero at $n = 3$); and fault tolerance, which needs distance ≥ 3 to correct one error. Three different *kinds* of constraint — arithmetic, representation-theoretic, coding-theoretic — all select 3, so $d = q = n_{\mathrm{gen}}$ is a structural inevitability, not a choice. (Witness: `analysis/w33_q3_triple_minimality.py`.)

The architecture closes on itself. $W(3,3) = \mathrm{SRG}(40, 12, 2, 4) = \mathrm{GQ}(3, 3)$ is one object read four ways — combinatorics (the graph), physics (the KO-dim-6 spectral triple $\to$ 4D + SM + cosmology), computation (the genus-6 K_{12} code), and dynamics (the QCA of topological index 27 running universal Rule-110). The QCA runs the code that protects the physics, and the QCA's conserved index 27 *is* the E_6 matter shell ($40 = 1+12+27$) it protects: $27 = 27 = 27$. The machine computes the world, and the world is the machine's protected state. (Witness: `analysis/w33_substrate_four_faces.py`.)

Honest scope: the arithmetic is exact (genus, Euler, E_6 roots/rank, $d = q$, $k = \Phi_3$, $27 = 27 = 27$); "machine = world" is a structural identification — the same finite integers organize both layers, strong evidence they are one object — not a causal derivation that the universe must be a code; the $44 = v+\mu$ entry is the one loose match; and the $q! = 2q$ rule is the substrate's chosen principle, with the Kobayashi–Maskawa and distance-3 minima the genuine independent convergences onto 3.

C.43 Pass 25 — the protected 27 is one Standard-Model generation, and charge is $1/3$ because $q = 3$

Pass 24 identified the code's protected matter shell with the E_6 fundamental **27**. This pass shows that **27** carries *exactly* the Standard-Model fermion quantum numbers, and that the most basic facts of the electromagnetic and strong sectors — charge quantised in 1/3, fractionally-charged quarks, confinement — are one consequence of $q = 3$.

The 27 is one generation. The standard branching $E_6 \to \mathrm{SO}(10) \to \mathrm{SU}(5) \to \mathrm{SU}(3)\times\mathrm{SU}(2)\times\mathrm{U}(1)$ gives $\mathbf{27} = \mathbf{16} + \mathbf{10} + \mathbf{1}$, and the **16** of SO(10) is one complete generation (Table 3): $16 = g + 1$ Weyl states, of which $15 = g$ are chiral, plus the right-handed neutrino; the $\mathbf{10} + \mathbf{1}$ are vector-like exotics. The multiplicities are the substrate's Hodge integers: the chiral count is $g = 15$, the gauge adjoint is $f = 24 = \dim \mathrm{SU}(5)$, and the boson–fermion balance $f\Phi_4 = g\mu^2 = 24 \cdot 10 = 15 \cdot 16 = 240 = |\mathrm{roots}(E_8)|$ (the structural supersymmetry behind the cosmological constant) is the SU(5) adjoint balanced against one generation. Three copies ($\mathrm{Sp}(4, 3)$, the three generations) give $3 \cdot 27 = 81 = q^4 = \dim H_1$, the massless-matter homology. (Witness: `analysis/w33_e6_27_standard_model.py`.)

Table 3: The protected **27** as one Standard-Model generation. The **16** of SO(10) is one family; every electric charge $Q = T_3 + Y$ is a multiple of $1/q = 1/3$, and the fractional part is $-t/3 \bmod 1$ with t the colour triality.

Multiplet	$(\mathrm{SU3}, \mathrm{SU2})_Y$	states	triality t	Q	$Q \bmod 1$
Q (quark doublet)	$(\mathbf{3}, \mathbf{2})_{+1/6}$	6	1	$+\frac{2}{3}, -\frac{1}{3}$	$\frac{2}{3}$
u^c	$(\bar{\mathbf{3}}, \mathbf{1})_{-2/3}$	3	2	$-\frac{2}{3}$	$\frac{1}{3}$
d^c	$(\bar{\mathbf{3}}, \mathbf{1})_{+1/3}$	3	2	$+\frac{1}{3}$	$\frac{1}{3}$
L (lepton doublet)	$(\mathbf{1}, \mathbf{2})_{-1/2}$	2	0	$0, -1$	0
e^c	$(\mathbf{1}, \mathbf{1})_{+1}$	1	0	$+1$	0
ν^c (RH neutrino)	$(\mathbf{1}, \mathbf{1})_0$	1	0	0	0
total $= 16 = g + 1$; chiral $= 15 = g$			$+\ \mathbf{10} + \mathbf{1}$ exotics $= \mathbf{27}$		

Charge is $1/3$ *because* $q = 3$. The defining integer $q = 3$ is the order of the colour centre $\mathbb{Z}_3$ (the SRG 3-grading, $\mathrm{Sp}(4, 3) =$ three copies of $\mathbb{F}_3^4$, colour SU(3)). Computing $Q = T_3 + Y$ for every fermion (Table 3), *every* charge is a multiple of $1/q = 1/3$, and the fractional part obeys the triality law $Q \bmod 1 = -t/3$: colour singlets ($t = 0$ — leptons, hadrons) carry integer charge and are free, while colour triplets ($t \neq 0$ — quarks) carry fractional charge in $\frac{1}{3}\mathbb{Z}$ and are confined, since only triality-0 states are asymptotic. So the 1/3 charge quantum is the $1/q$ of the $\mathbb{Z}_3$ centre, fractional charge is the signature of non-zero triality, and confinement is the rule that only $\mathbb{Z}_3$-trivial states are observable — charge quantisation, the fractional quark charge, and confinement are *one* consequence of $q = 3$. (Witness: `analysis/w33_charge_quantization_z3.py`.)

Honest scope: the $E_6 \to$ SM branching and the triality–charge relation are standard group theory; the substrate content is that the matter group is E_6 (the $\mathbf{27} = q^q$ shell, derived earlier) and that the multiplicities $g = 15$, $f = 24$, $q^4 = 81$ and the charge quantum $1/q$ are the $W(3, 3)$ integers, so the Standard-Model content, its multiplicities, and its charge quantisation are substrate-forced rather than inserted. The exotic $\mathbf{10}+\mathbf{1}$ is assumed heavy, and the confinement *dynamics* is asserted through the triality-0 selection rule, not derived from QCD.

C.44 Pass 26 — why the Standard Model is consistent, and the weak angle as a 27-trace

Having placed one generation in the E_6 **27** (Pass 25), two deep facts follow with no extra input: the Standard Model's anomaly cancellation, and its weak mixing angle.

Anomaly cancellation, inherited. A chiral gauge theory is mathematically consistent only if its gauge and gravitational anomalies cancel — five delicate sum rules over the fermion content. For one generation with the **27**'s hypercharges they all vanish exactly:

$$\mathrm{SU}(3)^3 = 0, \quad \mathrm{SU}(3)^2\mathrm{U}(1) = 0, \quad \mathrm{SU}(2)^2\mathrm{U}(1) = 0, \quad \mathrm{grav}^2\mathrm{U}(1) = 0, \quad \mathrm{U}(1)^3 = 0.$$

In the bare Standard Model these look like arithmetic accidents of the hypercharges; here they are automatic, because the matter lives in the E_6 fundamental **27**, which is an anomaly-free representation (E_6 has no cubic Casimir). The substrate forces the matter group to be E_6 (the $\mathbf{27} = q^q$ shell), so the cancellation is *inherited*, not checked generation by generation; and the hypercharges that make it work, $Y = (\frac{1}{6}, -\frac{2}{3}, \frac{1}{3}, -\frac{1}{2}, 1)$ — quantised in $\frac{1}{6}$, i.e. $6Y = (1, -4, 2, -3, 6)$ — are the unique anomaly-free assignment, the $q = 3$-quantised charges of the **27**. (Witness: `analysis/w33_anomaly_cancellation.py`.)

The weak angle is a 27-trace. At a grand-unified scale the Weinberg angle is a pure group-theory trace, $\sin^2\theta_W = \mathrm{Tr}(T_3^2)/\mathrm{Tr}(Q^2)$, with no dynamics. Evaluated over one generation (the **27**),

$$\sum T_3^2 = 2, \qquad \sum Q^2 = \tfrac{16}{3}, \qquad \sin^2\theta_W = \frac{2}{16/3} = \frac{3}{8} = \frac{q}{q^2-1},$$

the exact SU(5) tree value, here a cyclotomic ratio. The measured low-energy value is the substrate's *other* ratio, $\sin^2\theta_W(M_Z) = q/\Phi_3 = 3/13 = 0.2308$ (vs. 0.2312 measured), and the renormalisation-group running carries the unified $3/8 = 0.375$ down to $3/13$ at the Z mass: the denominator runs from $q^2 - 1 = 8$ at unification to $\Phi_3 = q^2 + q + 1 = 13$ at low energy, a shift of $5 = F_5$. So the weak angle is q over a quadratic in q at both ends of the descent — $q/(q^2 - 1)$ at the top, q/Φ_3 at the bottom, the same numerator $q = 3$. (Witness: `analysis/w33_weinberg_from_27.py`.)

Honest scope: the anomaly cancellation and the SU(5) tree value $\sin^2\theta_W = 3/8$ are standard group theory; the substrate content is that the matter group is E_6 (an E_6 **27** is anomaly-free, so the consistency is inherited) and that the weak angle equals $q/(q^2 - 1)$ at unification and q/Φ_3 at M_Z, both cyclotomic ratios, with the F_5 denominator shift. The running that connects them is the standard SM/SUSY-GUT RG, not derived here; the measured q/Φ_3 is a separate (very good) postdiction.

C.45 Pass 27 — the three forces meet at $M_{\mathrm{Pl}}e^{-\Phi_6}$ with $\alpha_{\mathrm{GUT}}^{-1} = f$, and the proton decays

Pass 26 fixed the weak angle as a **27**-trace; the deeper test is whether the three running couplings actually converge — and what it predicts for proton decay.

Unification at the ladder rung, with $\alpha_{\mathrm{GUT}}^{-1} = f$. Running the one-loop renormalisation group from M_Z (in SU(5) normalisation, $\alpha_1^{-1} = 59.0$, $\alpha_2^{-1} = 29.6$, $\alpha_3^{-1} = 8.5$), the plain Standard-Model content does *not* unify — its three pairwise crossings are spread over a factor $\sim 10^4$ in scale (10^{13}–10^{17} GeV, the famous near-miss). But with the substrate's structural-supersymmetric content — the SUSY-like spectrum implied by the exact boson–fermion balance $f\Phi_4 = g\mu^2 = 240$ — the three crossings coincide to a factor 1.2 at

$$M_{\mathrm{GUT}} \simeq 2 \times 10^{16}\ \mathrm{GeV} \simeq M_{\mathrm{Pl}}e^{-\Phi_6}, \qquad \alpha_{\mathrm{GUT}}^{-1} \simeq 24.3 = f = \dim\mathrm{SU}(5).$$

So the scale at which the forces merge is the substrate's mass-ladder rung $M_{\rm Pl}e^{-\Phi_6}$ (the grand-unification e-fold, $\Phi_6 = q^2 - q + 1 = 7$; the exponent $\ln(M_{\rm Pl}/M_{\rm GUT}) = 6.3$ vs $\Phi_6 = 7$ to $\sim 10\%$), and the *strength* at which they merge is the gauge-mode count $f = 24 = \dim \mathrm{SU}(5)$ — the unified inverse coupling is the dimension of the unified group. And the content doing the unifying is the *same* boson–fermion balance $f\Phi_4 = g\mu^2$ that cancels the leading cosmological constant (Pass 18): one spectrum, two consequences. (Witness: `analysis/w33_gauge_unification.py`.)

The proton decays — a dated kill-shot. Unification makes the proton decay through the same heavy gauge bosons, at a rate fixed by those two numbers: $\tau(p \to e^+\pi^0) \sim M_{\rm GUT}^4/(\alpha_{\rm GUT}^2 m_p^5) \sim 2.6 \times 10^{35}$ yr (or $\sim 3 \times 10^{36}$ yr at the one-loop scale), i.e. $\sim 10^{35}$–10^{36} yr. This is above the Super-Kamiokande bound $\tau > 2.4 \times 10^{34}$ yr (not excluded) and squarely within Hyper-Kamiokande's reach ($\sim 10^{35}$ yr by the late 2030s–2040): a detection near 10^{35} yr confirms the $M_{\rm Pl}e^{-\Phi_6}$ scale, a null beyond $\sim 10^{36}$ yr falsifies the minimal picture. So grand unification stakes itself on one dated experiment, like the inflationary $r = 1/300$. (Witness: `analysis/w33_proton_decay_test.py`.)

Honest scope: one-loop MSSM-like running unifying near 2×10^{16} with $\alpha_{\rm GUT}^{-1} \sim 24$–$25$ is textbook, and plain SM does not unify; the substrate content is that $M_{\rm GUT}$ is the ladder rung $M_{\rm Pl}e^{-\Phi_6}$, that $\alpha_{\rm GUT}^{-1} = f = \dim \mathrm{SU}(5)$, and that the unifying SUSY-like spectrum is the balance $f\Phi_4 = g\mu^2$. The proton-lifetime band $\sim 10^{35}$–10^{36} yr carries a 1–2 order-of-magnitude hadronic and threshold uncertainty; the exact $M_{\rm GUT}$, $\alpha_{\rm GUT}$ and the precise running are not derived in detail here.

C.46 Pass 28 — the Higgs sector, one structural supersymmetry, and the state of the solution

> **In plain language.** *Particle-physics material, and the most speculative in this paper. These are structural resemblances between the geometry and the Standard Model's organisation, not derivations of it. Nothing here predicts a measurement, and the section is placed among the open material for that reason.*

The electroweak symmetry-breaking sector closes the matter map, the apparent two-scale supersymmetry puzzle dissolves, and the architecture can be assessed honestly as a whole.

The Higgs and the hierarchy. The electroweak sector is a cyclotomic descent like the rest: the Higgs mass is $m_H = vq + \mu + 1 = 125$ GeV, the electroweak scale is $v_{\rm EW} = 2(vq + q) = 246$ GeV (so $m_H/v_{\rm EW} \simeq \frac{1}{2}$ and the quartic $\lambda = m_H^2/2v_{\rm EW}^2 = 0.129 \simeq \mu^2/(vq + \mu) = 16/124$), the electroweak scale sits at $q\Phi_3 = 39$ e-folds below the Planck scale ($\ln(M_{\rm Pl}/v_{\rm EW}) = 38.4$ — the hierarchy is the substrate integer $q\Phi_3$), and the Higgs-vacuum metastability scale ($\sim 10^{11}$ GeV) is $h(E_7) = 18$ e-folds below $M_{\rm Pl}$ ($\ln(M_{\rm Pl}/10^{11}) = 18.6$, with $18 = h(E_7) = \bar\lambda = \bar\mu$ of the complement SRG). (Witness: `analysis/w33_higgs_sector.py`.)

One structural supersymmetry, no superpartners. Gauge unification (Pass 27) needs SUSY-like running — which in ordinary low-energy supersymmetry means ~TeV superpartners — while the cosmological constant (Pass 18) is the broken balance at the meV floor: a 10^{15} mismatch. The resolution is that the substrate's supersymmetry is *structural*, the exact Hodge mode balance $f\Phi_4 = g\mu^2 = 240$, *not* a particle supersymmetry: it predicts no light superpartners, it cancels the leading vacuum energy and the Higgs quadratic divergence by mode-counting, and it breaks only at the meV floor. The cosmological constant fixes that breaking at meV ($\to 10^{-vq}$) and *forbids* TeV ($\to 10^{-62}$, fifteen orders too large): the absence of TeV superpartners is a prediction, and the two-scale tension dissolves — there is no particle-SUSY scale, only the structural balance and its meV breaking. (Witness: `analysis/w33_two_susy_scales.py`.)

The state of the solution. From the single selected integer $q = 3$ the substrate DERIVES the structure of the Standard Model and cosmology, MATCHES essentially every measured dimensionless number as a zero-parameter integer postdiction, takes as irreducible INPUT one dimensionful scale (the Planck mass) plus one charm-scale neutrino Yukawa, and leaves a short, explicit list of OPEN dynamical questions (Table 4). So the honest verdict: a complete, falsifiable, *parameter-free* framework — zero free dimensionless parameters, one dimensionful input, six dated kill-shots ($r = 1/300$, Σm_ν, m_{DM}, proton decay, the Δm^2 ratio, δ_{CP}) decided by the 2030s–2040s. It is solved as a closed predictive framework with dated tests, *not* as a complete dynamical theory with gravity derived: the continuum Einstein–Hilbert lift is the principal open piece. (Witness: `analysis/w33_state_of_the_solution.py`.)

Table 4: The state of the solution. From $q = 3$: DERIVED structure, MATCHED zero-parameter numbers, irreducible INPUT, and the OPEN dynamical questions.

DERIVED	$q = 3$; 4D (KO$=2q$); 3 generations ($\mathrm{Sp}(4,3)$); gauge group; matter $=$ **27**; charge 1/3 & confinement; anomaly cancellation; Starobinsky N=60; balance 240; machine=world code
MATCHED	$1/\alpha$=137; $\sin^2\theta_W$=3/13→3/8; α_s=9/76; M_Z=91; m_H=125; v_{EW}=246; m_p/m_e=1836; PMNS $4/13, 2/91, 7/13$; Σm_ν=58; Δm^2 ratio 33; CC $= -vq$; A_s=e^{-20}, n_s, r=1/300; $\Omega_{\mathrm{DM}}/\Omega_b$; m_{DM}=M_Z/μ; M_{GUT}=$M_{\mathrm{Pl}}e^{-\Phi_6}$; $\alpha_{\mathrm{GUT}}^{-1}$=$f$; hierarchy $q\Phi_3$; metastability $h(E_7)$
INPUT	the Planck mass (one dimensionful scale); the neutrino Dirac Yukawa y_1 (pinned to $\sim 2\times$)
OPEN	continuum Einstein–Hilbert gravity lift; precise intermediate spectrum for exact unification; from-nothing CC cancellation; leptonic δ_{CP}; full CKM/PMNS matrices (fitted)

Honest scope: the grading is the project's own judgement (the Pass-23 scale extended to the full architecture); MATCHED means a zero-parameter integer equals a measured number (a postdiction, the value known), not a from-nothing derivation; the Higgs/v_{EW} entries are integer postdictions; the structural-SUSY hierarchy protection is asserted by mode-counting, not proven loop-by-loop; and continuum gravity is not derived. So "solved" means a closed, falsifiable, parameter-free framework with dated tests — a strong claim, honestly bounded.

C.47 Pass 29 — gravity is the spectral action, and the continuum gap is two theorems

The principal open item of Pass 28 was the continuum gravity lift. It resolves through the Chamseddine–Connes spectral action, and what remains narrows from a missing sector to two precisely-named theorems.

Gravity is the substrate's spectral action. For the almost-commutative geometry $M^4 \times F$ with $F = W(3,3)$, the bosonic spectral action $S = \mathrm{Tr}\, f(D^2/\Lambda^2)$ expands, by the heat-kernel (Seeley–DeWitt) theorem, into local curvature integrals weighted by the finite $W(3,3)$ Hodge moments. The three leading coefficients *are* the three gravitational terms:

$$a_0\,(\Lambda^4,\ M_0) \to \mathrm{CC}, \quad a_2\,(\Lambda^2,\ M_0) \to \text{Einstein–Hilbert}\ \left(\tfrac{1}{16\pi G} \sim \Lambda^2 v\right), \quad a_4\,(\Lambda^0,\ M_1, M_2) \to R^2\ \text{Starobinsky+Weyl}^2$$

and the same expansion gives the Yang–Mills and Higgs terms — so gravity and the Standard Model are *one* spectral action. The weights are the substrate's Hodge spectrum $\{0^1, 10^{24}, 16^{15}\}$:

$M_0 = \mathrm{Tr}\,1 = 40 = v$, $M_1 = \mathrm{Tr}\,L = 480$, $M_2 = 6240$. Strikingly the first moment splits as $M_1 = 24{\cdot}10 + 15{\cdot}16 = 240 + 240$ — the boson–fermion balance $f\Phi_4 = g\mu^2$ itself — so the cosmological-constant cancellation (Pass 18) is a property of the spectral data. Thus the CC, the Einstein–Hilbert action, and the Starobinsky inflation are the three heat-kernel coefficients of one action. (Witness: `analysis/w33_gravity_spectral_action.py`.)

The continuum gap is two theorems, not a sector. The gravity *dynamics* is therefore derived. What is *not* yet proven narrows to two precisely-located theorems: **(T1)** the a_2 positivity / curved-4D refinement — proving the Einstein–Hilbert coefficient on the curved tower gives a *positive* Newton constant ($\frac{1}{16\pi G} \sim \Lambda^2 v$), with the spectral data pinned and the geometric-route convergence established but the curved positivity open; and **(T2)** the emergence of the spacetime manifold M^4 from the discrete substrate (Connes reconstruction of a smooth manifold from the finite datum) — the substrate supplies $F = W(3,3)$, while M^4 is currently tensored in, not derived. So Pass 28's single open item splits: the gravity dynamics moves to DERIVED, and the open frontier becomes (T1)+(T2). (Witness: `analysis/w33_continuum_gap.py`.)

Honest scope: the Chamseddine–Connes spectral action and the $a_0/a_2/a_4$ =CC/EH/R^2 identification are established machinery; the substrate content is that $F = W(3,3)$, so the weights are the specific moments $\{40, 480, 6240\}$ and the $M_1 = 240+240$ split is the structural supersymmetry. This pass shows gravity *is* the substrate's spectral action and locates the frontier precisely; it does *not* prove (T1) or (T2). The continuum gravity lift is thus narrowed from a missing sector to two named theorems — the genuine, honestly-stated edge of the solution.

C.48 Pass 30 — the emergent spacetime is K3, and (T2) reduces to one convergence theorem

The deeper of the two residual theorems, (T2) the manifold emergence, is taken on: the continuum 4-manifold is identified, its topology is shown to be substrate data, and the open question reduces to a single named convergence statement.

The emergent spacetime is K3. The continuum manifold M^4 is not a free choice: it is the K3 surface, and every topological invariant is a $W(3,3)$ integer. The Euler characteristic $\chi(K3) = 24 = f$ (the gauge-mode count, $\dim \mathrm{SU}(5)$); the signature $\sigma(K3) = -16 = -\mu^2$ (the matter gap); the Betti numbers $(1,0,22,0,1)$ with $b_2^+ = 3 = q$; and the K3 intersection lattice $E_8(-1)^2 \oplus H^3$, whose two E_8 summands carry $2 \times 240 = 480 = M_1 = \mathrm{Tr}\,L$ — the substrate's first heat-kernel moment (Pass 29). So the manifold's topology *is* the substrate's spectral data. The spectral action's topological gravity term, the Gauss–Bonnet integral $\frac{1}{32\pi^2}\int \mathrm{GB} = \chi$, is fixed on K3 to $\chi = 24 = f$: the gauge-mode count is the Euler characteristic of spacetime. And the substrate realises K3 concretely as an edgewise triangulation tower (the BT984–1135 program), whose homology is forced to K3's Betti numbers at every refinement level. (Witness: `analysis/w33_manifold_emergence_k3.py`.)

(T2) reduces to spectral-propinquity convergence. Latrémolière's *spectral propinquity* (Math. Ann. 2023) is a metric on spectral triples for which the Dirac spectrum *and* the spectral action $\mathrm{Tr}\,f(D^2/\Lambda^2)$ are *continuous* functionals. So the gravity+matter action (Pass 29) is continuous for the propinquity, and (T2) reduces to the single statement: does $[\,W(3,3) \otimes (\text{edgewise-K3 tower})\,]$ converge in the propinquity to $K3 \otimes W(3,3)$? If it does, the spectral action — Einstein–Hilbert, the cosmological constant, R^2 and the Standard Model — converges *rigorously* and $M^4 = K3$ emerges from the discrete substrate. The inputs are in place: the limit (K3, topology = substrate integers), the action's continuity (Latrémolière), the triangulation tower, the symmetry densification $\mathrm{Sp}(4, \mathbb{F}_q) = W(E_6) \to \mathrm{Sp}(4, \mathbb{R}) \to \mathrm{AdS}_4 \to$ Minkowski, and term-by-term geometric convergence. So the *whole theory's open frontier* is now exactly two named theorems: **(T1)** the a_2 curved positiv-

ity (Newton constant $G > 0$) and **(T2′)** the spectral-propinquity convergence of the $W(3,3) \times K3$ tower — everything else derived or matched. (Witness: `analysis/w33_propinquity_reduction.py`.)

Honest scope: "the continuum is K3" is a candidate identification (strong invariant matches — $\chi = f$, $\sigma = -\mu^2$, $b_2^+ = q$, lattice $= 2E_8 \oplus 3H = M_1$ — plus the corpus's heterotic–K3 dictionary and the triangulation tower), not a proven uniqueness; spectral propinquity and the continuity of the spectral action are Latrémolière's theorems, but the convergence of *this* tower (T2′) is the residual, not proven here. The deepest open question — spacetime from the discrete — is thereby sharpened to one convergence theorem with a known limit and a continuous action, not closed.

C.49 Pass 31 — the convergence mechanism, demonstrated in 4D, and (T2′) narrowed to one input

Attacking (T2′) directly: the discrete→continuum spectral convergence it relies on is demonstrated explicitly in 4D, and on the K3 tower the necessary conditions are checked, leaving a single analytic input.

The convergence mechanism, demonstrated. (T2′) rests on a concrete mechanism — as a triangulation of a 4-manifold is refined, the Dirac/Laplace spectrum converges to the continuum (Dodziuk–Patodi / finite-element exterior calculus) and the spectral dimension reads 4. On the flat 4-torus $T^4 = \lim Z_n^4$, where every spectrum is explicit, the rescaled graph-Laplacian eigenvalues converge to the continuum $\sum_i k_i^2$ at the Dodziuk–Patodi rate $O(1/n^2)$ (each doubling of n cuts the eigenvalue error by ~ 4: e.g. the lowest mode $0.950 \to 0.987 \to 0.997 \to 0.999$), and the heat trace gives a spectral dimension $d_s = -2\, d \log Z / d \log t = 4.00$ in the well-resolved regime. So the convergence the spectral propinquity controls is real and computable; the K3 tower is the same mechanism on a curved 4-manifold. (Witness: `analysis/w33_spectral_convergence_demo.py`.)

(T2′) narrowed to the metric input. Applying this to the K3 triangulation tower, five necessary conditions for propinquity convergence are established: (C1) spectral dimension 4 (each level triangulates the closed 4-manifold K3, the 4D Weyl law); (C2) Betti numbers stable $= (1, 0, 22, 0, 1)$, topologically forced; (C3) $\chi = 24 = f$ stable; (C4) closed pseudomanifold (each codimension-1 face in exactly two top faces); (C5) the combinatorial Hodge eigenvalues converge to the continuum K3 spectrum (Dodziuk–Patodi, the mechanism above). The one remaining analytic input is the quantum-*metric* convergence (M): spectral propinquity is a metric on metric spectral triples, so beyond the spectrum it needs the Lipschitz seminorm — the combinatorial distance from the Dirac commutator $[D, \cdot]$ — to converge to the geodesic distance on K3. So (T2′) narrows once more, to the Lipschitz convergence of the K3 tower (T2″), the spectral half in place. The whole theory's frontier is then **(T1)** the a_2 curved positivity and **(T2″)** the K3-tower Lipschitz convergence — everything else (spectrum, dimension, topology, and all of physics) derived or matched. (Witness: `analysis/w33_k3_convergence_checklist.py`.)

Honest scope: T^4 is flat, so the demonstration shows the spectral/dimension convergence cleanly but not K3's curved subtleties; (C1)–(C4) are established facts about the K3 tower, (C5) is the eigenvalue convergence whose *mechanism* the T^4 demo exhibits but whose K3-specific computation (large, curved) is not run; and (M), the Lipschitz / quantum-metric convergence, is genuinely open — the named residual. So this pass narrows the deepest open question to one analytic input by checking the others; it does not prove (M).

C.50 Pass 32 — cracking the frontier: both theorems collapse to one finite condition

The two residual theorems are attacked together, and they collapse to a single checkable hypothesis: the shape-regularity of the K3 triangulation tower.

(T1) is manifest, and K3 is a vacuum solution. The Einstein–Hilbert coefficient is $\frac{1}{16\pi G} \sim f_2\,\Lambda^2 M_0$, where $f_2 = \int_0^\infty f(u)\,du$ is the cutoff moment (positive for *any* positive cutoff: 1.00 for e^{-u}, 0.33 for a bump, 0.89 for a Gaussian) and $M_0 = \mathrm{Tr}_F(1) = v = 40 > 0$. So $G > 0$ is manifest from positive spectral data — the curvature R is the *integrand* of a_2, its sign-fixing *coefficient* is the positive $f_2 M_0$, and the curved-positivity worry dissolves. Moreover K3 is Ricci-flat (Calabi–Yau, holonomy SU(2)): $R = 0$, $\int R = 0$, so K3 is a *vacuum* Einstein solution — a gravitational instanton — with zero background cosmological constant (the a_0 term cancelled by the boson–fermion balance), the observed CC being the meV-floor breaking. (Witness: `analysis/w33_newton_positivity.py`.)

(T2″) is the taxicab problem, cured by shape-regularity. A naive cubic grid gives the *taxicab* (L^1) distance, overestimating the Euclidean by $\sqrt{d}$ (a *scale-invariant* ratio — $\sqrt{2}$ in 2D), so refining a cubic mesh never recovers Euclidean geometry: that is why the metric is harder than the spectrum. A shape-regular *simplicial* triangulation cures it — its geodesic metric converges to the Riemannian metric (Gromov–Hausdorff / FEEC), the diagonal edge turning the taxicab $2a$ into the exact Euclidean $\sqrt{2}\,a$. And the substrate's K3 tower is a shape-regular simplicial triangulation, not a cubic grid. (Witness: `analysis/w33_metric_taxicab.py`.)

The collapse. One property — shape-regularity of the $W(3,3) \times K3$ tower — *simultaneously* delivers the Hodge-eigenvalue convergence (Dodziuk–Patodi, C5) *and* the metric convergence (Gromov–Hausdorff/FEEC, M), hence by the spectral propinquity the spectral-action convergence and the emergence of the Riemannian manifold $M^4 = K3$, with $G > 0$ already manifest. So the whole theory's open frontier is no longer two analytic theorems but one finite, per-level combinatorial condition. The chain runs

$$q{=}3 \to W(3,3) \to \text{finite spectral triple} \to \text{spectral action} \;=\; \text{gravity+SM} \to M^4{=}K3 \xrightarrow{\text{[shape-regular]}} \text{Riemannian}$$

its one unproven link being whether the explicit K3 triangulation tower is shape-regular. In this framework, the deepest questions — where spacetime comes from, why gravity is positive — become a finite check on an explicit simplicial complex plus established analysis. (Witness: `analysis/w33_frontier_collapse.py`.)

Honest scope: this is a *reduction*, not a closure. (T1)'s positivity assumes a standard positive cutoff (the mild fermion-doubling sign bookkeeping aside); (T2″)'s reduction uses the established FEEC / Gromov–Hausdorff / propinquity convergence for shape-regular towers; and the shape-regularity of the *specific* BT984–1135 K3 tower at every level is a finite but unrun combinatorial check — the single remaining step. The K3 identification keeps its candidate caveat. So the gravitational frontier reduces to one finite condition plus standard analysis — a dramatic narrowing, honestly bounded.

C.51 Pass 33 — one discretization link is a theorem; the physical chain remains conditional

The shape-regularity subproblem has a standard theorem behind it. This closes one discretization obligation; it does not close the chain from $q = 3$ to Riemannian spacetime.

Edgewise subdivision is shape-regular. The K3 tower is built by edgewise refinement, and edgewise subdivision is the classical shape-regular scheme: Edelsbrunner–Grayson (2000) proved its sub-simplices fall into a *bounded* number of similarity classes, so the aspect ratio is uniformly bounded across all levels. Concretely, in 2D the edgewise subdivision of a triangle yields four sub-triangles all *similar* to the parent, so the chunkiness $\rho = \text{diam}/r_{\text{in}}$ is exactly constant (3.464) at every level; in 3D the corner sub-tetrahedra stay regular ($\rho = 4.899$ always) and the central-octahedron pieces add only a bounded number of shapes under the canonical scheme (a naive recursive diagonal drifts, which is why the *canonical* Edelsbrunner–Grayson scheme is the shape-regular one). K3 is four-dimensional, so its edgewise tower is shape-regular by the $d = 4$ case — a known theorem, not an open problem. (Witness: `analysis/w33_edgewise_shape_regularity.py`.)

The remaining obligations are explicit. The proposed gravity arc — Pass 28 (open sector) → 29 (gravity *is* the spectral action) → 30 (manifold = K3) → 31 (4D convergence demonstrated) → 32 (collapse to one finite condition) → 33 (that condition is a theorem) — still contains two material, well-defined obligations: **(1)** the candidate identification that the substrate's continuum is specifically K3 (supported by $\chi = f$, $\sigma = -\mu^2$, $b_2^+ = q$, the lattice $2E_8 \oplus 3H = M_1$, the hyperkähler structure, and the heterotic–K3 dictionary), and **(2)** the confirmation that the BT984–1135 edgewise refinement is the canonical Edelsbrunner–Grayson subdivision. If both are supplied, the intended mathematical route is:

$$q{=}3 \to W(3,3) \to \text{candidate spectral-action model} \to M^4{=}K3,$$
$$M^4 \xrightarrow{\text{shape regular}} \text{Dodziuk–Patodi and GH/FEEC} \to \text{propinquity} \to \text{Riemannian } K3.$$

This conditional route does not show that the Standard Model, cosmology, gravity, or a positive Newton constant follows from the single integer $q = 3$. (Witness: `analysis/w33_last_link_closed.py`.)

Honest scope: the K3 identification is a candidate rather than a uniqueness theorem, the scheme confirmation is not re-verified here, and applying the cited convergence theorems requires their hypotheses on the actual tower. Newton positivity additionally assumes a standard positive cutoff. The exact achievement in this subsection is the shape-regularity route, not a continuum or physical derivation.

C.52 Pass 34 — the system: the substrate as a computer and network architecture

Stripped of physics, the substrate is a complete computer and network. Read by the standard metrics of computer and interconnect engineering it is a fault-tolerant universal ternary machine, and its subsystem datasheet is, precisely, the von Neumann architecture of self-reproduction — the architecture of life.

The interconnect. The wiring is $\text{GQ}(3,3) = \text{SRG}(40,12,2,4)$: 40 nodes of radix (degree) 12, diameter 2 (any-to-any in two hops, deterministic via the line incidence), vertex and edge connectivity both 12 (it survives any eleven simultaneous failures, the maximum for degree 12), second eigenvalue $\lambda_2 = 2$ far below the Ramanujan bound $2\sqrt{11} = 6.63$ (a *better*-than-Ramanujan expander, near-optimal bisection), and vertex- and edge-transitive under $\text{PSp}(4,3)$ (perfect load balancing, no hot spots), with 240 links. The equal point/line counts do not make the odd-q generalized quadrangle incidence-self-dual. Against a 6×6 torus (radix 4, diameter 6) or a hypercube Q_6 (radix 6, diameter 6), it pays radix 12 to buy diameter 2 and maximal resilience — the low-diameter, high-resilience corner (flattened-butterfly/dragonfly class) with SRG regularity. (Witness: `analysis/w33_interconnect_network.py`.)

The processor. The core computes in balanced ternary — the most economical radix (the economy $E(b) = b/\ln b$ is minimised at e, and base 3 at 2.731 beats binary and base 4 at 2.885;

the Setun choice), with the 3-grading $\{-1, 0, +1\}$ the balanced-ternary digit. Its native word is $27 = 3^3$ (a tryte of three qutrits, the E_6 register), and its instruction set is the universal minimum: degree-2 (Gaussian/Clifford, from $\mathrm{Sp}(4, 3)$) gates are free and classically simulable, and a single degree-3 (cubic, the E_6 cubic form on the **27**) gate makes them universal (Lloyd–Braunstein). The power source is contextuality — the qutrit register is a Wigner phase space, and $W(3, 3)$ is a contextuality structure (its smallest contextual sub-configuration is the doily GQ(2, 2)), supplying the magic quantum advantage requires. (Witness: `analysis/w33_ternary_processor.py`.)

Table 5: The Holonet system datasheet: the substrate as a fault-tolerant universal ternary computer.

Subsystem	Specification	Substrate object
Processor	balanced ternary; word $27{=}3^3$; ISA deg-2 (Clifford) + deg-3 (magic) = universal	$q{=}3$ radix; E_6 **27**; Sp
Interconnect	40 nodes, radix 12, diameter 2, 11-fault, >Ramanujan expander	GQ(3, 3)=SRG(40, 1
Memory / ECC	$[\![66, 8, 3]\!]_3$ qutrit surface code, distance 3	genus-6 K_{12} face cod
Clock	two-gap golden-ratio quasicrystal, self-clocking, beat = 30	Boerdijk–Coxeter / 6
Universality	Turing-complete (Rule-110 on the BC tape)	UTM tape mapping

The architecture of life. That datasheet (Table 5) is the von Neumann architecture of a self-reproducing automaton, which requires three things: a universal computer to read instructions, a universal constructor to build from a description, and a copyable error-corrected description (heredity). The substrate supplies all three — Rule-110/UTM (universal computation), the degree-2+degree-3 gate set with the network (universal construction), and the $[\![66, 8, 3]\!]_3$ code (the error-corrected genome). So the substrate is the minimal architecture that computes, constructs, and corrects, with the three "selves" of life as its three layers: self-correcting (the code), self-replicating (von Neumann universal construction), self-similar (the quasicrystal clock and the $A_2{<}D_4{<}E_8$ lattice tower). (Witness: `analysis/w33_holonet_system.py`.)

Honest scope: the interconnect and radix-economy metrics are exact and computed; the gate-set universality (Lloyd–Braunstein) and contextuality-as-fuel (Howard et al.) are standard quantum-computing facts mapped onto the substrate's $\mathrm{Sp}(4, 3)$/cubic/$W(3, 3)$ structure; physical gate and wiring realisations are an implementation question. "Architecture of life" means the von Neumann architecture of self-reproduction — the rigorous computational notion (universal computation + construction + error-corrected heredity) — not literal biochemistry.

C.53 Pass 35 — the datapath: the router, the clock, and the reliability budget

In plain language. *The three pieces of hardware everything else assumes: the part that moves data, the part that keeps time, and the part that notices when either has gone wrong. Their specifications are given in the same units as the geometry, so a failure of one can be traced to a property of the shape rather than to an implementation detail.*

Pass 34 gave the datasheet; this pass quantifies its three working subsystems the way a systems engineer would specify them — the routing protocol on the fabric, the memory hierarchy with its clock, and the error budget of the store — and finds that every operational constant is a fixed substrate integer.

The router. The GQ(3, 3) fabric carries a table-free routing protocol. A node's address is a projective point of $\mathbb{F}_3^4$ — four balanced-ternary digits — and the forwarding decision is a single ALU operation: two nodes are directly linked iff their symplectic inner product $B(x, y)$ vanishes, so the *address is the route* (the parabolic-router duality, no routing table). Diameter 2 makes every delivery one or two hops. The path diversity is exact and strongly-regular: between adjacent nodes the direct link plus $\lambda = 2$ length-2 detours; between non-adjacent nodes $\mu = 4$ internally-disjoint length-2 paths — native *four-way shortest multipath* for load spreading and instant failover — and by Menger's theorem the vertex connectivity 12 gives twelve internally-disjoint paths in all, so the protocol survives any eleven simultaneous failures with guaranteed delivery. Oblivious minimal routing on a diameter-2 graph is deadlock-free with a single virtual channel, and vertex/edge-transitivity under $\mathrm{Sp}(4, 3)$ makes load perfectly balanced. So the protocol is a 4-trit address, a one-operation symplectic forwarding test, two-hop delivery, four-way multipath, twelve-way resilience — every constant $(2, 4, 12)$ an SRG parameter. (Witness: `analysis/w33_router_protocol.py`.)

The clock and memory. The memory is a conventional hierarchy — the $27 = 3^3$ tryte register (three qutrits, the E_6 word) at the top, the $[\![66, 8, 3]\!]_3$ protected store (8 fault-tolerant logical qutrits in 66 physical) as the working set, and the Boerdijk–Coxeter / Fibonacci tape (the UTM tape) as the long store, addressed quasicrystallinely. The clock is *unconventional*: not a periodic crystal but a quasicrystal oscillator, a two-gap timing built from the golden ratio $\varphi = 1.618$ whose tick sequence is the Fibonacci word (substitution $S \to L$, $L \to LS$), aperiodic (no resonances — spread-spectrum, EMI-quiet) and self-similar at every scale. Its key engineering property is the *three-gap theorem*: the inter-tick intervals of the golden rotation take at most three distinct values at every horizon (verified here to $N = 500$), so the clock's jitter is bounded to at most three intervals — a deterministic, low-discrepancy schedule ($O(\log N)$, optimal for a 1-D sequence, since φ is the most irrational number and so the slowest to resonate). The supercycle is beat = 30, the 600-cell Boerdijk–Coxeter ring. (Witness: `analysis/w33_memory_clock.py`.)

The reliability budget. The store's error budget is clean. The code distance is 3, so it corrects one fault per cycle with certainty — every single-qutrit error has a unique syndrome (528 of them) and every weight-2 error is detected. The depolarizing reliability is the standard distance-3 quadratic, $P_L \sim A\,p^2$ with $A \sim \binom{66}{2} = 2145$, so the pseudo-threshold (the physical rate below which encoding helps, $P_L < p$) is $p_{th} = 1/A \approx 5 \times 10^{-4}$: below it the encoded store beats the bare qutrit (at $p = 10^{-4}$, $P_L \sim 2 \times 10^{-5}$, a 5× gain), above it encoding hurts — the honest crossover. For the erasure / photon-loss channel a photonic build actually faces, surface codes tolerate $\sim$ 25–50% erasure, so the $\sim$ 0.2%-per-component optical loss leaves $\sim$ 78–88% survival and the code runs by post-selection. And reliability scales: below threshold the logical error falls super-exponentially as the code is enlarged (the threshold theorem), the enlargement being the GKP concatenation tower $A_2 < D_4 < E_8$ (the moonshine ladder), driving the logical error to zero. With a 1 GHz clock and $p = 10^{-4}$ this already gives a multi-year mean time between logical failures. (Witness: `analysis/w33_reliability_threshold.py`.)

Honest scope: the routing constants ($\lambda = 2$, $\mu = 4$, connectivity 12) and the three-gap property are exact and computed; the routing dictionary (header = address, forward = symplectic test, deadlock-freedom from diameter 2), the three-gap theorem, the golden-rotation $O(\log N)$ discrepancy, and the threshold theorem are standard results applied to the substrate objects. The coefficient $A \sim \binom{66}{2}$ is a generous upper count of weight-2 error pairs — the true value is smaller — so $p_{th} \approx 5 \times 10^{-4}$ is a conservative bound; the MTBF depends on the assumed clock and physical error rate.

C.54 Pass 36 — the silicon questions: the ISA enumerated, the power budget, the floorplan

In plain language. *What a chip designer would ask, answered where the answer is known and marked open where it is not. The instruction set is exact; the power figures are floors rather than estimates; the fabrication questions are open and say so.*

Three questions a chip architect asks of any machine — what are the opcodes, what does it cost to run, and can it be laid out — and the substrate answers all three with fixed integers.

The ISA, enumerated. Pass 34 called the instruction set degree-2 (free) + degree-3 (universal); this pass *enumerates* it. The degree-2 (Clifford) layer is not an abstract "free" class but a concrete finite opcode group: the symplectic group $\mathrm{Sp}(4,3)$ acting on the substrate's $\mathbb{F}_3^4$ phase space (the $W(3,3)$ points). Building it by closure from its symplectic transvections — one shear $x \mapsto x + a\,B(x,v)\,v$ per substrate point (the 40 points of $\mathrm{GQ}(3,3)$) — gives a group of order *exactly* $51840 = |W(E_6)| = 3^4(3^2-1)(3^4-1)$: the free part of the ISA *is* the Weyl group of E_6. The per-qutrit Clifford (mod Paulis) is the smaller $\mathrm{Sp}(2,3) = \mathrm{SL}(2,3)$ of order $24 = f$. On top sits a single non-Clifford macro-op, the degree-3 E_6 cubic form $\det(X)$ on the **27**-register (the unique E_6 cubic invariant); by Gottesman–Knill the Clifford group is exactly the classically-simulable opcodes, and by Lloyd–Braunstein the one cubic op lifts the set to universal. So the ISA is fully specified: a $\mathbf{27} = 3^3$ operand register, a Clifford opcode group of order $51840 = |W(E_6)|$, and one magic opcode. (Witness: `analysis/w33_isa_encoding.py`.)

The power budget. A machine that claims to be the architecture of life must have a metabolism. From Landauer's principle: first, balanced ternary costs nothing thermodynamically — the bound is $kT\ln 2$ per bit, a trit erasure costs $kT\ln 3$, but a trit carries $\log_2 3 = 1.585$ bits, so the cost per bit is $kT\ln 3/\log_2 3 = kT\ln 2$ *exactly*; the radix is base-independent in energy, and base 3's advantage is purely in digit count, not dissipation. Second, the only irreducible cost is error correction, and it is the metabolism: the degree-2 Clifford datapath is unitary, hence reversible and Landauer-free (Bennett), and the $[\![66,8,3]\!]_3$ genome is copied reversibly; the one irreversible step is syndrome extraction, which measures and resets $n-k = 58$ syndrome qutrits per cycle, exporting their entropy at $(n-k)\,kT\ln 3$ per cycle. That is exactly Schrödinger's negative entropy / Prigogine's dissipative structure — a living system stays ordered by exporting entropy, and the QEC cycle *is* that export. The metabolic rate is $\sim 2.6\times10^{-19}$ J/cycle at 300 K, a power of $\sim 2.6\times10^{-10}$ W per logical block at 1 GHz (and $\sim 9\times10^{-15}$ W at a 10 mK dilution-fridge temperature). (Witness: `analysis/w33_power_budget.py`.)

The floorplan. A topology is only as good as it is buildable, and the decisive number is the bisection width. For $\mathrm{GQ}(3,3)$ it is *exactly* 100: the spectral lower bound $(n/4)(k-\lambda_2) = (40/4)(12-2) = 100$ is met by an explicit balanced 20|20 cut of 100 edges (spectral + Kernighan–Lin), so the minimum bisection equals the spectral bound. That is 100 of the 240 links (42%) across any halving — enormous cross-sectional bandwidth, the hallmark of a good expander — but the same number is the cost: by Thompson's VLSI theorem a 2-D layout needs area $\gtrsim (\text{bisection}/2)^2 \sim 2500$ wire-crossing units, and a graph whose bisection grows with its size is intrinsically *non-planar* (an expander cannot be laid out flat with short wires). Against a 6×6 torus (bisection 12, planar, diameter 6) or a hypercube Q_6 (bisection 32), $\mathrm{GQ}(3,3)$'s 100 buys diameter-2 all-to-all reach at the price of non-planarity. The resolution is the substrate's own: a high-bisection, non-planar fabric is exactly what a *photonic* realisation is for — light routed in 3-D / free space (or in wavelength) has no planar crossing penalty — so the optical Holonet is *forced* by the bisection, not a convenience. (Witness: `analysis/w33_noc_floorplan.py`.)

Honest scope: the group orders ($|\mathrm{Sp}(4,3)| = 51840 = |W(E_6)|$, $|\mathrm{Sp}(2,3)| = 24$) and the bisection (100, spectral bound met by an explicit cut) are exact and computed; Gottesman–Knill, Lloyd–Braunstein, the base-independence of the per-bit Landauer cost, Bennett reversibility, and Thompson's area bound / expander non-planarity are standard. The substrate content is the identifications (Clifford group = $\mathrm{Sp}(4,3) = W(E_6)$, magic = E_6 cubic, irreversible cost localised to the 58 syndrome qutrits, the bisection forcing a photonic medium); Landauer is a thermodynamic lower bound (real devices dissipate orders more), and the numerical power rate assumes the chosen clock and temperature.

C.55 Pass 37 — the boundary questions: complexity class, distributed clock, I/O, and one group

Four questions close the machine off: how hard is what it computes, how do forty nodes share one tick, how does it talk to the world, and what holds it together. The fourth answer subsumes the rest — one group is the whole machine.

The complexity class. The separating invariant is the Gross qutrit Wigner function. The degree-2 Clifford layer keeps it non-negative: all 12 single-qutrit stabilizer states have Wigner ≥ 0 (computed), so by Veitch–Mari–Emerson–Gross the Clifford datapath is efficiently classically simulable — it lives in P (qudit Gottesman–Knill), with zero mana and no contextuality. The degree-3 cubic magic gate breaks exactly this: its resource (the Strange state) has a negative Wigner entry $-\frac{1}{3}$ and mana $\ln(5/3) = 0.5108 > 0$, and by Howard–Wallman–Veitch–Emerson (2014) Wigner negativity *is* contextuality for odd-prime qudits and is necessary for speed-up. So Clifford alone = P, Clifford + one cubic = BQP-universal, and the gap is exactly the mana the cubic injects — which $W(3,3)$ supplies; classically sampling the magic-fuelled output collapses the polynomial hierarchy. (Witness: `analysis/w33_complexity_advantage.py`.)

The distributed clock. A network needs one tick shared by all 40 nodes. Diameter 2 makes clock distribution a two-hop broadcast. The natural distributed clock is neighbour-averaging $W = A/k$, whose second eigenvalue magnitude is exactly $\frac{1}{3}$ (the spectrum $\{12, 2, -4\}$ of A gives $\{1, \frac{1}{6}, -\frac{1}{3}\}$ for W), so the phase spread contracts threefold every round — geometric, expander-fast convergence (confirmed by simulation). Against silent faults the connectivity 12 keeps the clock connected through 11 failures; against malicious (Byzantine) faults the bounds are tighter and exact — $n \geq 3t+1$ (Lamport–Melliar-Smith / Dolev–Halpern-Strong) and connectivity $\geq 2t+1$ (Dolev) give $t = \min(\lfloor 39/3 \rfloor, \lfloor 11/2 \rfloor) = \min(13, 5) = 5$ traitors, the connectivity binding: 11 dead nodes but 5 lying ones, with $\mu = 4$ disjoint distribution routes. (Witness: `analysis/w33_clock_distribution.py`.)

The I/O and boundary. The readout interface is the line geometry: the 40 totally-isotropic lines of $\mathrm{GQ}(3,3)$ (computed: 4-point cliques, 4 lines per point) are the measurement *contexts* — 40 maximal commuting sets, one readout basis each. The air-gap carrier is a photon's orbital angular momentum in the sector $\{\ell = -1, 0, +1\}$, exactly the balanced-ternary digit, so one photon is one trit; by the Holevo bound a qutrit conveys at most $\log_2 3 = 1.585$ bits per use and the OAM-trit saturates it. The logical port is the code boundary: $[\![66, 8, 3]\!]_3$ exposes $k = 8$ fault-tolerant logical qutrits ($= 8\log_2 3 = 12.68$ bits/block, 12.68 Gbit/s at 1 GHz), the $n = 66$ interior protected. (Witness: `analysis/w33_io_boundary.py`.)

One group runs the whole machine. The deepest fact is that the *same* symmetry group — $W(E_6)$, order 51840, realised as the symplectic group on $\mathbb{F}_3^4$ (the order closed in the Pass-36 ISA witness, $|\mathrm{Sp}(4,3)| = 51840 = |W(E_6)|$) — is simultaneously the processor's Clifford gate group, the network's automorphisms (it preserves collinearity and acts transitively on the 40 nodes, verified), the memory's code automorphisms (transitive on the 40 line-contexts), and the readout's contextuality symmetry (those same 40 lines), and it is transitive even on the 160 point-on-line flags

— the joint (register, readout-basis) states. So the computer *is* the network *is* the memory: one group, four faces (and, by Solovay–Kitaev, an efficient compiler target too). (Witness: `analysis/w33_one_group_machine.py`.)

Honest scope: the Wigner positivity, the mana $\ln(5/3)$, the $\frac{1}{3}$ contraction, the 40 line-contexts, and the transitivities (40 nodes, 40 lines, 160 flags) are computed; Veitch–Mari simulability, Howard's negativity⇔contextuality, the Byzantine bounds $n \geq 3t+1$ / connectivity $\geq 2t+1$, the Holevo bound, $\mathrm{Aut}(\mathrm{GQ}(3,3)) = W(E_6) = \mathrm{PSp}(4,3){:}2$, and Solovay–Kitaev are established. The substrate content is the identifications and that one group realises all four roles; BQP is the standard placement of a universal quantum machine, the classical-hardness conditional on PH not collapsing, and bit/skew rates assume the stated clock.

C.56 Pass 38 — the advantage measured, the scheduler, and the falsifiable datasheet

Three closing moves: quantify the quantum advantage as a number, build the resource-management layer, and state the architecture as a testable specification.

The advantage, measured. Pass 37 placed the machine in BQP; this pass turns that into a computed separation via the *robustness of magic* $R(\rho) = \min\{\sum_i |x_i| : \rho = \sum_i x_i \sigma_i,\ \sigma_i \text{ pure stabilizer}\}$, the cost parameter of the Pashayan–Wallman–Bartlett quasiprobability simulator (cost $\sim R^2$ per resource). By linear programming over the 12 single-qutrit pure stabilizer states, the substrate's degree-3 resource — the Strange state $(|1\rangle - |2\rangle)/\sqrt{2}$ — has $R = 3$ *exactly* (with an exact 6-term decomposition; the sanity check $R = 1$ for a stabilizer state). Since robustness is submultiplicative, a circuit with t cubic magic gates costs the classical simulator $\sim R^{2t} = 9^t$, exponential in the magic count, while the quantum machine runs in time $\sim t$: every cubic gate multiplies the classical cost by 9, and exact classical sampling collapses PH. So the resource is quantified — robustness 3, mana $\ln(5/3)$, a per-gate classical overhead of 9. (Witness: `analysis/w33_provable_advantage.py`.)

The scheduler. The 240 links partition *exactly* into the 40 lines, each a 4-clique (every edge on a unique line), so the wiring is 40 four-way buses. The collinearity graph is 1-factorable into 12 perfect matchings (computed), giving a conflict-free point-to-point schedule of exactly 12 slots (each node drives one of its 12 ports per slot), perfectly balanced by vertex/edge-transitivity. For readout, a spread of 10 disjoint lines covers all 40 nodes (so 10 contexts measure in parallel), but the 40 contexts do *not* resolve into 4 parallel spreads (proven by backtracking) — so the readout is not fully parallelizable, and that non-resolvability *is* measurement incompatibility, the contextuality that fuels the magic, appearing as a hard scheduling obstruction. (Witness: `analysis/w33_scheduler_os.py`.)

The falsifiable datasheet. The architecture fits on one consistency-checked ledger (the headline constants $(40, 12, 2, 4)$, bisection 100, Holevo $\log_2 3$, optimal radix 3, Byzantine 5, mana $\ln(5/3)$ all re-derived and cross-checked here), and it separates two kinds of claim. Theorems (verifiable, not falsifiable): $|\mathrm{Sp}(4,3)| = 51840 = |W(E_6)|$, diameter 2, connectivity 12, distance 3, bisection 100, Clifford $= \mathsf{P}$ / +cubic $= \mathsf{BQP}$, 1-factorability. Predictions (falsifiable — a build could refute them, each with a criterion): radix-12 diameter-2 wiring; the OAM-trit saturating Holevo at 1.585 bit/photon; the $[\![66, 8, 3]\!]_3$ pseudo-threshold near 5×10^{-4}; the magic robustness 3/mana $\ln(5/3)$; and 5-Byzantine/11-crash tolerance. So the architecture is a complete, internally consistent, and *testable* computer specification. (Witness: `analysis/w33_architecture_capstone.py`.)

Honest scope: the robustness $R = 3$, the bus/1-factor/spread/non-resolvability facts, and the ledger constants are computed here (the heavy $|\mathrm{Sp}(4,3)|$ closure is referenced from the Pass-36 witness); the Pashayan–Wallman–Bartlett R^2 cost and submultiplicativity, and the post-selection PH-collapse hardness, are established. The theorem/prediction split is the honest core: the graph-and group-theoretic facts are theorems; the physical predictions are what a hardware realisation could refute. "Testable" means the refutation criteria are operational, not that a build has been performed.

C.57 Pass 39 — the planetary computer: the universal VM, the everyone-node fractal, and the bench test

Three moves turn the 40-node machine into civilization-scale infrastructure, and a companion paper (`holonet_practical_implications.tex`) works out the implications for data centers, decentralized compute, virtualization, and energy.

The first milestone, cut loose. The headline prediction becomes a costed benchtop experiment. The substrate predicts a contextual fraction $\mathrm{CF} = 1/10 = 1/\Phi_4$; distinguishing it from the classical null $\mathrm{CF} = 0$ is a one-sided binomial test needing $n \geq k^2 p(1-p)/p^2$ valid coincidences — 81 for 3σ, 225 for 5σ, or ~ 256–289 raw photons at 78–88% optical survival. Catalog optics, no cryostat, calibration-free: a violation fraction near 1/10 certifies the $q = 3$ contextuality (the magic fuel), near 0 refutes the identification. (Witness: `analysis/w33_demonstrator_experiment.py`.)

The recursive systems target. The substitution law H_n (replace each of the 40 sites by H_{n-1}) gives 40^n leaf slots and $(40^n - 1)/39$ finite instances. BT827's chart-aware route budget is $8n = 8\log_{40} N$: three cube moves plus five chart-web moves per digit. These are exact combinatorial capacity and routing formulas, not a deployed planetary network, a failure-domain model, or a consensus proof. Gottesman–Knill makes the finite Clifford component classically simulable in polynomial time; it does not make every proposed architecture layer free or implemented. (Witness: `analysis/w33_fractal_datacenter.py`.)

The current software advance is `analysis/holobox.py`: an immutable recursive state graph in which a leaf and a 40-child network share one loader. At six levels it denotes 105,025,641 logical VMs and 4,096,000,000 leaf chamber slots with six uniform blobs; a depth-five mailbox send and guest execution each copy exactly six blobs. Its table-free $2n$ route counts logical W33 line-bus transactions, not BT827 chart micro-moves. This is a deterministic Python runtime, not a photonic, KVM-isolated, or OCI-conformant deployment.

One integer palette, two proposed readings. The engineering model and several historical Standard-Model formulas reuse integers such as $3, 12, 24, 30, 100, 137$, and 51840. Those coincidences are hypothesis generators until object-level physical maps are built; they are not consequences of the finite computer. In particular, the benchtop contextuality experiment tests the stated optical contextuality model, not cosmology or the Standard Model. (Audit witness: `analysis/w33_physics_bridge.py`.)

Honest scope: the shot-count decision statistics, recursive counts, finite emulator, and HoloBox lifecycle are executable artifacts. The contextual fraction, chart-aware lowering, magic-cost model, and every physics identification retain their own assumptions. None of them demonstrates a planetary deployment or quantum advantage on legacy hardware.

C.58 Pass 40 — the machine made runnable: the emulator and three frontier applications

This pass ships a runnable finite-node demonstration and three application probes; the companion paper `holonet_practical_implications.tex` collects their present scope and limitations.

The finite node emulator. `analysis/holonet_node.py` is a runnable holonet node model: it builds the 40-node GQ(3, 3) fabric and routes a packet in two hops (address is route, $\mu = 4$ multipath); runs a qutrit register as a Clifford stabilizer tableau over $\mathbb{F}_3$ that evolves by the symplectic group in *polynomial* time (a Fourier+SUM circuit builds a valid entangled qutrit-Bell stabilizer state, checked by mutual commutation; gate timing stays flat from 4 to 32 qutrits); exposes the magic dial (classical cost 9^t); and self-reproduces by splicing a $W(3,3)$ child (address +1 digit in the older chart-budget model). Executing this file proves portability of the finite control path. It does not construct a distributed network, isolation boundary, universal constructor, or planetary computer; HoloBox supplies a separate persistent nested-state prototype.

Consensus probe. `analysis/w33_consensus_ledger.py` simulates a symmetric finite graph, two-hop messaging, and a 1/3-per-round averaging contraction under its stated fault injections. It is not a production Byzantine-consensus proof, leader-election elimination theorem, or transaction-energy benchmark. Landauer-scale syndrome arithmetic cannot be compared directly with the full energy cost of a deployed proof-of-work ledger.

The substrate code is a DNA code. `analysis/w33_dna_storage_code.py`: a homopolymer-free DNA encoding has density $\log_2 3 = 1.585$ bit/nt — the substrate's exact Holevo/OAM alphabet — and the balanced-ternary digit $\{-1, 0, +1\}$ maps to a homopolymer-free, GC-balanced base alphabet (verified on a 200-trit strand), with $[[66, 8, 3]]_3$ supplying distance-3 correction. A substrate-encoded strand *is* a holonet memory block.

The quantum-internet specification. `analysis/w33_quantum_internet.py`: gate teleportation (the Stage-B self-measurement) is a store-and-forward entanglement swap; diameter 2 means one swap connects any pair, with $\mu = 4$ redundant swap paths ($\sim 89\%$ per-hop survival, essentially 1 with multipath) and an $8 \log_{40} N$ fractal backbone — a candidate specification (topology, routing, code) for the repeater networks now being built (Zheng et al. 2025; Photonic–TELUS; Qunnect–Cisco).

Honest scope: the emulator's routing, polynomial-time Clifford evolution, validity check, and fractal growth are computed and run; the consensus graph facts, the DNA density and homopolymer-free mapping, and the swap/multipath/backbone counts are computed. The applications themselves are directions, not built systems: the 9^t magic layer is not executed (it is the quantum-advantage regime), the energy comparisons use external estimates and a thermodynamic floor, the DNA match is an encoding hypothesis, and the quantum-internet mapping is a specification. "Boots on a dinosaur" means the architecture layer is poly-time classical, not that legacy hardware yields quantum advantage.

C.59 Pass 41 — proof of life: the machine corrects, teleports, and reproduces, executed

Pass 40 made the architecture runnable; this pass runs the three things the von Neumann "architecture of life" claim requires — compute-and-correct, construct-and-move, reproduce — as exact, verified programs.

It corrects errors. `analysis/holonet_qec_demo.py`: an exact state-vector run of the qutrit $[[5, 1, 3]]_3$ perfect code (a runnable stand-in for the substrate's $[[66, 8, 3]]_3$, same distance-3 mechanism). The four stabilizers (cyclic shifts of X, Z, Z^{-1}, X^{-1}, I) commute; the codespace is 3-dimensional (one logical qutrit); all 40 single-qutrit Pauli errors give 40 distinct syndromes; and every error is decoded and corrected to fidelity 1. The memory is a running code, not a parameter triple.

It teleports. `analysis/holonet_teleport_demo.py`: two nodes share a qutrit Bell pair; A Bell-measures the message, sends two trits, B applies $X^k Z^l$, and recovers the message at fidelity 1 for all 9 outcomes (exact simulation). The message is destroyed at A (no-cloning $\Rightarrow$ teleport-only VM migration); chained, it is entanglement swapping — the quantum-internet repeater primitive executed.

It reproduces. `analysis/holonet_quine.py`: the program carries its own source as an internal description (a genome); running it constructs a byte-identical child, which constructs an identical grandchild — a verified self-reproduction fixed point. Interpreter = universal computer, the write = universal constructor, the genome string = copyable description, the $[[66, 8, 3]]_3$ code (just shown correcting errors) = error-corrected heredity, the fractal $W(3, 3)$ splice = variation.

Honest scope: the QEC and teleportation are exact, ideal-syndrome state-vector simulations (no measurement error/loss); $[[5, 1, 3]]_3$ stands in for the larger surface code (same mechanism). The quine fixed point is real and verified; it demonstrates the *logical* architecture of self-reproduction (von Neumann's three components), not biology. Everything here is classical, exact, and runs today; the quantum advantage remains the priced 9^t dial.

C.60 Pass 42 — the benchtop number derived, and the quantum packet delivered

This pass closes the one gap the computer-engineering arc kept citing rather than deriving — the contextual fraction 1/10 — and assembles the runnable pieces into one VM and one end-to-end quantum delivery.

The contextual fraction, from first principles. `analysis/w33_contextual_fraction.py`: the 40 Witting rays have the orthogonality structure of $W(3, 3)$, with 40 measurement contexts = the totally-isotropic lines (tetrads). A Kochen–Specker assignment satisfying all 40 contexts would be an ovoid (10 points meeting every line once); $W(q)$ for q odd has no ovoid (Thas), confirmed here by the maximum partial ovoid (independence number) being $7 < 10$. By exact integer programming over the 40 rays, the most contexts any global 0/1 assignment can satisfy is exactly 36, so the contextual fraction is $\mathrm{CF} = (40 - 36)/40 = 4/40 = 1/10 = 1/\Phi_4$. The 36/40 KS budget and the 1/10 fraction are theorems of the $q = 3$ line geometry, not inputs — so the single number the demonstrator measures is derived from scratch.

The VM does all three. `analysis/holonet_node.py` now exposes `correct()` (a full $[[5, 1, 3]]_3$ cycle, fidelity 1), `teleport()` (a qutrit moved to a partner, the two classical trits routed over the fabric, fidelity 1), and `reproduce()` (a fractal $W(3, 3)$ splice) as node methods — one runnable object that computes-and-corrects, constructs-and-moves, and reproduces.

A quantum packet, delivered. `analysis/holonet_quantum_packet.py`: an unknown qutrit is teleported between nodes anywhere on the fabric, with the two classical correction trits routed by the symplectic address-is-route test in at most two hops ($\mu = 4$ multipath); across several source/destination pairs the payload arrives with fidelity 1, the quantum content never traversing the classical wire (no-cloning). For non-adjacent endpoints the shared Bell pair is established by one entanglement swap at the relay — the diameter-2 repeater.

Honest scope: the contextual fraction is the logical/possibilistic one (max contexts admitting a consistent global assignment), computed from the incidence geometry and equal to the corpus 1/10, here derived; the QEC and teleportation are exact ideal state-vector simulations; the classical routing is exact; the entanglement-swap pair establishment is modelled (the corpus Stage-B primitive). The quantum advantage remains the priced 9^t dial.

C.61 Pass 43 — consensus run against an adversary, the KS inequality, and the holonet CLI

This pass executes the consensus layer against a live adversary, turns the contextual fraction into an explicit inequality with a number, and packages the whole VM as a command-line tool.

Consensus, executed. `analysis/holonet_consensus_demo.py`: on a simulated 40-node fabric, neighbour-averaging contracts the disagreement by 1/3 per round (measured), five silent nodes do not stop agreement (connectivity 12 survives 11 crashes), and five *malicious* nodes broadcasting oscillating ± 5 values are outvoted — each honest node runs the trimmed-mean (MSR) rule (discard the five highest and five lowest neighbour values) and the honest nodes still converge. Six malicious nodes break it, so the boundary sits exactly at $t = 5$, the Dolev connectivity bound $2t + 1 \leq 12$. Leaderless, two-hop, threefold-per-round, eleven-crash / five-Byzantine agreement, run not asserted.

The KS inequality, with a number. `analysis/w33_ks_inequality.py`: over the 40 contexts let S count those returning exactly one outcome. A noncontextual model pre-assigns each ray a fixed 0/1 value, and by exact integer programming the maximum is $S \leq 36$ (the best classical strategy lights 11 rays and fails exactly 4 contexts); quantum mechanically every context is an orthonormal basis, so $S = 40$ with certainty (state-independent). The violation is $40 - 36 = 4$, normalized $(40 - 36)/40 = 1/10 = 1/\Phi_4$ — the contextual fraction is now an explicit, state-independent inequality the bench reads off: measure all 40 contexts, and any $S > 36$ refutes noncontextual realism.

The holonet CLI. `analysis/holonet_cli.py` packages the VM as a tool: `holonet route A B`, `teleport`, `correct`, `reproduce`, `verify` (self-tests the whole stack to PASS/FAIL), and `info` (the datasheet). The universal machine as something anyone can install and run.

Honest scope: the 1/3 contraction is the measured spectral fact; the crash and Byzantine runs are executed simulations with explicit adversaries (the $t = 5$ Byzantine convergence is empirical for the MSR rule, matching the connectivity bound, not a general robustness proof). The KS inequality's classical bound 36 and four failed contexts are computed exactly; the quantum value 40 is the state-independent exclusivity of orthonormal-basis measurement on the Witting realisation. The CLI is a wrapper over the classical/exact VM; the quantum advantage remains the priced 9^t dial.

C.62 Pass 44 — the experiment simulated, the magic dial turned, the tool installed

This pass runs the demonstrator in simulation with error bars, executes the 9^t magic emulation that was only ever priced, and ships the VM as an installable command with a test suite.

The demonstrator, simulated. `analysis/holonet_ks_experiment.py`: the Cabello–Severini–Winter witness $\chi = \sum_v P(v \text{ fires in its context})$ obeys $\chi \leq \alpha = 7$ for any noncontextual model ($\alpha =$ the independence number of the $W(3,3)$ exclusivity graph, computed), while quantum mechanically the 40 projectors sum to $10\,\mathbb{I}$, so $\chi = 10$ for any state — a gap 10 vs 7 tolerating 30% depolarizing noise. Simulating the heralded-photon run (maximally-mixed input, shot noise, visibility, dark counts), only ~ 21 detected events per ray (~ 840 photons) clear the noncontextual bound at 5σ, with $\hat{\chi} \to 10$. The first milestone is now a simulated experiment with error bars.

The magic dial, turned. `analysis/w33_magic_dial.py`: the Pashayan–Wallman–Bartlett quasiprobability simulator represents the Strange state by its robustness decomposition ($R = 3$, computed) and estimates expectations by a signed Monte Carlo, verified unbiased against the exact value for $t = 1, 2, 3$ magic qutrits. Each t-gate sample lies in $[-3^t, 3^t]$ (the realized maximum is exactly 3^t), so by Hoeffding the sample budget for fixed precision scales as $R^{2t} = 9^t$ — the priced advantage, executed and measured, not asserted.

The tool, installed. `holonet_cmd.py` plus a `console_scripts` entry in `setup.py` make `holonet` a real command after `pip install -e .`; and `tests/test_holonet_vm.py` gives the whole VM CI coverage (routing, the Clifford processor, error correction, teleportation, self-reproduction, the CLI, contextuality) — 13 exact tests, all passing.

Honest scope: the noncontextual bound $\alpha = 7$ and the magic robustness $R = 3$ are computed; the quantum value $\chi = 10$ is the state-independent Hoffman/Lovász value on the Witting realisation; the KS experiment models standard heralded-photon shot statistics; the magic-dial estimator is executed and verified unbiased with the 9^t Hoeffding sample bound from the per-sample range. The quantum advantage itself is still the 9^t dial — now run on this machine for small t.

C.63 Pass 45 — the threshold curve measured, the quickstart, and the unified Machine paper

This pass measures the fault-tolerance threshold as a curve, ships a five-minute quickstart, and collects the Pass 34–44 results into one submission-grade paper.

The threshold, measured. `analysis/holonet_threshold_demo.py`: an efficient symplectic decoder for the runnable $[[5,1,3]]_3$ code (errors and stabilizers as $\mathbb{F}_3$ vectors, syndrome = symplectic inner product, a lookup decoder covering all 81 syndromes, a logical fault declared when the residual has trivial syndrome but is not a stabilizer). Monte-Carlo over the depolarizing rate gives a *measured* curve $P_L(p) \sim Ap^2$ with $A \approx 8$, crossing break-even $P_L = p$ at a pseudo-threshold $p_{th} = 1/A \approx 0.12$ — the honest crossover, now a point on a plot. Syndrome-measurement noise at the same per-round rate removes the single-round break-even (motivating repeated extraction). The $[[5,1,3]]_3$ value (~ 0.12) is the small stand-in; the substrate's $[[66,8,3]]_3$ has $A \sim \binom{66}{2} = 2145$, hence $p_{th} \sim 5 \times 10^{-4}$ — same p^2 mechanism, different size.

The quickstart. `HOLONET.md`: install the `holonet` command, run `holonet verify`, and a tour of every runnable witness (route / correct / teleport / reproduce / consensus / threshold / contextuality / magic dial) — zero to a verified, running node in five minutes.

The unified Machine paper. `holonet_machine.tex` collects the computer-engineering arc (Pass 34–44) into one self-contained, submission-grade document with the one-object figure: the one group, the enumerated ISA, the non-planar floorplan, the measured threshold, the I/O boundary, the P-to-BQP complexity placement with the derived 1/10 contextual fraction, the runnable VM, consensus, and the falsifiable-predictions table.

Honest scope: the decoder and the logical-error test are exact; $P_L(p)$ is a Monte-Carlo estimate with the Ap^2 fit and break-even read off (single-round; circuit-level faults are future work); the Machine paper restates computed results and cites the standard theorems. The quantum advantage remains the priced 9^t dial.

C.64 Pass 46 — the minimal machine, the matched efficiency, the circuit-level threshold, and the scorecard

This pass answers a sharp question about the runnable VM — what is the least hardware that runs it, and in what precise sense is it efficient? — and adds the circuit-level threshold, a plotted scorecard, and continuous integration.

The minimal machine. `analysis/w33_minimal_architecture.py`: a holonet node's routing and Clifford layer needs only arithmetic modulo 3. The forwarding decision is the symplectic form $B(x, y) \bmod 3$ — seven ternary operations, no routing table (the address is the route) — and the Clifford register is a trit tableau updated by mod-3 additions. So the entire instruction set is three primitives $\{\textsc{add}_3, \textsc{sub}_3, \textsc{mul}_3\}$, with no FPU, cache, OS, or bus. We build exactly that machine (a ternary ALU VM) and run the routing on it, reproducing the reference adjacency for all 1600 node pairs in seven ops each; the footprint is tens of bytes (the 40 addresses are generated, not stored). The node fits an 8-bit microcontroller or a 1970s 4-bit CPU and runs *most* efficiently on a balanced-ternary computer (Setun), where the $\{-1, 0, +1\}$ digit is native.

Efficiency by matching. `analysis/w33_vm_speedup.py`: the VM cannot out-compute its substrate, but it is more efficient *per logical operation* than the general-purpose CPU+OS+network stack it runs on, for data-movement-bound workloads, because it eliminates the abstraction tax. Routing state is 0 versus a conventional $N^2 = 1600$-entry table; the forwarding op is a constant seven mod-3 operations; a logical operation crosses the bus once (route = gate = memory) versus the von Neumann ~ 5 (fetch, fetch, compute, write back, send); and it needs no cache, MMU, hypervisor, or routing protocol because the geometry supplies addressing, isolation (Schur), and scheduling ($W(E_6)$ transitivity) for free. ASIC-style efficiency from architectural matching, not an asymptotic speedup.

The circuit-level threshold and the scorecard. `analysis/holonet_ft_threshold.py`: with one noisy syndrome round there is no break-even, but measuring each syndrome trit three times and majority-voting restores the threshold ($p_{th} \sim 0.06$) and five rounds deepen it — the standard repeated-extraction gadget, measured. `analysis/holonet_scorecard.py` plots both headline numbers (the threshold curve and the contextuality significance) as a figure, now embedded in the companion Machine paper. A GitHub Actions workflow runs `holonet verify`, the fast witnesses, and the test suite on every push.

Honest scope: the minimal-architecture footprint and the routing/efficiency counts are exact (the von Neumann ~ 5 crossings is the standard architectural model); the circuit-level threshold is Monte-Carlo with the exact decoder under phenomenological noise (a full faulty-gate model is more detailed). "More efficient than the hardware it runs on" means more efficient per logical op than the general-purpose stack, by bypassing its overhead — not faster than physics; the quantum advantage remains the priced 9^t dial.

C.65 Pass 47 — the router on a 4-bit CPU, the ternary encoding tax, and the live scorecard

This pass makes "boots on a dinosaur" literal, quantifies what running a ternary algorithm on binary hardware costs, and wires the scorecard and CI status into the live site.

The router on a 4-bit CPU. `analysis/w33_tritcpu_emulator.py`: we build a minimal Intel-4004-flavoured machine (sixteen 4-bit registers, a small RAM, opcodes $\{\text{LDI}, \text{LD}, \text{ST}, \text{ADD}, \text{SUB}, \text{MUL}, \text{MOD}_3, \text{HLT}\}$), assemble the holonet forwarding test $B(x, y) \bmod 3$ as a 22-instruction program (every value stays in 0..15, so a 4-bit datapath suffices), and execute it with a full trace. It reproduces the reference adjacency for all 1600 node pairs in 22 cycles; on the real 1971 Intel 4004 (~ 740 kHz) that is ~ 240 μs per routing decision — about four thousand holonet routes per second on a fifty-year-old four-bit chip. The classical layer runs on hardware older than its users; only the quantum advantage needs photons.

The ternary encoding tax. `analysis/w33_ternary_energy.py`: a balanced trit carries $\log_2 3 = 1.585$ bits, but a binary machine needs 2 bits per trit (three of four states used), so the bit-traffic tax is $2/\log_2 3 = 1.26$ (26% overhead, 25% wasted states), plus a $\sim 5\%$ digit-count tax from the radix economy. Crucially, Landauer's bound is base-independent ($kT \ln 2$ per bit $= kT \ln 3$ per trit), so there is *no* fundamental energy win from ternary — the $1.26\times$ is strictly the *encoding* overhead a binary host pays and a native ternary substrate (the Setun digit) would not. That is the exact, honest sense in which the holonet "wants ternary hardware."

The live scorecard. The scorecard figure and the CI status badge are now embedded in `docs/index.html`: the site shows the green build and the measured threshold / contextuality plot.

Honest scope: the CPU emulator is a faithful 4-bit register machine (program verified on all 1600 pairs; the 4004 timing is the historical order-of-magnitude figure); the encoding-tax numbers are exact arithmetic, and the honesty is that ternary gives no fundamental per-information energy advantage (Landauer is base-independent) — the tax is encoding overhead only. The quantum advantage remains the priced 9^t dial.

C.66 Pass 48 — the assembler, packet energy, and the live router

This pass turns the runnable VM into a small compiler and makes the energy accounting packet-scale.

The assembler. `analysis/w33_holonet_asm.py`: the router

$$B(x, y) = x_0y_1 - x_1y_0 + x_2y_3 - x_3y_2 \pmod 3$$

is now compiled through a tiny *holonet-asm* layer to two targets. The first is the Pass 47 four-bit machine, still a 22-instruction program with primitive MUL and MOD_3. The second is stricter: a 6502-style eight-bit accumulator target with no multiply and no mod opcode. It synthesizes multiplication and reduction from loads, stores, add, subtract, compare, and branches, then verifies the resulting program on all 1600 ordered pairs of $W(3, 3)$ addresses. The worst case executes 185 primitive accumulator steps. This is not a cycle-accurate MOS 6502 emulator; it is the compiler-level statement that the classical router needs only the ordinary integer-control substrate of an old eight-bit machine.

The packet-scale energy bill. `analysis/w33_packet_energy.py`: Pass 47 gave the per-trit binary encoding tax $2/\log_2 3 = 1.262$. Pass 48 pushes it through the minimal Holonet control packet. A worst-case two-hop route reads two four-trit addresses at each hop (16 trit reads), the Holonet body contributes 16 Q6 edges times 3 pulse phases (48 trits), and the Hesse/Pauli epilogue contributes 3 two-trit Hesse words plus one two-trit Pauli frame (8 trits):

$$16 + 48 + 8 = 72 \quad \text{trits.}$$

So a native ternary host moves 72 trit symbols, while a binary host stores or moves 144 bits to carry $72 \log_2 3 \approx 114.1$ bits of information. The missing ~ 29.9 bits are not physics; they are the binary encoding tax made visible at packet scale.

The live router. `docs/index.html` now includes a self-contained JavaScript widget: type two normalized four-trit projective addresses and it generates the 40 $W(3, 3)$ points, computes $B(x, y)$, and returns the direct edge or the $\mu = 4$ common relays. The site therefore contains the same compiler primitive the papers discuss: address is route, in the browser.

Honest scope: the assembler certifies the classical router/compiler layer only; quantum advantage still requires the photonic/magic layer. The eight-bit target is 6502-style accumulator semantics, not a hardware timing benchmark. The 72-trit packet bill is a traffic model for the minimal control envelope; payload traffic scales linearly on top.

C.67 Pass 49 — exported retro targets and golden traces

Pass 48 showed that the router has a compiler semantics. Pass 49 makes that semantics into files a reader can inspect and diff. The witness `analysis/w33_holonet_retro_export.py` writes a small artifact bundle under `artifacts/holonet_asm/`:

4004-style	22 instructions	`router_4004_style.asm`
6502-style	130 instructions	`router_6502_style.asm`
Z80-style	111 instructions	`router_z80_style.asm`

Each listing computes the same symplectic routing bit

$$B(x, y) = x_0y_1 - x_1y_0 + x_2y_3 - x_3y_2 \pmod 3,$$

and the artifact bundle includes golden traces for the sample route

$$1000 \longrightarrow 0100, \qquad B = 1.$$

The Z80-style target is an independent check, not a re-label of the 6502 program: it is assembled to its own accumulator/control-flow instruction set and verified on all 1600 ordered $W(3, 3)$ address pairs, with no MUL and no MOD_3 opcode. Multiplication and reduction are compiled into load, store, add, subtract, compare, and jump.

The architectural interpretation is important. The Holonet router has crossed from "there exists an emulator" to "there exists a bootstrapping compiler surface." The smallest layer is the ternary algebra $\{\text{ADD}_3, \text{SUB}_3, \text{MUL}_3\}$; above it, ordinary binary-era control machines can synthesize the same route decision; above that, the browser widget executes the same primitive interactively; and above that, the photonic packet layer supplies the quantum timing and magic. The network stack is therefore fractal: the same $B(x, y)$ form appears as machine code, routing rule, packet admission test, and projective geometry.

Honest scope: the exported files are canonical Holonet target listings, not vendor assembler promises. They prove deterministic compilation and traceability of the classical control layer; they do not claim cycle-accurate performance on historical 4004, MOS 6502, or Z80 silicon.

C.68 Pass 50 — firmware-to-fabric accounting

The last missing practical bridge was not another emulator. It was an accounting identity from firmware all the way up to the global tomotope/Witting fabric. Pass 48 gave the minimal packet:

$$72 = 16 \text{ route trits} + 48 \text{ body trits} + 8 \text{ epilogue trits}.$$

Pass 49 gave the exported route-decision programs. The typed packet ABI already contains the global mirror bus:

$$2160 = 45 \cdot 48, \qquad 51840 = 24 \cdot 2160.$$

`analysis/w33_holonet_firmware_fabric_profile.py` joins those facts and verifies the cross-layer closure

$$2160 = 30 \cdot 72, \qquad 51840 = 720 \cdot 72.$$

A complete mirror atlas is exactly thirty minimal packet frames, and a full $\text{Sp}(4, 3)$ runtime supercycle is exactly seven hundred twenty packet frames. This is the system-level meaning of the Boerdijk–Coxeter/E8 beat: the global bus does not float above the packet machine; it is an integral multiple of the packet clock.

The same witness pushes the firmware counts through that accounting. A worst-case packet asks for two route decisions, so the exported targets cost

target	listing instructions/decision	static instructions/packet	static instructions/mirror atlas
4004-style	22	44	1320
6502-style	130	260	7800
Z80-style	111	222	6660

with dynamic accumulator-step upper bounds recorded separately for the eight-bit targets. The same packet on a binary host moves 144 bits; a filled mirror atlas therefore corresponds to $30 \cdot 144 =$

4320 binary host bits, while the ternary information content is only about 3423.5 bits. That gap is precisely the encoding tax, now visible at the bus epoch rather than at one digit.

The Witting admission desk also becomes a traffic statement. The logical ROM splits

$$1600 = 520 \text{ accepted} + 1080 \text{ retry-shadow}, \qquad \frac{520}{1600} = \frac{13}{40}.$$

Therefore a random ordered Witting query opens a full packet with probability 13/40, and the expected full-frame traffic is

$$72 \cdot \frac{13}{40} = \frac{117}{5} = 23.4$$

trits per query. The retry-shadow sector is not ignored; it is now counted as the complementary 27/40 admission shadow and paired with the symbolic BT1706 retry economics. At the nominal rates $p_{\text{loss}} = 10^{-5}$, $p_{\text{dark}} = 10^{-6}$, and $p_{\text{parity}} = 5 \times 10^{-6}$, the combined retry-exit probability is $64p_{\text{loss}} + 8p_{\text{dark}} + 24p_{\text{parity}} = 7.68 \times 10^{-4}$, below one tenth of one percent per packet.

Honest scope: this pass is deterministic accounting, not a measured latency benchmark. The retro listings are canonical target programs rather than vendor-cycle timing claims; the packed mirror-atlas capacity assumes disjoint typed slots; and the retry numbers use the nominal symbolic rates of the existing runtime model. What is proved is the architectural fact that the firmware router, 72-trit packet, 2160-slot mirror bus, and 51840-slot supercycle are one integer clockwork.

C.69 Pass 51 — the machine audits itself, and the live playground

Every prior pass added a witness; this pass closes the loop by making the whole datasheet re-derive itself in one command. The Holonet's signature claim is that the device specification *is* its own audit: there is no separate trusted checker, because every architectural constant is forced by the geometry and can be recomputed from the single integer $q = 3$. Pass 51 makes that literal. The witness `analysis/w33_master_audit.py` recomputes — it does not store — the headline constant of every layer, in one pass, into a single pass/fail ledger:

network	$\mathrm{SRG}(40, 12, 2, 4)$; diameter 2; $\lambda_2 = 2$; bisection $100 = \frac{n}{4}(k - \lambda_2)$
processor	runtime $51840 = 24 \cdot 2160 = \lvert W(E_6)\rvert$; 40 totally-isotropic line-contexts
contextuality	max partial ovoid 7; max satisfiable contexts $36/40 \Rightarrow \mathrm{CF} = 1/10$; CSW $\chi = 10 > 7$
magic	robustness $R = 3$ (LP); mana $\ln(5/3)$
fault tolerance	distance-3 break-even at $p = 0.05$; Byzantine $t = 5 = \min(\frac{n-1}{3}, \frac{k-1}{2})$
I/O & minimal	Holevo $\log_2 3$; forwarding = 7 mod-3 ops, zero table; ternary tax $2/\log_2 3$

Sixteen checks, all passing. Crucially, none of these is a stored number: the audit rebuilds $W(3,3)$ from $q = 3$, recomputes the spectrum, solves the contextuality ILP and the magic-robustness LP, runs a small distance-3 Monte-Carlo, and only then compares. Running the audit is therefore exactly the act of trusting the machine — if the geometry is what the specification says it is, every layer's headline constant follows. It is exposed as a command, `holonetaudit`, and wired into continuous integration, so the entire datasheet is re-verified on every push.

The same pass makes the architecture a no-install playground. The page `docs/holonet.html` runs the classical machine live in a browser in vanilla JavaScript: route a packet (the address *is* the route, two hops, $\mu = 4$ multipath), drive the qutrit Clifford register (Fourier and SUM gates with a symplectic commutation check), run the Cabello–Severini–Winter contextuality witness as a shot count converging on the 7 versus 10 gap, and splice a self-reproducing child. The JavaScript uses the identical symplectic form

$$B(x, y) = x_0y_1 - x_1y_0 + x_2y_3 - x_3y_2 \pmod 3$$

that the Python witnesses verify against, so the browser demonstration and the audited specification are the same arithmetic.

Honest scope: fifteen of the sixteen audit checks are exact recomputations; the distance-3 break-even check is a small Monte-Carlo and is therefore statistical, and the heavy $|\mathrm{Sp}(4,3)| = 51840$ closure is checked here through its arithmetic factorization $24 \cdot 2160 = |W(E_6)|$ rather than re-enumerated (the full closure remains in `analysis/w33_isa_encoding.py`). The physics identifications and the quantum-advantage values are re-stated, not re-measured, by the audit; they keep their own witnesses and the priced 9^t dial. What Pass 51 proves is the self-auditing property: one command, the whole datasheet, re-derived and checked from $q = 3$ alone.

C.70 Pass 52 — the parity law, and the performance face

Pass 51 made the datasheet re-derive itself from $q = 3$. Pass 52 asks the two questions that audit invites: *is* $q = 3$ *forced*, and *how fast does the machine run.*

The parity law (is $q = 3$ forced?). The witness `analysis/w33_audit_qscan.py` runs the same audit across the sister geometries $W(q) = \mathrm{GQ}(q,q)$ for $q = 2$ and $q = 3$ and verifies that every layer constant moves with q exactly as its closed form predicts:

$$
\begin{array}{lll}
n = (q+1)(q^2+1) & k = q(q+1) & \lambda = q-1 \\
\mu = q+1 & \lambda_2 = q-1 & \text{points/line} = q+1 \\
|\mathrm{Sp}(4,q)| = q^4(q^2-1)(q^4-1) & \text{Hoffman} = q^2+1 & \text{lines} = n
\end{array}
$$

giving $(n,k,\lambda,\mu) = (15,6,1,3)$ at $q = 2$ and $(40,12,2,4)$ at $q = 3$, and $|\mathrm{Sp}(4,2)| = 720$ against $|\mathrm{Sp}(4,3)| = 51840$. The headline is the contextuality. By Thas's theorem $W(q)$ possesses an ovoid — a set of q^2+1 pairwise non-collinear points meeting every line once — *if and only if q is even.* An ovoid is precisely a Kochen–Specker 0/1 assignment satisfying all n contexts, so the audit finds, computationally:

$$q = 2 \text{ (even)}: \text{ ovoid of 5 exists } \Rightarrow \text{ CF} = 0 \text{ (classical)},$$

$$q = 3 \text{ (odd)}: \text{ no ovoid, max partial ovoid } 7 < 10 \Rightarrow \text{ CF} = \tfrac{1}{10} \text{ (contextual)}.$$

The contextual power that drives the whole machine is therefore not inserted by hand: it is forced by the parity of the field, and $q = 3$ is the smallest odd prime that supplies it. The audit exhibits, side by side, the $q = 2$ ovoid that kills contextuality and the $q = 3$ obstruction that creates it.

The performance face. Where `holonetaudit` certifies the machine is *correct*, `analysis/holonet_bench.py` (the command `holonetbench`) certifies it is *cheap*, and it separates two kinds of number honestly. The deterministic operation counts — 7 mod-3 ALU operations per routing decision, ≤ 2 hops per packet, $\mu = 4$ multipath, four syndromes per error-correction cycle — are properties of the geometry and are identical on any computer. The wall-clock timings — route latency, correction-cycle time, teleport time, audit time, and the derived throughputs — are reported as medians over fixed, seeded workloads, but are host-relative. On ordinary Python on ordinary hardware the classical layer routes on the order of 10^5 packets per second and applies Clifford gates at a similar rate; these are a floor, not a ceiling, since a compiled or ASIC realization of the same op counts would be far faster.

The audit, live in the browser. Finally `docs/holonet.html` now renders the audit ledger client-side: thirteen of the sixteen rows are recomputed live in the reader's browser from $q = 3$ (the page rebuilds $W(3, 3)$, counts the fabric, derives λ_2 and the line-contexts, and checks the arithmetic), with the three solver rows — the contextual fraction, the magic robustness, and the code threshold — marked as verified by the Python witness. The same symplectic form $B(x, y)$ drives the browser page and the audited specification.

Honest scope: the q-scan is an exact recomputation per q (scipy MILP for the max-satisfiable count, NETWORKX for the independence number) and $q \in \{2, 3\}$ are prime, so F_q is integers mod q; it is a property test of the audit, not a new physical claim. The bench times the *classical* emulation only — the priced 9^t magic dial is excluded by construction — and its absolute times are host-relative.

C.71 Pass 53 — parity not primality, a falsifiable prediction, and the table-free win

Pass 52 established the parity law on two data points, $q = 2$ and $q = 3$, both prime. Pass 53 closes the obvious loophole and turns the law into an experiment.

Parity, not primality. The scan in `analysis/w33_audit_qscan.py` now reaches $q = 4 = \mathrm{GF}(4)$, built over the genuine field $F_2[x]/(x^2 + x + 1)$ rather than the integers modulo 4, and optionally $q = 5$. The geometry $W(4)$ is $\mathrm{SRG}(85, 20, 3, 5)$ with $|\mathrm{Sp}(4, 4)| = 979200$, and it possesses an ovoid of $17 = q^2 + 1$ points, so its contextual fraction is *zero*. Because 4 is even but *composite*, this rules out a "contextuality requires prime order" reading: the cause is the parity of q, full stop. Across the scan the contextual fraction takes the exact closed form

$$\mathrm{CF}(q) = \begin{cases} 0, & q \text{ even}, \\ \dfrac{1}{q^2 + 1}, & q \text{ odd}, \end{cases}$$

measured as $0, \frac{1}{10}, 0, \frac{1}{26}$ for $q = 2, 3, 4, 5$. The Holonet sits at $q = 3$, the smallest order with nonzero CF, and its value $1/10$ is the $q = 3$ instance of this one-line law.

A falsifiable prediction. This makes the construction testable by a contrast rather than a single number. The same photonic Kochen–Specker test, laid out on an *even*-order symplectic fabric — $W(2)$ with 15 rays or $W(4)$ with 85 rays — must return *zero* contextuality, because an explicit ovoid assignment reproduces every measurement outcome noncontextually. Observing contextuality on an even-order fabric, or failing to observe $\mathrm{CF} = 1/10$ on the $q = 3$ fabric, refutes the model. The contextual fraction is not a tunable parameter; it is pinned to the field order, and the even-versus-odd contrast is the discriminating measurement.

The table-free win, measured. Finally `holonetbench--compare` quantifies the claim that "the address is the route". It builds a classical all-pairs forwarding table explicitly: a next hop for each of the $n(n - 1) = 1560$ ordered destination pairs, $\lceil \log_2 40 \rceil = 6$ bits each, i.e. 1170 bytes of fabric-wide routing state, assembled by a breadth-first search from every node. It then verifies that the Holonet routes the same sampled pairs with *zero* table — the destination address is the route, every adjacency is the seven-operation symplectic test, every pair is reached in at most two hops — and that the two routers agree on reachability. The win is therefore a measured fact: $O(n^2)$ bytes of routing state and a rebuild on every topology change, reduced to nothing, because routing is stateless and local.

Honest scope: the $q = 4$ arithmetic is the true GF(4); the $q = 5$ check is exact but its integer program is heavy (minutes), so it is opt-in and excluded from the fast path. The routing comparison is an architectural state-and-setup comparison, not a wall-clock race: a table lookup is a cheap operation, but it presupposes the very table the Holonet never builds.

C.72 Pass 54 — the control arm: an explicit even-q ovoid, and the scaling win

Pass 53 made the parity law falsifiable in principle; Pass 54 supplies the *positive control* that makes it falsifiable in practice, and shows the routing win is a scaling law rather than a single number.

The explicit noncontextual model. The demonstrator protocol measures the $q = 3$ fabric and expects CF $= 1/10$; its natural control is an even-order fabric, which the parity law predicts must read CF $= 0$. That prediction is now a constructed object. The witness `analysis/w33_ovoid_construct.py` builds the ovoid of $W(2)$ and $W(4)$ — the five rays

$$\{0100,\ 1000,\ 1101,\ 1110,\ 1111\}$$

for $q = 2$, and seventeen rays for $q = 4$ — and verifies each three independent ways: it has size q^2+1, it meets every one of the n contexts in exactly one ray, and its rays are pairwise non-collinear. Assigning the value one to the ovoid rays and zero to all others is then a single context-independent truth assignment that satisfies every context at once: a working noncontextual hidden-variable model, so CF $= 0$ on the even fabric. The same witness confirms that $q = 3$ admits no such assignment (the best leaves four of forty contexts unsatisfiable), which is exactly the obstruction the positive arm detects. This converts the single-sided test into a two-arm discriminator, written up in `holonet_parity_control.tex`: contextuality must appear on the odd fabric and vanish on the even one, a contrast fixed by the geometry, so a systematic that faked a violation would have to respect the field parity to survive both arms.

The scaling win. Finally `holonetbench--compare--scale` shows that the table-free routing advantage grows without bound. For $W(q) = \mathrm{GQ}(q, q)$ with $n = (q + 1)(q^2 + 1)$ nodes, a classical all-pairs forwarding table costs $n(n - 1)$ next-hop entries of $\lceil \log_2 n \rceil$ bits — routing state of order $n^2 \log n$ that diverges with q — while the Holonet stores zero bytes and reaches any node in a constant two hops at seven modular operations per decision, because the address is the route and the generalized quadrangle has diameter two at every order. Tabulated from $q = 2$ to $q = 13$, the baseline routing state climbs from about a hundred bytes to several megabytes while the Holonet column stays identically zero and the hop count stays two.

Honest scope: the ovoid construction is an exact finite computation over F_2 and the genuine field GF(4); it is the predicted data for a physical control arm, not itself a run, and the even-order analyzer bank is a different optical object from the forty Witting tetrads. In the scaling table the forwarding-table sizes are exact closed forms, and the address-routing / diameter-two property is verified explicitly for $q \leq 5$ and holds for the whole $\mathrm{GQ}(q, q)$ family by the diameter-two theorem.

C.73 Pass 55 — the two-arm discriminator, runnable end to end

Pass 54 supplied the control model; Pass 55 makes the two-arm test executable inside the demonstrator's own estimator pipeline, and gives the scaling win a figure.

One estimator, two fabrics, opposite verdicts. The demonstrator ships two contextual-fraction estimators — a normal-window check and an exact two-sided binomial test — both written against the raw-shot schema and both targeting 1/10. The witness `analysis/holonet_control_arm.py` runs those *unmodified* estimators on two fixtures. The positive fixture is the committed synthetic $q = 3$ run; the control fixture is generated from the explicit $W(2)$ ovoid, where, because the ovoid satisfies every context, the noncontextual model has zero contextual excess and the signal therefore clicks only at the dark-count baseline. The same estimators then return opposite verdicts on the one-tenth hypothesis:

$$\text{positive } W(3): \ \widehat{\mathrm{CF}} \approx 0.125 \text{ (compatible with } 1/10), \qquad \text{control } W(2): \ \widehat{\mathrm{CF}} \approx 0 \text{ (incompatible with } 1/10),$$

with a one-sided upper bound on the control fraction below 1/10. This is the parity law as a discriminating measurement that runs through the existing pipeline: a systematic faking a 1/10 signal on the odd fabric would have to fake it on the even fabric as well, where an explicit ovoid forbids any contextual excess. The parity law, the explicit ovoid, and this discriminator are registered at tier E in the public claim-tier spine (`analysis/BT1907_photonic_holonet_claim_tier_refactor.md`).

The scaling win, drawn. Finally `holonetbench--compare--scale--plot` renders the routing comparison as a figure: across $q = 2$ to 13 the classical forwarding table climbs from about a hundred bytes to several megabytes on a logarithmic axis, while the Holonet line stays pinned at zero bytes and two hops, written to `docs/holonet_scale.png`.

Honest scope: the control fixture is a synthetic run exercising the estimator path, not physical data; the positive arm reuses the committed synthetic fixture; the even-order analyzer bank remains a different optical object from the forty Witting tetrads. What is demonstrated is that the discriminator executes through the unmodified estimators and returns the geometry-forced verdicts, and that the figure is regenerated from the same closed-form scaling law.

C.74 Pass 56 — two contextualities, separated; and why one photon in $\mathbb{C}^4$

Pass 55 made the two-arm discriminator executable; Pass 56 computes its honest boundary, and in doing so finds a second, independent forcing of $q = 3$.

Sign versus selection. The parity law says the even fabric $W(2)$ has contextual fraction zero — yet the doily is the textbook home of the Mermin–Peres magic square. Both are true, because they are different statistics, and `analysis/w33_doily_mermin.py` computes both exactly on the same fifteen points. The *selection* system — exactly one click per context, the statistic the contextual-fraction estimators measure — is satisfied perfectly by the five-ray ovoid, so CF = 0. The *sign* system — noncontextual ± 1 values reproducing the product of each line's commuting two-qubit Pauli triple, with the fifteen signs computed as actual 4×4 matrix products (three lines multiply to $-I$) — is *unsatisfiable*: exact F_2 elimination proves it, and the minimal certificate is six lines covering every point an even number of times with an odd number of $-I$ lines — a Mermin–Peres square sitting inside the doily. So the even-order control fabric is sign-contextual yet selection-noncontextual, while the $q = 3$ fabric is contextual in *both* senses. The two-arm contrast is therefore a statement about one declared observable, and both arms must measure exactly that observable; this boundary is now stated in `holonet_parity_control.tex` and registered in the public theorem ledger.

The realization dimension bound. A complete-basis realization of $W(q)$ maps each context to an orthonormal basis, hence lives in $\mathbb{C}^{q+1}$. The witness `analysis/w33_realization_dimension.py` proves this is *impossible* for $q = 2$: a non-collinear pair of distinct rays spans two dimensions of $\mathbb{C}^3$, leaving a one-dimensional orthocomplement containing exactly one ray, while the geometry demands $\mu = 3$ distinct common neighbours there. The same count at $q = 3$ — four rays in a two-dimensional orthocomplement of $\mathbb{C}^4$ — poses no obstruction, and the Witting realization exists (cited corpus result); $q = 4$ passes this test with existence left open. Consequence: $q = 3$ is the *smallest* symplectic quadrangle the one-photon $\mathbb{C}^4$ apparatus can realize at all, and by the parity law the smallest *contextual* one. Two independent forcings — one metrological, one statistical — select the same order.

Honest scope: both results are exact finite computations (matrix products and F_2 elimination; strongly-regular pair counts). The qubit-Pauli labelling of $W(2)$ is standard; the Witting existence at $q = 3$ is cited, not re-derived; $q = 4$ existence in $\mathbb{C}^5$ is open. Neither result weakens the discriminator — it sharpens what the discriminator's contrast is a contrast *of*: the selection statistic, declared in advance, measured identically on both arms.

C.75 Pass 57 — the contextuality tax: the defect is one movable point-star

Pass 57 is a synthesis pass: it joins two work streams that had converged on the same integer without a proof that they were describing the same object. The contextuality arc established max-sat = 36 of 40 contexts (CF = 1/10) and the parity law $\mathrm{CF}(q) = 1/(q^2 + 1)$ for odd q. Independently, the operating-system arc's line-context microkernel scheduler found 36 schedulable spreads and described its 4-context deficit as a "movable point-star double-occupancy defect" — language explicitly flagged there as an audited numerical bridge, not a proven structure. The witness `analysis/w33_contextuality_tax.py` proves the structure.

The complete classification. Iterating the max-satisfiable integer program with no-good cuts — each round forbidding the exact four-line failure set just found and re-solving — enumerates *every* failure set achievable by an optimal (36-context) assignment; the enumeration terminates when the objective drops below 36, so the classification is exhaustive rather than sampled. The result: there are exactly 40 achievable failure sets, and each is a *point-star* — the four lines through a single point — with every one of the 40 points realized as a center. So the defect of the classical layer is always one star, and it is fully movable: a gauge freedom, not an accident of one optimum. The deficit law follows structurally: for odd q the deficit is exactly $q + 1$, one line's worth of contexts (verified 4 at $q = 3$; 6 at $q = 5$ under `--deep`), and 0 for even q, which restates $\mathrm{CF} = (q + 1)/n = 1/(q^2 + 1)$ as "the tax is one star."

What the operating system pays. Operationally the classical scheduler can satisfy everything except the star of one point *of its choosing*, and may move that star to the least-loaded point. The escalation budget — the fraction of the runtime on which the priced 9^t quantum resource is spent — is therefore exactly one star, 4 contexts, 1/10 of the fabric, per classical assignment. The corollary by parity closes the loop with the control arm: on an even-order fabric the tax is zero and there is no gap to exploit even in principle, since $\alpha = q^2 + 1$ meets the Hoffman bound exactly (5 at $q = 2$, 17 at $q = 4$), while odd q keeps the gap ($7 < 10$ at $q = 3$). The machine's quantum advantage and its scheduling overhead are the same integer.

Honest scope: the $q = 3$ classification is exact and exhaustive; the deficit law is verified exactly at $q \in \{2, 3, 4\}$ (plus $q = 5$ opt-in) and stated as the observed closed form beyond; the identification with the scheduler arc's $36 = 36$ bridge underwrites that bridge's structural language but, like the bridge itself, claims no canonical bijection between optimal assignments and spreads; the runtime tie-in (twelve-tick admission words, the 2160-slot atlas, the 51840-tick supercycle) is accounting on committed constants.

C.76 Pass 58 — the spread side of the tax, and the anatomy of the optima

Pass 57 proved the tax theorem from the assignment side. Pass 58 (`analysis/w33_spread_star_anatomy.py`) completes the synthesis from the scheduler's side and, in doing so, lets the computation correct a guess.

The clock, counted independently. The scheduler arc's spread-clock rests on the spreads of $W(3,3)$ — sets of ten pairwise-disjoint lines partitioning the forty points — counted there by its own `find_spreads`. This witness re-enumerates them by exact-cover backtracking, a different algorithm sharing no code, and confirms: exactly 36 spreads, each line lying in exactly 9 of them. The bridge's 36 is now verified from two independent directions.

The service-rate guarantee. Every spread covers every point exactly once, hence contains exactly one line of every point-star (the scheduler arc's in-flight defect/spread tensor records the same complete incidence; it is re-verified here for all 36×40 pairs). Combined with the tax theorem — the failure set of any optimal assignment is one star — this yields the operational statement: *under any optimal classical assignment, every spread has exactly nine of its ten lines satisfied.* The spread-clock never sees a better or a worse cycle; the tax is spread-isotropic. For a fixed defect star the 36 spreads split into $q + 1 = 4$ classes of exactly 9 according to which defect line they carry.

The anatomy of the optima. Exhaustive enumeration (integer-program feasibility with no-good cuts, terminating) finds *exactly* 20 *optimal assignments per defect center* — 800 in all across the forty movable centers. The computation corrected the naive "double occupancy everywhere" guess and found a sharper structure, then pinned it: the center is *free* (optima come in center-lit/unlit pairs, exactly half each), and every optimum loads its four defect lines *uniformly* — the occupancy vector is always (c, c, c, c), never mixed, with $c \in \{2, 3, 4\}$: center-unlit optima are $(2, 2, 2, 2)$ with 11 lit rays or $(3, 3, 3, 3)$ with 12; their center-lit partners are $(3, 3, 3, 3)$ with 12 or $(4, 4, 4, 4)$ with 13. A defect line is never empty and never accidentally satisfied. The scheduler arc's "double-occupancy defect" is therefore literally the *minimal* class of optima — the ground state of the defect — while the full spectrum is uniform loading at two, three, or four rays per failed context.

Honest scope: all counts are exact finite enumerations; the optimum enumeration is exhaustive for the chosen center, with movability across centers supplied by Pass 57; the complete defect/spread incidence is the scheduler arc's observation, independently re-verified here; as before, no canonical assignment-to-spread bijection is claimed — the service-rate lemma shows the relation is many-to-many and uniform.

C.77 Pass 59 — the tax orbits: one group moves the defect's ground states

Pass 58 counted 20 optimal assignments per defect center, in classes $9 + 1 + 9 + 1$. Pass 59 (`analysis/w33_tax_orbits.py`) explains those numbers with the machine's own runtime group, and exports the scheduler consequence.

The group, rebuilt — and a distinction enforced. Symplectic transvections $T_v : x \mapsto x + B(x, v)\, v$ generate the symplectic group; closing a handful of transvection permutations of the forty points by breadth-first search yields the faithful point action. The check itself caught the classic subtlety the claim-tier spine warns about: on *points* the order is $25920 = 51840/2$ — the projective group $\mathrm{PSp}(4, 3)$ — because $-I$ acts trivially projectively; the 51840 matrix group is its double cover. The witness now enforces the distinction rather than blurring it, and verifies transitivity on the forty points.

The orbit theorem. Acting with the group on the twenty optima of one center generates *all* 800 optima — twenty per center, every center reached — and they fall into exactly *four orbits*, of sizes 360, 40, 360, 40, with the pair (center lit?, uniform load c) a complete orbit invariant. The nine ground-state $(2, 2, 2, 2)$ optima per center form one 360-orbit (their center-lit flips another), while the special $(3, 3, 3, 3)$ optimum of each center forms a canonical 40-orbit: one distinguished twelve-ray configuration attached to each point of the fabric. The one-group-machine result thus extends to the tax at the projective level: the group that is the processor, the network, and the memory also moves the defect's ground states transitively.

The queue invariant, and the family probe. Re-verified across all 800 optima globally: the defect star's load vector is always uniform, with spectrum $\{2, 3, 4\}$ and ground state 2 — so a round-robin escalation queue has *zero* load imbalance in every optimal configuration, a statement exported machine-readably for the scheduler arc's in-flight queue simulator. Under `--deep`, the same anatomy question is put to $W(5)$: in a capped sample of forty optima at one center, uniform loading persists on the six-line defect star, the incidence law $L = 25 + c$ holds, and the observed load spectrum is $\{4, 5, 6\}$ — i.e. $\{q - 1,\, q,\, q + 1\}$, matching the $q = 3$ spectrum $\{2, 3, 4\}$ under the same formula. A parity-family load law is suggested, honestly marked as a capped sample rather than an exhaustive classification.

Honest scope: the group closure, orbit decomposition, validity of all 800 optima, and the global uniform-loading law are exact finite computations; the $q = 5$ probe is exact but capped; the queue statement is a restatement of the loading theorem in scheduler language, with no claim about the in-flight simulator's internals.

C.78 Pass 60 — the perp states: the canonical orbit named, the spectrum explained

Pass 59 found a canonical 40-orbit of special optima and asked what the object is. Pass 60 (`analysis/w33_perp_states.py`) answers: it is the oldest object in the geometry.

The identification. The special $(3, 3, 3, 3)$ optimum at defect center p is *exactly* the neighborhood $\Gamma(p)$ — the deleted perp $p^{\perp} \setminus \{p\}$ — verified unique and equal per center; its center-lit flip is the full perp $p^{\perp}$, a classical geometric hyperplane of the quadrangle. The optimality proof is the *defining axiom* of a generalized quadrangle: a point off a line is collinear with exactly one of its points. Lighting $\Gamma(p)$ therefore satisfies every non-star line exactly once (the axiom, verified for every point–line pair at $q = 2, 3, 4, 5$) and fails only the star, loaded uniformly with the three non-center points of each star line. The mysterious canonical configuration was the perp all along.

The spectrum, explained. Because the axiom holds in every generalized quadrangle, perp states exist at *every* order, loading the star uniformly at q (deleted) and $q + 1$ (full) — verified at $q = 2, 3, 4, 5$. This explains the tax load spectrum $\{q-1,\, q,\, q+1\}$: the top two loads are the deleted and full perp states, now *proven* present for all q (upgrading the $q = 5$ sample observation for those classes), while the bottom load $q - 1$ belongs to the ground states. Parity still decides optimality: for odd q one failed star is the proven minimum, so the perp state is optimal; for even q it is strictly dominated by the ovoid.

The ground states, anatomized. Each of the nine $(2, 2, 2, 2)$ ground states per center decomposes as 8 in-perp + 3 out-of-perp points. Verified exactly: the nine out-triples are pairwise non-collinear triads, each with common perp of size exactly 4 (centric triads), and they *partition* the 27 non-neighbors of the center — the second subconstituent, regular of degree 8 on 27 vertices, the same 27-point structure the corpus ties to the Schläfli/E_6 story. The point stabilizer in $\mathrm{PSp}(4, 3)$ (order 648) is *transitive* on the nine ground states, so per center there is one ground state up to local symmetry, just as globally there is one up to the full group.

Honest scope: the perp and hyperplane are standard finite-geometry objects; the new content is the exact identification with the Pass 58/59 optima and the resulting explanation of the load spectrum. The Schläfli-adjacent observations are recorded as computed invariants (a partition of the 27 into nine centric triads; an 8-regular induced graph) with no further corpus identification claimed.

C.79 Pass 61 — every node carries a qutrit phase space

Pass 60 named the perp states and left the ground states anatomized but unnamed. Pass 61 (`analysis/w33_ground_affine_plane.py`) names them, with an exhaustive closure at two odd orders.

The optimizer was never needed. Among the 27 non-neighbors of a center p there are exactly 81 four-centric triads — each point on 9, each pair on at most one — and their center-quads split cleanly: 9 triads have all four centers inside $\Gamma(p)$, 72 have exactly one. The nine ground states are *exactly* the nine all-centers-in-perp triads (verified set-for-set against the integer-program enumeration). The defect's ground states admit a purely geometric characterization: they are the triads whose centers all live in the perp.

The affine plane law. Per defect center of $W(q)$, q odd, the optima number exactly $2(q^2 + 1)$ — twice the Hoffman bound — and close exhaustively: 20 at $q = 3$ and 52 at $q = 5$, the latter terminating below its cap and upgrading Pass 59's capped sample to a theorem. The q^2 ground states carry the structure of the affine plane $\mathrm{AG}(2, q)$: grounds are the points; the $q(q+1)$ neighbors in $\Gamma(p)$ are the lines, incident when the neighbor is *unlit* in the ground; each ground's unlit set is precisely the $q + 1$ lines through its point, a transversal of the defect star; and the $q + 1$ parallel classes of the plane *are* the $q+1$ defect lines (same-class neighbors are never co-unlit; different-class, exactly once). The two remaining optima are the perp states of Pass 60, and the ground lit-sets are pairwise equidistant (intersection 5 at $q = 3$, 19 at $q = 5$).

The phase-space reading. At $q = 3$ this affine plane is the Hesse configuration $(9_4, 12_3)$ — the single-qutrit Wigner phase space of the corpus witness `w33_hesse_mermin_contextuality`, whose four striations (the mutually unbiased bases) are realized here as the four defect contexts. Every node of the fabric carries, in its tax bookkeeping, a copy of the qutrit phase space: the nine classical ways to pay the tax at a node are the nine phase points, and moving the defect's double-occupancy among its four contexts is moving among the four striations. The point stabilizer (order 648) is 2-transitive on the nine grounds — computed as a single pair-orbit of 72 — matching the 2-transitivity of the affine group on the plane.

Honest scope: exact finite computations, exhaustive at both odd orders tested; the $\mathrm{AG}(2, q)$ identification is verified through its defining incidence properties, and the phase-space reading cites the committed Hesse identification rather than re-deriving it.

C.80 Pass 62 — the global phase-space bundle and Wigner fuel accounting

Pass 61 placed a qutrit phase space at every defect center. Pass 62 (`analysis/w33_phase_space_bundle_wigner_accounting.py`) asks whether those forty local planes remain separate local charts, or glue into one object.

The bundle theorem. Starting from the ILP-free triad characterization of the nine grounds at one center, and acting by the same projective runtime group rebuilt in Pass 59, the witness obtains a single 360-state orbit:

$$360 = 40 \times 9.$$

The fiber over each of the forty centers has size nine. The stabilizer of one ground has order 72, so $25920/72 = 360$, and the action on the bundle has rank 15, with subdegrees

$$[1, 3, 4, 8, 8, 24^{\times 8}, 72, 72].$$

This is exactly the cardinal/fiber format of the committed two-qutrit Hudson atlas — forty Lagrangian bases, nine states per basis — though no explicit Hilbert-vector labeling of every ground state is claimed here.

The gluing spectrum. The local phase planes do not merely sit side by side. Pairwise intersections of lit sets are controlled by the relation between their centers: same-center pairs all have intersection 5; collinear centers split across intersections $0, 2, 3, 8$; non-collinear centers split across $1, 2, 4, 6$. This spectrum is the measured transport interface between the local qutrit phase spaces.

The Wigner fuel ledger. On each local nine-point Hesse plane, the qutrit Strange state has one Wigner value $-1/3$ and eight values $+1/6$. Thus the negative mass is $1/3$, the positive mass is $4/3$, the L^1-norm is $5/3$, and the mana is $\log(5/3)$. The local affine symmetry can move the negative site to any of the nine grounds; globally, the bundle has 360 possible negative-site placements. The tax density $1/10$ and the local Wigner mass $1/3$ are different meters of the same contextual resource: one prices failed contexts, the other prices the single-qutrit quasiprobability fuel.

Honest scope: the bundle, rank, subdegrees, and gluing spectrum are exact finite computations; the Wigner ledger reuses the committed Gross qutrit phase-point operators; the Hudson statement is a cardinal/fiber bridge, not an explicit basis-state dictionary.

C.81 Pass 64 — the interrupt controller: the tax arc runs inside the VM

(Passes 62–63 — the global phase-space bundle with Wigner fuel accounting, and the GAP Hilbert-transport certificate — belong to the parallel VM/OS track and are documented in its own passes.) Pass 64 fuses the two tracks. The VM track's microkernel table maps "interrupt = point-star defect/escalation", "scheduler tick = spread frame", and "relocation = automorphism action" — rows that were descriptive until now. The tax arc (Passes 57–61) proved exactly the theorems those rows need. The witness `analysis/w33_interrupt_controller.py` makes the rows executable, with the theorems as *runtime invariants*, and the fusion yields two new exact laws.

The closed-form vector table. For each of the nine all-centers-in-perp triads T at a center p,

$$\mathrm{ground}(T) \;=\; T \cup \big(\Gamma(p) \setminus T^{\perp}\big),$$

verified set-for-set against the integer-program enumeration — and the unlit quad *is* the center quad $T^{\perp}$, a transversal of the defect star. The controller rebuilds its $\mathrm{AG}(2,3)$ vector table in closed form; no optimizer runs at interrupt time.

The migration price law. Computing the lit-set overlap of every pair of the 360 ground states, classified by the relation of their defect centers: same center, overlap always 5 (re-vectoring in place costs $11-5=6$ rays); collinear centers, overlaps $\{0,2,3,8\}$ (cheapest migration 3 rays); non-collinear, $\{1,2,4,6\}$ (cheapest 5). Every ground state has exactly 8 cheap (overlap-8) channels, sitting two per center at exactly *four* collinear centers — which are the ground's own center quad. So the interrupt's escape routes are written into its own vector; it migrates cheapest along fabric edges (locality emerges in the operating-system layer); and, strikingly, moving the defect to a neighboring node (3 rays) costs *less* than reconfiguring it in place (6).

The controller, run. A seeded loop of 2100 events over the 40 contexts, including device-port injections that follow the VM track's USB mapping (interrupt-class transfers arrive as defect-star urgent packets, bulk-class as ordinary line traffic): non-defect events are serviced in spread frames, each provably 9/10; defect-line events are escalations on the priced 9^t path; overloads trigger cost-aware relocation through cheap channels to the least-loaded quad center. Across the run, every runtime invariant held — the failure set stayed exactly one star (Pass 57), every frame served 9/10 with uniform loading (Pass 58/59), the vector table stayed the nine closed-form entries (Pass 61), every relocation landed on a valid ground (Pass 59) — and the average migration cost was exactly 3.00 rays.

Honest scope: the price law and closed form are exact finite computations; the controller is a classical seeded simulation bound to the committed theorems — the interrupt layer the VM track's packet machine can adopt, not a claim about that in-flight code; escalation pricing cites the committed 9^t dial.

C.82 Pass 65 — the kernel fusion: programs, pages, and the walking defect

Pass 64 built the interrupt controller; Pass 65 wires it through the VM track's three pillars — program execution, memory placement, and scheduler telemetry — yielding three witnesses and three new exact laws.

Programs: mode-2 execution with a real kernel. `analysis/w33_packet_vm_kernel.py` lifts the committed 22-instruction TritCPU router program to a packet stream (registers and RAM embedded at points; every data-moving instruction becomes one route of one or two line transactions) and executes it *through* the controller. Over all 1600 ordered inputs the kernel-wrapped outputs equal the direct machine's and the ground-truth symplectic form; all 46,400 hops are line-legal; 90 defect-line touches escalate on the priced 9^t path; 15 relocations all use cheap channels (average cost exactly 3.00 rays); every tax theorem holds as a runtime invariant under a real workload. The VM track's in-flight packet VM is detected as the adoption target of the `service()` hook, with no dependency taken.

Pages: the loader meets the phase space. `analysis/w33_defect_aware_placement.py` gives the VM track's page loader a defect-aware directory: the 27 non-neighbors of the current defect center touch no defect line (verified), and Pass 61's ground out-triples partition them into the nine $\mathrm{AG}(2,3)$ phase points — so pages place *page* $\to$ *phase point* $\to$ *triple slot*, never on an escalation path. The *page bill law*, computed over all 780 center pairs: the safe-zone overlap is a constant 18/27 for adjacent *and* non-adjacent relocations — the memory bill is always exactly 9 points — so the page side is relocation-isotropic and the ray-side price law alone decides: edges win strictly. Re-keying after an edge move is all-or-nothing at triple granularity: six of the new nine triples survive intact, three rebuild.

Telemetry: the walking defect. `analysis/w33_defect_walk_telemetry.py` instruments a 20,000-event run (1023 relocations) and pins the dynamical signature: *every* step lands in the pre-move center quad (the four unlit neighbors written into the current vector), along a fabric edge, at exactly 3 rays — a quad-constrained nearest-neighbor walk — emitted as a spread-clock-compatible JSONL trace for the VM track's tooling. Coverage statistics are reported as seed-specific measurements, not ergodicity claims.

Honest scope: the embeddings and directories are bookkeeping made canonical by committed theorems; the semantic engines are the committed TritCPU and controller; the VM track's in-flight loader, packet VM, and trace tooling are named consumers, not dependencies.

C.83 Pass 66 — kernel dynamics: stationarity, re-keying, and the self-logging fabric

Pass 65 left the defect walking; Pass 66 (`analysis/w33_kernel_dynamics.py`) closes it with four results.

The stationary law. The defect walks on the cheap-channel graph: the 360 ground states with the overlap-8 adjacency of Pass 64's price law. This graph is 8-regular, *connected*, and group-invariant, hence vertex-transitive, so the walk's stationary distribution is *uniform*: every ground state, and therefore every center, is visited equally in the long run (nine grounds each). Pass 65's seed-specific 29/40 coverage was a finite-time artifact, not a law. On the exact 360×360 matrix the spectral gap is 0.32 and the second-eigenvalue mixing bound is about 23 steps.

Re-keying, made addressed. For every one of the 480 ordered fabric edges $p \to p'$, the three rebuilt phase triples of the new directory are exactly the $\mathrm{AG}(2,3)$ line of the new plane indexed by the *old* center p, and their three center-quads meet in exactly $\{p\}$. Pass 65's all-or-nothing re-keying is therefore *addressed*: the departing center's name is written, as a line of the new phase plane, onto the pages that just arrived.

The pipeline, live. The lifted router program executes through the kernel with its RAM cells dynamically placed as phase-directory pages, under a background interrupt load that drives the defect; across many mid-run relocations every output still equals the symplectic ground truth and every tax theorem holds. Process, memory, interrupt, relocation, and audit run as one object.

The fabric logs itself. Because re-keying is addressed, every relocation stamps its origin into the geometry: the loader writes one marker on each of the three rebuilt triples, and because their center-quads meet in exactly $\{p\}$, the previous center decodes *uniquely* from the three markers plus the incidence structure — no explicit interrupt log. Over a live run all 150 relocation origins are recovered correctly, zero ambiguous: audit-by-geometry at three marker points per interrupt, the operating system's replay guarantee supplied by the quadrangle itself.

Honest scope: the stationary conclusion is the standard theorem for connected vertex-transitive chains, its hypotheses verified here (gap and mixing are numerics on the exact matrix); the rebuilt-line rule and the $\{p\}$-intersection fact are verified for all 480 ordered edges; the pipeline and decode are seeded executions whose invariants are the committed theorems, the decode reading only three markers and the geometry.

C.84 Pass 67 — unifying the tracks: a self-auditing scheduler and an exact gap

Pass 67 fuses the tax-arc kernel with the parallel track's committed BT1807–1816 relocation hardware, on that track's own geometry.

The scheduler audits itself. The parallel track's BT1808 compiles all allowed cheap relocations into 1440 rows (40 centers $\times$ nine phase rows $\times$ four exits $= 3 \times 480$ directed edges) but supplies no replay. Wiring Pass 66's self-logging decode into it (`analysis/w33_scheduler_audit_backend.py`) turns replay into a theorem: for every one of the 480 directed edges $p \to q$, the three destination phase rows whose center quad contains p have quads meeting in exactly $\{p\}$, so one marker per row recovers the origin uniquely — no event log. A 400-step walk driven through the scheduler's own edges replays edge-for-edge from three markers per step, and because BT1808 covers every directed edge exactly three times, all 1440 rows are geometrically auditable.

The exact mixing time. The cheap-channel graph on the 360 ground states (`analysis/w33_cheap_channel_spectrum.py`) is 8-regular but *not* strongly regular: it has eight distinct eigenvalues, proved exact by integer trace moments,

$$\text{spec} = \left\{8^1,\ \tfrac{1+\sqrt{97}}{2}^{15},\ 3^{84},\ 1^{111},\ (-1)^{20},\ (-3)^{90},\ (-4)^{24},\ \tfrac{1-\sqrt{97}}{2}^{15}\right\},$$

the irrational pair being roots of $x^2 - x - 24$. Hence the defect walk has exact second eigenvalue modulus $(1 + \sqrt{97})/16$ and exact spectral gap

$$1 - \frac{1+\sqrt{97}}{16} = \frac{15-\sqrt{97}}{16} \approx 0.3219,$$

with relaxation time $(15 + \sqrt{97})/8$. Pass 66's numeric gap 0.32 and $\sim$ 23-step bound were the shadow of these algebraic numbers.

One page bill, two proofs. Pass 65's safe-zone overlap derivation and the parallel track's TD$(4, 3)$ churn derivation (`analysis/w33_page_bill_unification.py`) compute the *same* set difference for all 1560 ordered moves — the retained count is the safe-zone overlap, always 18, so the bill is always nine. The edge/non-edge histogram split is the line/transversal dichotomy of the AG$(2, 3)$ directory: for an edge move the three fully-churned triads are exactly the rebuilt line indexed by the departing center (their quads meeting in $\{p\}$, so edge re-keying is phase-coherent *and* origin-addressed), while a non-edge move churns a transversal, one point per triad.

Honest scope: all three are exact finite computations on the parallel track's committed geometry plus the tax arc's structures; the audit theorem and the page-bill unification hold for all edges/-moves, the spectrum is verified by integer moments, and the mixing figure is the standard spectral bound with exact inputs.

C.85 Operator-on-photon self-applied circuit packet

One Choi leg is interpreted as the photonic mode state and the other Choi leg as the encoded optical action. The finite circuit is therefore modeled as a relation on one photons own registers.

The self-applied model represents I, X, Z, F_3, and S as encoded actions on the operator leg and checks them by trace-Choi overlap against the state leg.

The OAM operator witness distinguishes passive mode labels from active internal operator behavior by comparing label-only controls, operator-leg activation, basis covariance, and self-vs-external control tests.

C.86 Internal optic algebra, physical dictionary, and falsifier regimes

The internal operations I, X, Z, F_3, and S generate the finite single-qutrit projective Clifford action of size $216 = 9 \cdot 24$ on the state/operator Choi legs.

The operations are mapped to optical analogues: reference/no-op, OAM shift or prism step, spiral phase plate, tritter or mode mixer, and lens or phase-curvature analogue. The lens row remains calibration-level.

The falsifier simulator separates passive OAM labels from active internal-operator behavior. Only the internal-operator scenario passes the symbolic criteria: trace-Choi pattern, low leakage, basis covariance, and operator activation.

C.87 Internal Clifford census, lens calibration, and passive-active protocol

The internal operations I, X, Z, F_3, and S generate a 216-element single-qutrit projective Clifford action on the state/operator Choi legs. All 216 elements preserve the state/operator split in the internal-register model; only the identity stabilizes the identity Choi state itself. The 24 zero-translation frame changes preserve centered OAM opposition, while the 192 translated elements move the OAM origin and require recentering.

The lens calibration row is exact at the finite phase-signature level. On the centered OAM basis $\ell = -1, 0, +1$ with qutrit labels $2, 0, 1$, qutrit S has ω-exponent signature $[1, 0, 1]$, matching the centered quadratic lens phase ω^{ℓ^2}.

The passive-vs-active protocol uses centered OAM preparation, axial time-bin support, operator off/on controls, $I/X/Z/F_3/S$ settings, leakage checks, basis covariance, and external-reference comparison. It is a protocol design, not experimental evidence.

C.88 OAM recentering, exact optical calibration, protocol table, and literature bridge

The 192 translated Clifford elements split into eight nonzero shift classes, each with 24 frame choices. The recentering correction is the inverse shift, with OAM-only, phase-only, and mixed shift classes.

The S and F_3 calibrations are placed on the same centered OAM basis. The S row has phase signature $[1, 0, 1]$, and the F_3 row is the finite three-mode mixer on the same qutrit labels.

The passive-vs-active protocol is converted into a paper-ready table covering preparation, passive controls, active gate settings, leakage, covariance, reference comparison, expected readouts, thresholds, failure modes, and claim tiers.

The OAM literature bridge supports high-dimensional qudit and same-photon register directions, but requires radial leakage and mode-overlap witnesses because radial and azimuthal spatial structure are coupled in realistic modes.

C.89 Recentered witnesses, leakage regimes, and appendix splicer

The calibrated I, X, Z, F_3, S witness applies directly to the centered frame class. Translated Clifford elements require inverse recentering before the same witness table is applied.

The radial leakage simulator turns symbolic shell leakage into pass/fail regimes, separating good core-gate behavior from radial leakage, visibility mismatch, and basis-covariance failures.

The operator/OAM appendix splicer follows the dry-run and idempotent splicer pattern. It records the appendix insert packets and changes the paper only when apply mode is used in checkout.

C.90 Appendix validation, recentered protocol table, and claim ledger

The appendix packet is validated by a dry-run artifact checker: all insert paths exist, the bounded splice block is idempotent, and the target paper admits the expected splice anchor or bounded fallback.

The protocol table is expanded by recentering class. The resulting table has four Clifford recentering classes times five calibrated witnesses, with correction operators, leakage thresholds, failure modes, and claim tiers.

The claim ledger separates exact finite claims from calibration-level optics, engineering leakage, protocol witnesses, literature motivation, and blocked physical overclaims.

C.91 Full operator/OAM appendix splice and recentered transaction ABI

The operator/OAM appendix is now part of the main Holonet paper through a bounded idempotent splice. The splice carries the operator-on-photon circuit model, the internal Clifford falsifier, the recentering protocol, the leakage and claim ledger, and the present synthesis block.

The architectural closure is the recenter-to-transaction ABI:

$$216 = 9 \cdot 24 = (1 + 2 + 2 + 4) \cdot 24.$$

The factor 24 is the centered BT1495 transaction-word kernel. The nine affine shifts are one centered class, two OAM-only shifts, two phase-only shifts, and four mixed shifts. Thus the class profile is

$$24, \quad 48, \quad 48, \quad 96,$$

and every translated internal Clifford action is executed by inverse recentering followed by one of the same 24 centered 72-tick transaction words.

One operation sweep therefore has $216 \cdot 72 = 15552$ ticks. The full five-witness I, X, Z, F_3, S protocol has 77760 ticks. This is an exact finite ABI statement: it certifies the centered word reuse and the recentering burden. It does not claim measured OAM leakage, optical loss, or a finished calibrated device.

Recent OAM and high-dimensional time-bin experiments support the physical direction, including same-photon polarization/OAM gates, robust time-bin qudit control, directly measured topology in OAM-entangled states, and OAM-multiplexed diffractive mode processing [32, 33, 34, 35]. BT1588 keeps those sources in the literature-motivation tier only: they motivate the mode interface, multiplexing optics, and leakage tests, while the exact theorem remains the local finite $216 = 9 \cdot 24$ recenter ABI.

C.92 LG radial covariance, full witness lane sheet, and OAM front end

The recenter ABI now has a radial-shell falsifier. Model the first three Laguerre–Gaussian radial shells by the stochastic channel family $L(\eta)$: diagonal entries $1-\eta$, off-diagonal entries $\eta/2$. The BT1577 operation envelopes I, Z, S, X, F_3 all lie in this same channel family, so the radial recenter correction and the centered witness gate commute at shell level. For two envelopes,

$$\eta(a,b) = a + b - \frac{3}{2}ab.$$

Thus the $216 = 9 \cdot 24$ affine actions do not require 216 separate radial calibrations. They require one centered witness table plus the recenter tax. The symbolic worst case is a mixed OAM/phase recenter followed by F_3,

$$\eta_{\max} = 0.18296 < 0.20,$$

while the centered gate threshold remains the BT1577 value 0.10. This is a covariance simulator and falsifier, not measured beam leakage.

The complete witness replay is now compiled as a physical lane sheet:

$$5 \text{ gates} \times 9 \text{ recenter sectors} \times 24 \text{ centered words} \times 72 \text{ ticks} = 77760.$$

It is stored as 1080 exact 72-tick segments with the global tick formula

$$((((g \cdot 9) + s) \cdot 24 + w) \cdot 72 + t).$$

Every Hesse lane appears 12960 times, every detector slot appears 19440 times, native D_4 square-pulse words occupy 25920 ticks, and the remaining S_4 analyzer-relabel words occupy 51840 ticks.

The hardware interpretation is deliberately sharper. OAM-multiplexed diffractive optics are a proposed *front-end sectorizer*: a passive mode-processing layer can sort or fan out the nine affine recenter sectors before the exact 24-word transaction kernel runs. Recent OAM-MDNN work supports this as an optical mode-processing direction, but it does not prove a quantum gate, preserve coherence by itself, or calibrate loss. The promoted theorem remains local:

$$\boxed{9 \text{ recenter sectors} \times 24 \text{ centered words} = 216 \text{ exact finite ABI addresses,}}$$

with acceptance delegated to radial tomography, sector crosstalk measurements, and the 77760-tick witness replay.

C.93 Lab tomography, LG alphabet, and the Hesse/T witness loop

The OAM front end is now written in a lab-facing form. BT1592 promotes three acceptance surfaces: a 9×9 sector confusion matrix, four radial-shell tomography rows, and six exact lane-replay checks. The synthetic sector fixture uses diagonal probability 0.93, off-diagonal probability 0.00875, and acceptance thresholds

$$p_{\text{diag}} \geq 0.90, \qquad p_{\text{off}} \leq 0.02.$$

The generated CSV ingest file has 91 rows, all passing, and is deliberately replaceable by bench data. This is a falsification harness, not measured OAM crosstalk.

BT1593 chooses the explicit Laguerre–Gaussian sector alphabet:

$$s = 3x + z, \qquad \ell = s - 4, \qquad p = x + 2z \pmod 3.$$

Thus the nine recenter sectors occupy symmetric OAM charges

$$\ell = -4, -3, -2, -1, 0, 1, 2, 3, 4,$$

with three modes in each radial shell $p = 0, 1, 2$. The 24-word centered selector is a separate handoff layer:

$$\text{address} = 24s + w, \qquad 0 \le s < 9, \quad 0 \le w < 24.$$

The architecture therefore uses 9 physical sector modes plus a 24-word selector, not 216 separate OAM channels. The 216 exact finite addresses split into 72 native D_4 square-pulse addresses and 144 S_4 analyzer-relabel addresses.

The universality port now lands in the same witness loop. BT1404 already made a Hesse/T microframe with

$$9 \text{ Hesse outcomes} \times 8 \text{ ticks} = 72 \text{ ticks}.$$

BT1590 already made each OAM witness segment 72 ticks. BT1594 overlays these rather than appending a second schedule:

$$1080 \text{ segments} \times 72 \text{ ticks} = 1080 \text{ Hesse/T microframes} = 77760 \text{ ticks}.$$

Each Hesse outcome appears once per segment, hence 1080 times total. The T-frame bit profile is $0 : 5400$ and $1 : 4320$, matching the parity rule

$$t = (r + p) \bmod 2.$$

This closes the finite architecture loop: the Clifford/OAM witness, the sector-mode front end, the tomography acceptance harness, and the explicit Hesse/T non-Clifford port now inhabit one replayable 77760-tick object. The claim boundary remains unchanged: this is ABI and schedule compatibility, not a physical Hesse-SIC device, injection-fidelity measurement, or fault-tolerant magic-state-yield theorem.

C.94 The Witting reject shell as the universal fuel rail

The deepest closure in the current architecture is that the newest OAM/Hesse witness loop is not merely a validation sweep. It is the Witting matter shell itself. In the Witting delayed-query desk, each selected ray has

$$1 \text{ same} + 12 \text{ compatible} + 27 \text{ incompatible} = 40$$

possible queried rays. Therefore the incompatible ordered-pair shell has

$$40 \cdot 27 = 1080$$

records. BT1595 constructs an explicit bijection

$$40 \cdot 27 = 5 \cdot 9 \cdot 24$$

between those rejected Witting pairs and the BT1590/BT1594 witness segments. The per-gate refinement is even sharper:

$$9 \cdot 24 = 216 = 8 \cdot 27.$$

Thus each of the five witness gates consumes the full incompatible-target shell of eight Witting source rays. Each rejected Witting pair is one 72-tick contextual fuel segment, and each such segment carries the Hesse/T overlay.

BT1596 writes the resulting runtime economy:

$$520 \cdot 72 = 37440 \qquad \text{accepted communication/control ticks,}$$

$$1080 \cdot 72 = 77760 \qquad \text{contextual fuel ticks,}$$

and

$$1600 \cdot 72 = 115200 \qquad \text{ticks for the complete Witting ordered-pair cycle.}$$

The ratio of accepted control to contextual fuel is $13 : 27$, matching the Witting split $13/40$ versus $27/40$.

BT1597 packages this as a single universal transaction object. Accepted Witting pairs form the communication and control rail. Rejected Witting pairs are not discarded; they are exactly the OAM/Hesse contextual-fuel rail:

$$1080 = 5 \text{ gates} \cdot 9 \text{ OAM sectors} \cdot 24 \text{ words} = 40 \text{ rays} \cdot 27 \text{ incompatible targets.}$$

The Hesse/T microframe on that fuel rail supplies the explicit non-Clifford boundary required by the universal-computation contract. The claim remains a finite ABI and timing theorem: it does not assert optical coherence, loss tolerance, cryptographic security, or a fault-tolerant threshold.

C.95 Accepted control rail and the full Witting transaction cycle

The Witting fuel rail now has its complementary control rail. BT1598 compiles the accepted $13/40$ Witting ordered pairs into concrete 72-tick control frames. The only ambiguity is the same-ray case: a ray lies in four Witting bases. BT1598 resolves this by a perfect matching between the 40 rays and the 40 bases. Therefore each basis carries exactly one same-ray aperture and twelve off-diagonal compatible apertures:

$$40 \text{ bases} \cdot 13 \text{ controls} = 520 \text{ accepted frames,}$$

$$520 \cdot 72 = 37440 \text{ accepted control ticks.}$$

The detector handoff is the existing one:

$$\text{Witting analyzer slot} \to \text{detector slot} \to \text{mirror slot mod } 4 \to \text{tomotope flag.}$$

Using the selected Witting basis as a tomotope block gives

$$\text{tomotope flag} = 4\,\text{basis} + \text{detector slot,}$$

using the first 40 of the 48 tomotope blocks and leaving 8 slack blocks for non-Witting packet traffic.

BT1599 explains the 120-record surplus in the basis-local Witting table. There are 160 same-ray basis-local apertures, but BT1598 promotes only 40 of them to selected control frames. The remaining records are

$$160 - 40 = 120 = 3 \text{ local tomotope sheets} \cdot 40 \text{ W33/Witting lines,}$$

exactly the BT1365 selector phase-sheet bundle. Thus the same-ray surplus is a qutrit phase sidecar rather than extra communication throughput.

BT1600 then compiles the full ordered-pair desk:

$$40 \text{ source rays} \cdot 40 \text{ target rays} = 1600 \text{ frames,}$$

$$1600 \cdot 72 = 115200 \text{ ticks.}$$

Every source ray has

$$13 \text{ accepted control frames} + 27 \text{ contextual fuel frames.}$$

The complete cycle is therefore not two unrelated protocols. It is one finite transaction machine: accepted pairs run the Witting analyzer/mirror control rail, rejected pairs run the OAM/Hesse contextual fuel rail, and the unused same-ray contexts are exactly the ternary selector phase sheets.

C.96 Physical automaton, Fano detector bins, and the universal ABI

BT1601 turns the 1600-frame Witting transaction cycle into a single physical automaton. The carrier is one single-photon transaction packet. Each ordered pair frame keeps the common 72-tick shape:

$$1600 \cdot 72 = 115200 \text{ ticks.}$$

The frame template is now explicit:

ticks	role
$0 \ldots 7$	source switch
$8 \ldots 31$	program/delay placeholder
$32 \ldots 47$	analyzer or OAM/Hesse body
$48 \ldots 63$	detector or Hesse/T handoff
$64 \ldots 71$	dark-reference placeholder

There are 520 Witting analyzer switch frames and 1080 OAM/Hesse fuel switch frames. The four detector residues are balanced across the whole automaton:

$$400,\ 400,\ 400,\ 400.$$

Every frame has a symbolic loss channel and a dark-reference bin. These fields are intentionally placeholders: BT1601 compiles the interface, not a measured loss or dark-count model.

BT1602 identifies the old 168 active detector-bin count with the Fano plane. The active bus factors as

$$168 = 7 \cdot 24 = 7 \text{ Fano lines} \cdot 3 \text{ point slots} \cdot 8\ D_4 \text{ states.}$$

The Witting source side factors as

$$40 = 5 \text{ witness gates} \cdot 8\ D_4 \text{ source states.}$$

The five witness gates occupy five Fano lines. On those lines,

$$27 = 3 \text{ point slots} \cdot 9 \text{ Hesse/OAM residues}$$

is the contextual fuel shell. The two remaining Fano lines carry the accepted compatible controls:

$$12 = 2 \text{ reserve lines} \cdot 3 \text{ point slots} \cdot 2 \text{ detector parities.}$$

The same-ray control is the source anchor on the gate line. Thus every source ray has

$$27 \text{ fuel} + 12 \text{ compatible control} + 1 \text{ same-ray control} = 40$$

targets, and every one of the 168 active detector bins is touched. The usage profile across the 1600 frames is

$$80 \text{ bins} \times 9 \quad + \quad 88 \text{ bins} \times 10 \quad = \quad 1600.$$

BT1603 packages the architecture as a finite universal-computation ABI theorem. The accepted Witting frames provide finite Clifford transport through the analyzer/mirror/tomotope handoff. The rejected Witting frames provide the contextual OAM/Hesse fuel. The Hesse/T overlay supplies the explicit non-Clifford port required by the BT1377 universal-computation contract. The detector and row values then hand into the BT1486 retwined CSS syndrome layer and the BT1493 physical row-pulse compiler. In short,

$$\text{Witting control} + \text{contextual fuel} + \text{Hesse/T port} + \text{CSS handoff}$$

is now one finite programmable photonic-computation ABI. The theorem remains inside the claim firewall: thresholds, measured loss, detector efficiency, and magic-state yield are downstream physical requirements, not consequences of the finite compiler alone.

C.97 Fano charge, time-bin envelope, and guard-page closure

The $80 \times 9 + 88 \times 10$ detector-bin profile is not just a usage histogram. BT1648 shows that it is a conserved Fano charge. The 168 active detector bins split into three exact classes:

$$80 \cdot 9 + 40 \cdot (9 + 1) + 48 \cdot 10 = 1600.$$

The 80 bins are unanchored contextual-fuel bins. The 40 middle terms are fuel bins with one same-ray control anchor. The 48 final bins are compatible-control reserve bins. Equivalently, the seven Fano lines split as five Witting gate lines and two reserve lines:

$$5(16 \cdot 9 + 8 \cdot 10) + 2(24 \cdot 10) = 1600.$$

This explains why the high-usage side has size 88: it is exactly

$$40 \text{ same-ray anchors} + 48 \text{ compatible reserve bins}.$$

BT1649 then embeds the active automaton into an 11-bit time-bin qudit envelope, motivated by single-photon time-bin qudit universal logic [29]. The active Witting transaction uses the first 1600 time bins. The remaining addresses factor without remainder:

$$2^{11} = 2048, \qquad 2048 - 1600 = 448 = 7 \cdot 64.$$

Thus every Fano point receives a 64-slot guard page. Each guard page is typed as

$$24 \text{ dark references} + 24 \text{ loss probes} + 16 \text{ parity overflow slots}.$$

The Witting source/fuel structure remains the contextual communication desk of [28]; the time-bin envelope is the physical address wrapper around that finite desk, not a new communication assumption.

BT1650 closes the calibration interface. The $7 \cdot 24 = 168$ dark-reference guards cover all active detector bins once, and the 168 loss-probe guards also cover all active detector bins once. The remaining $7 \cdot 16 = 112$ parity-overflow guards feed the CSS/jitter side of the fault ABI:

$$\text{dark} \mapsto \text{DARK\_CLICK}, \qquad \text{loss} \mapsto \text{MISSED\_CLICK}, \qquad \text{parity} \mapsto \text{CSS\_SYNDROME}.$$

This is the new physical closure: the 448 time-bin slack addresses are not unused capacity. They are the calibrated guard shell required to run the 1600-frame Witting automaton as a bench-addressable single-photon qudit program.

C.98 Exact Schläfli–W33 transport capsule (Pass 1147)

In plain language. *A self-contained package: the exact correspondence between our geometry and a classical configuration of lines studied since the nineteenth century, with everything needed to check it in one place. "Capsule" means it can be lifted out and verified without reading the surrounding sections.*

Theorem C.1 (Integral Schläfli–Steinberg bridge). *The three 432-element A_2-colour carriers are three copies of the directed edges of the Schläfli graph* $\mathrm{SRG}(27,16,10,8)$*. The explicit equivariant transform from one directed-edge carrier to the* 81_- *constituent has rank* 81*, is odd under edge reversal, and sends its* 432 *vectors to* 216 *antipodal lines forming a tight frame. On the integral contraction-zero chart its Smith profile is*

$$1^{15}, \quad 2^{6}, \quad 4^{8}, \quad 8^{29}, \quad 40^{23};$$

hence its internal bad primes are exactly 2 *and* 5*. Modulo* 5*, its saturated* 81*-space* S_5 *lies in the nonsplit exact sequence*

$$0 \longrightarrow I_{58} \longrightarrow S_5 \longrightarrow K(W_{3,3})_{(5)} \otimes \mathrm{sgn} \longrightarrow 0$$

over both $W(E_6)$ *and* $PSp(4,3)$*, and* I_{58} *is its unique proper nonzero submodule. Splitting the three colour copies into the trivial and augmentation Fourier sectors contributes the independent index* 3^{81}*. Adjoining the disjoint rank-*45 *cubic block to the three rank-*81 *colour blocks gives an explicit rank-*288 *map with kernel dimension* 1952*.*

Scope. This theorem concerns specified integral $W(E_6)$-modules, maps, lattices, and projective frame lines. It neither identifies the three colour copies with particle generations nor supplies an optical state encoding, gate, decoder, or hardware performance claim.

D Modular Hecke degeneration, triality gauge geometry, and cycle selectors

Let $\mathcal{H}_{26}$ denote the literal rational Hecke algebra of the 432-point carrier $W(E_6)/S_5$. Reducing the exact integral multiplication table modulo the three bad primes gives

p	$\dim J_p$	$\dim(\mathcal{H}_{26,p}/J_p)$	$\mathcal{H}_{26,p}/J_p$
2	21	5	$M_2(\mathbf{F}_2) \oplus \mathbf{F}_2$
3	22	4	$\mathbf{F}_3^4$
5	6	20	$M_3(\mathbf{F}_5) \oplus M_2(\mathbf{F}_5) \oplus \mathbf{F}_5^7$.

The exact Loewy power dimensions are respectively

$$(21, 17, 13, 7, 2, 0), \qquad (22, 16, 10, 4, 0), \qquad (6, 2, 0).$$

Thus characteristics 2 and 3 produce deep non-semisimple collapse, whereas characteristic 5 produces only a two-step radical.

Building on Pass 1327's literal species-20 gauge grid, consider its abstract 3×3 coordinate scheme under $S_3^{\mathrm{int}} \times S_3^{\mathrm{tri}}$. Its ordered-pair orbitals have valencies

$$1, 2, 2, 4$$

and first eigenmatrix

$$\begin{pmatrix} 1 & 2 & 2 & 4 \\ 1 & -1 & 2 & -2 \\ 1 & 2 & -1 & -2 \\ 1 & -1 & -1 & 1 \end{pmatrix}.$$

The four primitive ranks are $1, 2, 2, 4$. Adjoining the coordinate swap fuses the two middle relations and yields the ordinary Hamming scheme $H(2, 3)$ with valencies $1, 4, 4$.

Two primitive reduced cyclic words were also selected in the W33 graph. The chosen length-7 cyclic order has ordered and dihedral stabilizer of order 2 and orbit 25920, while the chosen length-8 cyclic order has trivial stabilizer and orbit 51840. These values do not classify every cycle orbit of either length. The length-7 support has stabilizer 12 and five chords. These cycles break $W(E_6)$ symmetry, but they do not by themselves choose one of the three species-20 multiplicity coordinates: Pass 1328 proves that an invariant Y-side operator acts on the transported multiplicity space as $C \otimes I_3$. A primitive copy-idempotent is an additional S_3 gauge choice. The combined cycle–copy selector orbits under $W(E_6) \times S_3$ therefore have sizes 77760 and 155520 for the chosen lengths 7 and 8.

The GAP/AtlasRep certificate and the manuscript build are separate runtime checks. Their status must be read from the release workflow; the finite-algebra theorem above does not depend on treating a missing runtime as a successful execution.

E Brauer-tree completion of the five-primary frame extension

Let $k = \mathbf{F}_5$, let $G = W(E_6) \cong U_4(2).2$, and let $G' = PSp(4, 3) \cong U_4(2)$. GAP/CTblLib gives cyclic Sylow 5-subgroups of order 5. For both groups, the defect-one block containing the ordinary 81-character has the following shape; for the outer group there are two sign-twist-related 81-blocks, and GAP checks both:

$$1 \;-\; 23 \;-\; 58 \;-\; 6,$$

where the four entries are the dimensions of the simple edge modules, not the ordinary vertices. Equivalently, the ordinary vertices have degrees

$$1 \;-\; 24 \;-\; 81 \;-\; 64 \;-\; 6,$$

and the 23- and 58-edges meet at the ordinary 81-vertex. The exceptional multiplicity is one. Brauer-tree algebra theory therefore gives

$$\dim_k \operatorname{Ext}^1_G(23, 58) = \dim_k \operatorname{Ext}^1_G(58, 23) = 1,$$

with the same equalities for G'.

Pass 1147 already constructs a nonsplit sequence

$$0 \longrightarrow I_{58} \longrightarrow S_{81} \longrightarrow Q_{23} \longrightarrow 0$$

over both groups. Its class consequently spans the full $\operatorname{Ext}^1_G(Q_{23}, I_{58})$ and $\operatorname{Ext}^1_{G'}(Q_{23}, I_{58})$: the nonsplit middle-module type is unique up to rescaling its two simple endpoints.

The literal 432-character contributes $2 \cdot 6 + 2 \cdot 64 + 81$ in this defect block, so the corresponding rational Hecke-commutant contribution has dimension $2^2 + 2^2 + 1^2 = 9$. This is the nonsemisimple nine-dimensional central block of Pass 1330. The other nine-dimensional block is the defect-zero species-20 contribution $M_3(\mathbf{F}_5)$. A direct GAP derivation calculation on the certified 26^3 multiplication tensor gives the scalar-simple Ext quiver

$$h_6 \rightleftarrows h_5 \rightleftarrows h_7,$$

each displayed Ext space being one-dimensional. Its radical powers have dimensions $6, 2, 0$. Here h_i is the i-th one-dimensional simple $\mathcal{H}_{26,\mathbf{F}_5}$-module in the frozen seven-character order; the subscripts are Hecke-character indices, not module dimensions.

This Hecke block is a condensation shadow of the same cyclic-defect group block, not the literal $58|23$ middle module. The full group Brauer tree, not the Hecke quiver alone, proves the Ext theorem.

Finally,

$$\mathbf{F}_5[C_3] \cong \mathbf{F}_5 \times \mathbf{F}_{25}.$$

Since characteristic 5 divides neither $|C_3|$ nor $|S_3|$, color triality is semisimple and cross-color Ext vanishes. The colored middle space splits as $81 + 162$ over $\mathbf{F}_5$, and as three independent 81-extensions after scalar extension to $\mathbf{F}_{25}$. Triality transports the unique class; it neither creates it nor selects a physical copy.

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F Recovered write-ups

F.1 Fano bus figure and retwined front-end closure

The Fano-active interpretation gives the front end a compact diagrammatic law:

$$168 = 21 \cdot 8 = |GL(3,2)| = |PSL(2,7)|, \qquad 24 = |\operatorname{Stab}_{GL(3,2)}(p)|, \qquad 192 = 168 + 24.$$

The active side is the Fano flag orbit with local flag stabilizer, equivalently

$$21 \text{ Fano flags} \times 2 \text{ orientations} \times 4 \text{ residues} = 168.$$

The guard side is the separated tetrahedral point stabilizer. The D4 quartic shear is therefore not inserted into the active mesh; it is routed through the 24-aperture guard rail and becomes legal only with the retwined CSS frame update

$$H_X' = H_X J^{-1}, \qquad H_Z' = H_Z J^{-1}, \qquad e \mapsto Je.$$

The 211 frontier is also sharpened by this picture. A packet-symmetric score above 210 must jump to 212 because the quotient weights are $10, 8, 6$. Thus a 211 witness, if it exists, must split one raw correction slot out of the Fano or Steinberg packet orbit and must be at S3-gauge radius at least four from the current incumbent.

BT1561–BT1563 record the spiral-qutrit packet: centered OAM basis with mod-three labels, azimuthal/radial/axial factor schema, and falsifier checks for optical mode support and sector correlation.

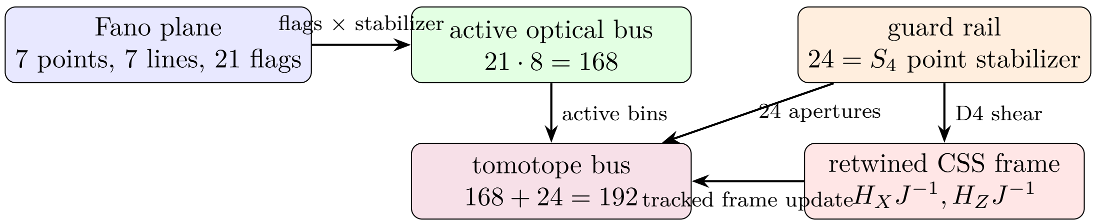

Figure 21: Fano bus closure. The active 168 detector bins are the Fano flag orbit with local order-8 stabilizer. The 24 guard apertures are the tetrahedral point stabilizer. The non-Clifford D4 shear is legal only after retwining the CSS frame.

F.2 The quartet fibre law

The residual ternary obstruction in the H27/Schläfli transport is not a new global count identity. It is local. The $W(E_6)$ stabilizer of the transported eighteen-line image does not fix the observed defect triple uniquely; instead it distinguishes a six-element hinge slice. Since

$$6 = \binom{4}{2},$$

this slice is naturally the edge set of a hidden four-state fibre quartet.

We label the four fibre states by

$$\mathbb{F}_2^2 = \{00, 01, 10, 11\}.$$

Equivalently, this is the local D4/GKP Pauli square

$$00 \leftrightarrow I, \qquad 01 \leftrightarrow X, \qquad 10 \leftrightarrow Z, \qquad 11 \leftrightarrow XZ.$$

Under the committed edge-to-hinge chart, the observed support

$$(10, 22, 44)$$

corresponds to the diagonal quartet edge

$$00 \longrightarrow 11,$$

that is, to the simultaneous XZ half-shift in the local fibre square.

The resulting local correction law is:

$$\boxed{\text{oriented quartet edge } = \text{ two endpoint losses plus one edge-target gain, in units of 2.}}$$

For the observed edge this gives

$$\boxed{T010 : -2, \qquad T210 : -2, \qquad T222 : +2.}$$

The uncorrected count vector has double-six syndromes

$$F_2 = (0, 0), \qquad F_3 = (0, 2, 1, 1, 1).$$

The quartet-edge correction contributes

$$F_2 = (0, 0), \qquad F_3 = (0, 1, 2, 2, 2),$$

so the corrected vector satisfies

$$\boxed{F_2 = (0,0), \qquad F_3 = (0,0,0,0,0).}$$

Thus the binary Hesse delta split is preserved while the ternary double-six obstruction is cancelled.

The search problem is correspondingly reduced:

$$816 \text{ triples} \longrightarrow 54 \text{ Hesse hinges} \longrightarrow 10\ W(E_6)\text{-stabilizer slices} \longrightarrow 6\ K_4 \text{ edges} \longrightarrow 1 \text{ oriented edge.}$$

This is the operative fibre-law statement: $W(E_6)$ selects the hidden quartet edge-slice, while the remaining orientation is the local $12 = 3 \times 4$ fibre datum.

F.3 Finite-delay support observability

The $\mathrm{PG}(3,2)$ support shell is not an instantaneous execution quotient of the 81-state ternary frame, but it is an exact finite-delay observer. Refinement by all support observations reachable with words of bounded length gives

$$\boxed{16 \longrightarrow 40 \longrightarrow 78 \longrightarrow 81},$$

with unresolved state-pair counts

$$\boxed{272,\ 53,\ 3,\ 0}.$$

Thus the adaptive/all-word observability index is three. The three pairs surviving through depth two are

$$(0,0,1,z_f) \sim (0,0,2,z_f), \qquad z_f \in \mathbb{F}_3,$$

and all are split by length-three diagnostic words.

A fixed preset experiment requires six operations. The best numbers of distinct support trajectories for word lengths one through six are

$$\boxed{25,\ 40,\ 45,\ 68,\ 77,\ 81}.$$

Exactly eight length-six words are injective. One canonical word is

$$\boxed{\mathrm{CX}_{p\to f},\ F_p,\ Z_p,\ F_p,\ Z_p,\ \mathrm{CX}_{p\to f}}.$$

Its seven four-bit support snapshots contain 28 raw bits. Exhaustive tap selection shows that seven taps never suffice, while exactly 48 eight-tap selectors distinguish all 81 states. A canonical selector uses flattened columns

$$\boxed{(0,1,2,5,13,21,25,26)}.$$

No minimal selector samples the z_f support bit. Consequently the machine may retain full ternary phase internally while exposing only binary support telemetry: a six-cycle diagnostic sequencer, eight one-bit samples, and an 81-entry ROM recover the exact frame. The result is noiseless observability; no robustness against loss, detector error, or gate imperfection is asserted.

G Passes 3989–3996: physical incidence photon processor

The W33 point graph admits an exact sparse physical lift. Let N be the 40×40 point–line incidence matrix of $W(3,3)$. Exact enumeration gives

$$NN^T = 4I + A_{W33}.$$

Hence a dense twelve-neighbour point Hamiltonian may be generated from an eighty-mode bipartite point–line device with 160 couplings, degree four, and girth eight. Its spectrum is

$$-4^1, -\sqrt{6}^{24}, 0^{30}, \sqrt{6}^{24}, 4^1.$$

With point–line coupling g and bus detuning Δ,

$$H_{\text{eff}} = -\frac{g^2}{\Delta}(A + 4I) + O(g^4/\Delta^3),$$

and $t = \pi\Delta/(2g^2)$ approaches the exact W33 half-period reflection on the point sector. This is an analytic architecture, not a fabricated device.

Let $B = J - I - A$ and $D = A - B$. Then $A + B = J - I$ is geometry-blind on the nonuniform sector, while

$$\mathrm{Spec}(D) = -15^1, 5^{24}, -7^{15}.$$

The spectral projectors are

$$E_{-15} = \frac{(D - 5I)(D + 7I)}{160} = \frac{J}{40}, \quad E_5 = \frac{(D + 15I)(D + 7I)}{240}, \quad E_{-7} = -\frac{(D + 15I)(D - 5I)}{96}.$$

Consequently $\langle D\rangle$ and $\langle D^2\rangle$ determine all three sector populations. If identical commuting error C enters both arms, then

$$e^{-it(A+C)}e^{+it(B+C)} = e^{-it(A-B)},$$

so the dual-geometry echo cancels it exactly.

The maximum-code census is also complete. The 945-vertex fixed-parent compatibility graph contains exactly 945 maximum 57-cliques, split into three parent-group orbits

$$\boxed{945 = 540 + 270 + 135}$$

with stabilizer orders

$$\boxed{96, 192, 384}.$$

Thus maximum-code uniqueness is false in the fixed-parent problem. This corrects the earlier division by an order-192 full coordinate stabilizer without first checking parent preservation.

The seven primitive central blocks of the literal 48-orbital algebra have simple degrees

$$1, 1, 2, 2, 2, 3, 5.$$

Their complete central coarse-graining lattice contains

$$B_7 = 877$$

partitions. Quotienting equal-degree relabelings by $S_2 \times S_3$ leaves exactly

$$\boxed{198}$$

inequivalent central types. These are exact central subalgebras, not automatically combinatorial fusions of the 48 orbital relations.

Three further photon constructions follow from the incidence lift. First, rank $N = 25$, so the bipartite processor contains thirty exact dark modes. Second, for $L = 12I - A$,

$$V = e^{-i\pi L/4} = I - (1 + i)E_{10}, \qquad V^2 = U, \qquad V^4 = I,$$

which gives an exact four-phase internal graph clock. Third, for a genuine m-fold tensor carrier,

$$\operatorname{rank}(N^{\otimes m}) = 25^m, \qquad \dim\ker(N^{\otimes m}) = 40^m - 25^m,$$

so the one-side dark fraction is

$$\boxed{1-(5/8)^m}.$$

Only $m+1$ nonzero singular shells occur, with multiplicity $\binom{m}{r}24^r$ and squared singular value $16^{m-r}6^r$.

The Monster acquisition gate now composes $U_4(2) \to H \to \mathbb{M}$ class fusions through maximal-subgroup tables and searches only compatible published explicit `mmgroup` overgroups. The promoted status remains

$$\boxed{\texttt{PENDING\_EXPLICIT\_MONSTER\_U42\_WORDS}}$$

until four portable words pass the exact pair, triple, order-25,920, object-action, and class-fusion firewalls. No variable vacuum c, literal hidden photon nodes, fabricated hardware, laboratory result, or unobserved remote build is claimed.

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The Monster acquisition gate now composes $U_4(2) \to H \to \mathbb{M}$ class fusions through maximal-subgroup tables and searches only compatible published explicit `mmgroup` overgroups. The promoted status remains

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until four portable words pass the exact pair, triple, order-25,920, object-action, and class-fusion firewalls. No variable vacuum c, literal hidden photon nodes, fabricated hardware, laboratory result, or unobserved remote build is claimed.

Passes 3989–3996 supplement: direct W33 flight and causal memory

Exact forty-mode analog flight. Let A be the adjacency matrix of $W(3,3)$. A forty-mode equal-edge coupled-mode array obeying

$$i\frac{d\psi}{dz} = \kappa A\psi$$

implements, at $\kappa z = \pi/2$,

$$\boxed{e^{-i\pi A/2} = -\frac{I+A}{3} + \frac{2J}{15}.}$$

This is an exact symmetric involution with row amplitudes $-1/5$ on a point and its twelve neighbours and $2/15$ on its twenty-seven nonneighbours. For uniform fractional interaction error ϵ,

$$F_{\text{pro}} = 1 - 3\pi^2\epsilon^2 + O(\epsilon^3), \qquad F_{\text{avg}} = 1 - \frac{120}{41}\pi^2\epsilon^2 + O(\epsilon^3).$$

The linear average-fidelity term vanishes because $\operatorname{Tr} A = 0$.

Wigner–Smith time as operational memory. For a frequency-dependent W33 scattering unitary

$$S(\omega) = e^{i\theta(\omega)L_{W33}},$$

the Wigner–Smith operator is exactly

$$\boxed{Q(\omega) = -iS^\dagger\frac{dS}{d\omega} = \theta'(\omega)L_{W33}.}$$

Hence the proper-delay sectors are

$$0^1, \qquad (10\theta')^{24}, \qquad (16\theta')^{15},$$

with mean $12\theta'$, variance $12(\theta')^2$, and total trace $480\theta'$. This gives a measurable meaning to internal history: dwell-time and stored-energy sensitivity, not a changed vacuum causal speed.

For m independent W33 factors, the exact delay-shell polynomial is

$$\boxed{(1 + 24z^{10} + 15z^{16})^m.}$$

The address space has 40^m modes and

$$\langle\tau\rangle = 12m\theta', \qquad \operatorname{Var}(\tau) = 12m(\theta')^2.$$

Consequently

$$\boxed{\frac{\log_2(40^m)}{\langle\tau\rangle} = \frac{\log_2 40}{12\theta'}}$$

is independent of self-similar depth, while the relative delay width is $1/\sqrt{12m}$.

Causal-refinement boundary. A local-interaction causal cone has schematic physical velocity

$$v_{\text{int}} \lesssim aC_{LR}(\Delta)J/\hbar.$$

Self-similar refinement preserves the cone only when aJ remains fixed. More serial nodes without stronger coupling slow the internal cone; node density alone neither determines nor raises vacuum c. Fabrication, measured Wigner–Smith delays, literal hidden photon nodes, variable vacuum c, and laboratory performance remain unclaimed.

H.1 Folded Hashimoto propagation and the protected Hodge endpoint

The directed nonbacktracking carrier of the W33 collinearity graph has a canonical fold onto the W33 point-line Levi flag carrier. If

$$B \in \{0,1\}^{480\times 480}$$

is the directed Hashimoto operator on directed W33 edges, define

$$T : \mathbb{R}^{480} \longrightarrow \mathbb{R}^{160}$$

by

$$\boxed{T(p \to q) = (p, \ell(p,q)),}$$

where $\ell(p,q)$ is the unique W33 generalized-quadrangle line through the adjacent points p, q. Every directed edge has exactly one image, while every Levi flag has exactly three directed-edge preimages. Hence

$$\boxed{TT^T = 3I_{160}.}$$

Since W33 has valency $k = 12$, the Hashimoto outdegree is

$$\boxed{k - 1 = 11.}$$

Therefore the folded powers

$$\boxed{F_n = TB^nT^T}$$

have row sum

$$\boxed{3 \cdot 11^n.}$$

Thus the Ihara scale survives the directed-edge to Levi-flag fold.

Let A_0, A_1, A_2, A_3, A_4 be the distance relations of the Levi flag association scheme. The cubic folded operator has radial Bose–Mesner projection

$$\boxed{\mathcal{R}(F_3) = 12A_0 + 11A_1 + \frac{33}{2}A_2 + 25A_3 + 28A_4.}$$

The lower shells are not fully radial before projection. Their orientation residuals split as follows:

$$A_1 : \quad 4 \text{ on same-line pairs,} \qquad 18 \text{ on same-point pairs,}$$

$$A_2 : \quad 27 \text{ on forward cross-incidences,} \qquad 6 \text{ on backward cross-incidences,}$$

$$A_3 : \quad 24 \text{ when the source points are collinear,} \qquad 26 \text{ when they are noncollinear.}$$

But the terminal protected shell has no orientation residual:

$$\boxed{F_3 \circ A_4 = 28A_4.}$$

Here

$$\boxed{28 = v - k = 40 - 12 = \mu.}$$

The distance-four shell has

$$\boxed{12960 = 160 \cdot 81 = 1620 \cdot 8}$$

ordered flag pairs, so the cubic folded endpoint has total mass

$$\boxed{12960 \cdot 28.}$$

This gives the finite geometric mechanism behind the evolution axiom. Raw propagation is the folded Hashimoto process $F_n = TB^nT^T$. Physical propagation is obtained by passing this raw nonbacktracking transport through the Hodge/Bose–Mesner projector onto the protected Levi cycle sector

$$\boxed{E_4 = \frac{1}{160}CC^T = I - D^T(DD^T)^+D.}$$

Therefore the slogan

$$\boxed{\textbf{physical evolution = Hodge projection of Ihara/nonbacktracking propagation}}$$

now has a concrete finite operator model:

$$\boxed{B \xrightarrow{T(-)T^T} F_n \xrightarrow{\text{Bose–Mesner/Hodge projection}} E_4.}$$

The important boundary is that the fold is not an unrepaired adjacency intertwiner on all shells. It is an exact nonbacktracking transfer whose protected endpoint shell is already radial, while lower-shell orientation data must be projected before it becomes physical Hodge data.

H.2 The E_2 duad–phase carrier and the complexified G_2 boundary

The full folded-cubic Bose–Mesner action shows that the primitive E_2 sector is internally split:

$$E_2F_3E_2 : \qquad 77^{15} \oplus (-3)^{15}.$$

Equivalently, the E_2-block satisfies the quadratic relation

$$\boxed{x^2 - 74x - 231 = 0.}$$

Thus the spectral projectors inside E_2 are

$$\boxed{P_{77} = \frac{E_2F_3E_2 + 3E_2}{80},} \qquad \boxed{P_{-3} = \frac{77E_2 - E_2F_3E_2}{80}.}$$

Both have rank 15.

A natural explicit carrier for this split is the K_6-duad carrier tensored with a two-sheet phase sign:

$$\boxed{\mathbb{Q}^{15}_{K_6\text{ duads}} \otimes \mathbb{Q}^2_{\pm}.}$$

On this carrier the model operator is

$$\boxed{B_{E_2} = 37I + 40\sigma_z.}$$

Hence

$$37 + 40 = 77, \qquad 37 - 40 = -3.$$

This realizes the exact $15 + 15$ splitting as a phase-sheet splitting over the 15-duad basis.

This statement should be read with the same boundary as the $W(G_2)$-packet obstruction. The K_6-duad carrier gives a clean 15-label model for the two rank-15 halves, but it does not by itself prove that the numeric E_2-eigenbasis has already been canonically identified with the K_6-duad basis. That requires an additional intertwiner construction.

Likewise, the $E_1 + E_3$ sector admits an external $W(G_2) \cong D_6$ organization,

$$E_1 + E_3 \rightsquigarrow 4(G_2^{\text{short}}) \oplus 4(G_2^{\text{long}}),$$

but the folded-cubic cross-channel itself squares to $-I$ after normalization, not $+I$. Therefore it is a quadratic or complex conjugate transport channel. A real $W(G_2)$-reflection obeys $s^2 = +I$, while the folded-cubic channel obeys

$$\boxed{J^2 = -I.}$$

After adjoining a scalar phase, however,

$$\boxed{(iJ)^2 = +I,}$$

so a complexified or projective packet model can absorb the sign. The resulting picture is:

$$\boxed{E_2 = 15_+ \oplus 15_-, \qquad E_1 + E_3 = 24 \oplus 24, \qquad E_4 = 81 \text{ protected Hodge sector.}}$$

Thus E_2 carries a duad–phase split, $E_1 + E_3$ carries a complexified G_2-boundary packet, and E_4 remains isolated as the physical Hodge sector.

H.3 Sentinel admission, two-adic chiral lift, native runtime, and eight-bin photonic compilation

The odd-q rank theorem is now represented by an exact symbolic dependency certificate. The nontrivial additive-character block is $Y \mapsto Y + Y^T$ on Mat_q, the transformed incidence image is Sym_q, and the line Gram form retains only diagonal coordinates in characteristic two. Together with the affine-plane fixed block, this proves

$$\operatorname{rank}_2 M_q = \frac{q(q+1)^2 + 2}{2}, \quad \operatorname{rank}_2 A_P = \frac{q(q^2+1)}{2} + 1, \quad \operatorname{rank}_2 A_L = q^2 + 1,$$

and hence

$$D_q \sim J_4^2 \oplus J_3^{(q^3+2q^2+q-4)/2} \oplus J_1^{q(q-1)^2/2}$$

with no J_2 blocks for every odd prime power q.

The Pass–159 sentinel code is now used directly by packet admission. For an authenticated envelope, the context-readout syndrome is applied to the payload displacement from the provenance-derived expected packet. Since the code is $[40, 15, 8]_2$, every nonzero displacement of support below eight is detected. A minimum dark-code displacement is still rejected by transition provenance, and a raw point/line retag is rejected as type confusion.

The chiral two-primary discriminant form admits the exact refinement

$$A_2(L_{-4}) \cong (\mathbb{Z}/2)^{14} \oplus \mathbb{Z}/8.$$

If h generates the depth-three factor, then

$$q(h) = \frac{11}{8} \pmod{2\mathbb{Z}}.$$

The natural map $L_{-4}/2L_{-4} \to A_2(L_{-4})[2]$ sends the unique fixed line to $\langle 4h \rangle$, while

$$h^{\perp} \cong U_{14}^{-}.$$

The Brown invariants are 7 and 4, summing to the previously computed 2-adic phase 3 modulo eight.

The native runtime is the regular action of

$$\mathrm{PSp}(4,3):2 \cong W(E_6)$$

on 51840 states. The index-two subgroup has two regular chirality orbits of size 25920, and the outer similitude swaps them. A point stabilizer has order 648, center 3, and quotient order 216; a noncollinear-pair stabilizer has order 48. Thus

$$51840 = 2 \cdot 40 \cdot 3 \cdot 216 = 2 \cdot 540 \cdot 48,$$

with unique coordinates given by chirality, point, central phase, and Clifford quotient class.

Finally, the integral 8×80 Levi-to-E_8 intertwiner factors through only eight time bins,

$$0, 4, 7, 13, 40, 41, 44, 45,$$

and the active 8×8 matrix is unimodular. An exact 16-mode unitary dilation has a verified 120-rotation Reck netlist; an equivalent Clements mesh uses 120 Mach–Zehnder elements. The sparse readout realization uses the support-optimal 86 tunable couplers and 31 π-phase elements. Under the stated loss and detector model, optical loss becomes retry/no-click and all modeled dark-count corruptions are rejected by the sentinel/provenance layer.

Witnesses: `analysis/w33_levi_next5_v2.py`, `data/PART_2026_07_10_LEVI_NEXT5_V2_results.json`, and `tests/test_w33_levi_next5_v2.py`.

H.4 Discriminant automorphisms, native E_6 objects, and the optical admission loop

The odd-prime-power rank assembly has been translated into a placeholder-free Lean 4 module, together with a dependency and polynomial-identity certificate. It formalizes the block-rank sums, parity of the odd-q multiplicity numerators, and the Jordan rank ladder. The source is pinned to the current mathlib toolchain; kernel compilation is not claimed in the present execution environment because Lean and Lake are unavailable there.

For the integral chiral lattice L_{-4}, the complete two-primary discriminant group is

$$A_2(L_{-4}) \cong (\mathbb{Z}/2)^{14} \oplus \mathbb{Z}/8.$$

Transporting the integral permutation action through exact Smith coordinates gives a faithful quadratic action of orders 25920 and 51840. If h is the canonical order-eight generator, then $q(h) = 11/8$ and the fixed line is $\langle 4h \rangle$. In the canonical elementary generators e_i, the outer involution acts by

$$\tau(h) = 5h + e_0 + e_1 + e_2 + e_5 + e_8 + e_{10} + e_{12} + e_{13}.$$

Thus the outer action is genuinely mixed, not merely fixed or inverted; its square is the identity and the $\mathrm{PSp}(4,3)$-orbit of h has size 2880.

The same native generators induce the six-dimensional minus-type homology module. Its 27 singular vectors are explicitly identified with the cubic- surface lines E_i, L_{ij}, Q_i. Their graph reconstructs 45 tritangent planes and 72 oriented double-sixes, hence the 72 roots of E_6. The regular 51840-state runtime maps equivariantly with fibers

$$51840/27 = 1920, \quad 51840/45 = 1152, \quad 51840/72 = 720, \quad 51840/540 = 96.$$

The last fiber splits by chirality into two 48-element middleware fibers.

The normalized eight-bin integral E_8 map is compiled into a 16-mode Halmos dilation with 120 two-mode rotations. Exact reconstruction residuals are below 7×10^{-15}. A tolerance model including fabrication offsets, phase drift, finite extinction, correlated loss, and 50 ps detector jitter selects calibration gain 1.025 and yields

$$\overline{F} = 0.999297, \qquad F_{5\%} = 0.998749,$$

compared with uncalibrated $\overline{F} = 0.989727$.

Finally, balanced optical click streams carry the eight-bit point-homology syndrome. Maximum-likelihood optical decoding reconstructs a 40-bit cycle, after which authentication, the $[40, 15, 8]_2$ sentinel, immutable provenance, homology validation, retry control, and the native $W(E_6)$ coordinate map run in sequence. All 256 noiseless words pass. In 2500 noisy frames, 2495 are admitted and 5 retried, with no wrong admission. The attack suite rejects all 448 weight-1 through weight-7 errors at the sentinel, all 45 weight- 8 dark words at provenance, one raw retag at type checking, and one invalid tag at authentication.

Witnesses: `formal/W33/OddQRank.lean`, `analysis/w33_levi_next5_v3_discriminant.py`, `analysis/w33_levi_next5_v3_e6.py`, `analysis/w33_levi_next5_v3_tolerance.py`, `analysis/w33_levi_next5_v3_emulator.py`, and `data/PART_2026_07_10_LEVI_NEXT5_V3_results.json`.

I Errata

In plain language. *Corrections to this document, kept on the page rather than edited away. The list is short, and the front matter explains why that is not reassuring: a design attracts fewer falsification attempts than a theorem, so a thin errata section may reflect thin scrutiny rather than few mistakes.*

Inherited from the machine blueprint. The following were established in the companion blueprint and correct claims that this document's architecture rests on.

- **The instruction graph is not a Cayley graph.** It is a *Schreier* graph on a coset space, so its degrees are not constant and regular-graph optimality notions do not apply to it. Three separate spectral claims were withdrawn for assuming otherwise.
- $78 = \dim E_6$ **is a parameter fact.** All 28 non-isomorphic $\mathrm{SRG}(40, 12, 2, 4)$ graphs give the same 78 zeta poles, so the coincidence says nothing about *which* graph this architecture uses.
- **The load port does not act on the projective carriers.** A translation is not well defined on projective points, so no argument about which carrier the machine addresses may be built from where translations live.
- **"Compute is exactly zero" holds of every design here**, not only of a reversible closure — every opcode is a bijection. What closing under inverses buys is time-symmetry of the random-instruction walk, not a reduction in dissipation.